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(19) **United States**(12) **Patent Application Publication**
SHIKU et al.(10) **Pub. No.: US 2025/0282889 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **BISPECIFIC ANTIBODY****C07K 16/28** (2006.01)(71) Applicants: **MIE UNIVERSITY**, Tsu-shi (JP);
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HASHIMOTO, Tokyo (JP); **Makiko**
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WATANABE, Tokyo (JP); **Shinji**
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C07K 2317/522 (2013.01); **C07K 2317/524**
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C07K 2317/55 (2013.01); **C07K 2317/56**
(2013.01); **C07K 2317/565** (2013.01); **C07K**
2317/622 (2013.01); **C07K 2317/73** (2013.01);
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(57)

ABSTRACT(22) Filed: **May 23, 2025****Related U.S. Application Data**(62) Division of application No. 17/799,648, filed on Aug.
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An antibody that can be used as an anti-tumor agent and an anti-tumor agent comprises a molecule containing such an antibody. Such an antibody or a binding fragment thereof comprises: a heavy chain CDRH1 consisting of SEQ ID NO: 54, a heavy chain CDRH2 consisting of SEQ ID NO: 55, a heavy chain CDRH3 consisting of SEQ ID NO: 56, a light chain CDRL1 consisting of SEQ ID NO: 57 or a light chain CDRL1 consisting of an amino acid sequence derived from SEQ ID NO: 57 in which amino acid 7 is W and/or amino acid 8 is K, a light chain CDRL2 consisting of SEQ ID NO: 58, and a light chain CDRL3 consisting of SEQ ID NO: 59 or a light chain CDRL3 consisting of an amino acid sequence derived from SEQ ID NO: 59 in which amino acid 2 is A or S.

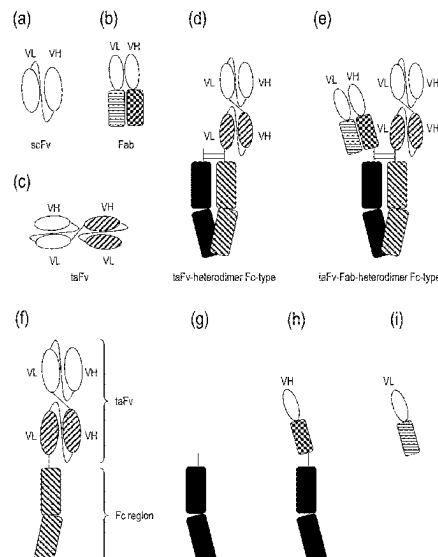
Specification includes a Sequence Listing.

Fig. 1

scFv	Substance added to T2											
	Solvent	Peptide										
	DMSO	NY-ESO	1F	2M	3A	4A	5A	6L	7F	8A	9A	gp100
NYA-0001	<u>1.04</u>	5.75	<u>0.60</u>	7.72	7.88	<u>0.83</u>	<u>0.78</u>	7.97	<u>0.97</u>	6.67	6.56	<u>0.43</u>
NYA-1143	<u>0.98</u>	12.90	<u>1.77</u>	11.89	10.20	<u>5.91</u>	<u>3.15</u>	12.92	<u>4.26</u>	8.74	15.12	<u>0.41</u>
NYA-1154	<u>1.02</u>	9.98	<u>3.98</u>	9.49	8.00	6.63	<u>1.12</u>	9.36	6.08	11.95	11.16	<u>0.46</u>
NYA-1163	<u>0.97</u>	3.67	<u>0.53</u>	1.85	<u>1.70</u>	<u>0.57</u>	<u>0.62</u>	<u>1.74</u>	<u>1.11</u>	6.00	4.01	<u>0.44</u>
NYA-2023	<u>0.94</u>	8.85	<u>0.72</u>	6.74	4.92	<u>2.48</u>	<u>1.21</u>	7.51	4.72	5.40	10.79	<u>0.40</u>
NYA-2027	<u>1.09</u>	5.75	<u>0.57</u>	7.84	9.91	<u>0.99</u>	<u>0.75</u>	7.85	5.41	9.47	6.90	<u>0.41</u>
NYA-2035	<u>1.03</u>	9.94	<u>0.60</u>	8.80	5.81	<u>2.16</u>	<u>2.27</u>	8.95	5.83	6.41	11.46	<u>0.38</u>
NYA-2044	<u>0.91</u>	12.93	<u>0.68</u>	13.53	10.90	<u>3.21</u>	<u>2.20</u>	13.39	<u>3.62</u>	10.04	15.61	<u>0.39</u>
NYA-2045	<u>0.81</u>	9.03	<u>0.53</u>	8.8	7.46	<u>1.17</u>	<u>0.94</u>	8.8	<u>1.99</u>	7.35	9.98	<u>0.39</u>
NYA-2047	<u>0.89</u>	7.99	<u>0.48</u>	7.34	6.09	<u>0.86</u>	<u>1.03</u>	7.07	<u>3.11</u>	6.61	9.54	<u>0.39</u>
NYA-2048	<u>0.90</u>	6.58	<u>0.51</u>	5.36	4.20	<u>0.63</u>	<u>0.67</u>	6.14	<u>1.46</u>	4.11	7.75	<u>0.41</u>
NYA-2060	<u>0.83</u>	8.78	<u>0.52</u>	6.93	5.51	<u>0.73</u>	<u>0.70</u>	7.94	<u>2.51</u>	6.18	9.48	<u>0.41</u>
NYA-2061	<u>0.85</u>	8.62	<u>0.54</u>	7.17	4.73	<u>0.59</u>	<u>0.66</u>	7.13	<u>2.04</u>	5.22	9.16	<u>0.40</u>
NYA-2143	<u>1.04</u>	13.06	<u>2.72</u>	13.19	10.28	7.19	<u>4.30</u>	12.87	<u>5.26</u>	9.69	14.50	<u>0.38</u>
NYC-0003	<u>0.92</u>	3.74	3.28	3.73	4.34	<u>0.76</u>	<u>0.69</u>	4.14	3.99	3.42	4.77	<u>0.39</u>
NYC-0004	<u>0.76</u>	6.02	3.56	5.66	4.75	<u>2.02</u>	<u>0.65</u>	3.78	5.07	3.69	7.99	<u>0.37</u>

Fig. 2A

Gene	Uniprot ID	Start position	Sequence	HLA-A0201 IC50 (nM)
DOLPP1	Q86YN1	104	SQFMWFFSV	2.64
IL20RB	Q6UXL0	14	SLFMWFFYA	2.95
PRKD1	Q15139	763	SLDMWSVG	14.74
PRKD3	O94806	756	SLDMWSVG	14.74
PRKD2	Q9BZL6	731	SLDMWSVG	14.74
CD163	Q86VB7	885	SIPMWVDN	145.66
P2RY8	Q86VZ1	186	SVAMWAVF	212.86
NY-ESO-1	P78358	157	SLLMWITQ	363.83

Fig. 2B

scFv	Substance added to T2							
	Solvent	Peptide	DOLPP1	IL20RB	PRKD2	CD163	P2RY8	gp100
	DMSO	NY-ESO						
NYA-0001	0.92	<u>5.12</u>	0.37	0.42	0.33	0.48	0.73	0.31
NYA-1143	0.94	<u>13.44</u>	0.37	0.42	0.32	0.46	0.69	0.30
NYA-1154	1.01	<u>13.76</u>	0.46	0.51	<u>2.72</u>	0.54	0.65	0.31
NYA-1163	0.88	<u>4.21</u>	0.36	0.41	0.31	0.46	0.69	0.30
NYA-2023	0.88	<u>10.11</u>	0.42	0.42	0.42	0.48	0.70	0.30
NYA-2027	0.96	<u>7.25</u>	0.39	0.42	0.32	0.51	0.70	0.31
NYA-2035	0.97	<u>11.14</u>	0.38	0.41	0.33	0.48	0.69	0.29
NYA-2044	0.95	<u>14.28</u>	0.38	0.43	0.33	0.47	0.72	0.32
NYA-2045	0.96	<u>13.84</u>	0.43	0.47	0.34	0.48	0.58	0.30
NYA-2047	0.98	<u>10.25</u>	0.40	0.43	0.35	0.47	0.72	0.30
NYA-2048	0.90	<u>8.82</u>	0.44	0.48	0.35	0.52	0.69	0.34
NYA-2060	0.85	<u>11.17</u>	0.42	0.45	0.33	0.47	0.62	0.32
NYA-2061	0.82	<u>10.37</u>	0.40	0.46	0.33	0.48	0.57	0.32
NYA-2143	0.90	<u>13.48</u>	0.38	0.42	0.32	0.46	0.69	0.31
NYC-0003	1.00	<u>5.22</u>	0.38	0.43	<u>1.53</u>	<u>1.08</u>	<u>2.01</u>	0.32
NYC-0004	1.01	<u>9.19</u>	0.39	0.43	<u>3.41</u>	0.49	<u>1.98</u>	0.32

Fig. 3

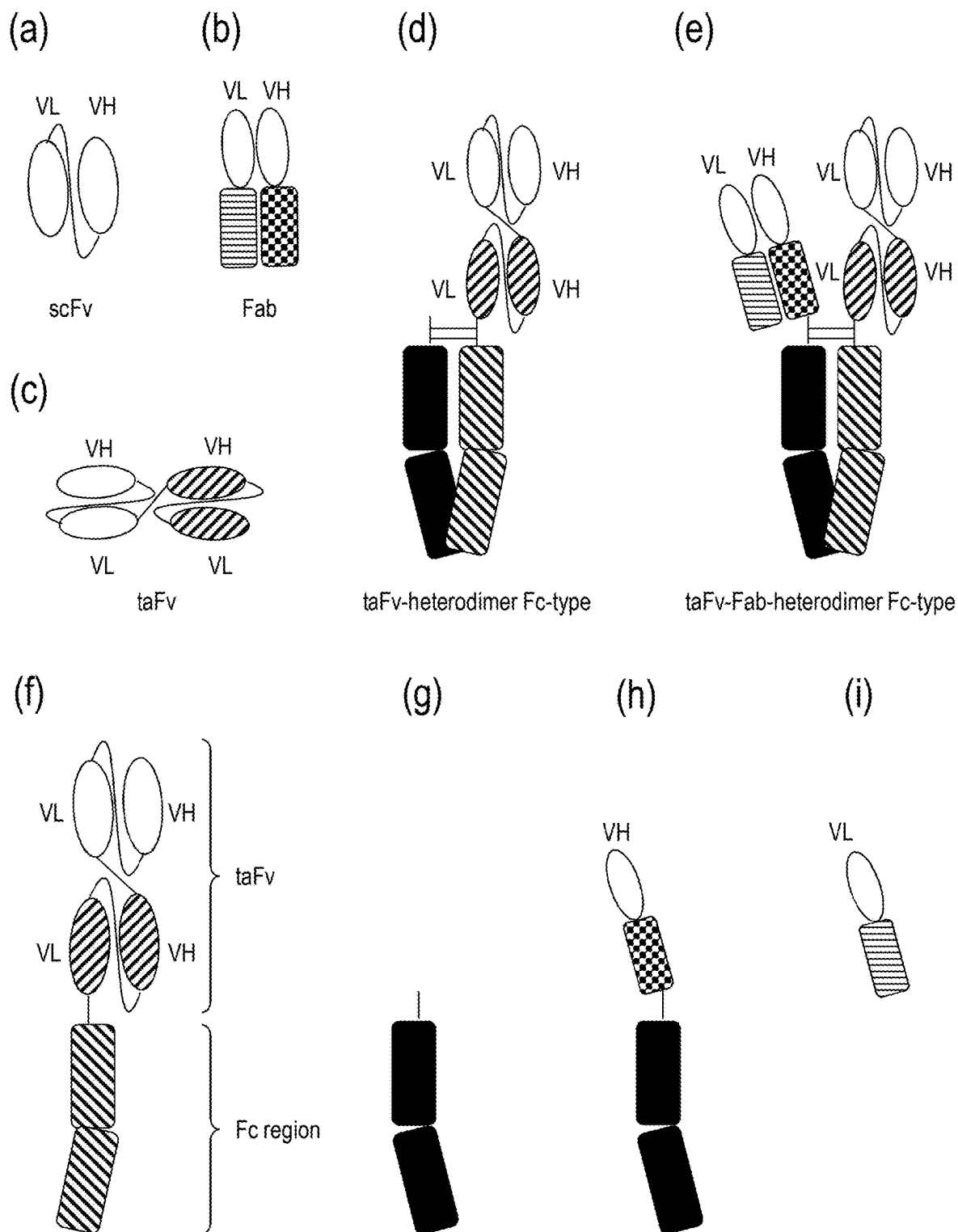


Fig. 4A

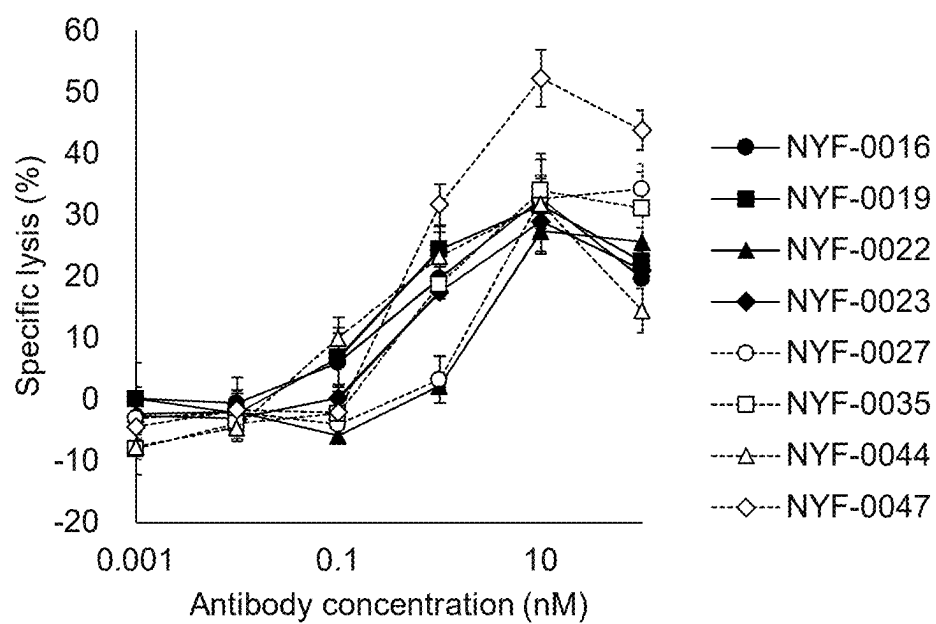


Fig. 4B

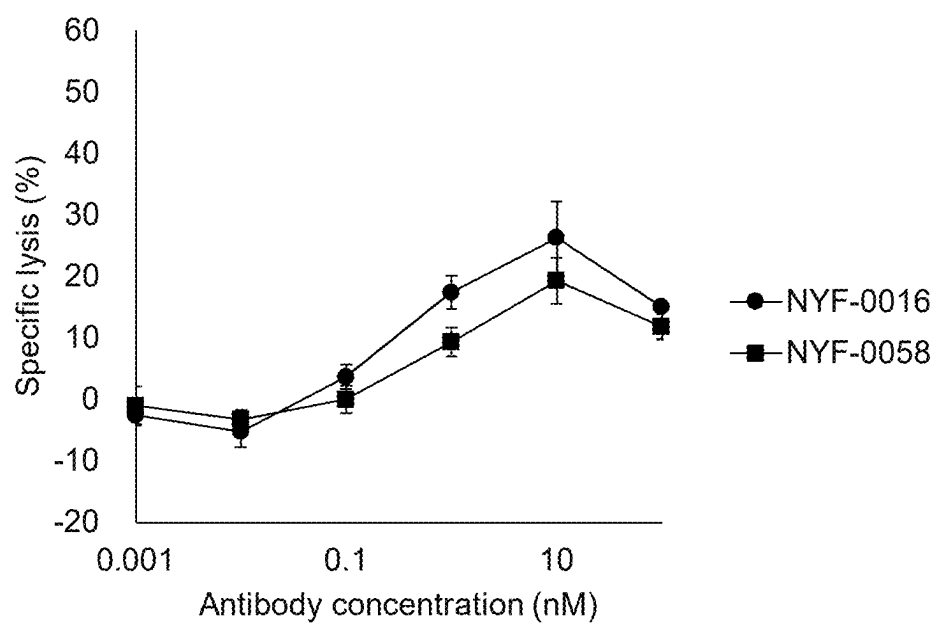


Fig. 4C

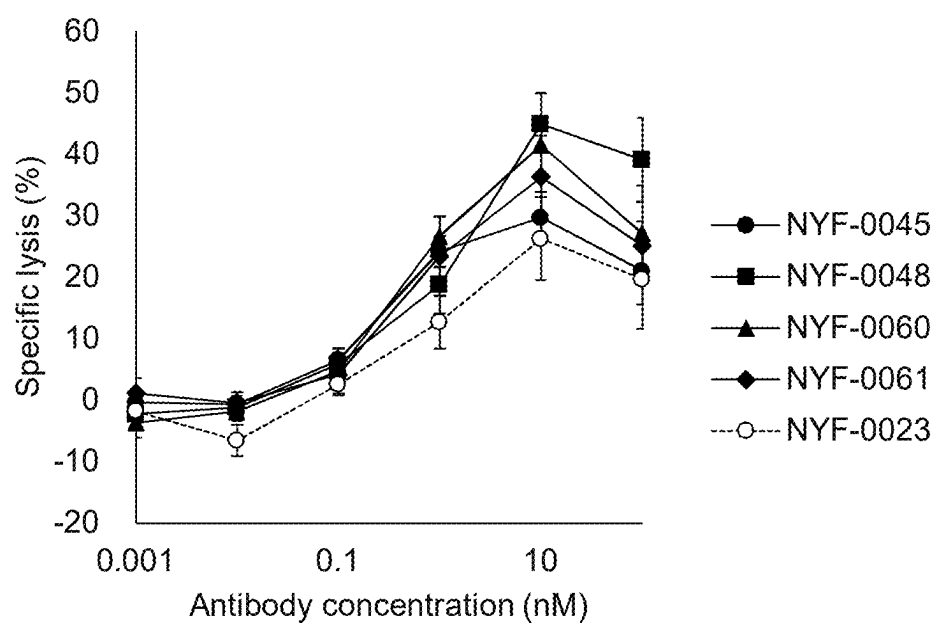


Fig. 4D

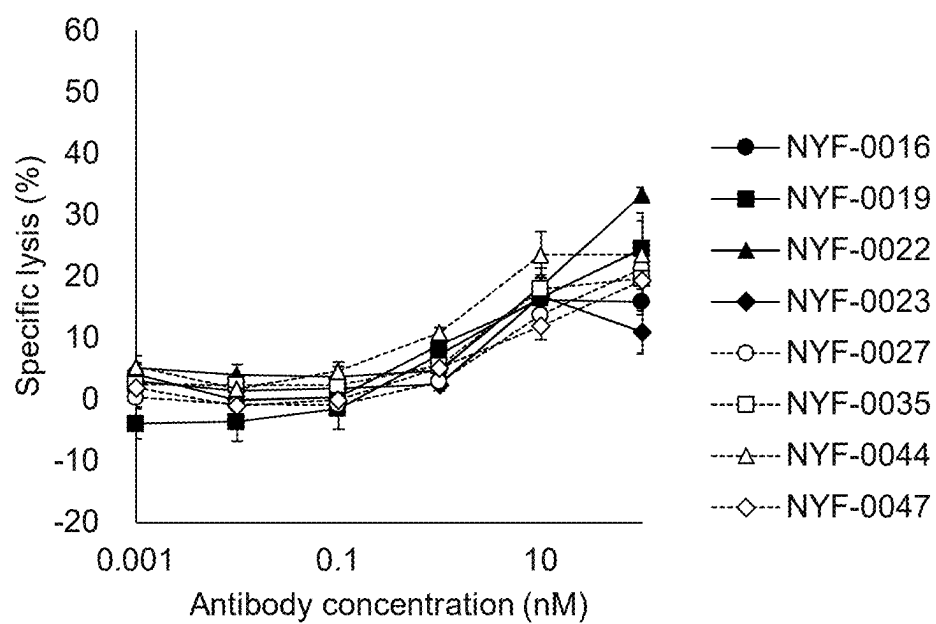


Fig. 4E

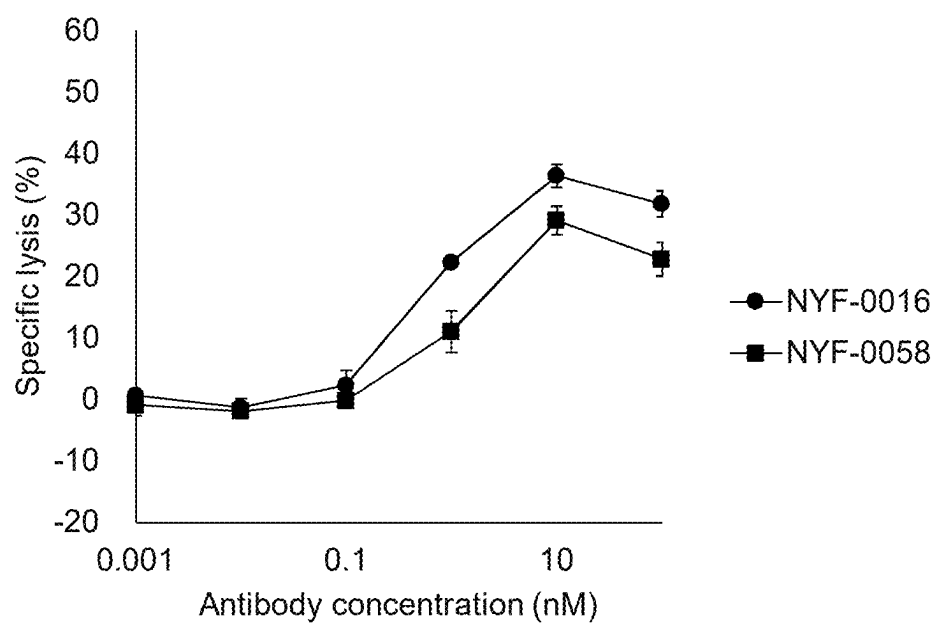


Fig. 4F

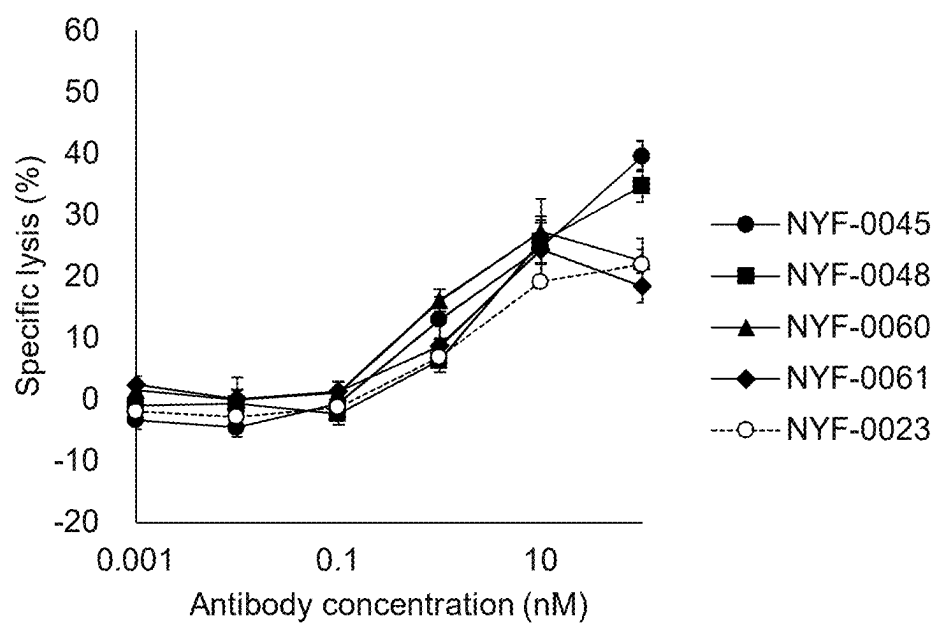


Fig. 4G

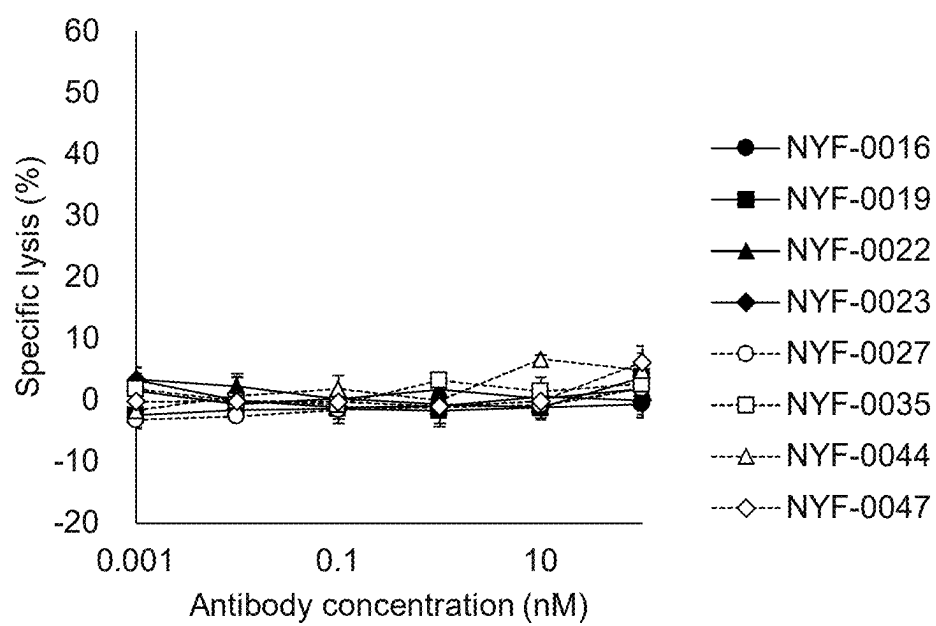


Fig. 4H

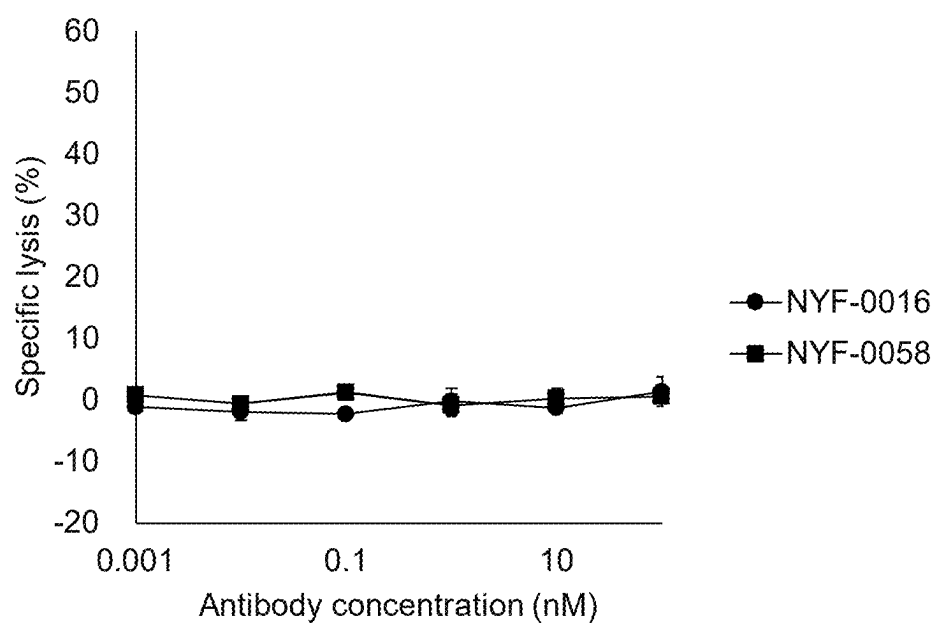


Fig. 4I

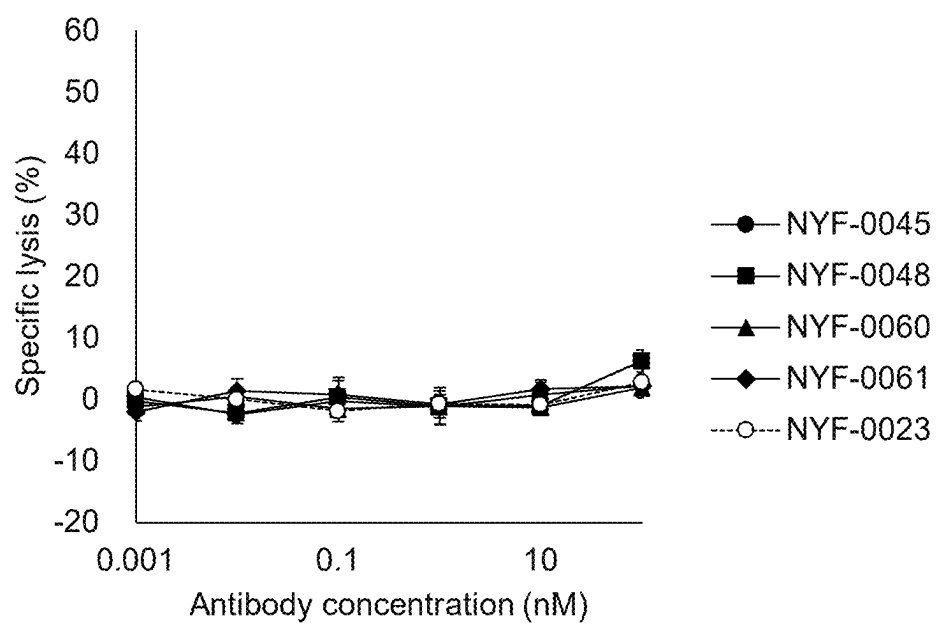


Fig. 4J

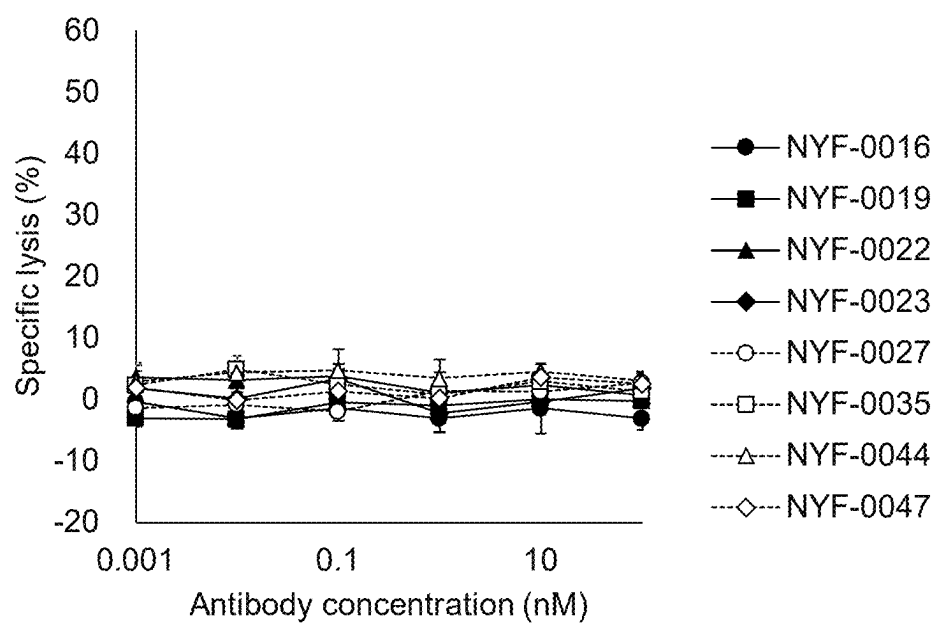


Fig. 4K

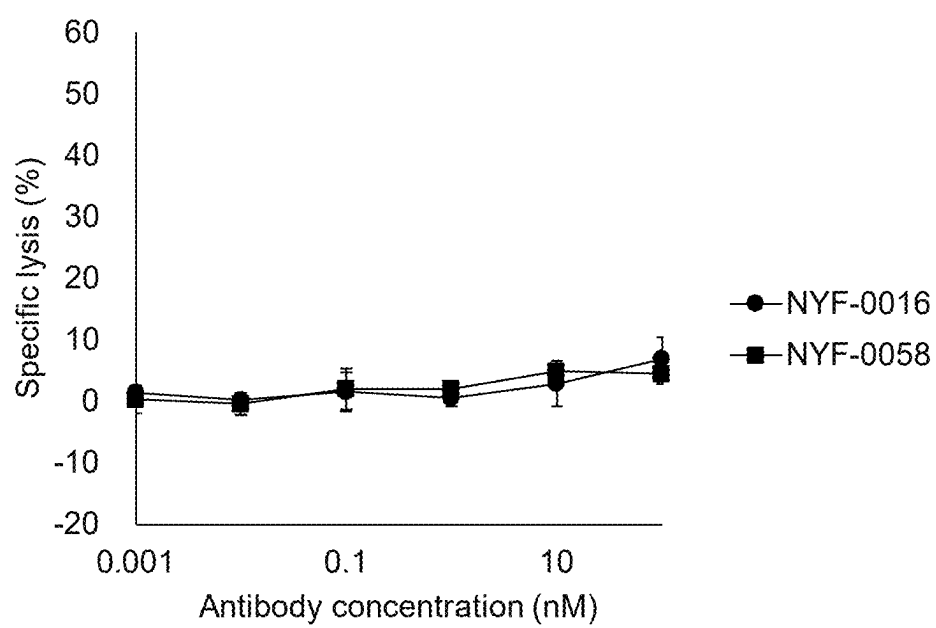


Fig. 4L

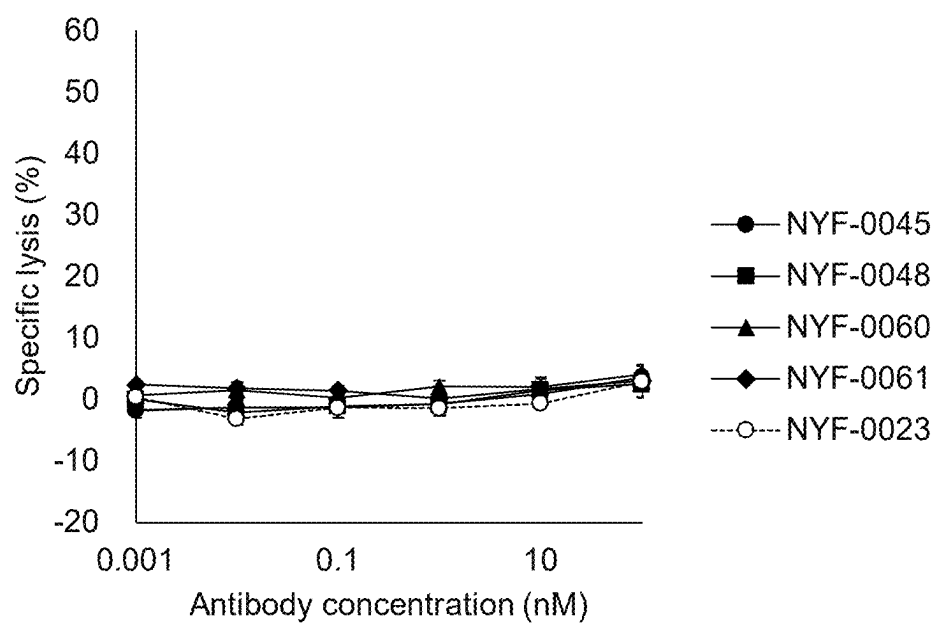


Fig. 5A

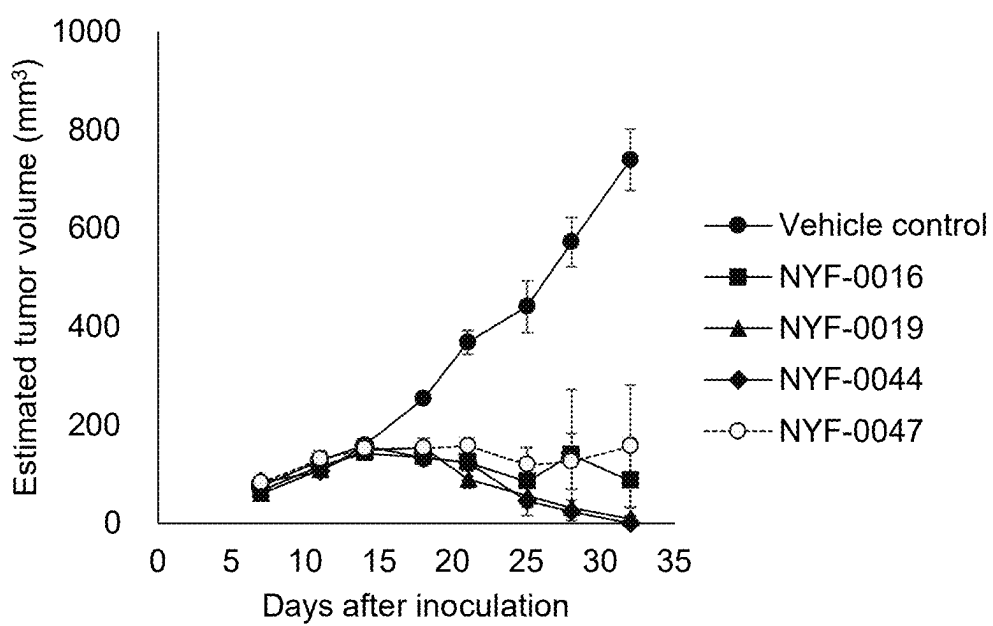


Fig. 5B

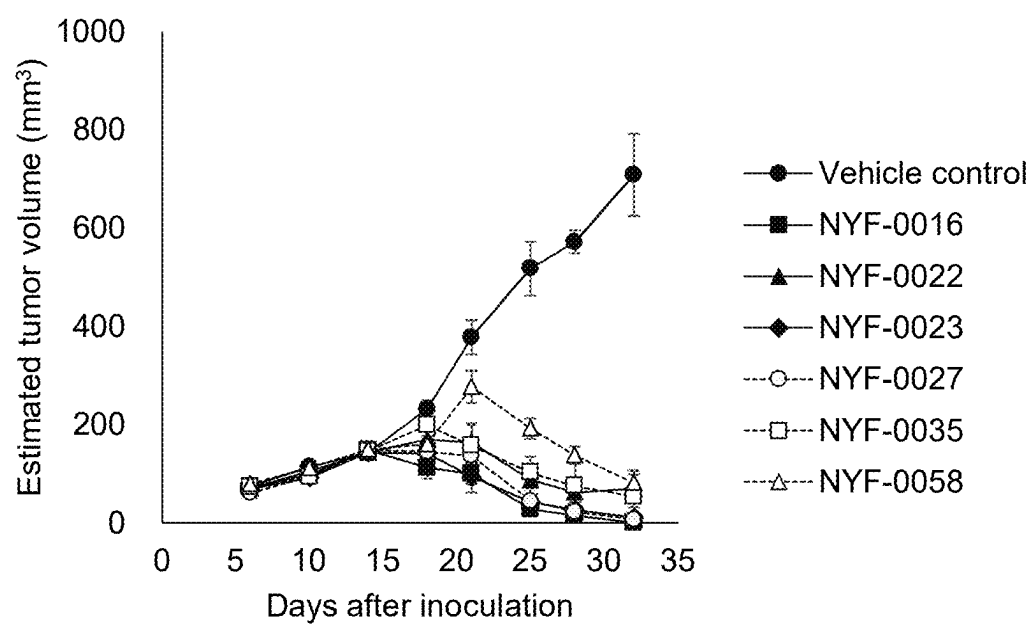


Fig. 5C

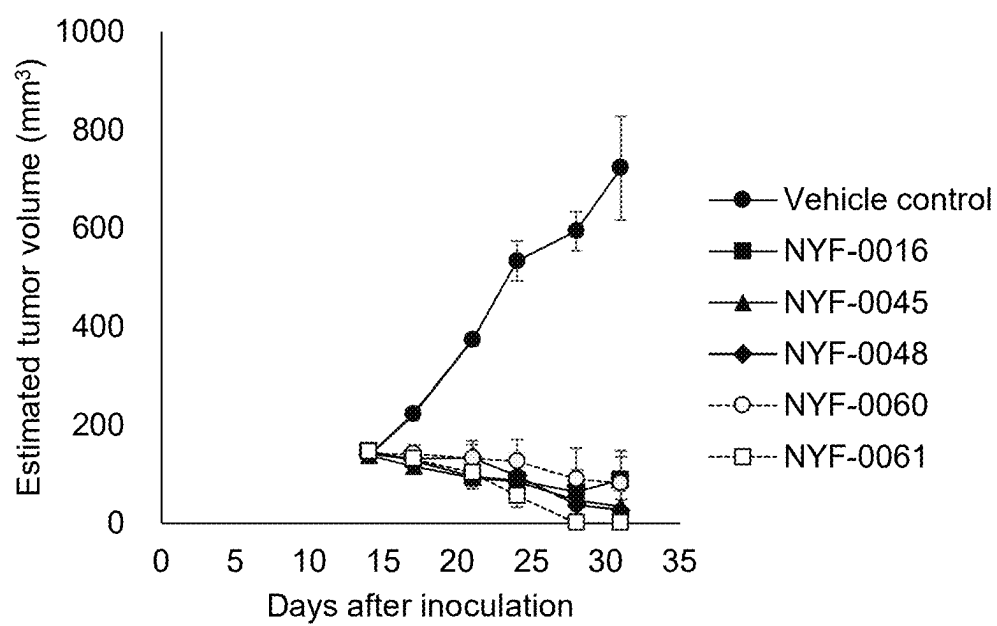


Fig. 6A

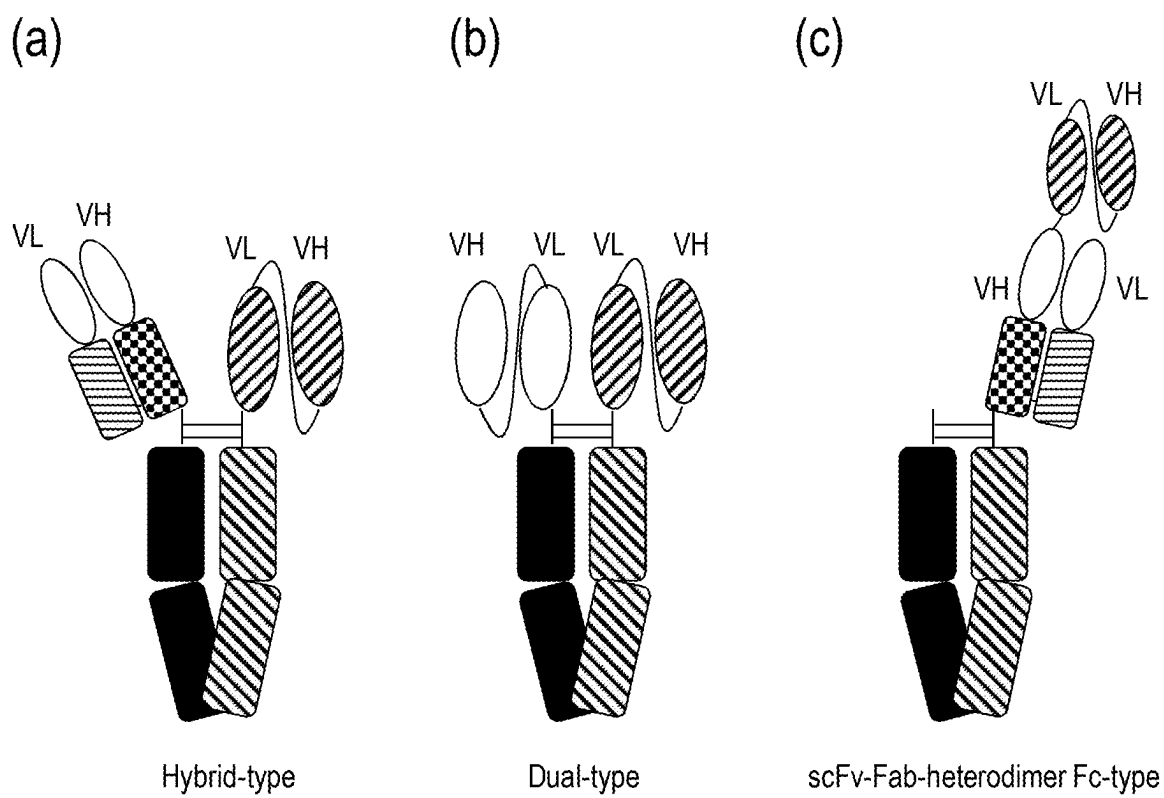
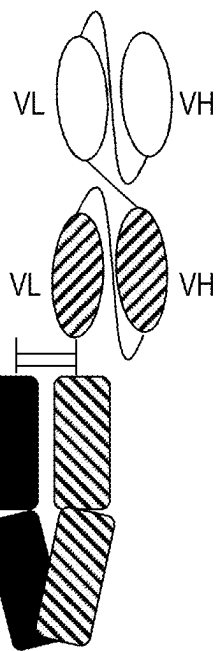


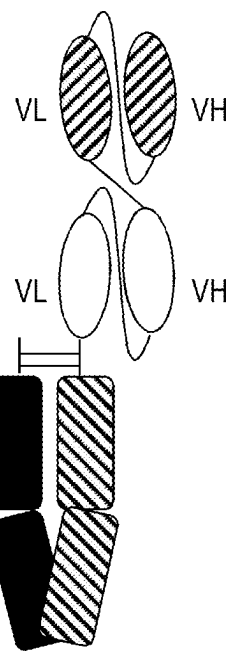
Fig. 6B

(a)



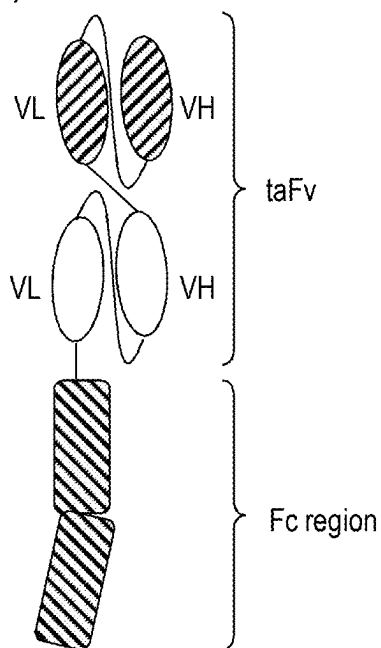
taFv-heterodimer Fc-type
(taFv-Fc-type)

(b)



taFv (inversed)-heterodimer Fc-type
(taFv (inversed)-Fc-type)

(c)



(d)



Fig. 7A

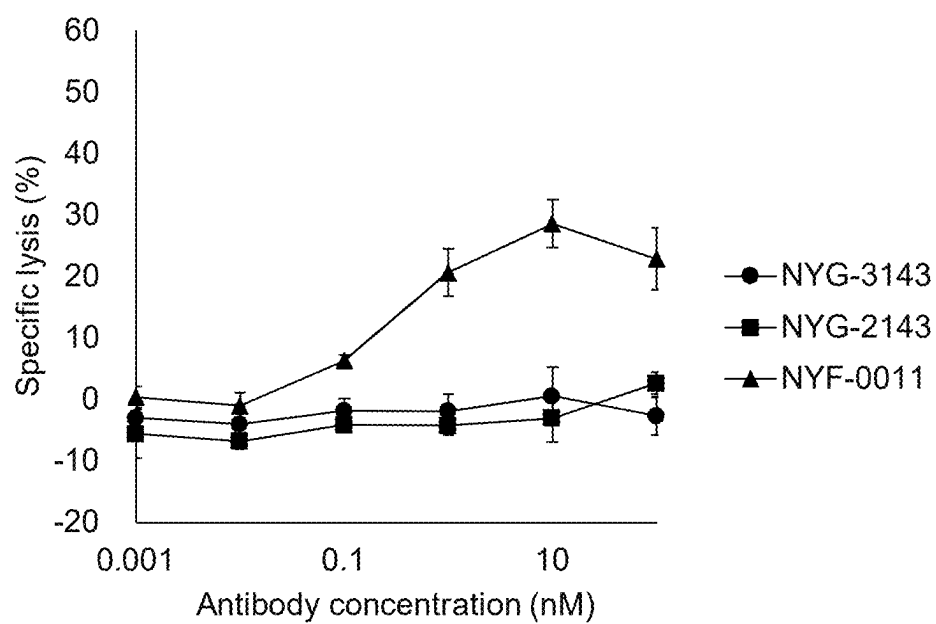


Fig. 7B

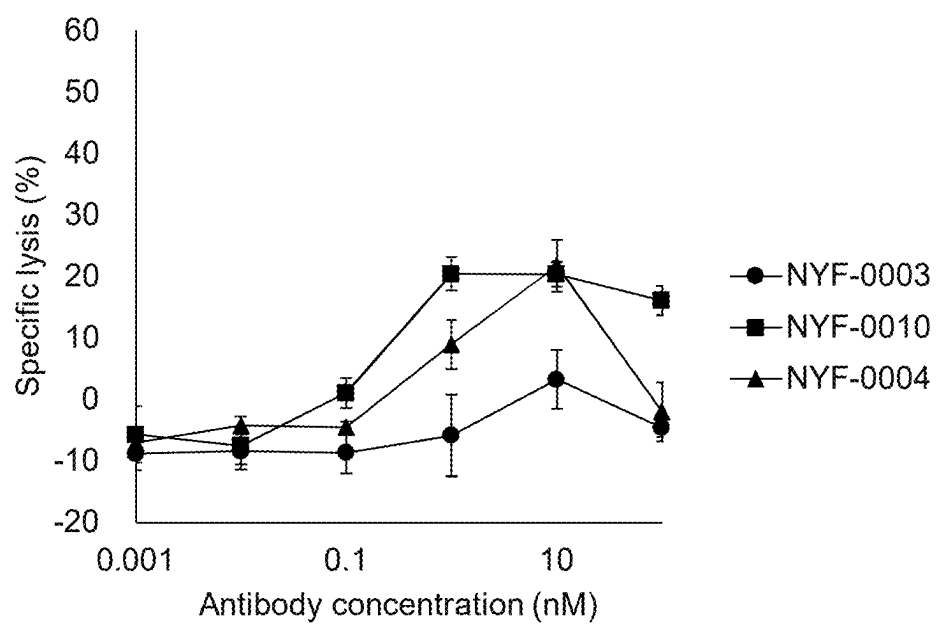


Fig. 8

NY-ESO Peptide
SLLMWITQC
(SEQ ID NO: 1)

Fig. 9

MAGEC-1 Peptide
ILFGISLREV
(SEQ ID NO: 2)

Fig. 10

scFv sequence analysis primer 1
CTCTTCGCTATTACGCCAGCTGGCGA
(SEQ ID NO: 3)

Fig. 11

scFv sequence analysis primer 2
ATAACAATTTACACAGGAAACAGCTATGA
(SEQ ID NO: 4)

Fig. 12

NYA-0001 heavy chain variable region

GAGGTGCAGCTGGTGGAGTCTGGGGGAGGCGTGGTCCAGCCTGGGAGGTCCCTGAC
ACCCTCCTGTTCAGCGTCTGGATTCTCCATCCGTAGTTATGATATACACTGGGTCC
GCCAGGCTCCAGGCAAGGGGCTAGAGTGGGTGGCCACTATATCATATGATGGAAGT
CAGAAGTACTTCGCAGACTCCGTGAAGGGCCGATTTACCATCTTCAGAGACGAATC
GGAGAACATGGTGTATCTGCAAATGAGCGGCCTGAGAGTTGAGGACACGGCTGTTT
ATCACTGTGCGAGAGGGAGTAGTGGTCATTATGAGGCTTTTGATATCTGGGGCCAA
GGGACAATGGTCACCGTCTCTTCA

(SEQ ID NO: 5)

Fig. 13

NYA-0001 heavy chain variable region
EVQLVESGGGVVQPGRSLTPSCSASGFSIRSYDIHWVRQAPGKGLEWVATISYDGS
QKYFADSVKGRFTTIFRDESENMVYLMQSGLRVEDTAVYHCARGSSGHYEAFDIWGQ
GTMVTVSS
(SEQ ID NO: 6)

Fig. 14

NYA-0001 light chain variable region

CAGTCTGTGTTGACGCAGCCGCCCTCAGTGTCTGCGGCCCCAGGACAGAAGGTCAC
CATCTCCTGCTCTGGAAGCAGCTCCAACATTGGGAATAATTATGTATCCTGGTACC
AGCAGCTCCCAGGAACAGCCCCCAAACCTCCTCATTTATGACAATAATAAGCGACCC
TCAGGGATTCTGACCGATTCTCTGGCTCCAAGTCTGGCACGTCAGCCACCCTGGG
CATCACCGGACTCCAGACTGGGGACGAGGCCGATTATTACTGCGGAACATGGGATA
GCAGCCTGAGTGCCCCCTTGGGTGTTTCGGCGGAGGGACCAAGGTCACCGTCCTA
(SEQ ID NO: 7)

Fig. 15

NYA-0001 light chain variable region
QSVLTQPPSVSAAPGQKVTISCSGSSSNIGNNYVSWYQQLPGTAPKLLIYDNNKRP
SGIPDRFSGSKSGTSATLGITGLQTGDEADYYCGTWDSSLSAPWVFGGGTKVTVL
(SEQ ID NO: 8)

Fig. 16

NYA-0060 heavy chain variable region

GAGGTGCAGCTGGTGGAGTCTGGGGGAGGCGTGGTCCAGCCTGGGAGGTCCCTGAC
ACCCTCCTGTTCAGCGTCTGGATTCTCCATCCGTAGCTATGATATACACTGGGTCC
GCCTGGCTCCAGGCAAGGGGCTAGAGTGGGTGGCCACTATATCATATGATGGAAGT
CAGAAGTACTTCGCAGACTCCGTGAAGGGCCGATTTACCATCTTCAGAGACGAATC
GGAGAACATGGTGTATCTGCAAATGAGCGGCCTGAGAGTTGAGGACACGGCTGTTT
ATCACTGTGCGAGAGGGAGTAGTGGTCATTATGAGGCTTTTGATATCTGGGGCCAA
GGGACAATGGTCACCGTCTCTTCA

(SEQ ID NO: 9)

Fig. 17

NYA-0060 heavy chain variable region
EVQLVESGGGVVQPGRSLTPSCSASGFSIRSYDIHWVRLAPGKGLEWVATISYDGS
QKYFADSVKGRFTTIFRDESENMVYLMQSGLRVEDTAVYHCARGSSGHYEAFDIWGQ
GTMVTVSS
(SEQ ID NO: 10)

Fig. 18

NYA-0060 light chain variable region

CAGTCTGTGTTGACGCAGCCGCCCTCAGTGTCTGCGGCCCCAGGACAGAAGGTCAC
CATCTCCTGCTCTGGAAGCAGCTCCAACATTGGGAATAATTATGTATCCTGGTACC
AGCAGCTCCCAGGAACAGCCCCCAAACCTCCTCATTTATGACAATAATAAGCGACCC
TCAGGGATTCTTGACCGATTCTCTGGCTCCAAGTCTGGCACGTCAGCCACCCTGGG
CATCACCGGACTCCAGACTGGGGACGAGGCCGATTATTACTGCGGAACATGGGATA
GCAGCCTGAGTGCCCCCTTGGGTGTTTCGGCGGAGGGACCAAGGTCACCGTCCTA
(SEQ ID NO: 11)

Fig. 19

NYA-0060 light chain variable region
QSVLTQPPSVSAAPGQKVTISCSGSSSNIGNNYVSWYQQLPGTAPKLLIYDNNKRP
SGIPDRFSGSKSGTSATLGITGLQTGDEADYYCGTWDSSLSAPWVFGGGTKVTVL
(SEQ ID NO: 12)

Fig. 20

NYA-0068 heavy chain variable region

GAGGTGCAGCTGGTGGAGTCTGGGGGAGGCGTGGTCCAGCCTGGGAGGTCCCTGAC
ACCCTCCTGTTCAGCGTCTGGATTCTCCATCCGTAGTTATGATATACACTGGGTCC
GCCAGGCTCCAGGCAAGGGGCTAGAGTGGGTGGCCACTATATCATATGATGGAAGT
CAGAAGTACTTCGCAGACTCCGTGAAGGGCCGATTTACCATCTTCAGAGACGAATC
GGAGAACATGGTGTATCTGCAAATGAGCGGCCTGAGAGTTGAGGACACGGCTGTTT
ATCACTGTGCGAGAGGGAGTAGTGGTCATTATGAGGCTTTTGATATCTGGGGCCAA
GGGACAATGGTCACCGTCTCTTCA

(SEQ ID NO: 13)

Fig. 21

NYA-0068 heavy chain variable region
EVQLVESGGGVVQPGRSLTPSCSASGFSIRSYDIHWVRQAPGKGLEWVATISYDGS
QKYFADSVKGRFTIFRDESENMVYLMQSGLRVEDTAVYHCARGSSGHYEAFDIWGQ
GTMVTVSS
(SEQ ID NO: 14)

Fig. 22

NYA-0068 light chain variable region

CAGTCTGTGTTGACGCAGCCGCCCTCAGTGTCTGCGGCCCCAGGACAGAAGGTCAC
CATCTCCTGCTCTGGAAGCAGCTCCAACATTGGGAATTGGTATGTATCCTGGTACC
AGCAGCTCCCAGGAACAGCCCCAAACTCCTCATTTATGACAATAATAAGCGACCC
TCAGGGATTCTTGACCGATTCTCTGGCTCCAAGTCTGGCACGTCAGCCACCCTGGG
CATCACCGGACTCCAGACTGGGGACGAGGCCGATTATTACTGCGGAACATGGGATA
GCAGCCTGAGTGCCCCCTTGGGTGTTTCGGCGGAGGGACCAAGGTCACCGTCCTA
(SEQ ID NO: 15)

Fig. 23

NYA-0068 light chain variable region

QSVLTQPPSVSAAPGQKVTISCSGSSSNIGNWYVSWYQQLPGTAPKLLIYDNNKRP
SGIPDRFSGSKSGTSATLGITGLQTGDEADYYCGTWDSSLSAPWVFGGGTKVTVL
(SEQ ID NO: 16)

Fig. 24

NYA-0082 heavy chain variable region

GAGGTGCAGCTGGTGGAGTCTGGGGGAGGCGTGGTCCAGCCTGGGAGGTCCCTGAC
ACCCTCCTGTTCAGCGTCTGGATTCTCCATCCGTAGTTATGATATACACTGGGTCC
GCCAGGCTCCAGGCAAGGGGCTAGAGTGGGTGGCCACTATATCATATGATGGAAGT
CAGAAGTACTTCGCAGACTCCGTGAAGGGCCGATTTACCATCTTCAGAGACGAATC
GGAGAACATGGTGTATCTGCAAATGAGCGGCCTGAGAGTTGAGGACACGGCTGTTT
ATCACTGTGCGAGAGGGAGTAGTGGTCATTATGAGGCTTTTGATATCTGGGGCCAA
GGGACAATGGTCACCGTCTCTTCA

(SEQ ID NO: 17)

Fig. 25

NYA-0082 heavy chain variable region
EVQLVESGGGVVQPGRSLTPSCSASGFSIRSYDIHWVRQAPGKGLEWVATISYDGS
QKYFADSVKGRFTTIFRDESENMVYLMQSGLRVEDTAVYHCARGSSGHYEAFDIWGQ
GTMVTVSS
(SEQ ID NO: 18)

Fig. 26

NYA-0082 light chain variable region

CAGTCTGTGTTGACGCAGCCGCCCTCAGTGTCTGCGGCCCCAGGACAGAAGGTCAC
CATCTCCTGCTCTGGAAGCAGCTCCAACATTGGGAATAATTATGTATCCTGGTACC
AGCGGCTCCCAGGAACAGCCCCCAAACCTCCTCATTTATGACAATAATAAGCGACCC
TCAGGGATTCTTGACCGATTCTCTGGCTCCAAGTCTGGCACGTCAGCCACCCTGGG
CATCACCGGACTCCAGACTGGGGACGAGGCCGATTATTACTGCGGATCATGGGATA
GCAGCCTGAGTGCCCCCTTGGGTGTTTCGGCGGAGGGACCAAGGTCACCGTCCTA
(SEQ ID NO: 19)

Fig. 27

NYA-0082 light chain variable region

QSVLTQPPSVSAAPGQKVTISCSGSSSNIGNNYVSWYQRLPGTAPKLLIYDNNKRP
SGIPDRFSGSKSGTSATLGITGLQTGDEADYYCGSWDSSLAPWVFGGGTKVTVL
(SEQ ID NO: 20)

Fig. 28

NYA-1163 tag adduct

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
CGGAGAGGTGCAGCTGGTGGAGTCTGGGGGAGGCGTGGTCCAGCCTGGGAGGTCCC
TGACACCCTCCTGTTCAGCGTCTGGATTCTCCATCCGTAGCTATGATATACTGG
GTCCGCCTGGCTCCAGGCAAGGGGCTAGAGTGGGTGGCCACTATATCATATGATGG
AAGTCAGAAGTACTTCGCAGACTCCGTGAAGGGCCGATTTACCATCTTCAGAGACG
AATCGGAGAACATGGTGTATCTGCAAATGAGCGGCCTGAGAGTTGAGGACACGGCT
GTTTATCACTGTGCGAGAGGGAGTAGTGGTCATTATGAGGCTTTTGATATCTGGGG
CCAAGGGACAATGGTCAACGTCTCTTCAGGTGGAGGCGGTTCAGGCGGAGGTGGCA
GCGGCGGTGGCGGGAGTCAGTCTGTGTTGACGCAGCCGCCCTCAGTGTCTGCGGCC
CCAGGACAGAAGGTCAACATCTCCTGCTCTGGAAGCAGCTCCAACATTGGGAATAA
TTATGTATCCTGGTACCAGCAGCTCCCAGGAACAGCCCCCAAACCTCCTCATTTATG
ACAATAATAAGCGACCCTCAGGGATTCTTGACCGATTCTCTGGCTCCAAGTCTGGC
ACGTCAGCCACCCTGGGCATCACCGGACTCCAGACTGGGGACGAGGCCGATTATTA
CTGCGGAACATGGGATAGCAGCCTGAGTGCCCTTGGGTGTTTCGGCGGAGGGACCA
AGGTCACCGTCCTAGGCGCGGCCGCAGGTGCAGGTGGTGATTACAAAGATGATGAC
GATAAAGGTGCAGCGGCGCATCACCATCATCACAC

(SEQ ID NO: 21), signal sequence (1-57), NYA-1163 (61-798), FLAG-His tag (799-876)

Fig. 29

NYA-2023 tag adduct

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
CGGAGAGGTGCAGCTGGTGGAGTCTGGGGGAGGCGTGGTCCAGCCTGGGAGGTCCC
TGACACCCTCCTGTTCAGCGTCTGGATTCTCCATCCGTAGTTATGATATACACTGG
GTCCGCCAGGCTCCAGGCAAGGGGCTAGAGTGGGTGGCCACTATATCATATGATGG
AAGTCAGAAGTACTTCGCAGACTCCGTGAAGGGCCGATTTACCATCTTCAGAGACG
AATCGGAGAACATGGTGTATCTGCAAATGAGCGGCCTGAGAGTTGAGGACACGGCT
GTTTATCACTGTGCGAGAGGGAGTAGTGGTCATTATGAGGCTTTTGATATCTGGGG
CCAAGGGACAATGGTCAACGTCTCTTCAGGTGGAGGCGGTTTCAGGCGGAGGTGGCA
GCGGCGGTGGCGGGAGTCAGTCTGTGTTGACGCAGCCGCCCTCAGTGTCTGCGGCC
CCAGGACAGAAGGTCAACATCTCCTGCTCTGGAAGCAGCTCCAACATTGGGAATTG
GTATGTATCCTGGTACCAGCAGCTCCCAGGAACAGCCCCCAAACCTCCTCATTTATG
ACAATAATAAGCGACCCTCAGGGATTCTTGACCGATTCTCTGGCTCCAAGTCTGGC
ACGTCAGCCACCCTGGGCATCACCGGACTCCAGACTGGGGACGAGGCCGATTATTA
CTGCGGAACATGGGATAGCAGCCTGAGTGCCCTTGGGTGTTTCGGCGGAGGGACCA
AGGTCACCGTCCTAGGCGCGGCCGCAGGTGCAGGTGGTGATTACAAAGATGATGAC
GATAAAGGTGCAGCGGCGCATCACCATCATCACAC

(SEQ ID NO: 22), signal sequence (1-57), NYA-2023 (61-798), FLAG-His tag (799-876)

Fig. 30

NYA-2027 tag adduct

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
CGGAGAGGTGCAGCTGGTGGAGTCTGGGGGAGGCGTGGTCCAGCCTGGGAGGTCCC
TGACACCCTCCTGTTCAGCGTCTGGATTCTCCATCCGTAGTTATGATATACACTGG
GTCCGCCAGGCTCCAGGCAAGGGGCTAGAGTGGGTGGCCACTATATCATATGATGG
AAGTCAGAAGTACTTCGCAGACTCCGTGAAGGGCCGATTTACCATCTTCAGAGACG
AATCGGAGAACATGGTGTATCTGCAAATGAGCGGCCTGAGAGTTGAGGACACGGCT
GTTTATCACTGTGCGAGAGGGAGTAGTGGTCATTATGAGGCTTTTGATATCTGGGG
CCAAGGGACAATGGTCAACGCTCTCTTCAGGTGGAGGCGGTTCAGGCGGAGGTGGCA
GCGGCGGTGGCGGGAGTCAGTCTGTGTTGACGCAGCCGCCCTCAGTGTCTGCGGCC
CCAGGACAGAAGGTCAACATCTCCTGCTCTGGAAGCAGCTCCAACATTGGGAATAA
TTATGTATCCTGGTACCAGCAGCTCCCAGGAACAGCCCCCAAACCTCCTCATTTATG
ACAATAATAAGCGACCCTCAGGGATTCTTGACCGATTCTCTGGCTCCAAGTCTGGC
ACGTCAGCCACCCTGGGCATCACCGGACTCCAGACTGGGGACGAGGCCGATTATTA
CTGCGGATCATGGGATAGCAGCCTGAGTGCCCTTGGGTGTTTCGGCGGAGGGACCA
AGGTCACCGTCCTAGGCGCGGCCGCAGGTGCAGGTGGTGATTACAAAGATGATGAC
GATAAAGGTGCAGCGGCGCATCACCATCATCACAC

(SEQ ID NO: 23), signal sequence (1-57), NYA-2027 (61-798), FLAG-His tag (799-876)

Fig. 31

NYA-1143 tag adduct

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
CGGAGAGGTGCAGCTGGTGGAGTCTGGGGGAGGCGTGGTCCAGCCTGGGAGGTCCC
TGACACCCTCCTGTTCAGCGTCTGGATTCTCCATCCGTAGCTATGATATACTGG
GTCCGCCTGGCTCCAGGCAAGGGGCTAGAGTGGGTGGCCACTATATCATATGATGG
AAGTCAGAAGTACTTCGCAGACTCCGTGAAGGGCCGATTTACCATCTTCAGAGACG
AATCGGAGAACATGGTGTATCTGCAAATGAGCGGCCTGAGAGTTGAGGACACGGCT
GTTTATCACTGTGCGAGAGGGAGTAGTGGTCATTATGAGGCTTTTGATATCTGGGG
CCAAGGGACAATGGTCAACGTCTCTTCAGGTGGAGGCGGTTCAGGCGGAGGTGGCA
GCGGCGGTGGCGGGAGTCAGTCTGTGTTGACGCAGCCGCCCTCAGTGTCTGCGGCC
CCAGGACAGAAGGTCAACATCTCCTGCTCTGGAAGCAGCTCCAACATTGGGAATTG
GTATGTATCCTGGTACCAGCAGCTCCCAGGAACAGCCCCCAAACCTCCTCATTTATG
ACAATAATAAGCGACCCTCAGGGATTCTTGACCGATTCTCTGGCTCCAAGTCTGGC
ACGTCAGCCACCCTGGGCATCACCGGACTCCAGACTGGGGACGAGGCCGATTATTA
CTGCGGAACATGGGATAGCAGCCTGAGTGCCCTTGGGTGTTTCGGCGGAGGGACCA
AGGTCACCGTCCTAGGCGCGGCCGCAGGTGCAGGTGGTGATTACAAAGATGATGAC
GATAAAGGTGCAGCGGCGCATCACCATCATCACAC

(SEQ ID NO: 24), signal sequence (1-57), NYA-1143 (61-
798), FLAG-His tag (799-876)

Fig. 32

NYA-2143 tag adduct

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
CGGAGAGGTGCAGCTGGTGGAGTCTGGGGGAGGCGTGGTCCAGCCTGGGAGGTCCC
TGACACCCTCCTGTTCAGCGTCTGGATTCTCCATCCGTAGCTATGATATACACTGG
GTCCGCCTGGCTCCAGGCAAGGGGCTAGAGTGGGTGGCCACTATATCATATGATGG
AAGTCAGAAGTACTTCGCAGACTCCGTGAAGGGCCGATTTACCATCTTCAGAGACG
AATCGGAGAACATGGTGTATCTGCAAATGAGCGGCCTGAGAGTTGAGGACACGGCT
GTTTATCACTGTGCGAGAGGGAGTAGTGGTCATTATGAGGCTTTTGATATCTGGGG
CCAAGGGACAATGGTCAACGTCTCTTCAGGTGGAGGCGGTTCAGGCGGAGGTGGCA
GCGGCGGTGGCGGGAGTCAGTCTGTGTTGACGCAGCCGCCCTCAGTGTCTGCGGCC
CCAGGACAGAAGGTCAACATCTCCTGCTCTGGAAGCAGCTCCAACATTGGGAATTG
GTATGTATCCTGGTACCAGCGGCTCCCAGGAACAGCCCCCAAACCTCCTCATTTATG
ACAATAATAAGCGACCCTCAGGGATTCTTGACCGATTCTCTGGCTCCAAGTCTGGC
ACGTCAGCCACCCTGGGCATCACCGGACTCCAGACTGGGGACGAGGCCGATTATTA
CTGCGGAACATGGGATAGCAGCCTGAGTGCCCTTGGGTGTTTCGGCGGAGGGACCA
AGGTCACCGTCCTAGGCGCGGCCGCAGGTGCAGGTGGTGATTACAAAGATGATGAC
GATAAAGGTGCAGCGGCGCATCACCATCATCACAC

(SEQ ID NO: 25), signal sequence (1-57), NYA-2143 (61-
798), FLAG-His tag (799-876)

Fig. 33

NYA-1163 tag adduct

MKHLWFFLLLVAAPRWVLSGEVQLVESGGGVVQPGRSLTPSCSASGFSIRSYDIHW
VRLAPGKGLEWVATISYDGSQKYFADSVKGRFTIFRDESENMVYLMQSGLRVEDTA
VYHCARGSSGHYEAFDIWGQGTMTVTVSSGGGGSGGGGSGGGGSQSVLTQPPSVSAA
PGQKVTISCSGSSSNIGNNYVSWYQQLPGTAPKLLIYDNNKRPSGIPDRFSGSKSG
TSATLGITGLQTGDEADYYCGTWDSSLSAPWVFGGGTKVTVLGAAAGAGGDYKDDD
DKGAAAHHHHHH

(SEQ ID NO: 26), signal sequence (1-19), NYA-1163 (21-266), FLAG-His tag (267-292)

Fig. 34

NYA-2023 tag adduct

MKHLWFFLLLVAAPRWVLSGEVQLVESGGGVVQPGRSLTPSCSASGFSIRSYDIHW
VRQAPGKGLEWVATISYDGSQKYFADSVKGRFTIFRDESENMVYLQMSGRLVEDTA
VYHCARGSSGHYEAFDIWGQGTMTVTVSSGGGGSGGGSGGGGSQSVLTQPPSVSAA
PGQKVTISCSGSSSNIGNWYVSWYQQLPGTAPKLLIYDNNKRPSGIPDRFSGSKSG
TSATLGITGLQTGDEADYYCGTWDSSLSAPWVFGGGTKVTVLGAAAGAGGDYKDDD
DKGAAAHHHHHH

(SEQ ID NO: 27), signal sequence (1-19), NYA-2023 (21-266), FLAG-His tag (267-292)

Fig. 35

NYA-2027 tag adduct

MKHLWFFLLLVAAPRWVLSGEVQLVESGGGVVQPGRSLTPSCSASGFSIRSYDIHW
VRQAPGKGLEWVATISYDGSQKYFADSVKGRFTIFRDESENMVYLMQSGLRVEDTA
VYHCARGSSGHYEAFDIWGQGTMTVTVSSGGGGSGGGGSGGGGSQSVLTQPPSVSAA
PGQKVTISCSGSSSNIGNNYVSWYQQLPGTAPKLLIYDNNKRPSGIPDRFSGSKSG
TSATLGITGLQTGDEADYYCGSWDSSLAPWVFGGGTKVTVLGAAAGAGGDYKDDD
DKGAAAHHHHHH

(SEQ ID NO: 28), signal sequence (1-19), NYA-2027 (21-266), FLAG-His tag (267-292)

Fig. 36

NYA-1143 tag adduct

MKHLWFFLLLVAAPRWVLSGEVQLVESGGGVVQPGRSLTPSCSASGFSIRSYDIHW
VRLAPGKGLEWVATISYDGSQKYFADSVKGRFTIFRDESENMVYLMQSGLRVEDTA
VYHCARGSSGHYEAFDIWGQGTMTVTVSSGGGGSGGGSGGGGSQSVLTQPPSVSAA
PGQKVTISCSGSSSNIGNWYVSWYQQLPGTAPKLLIYDNNKRPSGIPDRFSGSKSG
TSATLGITGLQTGDEADYYCGTWDSSLSAPWVFGGGTKVTVLGAAAGAGGDYKDDD
DKGAAAHHHHHH

(SEQ ID NO: 29), signal sequence (1-19), NYA-1143 (21-266), FLAG-His tag (267-292)

Fig. 37

NYA-2143 tag adduct

MKHLWFFLLLVAAPRWVLSGEVQLVESGGGVVQPGRSLTPSCSASGFSIRSYDIHW
VRLAPGKGLEWVATISYDGSQKYFADSVKGRFTIFRDESENMVYLQMSGRLVEDTA
VYHCARGSSGHYEAFDIWGQGTMTVTVSSGGGGSGGGGSGGGGSQSVLTQPPSVSAA
PGQKVTISCSGSSSNIGNWYVSWYQRLPGTAPKLLIYDNNKRPSGIPDRFSGSKSG
TSATLGITGLQTGDEADYYCGTWDSSLSAPWVFGGGTKVTVLGAAAGAGGDYKDDD
DKGAAAHHHHHH

(SEQ ID NO: 30), signal sequence (1-19), NYA-2143 (21-266), FLAG-His tag (267-292)

Fig. 38

NYA-1154 tag adduct

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
CGGAGAGGTGCAGCTGGTGGAGTCTGGGGGAGGCGTGGTCCAGCCTGGGAGGTCCC
TGACACCCTCCTGTTCAGCGTCTGGATTCTCCATCCGTAGTTATGATATGCACTGG
GTCCGCCAGGCTCCAGGCAAGGGGCTAGAGTGGGTGGCCACTATATCATATGATGG
AAGTCAGAAGTACTTCGCAGACTCCGTGAAGGGCCGATTTACCATCTTCAGAGACG
AATCGGAGAACATGGTGTATCTGCAAATGAGCGGCCTGAGAGTTGAGGACACGGCC
GTTTATCACTGTGCGAGAGGGAGTAGTAATCATTATGAGGCTTTTGATATCTGGGG
CCAAGGGACAATGGTCACCGTCTCTTCAGGTGGAGGCGGTTTCAGGCGGAGGTGGCA
GCGGCGGTGGCGGGAGTCAGTCTGTGTTGACGCAGCCGCCCTCAGTGTCTGCGGCC
CCAGGACAGAAGGTCACCATCTCCTGCTCTGGAAGCAGCTCCAACATTGGGAATAA
TTATGTATCCTGGTACCAGCGGCTCCCAGGAACAGCCCCCAAACCTCCTCATTTATG
ACAATAATAAGCGACCCTCAGGGATTCTGACCGATTCTCTGGCTCCAAGTCTGGC
ACGTCAGCCACCCTGGGCATCACCGGACTCCAGACTGGGGACGAGGCCGATTACTA
CTGCGGAGCATGGGATAGCAGCCTGAGTGCCCTTGGGTGTTTCGGCGGAGGGACCA
AGGTCACCGTCCTAGGCGCGGCCGCAGGTGCAGGTGGTGATTACAAAGATGATGAC
GATAAAGGTGCAGCGGCGCATCACCATCATCACCAC

(SEQ ID NO: 31), signal sequence (1-57), NYA-1154 (61-
798), FLAG-His tag (799-876)

Fig. 39

NYA-1154 tag adduct

MKHLWFFLLLVAAPRWVLSGEVQLVESGGGVVQPGRSLTPSCSASGFSIRSYDMHW
VRQAPGKGLEWVATISYDGSQKYFADSVKGRFTIFRDESENMVYLMQSGLRVEDTA
VYHCARGSSNHYEAFDIWGQGTMTVTVSSGGGGSGGGSGGGGSQSVLTQPPSVSAA
PGQKVTISCSGSSSNIGNNYVSWYQRLPGTAPKLLIYDNNKRPSGIPDRFSGSKSG
TSATLGITGLQTGDEADYYCGAWDSSLSAPWVFGGGTKVTVLGAAAGAGGDYKDDD
DKGAAAHHHHHH

(SEQ ID NO: 32), signal sequence (1-19), NYA-1154 (21-
266), FLAG-His tag (267-292)

Fig. 40

HLA-A*0201 (GenBank:ASA47534.1) truncate
MSHSMRYFFTSVSRPGRGEPRIAAGYVDDTQFVRFSDAASQRMPEPRAPWIEQEG
PEYWDGETRKVKAHSQTHRVDLGTLRGYYNQSEAGSHTVQRMYGCDVGSDWRFLRG
YHQYAYDGKDYIALKEDLRSWTAADMAAQTTKHKWEAAHVAEQLRAYLEGTCVEWL
RRYLENGKETLQRTDAPKTHMTHHAVSDHEATLRCWALSFPAEITLTWQRDGEDQ
TQDTELVETRPAGDGTFOKWAAVVPSGQEQRYTCHVQHEGLPKPLTLRWEGLNDI
FEAQKIEWHE
(SEQ ID NO: 33)

Fig. 41

β 2-microglobulin

MIQRTPKIQVYSRHPAENGKSNFLNCYVSGFHPSDIEVDLLKNGERIEKVEHSDLS

FSKDWSFYLLYYTEFTPTEKDEYACRVNHVTLSPKIVKWDRDM

(SEQ ID NO: 34)

Fig. 42

NYA-2035 tag adduct

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
CGGAGAGGTGCAGCTGGTGGAGTCTGGGGGAGGCGTGGTCCAGCCTGGGAGGTCCC
TGACACCCTCCTGTTCAGCGTCTGGATTCTCCATCCGTAGCTATGATATACTGG
GTCCGCCTGGCTCCAGGCAAGGGGCTAGAGTGGGTGGCCACTATATCATATGATGG
AAGTCAGAAGTACTTCGCAGACTCCGTGAAGGGCCGATTTACCATCTTCAGAGACG
AATCGGAGAACATGGTGTATCTGCAAATGAGCGGCCTGAGAGTTGAGGACACGGCT
GTTTATCACTGTGCGAGAGGGAGTAGTGGTCATTATGAGGCTTTTGATATCTGGGG
CCAAGGGACAATGGTCAACGTCTCTTCAGGTGGAGGCGGTTCAGGCGGAGGTGGCA
GCGGCGGTGGCGGGAGTCAGTCTGTGTTGACGCAGCCGCCCTCAGTGTCTGCGGCC
CCAGGACAGAAGGTCAACATCTCCTGCTCTGGAAGCAGCTCCAACATTGGGAATTG
GAAGGTATCCTGGTACCAGCAGCTCCCAGGAACAGCCCCCAAACCTCCTCATTTATG
ACAATAATAAGCGACCCTCAGGGATTCTTGACCGATTCTCTGGCTCCAAGTCTGGC
ACGTCAGCCACCCTGGGCATCACCGGACTCCAGACTGGGGACGAGGCCGATTATTA
CTGCGGAACATGGGATAGCAGCCTGAGTGCCCTTGGGTGTTTCGGCGGAGGGACCA
AGGTCACCGTCCTAGGCGCGGCCGCAGGTGCAGGTGGTGATTACAAAGATGATGAC
GATAAAGGTGCAGCGGCGCATCACCATCATCACAC

(SEQ ID NO: 35), signal sequence (1-57), NYA-2035 (61-
798), FLAG-His tag (799-876)

Fig. 43

NYA-2035 tag adduct

MKHLWFFLLLVAAPRWVLSGEVQLVESGGGVVQPGRSLTPSCSASGFSIRSYDIHW
VRLAPGKGLEWVATISYDGSQKYFADSVKGRFTIFRDESENMVYLQMSGRLVEDTA
VYHCARGSSGHYEAFDIWGQGTMTVTVSSGGGGSGGGSGGGGSQSVLTQPPSVSAA
PGQKVTISCSGSSSNIGNWKVSWYQQLPGTAPKLLIYDNNKRPSGIPDRFSGSKSG
TSATLGITGLQTGDEADYYCGTWDSSLSAPWVFGGGTKVTVLGAAAGAGGDYKDDD
DKGAAAHHHHHH

(SEQ ID NO: 36), signal sequence (1-19), NYA-2035 (21-266), FLAG-His tag (267-292)

Fig. 44

NYA-1143-VH01

EVQLVESGGGLVQPGGSLRLSCAASGFSIRSYDIHWVRLAPGKGLEWVATISYDGS
QKYYADSVKGRFTISRDESKNTLYLQMNSLRAEDTAVYHCARGSSGHYEAFDIWGQ
GTLVTVSS

(SEQ ID NO: 37)

Fig. 45

NYA-1143-VH02

QVQLVESGGGVVQPGRSLRLSCAASGFSIRSYDMHWVRQAPGKGLEWVATISYDGS
QKYYADSVKGRFTISRDN SKNTLYLQMSSLRAEDTAVYHCARGSSGHYEAFDIWGQ
GTLVTVSS

(SEQ ID NO: 38)

Fig. 46

NYA-1143-VH03

EVQLVESGGGLVQPGGSLRLSCAASGFSIRSYDIHWVRQAPGKGLEWVATISYDGS
QKYYADSVKGRFTISRDESKNTLYLQMNSLRAEDTAVYHCARGSSGHYEAFDIWGQ
GTLVTVSS

(SEQ ID NO: 39)

Fig. 47

NYA-1143-VL01

QSVLTQPPSASGTPGQRVTISCSGSSSNIGNWYVSWYQQLPGTAPKLLIYDNNKRP
SGVPDRFSGSKSGTSASLAISGLQSEDEADYYCGTWDSSLSAPWVFGGGTKLTVL
(SEQ ID NO: 40)

Fig. 48

NYA-2044 tag adduct

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
CGGTGAGGTACAGCTGGTGGGAATCCGGTGGAGGCCCTTGTGCAACCGGGAGGCTCCC
TGCGACTCTCTTGCGCCGCATCTGGCTTTTCCATACGATCCTACGATATTCATTGG
GTCCGACTCGCCCCCTGGCAAAGGTCTTGAATGGGTGCCCCTATCTCCTACGACGG
GTCTCAGAAATATTATGCGGATTCAGTGAAGGGGCGGTTTACAATTTACGAGACG
AGTCAAAGAATACACTGTACCTCCAAATGAATTCAGTGAAGAGCCGAGGATACTGCA
GTCTATCATTGTGCAAGAGGGTCCTCAGGCCACTACGAGGCCCTTTGATATATGGGG
CCAAGGCACCTTGGTAACCGTTAGTAGCGGAGGTGGAGGAAGCGGAGGCGGCGGTT
CCGGAGGAGGTGGAAGTCAATCCGTTCTGACTCAGCCACCCTCCGTGAGTGCAGCG
CCTGGACAGAAGGTCACAATATCATGTTCTGGCTCTTCAAGCAACATCGGAACTG
GTACGTGAGCTGGTACCAGCAGCTCCCCGGCACGGCGCCCAAGCTTCTGATATATG
ACAACAACAACCGGCCAGTGGGAATCCCTGACAGATTCAGTGGGTCTAAAAGTGGT
ACAAGCGCTACCCTGGGTATCACCGGATTGCAGACCGGAGATGAGGCGGACTATTA
TTGCGGAACCTGGGACTCCAGCCTGAGCGCTCCCTGGGTTTTCGGCGGAGGTACTA
AGGTTACCGTCTTGGGCGGGCCGCAGGTGCAGGTGGTGATTACAAAGATGATGAC
GATAAAGGTGCAGCGGCGCATCACCATCATCACCAC

(SEQ ID NO: 41), signal sequence (1-57), NYA-2044 (61-798), FLAG-His tag (799-876)

Fig. 49

NYA-2045 tag adduct

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
CGGTGAGGTACAGCTGGTGGGAATCCGGTGGAGGCCCTTGTGCAACCGGGAGGCTCCC
TGCGACTCTCTTGCGCCGCATCTGGCTTTTCCATACGATCCTACGATATTCATTGG
GTCCGACTCGCCCCCTGGCAAAGGTCTTGAATGGGTGCCACTATCTCCTACGACGG
GTCTCAGAAATATTATGCGGATTCAGTGAAGGGGCGGTTTACAATTTACGAGACG
AGTCAAAGAATACACTGTACCTCCAAATGAATTCAGTGAAGAGCCGAGGATACTGCA
GTCTATCATTGTGCAAGAGGGTCTCAGGCCACTACGAGGCCCTTTGATATATGGGG
CCAAGGCACCTTGGTAACCGTTAGTAGCGGAGGTGGAGGAAGCGGAGGCGGCGGTT
CCGGAGGAGGTGGAAGTCAATCCGTTCTGACTCAGCCACCCTCCGCCTCTGGCACC
CCGGGCCAACGGGTCACAATATCATGTTCTGGCTCTTCAAGCAACATCGGAACTG
GTACGTGAGCTGGTACCAGCAGCTCCCCGGCACGGCGCCCAAGCTTCTGATATATG
ACAACAACAAACGGCCCAGTGGAGTTCCTGACAGATTCAGTGGGTCTAAAAGTGGT
ACAAGCGCTAGCCTGGCCATAAGTGGTCTGCAGAGTGAAGATGAGGCGGACTATTA
TTGCGGAACCTGGGACTCCAGCCTGAGCGCTCCCTGGGTTTTCGGCGGAGGTACTA
AGCTGACCGTCTTGGGCGGGCCGCAGGTGCAGGTGGTGATTACAAAGATGATGAC
GATAAAGGTGCAGCGGCGCATCACCATCATCACCAC

(SEQ ID NO: 42), signal sequence (1-57), NYA-2045 (61-798), FLAG-His tag (799-876)

Fig. 50

NYA-2047 tag adduct

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
CGGTCAAGTACAGCTGGTGGGAATCCGGTGGAGGCGTGGTCCAGCCGGGACGCAGCT
TGAGACTGTCTGCGCTGCATCTGGCTTTTCCATACGATCCTACGATATGCACTGG
GTTTCGCCAAGCCCCTGGCAAAGGTCTTGAATGGGTGGCCACTATCTCCTACGACGG
GTCTCAGAAATATTACGCGGATTCAGTGAAGGGGCGGTTCACAATTTACGGGACA
ATTCAAAGAATACCTTGTATCTCCAGATGTCTTCACTGAGAGCCGAGGATACTGCA
GTCTATCATTGTGCAAGAGGGTCCTCAGGCCACTACGAGGCCTTTGATATATGGGG
CCAAGGCACCTTGGTAACCGTTAGTAGCGGAGGTGGAGGAAGCGGAGGCGGCGGT
CCGGAGGAGGTGGAAGTCAATCCGTTCTGACTCAGCCACCCTCCGTGAGTGCAGCG
CCTGGACAGAAGGTCACAATATCATGTTCTGGCTCTTCAAGCAACATCGGAAACTG
GTACGTGAGCTGGTACCAGCAGCTCCCCGGCACGGCGCCCAAGCTTCTGATATATG
ACAACAACAAACGGCCCAGTGGGAATCCCTGACAGATTCAGTGGGTCTAAAAGTGGT
ACAAGCGCTACCCTGGGTATCACCGGATTGCAGACCGGAGATGAGGCGGACTATTA
TTGCGGAACCTGGGACTCCAGCCTGAGCGCTCCCTGGGTTTTCGGCGGAGGTACTA
AGGTTACCGTCTTGGGCGGGCCGCAGGTGCAGGTGGTGATTACAAAGATGATGAC
GATAAAGGTGCAGCGGCGCATCACCATCATCACCAC

(SEQ ID NO: 43), signal sequence (1-57), NYA-2047 (61-798), FLAG-His tag (799-876)

Fig. 51

NYA-2048 tag adduct

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
CGGACAAGTTCAACTCGTGGAATCTGGTGGAGGGGTGGTACAACCAGGCCGGTCAC
TTAGACTGTCTGCGCTGCGAGTGGATTTTCCATCAGATCTTACGACATGCACTGG
GTTTCGCCAAGCACCCGGAAAGGGTTTGGAAATGGGTGGCTACGATCTCCTACGATGG
ATCCCAGAAGTATTACGCCGACAGCGTCAAGGGCCGATTTACAATATCACGGGACA
ATTCAAAGAATACCTTGTATCTCCAGATGTCTTCACTGAGAGCCGAGGATACTGCA
GTCTATCATTTGTGCAAGAGGGTCCTCAGGCCACTACGAGGCCTTTGATATATGGGG
CCAAGGCACCTTGGTAACCGTTAGTAGCGGAGGTGGAGGAAGCGGAGGCGGCGGTT
CCGGAGGAGGTGGAAGTCAATCCGTTCTGACTCAGCCACCCTCCGCCTCTGGCACC
CCGGGCCAACGGGTCACAATATCATGTTCTGGCTCTTCAAGCAACATCGGAAACTG
GTACGTGAGCTGGTACCAGCAGCTCCCCGGCACGGCGCCCAAGCTTCTGATATATG
ACAACAACAAACGGCCCAGTGGAGTTCTTGACAGATTCAAGTGGGTCTAAAAGTGGT
ACAAGCGCTAGCCTGGCCATAAGTGGTCTGCAGAGTGAAGATGAGGCGGACTATTA
TTGCGGAACCTGGGACTCCAGCCTGAGCGCTCCCTGGGTTTTTCGGCGGAGGTACTA
AGCTGACCGTCTTGGGCGGGCCGCAGGTGCAGGTGGTGATTACAAAGATGATGAC
GATAAAGGTGCAGCGGCGCATCACCATCATCACCAC

(SEQ ID NO: 44), signal sequence (1-57), NYA-2048 (61-798), FLAG-His tag (799-876)

Fig. 52

NYA-2060 tag adduct

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
CGGTGAGGTACAGCTGGTGGGAATCCGGTGGAGGCCCTTGTGCAACCGGGAGGCTCCC
TGCGACTCTCTTGCGCCGCATCTGGCTTTTCCATACGATCCTACGATATTCATTGG
GTCCGACAAGCCCCCTGGCAAAGGTCTTGAATGGGTGGCCACTATCTCCTACGACGG
GTCTCAGAAATATTATGCGGATTCAGTGAAGGGGCGGTTCAACAATTTACGAGACG
AGTCAAAGAATACACTGTACCTCCAAATGAATTCAGTGAAGAGCCGAGGATACTGCA
GTCTATCATTGTGCAAGAGGGTCCTCAGGCCACTACGAGGCCTTTGATATATGGGG
CCAAGGCACCTTGGTAACCGTTAGTAGCGGAGGTGGAGGAAGCGGAGGCGGCGGTT
CCGGAGGAGGTGGAAGTCAATCCGTTCTGACTCAGCCACCCTCCGTGAGTGCAGCG
CCTGGACAGAAGGTCACAATATCATGTTCTGGCTCTTCAAGCAACATCGGAAACTG
GTACGTGAGCTGGTACCAGCAGCTCCCCGGCACGGCGCCCAAGCTTCTGATATATG
ACAACAACAAACGGCCAGTGGGAATCCCTGACAGATTCAGTGGGTCTAAAAGTGGT
ACAAGCGCTACCCTGGGTATCACCGGATTGCAGACCGGAGATGAGGCGGACTATTA
TTGCGGAACCTGGGACTCCAGCCTGAGCGCTCCCTGGGTTTTCGGCGGAGGTACTA
AGGTTACCGTCTTGGGCGGGCCGCAGGTGCAGGTGGTGATTACAAAGATGATGAC
GATAAAGGTGCAGCGGCGCATCACCATCATCACCAC

(SEQ ID NO: 45), signal sequence (1-57), NYA-2060 (61-
798), FLAG-His tag (799-876)

Fig. 53

NYA-2061 tag adduct

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
CGGTGAGGTACAGCTGGTGGGAATCCGGTGGAGGCCCTTGTGCAACCGGGAGGCTCCC
TGCGACTCTCTTGCGCCGCATCTGGCTTTTCCATACGATCCTACGATATTCATTGG
GTCCGACAAGCCCCCTGGCAAAGGTCTTGAATGGGTGCCCCTATCTCCTACGACGG
GTCTCAGAAATATTATGCGGATTCAGTGAAGGGGCGGTTTACAATTTACGAGACG
AGTCAAAGAATACACTGTACCTCCAAATGAATTCAGTGAAGAGCCGAGGATACTGCA
GTCTATCATTGTGCAAGAGGGTCCTCAGGCCACTACGAGGCCCTTTGATATATGGGG
CCAAGGCACCTTGGTAACCGTTAGTAGCGGAGGTGGAGGAAGCGGAGGCGGCGGTT
CCGGAGGAGGTGGAAGTCAATCCGTTCTGACTCAGCCACCCTCCGCCTCTGGCACC
CCGGGCCAACGGGTCACAATATCATGTTCTGGCTCTTCAAGCAACATCGGAACTG
GTACGTGAGCTGGTACCAGCAGCTCCCCGGCACGGCGCCCAAGCTTCTGATATATG
ACAACAACAACGGCCAGTGGAGTTCCTGACAGATTCAGTGGGTCTAAAAGTGGT
ACAAGCGCTAGCCTGGCCATAAGTGGTCTGCAGAGTGAAGATGAGGCGGACTATTA
TTGCGGAACCTGGGACTCCAGCCTGAGCGCTCCCTGGGTTTTCGGCGGAGGTACTA
AGCTGACCGTCTTGGGCGGGCCGCAGGTGCAGGTGGTGATTACAAAGATGATGAC
GATAAAGGTGCAGCGGCGCATCACCATCATCACCAC

(SEQ ID NO: 46), signal sequence (1-57), NYA-2061 (61-798), FLAG-His tag (799-876)

Fig. 54

NYA-2044 tag adduct

MKHLWFFLLLVAAPRWVLSGEVQLVESGGGLVQPGGSLRLSCAASGFSIRSYDIHW
VRLAPGKGLEWVATISYDGSQKYYADSVKGRFTISRDESKNTLYLQMNSLRAEDTA
VYHCARGSSGHYEAFDIWGQGLVTVSSGGGGSGGGSGGGGSQSVLTQPPSVSAA
PGQKVTISCSGSSSNIGNWYVSWYQQLPGTAPKLLIYDNNKRPSGIPDRFSGSKSG
TSATLGITGLQTGDEADYYCGTWDSSLSAPWVFGGGTKVTVLGAAAGAGGDYKDDD
DKGAAAHHHHHH

(SEQ ID NO: 47), signal sequence (1-19), NYA-2044 (21-
266), FLAG-His tag (267-292)

Fig. 55

NYA-2045 tag adduct

MKHLWFFLLLVAAPRWVLSGEVQLVESGGGLVQPGGSLRLSCAASGFSIRSYDIHW
VRLAPGKGLEWVATISYDGSQKYYADSVKGRFTISRDESKNTLYLQMNSLRAEDTA
VYHCARGSSGHYEAFDIWGQGLVTVSSGGGGSGGGGSGGGGSQSVLTQPPSASGT
PGQRTISCSGSSSNIGNWYVSWYQQLPGTAPKLLIYDNNKRPSGVPDRFSGSKSG
TSASLAISGLQSEDEADYYCGTWDSSLSAPWVFGGGTKLTVLGAAAGAGGDYKDDD
DKGAAAHHHHHH

(SEQ ID NO: 48), signal sequence (1-19), NYA-2045 (21-
266), FLAG-His tag (267-292)

Fig. 56

NYA-0082

EVQLVESGGGVVQPGRSLTPSCSASGFSIRSYDIHWVRQAPGKGLEWVATISYDGS
QKYFADSVKGRFTIFRDESENMVYLQMSGRLVEDTAVYHCARGSSGHYEAFDIWGQ
GTMVTVSSGGGGSGGGGSGGGGSQSVLTQPPSVSAAPGQKVTISCSGSSSNIGNNY
VSWYQRLPGTAPKLLIYDNNKRPSGIPDRFSGSKSGTSATLGITGLQTGDEADYYC
GSWDSSLSAPWVFEGGGTKVTVL

(SEQ ID NO: 49)

Fig. 57

NYA-2047 tag adduct

MKHLWFFLLLVAAPRWVLSGQVQLVESGGGVVQPGRSLRLSCAASGFSIRSYDMHW
VRQAPGKGLEWVATISYDGSQKYYADSVKGRFTISRDN SKNTLYLQMSSLRAEDTA
VYHCARGSSGHYEAFDIWGQGLVTVSSGGGGSGGGSGGGGSQSVLTQPPSVSAA
PGQKVTISCSGSSSNIGNWYVSWYQQLPGTAPKLLIYDNNKRPSGIPDRFSGSKSG
TSATLGITGLQTGDEADYYCGTWDSSLSAPWVFGGGTKVTVLGAAAGAGGDYKDDD
DKGAAAHHHHHH

(SEQ ID NO: 50), signal sequence (1-19), NYA-2047 (21-
266), FLAG-His tag (267-292)

Fig. 58

NYA-2048 tag adduct

MKHLWFFLLLVAAPRWVLSGQVQLVESGGGVVQPGRSLRLSCAASGFSIRSYDMHW
VRQAPGKGLEWVATISYDGSQKYYADSVKGRFTISRDN SKNTLYLQMSSLRAEDTA
VYHCARGSSGHYEAFDIWGQGLVTVSSGGGGSGGGSGGGGSQSVLTQPPSASGT
PGQRTVISCSSSNIGNWYVSWYQQLPGTAPKLLIYDNNKRPSGVPDRFSGSKSG
TSASLAISGLQSEDEADYYCGTWDSSLSAPWVFGGGTKLTVLGAAAGAGGDYKDDD
DKGAAAHHHHHH

(SEQ ID NO: 51), signal sequence (1-19), NYA-2048 (21-
266), FLAG-His tag (267-292)

Fig. 59

NYA-2060 tag adduct

MKHLWFFLLLVAAPRWVLSGEVQLVESGGGLVQPGGSLRLSCAASGFSIRSYDIHW
VRQAPGKGLEWVATISYDGSQKYYADSVKGRFTISRDESKNTLYLQMNSLRAEDTA
VYHCARGSSGHYEAFDIWGQGLVTVSSGGGGSGGGGSGGGGSQSVLTQPPSVSAA
PGQKVTISCSGSSSNIGNWYVSWYQQLPGTAPKLLIYDNNKRPSGIPDRFSGSKSG
TSATLGITGLQTGDEADYYCGTWDSSLSAPWVFGGGTKVTVLGAAAGAGGDYKDDD
DKGAAAHHHHHH

(SEQ ID NO: 52), signal sequence (1-19), NYA-2060 (21-
266), FLAG-His tag (267-292)

Fig. 60

NYA-2061 tag adduct

MKHLWFFLLLVAAPRWVLSGEVQLVESGGGLVQPGGSLRLSCAASGFSIRSYDIHW
VRQAPGKGLEWVATISYDGSQKYYADSVKGRFTISRDESKNTLYLQMNSLRAEDTA
VYHCARGSSGHYEAFDIWGQGLVTVSSGGGGSGGGGSGGGGSQSVLTQPPSASGT
PGQRTVISCSSSNIGNWYVSWYQQLPGTAPKLLIYDNNKRPSGVPDRFSGSKSG
TSASLAISGLQSEDEADYYCGTWDSSLSAPWVFGGGTKLTVLGAAAGAGGDYKDDD
DKGAAAHHHHHH

(SEQ ID NO: 53), signal sequence (1-19), NYA-2061 (21-266), FLAG-His tag (267-292)

Fig. 61

NYA-0001-CDRH1	GFSIRSYD
NYA-0001-CDRH2	ISYDGSQK
NYA-0001-CDRH3	ARGSSGHYEAFDI
NYA-0001-CDRL1	SSNIGNNY
NYA-0001-CDRL2	DNN
NYA-0001-CDRL3	GTWDSSLSAPWV

(SEQ ID NOS: 54 to 57 and 59)

Fig. 62

NYA-2023-CDRL1
SSNIGNWY
(SEQ ID NO: 60)

Fig. 63

NYA-2027-CDRL3
GSWDSSLAPWV
(SEQ ID NO: 61)

Fig. 64

NYA-1154-CDRH3	ARGSSNHYEAFDI
NYA-1154-CDRL3	GAWDSSLAPWV
(SEQ ID NOs: 62 and 63)	

Fig. 65

NYA-0035-CDRL1
SSNIGNWK
(SEQ ID NO: 64)

Fig. 66

NYC-0003 tag adduct

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
CGGCGAAGTCCAGCTGTTGGAGTCAGGTGGCGGATTGGTGCAACCTGGCGGGTCAC
TGAGGCTGAGTTGTGCAGCTAGCGGCTTCACATTCTCTACGTATCAGATGAGCTGG
GTGAGACAGGCTCCAGGAAAGGGTCTGGAATGGGTGAGCGGGATTGTGTCTAGCGG
TGGCTCCACTGCCTATGCCGATAGCGTAAAAGGCCGCTTTACGATCTCTCGGGACA
ACTCTAAGAACACACTCTATCTGCAGATGAATTCCCTTAGAGCCGAAGATACCGCC
GTGTACTACTGTGCTGGGGAAGTCTCCCGTATTACGGTATGGACGTTTGGGGCCA
AGGCACCACTGTCACAGTGAGTTCCGGTGGAGGCGGGTCAGGCGGAGGCGGTAGTG
GAGGTGGAGGATCACAAAGCGAGCTGACACAGCCTAGGTCCGTATCCGGAAGTCCA
GGGCAGAGCGTCACCATCAGCTGCACTGGCACCTCTCGAGATGTCGGCGGATACAA
CTACGTGTCTTGGTATCAGCAGCATCCCGGCAAAGCGCCCAAACATCATACACG
ACGTGATTGAGCGGTCCAGTGGGGTTCCCGATCGTTTCAGCGGGTCAAAGTCCGGA
AATACCGCAAGCCTGACCATTTCTGGGCTTCAGGCAGAGGATGAGGCTGACTACTA
CTGCTGGTCCCTTTGCCGGGAGCTATTACGTGTTTGGGACAGGGACTGACGTGACTG
TTCTGGGCGCGGCCGAGGTGCAGGTGGTGATTACAAAGATGATGACGATAAAGGT
GCAGCGGCGCATCACCATCATCACCAC

(SEQ ID NO: 65), signal sequence (1-57), NYC-0003 (61-
789), FLAG-His tag (789-867)

Fig. 67

NYC-0004 tag adduct

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
CGGCGAAGTCCAGCTGTTGGAGTCAGGTGGCGGATTGGTGCAACCTGGCGGGTCAC
TGAGGCTGAGTTGTGCAGCTAGCGGCTTCACATTCTCTACGTATCAGATGAGCTGG
GTGAGACAGGCTCCAGGAAAGGGTCTGGAATGGGTGAGCGGGATTGTGTCTAGCGG
TGGCTCCACTGCCTATGCCGATAGCGTAAAAGGCCGCTTTACGATCTCTCGGGACA
ACTCTAAGAACACACTCTATCTGCAGATGAATTCCCTTAGAGCCGAAGATACCGCC
GTGTACTACTGTGCTGGGGAAGTCTCCCGTATTACGGTATGGACGTTTGGGGCCA
AGGCACCACTGTCACAGTGAGTTCCGGTGGAGGCGGGTCAGGCGGAGGCGGTAGTG
GAGGTGGAGGATCACAAAGCGAGCTGACACAGCCTAGGTCCGTATCCGGAAGTCCA
GGGCAGAGCGTCACCATCAGCTGCACTGGCACCGAGCGAGATGTCGGCGGATACAA
CTACGTGTCTTGGTATCAGCAGCATCCCGGCAAAGCGCCCAAACATCATACACG
ACGTGATTGAGCGGTCCAGTGGGGTTCCCGATCGTTTCAGCGGGTCAAAGTCCGGA
AATACCGCAAGCCTGACCATTTCTGGGCTTCAGGCAGAGGATGAGGCTGACTACTA
CTGCTGGTCCCTTTGCCGGGGGCTATTACGTGTTTGGGACAGGGACTGACGTGACTG
TTCTGGGCGCGGCCGAGGTGCAGGTGGTGATTACAAAGATGATGACGATAAAGGT
GCAGCGGCGCATCACCATCATCACCAC

(SEQ ID NO: 66), signal sequence (1-57), NYC-0004 (61-
789), FLAG-His tag (789-867)

Fig. 68

NYC-0003 tag adduct

MKHLWFFLLLVAAPRWVLSGEVQLLESGLVQPGGSLRLSCAASGFTFSTYQMSW
VRQAPGKGLEWVSGIVSSGGSTAYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTA
VYYCAGELLPYYGMDVWGQGTTVTVSSGGGGSGGGGSGGGGSQSELTQPRSVSGSP
GQSVTISCTGTSRDVGGYNYVSWYQQHPGKAPKLIHDVIERSSGVPDRFSGSKSG
NTASLTISGLQAEDEADYYCWSFAGSYVFGTGTDTVTLGAAAGAGGDYKDDDDKG
AAAHHHHHH

(SEQ ID NO: 67), signal sequence (1-19), NYC-0003 (21-
263), FLAG-His tag (264-289)

Fig. 69

NYC-0004 tag adduct

MKHLWFFLLLVAAPRWVLSGEVQLLESGLVQPGGSLRLSCAASGFTFSTYQMSW
VRQAPGKGLEWVSGIVSSGGSTAYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTA
VYYCAGELLPYYGMDVWGQGT TTVTVSSGGGGSGGGGSGGGGSQSELTQPRSVSGSP
GQSVTISCTGTERDVGGYNYVSWYQQHPGKAPKLIHDVIERSSGVPDRFSGSKSG
NTASLTISGLQAEDEADYYCWSFAGGYVFGTGTDTVTLGAAAGAGGDYKDDDDKG
AAAHHHHHH

(SEQ ID NO: 68), signal sequence (1-19), NYC-0004 (21-
263), FLAG-His tag (264-289)

Fig. 70

NYA-0001 tag adduct

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
CGGAGAGGTGCAGCTGGTGGAGTCTGGGGGAGGCGTGGTCCAGCCTGGGAGGTCCC
TGACACCCTCCTGTTCAGCGTCTGGATTCTCCATCCGTAGTTATGATATACACTGG
GTCCGCCAGGCTCCAGGCAAGGGGCTAGAGTGGGTGGCCACTATATCATATGATGG
AAGTCAGAAGTACTTCGCAGACTCCGTGAAGGGCCGATTTACCATCTTCAGAGACG
AATCGGAGAACATGGTGTATCTGCAAATGAGCGGCCTGAGAGTTGAGGACACGGCT
GTTTATCACTGTGCGAGAGGGAGTAGTGGTCATTATGAGGCTTTTGATATCTGGGG
CCAAGGGACAATGGTCAACGTCTCTTCAGGTGGAGGCGGTTTCAGGCGGAGGTGGCA
GCGGCGGTGGCGGGAGTCAGTCTGTGTTGACGCAGCCGCCCTCAGTGTCTGCGGCC
CCAGGACAGAAGGTCAACATCTCCTGCTCTGGAAGCAGCTCCAACATTGGGAATAA
TTATGTATCCTGGTACCAGCAGCTCCCAGGAACAGCCCCCAAACCTCCTCATTTATG
ACAATAATAAGCGACCCTCAGGGATTCTTGACCGATTCTCTGGCTCCAAGTCTGGC
ACGTCAGCCACCCTGGGCATCACCGGACTCCAGACTGGGGACGAGGCCGATTATTA
CTGCGGAACATGGGATAGCAGCCTGAGTGCCCTTGGGTGTTTCGGCGGAGGGACCA
AGGTCACCGTCCTAGGCGCGGCCGCAGGTGCAGGTGGTGATTACAAAGATGATGAC
GATAAAGGTGCAGCGGCGCATCACCATCATCACAC

(SEQ ID NO: 69), signal sequence (1-57), NYA-0001 (61-
798), FLAG-His tag (799-876)

Fig. 71

NYA-0001 tag adduct

MKHLWFFLLLVAAPRWVLSGEVQLVESGGGVVQPGRSLTPSCSASGFSIRSYDIHW
VRQAPGKGLEWVATISYDGSQKYFADSVKGRFTIFRDESENMVYLMQSGLRVEDTA
VYHCARGSSGHYEAFDIWGQGTMTVTVSSGGGGSGGGSGGGGSQSVLTQPPSVSAA
PGQKVTISCSGSSSNIGNNYVSWYQQLPGTAPKLLIYDNNKRPSGIPDRFSGSKSG
TSATLGITGLQTGDEADYYCGTWDSSLSAPWVFGGGTKVTVLGAAAGAGGDYKDDD
DKGAAAHHHHHH

(SEQ ID NO: 70), signal sequence (1-19), NYA-0001 (21-
266), FLAG-His tag (267-292)

Fig. 72

HC1

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
CGACAAAACCTCACACATGCCCACCCTGCCCAGCACCTGAAGCCGCAGGGGGACCCCT
CAGTCTTCCTCTTCCCCC AAAACCCAAGGACACCCTCATGATCTCCCGGACCCCT
GAGGTCACATGCGTGGTGGTGTGTCAGTGAGCCACGAAGACCCTGAGGTCAAGTTCAA
CTGGTACGTGGACGGCGTGGAGGTGCATAATGCCAAGACAAAGCCCCGGGAGGAGC
AGTACAACAGCACGTACCGGGTGGTCAGCGTCCTCACCGTCCTGCACCAGGACTGG
CTGAATGGCAAGGAGTACAAGTGCAAGGTCTCCAACAAAGCCCTCCCAGCCCCCAT
CGAGAAAACCATCTCCAAAGCCAAAGGCCAGCCCCGGGAACCACAGGTGTACGTGC
TGCCCCCATCCCGGGACGAGCTGACCAAGAACCAGGTGAGCCTGCTGTGCCTGGTC
AAAGGCTTCTATCCCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGCCAGCCCGA
GAACAACCTACCTGACCTGGCCTCCCGTGCTGGACTCCGACGGCTCCTTCTTCCTCT
ACAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGGCAACGTCTTCTCATGC
TCCGTGATGCATGAGGCTCTGCACAACCACTACACCCAGAAGAGCCTCTCCCTGTC
TCCCGGCAAG

(SEQ ID NO: 71), signal sequence (1-57), hinge (58-87),
CH2 (88-417), CH3 (418-738)

Fig. 73

NYF-0016-HC2

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
 CGGAGAGGTGCAGCTGGTGGAGTCTGGGGGAGGCGTGGTCCAGCCTGGGAGGTCCC
 TGACACCCTCCTGTTCAGCGTCTGGATTCTCCATCCGTAGCTATGATATACACTGG
 GTCCGCCTGGCTCCAGGCAAGGGGCTAGAGTGGGTGGCCACTATATCATATGATGG
 AAGTCAGAAGTACTTCGCAGACTCCGTGAAGGGCCGATTTACCATCTTCAGAGACG
 AATCGGAGAACATGGTGTATCTGCAAATGAGCGGCCTGAGAGTTGAGGACACGGCT
 GTTTATCACTGTGCGAGAGGGAGTAGTGGTCAATTATGAGGCTTTTGATATCTGGGG
 CCAAGGGACAATGGTCACCGTCTCTTCAGGTGGAGGCGGTTTCAGGCGGAGGTGGCA
 GCGGCGGTGGCGGGAGTCAGTCTGTGTTGACGCAGCCGCCCTCAGTGTCTGCGGCC
 CCAGGACAGAAGGTCACCATCTCCTGCTCTGGAAGCAGCTCCAACATTGGGAATTG
 GTATGTATCCTGGTACCAGCAGCTCCCAGGAACAGCCCCCAAACCTCTCATTTATG
 ACAATAATAAGCGACCCTCAGGGATTCTTGACCGATTCTCTGGCTCCAAGTCTGGC
 ACGTCAGCCACCTGGGCATCACCGGACTCCAGACTGGGGACGAGGCCGATTATTA
 CTGCGGAACATGGGATAGCAGCCTGAGTGCCCTTGGGTGTTGCGCGGAGGGACCA
 AGGTCAACCGTCTAGGGGGAGGCGGTTTCAGAAGTGCAGCTGGTGGAAATCCGGGGGG
 GGCTGGTGCAGCCTGGGGGGAGCCTGAGACTGAGTTGTGCCGCCCTCTGGGGTGAC
 ATTTAACTACTATGGCATGTCTTGATCCGCCAGGCACCTGGAAAGGGCCTGGAGT
 GGGTGGCCAGCATCACTAGGTCCGGCGGGCGAATCTACTATCCCGACAGCGTCAAG
 GGCAGGTTACAAATTTCCCGCGAGAACACACAGAAAACCTCTGTACCTGCAGATGAA
 TAGCCTGAGAGCCGAAGATACAGCTGTGTACTATTGCACTCTGGACGGCAGGGATG
 GGTGGGTCGCCTATTGGGGGGCAGGGAACCTGGTGACAGTCAGCTCCGGAGGAGGA
 GGATCTGGCGGAGGAGGCAGTGGGGGAGGCGGGTCAAACCTTATGCTCACTCAGCC
 GTCCTCTGTTTCTGGCGTACCTGGCCAACGGGTGACCATTAGCTGTACGGGTAATA
 CCGGGAATATCGGGTCTAACTACGTGAACTGGTATCAGCAGCTTCCAGGGACAGCT
 CCCAAGTTGCTGATCTATCGCGACGACAAAAGACCCTCAGGGGTCCCTGACCGATT
 TAGTGGCAGCAAAAGCGGTACTTCCGCTTCCCTGGCGATAACCGGCTTTCAGGCCG
 AAGATGAGGCAGACTACTATTGCCAGTCATATTCCAGCGGCTTCATCTTCGGAGGC
 GGAACATAAGCTGACAGTGTTGGCAGCCGAGCCCAAATCTTCTGACAAAACCTCACAC
 ATGCCACCCCTGCCCAGCACCTGAAGCCGAGGGGGACCCTCAGTCTTCTCTTCC
 CCCCCAAACCAAGGACACCCTCATGATCTCCCGGACCCCTGAGGTCACATGCGTG
 GTGGTGTCAGTGAGCCACGAAGACCCTGAGGTCAAGTTCAACTGGTACGTGGACGG
 CGTGGAGGTGCATAATGCCAAGACAAAGCCCCGGGAGGAGCAGTACAACAGCACGT
 ACCGGGTGGTCAGCGTCCCTACCGTCTGCACCAGGACTGGCTGAATGGCAAGGAG
 TACAAGTGCAAGGTCTCCAACAAAGCCCTCCCAGCCCCATCGAGAAAACCATCTC
 CAAAGCCAAAGGCCAGCCCCGGAACACAGGTGTACGTGTACCCCCATCCCGGG
 ACGAGCTGACCAAGAACCAGGTGAGCCTGACCTGCCTGGTCAAAGGCTTCTATCCC
 AGCGACATCGCCGTGGAGTGGGAGAGCAATGGCCAGCCGAGAACAACTACAAGAC
 CACCCCTCCCGTGCTGGACTCCGACGGCTCCTTCGCCCTCGTGAGCAAGCTCACCG
 TGGACAAGAGCAGGTGGCAGCAGGGCAACGTCTTCTCATGCTCCGTGATGCATGAG
 GCTCTGCACAACCACTACACCCAGAAGAGCCTCTCCCTGTCTCCCGGCAAG
 (SEQ ID NO: 72), signal sequence (1-57), NYA-1143 (61-
 798), linker (799-813), C3E-7085 (814-1533), linker
 (1534-1539), hinge (1540-1584), CH2 (1585-1914), CH3
 (1915-2235)

Fig. 74

NYF-0019-HC2

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
 CGGAGAGGTGCAGCTGGTGGAGTCTGGGGGAGGCGTGGTCCAGCCTGGGAGGTCCC
 TGACACCCTCCTGTTCAGCGTCTGGATTCTCCATCCGTAGCTATGATATACACTGG
 GTCCGCCTGGCTCCAGGCAAGGGGCTAGAGTGGGTGGCCACTATATCATATGATGG
 AAGTCAGAAGTACTTCGCAGACTCCGTGAAGGGCCGATTTACCATCTTCAGAGACG
 AATCGGAGAACATGGTGTATCTGCAAATGAGCGGCCTGAGAGTTGAGGACACGGCT
 GTTTATCACTGTGCGAGAGGGAGTAGTGGTCAATTATGAGGCTTTTGATATCTGGGG
 CCAAGGGACAATGGTCACCGTCTCTTCAGGTGGAGGCGGTTTCAGGCGGAGGTGGCA
 GCGGCGGTGGCGGGAGTCAGTCTGTGTTGACGCAGCCGCCCTCAGTGTCTGCGGCC
 CCAGGACAGAAGGTCACCATCTCCTGCTCTGGAAGCAGCTCCAACATTGGGAATTG
 GTATGTATCCTGGTACCAGCGGCTCCCAGGAACAGCCCCCAAACCTCTCATTTATG
 ACAATAATAAGCGACCCTCAGGGATTCTTGACCGATTCTCTGGCTCCAAGTCTGGC
 ACGTCAGCCACCTGGGCATCACCGGACTCCAGACTGGGGACGAGGCCGATTATTA
 CTGCGGAACATGGGATAGCAGCCTGAGTGCCCTTGGGTGTTGCGCGGAGGGACCA
 AGGTCAACCGTCTAGGGGGAGGCGGTTTCAGAAGTGCAGCTGGTGGAAATCCGGGGGG
 GGCTTGGTGCAGCCTGGGGGGAGCCTGAGACTGAGTTGTGCCGCTCTGGGGTGAC
 ATTTAACTACTATGGCATGTCTTGATCCGCCAGGCACCTGGAAAGGGCCTGGAGT
 GGGTGGCCAGCATCACTAGGTCCGGCGGGCGAATCTACTATCCCGACAGCGTCAAG
 GGCAGGTTACAAATTTCCCGCGAGAACACACAGAAAACCTCTGTACCTGCAGATGAA
 TAGCCTGAGAGCCGAAGATACAGCTGTGTACTATTGCACTCTGGACGGCAGGGATG
 GGTGGGTGCGCTATTGGGGGGCAGGGAACCTGGTGACAGTCAGCTCCGGAGGAGGA
 GGATCTGGCGGAGGAGGCAGTGGGGGAGGCGGGTCAAACCTTATGCTCACTCAGCC
 GTCCTCTGTTTCTGGCGTACCTGGCCAACGGGTGACCATTAGCTGTACGGGTAATA
 CCGGGAATATCGGGTCTAACTACGTGAACTGGTATCAGCAGCTTCCAGGGACAGCT
 CCCAAGTTGCTGATCTATCGCGACGACAAAAGACCCTCAGGGGTCCCTGACCGATT
 TAGTGGCAGCAAAAGCGGTACTTCCGCTTCCCTGGCGATAACCGGCTTTCAGGCCG
 AAGATGAGGCAGACTACTATTGCCAGTCATATTCCAGCGGCTTCATCTTCGGAGGC
 GGAACATAAGCTGACAGTGTTGGCAGCCGAGCCCAAATCTTCTGACAAAACCTCACAC
 ATGCCACCCCTGCCCAGCACCTGAAGCCGAGGGGGACCCTCAGTCTTCTCTTCC
 CCCCCAAACCAAGGACACCCTCATGATCTCCCGGACCCCTGAGGTCACATGCGTG
 GTGGTGTCAGTGAGCCACGAAGACCCTGAGGTCAAGTTCAACTGGTACGTGGACGG
 CGTGGAGGTGCATAATGCCAAGACAAAGCCCCGGGAGGAGCAGTACAACAGCACGT
 ACCGGGTGGTCAGCGTCCCTACCGTCTGACCAGGACTGGCTGAATGGCAAGGAG
 TACAAGTGCAAGGTCTCCAACAAAGCCCTCCCAGCCCCATCGAGAAAACCATCTC
 CAAAGCCAAAGGCCAGCCCCGGAACACAGGTGTACGTGTACCCCCATCCCGGG
 ACGAGCTGACCAAGAACCAGGTGAGCCTGACCTGCCTGGTCAAAGGCTTCTATCCC
 AGCGACATCGCCGTGGAGTGGGAGAGCAATGGCCAGCCGAGAACAACTACAAGAC
 CACCCCTCCCGTGCTGGACTCCGACGGCTCCTTCGCCCTCGTGAGCAAGCTCACCG
 TGGACAAGAGCAGGTGGCAGCAGGGCAACGTCTTCTCATGCTCCGTGATGCATGAG
 GCTCTGCACAACCACTACACCCAGAAGAGCCTCTCCCTGTCTCCCGGCAAG
 (SEQ ID NO: 73), signal sequence (1-57), NYA-2143 (61-
 798), linker (799-813), C3E-7085 (814-1533), linker
 (1534-1539), hinge (1540-1584), CH2 (1585-1914), CH3
 (1915-2235)

Fig. 75

NYF-0022-HC2

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
 CGGAGAGGTGCAGCTGGTGGAGTCTGGGGGAGGCGTGGTCCAGCCTGGGAGGTCCC
 TGACACCCTCCTGTTCAGCGTCTGGATTCTCCATCCGTAGCTATGATATACACTGG
 GTCCGCCTGGCTCCAGGCAAGGGGCTAGAGTGGGTGGCCACTATATCATATGATGG
 AAGTCAGAAGTACTTCGCAGACTCCGTGAAGGGCCGATTTACCATCTTCAGAGACG
 AATCGGAGAACATGGTGTATCTGCAAATGAGCGGCCTGAGAGTTGAGGACACGGCT
 GTTTATCACTGTGCGAGAGGGAGTAGTGGTCAATTATGAGGCTTTTGATATCTGGGG
 CCAAGGGACAATGGTCACCGTCTCTTCAGGTGGAGGCGGTTTCAGGCGGAGGTGGCA
 GCGGCGGTGGCGGGAGTCAGTCTGTGTTGACGCAGCCGCCCTCAGTGTCTGCGGCC
 CCAGGACAGAAGGTCACCATCTCCTGCTCTGGAAGCAGCTCCAACATTGGGAATAA
 TTATGTATCCTGGTACCAGCAGCTCCCAGGAACAGCCCCCAAACCTCTCATTTATG
 ACAATAATAAGCGACCCTCAGGGATTCTTGACCGATTCTCTGGCTCCAAGTCTGGC
 ACGTCAGCCACCTGGGCATCACCGGACTCCAGACTGGGGACGAGGCCGATTATTA
 CTGCGGAACATGGGATAGCAGCCTGAGTGCCCTTGGGTGTTGCGCGGAGGGACCA
 AGGTCAACCGTCTAGGGGGAGGCGGTTTCAGAAGTGCAGCTGGTGGAAATCCGGGGGG
 GGCTTGGTGCAGCCTGGGGGGAGCCTGAGACTGAGTTGTGCCGCTCTGGGGTGAC
 ATTTAACTACTATGGCATGTCTTGATCCGCCAGGCACCTGGAAAGGGCCTGGAGT
 GGGTGGCCAGCATCACTAGGTCCGGCGGGCGAATCTACTATCCCGACAGCGTCAAG
 GGCAGGTTACAAATTTCCCGCGAGAACACACAGAAAACCTCTGTACCTGCAGATGAA
 TAGCCTGAGAGCCGAAGATACAGCTGTGTACTATTGCACTCTGGACGGCAGGGATG
 GGTGGGTCGCCTATTGGGGGGCAGGGAACCTGGTGACAGTCAGCTCCGGAGGAGGA
 GGATCTGGCGGAGGAGGCAGTGGGGGAGGCGGGTCAAACCTTATGCTCACTCAGCC
 GTCCTCTGTTTCTGGCGTACCTGGCCAACGGGTGACCATTAGCTGTACGGGTAATA
 CCGGGAATATCGGGTCTAACTACGTGAACTGGTATCAGCAGCTTCCAGGGACAGCT
 CCCAAGTTGCTGATCTATCGCGACGACAAAAGACCCTCAGGGGTCCCTGACCGATT
 TAGTGGCAGCAAAAGCGGTACTTCCGCTTCCCTGGCGATAACCGGCTTTCAGGCCG
 AAGATGAGGCAGACTACTATTGCCAGTCATATTCCAGCGGCTTCATCTTCGGAGGC
 GGAACATAAGCTGACAGTGTTGGCAGCCGAGCCCAAATCTTCTGACAAAACCTCACAC
 ATGCCACCCCTGCCCAGCACCTGAAGCCGAGGGGGACCCTCAGTCTTCTCTTCC
 CCCCCAAACCAAGGACACCCTCATGATCTCCCGGACCCCTGAGGTCACATGCGTG
 GTGGTGTCAGTGAGCCACGAAGACCCTGAGGTCAAGTTCAACTGGTACGTGGACGG
 CGTGGAGGTGCATAATGCCAAGACAAAGCCCCGGGAGGAGCAGTACAACAGCACGT
 ACCGGGTGGTCAGCGTCCCTACCGTCTGACCAGGACTGGCTGAATGGCAAGGAG
 TACAAGTGCAAGGTCTCCAACAAAGCCCTCCCAGCCCCATCGAGAAAACCATCTC
 CAAAGCCAAAGGCCAGCCCCGGAACACAGGTGTACGTGTACCCCCATCCCGGG
 ACGAGCTGACCAAGAACCAGGTGAGCCTGACCTGCCTGGTCAAAGGCTTCTATCCC
 AGCGACATCGCCGTGGAGTGGGAGAGCAATGGCCAGCCGAGAACAACTACAAGAC
 CACCCCTCCCGTGCTGGACTCCGACGGCTCCTTCGCCCTCGTGAGCAAGCTCACCG
 TGGACAAGAGCAGGTGGCAGCAGGGCAACGTCTTCTCATGCTCCGTGATGCATGAG
 GCTCTGCACAACCACTACACCCAGAAGAGCCTCTCCCTGTCTCCCGGCAAG
 (SEQ ID NO: 74), signal sequence (1-57), NYA-1163 (61-
 798), linker (799-813), C3E-7085 (814-1533), linker
 (1534-1539), hinge (1540-1584), CH2 (1585-1914), CH3
 (1915-2235)

Fig. 76

NYF-0023-HC2

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
 CGGAGAGGTGCAGCTGGTGGAGTCTGGGGGAGGCGTGGTCCAGCCTGGGAGGTCCC
 TGACACCCTCCTGTTTCAGCGTCTGGATTCTCCATCCGTAGTTATGATATACACTGG
 GTCCGCCAGGCTCCAGGCAAGGGGCTAGAGTGGGTGGCCACTATATCATATGATGG
 AAGTCAGAAGTACTTCGCAGACTCCGTGAAGGGCCGATTTACCATCTTCAGAGACG
 AATCGGAGAACATGGTGTATCTGCAAATGAGCGGCCTGAGAGTTGAGGACACGGCT
 GTTTATCACTGTGCGAGAGGGAGTAGTGGTCAATTATGAGGCTTTTGATATCTGGGG
 CCAAGGGACAATGGTCACCGTCTCTTCAGGTGGAGGCGGTTTCAGGCGGAGGTGGCA
 GCGGCGGTGGCGGGAGTCAGTCTGTGTTGACGCAGCCGCCCTCAGTGTCTGCGGCC
 CCAGGACAGAAGGTCACCATCTCCTGCTCTGGAAGCAGCTCCAACATTGGGAATTG
 GTATGTATCCTGGTACCAGCAGCTCCCAGGAACAGCCCCAACTCCTCATTTATG
 ACAATAATAAGCGACCCTCAGGGATTCTTGACCGATTCTCTGGCTCCAAGTCTGGC
 ACGTCAGCCACCTGGGCATCACCGGACTCCAGACTGGGGACGAGGCCGATTATTA
 CTGCGGAACATGGGATAGCAGCCTGAGTGCCCTTGGGTGTTGCGCGGAGGGACCA
 AGGTCAACCGTCTAGGGGGAGGCGGTTTCAGAAGTGCAGCTGGTGAATCCGGGGGG
 GGCTTGGTGCAGCCTGGGGGGAGCCTGAGACTGAGTTGTGCCGCTCTGGGGTGAC
 ATTTAACTACTATGGCATGTCTTGATCCGCCAGGCACCTGGAAAGGGCCTGGAGT
 GGGTGGCCAGCATCACTAGGTCCGGCGGGCGAATCTACTATCCCGACAGCGTCAAG
 GGCAGGTTACAAATTTCCCGCGAGAACACACAGAAAACCTCTGTACCTGCAGATGAA
 TAGCCTGAGAGCCGAAGATACAGCTGTGTACTATTGCACTCTGGACGGCAGGGATG
 GGTGGGTCGCCTATTGGGGGGCAGGGAACCTGGTGACAGTCAGCTCCGGAGGAGGA
 GGATCTGGCGGAGGAGGCAGTGGGGGAGGCGGGTCAAACCTTATGCTCACTCAGCC
 GTCCTCTGTTTCTGGCGTACCTGGCCAACGGGTGACCATTAGCTGTACGGGTAATA
 CCGGGAATATCGGGTCTAACTACGTGAACTGGTATCAGCAGCTTCCAGGGACAGCT
 CCCAAGTTGCTGATCTATCGCGACGACAAAAGACCCTCAGGGGTCCCTGACCGATT
 TAGTGGCAGCAAAAGCGGTACTTCCGCTTCCCTGGCGATAACCGGCTTTCAGGCCG
 AAGATGAGGCAGACTACTATTGCCAGTCATATTCCAGCGGCTTCATCTTCGGAGGC
 GGAACATAAGCTGACAGTGTTGGCAGCCGAGCCCAAATCTTCTGACAAAACCTCACAC
 ATGCCACCCCTGCCCAGCACCTGAAGCCGAGGGGGACCCTCAGTCTTCTCTTCC
 CCCCCAAACCAAGGACACCCTCATGATCTCCCGGACCCCTGAGGTACATGCGTG
 GTGGTGTCAGTGAGCCACGAAGACCCTGAGGTCAAGTTCAACTGGTACGTGGACGG
 CGTGGAGGTGCATAATGCCAAGACAAAGCCCCGGGAGGAGCAGTACAACAGCACGT
 ACCGGGTGGTCAAGCTCCTCACCGTCTGCACCAGGACTGGCTGAATGGCAAGGAG
 TACAAGTGCAAGGTCTCCAACAAAGCCCTCCCAGCCCCATCGAGAAAACCATCTC
 CAAAGCCAAAGGCCAGCCCCGGGAACCACAGGTGTACGTGTACCCCCATCCCGGG
 ACGAGCTGACCAAGAACCAGGTGAGCCTGACCTGCCTGGTCAAAGGCTTCTATCCC
 AGCGACATCGCCGTGGAGTGGGAGAGCAATGGCCAGCCGAGAACAACTACAAGAC
 CACCCCTCCCGTGCTGGACTCCGACGGCTCCTTCGCCCTCGTGAGCAAGCTCACCG
 TGGACAAGAGCAGGTGGCAGCAGGGCAACGTCTTCTCATGCTCCGTGATGCATGAG
 GCTCTGCACAACCACTACACCCAGAAGAGCCTCTCCCTGTCTCCCGGCAAG
 (SEQ ID NO: 75), signal sequence (1-57), NYA-2023 (61-
 798), linker (799-813), C3E-7085 (814-1533), linker
 (1534-1539), hinge (1540-1584), CH2 (1585-1914), CH3
 (1915-2235)

Fig. 77

NYF-0027-HC2

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
 CGGAGAGGTGCAGCTGGTGGAGTCTGGGGGAGGCGTGGTCCAGCCTGGGAGGTCCC
 TGACACCCTCCTGTTCAGCGTCTGGATTCTCCATCCGTAGTTATGATATACACTGG
 GTCCGCCAGGCTCCAGGCAAGGGGCTAGAGTGGGTGGCCACTATATCATATGATGG
 AAGTCAGAAGTACTTCGCAGACTCCGTGAAGGGCCGATTTACCATCTTCAGAGACG
 AATCGGAGAACATGGTGTATCTGCAAATGAGCGGCCTGAGAGTTGAGGACACGGCT
 GTTTATCACTGTGCGAGAGGGAGTAGTGGTCAATTATGAGGCTTTTGATATCTGGGG
 CCAAGGGACAATGGTCACCGTCTCTTCAGGTGGAGGCGGTTTCAGGCGGAGGTGGCA
 GCGGCGGTGGCGGGAGTCAGTCTGTGTTGACGCAGCCGCCCTCAGTGTCTGCGGCC
 CCAGGACAGAAGGTCACCATCTCCTGCTCTGGAAGCAGCTCCAACATTGGGAATAA
 TTATGTATCCTGGTACCAGCAGCTCCCAGGAACAGCCCCCAAACCTCTCATTTATG
 ACAATAATAAGCGACCCTCAGGGATTCTTGACCGATTCTCTGGCTCCAAGTCTGGC
 ACGTCAGCCACCTGGGCATCACCGGACTCCAGACTGGGGACGAGGCCGATTATTA
 CTGCGGATCATGGGATAGCAGCCTGAGTGCCCTTGGGTGTTGCGCGGAGGGACCA
 AGGTCAACCGTCTAGGGGGAGGCGGTTTCAGAAGTGCAGCTGGTGAATCCGGGGGG
 GGCTTGGTGCAGCCTGGGGGGAGCCTGAGACTGAGTTGTGCCGCTCTGGGGTGAC
 ATTTAACTACTATGGCATGTCTTGATCCGCCAGGCACCTGGAAAGGGCCTGGAGT
 GGGTGGCCAGCATCACTAGGTCCGGCGGGCGAATCTACTATCCCGACAGCGTCAAG
 GGCAGGTTACAAATTTCCCGCGAGAACACACAGAAAACCTCTGTACCTGCAGATGAA
 TAGCCTGAGAGCCGAAGATACAGCTGTGTACTATTGCACTCTGGACGGCAGGGATG
 GGTGGGTCGCCTATTGGGGGGCAGGGAACCTGGTGACAGTCAGCTCCGGAGGAGGA
 GGATCTGGCGGAGGAGGCAGTGGGGGAGGCGGGTCAAACCTTATGCTCACTCAGCC
 GTCCTCTGTTTCTGGCGTACCTGGCCAACGGGTGACCATTAGCTGTACGGGTAATA
 CCGGGAATATCGGGTCTAACTACGTGAACTGGTATCAGCAGCTTCCAGGGACAGCT
 CCCAAGTTGCTGATCTATCGCGACGACAAAAGACCCTCAGGGGTCCCTGACCGATT
 TAGTGGCAGCAAAAGCGGTACTTCCGCTTCCCTGGCGATAACCGGCTTTCAGGCCG
 AAGATGAGGCAGACTACTATTGCCAGTCATATTCCAGCGGCTTCATCTTCGGAGGC
 GGAACATAAGCTGACAGTGTTGGCAGCCGAGCCCAAATCTTCTGACAAAACCTCACAC
 ATGCCACCCCTGCCCAGCACCTGAAGCCGAGGGGGACCCTCAGTCTTCTCTTCC
 CCCCCAAACCAAGGACACCCTCATGATCTCCCGGACCCCTGAGGTACATGCGTG
 GTGGTGTCAGTGAGCCACGAAGACCCTGAGGTCAAGTTCAACTGGTACGTGGACGG
 CGTGGAGGTGCATAATGCCAAGACAAAGCCCCGGGAGGAGCAGTACAACAGCACGT
 ACCGGGTGGTCAGCGTCCCTACCGTCTGACCAGGACTGGCTGAATGGCAAGGAG
 TACAAGTGCAAGGTCTCCAACAAAGCCCTCCCAGCCCCATCGAGAAAACCATCTC
 CAAAGCCAAAGGCCAGCCCCGGAACACAGGTGTACGTGTACCCCCATCCCGGG
 ACGAGCTGACCAAGAACCAGGTGAGCCTGACCTGCCTGGTCAAAGGCTTCTATCCC
 AGCGACATCGCCGTGGAGTGGGAGAGCAATGGCCAGCCGAGAACAACTACAAGAC
 CACCCCTCCCGTGCTGGACTCCGACGGCTCCTTCGCCCTCGTGAGCAAGCTCACCG
 TGGACAAGAGCAGGTGGCAGCAGGGCAACGTCTTCTCATGCTCCGTGATGCATGAG
 GCTCTGCACAACCACTACACCCAGAAGAGCCTCTCCCTGTCTCCCGGCAAG
 (SEQ ID NO: 76), signal sequence (1-57), NYA-2027(61-
 798), linker (799-813), C3E-7085 (814-1533), linker
 (1534-1539), hinge (1540-1584), CH2 (1585-1914), CH3
 (1915-2235)

Fig. 78

NYF-0035-HC2

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
 CGGAGAGGTGCAGCTGGTGGAGTCTGGGGGAGGCGTGGTCCAGCCTGGGAGGTCCC
 TGACACCCTCCTGTTCAGCGTCTGGATTCTCCATCCGTAGCTATGATATACACTGG
 GTCCGCCTGGCTCCAGGCAAGGGGCTAGAGTGGGTGGCCACTATATCATATGATGG
 AAGTCAGAAGTACTTCGCAGACTCCGTGAAGGGCCGATTTACCATCTTCAGAGACG
 AATCGGAGAACATGGTGTATCTGCAAATGAGCGGCCTGAGAGTTGAGGACACGGCT
 GTTTATCACTGTGCGAGAGGGAGTAGTGGTCAATTATGAGGCTTTTGATATCTGGGG
 CCAAGGGACAATGGTCACCGTCTCTTCAGGTGGAGGCGGTTTCAGGCGGAGGTGGCA
 GCGGCGGTGGCGGGAGTCAGTCTGTGTTGACGCAGCCGCCCTCAGTGTCTGCGGCC
 CCAGGACAGAAGGTCACCATCTCCTGCTCTGGAAGCAGCTCCAACATTGGGAATTG
 GAAGGTATCCTGGTACCAGCAGCTCCCAGGAACAGCCCCAACTCCTCATTTATG
 ACAATAATAAGCGACCCTCAGGGATTCTTGACCGATTCTCTGGCTCCAAGTCTGGC
 ACGTCAGCCACCTGGGCATCACCGGACTCCAGACTGGGGACGAGGCCGATTATTA
 CTGCGGAACATGGGATAGCAGCCTGAGTGCCCTTGGGTGTTGCGCGGAGGGACCA
 AGGTCAACCGTCTAGGGGGAGGCGGTTTCAGAAGTGCAGCTGGTGAATCCGGGGGG
 GGCTGGTGCAGCCTGGGGGGAGCCTGAGACTGAGTTGTGCCGCTCTGGGGTGAC
 ATTTAACTACTATGGCATGTCTTGATCCGCCAGGCACCTGGAAAGGGCCTGGAGT
 GGGTGGCCAGCATCACTAGGTCCGGCGGGCGAATCTACTATCCCGACAGCGTCAAG
 GGCAGGTTACAAATTTCCCGCGAGAACACACAGAAAACCTCTGTACCTGCAGATGAA
 TAGCCTGAGAGCCGAAGATACAGCTGTGTACTATTGCACTCTGGACGGCAGGGATG
 GGTGGGTGCGCTATTGGGGGGCAGGGAACCTGGTGACAGTCAGCTCCGGAGGAGGA
 GGATCTGGCGGAGGAGGCAGTGGGGGAGGCGGGTCAAACCTTATGCTCACTCAGCC
 GTCCTCTGTTTCTGGCGTACCTGGCCAACGGGTGACCATTAGCTGTACGGGTAATA
 CCGGGAATATCGGGTCTAACTACGTGAACTGGTATCAGCAGCTTCCAGGGACAGCT
 CCCAAGTTGCTGATCTATCGCGACGACAAAAGACCCTCAGGGGTCCCTGACCGATT
 TAGTGGCAGCAAAGCGGTACTTCCGCTTCCCTGGCGATAACCGGCTTTCAGGCCG
 AAGATGAGGCAGACTACTATTGCCAGTCATATTCCAGCGGCTTCATCTTCGGAGGC
 GGAACATAAGCTGACAGTGTTGGCAGCCGAGCCCAAATCTTCTGACAAAACCTCACAC
 ATGCCACCCCTGCCCAGCACCTGAAGCCGAGGGGGACCCTCAGTCTTCCTCTTCC
 CCCCCAAACCAAGGACACCCTCATGATCTCCCGGACCCCTGAGGTCACATGCGTG
 GTGGTGTCAGTGAGCCACGAAGACCCTGAGGTCAAGTTCAACTGGTACGTGGACGG
 CGTGGAGGTGCATAATGCCAAGACAAAGCCCCGGGAGGAGCAGTACAACAGCACGT
 ACCGGGTGGTCAGCGTCCCTACCGTCTGCACCAGGACTGGCTGAATGGCAAGGAG
 TACAAGTGCAAGGTCTCCAACAAAGCCCTCCCAGCCCCATCGAGAAAACCATCTC
 CAAAGCCAAAGGCCAGCCCCGGAACACAGGTGTACGTGTACCCCCATCCCGGG
 ACGAGCTGACCAAGAACCAGGTGAGCCTGACCTGCCTGGTCAAAGGCTTCTATCCC
 AGCGACATCGCCGTGGAGTGGGAGAGCAATGGCCAGCCGAGAACAACTACAAGAC
 CACCCCTCCCGTGCTGGACTCCGACGGCTCCTTCGCCCTCGTGAGCAAGCTCACCG
 TGGACAAGAGCAGGTGGCAGCAGGGCAACGTCTTCTCATGCTCCGTGATGCATGAG
 GCTCTGCACAACCACTACACCCAGAAGAGCCTCTCCCTGTCTCCCGGCAAG
 (SEQ ID NO: 77), signal sequence (1-57), NYA-2035 (61-
 798), linker (799-813), C3E-7085 (814-1533), linker
 (1534-1539), hinge (1540-1584), CH2 (1585-1914), CH3
 (1915-2235)

Fig. 79

NYF-0044-HC2

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
CGGTGAGGTACAGCTGGTGGGAATCCGGTGGAGGCCTTGTGCAACCGGGAGGCTCCC
TGCGACTCTCTTGCGCCGCATCTGGCTTTTCCATACGATCCTACGATATTCATTGG
GTCCGACTCGCCCCCTGGCAAAGGTCTTGAATGGGTGCGCACTATCTCCTACGACGG
GTCTCAGAAATATTATGCGGATTCAGTGAAGGGGCGGTTTACAATTTACGAGACG
AGTCAAAGAATACACTGTACCTCCAAATGAATTCAGTGAAGAGCCGAGGATACTGCA
GTCTATCATTGTGCAAGAGGGTCTCTCAGGCCACTACGAGGCCTTTGATATATGGGG
CCAAGGCACCTTGGTAACCGTTAGTAGCGGAGGTGGAGGAAGCGGAGGCGGCGGTT
CCGGAGGAGGTGGAAGTCAATCCGTTCTGACTCAGCCACCCTCCGTGAGTGCAGCG
CCTGGACAGAAGGTCACAATATCATGTTCTGGCTCTTCAAGCAACATCGGAAACTG
GTACGTGAGCTGGTACCAGCAGCTCCCCGGCACGGCGCCCAAGCTTCTGATATATG
ACAACAACAAACGGCCCAAGTGAATCCCTGACAGATTCAGTGGGTCTAAAAGTGGT
ACAAGCGCTACCTGGGTATCACCGGATTGCAGACCGGAGATGAGGCGGACTATTA
TTGCGGAACCTGGGACTCCAGCCTGAGCGCTCCCTGGGTTTTCGGCGGAGGTACTA
AGGTTACCGTCTTGGGAGGAGGCGGTTTCAAGTGCAGCTGGTGAATCCGGGGGG
GGCCTGGTGCAGCCTGGGGGGAGCCTGAGACTGAGTTGTGCCGCTCTGGGGTGAC
ATTTAACTACTATGGCATGTCTTGGATCCGCCAGGCACCTGGAAAGGGCCTGGAGT
GGGTGGCCAGCATCACTAGGTCCGGCGGGCGAATCTACTATCCCGACAGCGTCAAG
GGCAGGTTACAATTTCCCGCGAGAACACACAGAAAACCTCTGTACCTGCAGATGAA
TAGCCTGAGAGCCGAAGATACAGCTGTGTACTATTGCACTCTGGACGGCAGGGATG
GGTGGGTCGCTATTGGGGGGCAGGGAACCTGGTGACAGTCAGCTCCGGAGGAGGA
GGATCTGGCGGAGGAGGCAGTGGGGGAGGCGGGTCAAACCTTATGCTCACTCAGCC
GTCCTCTGTTTTCTGGCGTACCTGGCCAACGGGTGACCATTAGCTGTACGGGTAATA
CCGGGAATATCGGGTCTAACTACGTGAACTGGTATCAGCAGCTTCCAGGGACAGCT
CCCAAGTTGCTGATCTATCGCGACGACAAAAGACCCTCAGGGGTCCCTGACCGATT
TAGTGGCAGCAAAGCGGTACTTCCGCTTCCCTGGCGATAACCGGCTTTCAGGCCG
AAGATGAGGCAGACTACTATTGCCAGTCATATTCCAGCGGCTTCATCTTCGGAGGC
GGAACCTAAGCTGACAGTGTTGGCAGCCGAGCCCAAATCTTCTGACAAAACCTCACAC
ATGCCACCCCTGCCCAGCACCTGAAGCCGAGGGGGACCCTCAGTCTTCTCTTCC
CCCCAAAACCAAGGACACCCTCATGATCTCCCGGACCCCTGAGGTCACATGCGTG
GTGGTGTCAGTGAGCCACGAAGACCCTGAGGTCAAGTTCAACTGGTACGTGGACGG
CGTGGAGGTGCATAATGCCAAGACAAAGCCCCGGGAGGAGCAGTACAACAGCACGT
ACCGGGTGGTCAGCGTCCCTACCGTCTTGCACCAGGACTGGCTGAATGGCAAGGAG
TACAAGTGCAAGGTCTCCAACAAAGCCCTCCCAGCCCCATCGAGAAAACCATCTC
CAAAGCCAAAGGCCAGCCCCGGGAACACAGGTGTACGTGTACCCCCATCCCGGG
ACGAGCTGACCAAGAACCAGGTGAGCCTGACCTGCCTGGTCAAAGGCTTCTATCCC
AGCGACATCGCCGTGGAGTGGGAGAGCAATGGCCAGCCGAGAACAACTACAAGAC
CACCCCTCCCGTGCTGGACTCCGACGGCTCCTTCGCCCTCGTGAGCAAGCTCACCG
TGGACAAGAGCAGGTGGCAGCAGGGCAACGTCTTCTCATGCTCCGTGATGCATGAG
GCTCTGCACAACCACTACACCCAGAAGAGCCTCTCCCTGTCTCCCGGCAAG
(SEQ ID NO: 78), signal sequence (1-57), NYA-2044 (61-
798), linker (799-813), C3E-7085 (814-1533), linker
(1534-1539), hinge (1540-1584), CH2 (1585-1914), CH3
(1915-2235)

Fig. 80

NYF-0045-HC2

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
 CGGTGAGGTACAGCTGGTGGGAATCCGGTGGAGGCCTTGTGCAACCGGGAGGCTCCC
 TGCGACTCTCTTGCGCCGCATCTGGCTTTTCCATACGATCCTACGATATTCATTGG
 GTCCGACTCGCCCCCTGGCAAAGGTCTTGAATGGGTGCCACTATCTCCTACGACGG
 GTCTCAGAAATATTATGCGGATTCAGTGAAGGGGCGGTTTACAATTCACGAGACG
 AGTCAAAGAATACTGTACCTCCAAATGAATTCAGTGAAGGCCGAGGATACTGCA
 GTCTATCATTGTGCAAGAGGGTCTCAGGCCACTACGAGGCCTTTGATATATGGGG
 CCAAGGCACCTTGGTAACCGTTAGTAGCGGAGGTGGAGGAAGCGGAGGCGGCGGTT
 CCGGAGGAGGTGGAAGTCAATCCGTTCTGACTCAGCCACCCTCCGCTCTGGCACC
 CCGGGCCAACGGGTCAACAATATCATGTTCTGGCTCTTCAAGCAACATCGGAAACTG
 GTACGTGAGCTGGTACCAGCAGCTCCCCGGCACGGCGCCCAAGCTTCTGATATATG
 ACAACAACAAACGGCCAGTGGAGTTCCTGACAGATTCAGTGGGTCTAAAAGTGGT
 ACAAGCGCTAGCCTGGCCATAAGTGGTCTGCAGAGTGAAGATGAGGCGGACTATTA
 TTGCGGAACCTGGGACTCCAGCCTGAGCGCTCCCTGGGTTTTCGGCGGAGGTACTA
 AGCTGACCGTCTTGGGAGGAGGCGGTTTCAAGTGCAGCTGGTGGAAATCCGGGGGG
 GGCTTGGTGCAGCCTGGGGGGAGCCTGAGACTGAGTTGTGCCGCTCTGGGGTGAC
 ATTTAACTACTATGGCATGTCTTGGATCCGCCAGGCACCTGGAAAGGGCCTGGAGT
 GGGTGGCCAGCATCACTAGGTCCGGCGGGCGAATCTACTATCCCGACAGCGTCAAG
 GGCAGGTTCACAATTTCCCGCGAGAACACACAGAAAACCTCTGTACCTGCAGATGAA
 TAGCCTGAGAGCCGAAGATACAGCTGTGTACTATTGCACTCTGGACGGCAGGGATG
 GGTGGGTCGCCTATTGGGGGGCAGGGAACCTGGTGACAGTCAGCTCCGGAGGAGGA
 GGATCTGGCGGAGGAGGCAGTGGGGGAGGCGGGTCAAACCTTATGCTCACTCAGCC
 GTCCTCTGTTTTCTGGCGTACCTGGCCAACGGGTGACCATTAGCTGTACGGGTAATA
 CCGGGAATATCGGGTCTAACTACGTGAACTGGTATCAGCAGCTTCCAGGGACAGCT
 CCCAAGTTGCTGATCTATCGCGACGACAAAAGACCCTCAGGGGTCCCTGACCGATT
 TAGTGGCAGCAAAAGCGGTACTTCCGCTTCCCTGGCGATAACCGGCTTTCAGGCCG
 AAGATGAGGCAGACTACTATTGCCAGTCATATTCCAGCGGCTTCATCTTCGGAGGC
 GGAACCTAAGCTGACAGTGTGGCAGCCGAGCCCAAATCTTCTGACAAAACCTCACAC
 ATGCCACCCCTGCCAGCACCTGAAGCCGAGGGGGACCCTCAGTCTTCTCTTCC
 CCCCCAAACCAAGGACACCCTCATGATCTCCCGGACCCCTGAGGTCACATGCGTG
 GTGGTGTGAGTGAAGCACGAAGACCCTGAGGTCAAGTTCAACTGGTACGTGGACGG
 CGTGGAGGTGCATAATGCCAAGACAAAGCCCCGGGAGGAGCAGTACAACAGCACGT
 ACCGGGTGGTGCAGCGTCCCTACCGTCTGACCAGGACTGGCTGAATGGCAAGGAG
 TACAAGTGCAAGGTCTCCAACAAAGCCCTCCAGCCCCCATCGAGAAAACCATCTC
 CAAAGCCAAAGGCCAGCCCCGGGAACACAGGTGTACGTGTACCCCCATCCCGGG
 ACGAGCTGACCAAGAACCAGGTGAGCCTGACCTGCCTGGTCAAAGGCTTCTATCCC
 AGCGACATCGCCGTGGAGTGGGAGAGCAATGGCCAGCCGAGAACAACTACAAGAC
 CACCCCTCCCGTGCTGGACTCCGACGGCTCCTTCGCCCTCGTGAGCAAGCTCACCG
 TGGACAAGAGCAGGTGGCAGCAGGGCAACGTCTTCTCATGCTCCGTGATGCATGAG
 GCTCTGCACAACCACTACACCCAGAAGAGCCTCTCCCTGTCTCCCGGCAAG
 (SEQ ID NO: 79), signal sequence (1-57), NYA-2045 (61-
 798), linker (799-813), C3E-7085 (814-1533), linker
 (1534-1539), hinge (1540-1584), CH2 (1585-1914), CH3
 (1915-2235)

Fig. 81

NYF-0047-HC2

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
 CGGTCAAGTACAGCTGGTGGGAATCCGGTGGAGGCGTGGTCCAGCCGGGACGCAGCT
 TGAGACTGTCCTGCGCTGCATCTGGCTTTTCCATACGATCCTACGATATGCACTGG
 GTTCGCCAAGCCCCTGGCAAAGGTCTTGAATGGGTGCCACTATCTCCTACGACGG
 GTCTCAGAAATATTACGCGGATTCAGTGAAGGGGCGGTTTACAATTTACGGGACA
 ATTCAAAGAATACCTTGTATCTCCAGATGTCTTCACTGAGAGCCGAGGATACTGCA
 GTCTATCATTGTGCAAGAGGGTCTCAGGCCACTACGAGGCCCTTTGATATATGGGG
 CCAAGGCACCTTGGTAACCGTTAGTAGCGGAGGTGGAGGAAGCGGAGGCGGCGGTT
 CCGGAGGAGGTGGAAGTCAATCCGTTCTGACTCAGCCACCCTCCGTGAGTGCAGCG
 CCTGGACAGAAGGTCACAATATCATGTTCTGGCTCTTCAAGCAACATCGGAAACTG
 GTACGTGAGCTGGTACCAGCAGCTCCCCGGCACGGCGCCCAAGCTTCTGATATATG
 ACAACAACAAACGGCCCAAGTGAATCCCTGACAGATTCAGTGGGTCTAAAAGTGGT
 ACAAGCGCTACCCCTGGGTATCACCGGATTGCAGACCGGAGATGAGGCGGACTATTA
 TTGCGGAACCTGGGACTCCAGCCTGAGCGCTCCCTGGGTTTTCGGCGGAGGTACTA
 AGGTTACCGTCTTGGGAGGAGGCGGTTTCAAGTGCAGCTGGTGGAAATCCGGGGGG
 GGCTTGGTGCAGCCTGGGGGGAGCCTGAGACTGAGTTGTGCCGCTCTGGGGTGAC
 ATTTAACTACTATGGCATGTCTTGGATCCGCCAGGCACCTGGAAAGGGCCTGGAGT
 GGGTGGCCAGCATCACTAGGTCCGGCGGGCGAATCTACTATCCCGACAGCGTCAAG
 GGCAGGTTTACAATTTCCCGCGAGAACACACAGAAAACCTCTGTACCTGCAGATGAA
 TAGCCTGAGAGCCGAAGATACAGCTGTGTACTATTGCACTCTGGACGGCAGGGATG
 GGTGGGTCGCTATTGGGGGGCAGGGAACCTGGTGACAGTCAGCTCCGGAGGAGGA
 GGATCTGGCGGAGGAGGCAGTGGGGGAGGCGGGTCAAACCTTATGCTCACTCAGCC
 GTCCTCTGTTTTCTGGCGTACCTGGCCAACGGGTGACCATTAGCTGTACGGGTAATA
 CCGGGAATATCGGGTCTAACTACGTGAACTGGTATCAGCAGCTTCCAGGGACAGCT
 CCCAAGTTGCTGATCTATCGCGACGACAAAAGACCCTCAGGGGTCCCTGACCGATT
 TAGTGGCAGCAAAAGCGGTACTTCCGCTTCCCTGGCGATAACCGGCTTTCAGGCCG
 AAGATGAGGCAGACTACTATTGCCAGTCATATTCCAGCGGCTTCATCTTCGGAGGC
 GGAACCTAAGCTGACAGTGTTGGCAGCCGAGCCCAAATCTTCTGACAAAACCTCACAC
 ATGCCACCCCTGCCCAGCACCTGAAGCCGAGGGGGACCCTCAGTCTTCTCTTCC
 CCCCCAAACCAAGGACACCCTCATGATCTCCCGGACCCCTGAGGTCACATGCGTG
 GTGGTGTCAGTGAGCCACGAAGACCCTGAGGTCAAGTTCAACTGGTACGTGGACGG
 CGTGGAGGTGCATAATGCCAAGACAAAGCCCCGGGAGGAGCAGTACAACAGCACGT
 ACCGGGTGGTCAAGCTCCTCACCGTCTGCACCAGGACTGGCTGAATGGCAAGGAG
 TACAAGTGCAAGGTCTCCAACAAAGCCCTCCCAGCCCCATCGAGAAAACCATCTC
 CAAAGCCAAAGGCCAGCCCCGGGAACACAGGTGTACGTGTACCCCCATCCCGGG
 ACGAGCTGACCAAGAACCAGGTGAGCCTGACCTGCCTGGTCAAAGGCTTCTATCCC
 AGCGACATCGCCGTGGAGTGGGAGAGCAATGGCCAGCCGAGAACAACTACAAGAC
 CACCCCTCCCGTGCTGGACTCCGACGGCTCCTTCGCCCTCGTGAGCAAGCTCACCG
 TGGACAAGAGCAGGTGGCAGCAGGGCAACGTCTTCTCATGCTCCGTGATGCATGAG
 GCTCTGCACAACCACTACACCCAGAAGAGCCTCTCCCTGTCTCCCGGCAAG
 (SEQ ID NO: 80), signal sequence (1-57), NYA-2047 (61-
 798), linker (799-813), C3E-7085 (814-1533), linker
 (1534-1539), hinge (1540-1584), CH2 (1585-1914), CH3
 (1915-2235)

Fig. 82

NYF-0048-HC2

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
CGGACAAGTTCAACTCGTGGAATCTGGTGGAGGGGTGGTACAACCAGGCCGGTCAC
TTAGACTGTCCTGCGCTGCGAGTGGATTTTCCATCAGATCTTACGACATGCACTGG
GTTTCGCCAAGCACCCGGAAAGGGTTTGGAAATGGGTGGCTACGATCTCCTACGATGG
ATCCCAGAAGTATTACGCCGACAGCGTCAAGGGCCGATTTACAATATCACGGGACA
ATTCAAAGAATACCTTGTATCTCCAGATGTCTTCACTGAGAGCCGAGGATACTGCA
GTCTATCATTGTGCAAGAGGGTCTCAGGCCACTACGAGGCCCTTTGATATATGGGG
CCAAGGCACCTTGGTAACCGTTAGTAGCGGAGGTGGAGGAAGCGGAGGCCGGCGGTT
CCGGAGGAGGTGGAAGTCAATCCGTTCTGACTCAGCCACCCTCCGCCCTCTGGCACC
CCGGGCCAACGGGTCAACAATATCATGTTCTGGCTCTTCAAGCAACATCGGAAACTG
GTACGTGAGCTGGTACCAGCAGCTCCCCGGCACGGCGCCCAAGCTTCTGATATATG
ACAACAACAAACGGCCCAAGTGGAGTTCCTGACAGATTCAGTGGGTCTAAAAGTGGT
ACAAGCGCTAGCCTGGCCATAAGTGGTCTGCAGAGTGAAGATGAGGCGGACTATTA
TTGCGGAACCTGGGACTCCAGCCTGAGCGCTCCCTGGGTTTTTCGGCGGAGGTACTA
AGCTGACCGTCTTGGGAGGAGGCGGTTTCAAGTGCAGCTGGTGGAAATCCGGGGGG
GGCCTGGTGCAGCCTGGGGGGAGCCTGAGACTGAGTTGTGCCGCCCTCTGGGGTGAC
ATTTAACTACTATGGCATGTCTTGGATCCGCCAGGCACCTGGAAAGGGCCTGGAGT
GGGTGGCCAGCATCACTAGGTCCGGCGGGCGAATCTACTATCCCGACAGCGTCAAG
GGCAGGTTCACAATTTCCCGCGAGAACACACAGAAAACCTCTGTACCTGCAGATGAA
TAGCCTGAGAGCCGAAGATACAGCTGTGTACTATTGCACTCTGGACGGCAGGGATG
GGTGGGTGCGCTATTGGGGGGCAGGGAACCTGGTGACAGTCAGCTCCGGAGGAGGA
GGATCTGGCGGAGGAGGCAGTGGGGGAGGCGGGTCAAACCTTATGCTCACTCAGCC
GTCCTCTGTTTTCTGGCGTACCTGGCCAACGGGTGACCATTAGCTGTACGGGTAATA
CCGGGAATATCGGGTCTAACTACGTGAACTGGTATCAGCAGCTTCCAGGGACAGCT
CCCAAGTTGCTGATCTATCGCGACGACAAAAGACCCTCAGGGGTCCCTGACCGATT
TAGTGGCAGCAAAAGCGGTACTTCCGCTTCCCTGGCGATAACCGGCTTTCAGGCCG
AAGATGAGGCAGACTACTATTGCCAGTCATATTCCAGCGGCTTCATCTTCGGAGGC
GGAACCTAAGCTGACAGTGTGGCAGCCGAGCCCAAATCTTCTGACAAAACCTCACAC
ATGCCACCCCTGCCCAGCACCTGAAGCCGAGGGGGACCCTCAGTCTTCTCTTCC
CCCCAAAACCCAAGGACACCCTCATGATCTCCCGGACCCCTGAGGTCACATGCGTG
GTGGTGTCAGTGAGCCACGAAGACCCTGAGGTCAAGTTCAACTGGTACGTGGACGG
CGTGGAGGTGCATAATGCCAAGACAAAGCCCCGGGAGGAGCAGTACAACAGCACGT
ACCGGGTGGTCAGCGTCCCTACCGTCTTGCACCAGGACTGGCTGAATGGCAAGGAG
TACAAGTGCAAGGTCTCCAACAAAGCCCTCCCAGCCCCATCGAGAAAACCATCTC
CAAAGCCAAAGGCCAGCCCCGGGAACCACAGGTGTACGTGTACCCCCATCCCGGG
ACGAGCTGACCAAGAACCAGGTGAGCCTGACCTGCCTGGTCAAAGGCTTCTATCCC
AGCGACATCGCCGTGGAGTGGGAGAGCAATGGCCAGCCGAGAACAACTACAAGAC
CACCCCTCCCGTGCTGGACTCCGACGGCTCCTTCGCCCTCGTGAGCAAGCTCACCG
TGGACAAGAGCAGGTGGCAGCAGGGCAACGTCTTCTCATGCTCCGTGATGCATGAG
GCTCTGCACAACCACTACACCCAGAAGAGCCTCTCCCTGTCTCCCGGCAAG
(SEQ ID NO: 81), signal sequence (1-57), NYA-2048 (61-
798), linker (799-813), C3E-7085 (814-1533), linker
(1534-1539), hinge (1540-1584), CH2 (1585-1914), CH3
(1915-2235)

Fig. 83

NYF-0060-HC2

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
CGGTGAGGTACAGCTGGTGGGAATCCGGTGGAGGCCTTGTGCAACCGGGAGGCTCCC
TGCGACTCTCTTGCGCCGCATCTGGCTTTTCCATACGATCCTACGATATTCATTGG
GTCCGACAAGCCCCTGGCAAAGGTCTTGAATGGGTGCCACTATCTCCTACGACGG
GTCTCAGAAATATTATGCGGATTCAGTGAAGGGGCGGTTTACAATTCACGAGACG
AGTCAAAGAATACACTGTACCTCCAAATGAATTCAGTGAAGGCCGAGGATACTGCA
GTCTATCATTGTGCAAGAGGGTCCCTCAGGCCACTACGAGGCCCTTTGATATATGGGG
CCAAGGCACCTTGGTAACCGTTAGTAGCGGAGGTGGAGGAAGCGGAGGCGGCGGTT
CCGGAGGAGGTGGAAGTCAATCCGTTCTGACTCAGCCACCCTCCGTGAGTGCAGCG
CCTGGACAGAAGGTCACAATATCATGTTCTGGCTCTTCAAGCAACATCGGAAACTG
GTACGTGAGCTGGTACCAGCAGCTCCCCGGCACGGCGCCCAAGCTTCTGATATATG
ACAACAACAAACGGCCCAAGTGAATCCCTGACAGATTCAGTGGGTCTAAAAGTGGT
ACAAGCGCTACCCCTGGGTATCACCGGATTGCAGACCGGAGATGAGGCGGACTATTA
TTGCGGAACCTGGGACTCCAGCCTGAGCGCTCCCTGGGTTTTCGGCGGAGGTACTA
AGGTTACCGTCTTGGGAGGAGGCGGTTTCAAGTGCAGCTGGTGAATCCGGGGGG
GGCCTGGTGCAGCCTGGGGGGAGCCTGAGACTGAGTTGTGCCGCTCTGGGGTGAC
ATTTAACTACTATGGCATGTCTTGGATCCGCCAGGCACCTGGAAAGGGCCTGGAGT
GGGTGGCCAGCATCACTAGGTCCGGCGGGCGAATCTACTATCCCGACAGCGTCAAG
GGCAGGTTTACAATTTCCCGCGAGAACACACAGAAAACCTCTGTACCTGCAGATGAA
TAGCCTGAGAGCCGAAGATACAGCTGTGTACTATTGCACTCTGGACGGCAGGGATG
GGTGGGTCGCTATTGGGGGGCAGGGAACCCCTGGTGACAGTCAGCTCCGGAGGAGGA
GGATCTGGCGGAGGAGGCAGTGGGGGAGGCGGGTCAAACCTTATGCTCACTCAGCC
GTCCTCTGTTTTCTGGCGTACCTGGCCAACGGGTGACCATTAGCTGTACGGGTAATA
CCGGGAATATCGGGTCTAACTACGTGAACTGGTATCAGCAGCTTCCAGGGACAGCT
CCCAAGTTGCTGATCTATCGCGACGACAAAAGACCCTCAGGGGTCCCTGACCGATT
TAGTGGCAGCAAAGCGGTACTTCCGCTTCCCTGGCGATAACCGGCTTTCAGGCCG
AAGATGAGGCAGACTACTATTGCCAGTCATATTCCAGCGGCTTCATCTTCGGAGGC
GGAACCTAAGCTGACAGTGTTGGCAGCCGAGCCCAAATCTTCTGACAAAACCTCACAC
ATGCCACCCCTGCCCAGCACCTGAAGCCGAGGGGGACCCTCAGTCTTCTCTTCC
CCCCAAAACCCAAGGACACCCTCATGATCTCCCGGACCCCTGAGGTACATGCGTG
GTGGTGTCAGTGAGCCACGAAGACCCTGAGGTCAAGTTCAACTGGTACGTGGACGG
CGTGGAGGTGCATAATGCCAAGACAAAGCCCCGGGAGGAGCAGTACAACAGCACGT
ACCGGGTGGTCAGCGTCCCTACCGTCTTGCACCAGGACTGGCTGAATGGCAAGGAG
TACAAGTGCAAGGTCTCCAACAAAGCCCTCCCAGCCCCATCGAGAAAACCATCTC
CAAAGCCAAAGGCCAGCCCCGGGAACCACAGGTGTACGTGTACCCCCATCCCGGG
ACGAGCTGACCAAGAACCAGGTGAGCCTGACCTGCCTGGTCAAAGGCTTCTATCCC
AGCGACATCGCCGTGGAGTGGGAGAGCAATGGCCAGCCGAGAACAACTACAAGAC
CACCCCTCCCGTGCTGGACTCCGACGGCTCCTTCGCCCTCGTGAGCAAGCTCACCG
TGGACAAGAGCAGGTGGCAGCAGGGCAACGTCTTCTCATGCTCCGTGATGCATGAG
GCTCTGCACAACCACTACACCCAGAAGAGCCTCTCCCTGTCTCCCGGCAAG

(SEQ ID NO: 82), signal sequence (1-57), NYA-2060 (61-798), linker (799-813), C3E-7085 (814-1533), linker (1534-1539), hinge (1540-1584), CH2 (1585-1914), CH3 (1915-2235)

Fig. 84

NYF-0061-HC2

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
CGGTGAGGTACAGCTGGTGGGAATCCGGTGGAGGCCTTGTGCAACCGGGAGGCTCCC
TGCGACTCTCTTGCGCCGCATCTGGCTTTTCCATACGATCCTACGATATTCATTGG
GTCCGACAAGCCCCTGGCAAAGGTCTTGAATGGGTGCCACTATCTCCTACGACGG
GTCTCAGAAATATTATGCGGATTCAGTGAAGGGGCGGTTTACAATTCACGAGACG
AGTCAAAGAATACTGTACCTCCAAATGAATTCAGTGAAGGCCGAGGATACTGCA
GTCTATCATTGTGCAAGAGGGTCTCAGGCCACTACGAGGCCCTTTGATATATGGGG
CCAAGGCACCTTGGTAACCGTTAGTAGCGGAGGTGGAGGAAGCGGAGGCGGCGGTT
CCGGAGGAGGTGGAAGTCAATCCGTTCTGACTCAGCCACCCTCCGCCCTCTGGCACC
CCGGGCCAACGGGTCAACAATATCATGTTCTGGCTCTTCAAGCAACATCGGAAACTG
GTACGTGAGCTGGTACCAGCAGCTCCCCGGCACGGCGCCCAAGCTTCTGATATATG
ACAACAACAAACGGCCCAAGTGGAGTTCCTGACAGATTCAGTGGGTCTAAAAGTGGT
ACAAGCGCTAGCCTGGCCATAAGTGGTCTGCAGAGTGAAGATGAGGCGGACTATTA
TTGCGGAACCTGGGACTCCAGCCTGAGCGCTCCCTGGGTTTTCGGCGGAGGTACTA
AGCTGACCGTCTTGGGAGGAGGCGGTTTCAAGTGCAGCTGGTGGAAATCCGGGGGG
GGCCTGGTGCAGCCTGGGGGGAGCCTGAGACTGAGTTGTGCCGCCCTCTGGGGTGAC
ATTTAACTACTATGGCATGTCTTGGATCCGCCAGGCACCTGGAAAGGGCCTGGAGT
GGGTGGCCAGCATCACTAGGTCCGGCGGGCGAATCTACTATCCCGACAGCGTCAAG
GGCAGGTTCACAATTTCCCGCGAGAACACACAGAAAACCTCTGTACCTGCAGATGAA
TAGCCTGAGAGCCGAAGATACAGCTGTGTACTATTGCACTCTGGACGGCAGGGATG
GGTGGGTGCGCTATTGGGGGGCAGGGAACCTGGTGACAGTCAGCTCCGGAGGAGGA
GGATCTGGCGGAGGAGGCAGTGGGGGAGGCGGGTCAAACCTTATGCTCACTCAGCC
GTCCTCTGTTTCTGGCGTACCTGGCCAACGGGTGACCATTAGCTGTACGGGTAATA
CCGGGAATATCGGGTCTAACTACGTGAACTGGTATCAGCAGCTTCCAGGGACAGCT
CCCAAGTTGCTGATCTATCGCGACGACAAAAGACCCTCAGGGGTCCCTGACCGATT
TAGTGGCAGCAAAGCGGTACTTCCGCTTCCCTGGCGATAACCGGCTTTCAGGCCG
AAGATGAGGCAGACTACTATTGCCAGTCATATTCCAGCGGCTTCATCTTCGGAGGC
GGAACCTAAGCTGACAGTGTGGCAGCCGAGCCCAAATCTTCTGACAAAACCTCACAC
ATGCCCACCCTGCCCAGCACCTGAAGCCGAGGGGGACCCTCAGTCTTCTCTTCC
CCCCAAAACCAAGGACACCCTCATGATCTCCCGGACCCCTGAGGTCACATGCGTG
GTGGTGTGAGTGAAGCACGAAGACCCTGAGGTCAAGTTCAACTGGTACGTGGACGG
CGTGGAGGTGCATAATGCCAAGACAAAGCCCCGGGAGGAGCAGTACAACAGCACGT
ACCGGGTGGTCAAGCTCCTCACCGTCTGCACCAGGACTGGCTGAATGGCAAGGAG
TACAAGTGCAAGGTCTCCAACAAAGCCCTCCAGCCCCCATCGAGAAAACCATCTC
CAAAGCCAAAGGCCAGCCCCGGGAACCACAGGTGTACGTGTACCCCCATCCCGGG
ACGAGCTGACCAAGAACCAGGTGAGCCTGACCTGCCTGGTCAAAGGCTTCTATCCC
AGCGACATCGCCGTGGAGTGGGAGAGCAATGGCCAGCCGAGAACAACTACAAGAC
CACCCCTCCCGTGCTGGACTCCGACGGCTCCTTCGCCCTCGTGAGCAAGCTCACCG
TGGACAAGAGCAGGTGGCAGCAGGGCAACGTCTTCTCATGCTCCGTGATGCATGAG
GCTCTGCACAACCACTACACCCAGAAGAGCCTCTCCCTGTCTCCCGGCAAG
(SEQ ID NO: 83), signal sequence (1-57), NYA-2061 (61-
798), linker (799-813), C3E-7085 (814-1533), linker
(1534-1539), hinge (1540-1584), CH2 (1585-1914), CH3
(1915-2235)

Fig. 85

HC1

MKHLWFFLLLVAAPRWVLSDKTHTCPPCPAPEAAGGPSVFLFPPKPKDTLMISRTPEVTCVTVSVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYVLPISRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYLTWPPVLDSDGSFFLYSKLTVDKSRWQQGNVSCSVMHEALHNHYTQKSLSLSPGK

(SEQ ID NO: 84), signal sequence (1-19), hinge (20-29), CH2 (30-139), CH3 (140-246)

Fig. 86

NYF-0016-HC

MKHLWFFLLLVAAAPRWVLSGEVQLVESGGGVVQPGRSLTPSCSASGFSIRSYDIHW
VRLAPGKGLEWVATISYDGSQKYFADSVKGRFTIFRDESENMVYLQMSGRLVEDTA
VYHCARGSSGHYEAFDIWGQGTMTVTVSSGGGGSGGGSGGGGSQSVLTQPPSVSAA
PGQKVTISCSGSSSNIGNWYVSWYQQLPGTAPKLLIYDNNKRPSGIPDRFSGSKSG
TSATLGITGLQTGDEADYYCGTWDSSLSAPWVFGGGTKVTVLGGGGSEVQLVESGG
GLVQPGGSLRLSCAASGVTFNYYGMSWIRQAPGKGLEWVASITRSGGRIIYPDSVK
GRFTISRENTQKTLYLQMNSLRAEDTAVYYCTLDGRDGWVAYWGQGTTLVTVSSGGG
GSGGGGSGGGGSNFMLTQPSSVSGVPGQRTISCTGNTGNIGSNYVNWYQQLPGTA
PKLLIYRDDKRPSGVPDRFSGSKSGTSASLAITGFQAEDEADYYCQSYSSGFIFGG
GTKLTVLAAEPKSSDKTHTCPPCPAPEAAGGPSVFLFPPKPKDTLMISRTPEVTCV
VVSVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKE
YKCKVSNKALPAPIEKTISKAKGQPREPQVYVPPSRDELTKNQVSLTCLVKGFYP
SDIAVEWESNGQPENNYKTTTPVLDSDGSFALVSKLTVDKSRWQQGNVFSQVMHE
ALHNHYTQKSLSLSPGK

(SEQ ID NO: 85), signal sequence (1-19), NYA-1143 (21-266), linker (267-271), C3E-7085 (272-511), linker (512-513), hinge (514-528), CH2 (529-638), CH3 (639-745)

Fig. 87

NYF-0019-HC2

MKHLWFFLLLVAAAPRWVLSGEVQLVESGGGVVQPGRSLTPSCSASGFSIRSYDIHW
VRLAPGKGLEWVATISYDGSQKYFADSVKGRFTIFRDESENMVYLQMSGRLVEDTA
VYHCARGSSGHYEAFDIWGQGTMTVTVSSGGGGSGGGGSGGGGSQSVLTQPPSVSAA
PGQKVTISCSGSSSNIGNWYVSWYQRLPGTAPKLLIYDNNKRPSGIPDRFSGSKSG
TSATLGITGLQTGDEADYYCGTWDSSLSAPWVFGGGTKVTVLGGGGSEVQLVESGG
GLVQPGGSLRLSCAASGVTFNYYGMSWIRQAPGKGLEWVASITRSGGRIIYPDSVK
GRFTISRENTQKTLYLQMNSLRAEDTAVYYCTLDGRDGWVAYWGQGTTLVTVSSGGG
GSGGGGSGGGGSNFMLTQPSSVSGVPGQRTISCTGNTGNIGSNYVNWYQQLPGTA
PKLLIYRDDKRPSGVPDRFSGSKSGTSASLAITGFQAEDEADYYCQSYSSGFIFGG
GTKLTVLAAEPKSSDKTHTCPPCPAPEAAGGPSVFLFPPKPKDTLMISRTPEVTCV
VVSVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKE
YKCKVSNKALPAPIEKTISKAKGQPREPQVYVPPSRDELTKNQVSLTCLVKGFYP
SDIAVEWESNGQPENNYKTTTPVLDSDGSFALVSKLTVDKSRWQQGNVFSQVMHE
ALHNHYTQKSLSLSPGK

(SEQ ID NO: 86), signal sequence (1-19), NYA-2143 (21-266), linker (267-271), C3E-7085 (272-511), linker (512-513), hinge (514-528), CH2 (529-638), CH3 (639-745)

Fig. 88

NYF-0022-HC2

MKHLWFFLLLVAAAPRWVLSGEVQLVESGGGVVQPGRSLTPSCSASGFSIRSYDIHW
VRLAPGKGLEWVATISYDGSQKYFADSVKGRFTIFRDESENMVYLMQMSGLRVEDTA
VYHCARGSSGHYEAFDIWGQGTMTVTVSSGGGGSGGGGSGGGGSQSVLTQPPSVSAA
PGQKVTISCSGSSSNIGNNYVSWYQQLPGTAPKLLIYDNNKRPSGIPDRFSGSKSG
TSATLGITGLQTGDEADYYCGTWDSSLSAPWVFGGGTKVTVLGGGGSEVQLVESGG
GLVQPPGSLRLSCAASGVTFNYYGMSWIRQAPGKGLEWVASITRSGGRIIYPDSVK
GRFTISRENTQKTLYLQMNSLRAEDTAVYYCTLDGRDGVVAYWGQGTILVTVSSGGG
GSGGGGSGGGGSNFMLTQPSSVSGVPGQRTISCTGNTGNIGSNYVNWYQQLPGTA
PKLLIYRDDKRPSGVPDRFSGSKSGTSASLAITGFQAEDEADYYCQSYSSGFIFGG
GTKLTVLAAEPKSSDKTHTCPPCPAPEAAGGPSVFLFPPKPKDTLMISRTPEVTCV
VVSVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKE
YKCKVSNKALPAPIEKTISKAKGQPREPQVYVYPPSRDELTKNQVSLTCLVKGFYP
SDIAVEWESNGQPENNYKTTTPVLDSDGSFALVSKLTVDKSRWQQGNVFSQVMHE
ALHNHYTQKSLSLSPGK

(SEQ ID NO: 87), signal sequence (1-19), NYA-1163 (21-266), linker (267-271), C3E-7085 (272-511), linker (512-513), hinge (514-528), CH2 (529-638), CH3 (639-745)

Fig. 89

NYF-0023-HC2

MKHLWFFLLLVAAAPRWVLSGEVQLVESGGGVVQPGRSLTPSCSASGFSIRSYDIHW
VRQAPGKGLEWVATISYDGSQKYFADSVKGRFTIFRDESENMVYLMQSGLRVEDTA
VYHCARGSSGHYEAFDIWGQGTMTVTVSSGGGGSGGGGSGGGGSQSVLTQPPSVSAA
PGQKVTISCSGSSSNIGNWYVSWYQQLPGTAPKLLIYDNNKRPSGIPDRFSGSKSG
TSATLGITGLQTGDEADYYCGTWDSSLSAPWVFGGGTKVTVLGGGGSEVQLVESGG
GLVQPGGSLRLSCAASGVTFNYYGMSWIRQAPGKGLEWVASITRSGGRIIYPDSVK
GRFTISRENTQKTLYLQMNSLRAEDTAVYYCTLDGRDGWVAYWGQGTTLVTVSSGGG
GSGGGGSGGGGSNFMLTQPSSVSGVPGQRTISCTGNTGNIGSNYVNWYQQLPGTA
PKLLIYRDDKRPSGVPDRFSGSKSGTSASLAITGFQAEDEADYYCQSYSSGFIFGG
GTKLTVLAAEPKSSDKTHTCPPCPAPEAAGGPSVFLFPPKPKDTLMISRTPEVTCV
VVSVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKE
YKCKVSNKALPAPIEKTISKAKGQPREPQVYVYPPSRDELTKNQVSLTCLVKGFYP
SDIAVEWESNGQPENNYKTTTPVLDSDGSFALVSKLTVDKSRWQQGNVFSQVMHE
ALHNHYTQKSLSLSPGK

(SEQ ID NO: 88), signal sequence (1-19), NYA-2023 (21-266), linker (267-271), C3E-7085 (272-511), linker (512-513), hinge (514-528), CH2 (529-638), CH3 (639-745)

Fig. 90

NYF-0027-HC2

MKHLWFFLLLVAAAPRWVLSGEVQLVESGGGVVQPGRSLTPSCSASGFSIRSYDIHW
VRQAPGKGLEWVATISYDGSQKYFADSVKGRFTIFRDESENMVYLQMSGRLVEDTA
VYHCARGSSGHYEAFDIWGQGTMTVTVSSGGGGSGGGSGGGGSQSVLTQPPSVSAA
PGQKVTISCSGSSSNIGNNYVSWYQQLPGTAPKLLIYDNNKRPSGIPDRFSGSKSG
TSATLGITGLQTGDEADYYCGSWDSSLSAPWVFGGGTKVTVLGGGGSEVQLVESGG
GLVQPPGSLRLSCAASGVTFNYYGMSWIRQAPGKGLEWVASITRSGGRIIYPDSVK
GRFTISRENTQKTLYLQMNSLRAEDTAVYYCTLDGRDGWVAYWGQGTTLVTVSSGGG
GSGGGGSGGGGSNFMLTQPSSVSGVPGQRTISCTGNTGNIGSNYVNWYQQLPGTA
PKLLIYRDDKRPSGVPDRFSGSKSGTSASLAITGFQAEDEADYYCQSYSSGFIFGG
GTKLTVLAAEPKSSDKTHTCPPCPAPEAAGGPSVFLFPPKPKDTLMISRTPEVTCV
VVSVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKE
YKCKVSNKALPAPIEKTISKAKGQPREPQVYVYPPSRDELTKNQVSLTCLVKGFYP
SDIAVEWESNGQPENNYKTTTPVLDSDGSFALVSKLTVDKSRWQQGNVFSCSVMHE
ALHNHYTQKSLSLSPGK

(SEQ ID NO: 89), signal sequence (1-19), NYA-2027 (21-
266), linker (267-271), C3E-7085 (272-511), linker (512-
513), hinge (514-528), CH2 (529-638), CH3 (639-745)

Fig. 91

NYF-0035-HC2

MKHLWFFLLLVAAAPRWVLSGEVQLVESGGGVVQPGRSLTPSCSASGFSIRSYDIHW
VRLAPGKGLEWVATISYDGSQKYFADSVKGRFTIFRDESENMVYLQMSGRLVEDTA
VYHCARGSSGHYEAFDIWGQGTMTVTVSSGGGGSGGGSGGGGSQSVLTQPPSVSAA
PGQKVTISCSGSSSNIGNWKVSWYQQLPGTAPKLLIYDNNKRPSGIPDRFSGSKSG
TSATLGITGLQTGDEADYYCGTWDSSLSAPWVFGGGTKVTVLGGGGSEVQLVESGG
GLVQPPGSLRLSCAASGVTFNYYGMSWIRQAPGKGLEWVASITRSGGRIIYPDSVK
GRFTISRENTQKTLYLQMNSLRAEDTAVYYCTLDGRDGVVAYWGQGTILVTVSSGGG
GSGGGGSGGGGSNFMLTQPSSVSGVPGQRTISCTGNTGNIGSNYVNWYQQLPGTA
PKLLIYRDDKRPSGVPDRFSGSKSGTSASLAITGFQAEDEADYYCQSYSSGFIFGG
GTKLTVLAAEPKSSDKTHTCPPCPAPEAAGGPSVFLFPPKPKDTLMISRTPEVTCV
VVSVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKE
YKCKVSNKALPAPIEKTISKAKGQPREPQVYVYPPSRDELTKNQVSLTCLVKGFYP
SDIAVEWESNGQPENNYKTTTPVLDSDGSFALVSKLTVDKSRWQQGNVFSVSMHE
ALHNHYTQKSLSLSPGK

(SEQ ID NO: 90), signal sequence (1-19), NYA-2035 (21-266), linker (267-271), C3E-7085 (272-511), linker (512-513), hinge (514-528), CH2 (529-638), CH3 (639-745)

Fig. 92

NYF-0044-HC2

MKHLWFFLLLVAAAPRWVLSGEVQLVESGGGLVQPGGSLRLSCAASGFSIRSYDIHW
VRLAPGKGLEWVATISYDGSQKYYADSVKGRFTISRDESKNTLYLQMNSLRAEDTA
VYHCARGSSGHYEAFDIWGQGTTLVTVSSGGGGSGGGSGGGGSQSVLTQPPSVSAA
PGQKVTISCSGSSSNIGNWYVSWYQQLPGTAPKLLIYDNNKRPSGIPDRFSGSKSG
TSATLGITGLQTGDEADYYCGTWDSSLSAPWVFGGGTKVTVLGGGGSEVQLVESGG
GLVQPGGSLRLSCAASGVTFNYYGMSWIRQAPGKGLEWVASITRSGGRIIYPDSVK
GRFTISRENTQKTLYLQMNSLRAEDTAVYYCTLDGRDGVVAYWGQGTTLVTVSSGGG
GSGGGSGGGGSNFMLTQPSSVSGVPGQRTISCTGNTGNIGSNYVNWYQQLPGTA
PKLLIYRDDKRPSGVPDRFSGSKSGTSASLAITGFQAEDEADYYCQSYSSGFIFGG
GTKLTVLAAEPKSSDKTHTCPPCPAPEAAGGPSVFLFPPKPKDTLMISRTPEVTCV
VVSVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKE
YKCKVSNKALPAPIEKTISKAKGQPREPQVYVYPPSRDELTKNQVSLTCLVKGFYP
SDIAVEWESNGQPENNYKTTTPVLDSDGSFALVSKLTVDKSRWQQGNVFSQVMHE
ALHNHYTQKSLSLSPGK

(SEQ ID NO: 91), signal sequence (1-19), NYA-2044 (21-
266), linker (267-271), C3E-7085 (272-511), linker (512-
513), hinge (514-528), CH2 (529-638), CH3 (639-745)

Fig. 93

NYF-0045-HC2

MKHLWFFLLLVAAAPRWVLSGEVQLVESGGGLVQPGGSLRLSCAASGFSIRSYDIHW
VRLAPGKGLEWVATISYDGSQKYYADSVKGRFTISRDESKNTLYLQMNSLRAEDTA
VYHCARGSSGHYEAFDIWGQGTTLVTVSSGGGGSGGGGSGGGGSQSVLTQPPSASGT
PGQRTVISCSSSSNIGNWYVSWYQQLPGTAPKLLIYDNNKRPSGVPDRFSGSKSG
TSASLAISGLQSEDEADYYCGTWDSSLSAPWVFGGGTKLTVLGGGGSEVQLVESGG
GLVQPGGSLRLSCAASGVTFNYYGMSWIRQAPGKGLEWVASITRSGGRIIYPDSVK
GRFTISRENTQKTLYLQMNSLRAEDTAVYYCTLDGRDGWVAYWGQGTTLVTVSSGGG
GSGGGGSGGGGSNFMLTQPSSVSGVPGQRTVISCSTGNTGNIGSNYVNWYQQLPGTA
PKLLIYRDDKRPSGVPDRFSGSKSGTSASLAITGFQAEDEADYYCQSYSSGFIFGG
GTKLTVLAAEPKSSDKTHTCPPCPAPEAAGGPSVFLFPPKPKDTLMISRTPEVTCV
VVSVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKE
YKCKVSNKALPAPIEKTISKAKGQPREPQVYVPPSRDELTKNQVSLTCLVKGFYP
SDIAVEWESNGQPENNYKTTTPVLDSDGSFALVSKLTVDKSRWQQGNVFSQVMHE
ALHNHYTQKSLSLSPGK

(SEQ ID NO: 92), signal sequence (1-19), NYA-2045 (21-266), linker (267-271), C3E-7085 (272-511), linker (512-513), hinge (514-528), CH2 (529-638), CH3 (639-745)

Fig. 94

NYF-0047-HC2

MKHLWFFLLLVAAAPRWVLSGQVQLVESGGGVVQPGRSLRLSCAASGFSIRSYDMHW
VRQAPGKGLEWVATISYDGSQKYYADSVKGRFTISRDN SKNTLYLQMSSLRAEDTA
VYHCARGSSGHYEAFDIWGQGT LVT VSSGGGGSGGGSGGGGSQSVLTQPPSVSAA
PGQKVTISCSGSSSNIGNWYVSWYQQLPGTAPKLLIYDNNKRPSGIPDRFSGSKSG
TSATLGITGLQTGDEADYYCGTWDSSLSAPWVFGGGTKVTVLGGGGSEVQLVESGG
GLVQPPGSLRLSCAASGVTFNYYGMSWIRQAPGKGLEWVASITRSGGRIIYPDSVK
GRFTISRENTQKTLYLQMNSLRAEDTAVYYCTLDGRDGWVAYWGQGT LVT VSSGGG
GSGGGSGGGGSNFMLTQPSSVSGVPGQRTISCTGNTGNIGSNYVNWYQQLPGTA
PKLLIYRDDKRPSGVPDRFSGSKSGTSASLAITGFQAEDEADYYCQSYSSGFIFGG
GTKLTVLAAEPKSSDKTHTCPPCPAPEAAGGPSVFLFPPKPKDTLMISRTPEVTCV
VVSVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKE
YKCKVSNKALPAPIEKTISKAKGQPREPQVYVYPPSRDELTKNQVSLTCLVKGFYP
SDIAVEWESNGQPENNYKTTTPVLDSDGSFALVSKLTVDKSRWQQGNVFSQVMHE
ALHNHYTQKSLSLSPGK

(SEQ ID NO: 93), signal sequence (1-19), NYA-2047 (21-
266), linker (267-271), C3E-7085 (272-511), linker (512-
513), hinge (514-528), CH2 (529-638), CH3 (639-745)

Fig. 95

NYF-0048-HC2

MKHLWFFLLLVAAAPRWVLSGQVQLVESGGGVVQPGRSLRLSCAASGFSIRSYDMHW
VRQAPGKGLEWVATISYDGSQKYYADSVKGRFTISRDN SKNTLYLQMSSLRAEDTA
VYHCARGSSGHYEAFDIWGQGT LVT VSSGGGGSGGGSGGGGSQSVLTQPPSASGT
PGQRTVISCSSSSNIGNWYVSWYQQLPGTAPKLLIYDNNKRPSGVPDRFSGSKSG
TSASLAISGLQSEDEADYYCGTWDSSLSAPWVFGGGTKLTVLGGGGSEVQLVESGG
GLVQPPGSLRLSCAASGVTFNYYGMSWIRQAPGKGLEWVASITRSGGRIIYPDSVK
GRFTISRENTQKTLYLQMNSLRAEDTAVYYCTLDGRDGWVAYWGQGT LVT VSSGGG
GSGGGSGGGGSNFMLTQPSSVSGVPGQRTVISC TGNTGNIGSNYVNWYQQLPGTA
PKLLIYRDDKRPSGVPDRFSGSKSGTSASLAITGFQAEDEADYYCQSYSSGFIFGG
GTKLTVLAAEPKSSDKTHTCPPCPAPEAAGGPSVFLFPPKPKDTLMISRTPEVTCV
VVSVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKE
YKCKVSNKALPAPIEKTISKAKGQPREPQVYVYPPSRDELTKNQVSLTCLVKGFYP
SDIAVEWESNGQPENNYKTTTPVLDSDGSFALVSKLTVDKSRWQQGNV FSCSV MHE
ALHNHYTQKSLSLSPGK

(SEQ ID NO: 94), signal sequence (1-19), NYA-2048 (21-
266), linker (267-271), C3E-7085 (272-511), linker (512-
513), hinge (514-528), CH2 (529-638), CH3 (639-745)

Fig. 96

NYF-0060-HC2

MKHLWFFLLLVAAAPRWVLSGEVQLVESGGGLVQPGGSLRLSCAASGFSIRSYDIHW
VRQAPGKGLEWVATISYDGSQKYYADSVKGRFTISRDESKNTLYLQMNSLRAEDTA
VYHCARGSSGHYEAFDIWGQGTTLVTVSSGGGGSGGGGSGGGGSQSVLTQPPSVSAA
PGQKVTISCSGSSSNIGNWYVSWYQQLPGTAPKLLIYDNNKRPSGIPDRFSGSKSG
TSATLGITGLQTGDEADYYCGTWDSSLSAPWVFGGGTKVTVLGGGGSEVQLVESGG
GLVQPGGSLRLSCAASGVTFNYYGMSWIRQAPGKGLEWVASITRSGGRIIYPDSVK
GRFTISRENTQKTLYLQMNSLRAEDTAVYYCTLDGRDGVVAYWGQGTTLVTVSSGGG
GSGGGGSGGGGSNFMLTQPSSVSGVPGQRTISCTGNTGNIGSNYVNWYQQLPGTA
PKLLIYRDDKRPSGVPDRFSGSKSGTSASLAITGFQAEDEADYYCQSYSSGFIFGG
GTKLTVLAAEPKSSDKTHTCPPCPAPEAAGGPSVFLFPPKPKDTLMISRTPEVTCV
VVSVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKE
YKCKVSNKALPAPIEKTISKAKGQPREPQVYVYPPSRDELTKNQVSLTCLVKGFYP
SDIAVEWESNGQPENNYKTTTPVLDSDGSFALVSKLTVDKSRWQQGNVFSQVMHE
ALHNHYTQKSLSLSPGK

(SEQ ID NO: 95), signal sequence (1-19), NYA-2060 (21-
266), linker (267-271), C3E-7085 (272-511), linker (512-
513), hinge (514-528), CH2 (529-638), CH3 (639-745)

Fig. 97

NYF-0061-HC2

MKHLWFFLLLVAAAPRWVLSGEVQLVESGGGLVQPGGSLRLSCAASGFSIRSYDIHW
VRQAPGKGLEWVATISYDGSQKYYADSVKGRFTISRDESKNTLYLQMNSLRAEDTA
VYHCARGSSGHYEAFDIWGQGTTLVTVSSGGGGSGGGSGGGGSQSVLTQPPSASGT
PGQRTVISCSSSSNIGNWYVSWYQQLPGTAPKLLIYDNNKRPSGVPDRFSGSKSG
TSASLAISGLQSEDEADYYCGTWDSSLSAPWVFGGGTKLTVLGGGGSEVQLVESGG
GLVQPGGSLRLSCAASGVTFNYYGMSWIRQAPGKGLEWVASITRSGGRIIYPDSVK
GRFTISRENTQKTLYLQMNSLRAEDTAVYYCTLDGRDGWVAYWGQGTTLVTVSSGGG
GSGGGSGGGGSNFMLTQPSSVSGVPGQRTVISCSTGNTGNIGSNYVNWYQQLPGTA
PKLLIYRDDKRPSGVPDRFSGSKSGTSASLAITGFQAEDEADYYCQSYSSGFIFGG
GTKLTVLAAEPKSSDKTHTCPPCPAPEAAGGPSVFLFPPKPKDTLMISRTPEVTCV
VVSVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKE
YKCKVSNKALPAPIEKTISKAKGQPREPQVYVPPSRDELTKNQVSLTCLVKGFYP
SDIAVEWESNGQPENNYKTTTPVLDSDGSFALVSKLTVDKSRWQQGNVFSQVMHE
ALHNHYTQKSLSLSPGK

(SEQ ID NO: 96), signal sequence (1-19), NYA-2061 (21-266), linker (267-271), C3E-7085 (272-511), linker (512-513), hinge (514-528), CH2 (529-638), CH3 (639-745)

Fig. 98

NYA-0001-Fab-HC1-k delete

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
CGAGGTGCAGCTGGTGGAGTCTGGGGGAGGCGTGGTCCAGCCTGGGAGGTCCCTGA
CACCCTCCTGTTTACGCTCTGGATTCTCCATCCGTAGTTATGATATACACTGGGTC
CGCCAGGCTCCAGGCAAGGGGCTAGAGTGGGTGGCCACTATATCATATGATGGAAG
TCAGAAGTACTTCGCAGACTCCGTGAAGGGCCGATTTACCATCTTCAGAGACGAAT
CGGAGAACATGGTGTATCTGCAAATGAGCGGCCTGAGAGTTGAGGACACGGCTGTT
TATCACTGTGCGAGAGGGAGTAGTGGTCAATTATGAGGCTTTTGATATCTGGGGCCA
AGGGACAATGGTCAACGTCTCTTCAGCCTCCACCAAGGGCCCAAGCGTCTTCCCCC
TGGCACCCCTCCTCCAAGAGCACCTCTGGCGGCACAGCCGCCCTGGGCTGCCTGGTC
AAGGACTACTTCCCCGAACCCGTGACCGTGAGCTGGAACCTCAGGCGCCCTGACCAG
CGGCGTGACACCTTCCCCGCTGTCTTGAGTCCCTCAGGACTCTACTCCCTCAGCA
GCGTGGTGACCGTGCCCTCCAGCAGCTTGGGCACCCAGACCTACATCTGCAACGTG
AATCACAAGCCCAGCAACACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTA
CAAACTCACACATGCCACCCCTGCCAGCACCTGAAGCCGCAGGGGGACCCCTCAG
TCTTCCTCTTCCCCCAAACCCAAAGGACACCCTCATGATCTCCCGGACCCCTGAG
GTCACATGCGTGGTGGTGTGAGTGAGCCACGAAGACCCTGAGGTCAAGTTCAACTG
GTACGTGGACGGCGTGGAGGTGCATAATGCCAAGACAAAGCCCCGGGAGGAGCAGT
ACAACAGCACGTACCGGGTGGTCAGCGTCCTCACCGTCCTGCACCAGGACTGGCTG
AATGGCAAGGAGTACAAGTGCAAGGTCTCCAACAAAGCCCTCCCAGCCCCCATCGA
GAAAACCATCTCCAAAGCCAAAGGCCAGCCCCGGGAACCACAGGTGTACGTGCTGC
CCCCATCCCGGGACGAGCTGACCAAGAACCAGGTGAGCCTGCTGTGCCTGGTCAA
GGCTTCTATCCCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGCCAGCCCGAGAA
CAACTACCTGACCTGGCCTCCCGTGCTGGACTCCGACGGCTCCTTCTCCTCTACA
GCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGGCAACGTCTTCTCATGCTCC
GTGATGCATGAGGCTCTGCACAACCACTACACCCAGAAGAGCCTCTCCCTGTCTCC
CGGC

(SEQ ID NO: 97), signal sequence (1-57), NYA-0001_VH
(58-417), CH1 (418-711), hinge (712-756), CH2 (757-1086),
CH3 (1087-1404)

Fig. 99

NYA-0001-LC

ATGGTGCTGCAGACCCAGGTGTTTCATCTCCCTGCTGCTGTGGATCTCCGGCGCGTA
CGGCCAGTCTGTGTTGACGCAGCCGCCCTCAGTGTCTGCGGCCCCAGGACAGAAGG
TCACCATCTCCTGCTCTGGAAGCAGCTCCAACATTGGGAATAATTATGTATCCTGG
TACCAGCAGCTCCCAGGAACAGCCCCAAACTCCTCATTTATGACAATAATAAGCG
ACCCTCAGGGATTCTGACCGATTCTCTGGCTCCAAGTCTGGCACGTCAGCCACCC
TGGGCATCACCGGACTCCAGACTGGGGACGAGGCCGATTATTACTGCGGAACATGG
GATAGCAGCCTGAGTGCCCCCTGGGTGTTTCGGCGGAGGGACCAAGGTCACCGTCCT
AGGCCAGCCTAAGGCTGCCCCCTAGCGTGACCCTGTTCCCTCCTTCCAGCGAGGAGC
TTCAAGCTAACAAGGCCACCCTGGTGTGTCTTATCTCTGACTTCTACCCTGGCGCT
GTGACCGTGGCCTGGAAGGCTGACAGCTCCCCTGTGAAGGCCGGAGTGGAGACCAC
CACACCTAGCAAGCAGTCTAACAACAAGTACGCTGCCAGCTCCTACCTGAGCCTTA
CCCCTGAGCAGTGGAAGTCTCACAGAAGCTACTCCTGTCAAGTGACCCACGAGGGC
AGCACCGTGGAGAAGACCGTGGCTCCTACCGAGTGTTCC

(SEQ ID NO: 98), signal sequence (1-60), NYA-0001_VL
(61-393), CL (394-711)

Fig. 100

NYA-0001-Fab-HC1-k delete

MKHLWFFLLLVAAAPRWVLSEVQLVESGGGVVQPGRSLTPSCSASGFSIRSYDIHWV
RQAPGKGLEWVATISYDGSQKYFADSVKGRFTIFRDESENMVYLQMSGLRVEDTAV
YHCARGSSGHYEAFDIWGQGMVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLV
KDYPPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNV
NKKPSNTKVDKRVEPKSCDKTHTCPPCPAPEAAGGPSVFLFPPKPKDTLMISRTPE
VTCVVVSVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWL
NGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYVLPFSRDELTKNQVSLTCLVK
GFYPSDIAVEWESNGQPENNYLTWPPVLDSDGSFFLYSKLTVDKSRWQQGNVFNCS
VMHEALHNHYTQKSLSLSPG

(SEQ ID NO: 99), signal sequence (1-19), NYA-0001_VH
(20-139), CH1 (140-237), hinge (238-252), CH2 (253-362),
CH3 (363-468)

Fig. 101

NYA-0001-LC

MVLQTQVFISLLLWISGAYGQSVLTQPPSVSAAPGQKVTISCSGSSSNIGNNYVSW
YQQLPGTAPKLLIYDNNKRPSGIPDRFSGSKSGTSATLGITGLQTGDEADYYCGTW
DSSLSAPWVFGGGTKVTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGA
VTVAWKADSSPVKAGVETTTTPSKQSNNKYAASSYLSLTPEQWKSHRSYSCQVTHEG
STVEKTVAPTECS

(SEQ ID NO: 100), signal sequence (1-20), NYA-0001_VL
(21-131), CL (132-237)

Fig. 102

NYA-1143-Fab-HC1-k delete

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
CGAGGTGCAGCTGGTGGAGTCTGGGGGAGGCGTGGTCCAGCCTGGGAGGTCCCTGA
CACCCTCCTGTTTACGCTCTGGATTCTCCATCCGTAGCTATGATATACTGGGTC
CGCCTGGCTCCAGGCAAGGGGCTAGAGTGGGTGGCCACTATATCATATGATGGAAG
TCAGAAGTACTTCGCAGACTCCGTGAAGGGCCGATTTACCATCTTCAGAGACGAAT
CGGAGAACATGGTGTATCTGCAAATGAGCGGCCTGAGAGTTGAGGACACGGCTGTT
TATCACTGTGCGAGAGGGAGTAGTGGTCAATTATGAGGCTTTTGATATCTGGGGCCA
AGGGACAATGGTCAACGTCTCTTCAGCCTCCACCAAGGGCCCAAGCGTCTTCCCCC
TGGCACCCCTCCTCCAAGAGCACCTCTGGCGGCACAGCCGCCCTGGGCTGCCTGGTC
AAGGACTACTTCCCCGAACCCGTGACCGTGAGCTGGAACCTCAGGCGCCCTGACCAG
CGGCGTGACACCTTCCCCGCTGTCTGTCAGTCCCTCAGGACTCTACTCCCTCAGCA
GCGTGGTGACCGTGCCCTCCAGCAGCTTGGGCACCCAGACCTACATCTGCAACGTG
AATCACAAGCCCAGCAACACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTA
CAAACTCACACATGCCACCCCTGCCAGCACCTGAAGCCGCAGGGGGACCCCTCAG
TCTTCCTCTTCCCCCAAACCCAAAGGACACCCTCATGATCTCCCGGACCCCTGAG
GTCACATGCGTGGTGGTGTGTCAGTGAGCCACGAAGACCCTGAGGTCAAGTTCAACTG
GTACGTGGACGGCGTGGAGGTGCATAATGCCAAGACAAAGCCCCGGGAGGAGCAGT
ACAACAGCACGTACCGGGTGGTCAGCGTCCTCACCGTCCTGCACCAGGACTGGCTG
AATGGCAAGGAGTACAAGTGCAAGGTCTCCAACAAAGCCCTCCCAGCCCCCATCGA
GAAAACCATCTCCAAAGCCAAAGGCCAGCCCCGGGAACCACAGGTGTACGTGCTGC
CCCCATCCCGGGACGAGCTGACCAAGAACCAGGTGAGCCTGCTGTGCCTGGTCAA
GGCTTCTATCCCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGCCAGCCGAGAA
CAACTACCTGACCTGGCCTCCCGTGCTGGACTCCGACGGCTCCTTCTCCTCTACA
GCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGGCAACGTCTTCTCATGCTCC
GTGATGCATGAGGCTCTGCACAACCACTACACCCAGAAGAGCCTCTCCCTGTCTCC
CGGC

(SEQ ID NO: 101), signal sequence (1-57), NYA-1143_VH
(58-417), CH1 (418-711), hinge (712-756), CH2 (757-1086),
CH3 (1087-1404)

Fig. 103

NYA-1143-LC

ATGGTGCTGCAGACCCAGGTGTTTCATCTCCCTGCTGCTGTGGATCTCCGGCGCGTA
CGGCCAGTCTGTGTTGACGCAGCCGCCCTCAGTGTCTGCGGCCCCAGGACAGAAGG
TCACCATCTCCTGCTCTGGAAGCAGCTCCAACATTGGGAATTGGTATGTATCCTGG
TACCAGCAGCTCCCAGGAACAGCCCCAAACTCCTCATTTATGACAATAATAAGCG
ACCCTCAGGGATTCTGACCGATTCTCTGGCTCCAAGTCTGGCACGTCAGCCACCC
TGGGCATCACCGGACTCCAGACTGGGGACGAGGCCGATTATTACTGCGGAACATGG
GATAGCAGCCTGAGTGCCCCCTGGGTGTTTCGGCGGAGGGACCAAGGTCACCGTCCT
AGGCCAGCCTAAGGCTGCCCCCTAGCGTGACCCTGTTCCCTCCTTCCAGCGAGGAGC
TTCAAGCTAACAAGGCCACCCTGGTGTGTCTTATCTCTGACTTCTACCCTGGCGCT
GTGACCGTGGCCTGGAAGGCTGACAGCTCCCCTGTGAAGGCCGGAGTGGAGACCAC
CACACCTAGCAAGCAGTCTAACAACAAGTACGCTGCCAGCTCCTACCTGAGCCTTA
CCCCTGAGCAGTGGAAGTCTCACAGAAGCTACTCCTGTCAAGTGACCCACGAGGGC
AGCACCGTGGAGAAGACCGTGGCTCCTACCGAGTGTTCC

(SEQ ID NO: 102), signal sequence (1-60), NYA-1143_VL
(61-393), CL (394-711)

Fig. 104

C3E-7085-HC2-k delete

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
CGGAGAAGTGCAGCTGGTGGAAATCCGGGGGGGGCCTGGTGCAGCCTGGGGGGAGCC
TGAGACTGAGTTGTGCCGCTCTGGGGTGACATTTAACTACTATGGCATGTCTTGG
ATCCGCCAGGCACCTGGAAAGGGCCTGGAGTGGGTGGCCAGCATCACTAGGTCCGG
CGGGCGAATCTACTATCCCGACAGCGTCAAGGGCAGGTTCAACAATTTCCCGCGAGA
ACACACAGAAAACCTCTGTACCTGCAGATGAATAGCCTGAGAGCCGAAGATACAGCT
GTGTACTATTGCACTCTGGACGGCAGGGATGGGTGGGTGCGCTATTGGGGGCAGGG
AACCCTGGTGACAGTCAGCTCCGGAGGAGGAGGATCTGGCGGAGGAGGCAGTGGGG
GAGGCGGGTCAAACCTTTATGCTCACTCAGCCGTCTCTGTTTCTGGCGTACCTGGC
CAACGGGTGACCATTAGCTGTACGGGTAATACCGGGAATATCGGGTCTAACTACGT
GAACTGGTATCAGCAGCTTCCAGGGACAGCTCCCAAGTTGCTGATCTATCGCGACG
ACAAAAGACCCTCAGGGGTCCCTGACCGATTTAGTGGCAGCAAAAGCGGTACTTCC
GCTTCCCTGGCGATAACCGGCTTTTCAGGCCGAAGATGAGGCAGACTACTATTGCCA
GTCATATTCCAGCGGCTTCATCTTCGGAGGCGGAACTAAGCTGACAGTGTTGGCAG
CCGAGCCCAAATCTTCTGACAAAACCTCACACATGCCCACCCTGCCCAGCACCTGAA
GCCGCAGGGGGACCCTCAGTCTTCTCTTCCCCCAAACCCAAGGACACCCTCAT
GATCTCCCGGACCCCTGAGGTCACATGCGTGGTGGTGTGAGTGAGCCACGAAGACC
CTGAGGTCAAGTTCAACTGGTACGTGGACGGCGTGGAGGTGCATAATGCCAAGACA
AAGCCCCGGGAGGAGCAGTACAACAGCACGTACCGGGTGGTCAGCGTCCTCACCGT
CCTGCACCAGGACTGGCTGAATGGCAAGGAGTACAAGTGCAAGGTCTCCAACAAAG
CCCTCCCAGCCCCCATCGAGAAAACCATCTCCAAAGCCAAAGGCCAGCCCCGGGAA
CCACAGGTGTACGTGTACCCCCCATCCCGGGACGAGCTGACCAAGAACCAGGTCAG
CCTGACCTGCCTGGTCAAAGGCTTCTATCCCAGCGACATCGCCGTGGAGTGGGAGA
GCAATGGCCAGCCCGAGAACCACTACAAGACCACCCCTCCCGTGCTGGACTCCGAC
GGCTCCTTCGCCCTCGTGAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGG
CAACGTCTTCTCATGCTCCGTGATGCATGAGGCTCTGCACAACCACTACACCCAGA
AGAGCCTCTCCCTGTCTCCC

(SEQ ID NO: 103), signal sequence (1-57), C3E-7085 (61-780), linker (781-786), hinge (787-831), CH2 (832-1161), CH3 (1162-1479)

Fig. 105

NYA-1143-Fab-HC1-k delete

MKHLWFFLLLVAAPRWVLSEVQLVESGGGVVQPGRSLTPSCSASGFSIRSYDIHWV
RLAPGKGLEWVATISYDGSQKYFADSVKGRFTIFRDESENMVYLQMSGLRVEDTAV
YHCARGSSGHYEAFDIWGQGMVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLV
KDYPPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNV
N HKPSNTKVDKRVEPKSCDKTHTCPPCPAPEAAGGPSVFLFPPKPKDTLMISRTPE
VTCVVVSVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWL
NGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYVLPFSRDELTKNQVSLTCLVK
GFYPSDIAVEWESNGQPENNYLTWPPVLDSDGSFFLYSKLTVDKSRWQQGNVFNCS
VMHEALHNHYTQKSLSLSPG

(SEQ ID NO: 104), signal sequence (1-19), NYA-1143_VH
(20-139), CH1 (140-237), hinge (238-252), CH2 (253-362),
CH3 (363-468)

Fig. 106

NYA-1143-LC

MVLQTQVFISLLLWISGAYGQSVLTQPPSVSAAPGQKVTISCSGSSSNIGNWYVSW
YQQLPGTAPKLLIYDNNKRPSGIPDRFSGSKSGTSATLGITGLQTGDEADYYCGTW
DSSLAPWVFGGGTKVTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGA
VTVAWKADSSPVKAGVETTTTPSKQSNNKYAASSYLSLTPEQWKSHRSYSCQVTHEG
STVEKTVAPTECS

(SEQ ID NO: 105), signal sequence (1-20), NYA-1143_VL
(21-131), CL (132-237)

Fig. 107

C3E-7085-HC2-k delete

MKHLWFFLLLVAAAPRWVLSGEVQLVESGGGLVQPGGSLRLSCAASGVTFNYYGMSW
IRQAPGKGLEWVASITRSGGRIYYPD SVKGRFTISRENTQKTLYLQMNSLRAEDTA
VYYCTLDGRDGWVAYWGQGLTVTVSSGGGGSGGGGSGGGGSNFMLTQPSSVSGVPG
QRVTISCTGNTGNIGSNYVNWYQQLPGTAPKLLIYRDDKRPSGVPDRFSGSKSGTS
ASLAITGFQAEDEADYYCQSYSSGFIFGGGTKLTVLAAEPKSSDKTHTCPPCPAPE
AAGGPSVFLFPPKPKDTLMISRTPEVTCVTVSVSHEDPEVKFNWYVDGVEVHNAKT
KPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPRE
PQVYVYPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSD
GSFALVSKLTVDKSRWQQGNV FSCSV MHEALHNHYTQKSLSLSPG

(SEQ ID NO: 106), signal sequence (1-19), C3E-7085 (21-260), linker (261-262), hinge (263-277), CH2 (278-387), CH3 (388-493)

Fig. 108

NYA-1143-HC1-k delete

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
CGGAGAGGTGCAGCTGGTGGAGTCTGGGGGAGGCGTGGTCCAGCCTGGGAGGTCCC
TGACACCCTCCTGTTTACGCGTCTGGATTCTCCATCCGTAGCTATGATATACACTGG
GTCCGCCTGGCTCCAGGCAAGGGGCTAGAGTGGGTGGCCACTATATCATATGATGG
AAGTCAGAAGTACTTCGCAGACTCCGTGAAGGGCCGATTTACCATCTTCAGAGACG
AATCGGAGAACATGGTGTATCTGCAAATGAGCGGCCTGAGAGTTGAGGACACGGCT
GTTTATCACTGTGCGAGAGGGAGTAGTGGTCATTATGAGGCTTTTGATATCTGGGG
CCAAGGGACAATGGTCAACCGTCTCTTCAGGTGGAGGCGGTTTCAGGCGGAGGTGGCA
GCGGCGGTGGCGGGAGTCAGTCTGTGTTGACGCAGCCGCCCTCAGTGTCTGCGGCC
CCAGGACAGAAGGTCACCATCTCCTGCTCTGGAAGCAGCTCCAACATTGGGAATTG
GTATGTATCCTGGTACCAGCAGCTCCCAGGAACAGCCCCCAAACCTCCTCATTTATG
ACAATAATAAGCGACCCTCAGGGATTCTGACCGATTCTCTGGCTCCAAGTCTGGC
ACGTCAGCCACCCTGGGCATCACCGGACTCCAGACTGGGGACGAGGCCGATTATTA
CTGCGGAACATGGGATAGCAGCCTGAGTGCCCTTGGGTGTTGGGCGGAGGGACCA
AGGTCACCGTCTTAGCAGCCGAGCCCAAATCTTCTGACAAAACCTCACACATGCCCA
CCCTGCCCAGCACCTGAAGCCGCAGGGGGACCCTCAGTCTTCCTCTTCCCCCAA
ACCCAAGGACACCCTCATGATCTCCCGGACCCCTGAGGTCACATGCGTGGTGGTGT
CAGTGAGCCACGAAGACCCTGAGGTCAAGTTCAACTGGTACGTGGACGGCGTGGAG
GTGCATAATGCCAAGACAAAGCCCCGGGAGGAGCAGTACAACAGCACGTACCGGGT
GGTCAGCGTCTCACCCTGCTGCAACAGGACTGGCTGAATGGCAAGGAGTACAAGT
GCAAGGTCTCCAACAAAGCCCTCCCAGCCCCCATCGAGAAAACCATCTCCAAGCC
AAAGGCCAGCCCCGGGAACCACAGGTGTACGTGCTGCCCCCATCCCGGGACGAGCT
GACCAAGAACCAGGTCAGCCTGCTGTGCCTGGTCAAAGGCTTCTATCCCAGCGACA
TCGCCGTGGAGTGGGAGAGCAATGGCCAGCCCGAGAACAACTACCTGACCTGGCCT
CCCGTGCTGGACTCCGACGGCTCCTTCTTCTCTACAGCAAGCTCACCGTGGACAA
GAGCAGGTGGCAGCAGGGCAACGTCTTCTCATGCTCCGTGATGCATGAGGCTCTGC
ACAACCACTACACCCAGAAGAGCCTCTCCCTGTCTCCCGGC

(SEQ ID NO: 107), signal sequence (1-57), NYA-1143 (61-798), linker799-800), hinge (801-849), CH2 (850-1179), CH3 (1180-1497)

Fig. 109

NYA-1143-HC1-k delete

MKHLWFFLLLVAAPRWVLSGEVQLVESGGGVVQPGRSLTPSCSASGFSIRSYDIHW
VRLAPGKGLEWVATISYDGSQKYFADSVKGRFTIFRDESENMVYLMQMSGLRVEDTA
VYHCARGSSGHYEAFDIWGQGTMTVTVSSGGGGSGGGGSGGGGSQSVLTQPPSVSAA
PGQKVTISCSGSSSNIGNWYVSWYQQLPGTAPKLLIYDNNKRPSGIPDRFSGSKSG
TSATLGITGLQTGDEADYYCGTWDSLSAPWVFGGGTKVTVLAAEPKSSDKTHTCP
PCPAPEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVSVSHEDPEVKFNWYVDGVE
VHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKA
KGQPREPQVYVLPISRDELTKNQVSLCLVKGFYPSDIAVEWESNGQPENNYLTWP
PVLDSDGSEFFLYSKLTVDKSRWQQGNVFSVMHEALHNHYTQKSLSLSPG
(SEQ ID NO: 108), signal sequence (1-19), NYA-1143 (21-
266), linker (267-268), hinge (269-283), CH2 (284-393),
CH3 (394-499)

Fig. 110

C3E-7085-NYA-1154-Fab-HC2-k delete

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
CGGAGAAGTGCAGCTGGTGAATCCGGGGGGGGCCTGGTGCAGCCTGGGGGGAGCC
TGAGACTGAGTTGTGCCGCTCTGGGGTGACATTTAACTACTATGGCATGTCTTGG
ATCCGCCAGGCACCTGGAAAGGGCCTGGAGTGGGTGGCCAGCATCACTAGGTCCGG
CGGGCGAATCTACTATCCCGACAGCGTCAAGGGCAGGTTTACAATTTCCCGCGAGA
ACACACAGAAAACCTCTGTACCTGCAGATGAATAGCCTGAGAGCCGAAGATACAGCT
GTGTACTATTGCACTCTGGACGGCAGGGATGGGTGGGTGCGCTATTGGGGGCAGGG
AACCCTGGTGACAGTCAGCTCCGGAGGAGGAGGATCTGGCGGAGGAGGCAGTGGGG
GAGGCGGGTCAAACCTTTATGCTCACTCAGCCGTCTCTGTTTCTGGCGTACCTGGC
CAACGGGTGACCATTAGCTGTACGGGTAATACCGGGAATATCGGGTCTAACTACGT
GAACTGGTATCAGCAGCTTCCAGGGACAGCTCCCAAGTTGCTGATCTATCGCGACG
ACAAAAGACCCTCAGGGGTCCCTGACCGATTTAGTGGCAGCAAAAGCGGTACTTCC
GCTTCCCTGGCGATAACCGGCTTTTCAGGCCGAAGATGAGGCAGACTACTATTGCCA
GTCATATTCCAGCGGCTTCATCTTCGGAGGCGGAACCTAAGCTGACAGTGTTGGGGG
GAGGCGGTTTCAGAGGTGCAGCTGGTGGAGTCTGGGGGAGGCGTGGTCCAGCCTGGG
AGGTCCCTGACACCCTCCTGTTTCAGCGTCTGGATTCTCCATCCGTAGTTATGATAT
GCACTGGGTCCGCCAGGCTCCAGGCAAGGGGCTAGAGTGGGTGGCCACTATATCAT
ATGATGGAAGTCAGAAGTACTTCGCAGACTCCGTGAAGGGCCGATTTACCATCTTC
AGAGACGAATCGGAGAACATGGTGTATCTGCAAATGAGCGGCCTGAGAGTTGAGGA
CACGGCCGTTTATCACTGTGCGAGAGGGAGTAGTAATCATATGAGGCTTTTGATA
TCTGGGGCCAAGGGACAATGGTCAACCGTCTCTTCAGCCTCCACCAAGGGCCCAAGC
GTCTTCCCCCTGGCACCTCCTCCAAGAGCACCTCTGGCGGCACAGCCGCCCTGGG
CTGCCCTGGTCAAGGACTACTTCCCCGAACCCGTGACCGTGAGCTGGAACCTCAGGCG
CCCTGACCAGCGGCGTGCACACCTTCCCCGCTGTCCTGCAGTCTCTCAGGACTCTAC
TCCCTCAGCAGCGTGGTGACCGTGCCCTCCAGCAGCTTGGGCACCCAGACCTACAT
CTGCAACGTGAATCACAAGCCCAGCAACACCAAGGTGGACAAGAGAGTTGAGCCCA
AATCTTGTGACAAAACCTCACACATGCCACCCCTGCCAGCACCTGAAGCCGCAGGG
GGACCCTCAGTCTTCTCTTCCCCCAAAACCAAGGACACCCTCATGATCTCCCCG
GACCCCTGAGGTCACATGCGTGGTGGTGTGAGTGGTGGTGGTGGTGGTGGTGGTGGT
AGTTCAACTGGTACGTGGACGGCGTGGAGGTGCATAATGCCAAGACAAAGCCCCGG
GAGGAGCAGTACAACAGCACGTACCGGGTGGTGCAGCGTCTCACCGTCTCTGCACCA
GGACTGGCTGAATGGCAAGGAGTACAAGTGCAAGGTCTCCAACAAAGCCCTCCAG
CCCCATCGAGAAAACCATCTCCAAAGCCAAAGGCCAGCCCCGGGAACACAGGTG
TACGTGTACCCCCCATCCCGGGACGAGCTGACCAAGAACCAGGTGAGCCTGACCTG
CCTGGTCAAAGGCTTCTATCCCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGCC
AGCCCCGAGAACAACTACAAGACCACCCCTCCCGTGCTGGACTCCGACGGCTCCTTC
GCCCTCGTGAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGGCAACGTCTT
CTCATGCTCCGTGATGCATGAGGCTCTGCACAACCACTACACCCAGAAGAGCCTCT
CCCTGTCTCCCGGC

(SEQ ID NO: 109), signal sequence (1-57), C3E-7085 (61-780), linker (781-795), NYA-1154-VH (796-1155), CH1 (1156-1470), hinge (1471-1500), CH2 (1501-1830), CH3 (1831-2142)

Fig. 111

NYA-1154-LC

ATGGTGCTGCAGACCCAGGTGTTTCATCTCCCTGCTGCTGTGGATCTCCGGCGCGTA
CGGCCAGTCTGTGTTGACGCAGCCGCCCTCAGTGTCTGCGGCCCCAGGACAGAAGG
TCACCATCTCCTGCTCTGGAAGCAGCTCCAACATTGGGAATAATTATGTATCCTGG
TACCAGCGGCTCCCAGGAACAGCCCCAAACTCCTCATTTATGACAATAATAAGCG
ACCCTCAGGGATTCTGACCGATTCTCTGGCTCCAAGTCTGGCACGTCAGCCACCC
TGGGCATCACCGGACTCCAGACTGGGGACGAGGCCGATTACTACTGCGGAGCATGG
GATAGCAGCCTGAGTGCCCCCTGGGTGTTTCGGCGGAGGGACCAAGGTCACCGTCCT
AGGCCAGCCTAAGGCTGCCCCCTAGCGTGACCCTGTTCCCTCCTTCCAGCGAGGAGC
TTCAAGCTAACAAGGCCACCCTGGTGTGTCTTATCTCTGACTTCTACCCTGGCGCT
GTGACCGTGGCCTGGAAGGCTGACAGCTCCCCTGTGAAGGCCGGAGTGGAGACCAC
CACACCTAGCAAGCAGTCTAACAACAAGTACGCTGCCAGCTCCTACCTGAGCCTTA
CCCCTGAGCAGTGGAAGTCTCACAGAAGCTACTCCTGTCAAGTGACCCACGAGGGC
AGCACCGTGGAGAAGACCGTGGCTCCTACCGAGTGTTCC

(SEQ ID NO: 110), signal sequence (1-60), NYA-1154_VL
(61-393), CL (394-711)

Fig. 112

OAA-HC1-k delete

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
CGACAAAACCTCACACATGCCCACCCTGCCCAGCACCTGAAGCCGCAGGGGGACCCT
CAGTCTTCCTCTTCCCCCAAAACCCAAGGACACCCTCATGATCTCCCGGACCCCT
GAGGTCACATGCGTGGTGGTGTGAGTGAGCCACGAAGACCCTGAGGTCAAGTTCAA
CTGGTACGTGGACGGCGTGGAGGTGCATAATGCCAAGACAAAGCCCCGGGAGGAGC
AGTACAACAGCACGTACCGGGTGGTCAGCGTCCTCACCGTCCTGCACCAGGACTGG
CTGAATGGCAAGGAGTACAAGTGCAAGGTCTCCAACAAAGCCCTCCCAGCCCCCAT
CGAGAAAACCATCTCCAAGCCAAAGGCCAGCCCCGGGAACACAGGTGTACGTGC
TGCCCCCATCCCGGGACGAGCTGACCAAGAACCAGGTGAGCCTGCTGTGCCTGGTC
AAAGGCTTCTATCCCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGCCAGCCCGA
GAACAACCTACCTGACCTGGCCTCCCGTGCTGGACTCCGACGGCTCCTTCTTCCTCT
ACAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGGCAACGTCTTCTCATGC
TCCGTGATGCATGAGGCTCTGCACAACCACTACACCCAGAAGAGCCTCTCCCTGTC
TCCCGGC

(SEQ ID NO: 111), signal sequence (1-57), hinge (58-87),
CH2 (88-417), CH3 (418-735)

Fig. 113

C3E-7085-NYA-1154-Fab-HC2-k delete
MKHLWFFLLLVAAPRWVLSGEVQLVESGGGLVQPGGSLRLSCAASGVTFNYYGMSW
IRQAPGKGLEWVASITRSGGRIYYPD SVKGRFTISRENTQKTLYLQMNSLRAEDTA
VYYCTLDGRDGVAYWGQGLTVTVSSGGGGSGGGGSGGGGSNFMLTQPSSVSGVPG
QRTVISCTGNTGNIGSNYVNWYQQLPGTAPKLLIYRDDKRPSGVPDRFSGSKSGTS
ASLAITGFQAEDEADYYCQSYSSGFIFGGGTKLTVLGGGGSEVQLVESGGGVVQPG
RSLTPSCSASGFSIRS YDMHWVRQAPGKGLEWVATISYDGSQKYFADSVKGRFTIF
RDESENMVYLQMSGLRVEDTAVYHCARGSSNHYEAFDIWGQGTMTVTVSSASTKGPS
VFPLAPSSKSTSGGTAALGCLVKDYFPEPVTFSWNSGALTSGVHTFPAVLQSSGLY
SLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRVEPKSCDKTHTCPPCPAPEAAG
GPSVFLFPPKPKDTLMISRTPEVTCVVS VSHEDPEVKFNWYVDGVEVHNAKTKPR
EEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQV
YVYPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSDGSF
ALVSKLTVDKSRWQQGNV FSCSV MHEALHNHYTQKSLSLSPG
(SEQ ID NO: 112), signal sequence (1-19), C3E-7085 (21-
260), linker (261-265), NYA-1154-VH (266-385), CH1 (386-
490), hinge (491-500), CH2 (501-610), CH3 (611-714)

Fig. 114

NYA-1154-LC

MVLQTQVFISLLLWISGAYGQSVLTQPPSVSAAPGQKVTISCSGSSSNIGNNYVSW
YQRLPGTAPKLLIYDNNKRPSGIPDRFSGSKSGTSATLGITGLQTGDEADYYCGAW
DSSLAPWVFGGGTKVTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGA
VTVAWKADSSPVKAGVETTTTPSKQSNNKYAASSYLSLTPEQWKSHRSYSCQVTHEG
STVEKTVAPTECS

(SEQ ID NO: 113), signal sequence (1-20), NYA-1154_VL
(21-131), CL (132-237)

Fig. 115

OAA-HC1-k delete

MKHLWFFLLLVAAPRWVLSDKTHTCPPCPAPEAAGGPSVFLFPPKPKDTLMISRTPEVTCVTVSVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYVLPISRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYLTWPPVLDSDGSFFLYSKLTVDKSRWQQGNVSCSVMHEALHNHYTQKSLSLSPG

(SEQ ID NO: 114), signal sequence (1-19), hinge (20-29), CH2 (30-139), CH3 (140-245)

Fig. 116

NYF-0010-HC2-k delete

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
CGGAGAGGTGCAGCTGGTGGAGTCTGGGGGAGGCGTGGTCCAGCCTGGGAGGTCCC
TGACACCCTCCTGTTTCAGCGTCTGGATTCTCCATCCGTAGTTATGATATGCACTGG
GTCCGCCAGGCTCCAGGCAAGGGGCTAGAGTGGGTGGCCACTATATCATATGATGG
AAGTCAGAAGTACTTCGCAGACTCCGTGAAGGGCCGATTTACCATCTTCAGAGACG
AATCGGAGAACATGGTGTATCTGCAAATGAGCGGCCTGAGAGTTGAGGACACGGCC
GTTTATCACTGTGCGAGAGGGAGTAGTAATCATTATGAGGCTTTTGATATCTGGGG
CCAAGGGACAATGGTCACCGTCTCTTCAGGTGGAGGCGGTTTCAGGCGGAGGTGGCA
GCGGCGGTGGCGGGAGTCAGTCTGTGTTGACGCAGCCGCCCTCAGTGTCTGCGGCC
CCAGGACAGAAGGTCACCATCTCCTGCTCTGGAAGCAGCTCCAACATTGGGAATAA
TTATGTATCCTGGTACCAGCGGCTCCCAGGAACAGCCCCCAAACCTCTCATTTATG
ACAATAATAAGCGACCCTCAGGGATTCTTGACCGATTCTCTGGCTCCAAGTCTGGC
ACGTCAGCCACCTGGGCATCACCGGACTCCAGACTGGGGACGAGGCCGATTACTA
CTGCGGAGCATGGGATAGCAGCCTGAGTGCCCTTGGGTGTTGCGCGGAGGGACCA
AGGTCACCGTCTTAGGGGGAGGCGGTTTCAGAAGTGCAGCTGGTGAATCCGGGGGG
GGCCTGGTGCAGCCTGGGGGGAGCCTGAGACTGAGTTGTGCCGCTCTGGGGTGAC
ATTTAACTACTATGGCATGTCTTGATCCGCCAGGCACCTGGAAAGGGCCTGGAGT
GGGTGGCCAGCATCACTAGGTCCGGCGGGCGAATCTACTATCCCGACAGCGTCAAG
GGCAGGTTACAAATTTCCCGCGAGAACACACAGAAAACCTCTGTACCTGCAGATGAA
TAGCCTGAGAGCCGAAGATACAGCTGTGTACTATTGCACTCTGGACGGCAGGGATG
GGTGGGTCGCCTATTGGGGGGCAGGGAACCTGGTGACAGTCAGCTCCGGAGGAGGA
GGATCTGGCGGAGGAGGCAGTGGGGGAGGCGGGTCAAACCTTATGCTCACTCAGCC
GTCCTCTGTTTCTGGCGTACCTGGCCAACGGGTGACCATTAGCTGTACGGGTAATA
CCGGGAATATCGGGTCTAACTACGTGAACTGGTATCAGCAGCTTCCAGGGACAGCT
CCCAAGTTGCTGATCTATCGCGACGACAAAAGACCCTCAGGGGTCCCTGACCGATT
TAGTGGCAGCAAAGCGGTACTTCCGCTTCCCTGGCGATAACCGGCTTTCAGGCCG
AAGATGAGGCAGACTACTATTGCCAGTCATATTCCAGCGGCTTCATCTTCGGAGGC
GGAACATAAGCTGACAGTGTTGGCAGCCGAGCCCAAATCTTCTGACAAAACCTCACAC
ATGCCCACCCTGCCCAGCACCTGAAGCCGAGGGGGACCCTCAGTCTTCCTCTTCC
CCCCAAAACCAAGGACACCCTCATGATCTCCCGGACCCCTGAGGTCACATGCGTG
GTGGTGTCAGTGAGCCACGAAGACCCTGAGGTCAAGTTCAACTGGTACGTGGACGG
CGTGGAGGTGCATAATGCCAAGACAAAGCCCCGGGAGGAGCAGTACAACAGCACGT
ACCGGGTGGTCAGCGTCCCTACCGTCTGCACCAGGACTGGCTGAATGGCAAGGAG
TACAAGTGCAAGGTCTCCAACAAAGCCCTCCCAGCCCCATCGAGAAAACCATCTC
CAAAGCCAAAGGCCAGCCCCGGAACACAGGTGTACGTGTACCCCCATCCCGGG
ACGAGCTGACCAAGAACCAGGTGAGCCTGACCTGCCTGGTCAAAGGCTTCTATCCC
AGCGACATCGCCGTGGAGTGGGAGAGCAATGGCCAGCCGAGAACAACTACAAGAC
CACCCCTCCCGTGCTGGACTCCGACGGCTCCTTCGCCCTCGTGAGCAAGCTCACCG
TGGACAAGAGCAGGTGGCAGCAGGGCAACGTCTTCTCATGCTCCGTGATGCATGAG
GCTCTGCACAACCACTACACCCAGAAGAGCCTCTCCCTGTCTCCCGGC

(SEQ ID NO: 115), signal sequence (1-57), NYA-1154 (61-798), linker (799-813), C3E-7085 (814-1533), linker (1534-1539), hinge (1540-1584), CH2 (1585-1914), CH3 (1915-2232)

Fig. 117

NYF-0004-HC2-k delete

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
CGGAGAAGTGCAGCTGGTGAATCCGGGGGGGGCCTGGTGCAGCCTGGGGGGAGCC
TGAGACTGAGTTGTGCCGCTCTGGGGTGACATTTAACTACTATGGCATGTCTTGG
ATCCGCCAGGCACCTGGAAAGGGCCTGGAGTGGGTGGCCAGCATCACTAGGTCCGG
CGGGCGAATCTACTATCCCGACAGCGTCAAGGGCAGGTTTACAATTTCCCGCGAGA
ACACACAGAAAACCTCTGTACCTGCAGATGAATAGCCTGAGAGCCGAAGATACAGCT
GTGTACTATTGCACTCTGGACGGCAGGGATGGGTGGGTTCGCTATTGGGGGCAGGG
AACCTTGGTGCAGTCAGCTCCGGAGGAGGAGGATCTGGCGGAGGAGGCAGTGGGG
GAGGCGGGTCAAACCTTTATGCTCACTCAGCCGTCCTCTGTTTCTGGCGTACCTGGC
CAACGGGTGACCATTAGCTGTACGGGTAATACCGGGAATATCGGGTCTAACTACGT
GAACTGGTATCAGCAGCTTCCAGGGACAGCTCCCAAGTTGCTGATCTATCGCGACG
ACAAAAGACCCTCAGGGGTCCCTGACCGATTTAGTGGCAGCAAAGCGGTACTTCC
GCTTCCCTGGCGATAACCGGCTTTCAGGCCGAAGATGAGGCAGACTACTATTGCCA
GTCATATTCCAGCGGCTTCATCTTCGGAGGCGGAACTAAGCTGACAGTGTTGGGGG
GAGGCGGTTTCAGAGGTGCAGCTGGTGGAGTCTGGGGGAGGCGTGGTCCAGCCTGGG
AGGTCCCTGACACCCTCCTGTTTCAGCGTCTGGATTCTCCATCCGTAGTTATGATAT
GCACTGGGTCCGCCAGGCTCCAGGCAAGGGGCTAGAGTGGGTGGCCACTATATCAT
ATGATGGAAGTCAGAAGTACTTCGCAGACTCCGTGAAGGGCCGATTTACCATCTTC
AGAGACGAATCGGAGAACATGGTGTATCTGCAAATGAGCGGCCTGAGAGTTGAGGA
CACGGCCGTTTATCACTGTGCGAGAGGGAGTAGTAATCATTATGAGGCTTTTGATA
TCTGGGGCCAAGGGACAATGGTCACCGTCTCTTCAGGTGGAGGCGGTTTCAGGCGGA
GGTGGCAGCGGCGGTGGCGGGAGTCAGTCTGTGTTGACGCAGCCGCCCTCAGTGTC
TGCGGCCCCAGGACAGAAGGTCACCATCTCCTGCTCTGGAAGCAGCTCCAACATTG
GGAATAATTATGTATCCTGGTACCAGCGGCTCCAGGAACAGCCCCCAAACCTCCTC
ATTTATGACAATAATAAGCGACCCCTCAGGGATTCTTGACCGATTCTCTGGCTCCAA
GTCTGGCACGTGAGCCACCCTGGGCATCACCGGACTCCAGACTGGGGACGAGGCCG
ATTACTACTGCGGAGCATGGGATAGCAGCCTGAGTGCCCTTGGGTGTTTCGGCGGA
GGGACCAAGGTCACCGTCCTAGCAGCCGAGCCCAAATCTTCTGACAAAACCTCACAC
ATGCCACCCCTGCCCAGCACCTGAAGCCGAGGGGGACCCTCAGTCTTCCTCTTCC
CCCCAAAACCAAGGACACCCTCATGATCTCCCGGACCCCTGAGGTACATGCGTG
GTGGTGTGAGTGCAGCACGAAGACCCTGAGGTCAAGTTCAACTGGTACGTGGACGG
CGTGGAGGTGCATAATGCCAAGACAAAGCCCCGGGAGGAGCAGTACAACAGCACGT
ACCGGGTGGTCAGCGTCCTCACCGTCCTGCACCAGGACTGGCTGAATGGCAAGGAG
TACAAGTGCAAGGTCTCCAACAAAGCCCTCCAGCCCCCATCGAGAAAACCATCTC
CAAAGCCAAAGGCCAGCCCCGGGAACACAGGTGTACGTGTACCCCCATCCCGGG
ACGAGCTGACCAAGAACCAGGTGAGCCTGACCTGCCTGGTCAAAGGCTTCTATCCC
AGCGACATCGCCGTGGAGTGGGAGAGCAATGGCCAGCCGAGAACAACTACAAGAC
CACCCCTCCCGTGCTGGACTCCGACGGCTCCTTCGCCCTCGTGAGCAAGCTCACCG
TGGACAAGAGCAGGTGGCAGCAGGGCAACGTCTTCTCATGCTCCGTGATGCATGAG
GCTCTGCACAACCACTACACCCAGAAGAGCCTCTCCCTGTCTCCCGGC

(SEQ ID NO: 116), signal sequence (1-57), C3E-7085 (61-780), linker (781-795), NYA-1154(796-1533), linker (1534-1539), hinge (1540-1584), CH2 (1585-1914), CH3 (1915-2232)

Fig. 118

NYF-0011-HC2-k delete

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
CGGAGAGGTGCAGCTGGTGGAGTCTGGGGGAGGCGTGGTCCAGCCTGGGAGGTCCC
TGACACCCTCCTGTTCAGCGTCTGGATTCTCCATCCGTAGCTATGATATACACTGG
GTCCGCCTGGCTCCAGGCAAGGGGCTAGAGTGGGTGGCCACTATATCATATGATGG
AAGTCAGAAGTACTTCGCAGACTCCGTGAAGGGCCGATTTACCATCTTCAGAGACG
AATCGGAGAACATGGTGTATCTGCAAATGAGCGGCCTGAGAGTTGAGGACACGGCT
GTTTATCACTGTGCGAGAGGGAGTAGTGGTCAATTATGAGGCTTTTGATATCTGGGG
CCAAGGGACAATGGTCACCGTCTCTTCAGGTGGAGGCGGTTTCAGGCGGAGGTGGCA
GCGGCGGTGGCGGGAGTCAGTCTGTGTTGACGCAGCCGCCCTCAGTGTCTGCGGCC
CCAGGACAGAAGGTCACCATCTCCTGCTCTGGAAGCAGCTCCAACATTGGGAATTG
GTATGTATCCTGGTACCAGCAGCTCCCAGGAACAGCCCCCAAACCTCTCATTTATG
ACAATAATAAGCGACCCTCAGGGATTCTTGACCGATTCTCTGGCTCCAAGTCTGGC
ACGTCAGCCACCTGGGCATCACCGGACTCCAGACTGGGGACGAGGCCGATTATTA
CTGCGGAACATGGGATAGCAGCCTGAGTGCCCTTGGGTGTTGCGCGGAGGGACCA
AGGTCACCGTCTTAGGGGGAGGCGGTTTCAGAAGTGCAGCTGGTGGAAATCCGGGGGG
GGCCTGGTGCAGCCTGGGGGGAGCCTGAGACTGAGTTGTGCCGCTCTGGGGTGAC
ATTTAACTACTATGGCATGTCTTGATCCGCCAGGCACCTGGAAAGGGCCTGGAGT
GGGTGGCCAGCATCACTAGGTCCGGCGGGCGAATCTACTATCCCGACAGCGTCAAG
GGCAGGTTACAAATTTCCCGCGAGAACACACAGAAAACCTCTGTACCTGCAGATGAA
TAGCCTGAGAGCCGAAGATACAGCTGTGTACTATTGCACTCTGGACGGCAGGGATG
GGTGGGTCGCCTATTGGGGGGCAGGGAACCTGGTGACAGTCAGCTCCGGAGGAGGA
GGATCTGGCGGAGGAGGCAGTGGGGGAGGCGGGTCAAACCTTATGCTCACTCAGCC
GTCCTCTGTTTCTGGCGTACCTGGCCAACGGGTGACCATTAGCTGTACGGGTAATA
CCGGGAATATCGGGTCTAACTACGTGAACTGGTATCAGCAGCTTCCAGGGACAGCT
CCCAAGTTGCTGATCTATCGCGACGACAAAAGACCCTCAGGGGTCCCTGACCGATT
TAGTGGCAGCAAAGCGGTACTTCCGCTTCCCTGGCGATAACCGGCTTTCAGGCCG
AAGATGAGGCAGACTACTATTGCCAGTCATATTCCAGCGGCTTCATCTTCGGAGGC
GGAAC TAAGCTGACAGTGTGGCAGCCGAGCCCAAATCTTCTGACAAAAC TCACAC
ATGCCACCCCTGCCCAGCACCTGAAGCCGAGGGGGACCCTCAGTCTTCCTCTTCC
CCCCAAAACCCAAGGACACCCTCATGATCTCCCGGACCCCTGAGGTCACATGCGTG
GTGGTGTCAGTGAGCCACGAAGACCCTGAGGTCAAGTTCAACTGGTACGTGGACGG
CGTGGAGGTGCATAATGCCAAGACAAAGCCCCGGGAGGAGCAGTACAACAGCACGT
ACCGGGTGGTCAGCGTCCCTACCGTCTGCACCAGGACTGGCTGAATGGCAAGGAG
TACAAGTGCAAGGTCTCCAACAAAGCCCTCCCAGCCCCATCGAGAAAACCATCTC
CAAAGCCAAAGGCCAGCCCCGGAACACAGGTGTACGTGTACCCCCATCCCGGG
ACGAGCTGACCAAGAACCAGGTGAGCCTGACCTGCCTGGTCAAAGGCTTCTATCCC
AGCGACATCGCCGTGGAGTGGGAGAGCAATGGCCAGCCGAGAACAACTACAAGAC
CACCCCTCCCGTGCTGGACTCCGACGGCTCCTTCGCCCTCGTGAGCAAGCTCACCG
TGGACAAGAGCAGGTGGCAGCAGGGCAACGTCTTCTCATGCTCCGTGATGCATGAG
GCTCTGCACAACCACTACACCCAGAAGAGCCTCTCCCTGTCTCCCGGC

(SEQ ID NO: 117), signal sequence (1-57), NYA-1143 (61-798), linker (799-813), C3E-7085 (814-1533), linker (1534-1539), hinge (1540-1584), CH2 (1585-1914), CH3 (1915-2232)

Fig. 119

NYF-0010-HC2-k delete

MKHLWFFLLLVAAAPRWVLSGEVQLVESGGGVVQPGRSLTPSCSASGFSIRSYDMHW
VRQAPGKGLEWVATISYDGSQKYFADSVKGRFTIFRDESENMVYLQMSGRLVEDTA
VYHCARGSSNHYEAFDIWGQGTMTVTVSSGGGGSGGGSGGGGSQSVLTQPPSVSAA
PGQKVTISCSGSSSNIGNNYVSWYQRLPGTAPKLLIYDNNKRPSGIPDRFSGSKSG
TSATLGITGLQTGDEADYYCGAWDSSLSAPWVFGGGTKVTVLGGGGSEVQLVESGG
GLVQPGGSLRLSCAASGVTFNYYGMSWIRQAPGKGLEWVASITRSGGRIIYPDSVK
GRFTISRENTQKTLYLQMNSLRAEDTAVYYCTLDGRDGWVAYWGQGTTLVTVSSGGG
GSGGGGSGGGGSNFMLTQPSSVSGVPGQRTISCTGNTGNIGSNYVNWYQQLPGTA
PKLLIYRDDKRPSGVPDRFSGSKSGTSASLAITGFQAEDEADYYCQSYSSGFIFGG
GTKLTVLAAEPKSSDKTHTCPPCPAPEAAGGPSVFLFPPKPKDTLMISRTPEVTCV
VVSVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKE
YKCKVSNKALPAPIEKTISKAKGQPREPQVYVYPPSRDELTKNQVSLTCLVKGFYP
SDIAVEWESNGQPENNYKTTTPVLDSDGSFALVSKLTVDKSRWQQGNVFSQVMHE
ALHNHYTQKSLSLSPG

(SEQ ID NO: 118), signal sequence (1-19), NYA-1154 (21-266), linker (267-271), C3E-7085 (272-511), linker (512-513), hinge (514-528), CH2 (529-638), CH3 (639-744)

Fig. 120

NYF-0004-HC2-k delete

MKHLWFFLLLVAAAPRWVLSGEVQLVESGGGLVQPGGSLRLSCAASGVTFNYYGMSW
IRQAPGKGLEWVASITRSGGRIIYPDSVKGRFTISRENTQKTLYLQMNSLRAEDTA
VYYCTLDGRDGWVAYWGQGLTVTVSSGGGGSGGGGSGGGGSNFMLTQPSSVSGVPG
QRTVISCTGNTGNIGSNYVNWYQQLPGTAPKLLIYRDDKRPSGVPDRFSGSKSGTS
ASLAITGFQAEDEADYYCQSYSSGFIFGGGTKLTVLGGGGSEVQLVESGGGVVQPG
RSLTPSCSASGFSIRS YDMHWVRQAPGKGLEWVATISYDGSQKYFADSVKGRFTIF
RDESENMVYLQMSGLRVEDTAVYHCARGSSNHYEAFDIWGQGTMTVTVSSGGGGSGG
GGSGGGGSQSVLTQPPSVSAAPGQKVTISCSGSSSNIGNNYVSWYQRLPGTAPKLL
IYDNNKRPSGIPDRFSGSKSGTSATLGITGLQTGDEADYYCGAWDSSLAPWVFGG
GTKVTVLAAEPKSSDKTHTCPPCPAPEAAGGPSVFLFPPKPKDTLMISRTPEVTCV
VVSVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKE
YKCKVSNKALPAPIEKTISKAKGQPREPQVYVYPPSRDELTKNQVSLTCLVKGFYP
SDIAVEWESNGQPENNYKTTTPVLDSDGSFALVSKLTVDKSRWQQGNVFSCSVMHE
ALHNHYTQKSLSLSPG

(SEQ ID NO: 119), signal sequence (1-19), C3E-7085 (21-260), linker (261-265), NYA-1154 (272-511), linker (512-513), hinge (514-528), CH2 (529-638), CH3 (639-744)

Fig. 121

NYF-0011-HC2-k delete

MKHLWFFLLLVAAAPRWVLSGEVQLVESGGGVVQPGRSLTPSCSASGFSIRSYDIHW
VRLAPGKGLEWVATISYDGSQKYFADSVKGRFTIFRDESENMVYLQMSGRLVEDTA
VYHCARGSSGHYEAFDIWGQGTMTVTVSSGGGGSGGGSGGGGSQSVLTQPPSVSAA
PGQKVTISCSGSSSNIGNWYVSWYQQLPGTAPKLLIYDNNKRPSGIPDRFSGSKSG
TSATLGITGLQTGDEADYYCGTWDSSLSAPWVFGGGTKVTVLGGGGSEVQLVESGG
GLVQPGGSLRLSCAASGVTFNYYGMSWIRQAPGKGLEWVASITRSGGRIIYPDSVK
GRFTISRENTQKTLYLQMNSLRAEDTAVYYCTLDGRDGWVAYWGQGTTLVTVSSGGG
GSGGGGSGGGGSNFMLTQPSSVSGVPGQRVTISCTGNTGNIGSNYVNWYQQLPGTA
PKLLIYRDDKRPSGVPDRFSGSKSGTSASLAITGFQAEDEADYYCQSYSSGFIFGG
GTKLTVLAAEPKSSDKTHTCPPCPAPEAAGGPSVFLFPPKPKDTLMISRTPEVTCV
VVSVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKE
YKCKVSNKALPAPIEKTISKAKGQPREPQVYVYPPSRDELTKNQVSLTCLVKGFYP
SDIAVEWESNGQPENNYKTTTPVLDSDGSFALVSKLTVDKSRWQQGNVFSQVMHE
ALHNHYTQKSLSLSPG

(SEQ ID NO: 120), signal sequence (1-19), NYA-1143 (21-266), linker (267-271), C3E-7085 (272-511), linker (512-513), hinge (514-528), CH2 (529-638), CH3 (639-744)

Fig. 122

Point mutant NY-ESO peptide 1F
FLLMWITQC
(SEQ ID NO: 121)

Fig. 123

Point mutant NY-ESO peptide 2M
SMLMWITQC
(SEQ ID NO: 122)

Fig. 124

Point mutant NY-ESO peptide 3A
SLAMWITQC
(SEQ ID NO: 123)

Fig. 125

Point mutant NY-ESO peptide 4A
SLLAWITQC
(SEQ ID NO: 124)

Fig. 126

Point mutant NY-ESO peptide 5A
SLLMAITQC
(SEQ ID NO: 125)

Fig. 127

Point mutant NY-ESO peptide 6L
SLLMWLTQC
(SEQ ID NO: 126)

Fig. 128

Point mutant NY-ESO peptide 7F
SLLMWIFQC
(SEQ ID NO: 127)

Fig. 129

Point mutant NY-ESO peptide 8A
SLLMWITAC
(SEQ ID NO: 128)

Fig. 130

Point mutant NY-ESO peptide 9A
SLLMWITQA
(SEQ ID NO: 129)

Fig. 131

gp100 Peptide
IMDQVPFSV
(SEQ ID NO: 130)

Fig. 132

Homologous peptide DOLPP1
SQFMWFFSV
(SEQ ID NO: 131)

Fig. 133

Homologous peptide IL20RB
SLFMWFFYA
(SEQ ID NO: 132)

Fig. 134

Homologous peptide PRKD2
SLDMWSVG
(SEQ ID NO: 133)

Fig. 135

Homologous peptide CD163
SIPMWVDNV
(SEQ ID NO: 134)

Fig. 136

Homologous peptide P2RY8
SVAMWAVFL
(SEQ ID NO: 135)

Fig. 137

C3E-7034

GEVQLVESGGGLVQPGGSLRLSCAASGVTFNYYGMSWIRQAPGKGLEWVASITNSG
GRIYYPDSVKGRFTISRENTQKTLYLQMNSLRAEDTAVYYCTLDGRDGWVAYWGQG
TLVTVSSGGGGSGGGGSGGGGSNFMLTQPHSVSESPGKTVTISCKRNTGNIGSNYV
NWIQQHEGSSPTTIIYRDDKRPDGVSDRFSGSIDRSSKSASLTISNLKTEDEADYF
CQSYSSGFIFGGGTKLTVLGAAAGAGGDYKDDDDKGAHHHHHHH

(SEQ ID NO: 136), C3E-7034 (1-269), C3E-7034 scFv (2-
243), FLAG-His tag (244-269)

Fig. 138

C3E-7036

GEVQLVESGGGLVQPGGSLRLSCAASGVTFNYYGMSWIRQAPGKGLEWVASITNSG
GRIYYPDSVKGRFTISRENTQKTLYLQMNSLRAEDTAVYYCTLDGRDGWVAYWGQG
TLVTVSSGGGGSGGGSGGGGSNFMLTQPSSVSGVPGQRVTISCTGNTGNIGSNYV
NWIYQQLPGTAPKLLIYRDDKRPSGVPDRFSGSKSGTSASLAIITGFAEDEADYYCQ
SYSSGFIFGGGTKLTVLGAAAGAGGDYKDDDDKGAAHHHHHH

(SEQ ID NO: 137), C3E-7036 (1-267), C3E-7036 scFv (2-
241), FLAG-His tag (242-267)

Fig. 139

C3E-7085

GEVQLVESGGGLVQPGGSLRLSCAASGVTFNYYGMSWIRQAPGKGLEWVASITRSG
GRIYYPDSVKGRFTISRENTQKTLYLQMNSLRAEDTAVYYCTLDGRDGWVAYWGQG
TLVTVSSGGGGSGGGGSGGGGSNFMLTQPSSVSGVPGQRVTISCTGNTGNIGSNYV
NWIQQLPGTAPKLLIYRDDKRPSGVPDRFSGSKSGTSASLAITGFAEDEADYYCQ
SYSSGFIFGGGTKLTVLGAAAGAGGDYKDDDDKGAAHHHHHH

(SEQ ID NO: 138), C3E-7085 (1-267), C3E-7085 scFv (2-
241), FLAG-His tag (242-267)

Fig. 140

C3E-7088

GEVQLVESGGGLVQPGGSLRLSCAASGVTFNYYGMSWIRQAPGKGLEWVASITRSG
GRIYYPDSVKGRFTISRENTQKTLYLQMNSLRAEDTAVYYCTLDGRDGWVAYWGQG
TLVTVSSGGGGSGGGSGGGGSNFMLTQPHSVSESPGKTVTISCKRNTGNIGSNYV
NWKQHEGSSPTTIIYRNDKRPDGVSDRFSGSIDRSSKSASLTISNLKTEDEADYF
CQSYSSGFIFGGGTKLTVLGAAAGAGGDYKDDDDKGAAAHHHHHH
(SEQ ID NO: 139), C3E-7088(1-269), C3E-7088 scFv (2-243),
FLAG-His tag (244-269)

Fig. 141

C3E-7093

GEVQLVESGGGLVQPGGSLRLSCAASGVTFNYYGMSWIRQAPGKGLEWVASITSSG
GRIYYPDSVKGRFTISRENTQKTLYLQMNSLRAEDTAVYYCTLDGRDGWVAYWGQG
TLVTVSSGGGGSGGGGSGGGGSNFMLTQPHSVSESPGKTVTISCKRNTGNIGSNYV
NWIQQHEGSSPTTIIYRNDKRPDGVSDRFSGSIDRSSKSASLTISNLKTEDEADYF
CQSYSSGFIFGGGTKLTVLGAAAGAGGDYKDDDDKGAHHHHHHH

(SEQ ID NO: 140), C3E-7093 (1-269), C3E-7093 scFv (2-
243), FLAG-His tag (244-269)

Fig. 142

C3E-7085 heavy chain CDRH1	GVTFNYYG
C3E-7085 heavy chain CDRH2	ITRSGGRI
C3E-7085 heavy chain CDRH3	TLDGRDGWVAY
C3E-7085 light chain CDRL1	TGNIGSNY
C3E-7085 light chain CDRL2	RDD
C3E-7085 light chain CDRL3	QSYSSGFI

(SEQ ID NOS: 141 to 144 and 146)

Fig. 143

C3E-7078

GEVQLVESGGGLVQPGGSLRLSCAASGVTFNYYGMSWIRQAPGKGLEWVASITRSG
GRIYYPDSVKGRFTISRENTQKTLYLQMNSLRAEDTAVYYCTLDGRDGWVAYWGQG
TLVTVSSGGGGSGGGSGGGGSNFMLTQPHSVSESPGKTVTISCKRNTGNIGSNYV
NWXQQHEGSSPTTIIYRDDKRPDGVSDRFSGSIDRSSKSASLTISNLKTEDEADYF
CQSYSSGFIFGGGTKLTVLGAAAGAGGDYKDDDDKGAAAHHHHHH

(SEQ ID NO: 147), C3E-7078(1-269), C3E-7078 scFv (2-243),
FLAG-His tag (244-269)

Fig. 144

NYF-0014-HC2

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
CGGAGAGGTGCAGCTGGTGGAGTCTGGGGGAGGCGTGGTCCAGCCTGGGAGGTCCC
TGACACCCCTCCTGTTTCAGCGTCTGGATTCTCCATCCGTAGTTATGATATACACTGG
GTCCGCCAGGCTCCAGGCAAGGGGCTAGAGTGGGTGGCCACTATATCATATGATGG
AAGTCAGAAGTACTTCGCAGACTCCGTGAAGGGCCGATTTACCATCTTCAGAGACG
AATCGGAGAACATGGTGTATCTGCAAATGAGCGGCCTGAGAGTTGAGGACACGGCT
GTTTATCACTGTGCGAGAGGGAGTAGTGGTCATTATGAGGCTTTTGATATCTGGGG
CCAAGGGACAATGGTCCACCGTCTCTTCAGGTGGAGGCGGTTTCAGGCGGAGGTGGCA
GCGGCGGTGGCGGGAGTCAGTCTGTGTTGACGCAGCCGCCCTCAGTGTCTGCGGCC
CCAGGACAGAAGGTCACCATCTCCTGCTCTGGAAGCAGCTCCAACATTGGGAATAA
TTATGTATCCTGGTACCAGCAGCTCCCAGGAACAGCCCCCAAACCTCCTCATTTATG
ACAATAATAAGCGACCCTCAGGGATTCTTGACCGATTCTCTGGCTCCAAGTCTGGC
ACGTCAGCCACCCTGGGCATCACCGGACTCCAGACTGGGGACGAGGCCGATTATTA
CTGCGGAACATGGGATAGCAGCCTGAGTGCCCTTGGGTGTTTCGGCGGAGGGACCA
AGGTCACCGTCCTAGGGGGAGGCGGTTTCAGAAGTGCAGCTGGTGAATCCGGGGGG
GGCCTGGTGCAGCCTGGGGGGAGCCTGAGACTGAGTTGTGCCGCTCTGGGGTGAC
ATTTAACTACTATGGCATGTCTTGATCCGCCAGGCACCTGGAAAGGGCCTGGAGT
GGGTGGCCAGCATCACTAGGTCCGGCGGGCGAATCTACTATCCCGACAGCGTCAAG
GGCAGGTTCACAATTTCCCGCGAGAACACACAGAAAACCTCTGTACCTGCAGATGAA
TAGCCTGAGAGCCGAAGATACAGCTGTGTACTATTGCACTCTGGACGGCAGGGATG
GGTGGGTGCGCTATTGGGGGCAGGGAACCCTGGTGACAGTCAGCTCCGGAGGAGGA
GGATCTGGCGGAGGAGGCAGTGGGGGAGGCGGGTCAAACCTTTATGCTCACTCAGCC
GTCCTCTGTTTCTGGCGTACCTGGCCAACGGGTGACCATTAGCTGTACGGGTAATA
CCGGGAATATCGGGTCTAACTACGTGAACTGGTATCAGCAGCTTCCAGGGACAGCT
CCCAAGTTGCTGATCTATCGCGACGACAAAAGACCCTCAGGGGTCCCTGACCGATT
TAGTGGCAGCAAAAGCGGTACTTCCGCTTCCCTGGCGATAACCGGCTTTTCAGGCCG
AAGATGAGGCAGACTACTATTGCCAGTCATATTCCAGCGGCTTCATCTTCGGAGGC
GGAACATAAGCTGACAGTGTTGGCAGCCGAGCCCAAATCTTCTGACAAAACCTCACAC
ATGCCCACCCTGCCCAGCACCTGAAGCCGAGGGGGACCCTCAGTCTTCCTCTTCC
CCCCAAAACCCAAGGACACCCTCATGATCTCCCGGACCCCTGAGGTACATGCGTG
GTGGTGTGAGTGAGCCACGAAGACCCTGAGGTCAAGTTCAACTGGTACGTGGACGG
CGTGGAGGTGCATAATGCCAAGACAAAGCCCCGGGAGGAGCAGTACAACAGCACGT
ACCGGGTGGTTCAGCGTCCCTACCGTCTGACCAGGACTGGCTGAATGGCAAGGAG
TACAAGTGCAAGGTCTCCAACAAAGCCCTCCCAGCCCCCATCGAGAAAACCATCTC
CAAAGCCAAAGGCCAGCCCCGGGAACCACAGGTGTACGTGTACCCCCCATCCCGGG
ACGAGCTGACCAAGAACCAGGTGAGCCTGACCTGCCTGGTCAAAGGCTTCTATCCC
AGCGACATCGCCGTGGAGTGGGAGAGCAATGGCCAGCCGAGAACAACTACAAGAC
CACCCCTCCCGTGCTGGACTCCGACGGCTCCTTCGCCCTCGTGAGCAAGCTCACCG
TGGACAAGAGCAGGTGGCAGCAGGGCAACGTCTTCTCATGCTCCGTGATGCATGAG
GCTCTGCACAACCACTACACCCAGAAGAGCCTCTCCCTGTCTCCCGGCAAG

(SEQ ID NO: 148), signal sequence (1-57), NYA-0001 (61-798), linker (799-813), C3E-7085 (814-1533), linker (1534-1539), hinge (1540-1584), CH2 (1585-1914), CH3 (1915-2235)

Fig. 145

NYF-0014-HC2

MKHLWFFLLLVAAAPRWVLSGEVQLVESGGGVVQPGRSLTPSCSASGFSIRSYDIHW
VRQAPGKGLEWVATISYDGSQKYFADSVKGRFTIFRDESENMVYLQMSGRLVEDTA
VYHCARGSSGHYEAFDIWGQGTMTVTVSSGGGGSGGGSGGGGSQSVLTQPPSVSAA
PGQKVTISCSGSSSNIGNNYVSWYQQLPGTAPKLLIYDNNKRPSGIPDRFSGSKSG
TSATLGITGLQTGDEADYYCGTWDSSLSAPWVFGGGTKVTVLGGGGSEVQLVESGG
GLVQPPGSLRLSCAASGVTFNYYGMSWIRQAPGKGLEWVASITRSGGRIIYPDSVK
GRFTISRENTQKTLYLQMNSLRAEDTAVYYCTLDGRDGVVAYWGQGTTLVTVSSGGG
GSGGGGSGGGGSNFMLTQPSSVSGVPGQRTISCTGNTGNIGSNYVNWYQQLPGTA
PKLLIYRDDKRPSGVPDRFSGSKSGTSASLAITGFQAEDEADYYCQSYSSGFIFGG
GTKLTVLAAEPKSSDKTHTCPPCPAPEAAGGPSVFLFPPKPKDTLMISRTPEVTCV
VVSVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKE
YKCKVSNKALPAPIEKTISKAKGQPREPQVYVYPPSRDELTKNQVSLTCLVKGFYP
SDIAVEWESNGQPENNYKTTTPVLDSDGSFALVSKLTVDKSRWQQGNVFSQVMHE
ALHNHYTQKSLSLSPGK

(SEQ ID NO: 149), signal sequence (1-19), NYA-0001 (21-266), linker (267-271), C3E-7085 (272-511), linker (512-513), hinge (514-528), CH2 (529-638), CH3 (639-745)

Fig. 146

NYF-0082-HC2

MKHLWFFLLLVAAAPRWVLSGEVQLVESGGGVVQPGRSLTPSCSASGFSIRSYDIHW
VRQAPGKGLEWVATISYDGSQKYFADSVKGRFTIFRDESENMVYLQMSGRLVEDTA
VYHCARGSSGHYEAFDIWGQGTMTVTVSSGGGGSGGGSGGGGSQSVLTQPPSVSAA
PGQKVTISCSGSSSNIGNNYVSWYQRLPGTAPKLLIYDNNKRPSGIPDRFSGSKSG
TSATLGITGLQTGDEADYYCGSWDSSLSAPWVFGGGTKVTVLGGGGSEVQLVESGG
GLVQPPGSLRLSCAASGVTFNYYGMSWIRQAPGKGLEWVASITRSGGRIIYPDSVK
GRFTISRENTQKTLYLQMNSLRAEDTAVYYCTLDGRDGVVAYWGQGTTLVTVSSGGG
GSGGGSGGGGSNFMLTQPSSVSGVPGQRTISCTGNTGNIGSNYVNWYQQLPGTA
PKLLIYRDDKRPSGVPDRFSGSKSGTSASLAITGFQAEDEADYYCQSYSSGFIFGG
GTKLTVLAAEPKSSDKTHTCPPCPAPEAAGGPSVFLFPPKPKDTLMISRTPEVTCV
VVSVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKE
YKCKVSNKALPAPIEKTISKAKGQPREPQVYVPPSRDELTKNQVSLTCLVKGFYP
SDIAVEWESNGQPENNYKTTTPVLDSDGSFALVSKLTVDKSRWQQGNVFSQVMHE
ALHNHYTQKSLSLSPGK

(SEQ ID NO: 150), signal sequence (1-19), NYA-0082 (21-266), linker (267-271), C3E-7085 (272-511), linker (512-513), hinge (514-528), CH2 (529-638), CH3 (639-745)

Fig. 147

Human CD3ε amino acid sequence

MQSGTHWRVLGLCLLSVGWVGQDGNEEMGGITQTPYKVSISGTTVILTCPQYPGSE
ILWQHNDKNIGGDEDDKNIGSDEDHLSLKEFSELEQSGYYVCYPRGSKPEDANFYL
YLRARVCENCMEMDVMSVATIVIVDICITGGLLLLVYYWSKNRKAKAKPVTRGAGA
GGRQRGQNKERPPVPNPDYEPPIRKGQDLYSGLNQRI
(SEQ ID NO: 151: NCBI Reference Sequence: NP_000724.1)

Fig. 148

NYZ-0038-HC2

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
CGGTGAGGTACAGCTGGTGGGAATCCGGTGGAGGCCCTTGTCGAACCGGGAGGCTCCC
TGCGACTCTCTTGCGCCGCATCTGGCTTTTCCATACGATCCTACGATATTCATTGG
GTCCGACAAGCCCCTGGCAAAGGTCTTGAATGGGTGGCCACTATCTCCTACGACGG
GTCTCAGAAATATTATGCGGATTCACTGAAGGGGCGGTTTACAATTTACGAGACG
AGTCAAAGAATACACTGTACCTCCAAATGAATTCAGTGAAGAGCCGAGGATACTGCA
GTCTATCATTTGTGCAAGAGGGTCCCTCAGGCCACTACGAGGCCCTTTGATATATGGGG
CCAAGGCACCTTGGTAACCGTTAGTAGCGGAGGTGGAGGAAGCGGAGGCGGCGGTT
CCGGAGGAGGTGGAAGTCAATCCGTTCTGACTCAGCCACCCCTCCGCCTCTGGCACC
CCGGGCCAACGGGTCACAATATCATGTTCTGGCTCTTCAAGCAACATCGGAAACTG
GTACGTGAGCTGGTACCAGCAGCTCCCCGGCAGCGCGCCCAAGCTTCTGATATATG
ACAACAACAAACGGCCCAGTGGAGTTCCTGACAGATTCAGTGGGTCTAAAAGTGGT
ACAAGCGCTAGCCTGGCCATAAGTGGTCTGCAGAGTGAAGATGAGGCGGACTATTA
TTGCGGAACCTGGGACTCCAGCCTGAGCGCTCCCTGGGTTTTCGGCGGAGGTACTA
AGCTGACCGTCTTGGGAGGAGGCGGTTTCAAGAGTGCAGCTGGTGGATCCGGGGGG
GGCCTGGTGCAGCCTGGGGGGAGCCTGAGACTGAGTTGTGCCGCTCTGGGGTGAC
ATTTAACTACTATGGCATGTCTTGGATCCGCCAGGCACCTGGAAAGGGCCTGGAGT
GGGTGGCCAGCATCACTAGGTCCGGCGGGCGAATCTACTATCCCGACAGCGTCAAG
GGCAGGTTCACAATTTCCCGCGAGAACACACAGAAAACCTCTGTACCTGCAGATGAA
TAGCCTGAGAGCCGAAGATACAGCTGTGTACTATTGCACTCTGGACGGCAGGGATG
GGTGGGTGCGCTATTGGGGGTGTGGAACCCCTGGTGACAGTCAGCTCCGGAGGAGGA
GGATCTGGCGGAGGAGGCAGTGGGGGAGGCGGGTCAAACCTTTATGCTCACTCAGCC
GTCCTCTGTTTCTGGCGTACCTGGCCAACGGGTGACCATTAGCTGTACGGGTAATA
CCGGGAATATCGGGTCTAACTACGTGAACTGGTATCAGCAGCTTCCAGGGACATGT
CCCAAGTTGCTGATCTATCGCGACGACAAAAGACCCCTCAGGGGTCCCTGACCGATT
TAGTGGCAGCAAAAGCGGTACTTCCGCTTCCCTGGCGATAACCGGCTTTTACGGCCG
AAGATGAGGCAGACTACTATTGCCAGTCATATTCCAGCGGCTTCATCTTCCGAGGC
GGAACCTAAGCTGACAGTGTGGCAGCCGAGCCCAAATCTTCTGACAAAACCTCACAC
ATGCCCACCCTGCCCAGCACCTGAAGCCGAGGGGGACCCCTCAGTCTTCTCTTCC
CCCCAAAACCAAGGACACCCTCATGATCTCCCGGACCCCTGAGGTACATGCGTG
GTGGTGTGAGTGAGCCACGAAGACCCCTGAGGTCAAGTTCAACTGGTACGTGGACGG
CGTGGAGGTGCATAATGCCAAGACAAAGCCCCGGGAGGAGCAGTACAACAGCACGT
ACCGGGTGGTCAAGCTCCTCACCCTCCTGCACCAGGACTGGCTGAATGGCAAGGAG
TACAAGTGCAAGGTCTCCAACAAAGCCCTCCAGCCCCCATCGAGAAAACCATCTC
CAAAGCCAAAGGCCAGCCCCGGGAACACAGGTGTACGTGTACCCCCCATCCCGGG
ACGAGCTGACCAAGAACCAGGTGAGCCTGACCTGCCTGGTCAAAGGCTTCTATCCC
AGCGACATCGCCGTGGAGTGGGAGAGCAATGGCCAGCCGAGAACAACTACAAGAC
CACCCCTCCCGTGCTGGACTCCGACGGCTCCTTCGCCCTCGTGAGCAAGCTCACCG
TGGACAAGAGCAGGTGGCAGCAGGGCAACGTCTTCTCATGCTCCGTGATGCATGAG
GCTCTGCACAACCACTACACCCAGAAGAGCCTCTCCCTGTCTCCCGGCAAG

(SEQ ID NO: 152), signal sequence (1-57), NYA-2061 (61-798), linker (799-813), C3E-7096(814-1533), linker (1534-1539), hinge (1540-1584), CH2 (1585-1914), CH3 (1915-2235)

Fig. 149

NYZ-0082-HC2

ATGAAACACCTTGGTGTCTTCTTCTCTGCTGGTGGGACGCTCCAGATGGGTGCTGAG
CGGTGAGGTACAGCTGGTGGGAATCCGGTGGAGGCCCTTGTGCAACCGGGAGGCTCCC
TGCGACTCTCTTGCGCCGCATCTGGCTTTTCCATACGATCCTACGATATTATTGG
GTCCGACAAGCCCCCTGGCAAAGGTCTTGAATGGGTTGCCACTATCTCCTACGACGG
GTCTCAGAAATATTATGCGGATTTCAGTGAAGGGGCGGTTTACAATTTACAGAGACG
AGTCAAAGAATACACTGTACCTCCAAATGAATTCAGTGAAGCCGAGGATACTGCA
GTCTATCATTTGTGCAAGAGGGTCTCAGGCCACTACGAGGCCCTTTGATATATGGGG
CCAAGGCACCTTGGTAACCGTTAGTAGCGGAGGTGGAGGAAGCGGAGGCGGCGGTT
CCGGAGGAGGTGGAAGTGGAGGAGGTGGATCTCAATCCGTTCTGACTCAGCCACCC
TCCGCCTCTGGCACCCCGGGCCAACGGGTACAAATATCATGTTCTGGCTCTTCAAG
CAACATCGGAAACTGGTACGTGAGCTGGTACCAGCAGCTCCCCGGGCACGGCGCCCCA
AGCTTCTGATATATGACAACAACAACCGCCCCAGTGGAGTTTCTGACAGATTTCAGT
GGGTCTAAAAGTGGTACAAGCGCTAGCCTGGCCATAAGTGGTCTGCAGAGTGAAGA
TGAGGCGGACTATTATTGCGGAACCTGGGACTCCAGCCTGAGCGCTCCCTGGGTTT
TCGGCGGAGGTACTAAGCTGACCGTCTTGGGAGGAGGCGGTTTCAAGTGCAGCTG
GTGGAATCCGGGGGGGGCCTGGTGCAGCCTGGGGGGAGCCTGAGACTGAGTTGTGC
CGCCTCTGGGGTGACATTTAACTACTATGGCATGTCTTGGATCCGCCAGGCACCTG
GAAAGGGCCTGGAGTGGGTGGCCAGCATCACTAGGTCCGGCGGGCGAATCTACTAT
CCCGACAGCGTCAAGGGCAGGTTTACAATTTCCCGCGAGAACACACAGAAAACCTCT
GTACCTGCAGATGAATAGCCTGAGAGCCGAAGATACAGCTGTGTACTATTGCACTC
TGGACGGCAGGGATGGGTGGGTGCCTATTGGGGGTGTGGAACCCTGGTGCAGTC
AGCTCCGGAGGAGGAGGATCTGGCGGAGGAGGCAGTGGGGGAGGCGGGTCAAACCTT
TATGCTCACTCAGCCGTCCTCTGTTTCTGGCGTACCTGGCCAACGGGTGACCATTA
GCTGTACGGGTAATACCGGGAATATCGGGTCTAACTACGTGAACTGGTATCAGCAG
CTTCCAGGGACATGTCCCAAGTTGCTGATCTATCGCGACGACAAAAGACCCTCAGG
GGTCCCTGACCGATTTAGTGGCAGCAAAGCGGTACTTCCGCTTCCCTGGCGATAA
CCGGCTTTTCAGGCCGAAGATGAGGCAGACTACTATTGCCAGTCATATTCCAGCGGC
TTCATCTTCGGAGGCGGAACCTAAGCTGACAGTGTTGGCAGCCGAGCCCCAATCTTC
TGACAAAACCTCACACATGCCACCCTGCCAGCACCTGAAGCCGAGGGGGGACCCCT
CAGTCTTCTCTTCCCCCAGAACCCAAAGGACACCCTCATGATCTCCCGGACCCCT
GAGGTACATGCGTGGTGGTGTGAGTGAGCCACGAAGACCCTGAGGTCAAGTTCAA
CTGGTACGTGGACGGCGTGGAGGTGCATAATGCCAAGACAAAGCCCCGGGAGGAGC
AGTACAACAGCACGTACCGGGTGGTACGCGTCCTACCGTCTTGCACCAGGACTGG
CTGAATGGCAAGGAGTACAAGTGCAAGGTCTCCAACAAAGCCCTCCAGCCCCCAT
CGAGAAAACCATCTCCAAAGCCAAAGGCCAGCCCCGGGAACCACAGGTGTACGTGT
ACCCCCCATCCCGGGACGAGCTGACCAAGAACCAGGTGACCTGACCTGCCTGGTC
AAAGGCTTCTATCCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGCCAGCCCCGA
GAACAACCTACAAGACCACCCTCCCGTGCTGGACTCCGACGGCTCTTTCGCCCTCG
TGAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGGCAACGTCTTCTCATGC
TCCGTGATGCATGAGGCTCTGCACAACCACTACACCAGAGAGCCTCTCCCTGTC
TCCCGGCAAG

(SEQ ID NO: 153), signal sequence (1-57), NYA-3061 (61-813), linker (814-828), C3E-7096(829-1548), linker (1549-1554), hinge (1555-1599), CH2 (1600-1929), CH3 (1930-2250)

Fig. 150

NYZ-0083-HC2

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
CGGTGAGGTACAGCTGGTGGGAATCCGGTGGAGGCCTTGTGCAACCGGGAGGCTCCC
TGCGACTCTCTTGCGCCGCATCTGGCTTTTCCATACGATCCTACGATATTCATTGG
GTCCGACAAGCCCCCTGGCAAAGGTCTTGAATGGGTGGCACTATCTCCTACGACGG
GTCTCAGAAATATTATGCGGATTCACTGAAGGGGCGGTTACAAATTTACAGAGACG
AGTCAAAGAATACACTGTACCTCCAAATGAATTCAGTGAAGCCGAGGATACTGCA
GTCTATCATTGTGCAAGAGGGTCTCAGGCCACTACGAGGCCTTTGATATATGGGG
CCAAGGCACCTTGGTAACCGTTAGTAGCGGAGGTGGAGGAAGCGGAGGCGGCGGTT
CCGGAGGAGGTGGAAGTGGAGGAGGTGGATCTCAATCCGTTCTGACTCAGCCACCC
TCCGCCTCTGGCACCCCGGGCCAACGGGTCACAATATCATGTTCTGGCTCTTCAAG
CAACATCGGAACTGGTACGTGAGCTGGTACCAGCAGCTCCCCGGCACGGCGCCCA
AGCTTCTGATATATGACAACAACAAACGGCCCAAGTGGAGTTCCTGACAGATTCAGT
GGGTCTAAAAGTGGTACAAGCGCTAGCCTGGCCATAAGTGGTCTGCAGAGTGAAGA
TGAGGCGGACTATTATTGCGGAACCTGGGACTCCAGCCTGAGCGCTCCCTGGGTTT
TCGGCGGAGGTACTAAGCTGACCGTCTTGGGAGGAGGCGGTTGAGAAGTGCAGCTG
GTGGAATCCGGGGGGGGCCTGGTGCAGCCTGGGGGGAGCCTGAGACTGAGTTGTGC
CGCCTCTGGGGTGACATTTAACTACTATGGCATGTCTTGGATCCGCCAGGCACCTG
GAAAGTGTCTGGAGTGGGTGGCCAGCATCACTAGGTCCGGCGGGCGAATCTACTAT
CCCGACAGCGTCAAGGGCAGGTTACAAATTTCCCGCGAGAACACACAGAAAACCTCT
GTACCTGCAGATGAATAGCCTGAGAGCCGAAGATACAGCTGTGTACTATTGCACTC
TGGACGGCAGGGATGGGTGGGTGCGCTATTGGGGGCAGGGAACCCTGGTGACAGTC
AGCTCCGGAGGAGGAGGATCTGGCGGAGGAGGAGTGGGGGAGGCGGGTCAAACCT
TATGCTCACTCAGCCGTCCTCTGTTTCTGGCGTACCTGGCCAACGGGTGACCATTA
GCTGTACGGGTAATACCGGGAATATCGGGTCTAACTACGTGAACCTGGTATCAGCAG
CTTCCAGGGACAGCTCCCAAGTTGCTGATCTATCGCGACGACAAAAGACCCTCAGG
GGTCCCTGACCGATTTAGTGGCAGCAAAGCGGTACTTCCGCTTCCCTGGCGATAA
CCGGCTTTTCAAGCCGAAGATGAGGCAGACTACTATTGCCAGTCATATTCCAGCGGC
TTCATCTTCCGATGTGGAACCTAAGCTGACAGTGTGGCAGCCGAGCCCAATCTTC
TGACAAAACCTCACACATGCCACCCCTGCCCAGCACCTGAAGCCGCAGGGGGACCCCT
CAGTCTTCTCTTCCCCCCTAAACCCAAAGGACACCCCTCATGATCTCCCGGACCCCT
GAGGTCACATGCGTGGTGGTGTGAGTGAAGCCACGAAGACCCTGAGGTCAAGTTCAA
CTGGTACGTGGACGGCGTGGAGGTGCATAATGCCAAGACAAAGCCCCGGGAGGAGC
AGTACAACAGCACGTACCGGGTGGTCAAGCTCCTCACCCTCCTGCACCAGGACTGG
CTGAATGGCAAGGAGTACAAGTGCAAGGTCTCCAACAAAGCCCTCCCAGCCCCCAT
CGAGAAAACCATCTCCAAAGCCAAAGGCCAGCCCCGGGAACCACAGGTGTACGTGT
ACCCCCCATCCCGGGACGAGCTGACCAAGAACCAGGTGAGCCTGACCTGCCTGGTC
AAAGGCTTCTATCCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGCCAGCCCCGA
GAACAACCTACAAGACCACCCCTCCCGTGTGACTCCGACGGCTCCTTCGCCCTCG
TGAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGGCAACGTCTTCTCATGC
TCCGTGATGCATGAGGCTCTGCACAACCACTACACCCAGAAGAGCCTCTCCCTGTC
TCCCGGCAAG

(SEQ ID NO: 154), signal sequence (1-57), NYA-3061 (61-813), linker (814-828), C3E-7097(829-1548), linker (1549-1554), hinge (1555-1599), CH2 (1600-1929), CH3 (1930-2250)

Fig. 151

NYZ-0038-HC2

MKHLWFFLLLVAAAPRWVLSGEVQLVESGGGLVQPGGSLRLSCAASGFSIRSYDIHW
VRQAPGKGLEWVATISYDGSQKYYADSVKGRFTISRDESKNTLYLQMNSLRAEDTA
VYHCARGSSGHYEAFDIWGQGTTLVTVSSGGGGSGGGSGGGGSQSVLTQPPSASGT
PGQRTVISCSSSSNIGNWYVSWYQQLPGTAPKLLIYDNNKRPSGVPDRFSGSKSG
TSASLAISGLQSEDEADYYCGTWDSSLSAPWVFGGGTKLTVLGGGGSEVQLVESGG
GLVQPGGSLRLSCAASGVTFNYYGMSWIRQAPGKGLEWVASITRSGGRIIYPDSVK
GRFTISRENTQKTLYLQMNSLRAEDTAVYYCTLDGRDGWVAYWCGTTLVTVSSGGG
SGGGGGSGGGGSNFMLTQPSSVSGVPGQRTVISCSTGNTGNIGSNYVNWYQQLPGTC
PKLLIYRDDKRPSGVPDRFSGSKSGTSASLAITGFQAEDEADYYCQSYSSGFIFGG
GTKLTVLAAEPKSSDKTHTCPPCPAPEAAGGPSVFLFPPKPKDTLMISRTPEVTCV
VVSVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKE
YKCKVSNKALPAPIEKTISKAKGQPREPQVYVPPSRDELTKNQVSLTCLVKGFYP
SDIAVEWESNGQPENNYKTTTPVLDSDGSFALVSKLTVDKSRWQQGNVFSQVMHE
ALHNHYTQKSLSLSPGK

(SEQ ID NO: 155), signal sequence (1-19), NYA-2061 (21-266), linker (267-271), C3E-7096(272-511), linker (512-513), hinge (514-528), CH2 (529-638), CH3 (639-745)

Fig. 152

NYZ-0082-HC2

MKHLWFFLLLVAAAPRWVLSGEVQLVESGGGLVQPGGSLRLSCAASGFSIRSYDIHW
VRQAPGKGLEWVATISYDGSQKYYADSVKGRFTISRDESKNTLYLQMNSLRAEDTA
VYHCARGSSGHYEAFDIWGQGTLLTVSSGGGGSGGGGSGGGGSGGGGSSQSVLTQPP
SASGTPGQRTISCSGSSSNIGNWYVSWYQQLPGTAPKLLIYDNNKRPSGVPDRFS
GSKSGTSASLAISGLQSEDEADYYCGTWDSSLSAPWVFGGGTKLTVLGGGGSEVQL
VESGGGLVQPGGSLRLSCAASGVTFNYYGMSWIRQAPGKGLEWVASITRSGGRIYY
PDSVKGRFTISRENTQKTLYLQMNSLRAEDTAVYYCTLDGRDGVVAYWGCGLVTV
SSGGGGSGGGGSGGGGSGNFMLTQPSSVSGVPGQRTISCTGNTGNIGSNYVNWYQQ
LPGTCPKLLIYRDDKRPSGVPDRFSGSKSGTSASLAITGEQAEDEADYYCQSYSSG
FIFGGGTKLTVLAAEPKSSDKTHTCPPCPAPEAAGGPSVFLFPPKPKDTLMISRTPE
VTCVTVSVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDW
LNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYVPPSRDELTKNQVSLTCLV
KGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFALVSKLTVDKSRWQQGNVFC
SVMHEALHNHYTQKSLSLSPGK

(SEQ ID NO: 156), signal sequence (1-19), NYA-3061 (21-271), linker (272-276), C3E-7096(277-516), linker (517-518), hinge (519-533), CH2 (534-643), CH3 (644-750)

Fig. 153

NYZ-0083-HC2

MKHLWFFLLLVAAAPRWVLSGEVQLVESGGGLVQPGGSLRLSCAASGFSIRSYDIHW
VRQAPGKGLEWVATISYDGSQKYYADSVKGRFTISRDESKNTLYLQMNSLRAEDTA
VYHCARGSSGHYEAFDIWGQGTTLVTVSSGGGGSGGGSGGGSGGGGSQSVLTQPP
SASGTPGQRTISCSGSSSNIGNWYVSWYQQLPGTAPKLLIYDNNKRPSGVPDRFS
GSKSGTSASLAISGLQSEDEADYYCGTWDSSLSAPWVFGGGTKLTVLGGGGSEVQL
VESGGGLVQPGGSLRLSCAASGVTFNYYGMSWIRQAPGKCLEWVASITRSGGRIYY
PDSVKGRFTISRENTQKTLYLQMNSLRAEDTAVYYCTLDGRDGWVAYWGQGTTLVTV
SSGGGGSGGGSGGGGSNFMLTQPSSVSGVPGQRTISCTGNTGNIGSNYVNWYQQ
LPGTAPKLLIYRDDKRPSGVPDRFSGSKSGTSASLAITGFQAEDEADYYCQSYSSG
FIFGCGTKLTVLAAEPKSSDKTHTCPPCPAPEAAGGPSVFLFPPKPKDITLMISRTF
EVTCTVVSVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDW
LNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYVYPPSRDELTKNQVSLTCLV
KGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFALVSKLTVDKSRWQQGNVFC
SVMHEALHNHYTQKSLSLSPGK

(SEQ ID NO: 157), signal sequence (1-19), NYA-3061 (21-271), linker (272-276), C3E-7097 (277-516), linker (517-518), hinge (519-533), CH2 (534-643), CH3 (644-750)

Fig. 154

NYZ-1010-HC2

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
CGGTGAGGTACAGCTGGTGGGAATCCGGTGGAGGCCTTGTGCAACCGGGAGGCTCCC
TGCGACTCTCTTGCGCCGCATCTGGCTTTTCCATACGATCCTACGATATTCATTGG
GTCCGACAAGCCCCTGGCAAAGGTCTTGAATGGGTGGCCACTATCTCCTACGACGG
GTCTCAGAAATATTATGCGGATTCAGTGAAGGGGCGGTTTCAAAATTTACGAGACG
AGTCAAAGAATACTGTACCTCCAAATGAATTCAGTGAAGAGCCGAGGATACTGCA
GTCTATCATTGTGCAAGAGGGTCCCTCAGGCCACTACGAGGCCTTTGATATATGGGG
CCAAGGCACCTTGGTAACCGTTAGTAGCGGAGGTGGAGGAAGCGGAGGCGGCGGTT
CCGGAGGAGGTGGAAGTGGAGGAGGTGGATCTCAATCCGTTCTGACTCAGCCACCC
TCCGCCTCTGGCACCCCGGGCCAACGGGTCAACAATATCATGTTCTGGCTCTTCAAG
CAACATCGGAACTGGTACGTGAGCTGGTACCAGCAGCTCCCCGGCACGGCGCCCA
AGCTTCTGATATATGACAACAACAACGGCCCAAGTGGAGTTCCTGACAGATTTCAGT
GGGTCTAAAAGTGGTACAAGCGCTAGCCTGGCCATAAGTGGTCTGCAGAGTGAAGA
TGAGGCGGACTATTATTGCGGAACCTGGGACTCCAGCCTGAGCGCTCCCTGGGTTT
TCGGCGGAGGTACTAAGCTGACCGTCTTGGGAGGAGGCGGTTTCAAGTGCAGCTG
GTGGAATCCGGGGGGGGCCTGGTGCAGCCTGGGGGGAGCCTGAGACTGAGTTGTGC
CGCCTCTGGGGTGACATTTAACTACTATGGCATGTCTTGGATCCGCCAGGCACCTG
GAAAGGGCCTGGAGTGGGTGGCCAGCATCACTAGGTCCGGCGGGCGAATCTACTAT
CCCGACAGCGTCAAGGGCAGGTTTCAAAATTTCCCGCGAGAACACACAGAAAATCT
GTACCTGCAGATGAATAGCCTGAGAGCCGAAGATACAGCTGTGTACTATTGCACTC
TGGACGGCAGGGATGGGTGGGTGCCTATTGGGGTCAGGGAACCCTGGTGACAGTC
AGCTCCGCCTCCACCAAGGGCCCAAGCGTCTTCCCCCTGGCACCCCTCCTCCAAGAG
CACCTCTGGCGGCACAGCCGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAAC
CCGTGACCGTGAGCTGGAACCTCAGGCGCCCTGACCAGCGGCGTGACACCTTCCCC
GCTGTCTTGCAGTCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCTC
CAGCAGCTTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAGCCAGCAACA
CCAAGGTGGACAAGAGAGTTGAGCCCAAGTCTTGGCAGAAAATCACAATGCCCCA
CCCTGCCCAGCACCTGAAGCCGAGGGGGACCCTCAGTCTTCTCTTCCCCCAAA
ACCCAAGGACACCCTCATGATCTCCCGGACCCCTGAGGTACATGCGTGGTGGTGT
CAGTGAGCCACGAAGACCCTGAGGTCAAGTTCAACTGGTACGTGGACGGCGTGGAG
GTGCATAATGCCAAGACAAAGCCCCGGGAGGAGCAGTACAACAGCACGTACCGGGT
GGTCAGCGTCTCACTGTCTTGCACCAGGACTGGCTGAATGGCAAGGAGTACAAGT
GCAAGGTCTCCAACAAAGCCCTCCCAGCCCCCATCGAGAAAACCATCTCCAAGCC
AAAGGCCAGCCCCGGGAACCACAGGTGTACGTGTACCCCCCATCCGGGACGAGCT
GACCAAGAACCAGGTCAGCCTGACCTGCCTGGTCAAAGGCTTCTATCCCAGCGACA
TCGCCGTGGAGTGGGAGAGCAATGGCCAGCCCGAGAACAACTACAAGACCACCCCT
CCCGTGCTGGACTCCGACGGCTCCTTCGCCCTCGTGAGCAAGCTCACCGTGGACAA
GAGCAGGTGGCAGCAGGGCAACGTCTTCTCATGCTCCGTGATGCATGAGGCTCTGC
ACAACCACTACACCCAGAAGAGCCTCTCCCTGTCTCCCGGCAAG

(SEQ ID NO: 158), signal sequence (1-57), NYA-3061 (61-813), linker (814-828), C3E-7085-VH (829-1182), CH1 (1183-1476), hinge (1477-1521), CH2 (1522-1851), CH3 (1831-2172)

Fig. 155

C3E-7085-LC

ATGGTGCTGCAGACCCAGGTGTTTCATCTCCCTGCTGCTGTGGATCTCCGGCGCGTA
CGGCAACTTTATGCTCACTCAGCCGTCCTCTGTTTCTGGCGTACCTGGCCAACGGG
TGACCATTAGCTGTACGGGTAATACCGGGAATATCGGGTCTAACTACGTGAACTGG
TATCAGCAGCTTCCAGGGACAGCTCCCAAGTTGCTGATCTATCGCGACGACAAAAG
ACCCTCAGGGGTCCCTGACCGATTTAGTGGCAGCAAAAGCGGTACTTCCGCTTCCC
TGGCGATAACCGGCTTTCAGGCCGAAGATGAGGCAGACTACTATTGCCAGTCATAT
TCCAGCGGCTTCATCTTCGGAGGCGGAATAAGCTGACAGTGCTGGGCCAGCCTAA
GGCTGCCCCCTAGCGTGACCCTGTTCCCTCCTTCCAGCGAGGAGCTTCAAGCTAACA
AGGCCACCCTGGTGTGTCTTATCTCTGACTTCTACCCTGGCGCTGTGACCGTGGCC
TGGAAGGCTGACAGCTCCCCTGTGAAGGCCGGAGTGAGACCACCACACCTAGCAA
GCAGTCTAACAACAAGTACGCTGCCAGCTCCTACCTGAGCCTTACCCCTGAGCAGT
GGAAGTCTCACAGAAGCTACTCCTGTCAAGTGACCCACGAGGGCAGCACCGTGGAG
AAGACCGTGGCTCCTACCGAGTGTTCC

(SEQ ID NO: 159), signal sequence (1-60), C3E-7085_VL
(61-381), CL (382-699)

Fig. 156

NYZ-1010-HC2

MKHLWFFLLLVAAPRWVLSGEVQLVESGGGLVQPGGSLRLSCAASGFSIRSYDIHW
VRQAPGKGLEWVATISYDGSQKYADSVKGRFTISRDESKNTLYLQMNSLRAEDTA
VYHCARGSSGHYEAFDIWGQGTLVTVSSGGGGSGGGGSGGGGSGGGGSQSVLTQPP
SASGTPGQRVTISCSGSSSNIGNWYVSWYQQLPGTAPKLLIYDNNKRPSGVPDRFS
GSKSGTSASLAISGLQSEDEADYYCGTWDSLSAPWVFGGGTKLTVLGGGGSEVQL
VESGGGLVQPGGSLRLSCAASGVTFNYYGMSWIRQAPGKGLEWVASITRSGGRIYY
PDSVKGRFTISRENTQKTLYLQMNSLRAEDTAVYYCTLDGRDGVVAYWGQGTLLTV
SSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTFSWNSGALTSGVHTFP
AVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKRVEPKSCDKTHTCP
PCPAPEAAAGGPSVFLEFPPKPKDTLMISRTPEVTCVVSFVSHEDPEVKFNWYVDGVE
VHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKA
KGQPREPQVYVYPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTT
PVLDSGDSFALVSKLTVDKSRWQQGNVFSQVMHEALHNHYTQKSLSLSPGK

(SEQ ID NO: 160), signal sequence (1-19), NYA-3061 (21-271), linker (272-276), C3E-7085-VH (277-394), CH1 (395-492), hinge (493-507), CH2 (508-617), CH3 (618-724)

Fig. 157

C3E-7085-LC

MVLQTQVFISLLLWISGAYGNFMLTQPSSVSGVPGQRVTTISCTGNTGNIGSNYVNW
YQQLPGTAPKLLIYRDDKRPSGVPDRFSGSKSGTSASLAITGFQAEDEADYYCQSY
SSGFIFGGGTKLTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTVA
WKADSSPVKAGVETTTTPSKQSNNKYAASSYLSLTPEQWKSHRSYSCQVTHEGSTVE
KTVAPTECS

(SEQ ID NO: 161), signal sequence (1-20), C3E-7085_VL
(21-127), CL (128-233)

Fig. 158A

Antibody	Substance added to T2											
	Solvent	Peptide										
	DMSO	NY-ESO	1F	2M	3A	4A	5A	6L	7F	8A	9A	gp100
NYZ-0038	<u>1.02</u>	57.13	<u>0.66</u>	45.71	29.72	<u>1.39</u>	<u>1.13</u>	47.00	<u>5.56</u>	<u>19.15</u>	58.37	<u>0.37</u>
NYZ-0082	<u>1.02</u>	54.49	<u>0.79</u>	41.13	<u>27.09</u>	<u>2.26</u>	<u>1.74</u>	45.35	<u>10.08</u>	<u>19.93</u>	58.10	<u>0.36</u>
NYZ-0083	<u>1.01</u>	53.60	<u>0.96</u>	41.34	28.94	<u>3.05</u>	<u>2.15</u>	45.85	<u>12.76</u>	<u>22.46</u>	56.72	<u>0.34</u>
NYZ-1010	<u>1.15</u>	43.32	<u>0.86</u>	39.52	<u>21.30</u>	<u>1.05</u>	<u>1.02</u>	37.59	<u>3.86</u>	<u>17.38</u>	51.90	<u>0.53</u>

Fig. 158B

Antibody	Substance added to T2							
	Solvent	Peptide						
	DMSO	NY-ESO	DOLPP1	IL20RB	PRKD2	CD163	P2RY8	gp100
NYZ-0038	1.07	<u>51.65</u>	0.71	0.70	0.41	0.66	0.89	0.44
NYZ-0082	1.11	<u>47.54</u>	0.69	0.69	0.40	0.62	0.87	0.42
NYZ-0083	1.01	<u>47.17</u>	0.63	0.62	0.37	0.55	0.76	0.38
NYZ-1010	1.19	<u>44.35</u>	0.70	0.88	0.54	0.67	0.95	0.54

Fig. 159A

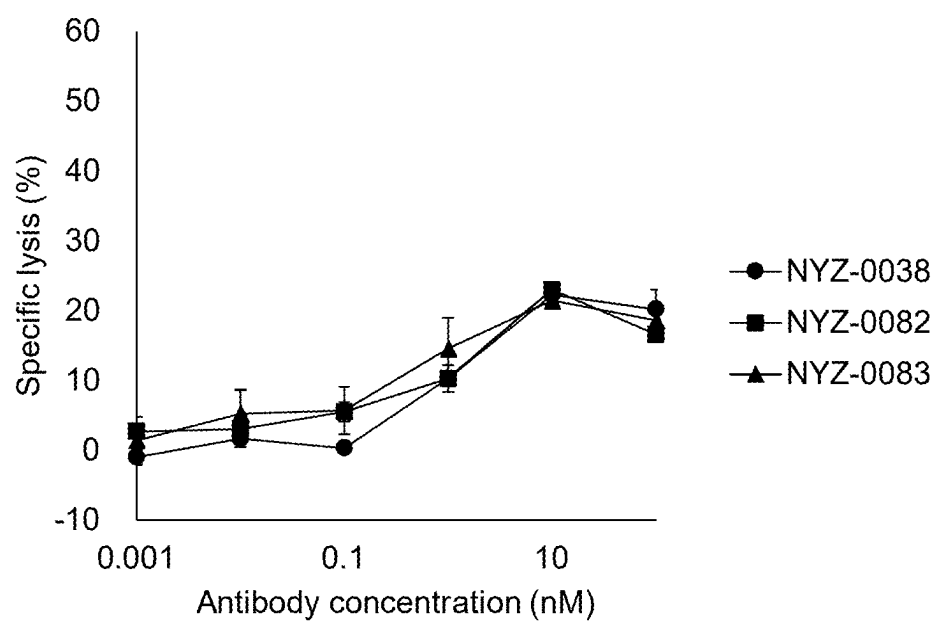


Fig. 159B

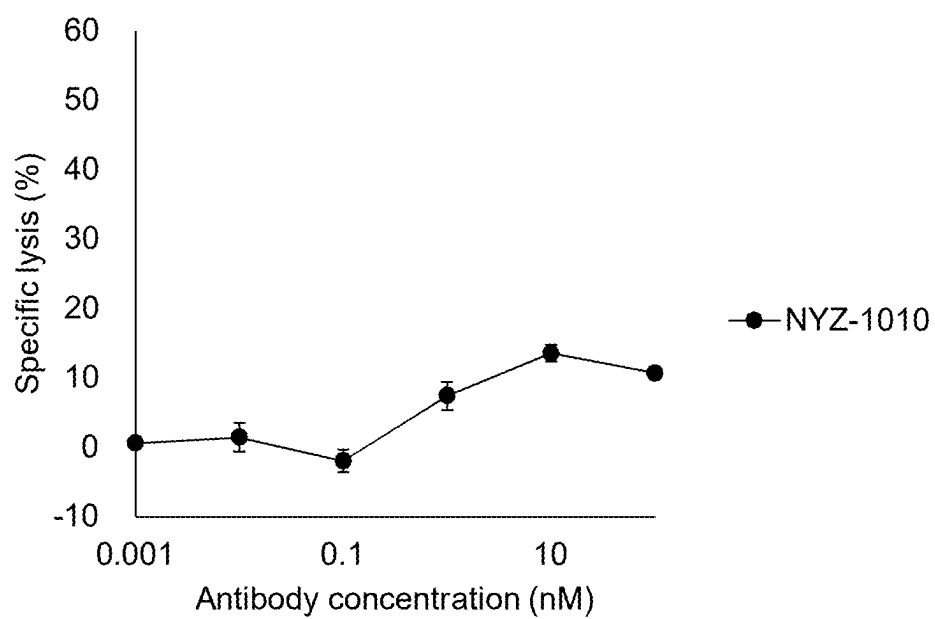


Fig. 159C

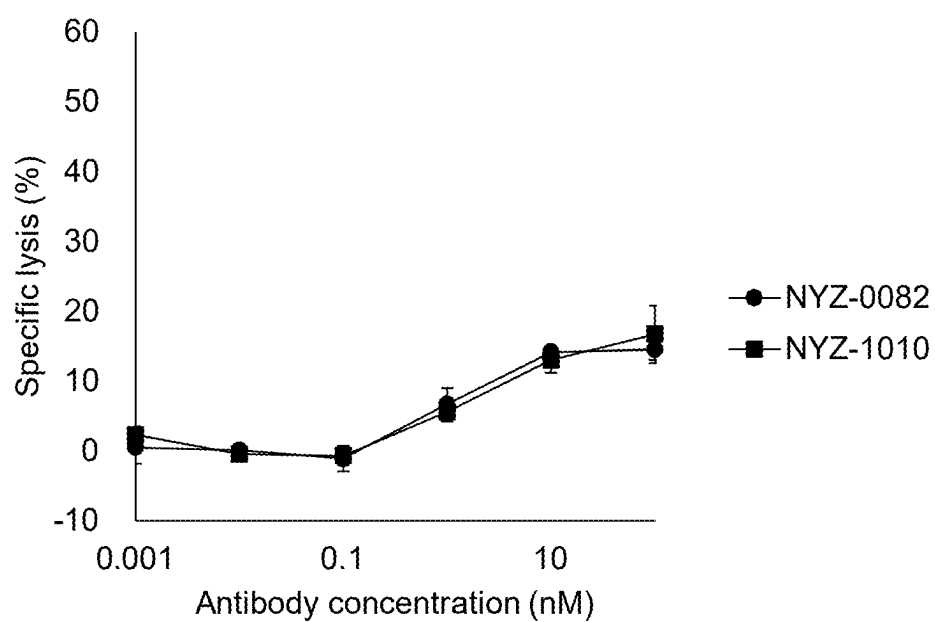


Fig. 159D

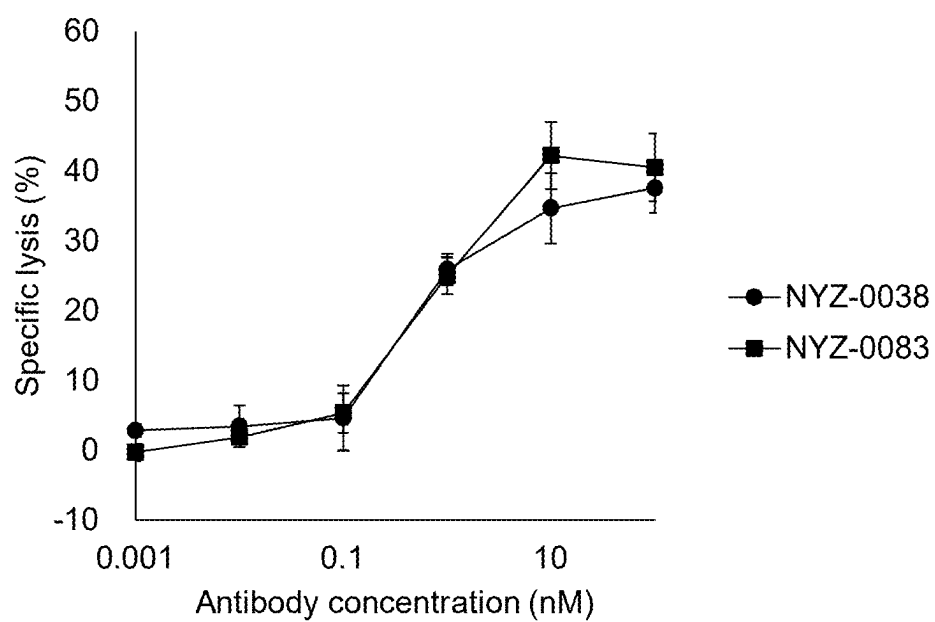


Fig. 159E

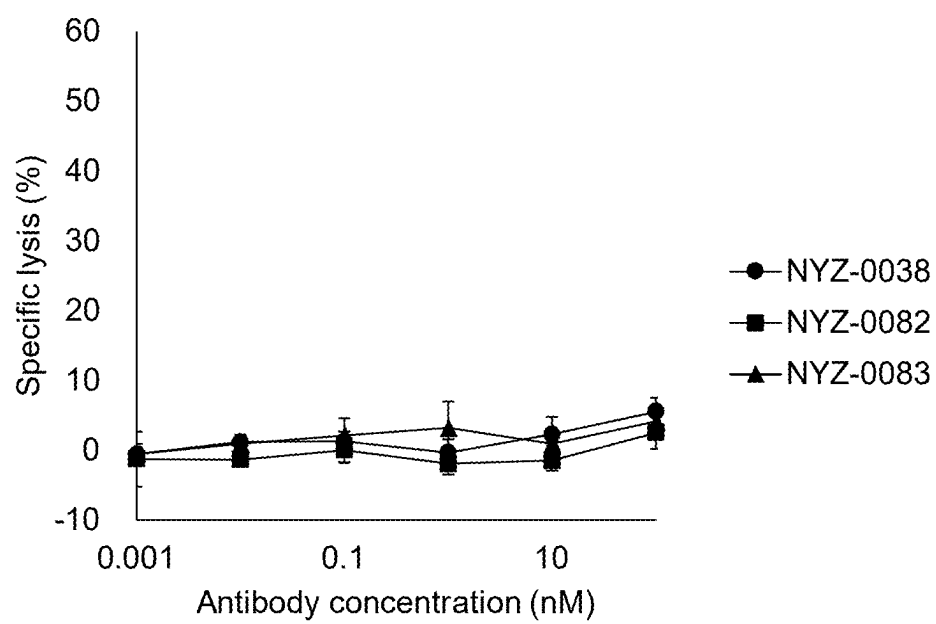


Fig. 159F

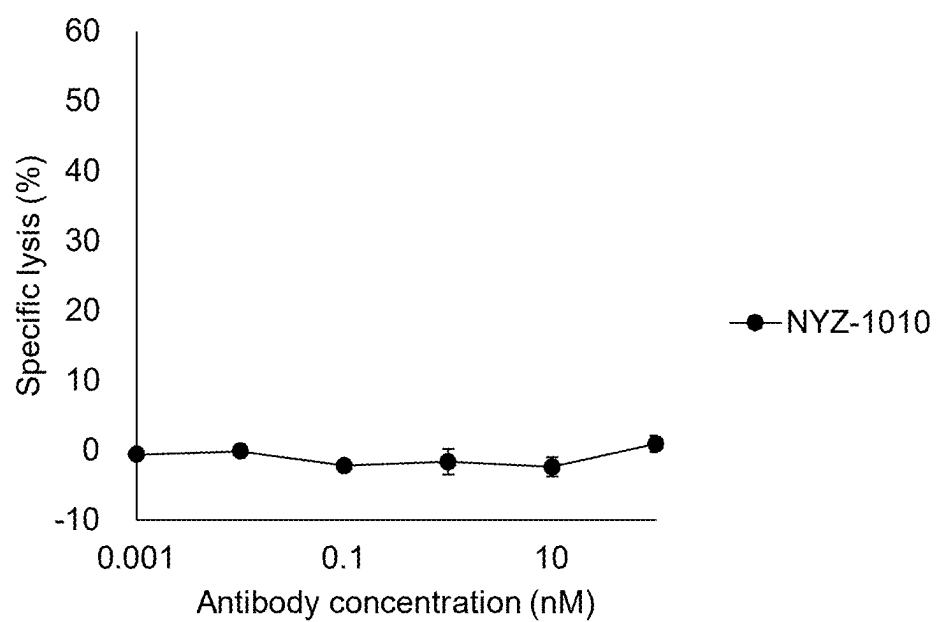


Fig. 159G

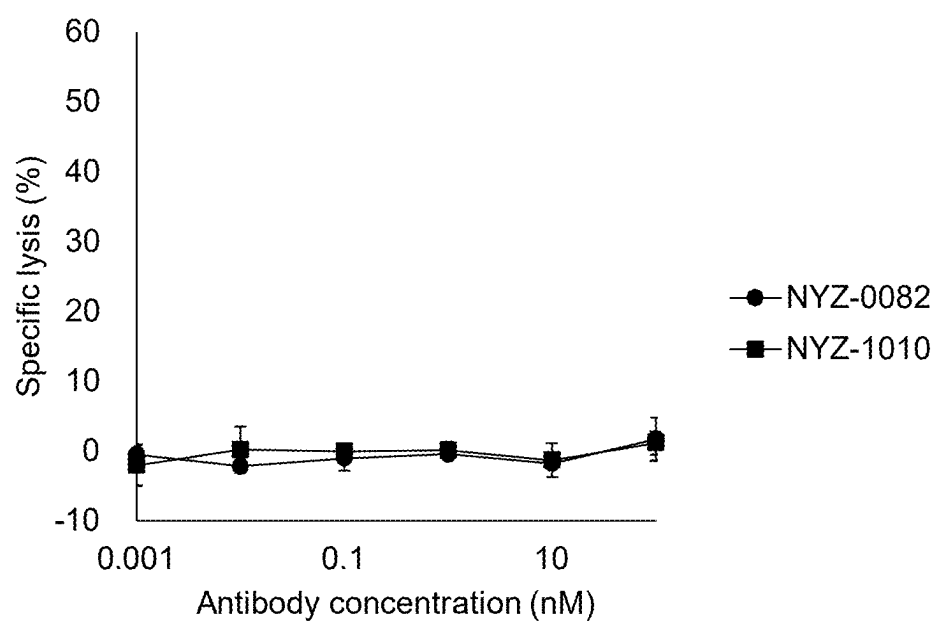


Fig. 159H

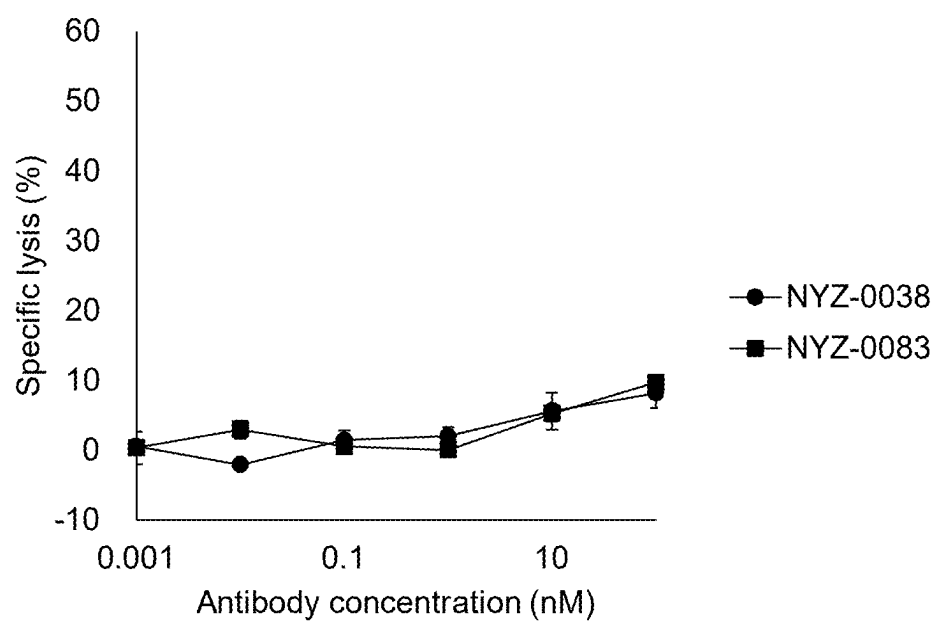


Fig. 160A

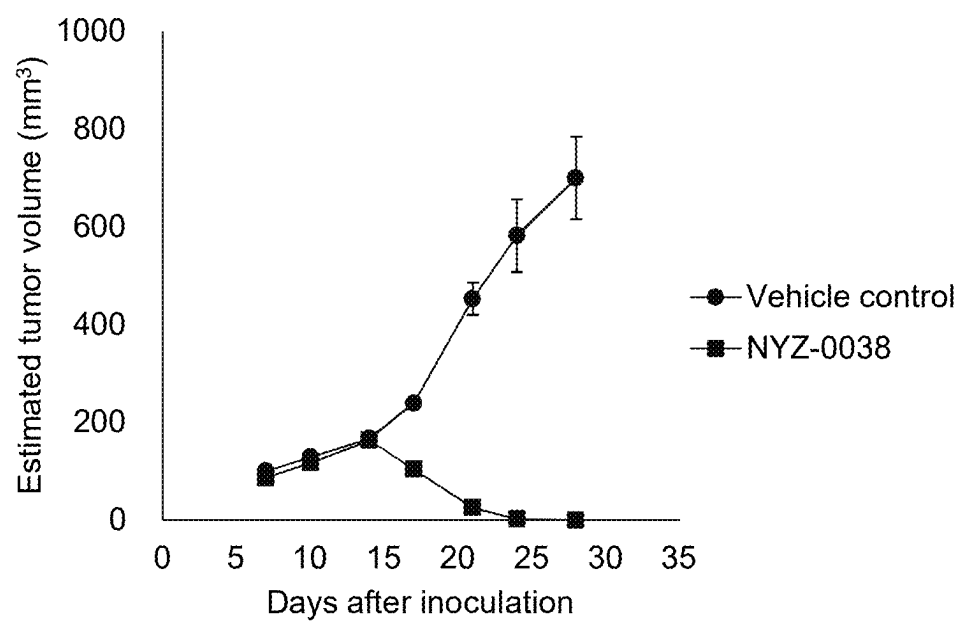


Fig. 160B

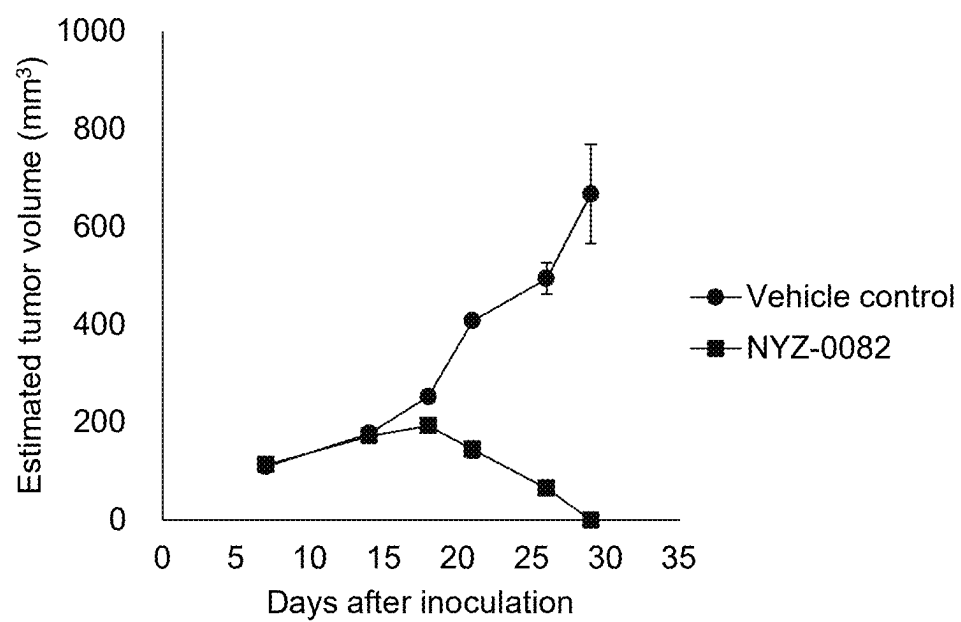


Fig. 160C

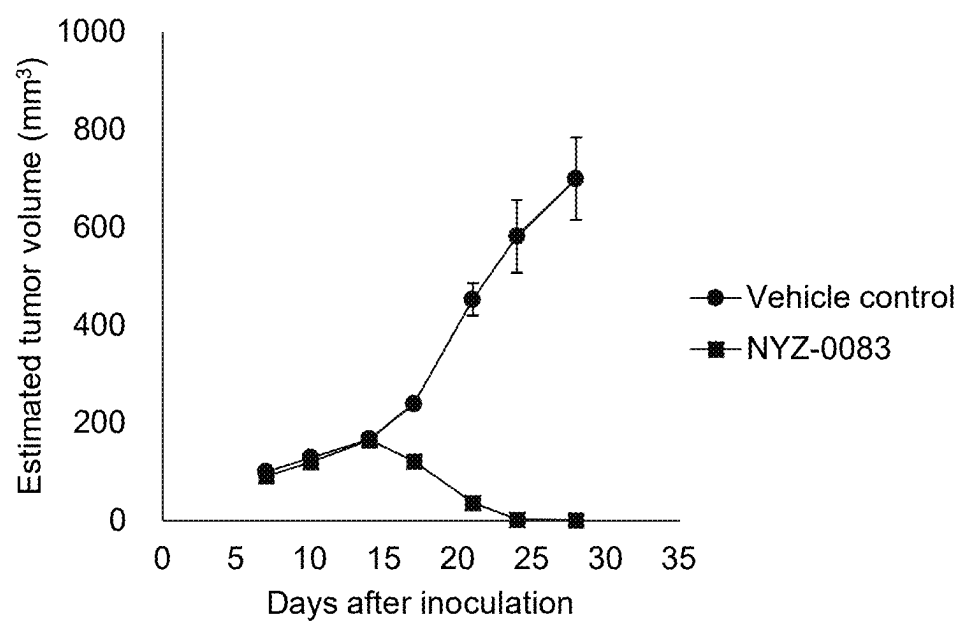


Fig. 160D

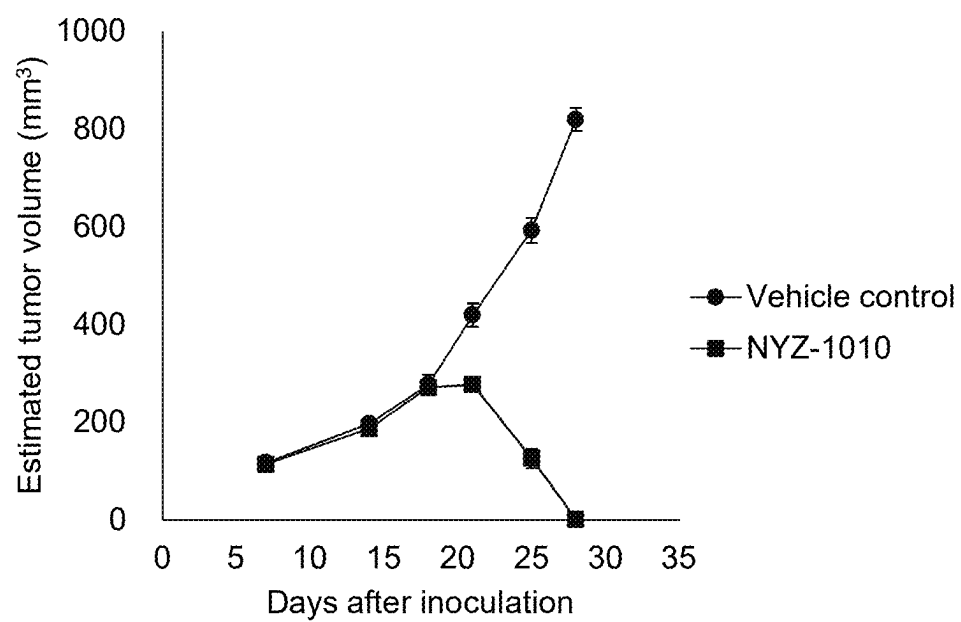


Fig. 161

GGGGS linker
GGGGS
(SEQ ID NO: 162)

Fig. 162

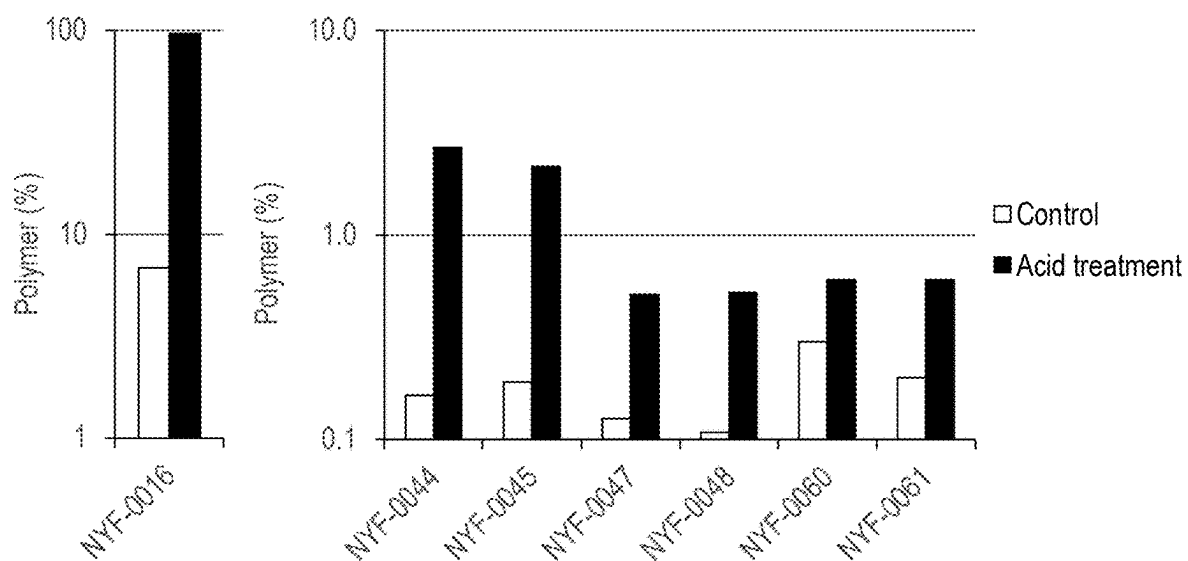


Fig. 163

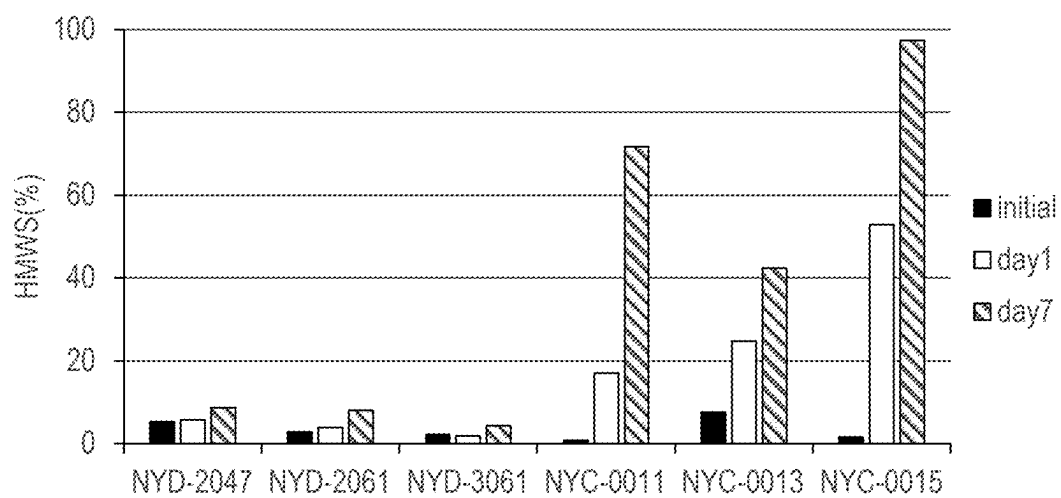


Fig. 164

NYA-3061

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
CGGTGAGGTACAGCTGGTGGGAATCCGGTGGAGGCCCTTGTGCAACCGGGAGGCTCCC
TGCGACTCTCTTGCGCCGCATCTGGCTTTTCCATACGATCCTACGATATTCATTGG
GTCCGAcaaGCCCCCTGGCAAAGGTCTTGAATGGGTGGCCACTATCTCCTACGACGG
GTCTCAGAAATATTATGCGGATTCAGTGAAGGGGCGGTTCAAAATTTACGAGACG
AGTCAAAGAATACACTGTACCTCCAAATGAATTCAGTGAAGAGCCGAGGATACTGCA
GTCTATCATTGTGCAAGAGGGTCCTCAGGCCACTACGAGGCCCTTTGATATATGGGG
CCAAGGCACCTTGGTAACCGTTAGTAGCGGAGGTGGAGGAAGCGGAGGCGGCGGTT
CCGGAGGAGGTGGAAGTGGAGGAGGTGGATCTCAATCCGTTCTGACTCAGCCACCC
TCCGCCTCTGGCACCCCGGGCCAACGGGTCACAATATCATGTTCTGGCTCTTCAAG
CAACATCGGAAACTGGTACGTGAGCTGGTACCAGCAGCTCCCCGGCACGGCGCCCA
AGCTTCTGATATATGACAACAACAACGGCCAGTGGAGTTCCTGACAGATTCAGT
GGGTCTAAAAGTGGTACAAGCGCTAGCCTGGCCATAAGTGGTCTGCAGAGTGAAGA
TGAGGCGGACTATTATTGCGGAACCTGGGACTCCAGCCTGAGCGCTCCCTGGGTTT
TCGGCGGAGGTACTAAGCTGACCGTCTTGGGCGCGGCCGAGGTGCAGGTGGTGAT
TACAAAGATGATGACGATAAAGGTGCAGCGGCGCATCACCATCATCACCAC

(SEQ ID NO: 163), signal sequence (1-57), NYA-3061 (61-813), FLAG-His tag (814-891)

Fig. 165

NYA-3061

MKHLWFFLLLVAAPRWVLSGEVQLVESGGGLVQPGGSLRLSCAASGFSIRSYDIHW
VRQAPGKGLEWVATISYDGSQKYYADSVKGRFTISRDESKNTLYLQMNSLRAEDTA
VYHCARGSSGHYEAFDIWGQGLVTVSSGGGGSGGGSGGGSGGGGSQSVLTQPP
SASGTPGQRVTISCSGSSSNIGNWYVSWYQQLPGTAPKLLIYDNNKRPSGVPDRFS
GSKSGTSASLAISGLQSEDEADYYCGTWDSLSAPWVFGGGTKLTVLGAAAGAGGD
YKDDDDKGAAAHHHHHH

(SEQ ID NO: 164), signal sequence (1-19), NYA-3061 (21-
271), FLAG-His tag (272-297)

Fig. 166

NYC-0005

ATGAAGCACCTGTGGTTCTTTCTGCTGCTGGTGGCCGCTCCTAGATGGGTGTTGTC
TGGCGAAGTGACAGCTGGTGGGAATCTGGTGGCGGAGTTGTTACGCCTGGCGGCTCTC
TGAGACTGAGCTGTGTGGCCAGCGGCTTCACCTTCGACGACTACGCCATGTACTGG
GTCCGACAGGTGCCAGGCAAGGACCTGGAATGGGTGTCCCTGATTTCTGGCGACGG
CGACATCACCTACTACGTGGACTCTGTGAAGGGCAGATTACCGTGTCCAGAGACA
ACAACAAGAACAGCCTGTACCTGCAGATGAAGTCCCTGCGCGTGGAAGATACAGCC
CTGTACTACTGCGCCAAGGACATGATCTACTACGCCTCTTGGAGCGGCTACGGCAG
CAGCGATTACTACTACTATGTGATGGACGTGTGGGGCCAGGGCACCACCGTTACAG
TTTCTAGCGGAGGCGGAGGAAGTGGCGGCGGAGGATCTGGCGGTGGTGGTTCTGAT
ATCCAGATGACACAGAGCCCCAGCAGCCTGTCTGCCTCTGTGGGAGACAGAGTGAC
CATCACCTGTAGAGCCAGCCAGAGCATCAGCAGCTACCTGAACTGGTATCAGCAGA
AGCCCGGCAAGGCCCTAAACTGCTGATCTATGCCGCCTCCAGTCTGCAGAGCGGA
GTGCCTAGCAGATTTTCTGGCAGCGGCTCCGGCACCATTTCACCCTGACCATATC
TAGCCTGCAGCCTGAGGACTTCGCCACCTACTATTGCCAGCAGAGCTACAGCACCC
CTCCTATCACCTTTGGCCAGGGAACCAGACTGGAAATCAAAGGCGCTGCTGCAGGC
GCTGGCGGCGACTACAAAGACGATGATGATAAGGGCGCTGCCGCTCACCACCACCA
TCACCAT

(SEQ ID NO: 165), signal sequence (1-57), NYC-0005 (61-825), FLAG-His tag (826-903)

Fig. 167

NYC-0005

MKHLWFFLLLVAAPRWVLSGEVQLVESGGGVVQPGGSLRLSCVASGFTFDDYAMYW
VRQVPGKDLEWVSLISGDGDITYYVDSVKGRFTVSRDNNKNSLYLQMKSLRVEDTA
LYYCAKDMIYYASWSGYGSSDYYYYVMDVWGQGTTVTVSSGGGGSGGGGSGGGGSD
IQMTQSPSSLSASVGDRVTITCRASQSISSYLNWYQQKPGKAPKLLIYAASSLQSG
VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPPITFGQGTRLEIKGAAAG
AGGDYKDDDDKGAHHHHHHH

(SEQ ID NO: 166), signal sequence (1-19), NYC-0005 (21-
275), FLAG-His tag (276-301)

Fig. 168

NYC-0006

ATGAAGCACCTGTGGTTCTTTCTGCTGCTGGTGGCCGCTCCTAGATGGGTGTTGTC
TGGCGAAGTGACAGCTGGTGGGAATCTGGTGGCGGAGTTGTTACGCCTGGCGGCTCTC
TGAGACTGTCTTGTGCCGCCAGCGGCTTCACCTTCGACGACTACGCCATGTACTGG
GTCCGACAGGCCCTGGCAAAGGCCCTTGAATGGGTGTCCCTGATCAGCGGAGATGG
CGGCAGCATGTACTACGCCGATAGCGTGAAGGGCAGATTACCATCAGCCGGGACA
ACAGCAAGAACAGCCTGTACCTGCAGATGAACTCCCTGCGGACCGAGGACACAGCC
CTGTACTACTGCGCCAAGGATATGATCTTCTACGCCTTTTGGAGCGGCTACGGCAG
CAGCGATTACTACTACTACGTGATGGACGTGTGGGGCCAGGGCACCACAGTGACAG
TTTATTCTGGCGGCGGAGGATCTGGCGGAGGTGGAAGCGGAGGCGGTGGATCTGAT
ATCCAGATGACACAGAGCCCCAGCAGCCTGTCTGCCTCTGTGGGAGACAGAGTGAC
CATCACCTGTCTGGGCCAGCCAGAGAATCAGCACCTACCTGAACTGGTATCAGCAGA
AGCCCGGCAAGGCCCTAAGCTGCTGATCTATGCTGCCTCCAGTCTGCAGAGCGGC
GTGCCAAGCAGATTTTCTGGCAGCGGCTCTGGCACCGACTTCACCCTGACCATATC
TAGCCTGCAGCCTGAGGACTTCGCCACCTACTACTGCCAGCAGAGCTACAGCACCC
CTCCTATCACCTTTGGCCAGGGAACCAGACTGGAAATCAAAGGCGCTGCTGCAGGC
GCAGGCGGCGACTACAAAGACGATGATGATAAGGGCGCTGCCGCTCACCACCACCA
TCACCAT

(SEQ ID NO: 167), signal sequence (1-57), NYC-0006(61-825), FLAG-His tag (826-903)

Fig. 169

NYC-0006

MKHLWFFLLLVAAPRWVLSGEVQLVESGGGVVQPGGSLRLSCAASGFTFDDYAMYW
VRQAPGKGLEWVSLISGDGGS MYYADSVKGRFTISRDN SKNSLYLQMNSLRTE
D
TALYYCAKDMIFYAFWSGYGSSDYYYYVMDVWGQGTTVTVYSGGGGSGGGG
SGGGGSD
IQMTQSPSSLSASVGDRVTITCRASQRISTYLNWYQQKPKAPKLLIYAASSLQSG
VPSRFSGSGSGTDFTLTIS SLQPEDFATYYCQQSYSTPPITFGQGRLEIKGAAAG
AGGDYKDDDDKGA AHHHHH

(SEQ ID NO: 168), signal sequence (1-19), NYC-0006(21-
275), FLAG-His tag (276-301)

Fig. 170

NYC-0007

ATGAAGCACCTGTGGTTCTTTCTGCTGCTGGTGGCCGCTCCTAGATGGGTGTTGTC
TGGCGAAGTGCAGCTGGTGAATCTGGTGGCGGAGTTGTTTCAGCCTGGCGGCTCTC
TGAGACTGTCTTGTGCCGCCAGCGGCTTCACCTTCGACGATTATGCCATGCACTGG
GTCCGACAGGCCCCCTGGCAAAGGACTTGAATGGGTGTCCCTGATCAGCGGCGACGG
CGACAATACCTACTACGCCGATAGCGTGAAGGGCAGATTACCATCAGCCGGGACA
ACAACAAGAACAGCCTGTACCTGCAGATGAACTCCCTGCGGACCGAGGACACCGCC
TTCTACTACTGTGCCAAAGAGCTGATCTTCGGCAAGGTGCTGCACGACTTTTACTA
CTACGTGATGGACGTGTGGGGCCAGGGCACCACAGTGACAGTTTCTAGCGGAGGCG
GAGGAAGTGGCGGCGGAGGATCTGGCGGTGGTGGTTCTGATATCCAGATGACACAG
AGCCCCAGCAGCCTGTCTGCCTCTGTGGGAGACAGAGTGACCATCACCTGTAGAGC
CAGCCAGAGCATCAGCAGCTACCTGAACTGGTATCAGCAGAAGCCCGGCAAGGCC
CTAAACTGCTGATCTATGCCGCCTCCAGTCTGCAGAGCGGAGTGCCTAGCAGATTT
TCTGGCAGCGGCTCCGGCACCGATTTCAACCTGACCATATCTAGCCTGCAGCCTGA
GGACTTCGCCACCTACTACTGCCAGCAGAGCTACAGCACCCCTCCTATCACCTTTG
GCCAGGGAACCAGACTGGAAATCAAAGGCGCTGCTGCAGGCGCAGGCGGCGACTAC
AAAGACGATGATGATAAGGGCGCTGCCGCTCACCACCACCATCACCAT

(SEQ ID NO: 169), signal sequence (1-57), NYC-0007(61-810), FLAG-His tag (811-888)

Fig. 171

NYC-0007

MKHLWFFLLLVAAPRWVLSGEVQLVESGGGVVQPGGSLRLSCAASGFTFDDYAMHW
VRQAPGKGLEWVSLISGDGDNITYYADSVKGRFTISRDNKNSLYLQMNSLRTE
DYA
FYCAKELIFGKVLHDFYYYVMDVWGQGTTVTVSSGGGGSGGGGSGGGGSDIQMTQ
SPSSLSASVGDRTITCRASQSISSYLNWYQQKPKAPKLLIYAASSLQSGVPSRF
SGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPPITFGQGRLEIKGAAAGAGGDY
KDDDDKGAHHHHHHH

(SEQ ID NO: 170), signal sequence (1-19), NYC-0007 (21-270), FLAG-His tag (271-296)

Fig. 172

NYC-0008

ATGAAGCACCTGTGGTTCTTTCTGCTGCTGGTGGCCGCTCCTAGATGGGTGTTGTC
TGGCGAAGTGCAGCTGGTGAATCTGGTGGCGGAGTTGTTTCAGCCTGGCGGCTCTC
TGAGACTGAGCTGTGAAGCCTCCGGCTTCATCTTCGACGACTACGCCATGCACTGG
GTCCGACAGGCACCTGGCAAAGGCCTTGAATGGGTGTCCCTGATCTCTGGCGACGG
CGACATCATCTACTACGCCGACTCTGTGAAGGGCAGATTACCATCAGCCGGGACA
ACAGCAAGAACAGCCTGTACCTGCAGATGAACTCCCTGATCATCGAGGACACAGCC
CTGTACTACTGCGCCAAGGATTGGGTGTTTCGGCGTCGTGATGACCCACTACTGGTA
CTTCGGCCTGGATGTGTGGGGCCAGGGAACCACAGTGACAGTTTCTAGCGGAGGCG
GAGGAAGTGGCGGCGGAGGATCTGGCGGTGGTGGTTCTGATATCCAGATGACACAG
AGCCCCAGCAGCCTGTCTGCCTCTGAAGGCGACAGAGTGACCATCACCTGTAGAGC
CAGCCAGAGCATCAGCACCTACCTGAACTGGTATCAGCAGAAGCCCGGCAAGGCC
CTAAGCTGCTGATCTATGGTGCCTCCAGTCTGCAGAGCGGCGTGCCAAGCAGATTT
TCTGGCAGCGGCTCTGGCACCGACTTCACCCTGACCATATCTAGCCTGCAGCCTGA
GGACTTCGCCACCTACTACTGCCAGCAGTCCTACAGCACCCCTCCTATCACATTTG
GCCAGGGCACCAAGGTGGAAATCAAAGGCGCTGCTGCAGGCGCTGGCGGCGACTAC
AAAGACGATGATGATAAGGGCGCTGCCGCTCACCACCACCATCACCAT

(SEQ ID NO: 171), signal sequence (1-57), NYC-0008(61-
810), FLAG-His tag (811-888)

Fig. 173

NYC-0008

MKHLWFFLLLVAAPRWVLSGEVQLVESGGGVVQPGGSLRLSCEASGFIFDDYAMHW
VRQAPGKGLEWVSLISGDGDIYYADSVKGRFTISRDNKNSLYLQMNSLIIEDTA
LYYCAKDWVFGVVMTHYWYFGLDVWGQGTTVTVSSGGGGSGGGGSGGGGSDIQMTQ
SPSSLSASEGDRVTITCRASQSISTYLNWYQQKPGKAPKLLIYGASSLQSGVPSRF
SGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPPITFGQGTKVEIKGAAAGAGGDY
KDDDDKGAHHHHHHH

(SEQ ID NO: 172), signal sequence (1-19), NYC-0008 (21-270), FLAG-His tag (271-296)

Fig. 174

NYC-0009

ATGAAGCACCTGTGGTTCTTTCTGCTGCTGGTGGCCGCTCCTAGATGGGTGTTGTC
TGGCGAAGTGCAGCTGGTGAATCTGGTGGCGGAGTTGTTTCAGCCTGGCGGCTCTC
TGAGACTGTCTTGTGCCGCCAGCGGCTTCACCTTCGACGATTATGCCATGCACTGG
GTCCGACAGGCCCCCTGGCAAAGGACTTGAATGGGTGTCCCTGATCTCTGGCGGAGG
CGGCGGAACCTACTACAGCGATTCTGTGAAGGGCAGATTCACCATCAGCCGGGACA
CCAGCAAGGACAGCCTGTACCTGCAGATGAACAGCCTGCGGACCGAGGACACAGCC
CTGTACTACTGCGCCAAGGACATGGTGTTCGGCGTGGTCACCCCTTACTACTACTT
CGCCCTGGATGTGTGGGGCCAGGGCACAACAGTGACAGTCTCTTCTGGCGGCGGAG
GAAGCGGAGGCGGAGGATCCGGTGGTGGTGGATCTGACATCCAGATGACACAGAGC
CCCAGCAGCCTGTCTGCCTCTGTGGGAGACAGAGTGACCATCACCTGTAGAGCCAG
CCAGAGCATCAACAGCTACCTGAACTGGTATCAGCAGAAGCCCGGCAAGGCCCTA
AGCTGCTGATCTATGCTGCCTCCAGTCTGCAGAGCGGCGTGCCAAGCAGATTTTCT
GGCAGCGGCTCTGGCACCGACTTCACCCTGACCATATCTAGCCTGCAGCCTGAGGA
CTTCGCCACCTACTACTGCCAGCAGAGCTACAGCGCCCCTCCTATCACATTTGGCC
AGGGAACCAGACTGGAAATCAAAGGCGCTGCTGCAGGCGCAGGCGGCGACTACAAA
GACGATGATGATAAGGGCGCTGCCGCTCACCACCATCACCATCAT

(SEQ ID NO: 173), signal sequence (1-57), NYC-0009(61-807), FLAG-His tag (808-885)

Fig. 175

NYC-0009

MKHLWFFLLLVAAPRWVLSGEVQLVESGGGVVQPGGSLRLSCAASGFTFDDYAMHW
VRQAPGKGLEWVSLISGGGGGTYYSDSVKGRFTISRDTSKDSLYLQMNSLRTE
DLYYCAKDMVFGVVTPYYYFALDVWGQGTTVTVSSGGGGSGGGGSGGGGSDIQMTQS
PSSLSASVGDRVTITCRASQSINSYLNWYQQKPGKAPKLLIYAASSLQSGVPSRFS
GSGSGTDFTLTISSLQPEDFATYYCQQSYSAAPPITFGQGRLEIKGAAAGAGGDYK
DDDDKGAAAHHHHHH

(SEQ ID NO: 174), signal sequence (1-19), NYC-0009(21-269), FLAG-His tag (270-295)

Fig. 176

NYC-0010

ATGAAGCACCTGTGGTTCTTTCTGCTGCTGGTGGCCGCTCCTAGATGGGTGTTGTC
TGGCGAAGTGCAGCTGGTGGAAATCTGGTGGCGGAGTTGTTTCAGCCTGGCGGCTCTC
TGAGACTGAGCTGTGCTGTGTCCGGCTTCACCTTCGACGACTACGCCATGCACTGG
GTCCGACAGGCACCTGGCAAAGGCCCTTGAATGGGTGTCCCTGATCAGCGGAGATGG
CGGCAGCACACACTACGCCGATTCTGTGAAGGGCAGATTACCATCAGCCGGGACA
ACAGCAAGAACAGCCTGTACCTGCAGATGAACTCCCTGCGGACAGGCGACACAGCC
CTGTACTACTGCGCCAAGGACATGATCTTCGCCGTGGTCATCACCGACTACCACTA
CTACGGCATGGACGTGTGGGGCCAGGGAACCACAGTGACAGTTTCTAGCGGAGGCG
GAGGAAGTGGCGGCGGAGGATCTGGCGGTGGTGGTTCTGATATCCAGATGACACAG
AGCCCCAGCAGCCTGTCTGCCTCTGTGGGAGACAGAGTGACCATCACCTGTAGAGC
CAGCCAGAGCATCAGCAGCTACCTGAACTGGTATCAGCAGAAGCCCGGCAAGGCC
CTAAACTGCTGATCTATGCCGCCTCCAGTCTGCAGAGCGGAGTGCCTAGCAGATTT
TCTGGCAGCGGCTCCGGCACCGATTTCAACCCTGACCATATCTAGCCTGCAGCCTGA
GGACTTCGCCACCTACTACTGCCAGCAGAGCTACAGCACCCCTCCTATCACCTTTG
GCCAGGGCACCAGACTGGAAATCAAAGGCGCTGCTGCAGGCGCTGGCGGCGACTAC
AAAGACGATGATGATAAGGGCGCTGCCGCTCACCACCACCATCACCAT

(SEQ ID NO: 175), signal sequence (1-57), NYC-0010 (61-810), FLAG-His tag (811-888)

Fig. 177

NYC-0010

MKHLWFFLLLVAAPRWVLSGEVQLVESGGGVVQPGGSLRLSCAVSGFTFDDYAMHW
VRQAPGKGLEWVSLISGDGGSTHYADSVKGRFTISRDNKNSLYLQMNSLRTGDTA
LYYCAKDMIFAVVITDYHYYGMDVWGQGTTVTVSSGGGGSGGGGSGGGGSDIQMTQ
SPSSLSASVGDRVTITCRASQSISSYLNWYQQKPKAPKLLIYAASSLQSGVPSRF
SGSGSGTDFTLTISLQPEDFATYYCQQSYSTPPITFGQGRLEIKGAAAGAGGDY
KDDDDKGAHHHHHHH

(SEQ ID NO: 176), signal sequence (1-19), NYC-0010 (21-
270), FLAG-His tag (271-296)

Fig. 178

HC-h

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTGCTGAG
CGACAAAACCTCACACATGCCCACCCTGCCCAGCACCTGAAGCCGCCGGCGGACCCT
CAGTCTTCCTCTTCCCCC AAAACCCAAGGACACCCTCATGATCTCCCGGACCCCT
GAGGTCACATGCGTGGTGGTGGACGTGAGCCACGAAGACCCTGAGGTCAAGTTCAA
CTGGTACGTGGACGGCGTGGAGGTGCATAATGCCAAGACCAAGCCTAGAGAGGAAC
AGTACAACAGCACCTACCGGGTGGTGTCCGTGCTGACAGTGCTGCACCAGGACTGG
CTGAACGGCAAAGAGTACAAGTGCAAGGTGTCCAACAAGGCCCTGCCTGCCCCCAT
CGAGAAAACCATCAGCAAGGCCAAGGGCCAGCCCCGGAACCTCAAGTGTGCACCC
TGCCACCCTCCCGGGATGAGCTGACCAAGAACCAGGTGTCCCTGAGCTGTGCCGTG
AAGGGCTTCTACCCCTCCGATATCGCCGTGGAATGGGAGAGCAACGGCCAGCCCGA
GAACAAC TACAAGACCACCCCCCTGTGCTGGACAGCGACGGCTCATTCTTCCTGG
TGTCCAAGCTGACCGTGGACAAGTCCAGATGGCAGCAGGGCAACGTCTTCTCATGC
TCCGTGATGCATGAGGCTCTGCACAACCACTACACCCAGAAGAGCCTCTCCCTGTC
TCCCGGCAAA

(SEQ ID NO: 177), signal sequence (1-57), constant
region (58-738)

Fig. 179

HC-h

MKHLWFFLLLVAAPRWVLSDKTHTCPPCPAPEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVCTLPSPSRDELTKNQVSLSCAVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLVSKLTVDKSRWQQGNVFC
SVMHEALHNHYTQKSLSLSPGK

(SEQ ID NO: 178), signal sequence (1-19), constant region (20-246)

Fig. 180

NYD-2047-HC-k

ATGAAGCACCTGTGGTTCCTTTCTGCTGCTGGTGGCCGCTCCTAGATGGGTGCTGTC
TGGTCAAGTACAGCTGGTGGAAATCCGGTGGAGGCGTGGTCCAGCCGGGACGCAGCT
TGAGACTGTCCTGCGCTGCATCTGGCTTTTCCATACGATCCTACGATATGCACTGG
GTTTCGCCAAGCCCCTGGCAAAGGTCTTGAATGGGTGGCCACTATCTCCTACGACGG
GTCTCAGAAATATTACGCGGATTCAGTGAAGGGGCGGTTTCACAATTTACGGGACA
ATTCAAAGAATACCTTGTATCTCCAGATGTCTTCACTGAGAGCCGAGGATACTGCA
GTCTATCATTTGTGCAAGAGGGTCCCTCAGGCCACTACGAGGCCTTTGATATATGGGG
CCAAGGCACCTTGGTAACCGTTAGTAGCGGAGGTGGAGGAAGCGGAGGCGGCGGTT
CCGGAGGAGGTGGAAGTCAATCCGTTCTGACTCAGCCACCCTCCGTGAGTGCAGCG
CCTGGACAGAAGGTCACAATATCATGTTCTGGCTCTTCAAGCAACATCGGAAACTG
GTACGTGAGCTGGTACCAGCAGCTCCCCGGCAGGCGCCCAAGCTTCTGATATATG
ACAACAACAAACGGCCCCAGTGGAAATCCCTGACAGATTCAGTGGGTCTAAAAGTGGT
ACAAGCGCTACCCTGGGTATCACCGGATTGCAGACCGGAGATGAGGCGGACTATTA
TTGCGGAACCTGGGACTCCAGCCTGAGCGCTCCCTGGGTTTTCGGCGGAGGTACTA
AGGTTACCGTCTTGGGCGGAGAACCCAAGAGCgcaGACAAGACCCACACCTGTCCT
CCATGTCCTGCTCCAGAAGCTGCAGGCGGCCCTTCCGTGTTTCTGTTCCCTCCAAA
GCCTAAGGACACCCTGATGATCAGCCGGACACCTGAAGTGACCTGCGTGGTGGTGG
ATGTGTCCACGAGGATCCCGAAGTGAAGTTCAATTGGTACGTGGACGGCGTGGAA
GTGCACAACGCCAAGACCAAGCCTAGAGAGGAACAGTACAACAGCACCTACAGAGT
GGTGTCTGTGCTGACAGTGCTGCACCAGGACTGGCTGAACGGCAAAGAGTACAAGT
GCAAGGTGTCCAACAAGGCCCTGCCTGCTCCTATCGAGAAAACCATCAGCAAGGCC
AAGGGCCAGCCTAGGGAACCCCAGGTTTACACCCTGCCTCCATGCAGGGATGAGCT
GACCAAGAACCAGGTGTCCCTGTGGTGCCTGGTTAAGGGCTTCTACCCCTCCGATA
TCGCCGTGGAATGGGAGAGCAATGGCCAGCCAGAGAACAACACTACAAGACAACCCCT
CCTGTGCTGGACTCCGACGGCTCATTCTTCTGTACAGCAAGCTGACCGTGGACAA
GAGCAGATGGCAGCAGGGCAACGTGTTTCACTGTCAGCGTGATGCACGAGGCCCTGC
ACAACCACTACACACAGAAGTCCCTGTCTCTGAGCCCCGGCAA

(SEQ ID NO: 179), signal sequence (1-57), NYA-2047(61-798), linker (799-804), constant region (805-1500)

Fig. 181

NYD-2047-HC-k

MKHLWFFLLLVAAPRWVLSGQVQLVESGGGVVQPGRSLRLSCAASGFSIRSYDMHW
VRQAPGKGLEWVATISYDGSQKYYADSVKGRFTISRDN SKNTLYLQMSSLRAEDTA
VYHCARGSSGHYEAFDIWGQGT LVT VSSGGGGSGGGSGGGGSQSVLTQPPSVSAA
PGQKVTISCSGSSSNIGNWYVSWYQQLPGTAPKLLIYDNNKRPSGIPDRFSGSKSG
TSATLGITGLQTGDEADYYCGTWDSLSAPWVFGGGTKVTVLGGEPKSADKTHTCP
PCPAPEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVE
VHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKA
KGQPREPQVYTLPPCRDELTKNQVSLWCLVKGFYPSDIAVEWESNGQPENNYKTTTP
PVLDS DGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK

(SEQ ID NO: 180), signal sequence (1-19), NYA-2047(21-266), linker (267-268), constant region (269-500)

Fig. 182

NYD-2061-HC-k

ATGAAGCACCTGTGGTTCCTTCTGCTGCTGGTGGCCGCTCCTAGATGGGTGCTGTC
TGGTGAGGTACAGCTGGTGGAATCCGGTGGAGGCCTTGTGCAACCGGGAGGCTCCC
TGCGACTCTCTTGCGCCGCATCTGGCTTTTCCATACGATCCTACGATATTCATTGG
GTCCGAcaaGCCCCCTGGCAAAGGTCTTGAATGGGTGGCCACTATCTCCTACGACGG
GTCTCAGAAATATTATGCGGATTCAGTGAAGGGGCGGTTTACAATTTACGAGACG
AGTCAAAGAATACACTGTACCTCCAAATGAATTCAGTGAAGAGCCGAGGATACTGCA
GTCTATCATTGTGCAAGAGGGTCCCTCAGGCCACTACGAGGCCTTTGATATATGGGG
CCAAGGCACCTTGGTAACCGTTAGTAGCGGAGGTGGAGGAAGCGGAGGCGGCGGTT
CCGGAGGAGGTGGAAGTCAATCCGTTCTGACTCAGCCACCCTCCGCCTCTGGCACC
CCGGGCCAACGGGTCACAATATCATGTTCTGGCTCTTCAAGCAACATCGGAAACTG
GTACGTGAGCTGGTACCAGCAGCTCCCCGGCACGGCGCCCAAGCTTCTGATATATG
ACAACAACAAACGGCCCCAGTGGAGTTCCCTGACAGATTCAGTGGGTCTAAAAGTGGT
ACAAGCGCTAGCCTGGCCATAAGTGGTCTGCAGAGTGAAGATGAGGCGGACTATTA
TTGCGGAACCTGGGACTCCAGCCTGAGCGCTCCCTGGGTTTTCGGCGGAGGTACTA
AGCTGACCGTCTTGGGCGGAGAACCCAAGAGCgcaGACAAGACCCACACCTGTCCT
CCATGTCCTGCTCCAGAAGCTGCAGGCGGCCCTTCCGTGTTTCTGTTCCCTCCAAA
GCCTAAGGACACCCTGATGATCAGCCGGACACCTGAAGTGACCTGCGTGGTGGTGG
ATGTGTCCACGAGGATCCCGAAGTGAAGTTCAATTGGTACGTGGACGGCGTGGA
GTGCACAACGCCAAGACCAAGCCTAGAGAGGAACAGTACAACAGCACCTACAGAGT
GGTGTCTGTGCTGACAGTGCTGCACCAGGACTGGCTGAACGGCAAAGAGTACAAGT
GCAAGGTGTCCAACAAGGCCCTGCCTGCTCCTATCGAGAAAACCATCAGCAAGGCC
AAGGGCCAGCCTAGGGAACCCCAGGTTTACACCCTGCCTCCATGCAGGGATGAGCT
GACCAAGAACCAGGTGTCCCTGTGGTGCCTGGTTAAGGGCTTCTACCCCTCCGATA
TCGCCGTGGAATGGGAGAGCAATGGCCAGCCAGAGAACAACACTACAAGACAACCCCT
CCTGTGCTGGACTCCGACGGCTCATTCTTCTGTACAGCAAGCTGACCGTGAGACAA
GAGCAGATGGCAGCAGGGCAACGTGTTTCACTGTCAGCGTGATGCACGAGGCCCTGC
ACAACCACTACACACAGAAGTCCCTGTCTCTGAGCCCCGGCAA

(SEQ ID NO: 181), signal sequence (1-57), NYA-2061 (61-798), linker (799-804), constant region (805-1500)

Fig. 183

NYD-2061-HC-k

MKHLWFFLLLVAAPRWVLSGEVQLVESGGGLVQPGGSLRLSCAASGFSIRSYDIHW
VRQAPGKGLEWVATISYDGSQKYYADSVKGRFTISRDESKNTLYLQMNSLRAEDTA
VYHCARGSSGHYEAFDIWGQGTLLVTVSSGGGGSGGGGSGGGGSQSVLTQPPSASGT
PGQRTVISCSSSSNIGNWYVSWYQQLPGTAPKLLIYDNNKRPSGVPDRFSGSKSG
TSASLAISGLQSEDEADYYCGTWDSLSAPWVFGGGTKLTVLGGEPKSADKTHTCP
PCPAPEAAGGPSVFLFPPKPKDTLMI SRTPEVTCVVVDVSHEDPEVKFNWYVDGVE
VHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKA
KGQPREPQVYTLPPCRDELTKNQVSLWCLVKGFYPSDIAVEWESNGQPENNYKTTTP
PVLDS DGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK

(SEQ ID NO: 182), signal sequence (1-19), NYA-2061 (21-266), linker (267-268), constant region (269-500)

Fig. 184

NYD-3061-HC-k

ATGAAGCACCTGTGGTTCTTTCTGCTGCTGGTGGCCGCTCCTAGATGGGTGCTGTC
TGGTGAGGTACAGCTGGTGGAAATCCGGTGGAGGCCCTTG TGCAACCGGGAGGCTCCC
TGCGACTCTCTTGCGCCGCATCTGGCTTTTCCATACGATCCTACGATATTCATTGG
GTCCGAc aaGCCCCCTGGCAAAGGTCTTGAATGGGTGGCCACTATCTCCTACGACGG
GTCTCAGAAATATTATGCGGATTTCAGTGAAGGGGCGGTT CACAATTTACGAGACG
AGTCAAAGAATACTGTACCTCCAAATGAATTCAGTGAAGAGCCGAGGATACTGCA
GTCTATCATTGTGCAAGAGGGTCCTCAGGCCACTACGAGGCCCTTTGATATATGGGG
CCAAGGCACCTTGGTAAACGTTAGTAGCGGAGGTGGAGGAAGCGGAGGCGGCGGTT
CCGGAGGAGGTGGAAGTGGAGGAGGTGGATCTCAATCCGTTCTGACTCAGCCACCC
TCCGCCTCTGGCACCCCCGGGCCAACGGGTCACAATATCATGTTCTGGCTCTTCAAG
CAACATCGGAACTGGTACGTGAGCTGGTACCAGCAGCTCCCCGGCACGGCGCCCCA
AGCTTCTGATATATGACAACAACAAACGGCCCCAGTGGAGTTCCTGACAGATTTCAGT
GGGTCTAAAAGTGGTACAAGCGCTAGCCTGGCCATAAGTGGTCTGCAGAGTGAAGA
TGAGGCGGACTATTATTGCGGAACCTGGGACTCCAGCCTGAGCGCTCCCTGGGT TT
TCGGCGGAGGTACTAAGCTGACCGTCTTGGGCGGAGAACCCAAGAGCGCAGACAAG
ACCCACACCTGTCCTCCATGTCCTGCTCCAGAAGCTGCAGGCGGCCCTTCCGTGTT
TCTGTTCCCTCCAAAGCCTAAGGACACCCTGATGATCAGCCGGACACCTGAAGTGA
CCTGCGTGGTGGTGGATGTGTCCACGAGGATCCCGAAGTGAAGTTCAATTGGTAC
GTGGACGGCGTGGAAGTGCAACGCCAAGACCAAGCCTAGAGAGGAACAGTACAA
CAGCACCTACAGAGTGGTGTCTGTGCTGACAGTGTCTGCACCAGGACTGGCTGAACG
GCAAAGAGTACAAGTGCAAGGTGTCCAACAAGGCCCTGCCTGCTCCTATCGAGAAA
ACCATCAGCAAGGCCAAGGGCCAGCCTAGGGAACCCCAGGTTTACACCCTGCCTCC
ATGCAGGGATGAGCTGACCAAGAACCAGGTGTCCCTGTGGTGCCTGGTTAAGGGCT
TCTACCCCTCCGATATCGCCGTGGAATGGGAGAGCAATGGCCAGCCAGAGAACAAAC
TACAAGACAACCCCTCCTGTGCTGGACTCCGACGGCTCATTCTTCCTGTACAGCAA
GCTGACCGTGGACAAGAGCAGATGGCAGCAGGGCAACGTGTT CAGCTGCAGCGTGA
TGCACGAGGCCCTGCACAACCACTACACACAGAAGTCCCTGTCTCTGAGCCCCGGC
AAA

(SEQ ID NO: 183), signal sequence (1-57), NYA-3061 (61-813), linker (814-819), constant region (820-1515)

Fig. 185

NYD-3061-HC-k

MKHLWFFLLLVAAPRWVLSGEVQLVESGGGLVQPGGSLRLSCAASGFSIRSYDIHW
VRQAPGKGLEWVATISYDGSQKYYADSVKGRFTISRDESKNTLYLQMNSLRAEDTA
VYHCARGSSGHYEAFDIWGQGTLVTVSSGGGGSGGGSGGGSGGGGSQSVLTQPP
SASGTPGQRVTISCSGSSSNIGNWYVSWYQQLPGTAPKLLIYDNNKRPSGVPDRFS
GSKSGTSASLAISGLQSEDEADYYCGTWDSSLSAPWVFGGGTKLTVLGGEPKSADK
THTCPPCPAPEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEK
TISKAKGQPREPQVYTLPPCRDELTKNQVSLWCLVKGFYPSDIAVEWESNGQPENN
YKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPG
K

(SEQ ID NO: 184), signal sequence (1-19), NYA-3061 (21-271), linker (272-273), constant region (274-505)

Fig. 186

NYC-0011-HC-k

ATGAAGCACCTGTGGTTCTTTCTGCTGCTGGTGGCCGCTCCTAGATGGGTGCTGTC
TGGCGAAGTGCAGCTGGTGGAAATCTGGTGGCGGAGTTGTTTCTAGCCTGGCGGCTCTC
TGAGACTGAGCTGTGTGGCCAGCGGCTTCACCTTCGACGACTACGCCATGTACTGG
GTCCGACAGGTGCCAGGCAAGGACCTGGAATGGGTGTCCCTGATTCTGGCGACGG
CGACATCACCTACTACGTGGACTCTGTGAAGGGCAGATTCACCGTGTCCAGAGACA
ACAACAAGAACAGCCTGTACCTGCAGATGAAGTCCCTGCGCGTGGAAGATACAGCC
CTGTACTACTGCGCCAAGGACATGATCTACTACGCCTCTTGGAGCGGCTACGGCAG
CAGCGATTACTACTACTATGTGATGGACGTGTGGGGCCAGGGCACCACCGTTACAG
TTTCTAGCGGAGGCGGAGGAAGTGGCGGCGGAGGATCTGGCGGTGGTGGTTCTGAT
ATCCAGATGACACAGAGCCCCAGCAGCCTGTCTGCCTCTGTGGGAGACAGAGTGAC
CATCACCTGTAGAGCCAGCCAGAGCATCAGCAGCTACCTGAACTGGTATCAGCAGA
AGCCCCGGCAAGGCCCTAAACTGCTGATCTATGCCGCCTCCAGTCTGCAGAGCGGA
GTGCCTAGCAGATTTTCTGGCAGCGGCTCCGGCACCGATTTACCCCTGACCATATC
TAGCCTGCAGCCTGAGGACTTCGCCACCTACTATTGCCAGCAGAGCTACAGCACCC
CTCCTATCACCTTTGGCCAGGGAACCAGACTGGAAATCAAAGGCGGAGAACCCAAG
AGCgcaGACAAGACCCACACCTGTCCTCCATGTCTCTGCTCCAGAAGCTGCAGGCGG
CCCTTCCGTGTTTCTGTTCCCTCCAAAGCCTAAGGACACCCTGATGATCAGCCGGA
CACCTGAAGTGACCTGCGTGGTGGTGGATGTGTCCCACGAGGATCCCGAAGTGAAG
TTCAATTGGTACGTGGACGGCGTGGAAGTGACAACGCCAAGACCAAGCCTAGAGA
GGAACAGTACAACAGCACCTACAGAGTGGTGTCTGTGCTGACAGTGCTGCACCAGG
ACTGGCTGAACGGCAAAGAGTACAAGTGCAAGGTGTCCAACAAGGCCCTGCCTGCT
CCTATCGAGAAAACCATCAGCAAGGCCAAGGGCCAGCCTAGGGAACCCAGGTTTA
CACCCCTGCCTCCATGCAGGGATGAGCTGACCAAGAACCAGGTGTCCCTGTGGTGCC
TGGTTAAGGGCTTCTACCCCTCCGATATCGCCGTGGAATGGGAGAGCAATGGCCAG
CCAGAGAACAACCTACAAGACAACCCCTCCTGTGCTGGACTCCGACGGCTCATTTCTT
CCTGTACAGCAAGCTGACCGTGGACAAGAGCAGATGGCAGCAGGGCAACGTGTTCA
GCTGCAGCGTGATGCACGAGGCCCTGCACAACCACTACACACAGAAGTCCCTGTCT
CTGAGCCCCGGCAA

(SEQ ID NO: 185), signal sequence (1-57), NYC-0005 (61-825), linker (826-831), constant region (832-1527)

Fig. 187

NYC-0011-HC-k

MKHLWFFLLLVAAAPRWVLSGEVQLVESGGGVVQPGGSLRLSCVASGFTFDDYAMYW
VRQVPGKDLEWVSLISGDGDITYYVDSVKGRFTVSRDNNKNSLYLQMKSLRVEDTA
LYYCAKDMIYYASWSGYGSSDYVMDVWGQGTTVTVSSGGGGSGGGGSGGGGSD
IQMTQSPSSLSASVGDRVITICRASQSISSYLNWYQQKPGKAPKLLIYAASSLQSG
VPSRFSGSGSGTDFTLTISSLPEDFATYYCQQSYSTPPITFGQGRLEIKGGEPK
SADKTHTCPPCPAPEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVK
FNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPA
PIEKTISKAKGQPREPQVYTLPPCRDELTKNQVSLWCLVKGFYPSDIAVEWESNGQ
PENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSQSVMEALHNHYTQKSLS
LSPGK

(SEQ ID NO: 186), signal sequence (1-19), NYC-0005 (21-275), linker (276-277), constant region (278-509)

Fig. 188

NYC-0012-HC-k

ATGAAGCACCTGTGGTTCTTTCTGCTGCTGGTGGCCGCTCCTAGATGGGTGCTGTC
TGGCGAAGTGCAGCTGGTGAATCTGGTGGCGGAGTTGTTTCAGCCTGGCGGCTCTC
TGAGACTGTCTTGTGCCGCCAGCGGCTTCACCTTCGACGACTACGCCATGTACTGG
GTCCGACAGGCCCCCTGGCAAAGGCCCTTGAATGGGTGTCCCTGATCAGCGGAGATGG
CGGCAGCATGTACTACGCCGATAGCGTGAAGGGCAGATTACCATCAGCCGGGACA
ACAGCAAGAACAGCCTGTACCTGCAGATGAACTCCCTGCGGACCGAGGACACAGCC
CTGTACTACTGCGCCAAGGATATGATCTTCTACGCCCTTTTGGAGCGGCTACGGCAG
CAGCGATTACTACTACTACGTGATGGACGTGTGGGGCCAGGGCACCACAGTGACAG
TTTATTCTGGCGGCGGAGGATCTGGCGGAGGTGGAAGCGGAGGCGGTGGATCTGAT
ATCCAGATGACACAGAGCCCCAGCAGCCTGTCTGCCTCTGTGGGAGACAGAGTGAC
CATCACCTGTCTGGGCCAGCCAGAGAATCAGCACCTACCTGAACTGGTATCAGCAGA
AGCCCCGGCAAGGCCCTAAGCTGCTGATCTATGCTGCCTCCAGTCTGCAGAGCGGC
GTGCCAAGCAGATTTTCTGGCAGCGGCTCTGGCACCGACTTCACCCTGACCATATC
TAGCCTGCAGCCTGAGGACTTCGCCACCTACTACTGCCAGCAGAGCTACAGCACCC
CTCCTATCACCTTTTGGCCAGGGAACCAGACTGGAAATCAAAGGCGGAGAACCCAAG
AGCgcaGACAAGACCCACACCTGTCCTCCATGTCTGCTCCAGAAGCTGCAGGCGG
CCCTTCCGTGTTTCTGTTCCCTCCAAAGCCTAAGGACACCCTGATGATCAGCCGGA
CACCTGAAGTGACCTGCGTGGTGGTGGATGTGTCCCACGAGGATCCCGAAGTGAAG
TTCAATTGGTACGTGGACGGCGTGGAAGTGACAACGCCAAGACCAAGCCTAGAGA
GGAACAGTACAACAGCACCTACAGAGTGGTGTCTGTGCTGACAGTGCTGCACCAGG
ACTGGCTGAACGGCAAAGAGTACAAGTGCAAGGTGTCCAACAAGGCCCTGCCTGCT
CCTATCGAGAAAACCATCAGCAAGGCCAAGGGCCAGCCTAGGGAACCCAGGTTTA
CACCCCTGCCTCCATGCAGGGATGAGCTGACCAAGAACCAGGTGTCCCTGTGGTGCC
TGGTTAAGGGCTTCTACCCCTCCGATATCGCCGTGGAATGGGAGAGCAATGGCCAG
CCAGAGAACAACCTACAAGACAACCCCTCCTGTGCTGGACTCCGACGGCTCATTTCTT
CCTGTACAGCAAGCTGACCGTGGACAAGAGCAGATGGCAGCAGGGCAACGTGTTCA
GCTGCAGCGTGATGCACGAGGCCCTGCACAACCACTACACACAGAAGTCCCTGTCT
CTGAGCCCCGGCAA

(SEQ ID NO: 187), signal sequence (1-57), NYC-0006 (61-825), linker (826-831), constant region (832-1527)

Fig. 189

NYC-0012-HC-k

MKHLWFFLLLVAAPRWVLSGEVQLVESGGGVVQPGGSLRLSCAASGFTFDDYAMYW
VRQAPGKGLEWVSLISGDGGSMTYADSVKGRFTISRDN SKNSLYLQMNSLRTE
DYA
LYYCAKDMIFYAFWSGYGSSDYVMDVWGQGTTVTVYSGGGGSGGGGSGGGGSD
IQMTQSPSSLSASVGDRVTITCRASQRISTYLNWYQQKPGKAPKLLIYAASSLQSG
VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPPITFGQGTRLEIKGGE
PK
SADKTHTCPPCPAPEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVK
FNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALP
A
PIEKTISKAKGQPREPQVYTLPPCRDELTKNQVSLWCLVKGFYPSDIAVEWESNGQ
PENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNV FSCSV MHEALHNHYTQKSLS
LSPGK

(SEQ ID NO: 188), signal sequence (1-19), NYC-0006 (21-275), linker (276-277), constant region (278-509)

Fig. 190

NYC-0013-HC-k

ATGAAGCACCTGTGGTTCCTTCTGCTGCTGGTGGCCGCTCCTAGATGGGTGCTGTC
TGGCGAAGTGCAGCTGGTGGAAATCTGGTGGCGGAGTTGTTTCAGCCTGGCGGCTCTC
TGAGACTGTCTTGTGCCGCCAGCGGCTTCACCTTCGACGATTATGCCATGCACTGG
GTCCGACAGGCCCTGGCAAAGGACTTGAATGGGTGTCCCTGATCAGCGGCGACGG
CGACAATACCTACTACGCCGATAGCGTGAAGGGCAGATTACCATCAGCCGGGACA
ACAACAAGAACAGCCTGTACCTGCAGATGAACTCCCTGCGGACCGAGGACACCGCC
TTCTACTACTGTGCCAAAGAGCTGATCTTCGGCAAAGGTGCTGCACGACTTTTACTA
CTACGTGATGGACGTGTGGGGCCAGGGCACCACAGTGACAGTTTCTAGCGGAGGCG
GAGGAAGTGGCGGCGGAGGATCTGGCGGTGGTGGTTCTGATATCCAGATGACACAG
AGCCCCAGCAGCCTGTCTGCCTCTGTGGGAGACAGAGTGACCATCACCTGTAGAGC
CAGCCAGAGCATCAGCAGCTACCTGAACTGGTATCAGCAGAAGCCCGGCAAGGCC
CTAAACTGCTGATCTATGCCGCTCCAGTCTGCAGAGCGGAGTGCCTAGCAGATTT
TCTGGCAGCGGCTCCGGCACCGATTTACCCCTGACCATATCTAGCCTGCAGCCTGA
GGACTTCGCCACCTACTACTGCCAGCAGAGCTACAGCACCCCTCCTATCACCTTTG
GCCAGGGAACCAGACTGGAAATCAAAGGCGGAGAACCCAAGAGCgcaGACAAGACC
CACACCTGTCTCCATGTCTTGCTCCAGAAGCTGCAGGCGGCCCTTCCGTGTTTCT
GTTCCCTCCAAAGCCTAAGGACACCCCTGATGATCAGCCGGACACCTGAAGTGACCT
GCGTGGTGGTGGATGTGTCCACGAGGATCCCGAAGTGAAGTTCAATTGGTACGTG
GACGGCGTGGAAGTGCACAACGCCAAGACCAAGCCTAGAGAGGAACAGTACAACAG
CACCTACAGAGTGGTGTCTGTGCTGACAGTGCTGCACCAGGACTGGCTGAACGGCA
AAGAGTACAAGTGCAAGGTGTCCAACAAGGCCCTGCCTGCTCCTATCGAGAAAACC
ATCAGCAAGGCCAAGGGCCAGCCTAGGGAACCCAGGTTTACACCCTGCCTCCATG
CAGGGATGAGCTGACCAAGAACCAGGTGTCCCTGTGGTGCCTGGTTAAGGGCTTCT
ACCCCTCCGATATCGCCGTGGAATGGGAGAGCAATGGCCAGCCAGAGAACAACTAC
AAGACAACCCCTCCTGTGCTGGACTCCGACGGCTCATTCCTTCTGTACAGCAAGCT
GACCGTGGACAAGAGCAGATGGCAGCAGGGCAACGTGTTTCAGCTGCAGCGTGATGC
ACGAGGCCCTGCACAACCACTACACACAGAAGTCCCTGTCTCTGAGCCCCGGCAA
(SEQ ID NO: 189), signal sequence (1-57), NYC-0007 (61-
810), linker (811-816), constant region (817-1512)

Fig. 191

NYC-0013-HC-k

MKHLWFFLLLVAAPRWVLSGEVQLVESGGGVVQPGGSLRLSCAASGFTFDDYAMHW
VRQAPGKGLEWVSLISGDGDNTYYADSVKGRFTISRDNKNKSLYLQMNSLRTEDTA
FYCAKELIFGKVLHDFYYYVMDVWGQGTTVTVSSGGGGSGGGGSGGGGSDIQMTQ
SPSSLSASVGDRVTITCRASQSISSYLNWYQQKPGKAPKLLIYAASSLQSGVPSRF
SGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPPITFGQGRLEIKGGEPKSADKT
HTCPPCPAPEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYV
DGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKT
ISKAKGQPREPQVYTLPPCRDELTKNQVSLWCLVKGFYPSDIAVEWESNGQPENNY
KTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPGK
(SEQ ID NO: 190), signal sequence (1-19), NYC-0007 (21-
270), linker (271-272), constant region (273-504)

Fig. 192

NYC-0014-HC-k

ATGAAGCACCTGTGGTTCCTTCTGCTGCTGGTGGCCGCTCCTAGATGGGTGCTGTC
TGGCGAAGTGCAGCTGGTGGAAATCTGGTGGCGGAGTTGTTTCAGCCTGGCGGCTCTC
TGAGACTGAGCTGTGAAGCCTCCGGCTTCATCTTCGACGACTACGCCATGCACTGG
GTCCGACAGGCACCTGGCAAAGGCCCTGAATGGGTGTCCCTGATCTCTGGCGACGG
CGACATCATCTACTACGCCGACTCTGTGAAGGGCAGATTACCATCAGCCGGGACA
ACAGCAAGAACAGCCTGTACCTGCAGATGAACTCCCTGATCATCGAGGACACAGCC
CTGTACTACTGCGCCAAGGATTGGGTGTTTCGGCGTCTGTGATGACCCACTACTGGTA
CTTCGGCCTGGATGTGTGGGGCCAGGGAACACAGTGACAGTTTCTAGCGGAGGCG
GAGGAAGTGGCGGCGGAGGATCTGGCGGTGGTGGTTCTGATATCCAGATGACACAG
AGCCCCAGCAGCCTGTCTGCCTCTGAAGGCGACAGAGTGACCATCACCTGTAGAGC
CAGCCAGAGCATCAGCACCTACCTGAACTGGTATCAGCAGAAGCCCGGCAAGGCC
CTAAGCTGCTGATCTATGGTGCCTCCAGTCTGCAGAGCGGCGTGCCAAGCAGATTT
TCTGGCAGCGGCTCTGGCACCGACTTCACCCTGACCATATCTAGCCTGCAGCCTGA
GGACTTCGCCACCTACTACTGCCAGCAGTCCTACAGCACCCCTCCTATCACATTTG
GCCAGGGCACCAAGGTGGAAATCAAAGGCGGAGAACCCAAGAGCgcaGACAAGACC
CACACCTGTCTCCATGTCTCTGCTCCAGAAGCTGCAGGCGGCCCTTCCGTGTTTCT
GTTCCCTCCAAAGCCTAAGGACACCCCTGATGATCAGCCGGACACCTGAAGTGACCT
GCGTGGTGGTGGATGTGTCCCACGAGGATCCCGAAGTGAAGTTCAATTGGTACGTG
GACGGCGTGGAAGTGCACAACGCCAAGACCAAGCCTAGAGAGGAACAGTACAACAG
CACCTACAGAGTGGTGTCTGTGCTGACAGTGCTGCACCAGGACTGGCTGAACGGCA
AAGAGTACAAGTGCAAGGTGTCCAACAAGGCCCTGCCTGCTCCTATCGAGAAAACC
ATCAGCAAGGCCAAGGGCCAGCCTAGGGAACCCAGGTTTACACCCTGCCTCCATG
CAGGGATGAGCTGACCAAGAACCAGGTGTCCCTGTGGTGCCTGGTTAAGGGCTTCT
ACCCCTCCGATATCGCCGTGGAATGGGAGAGCAATGGCCAGCCAGAGAACAACTAC
AAGACAACCCCTCCTGTGCTGGACTCCGACGGCTCATTCCTTCTGTACAGCAAGCT
GACCGTGGACAAGAGCAGATGGCAGCAGGGCAACGTGTTTCAGCTGCAGCGTGATGC
ACGAGGCCCTGCACAACCACTACACACAGAAGTCCCTGTCTCTGAGCCCCGGCAA
(SEQ ID NO: 191), signal sequence (1-57), NYC-0008 (61-
810), linker (811-816), constant region (817-1512)

Fig. 193

NYC-0014-HC-k

MKHLWFFLLLVAAPRWVLSGEVQLVESGGGVVQPGGSLRLSCEASGFIFDDYAMHW
VRQAPGKGLEWVSLISGDGDIYYADSVKGRFTISRDNKNSLYLQMNSLIIEDTA
LYYCAKDWFVFGVVMTHYWFGLDVWGQGTTVTVSSGGGGSGGGSGGGGSDIQMTQ
SPSSLSASEGDRVTITCRASQSISTYLNWYQQKPGKAPKLLIYGASSLQSGVPSRF
SGSGSGTDFTLTISLQPEDFATYYCQQSYSTPPITFGQGTKVEIKGGEPKSADKT
HTCPPCPAPEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYV
DGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKT
ISKAKGQPREPQVYTLPPCRDELTKNQVSLWCLVKGFYPSDIAVEWESNGQPENNY
KTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK
(SEQ ID NO: 192), signal sequence (1-19), NYC-0008 (21-
270), linker (271-272), constant region (273-504)

Fig. 194

NYC-0015-HC-k

ATGAAGCACCTGTGGTTCTTTCTGCTGCTGGTGGCCGCTCCTAGATGGGTGCTGTC
TGGCGAAGTGCAGCTGGTGGAAATCTGGTGGCGGAGTTGTTTCAGCCTGGCGGCTCTC
TGAGACTGTCTTGTGCCGCCAGCGGCTTCACCTTCGACGATTATGCCATGCACTGG
GTCCGACAGGCCCTGGCAAAGGACTTGAATGGGTGTCCCTGATCTCTGGCGGAGG
CGGCGGAACCTACTACAGCGATTCTGTGAAGGGCAGATTCACCATCAGCCGGGACA
CCAGCAAGGACAGCCTGTACCTGCAGATGAACAGCCTGCGGACCGAGGACACAGCC
CTGTACTACTGCGCCAAGGACATGGTGTTCGGCGTGGTCACCCCTTACTACTACTT
CGCCCTGGATGTGTGGGGCCAGGGCACAACAGTGACAGTCTCTTCTGGCGGCGGAG
GAAGCGGAGGCGGAGGATCCGGTGGTGGTGGATCTGACATCCAGATGACACAGAGC
CCCAGCAGCCTGTCTGCCTCTGTGGGAGACAGAGTGACCATCACCTGTAGAGCCAG
CCAGAGCATCAACAGCTACCTGAACTGGTATCAGCAGAAGCCCGGCAAGGCCCTA
AGCTGCTGATCTATGCTGCCTCCAGTCTGCAGAGCGGCGTGCCAAGCAGATTTTCT
GGCAGCGGCTCTGGCACCGACTTCACCCTGACCATATCTAGCCTGCAGCCTGAGGA
CTTCGCCACCTACTACTGCCAGCAGAGCTACAGCGCCCTCCTATCACATTTGGCC
AGGGAACCAGACTGGAAATCAAAGGCGGAGAACCCAAGAGCgcaGACAAGACCCAC
ACCTGTCCTCCATGTCTCTGCTCCAGAAGCTGCAGGCGGCCCTTCCGTGTTTCTGTT
CCCTCCAAAGCCTAAGGACACCCTGATGATCAGCCGGACACCTGAAGTGACCTGCG
TGGTGGTGGATGTGTCCCACGAGGATCCCGAAGTGAAGTTCAATTGGTACGTGGAC
GGCGTGGAAGTGCACAACGCCAAGACCAAGCCTAGAGAGGAACAGTACAACAGCAC
CTACAGAGTGGTGTCTGTGCTGACAGTGCTGCACCAGGACTGGCTGAACGGCAAAG
AGTACAAGTGCAAGGTGTCCAACAAGGCCCTGCCTGCTCCTATCGAGAAAACCATC
AGCAAGGCCAAGGGCCAGCCTAGGGAACCCAGGTTTACACCCTGCCTCCATGCAG
GGATGAGCTGACCAAGAACCAGGTGTCCCTGTGGTGCCTGGTTAAGGGCTTCTACC
CCTCCGATATCGCCGTGGAATGGGAGAGCAATGGCCAGCCAGAGAACAACACTACAAG
ACAACCCCTCCTGTGCTGGACTCCGACGGCTCATTCTTCCTGTACAGCAAGCTGAC
CGTGGACAAGAGCAGATGGCAGCAGGGCAACGTGTTCAGCTGCAGCGTGATGCACG
AGGCCCTGCACAACCACTACACACAGAAGTCCCTGTCTCTGAGCCCCGGCAA
(SEQ ID NO: 193), signal sequence (1-57), NYC-0009 (61-
807), linker (808-813), constant region (814-1509)

Fig. 195

NYC-0015-HC-k

MKHLWFFLLLVAAPRWVLSGEVQLVESGGGVVQPGGSLRLSCAASGFTFDDYAMHW
VRQAPGKGLEWVSLISGGGGGTYYSDSVKGRFTISRDTSKDSLYLQMNSLRTE
DLYYCAKDMVFGVVTPYYYFALDVWGQGTTVTVSSGGGGSGGGSGGGGSDIQMTQS
PSSLSASVGDRVTITCRASQSINSYLNWYQQKPGKAPKLLIYAASSLQSGVPSRFS
GSGSGTDFTLTISSLQPEDFATYYCQQSYSAAPPITFGQGRLEIKGGEPKSADKTH
TCPPCPAPEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVD
GVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTI
SKAKGQPREPQVYTLPPCRDELTKNQVSLWCLVKGFYPSDIAVEWESNGQPENNYK
TTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFSQSVMEALHNHYTQKSLSLSPGK
(SEQ ID NO: 194), signal sequence (1-19), NYC-0009 (21-
269), linker (270-271), constant region (272-503)

Fig. 196

NYC-0016-HC-k

ATGAAGCACCTGTGGTTCCTTCTGCTGCTGGTGGCCGCTCCTAGATGGGTGCTGTC
TGGCGAAGTGCAGCTGGTGGAAATCTGGTGGCGGAGTTGTTTCAGCCTGGCGGCTCTC
TGAGACTGAGCTGTGCTGTGTCCGGCTTCACCTTCGACGACTACGCCATGCACTGG
GTCCGACAGGCACCTGGCAAAGGCCCTTGAATGGGTGTCCCTGATCAGCGGAGATGG
CGGCAGCACACACTACGCCGATTCTGTGAAGGGCAGATTCACCATCAGCCGGGACA
ACAGCAAGAACAGCCTGTACCTGCAGATGAACTCCCTGCGGACAGGCGACACAGCC
CTGTACTACTGCGCCAAGGACATGATCTTCGCCGTGGTCATCACCGACTACCACTA
CTACGGCATGGACGTGTGGGGCCAGGGAACCACAGTGACAGTTTCTAGCGGAGGCG
GAGGAAGTGGCGGCGGAGGATCTGGCGGTGGTGGTTCTGATATCCAGATGACACAG
AGCCCCAGCAGCCTGTCTGCCTCTGTGGGAGACAGAGTGACCATCACCTGTAGAGC
CAGCCAGAGCATCAGCAGCTACCTGAACTGGTATCAGCAGAAGCCCGGCAAGGCC
CTAAACTGCTGATCTATGCCGCTCCAGTCTGCAGAGCGGAGTGCCTAGCAGATTT
TCTGGCAGCGGCTCCGGCACCGATTTACCCCTGACCATATCTAGCCTGCAGCCTGA
GGACTTCGCCACCTACTACTGCCAGCAGAGCTACAGCACCCCTCCTATCACCTTTG
GCCAGGGCACCAGACTGGAAATCAAAGGCGGAGAACCCAAGAGCgcaGACAAGACC
CACACCTGTCTCCATGTCTCTGCTCCAGAAGCTGCAGGCGGCCCTTCCGTGTTTCT
GTTCCCTCCAAAGCCTAAGGACACCCCTGATGATCAGCCGGACACCTGAAGTGACCT
GCGTGGTGGTGGATGTGTCCACGAGGATCCCGAAGTGAAGTTCAATTGGTACGTG
GACGGCGTGGAAGTGCACAACGCCAAGACCAAGCCTAGAGAGGAACAGTACAACAG
CACCTACAGAGTGGTGTCTGTGCTGACAGTGCTGCACCAGGACTGGCTGAACGGCA
AAGAGTACAAGTGCAAGGTGTCCAACAAGGCCCTGCCTGCTCCTATCGAGAAAACC
ATCAGCAAGGCCAAGGGCCAGCCTAGGGAACCCAGGTTTACACCCTGCCTCCATG
CAGGGATGAGCTGACCAAGAACCAGGTGTCCCTGTGGTGCCTGGTTAAGGGCTTCT
ACCCCTCCGATATCGCCGTGGAATGGGAGAGCAATGGCCAGCCAGAGAACAACTAC
AAGACAACCCCTCCTGTGCTGGACTCCGACGGCTCATTCTTCTGTACAGCAAGCT
GACCGTGGACAAGAGCAGATGGCAGCAGGGCAACGTGTTTCAGCTGCAGCGTGATGC
ACGAGGCCCTGCACAACCACTACACACAGAAGTCCCTGTCTCTGAGCCCCGGCAA
(SEQ ID NO: 195), signal sequence (1-57), NYC-0010 (61-
810), linker (811-816), constant region (817-1512)

Fig. 197

NYC-0016-HC-k

MKHLWFFLLLVAAPRWVLSGEVQLVESGGGVVQPGGSLRLSCAVSGFTFDDYAMHW
VRQAPGKGLEWVSLISGDGGSTHYADSVKGRFTISRDNKNSLYLQMNSLRTGDTA
LYYCAKDMIFAVVITDYHYYGMDVWGQGTTVTVSSGGGGSGGGGSGGGGSDIQMTQ
SPSSLSASVGDRVTITCRASQSISSYLNWYQQKPGKAPKLLIYAASSLQSGVPSRF
SGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPPITFGQGRLEIKGGEPKSADKT
HTCPPCPAPEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYV
DGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKT
ISKAKGQPREPQVYTLPPCRDELTKNQVSLWCLVKGFYPSDIAVEWESNGQPENNY
KTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSQSVMEALHNHYTQKSLSLSPGK
(SEQ ID NO: 196), signal sequence (1-19), NYC-0010 (21-
270), linker (271-272), constant region (273-504)

Fig. 198

NYZ-1007-HC2

MKHLWFFLLLVAAPRWVLSGEVQLVESGGGLVQPGGSLRLSCAASGFSIRSYDIHW
VRQAPGKGLEWVATISYDGSQKYYADSVKGRFTISRDESKNTLYLQMNSLRAEDTA
VYHCARGSSGHYEAFDIWGQGTTLVTVSSGGGGSGGGGSGGGGSQSVLTQPPSASGT
PGQRVTISCSGSSSNIGNWYVSWYQQLPGTAPKLLIYDNNKRPSGVPDRFSGSKSG
TSASLAISGLQSEDEADYYCGTWDSSLSAPWVFGGGTKLTVLGGGGSEVQLVESGG
GLVQPGGSLRLSCAASGVTFNYYGMSWIRQAPGKGLEWVASITRSGGRIYYPDSVK
GRFTISRENTQKTLYLQMNSLRAEDTAVYYCTLDGRDGWVAYWGQGTTLVTVSSAST
KGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTFSWNSGALTSGVHTFPAVLQS
SGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRVEPKSCDKTHTCPPCPAP
EAAGGPSVFLEFPPKPKDTLMISRTPEVTCVVSFSVSHEDPEVKFNWYVDGVEVHNAK
TKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR
EPQVYVYPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSD
DGSFALVSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK

(SEQ ID NO: 197), signal sequence (1-19), NYA-2061 (21-266), linker (267-271), C3E-7085-VH (272-389), CH1 (390-487), hinge (488-502), CH2 (503-612), CH3 (613-719)

Fig. 199

NYZ-1017-HC2

MKHLWFFLLLVAAPRWVLSGQVQLVESGGGVVQPGRSLRLSCAASGFSIRSYDMHW
VRQAPGKGLEWVATISYDGSQKYYADSVKGRFTISRDN SKNTLYLQMSSLRAEDTA
VYHCARGSSGHYEAFDIWGQGT LVT VSSGGGGSGGGSGGGGSQSVLTQPPSVSAA
PGQKVTISCSGSSSNIGNWYVSWYQQLPGTAPKLLIYDNNKRPSGIPDRFSGSKSG
TSATLGITGLQTGDEADYYCGTWDSLSAPWVFGGGTKVTVLGGGGSEVQLVESGG
GLVQPGGSLRLSCAASGVTFNYYGMSWIRQAPGKGLEWVASITRSGGRIYYPDSVK
GRFTISRENTQKTLYLQMNSLRAEDTAVYYCTLDGRDGWVAYWGQGT LVT VSSAST
KGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTFSWNSGALTSGVHTFPAVLQS
SGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRVEPKSCDKTHTCPPCPAP
EAAGGPSVFLEFPPKPKDTLMISRTPEVTCVVS VSHEDPEVKFNWYVDGVEVHNAK
TKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR
EPQVYVYPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSD
DGSFALVSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK

(SEQ ID NO: 198), signal sequence (1-19), NYA-2047 (21-
266), linker (267-271), C3E-7085-VH (272-389) , CH1
(390-487), hinge (488-502), CH2 (503-612), CH3 (613-719)

BISPECIFIC ANTIBODY**CROSS-REFERENCE TO RELATED APPLICATIONS**

[0001] The present application is a Divisional of U.S. patent application Ser. No. 17/799,648, filed on Aug. 12, 2022, which claims priority under 35 U.S.C. § 371 to International Patent Application No. PCT/JP2021/013378, filed Mar. 29, 2021, which claims priority to and the benefit of Japanese Patent Application No. 2020-061476, filed on Mar. 30, 2020. The contents of these applications are hereby incorporated by reference in their entireties.

SEQUENCE LISTING

[0002] The instant application contains a Sequence Listing which has been submitted via USPTO Patent Center is hereby incorporated by reference in its entirety. Said electronic copy, is named 127565-0109_SL.xml and is 372 kb in size.

TECHNICAL FIELD

[0003] The present invention relates to an antibody, a bispecific antibody, and the like that are useful for the treatment of tumors.

BACKGROUND ART

[0004] NY-ESO-1 is a molecule that has been identified in esophageal cancer by a method of serological analysis of recombinant cDNA expression libraries (SEREX) (Non-Patent Literature 1), and LAGE-1 is also referred to as “NY-ESO-2,” which is a molecule identified via representational difference analysis of the tumor cDNA library (Non-Patent Literature 2). While expression of such molecules is known to be localized in the testis in the case of normal tissue, the mechanism thereof remains unknown. Since expression of NY-ESO-1 and LAGE-1 has been reported to occur in a wide variety of cancer species, such as melanoma, lung cancer, bladder cancer, ovarian cancer, soft-tissue sarcoma, and myeloma (Non-Patent Literature 3), association thereof with cancer is also suggested. In addition, reports have been made concerning the correlation with the degree of malignancy. For example, the NY-ESO-1 expression level is higher in the metastatic focus of melanoma than in the primary focus thereof (Non-Patent Literature 4), the NY-ESO-1 and LAGE-1 expression levels are higher in the advanced urothelial carcinoma than in the early stage urothelial carcinoma (Non-Patent Literature 5), and the NY-ESO-1 expression level is higher in high-risk myeloma with chromosomal abnormalities than in myeloma without chromosomal abnormalities (Non-Patent Literature 6). On the basis of such information, NY-ESO-1 and LAGE-1 have drawn attention as molecules with high cancer-specificity, and a great deal of research and development has been made aimed at drug discovery in cancer vaccine therapy. To date, however, there have been no pharmaceutical products approved in this respect.

[0005] The 9-mer NY-ESO peptides of NY-ESO-1 and of LAGE-1 each comprising residues 157 to 165 (i.e., SLL-MWITQC) are known to form a complex with HLA (histocompatibility leukocyte antigen)-A2 (i.e., the HLA/NY-ESO peptide complex) and present the complex extracellularly (Non-Patent Literature 7). As described above, expression of NY-ESO-1 and LAGE-1 is cancer-

specific. This indicates that the HLA/NY-ESO peptide complex is a therapeutic target specific to cancer that is selectively present on HLA-A2-positive and NY-ESO-1- or LAGE-1-positive cancer cells (Non-Patent Literature 8). In addition, molecules that bind to the HLA/NY-ESO peptide complex, such as TCR (Patent Literature 1 and Patent Literature 2) and antibodies (Patent Literature 3 and Patent Literature 8) have been reported.

[0006] An application of a molecule binding to a cancer-targeting molecule is a CD3 bispecific antibody that functions on the basis of the T cell redirecting mechanism whereby it recruits T cells to cancer cells and thereby exerts antitumor effects by cytotoxicity (Non-Patent Literature 9 and Non-Patent Literature 10). An example of a currently available CD3 bispecific antibody medicine is Blinatumomab, which is a CD19 bispecific T-cell engager (BiTE). Such antibody medicine is approved for acute lymphoblastic leukemia (ALL), and clinical trials targeting other blood cancers have been in progress. However, the bispecific antibody format thereof is tandem scFv (taFv) without the Fc region, and the blood half-life upon administration thereof to a patient is significantly shorter than that attained with the use of the IgG type antibody that is commonly used as a therapeutic antibody (Non-Patent Literature 11).

[0007] Concerning bispecific antibodies having the heterodimer Fc region exhibiting the blood half-life equivalent to that of the IgG type antibody, research and clinical trials on CD3 bispecific antibodies with a variety of antibody formats, such as knobs-into-holes, CrossMAb, and DuoBody® (Patent Literatures 4, 5, 6, and 7), have been in progress, and a CD3 bispecific antibody utilizing an antibody against the HLA/NY-ESO peptide complex is reported (Non-Patent Literature 12).

CITATION LIST**Patent Literature**

- [0008]** Patent Literature 1: WO 2005/113595
- [0009]** Patent Literature 2: WO 2017/109496
- [0010]** Patent Literature 3: WO 2010/106431
- [0011]** Patent Literature 4: WO 1998/050431
- [0012]** Patent Literature 5: WO 2006/106905
- [0013]** Patent Literature 6: WO 2011/028952
- [0014]** Patent Literature 7: WO 2011/131746
- [0015]** Patent Literature 8: WO 2021/003357

Non-Patent Literature

- [0016]** Non-Patent Literature 1: Proc. Natl. Acad. Sci., U.S.A., 94 (5), 1914-8, 1997
- [0017]** Non-Patent Literature 2: Int. J. Cancer, 76 (6), 903-8, 1998
- [0018]** Non-Patent Literature 3: Immunol. Cell Biol., 84 (3), 303-17, 2006
- [0019]** Non-Patent Literature 4: J. Surg. Res., 98 (2), 76-80, 2001
- [0020]** Non-Patent Literature 5: Cancer Res., 61 (12), 4671-4, 2001
- [0021]** Non-Patent Literature 6: Blood, 105 (10), 3939-44, 2005
- [0022]** Non-Patent Literature 7: J. Exp. Med., 187 (2), 265-70, 1998
- [0023]** Non-Patent Literature 8: J. Immunol., 176 (12), 7308-16, 2006

- [0024] Non-Patent Literature 9: Nature, 314 (6012), 628-31, 1985
- [0025] Non-Patent Literature 10: Int. Rev. Immunol., 4 (2), 159-73, 1989
- [0026] Non-Patent Literature 11: Drug Des. Devel. Ther., 10, 757-765, 2016
- [0027] Non-Patent Literature 12: Abstracts of the 21st Annual Meeting of the Japanese Association of Cancer Immunology, 013-4

SUMMARY OF INVENTION

Technical Problem

[0028] The present invention provides a novel anti-HLA-A2/NY-ESO antibody that can be used as an anti-tumor agent and an anti-tumor agent comprising, as an active ingredient, a molecule that binds to HLA-A2/NY-ESO comprising such antibody and the like.

Solution to Problem

[0029] The present inventors have conducted concentrated studies in order to solve the above problem. As a result, they discovered a novel anti-HLA-A2/NY-ESO antibody and a molecule that binds to HLA-A2/NY-ESO comprising such an antibody and the like. This has led to the completion of the present invention.

[0030] Specifically, the present invention includes the following.

[0031] [1] An antibody that binds specifically to human HLA/NY-ESO or a binding fragment thereof comprising:

[0032] a heavy chain CDRH1 consisting of the amino acid sequence as shown in SEQ ID NO: 54,

[0033] a heavy chain CDRH2 consisting of the amino acid sequence as shown in SEQ ID NO: 55,

[0034] a heavy chain CDRH3 consisting of the amino acid sequence as shown in SEQ ID NO: 56,

[0035] a light chain CDRL1 consisting of the amino acid sequence as shown in SEQ ID NO: 57 or a light chain CDRL1 consisting of an amino acid sequence derived from the amino acid sequence as shown in SEQ ID NO: 57 in which amino acid 7 is W and/or amino acid 8 is K,

[0036] a light chain CDRL2 consisting of the amino acid sequence as shown in SEQ ID NO: 58, and

[0037] a light chain CDRL3 consisting of the amino acid sequence as shown in SEQ ID NO: 59 or a light chain CDRL3 consisting of an amino acid sequence derived from the amino acid sequence as shown in SEQ ID NO: 59 in which amino acid 2 is A or S.

[0038] [2] The antibody according to [1] comprising CDRH1 to CDRH3 and CDRL1 to CDRL3 of one or more groups selected from the group consisting of the groups (i) to (v) below or a binding fragment thereof:

[0039] (i) a heavy chain CDRH1 consisting of the amino acid sequence as shown in SEQ ID NO: 54,

[0040] a heavy chain CDRH2 consisting of the amino acid sequence as shown in SEQ ID NO: 55,

[0041] a heavy chain CDRH3 consisting of the amino acid sequence as shown in SEQ ID NO: 56,

[0042] a light chain CDRL1 consisting of the amino acid sequence as shown in SEQ ID NO: 57,

[0043] a light chain CDRL2 consisting of the amino acid sequence as shown in SEQ ID NO: 58, and

[0044] a light chain CDRL3 consisting of the amino acid sequence as shown in SEQ ID NO: 59;

[0045] (ii) a heavy chain CDRH1 consisting of the amino acid sequence as shown in SEQ ID NO: 54,

[0046] a heavy chain CDRH2 consisting of the amino acid sequence as shown in SEQ ID NO: 55,

[0047] a heavy chain CDRH3 consisting of the amino acid sequence as shown in SEQ ID NO: 56,

[0048] a light chain CDRL1 consisting of an amino acid sequence derived from the amino acid sequence as shown in SEQ ID NO: 57 in which amino acid 7 is W,

[0049] a light chain CDRL2 consisting of the amino acid sequence as shown in SEQ ID NO: 58, and

[0050] a light chain CDRL3 consisting of the amino acid sequence as shown in SEQ ID NO: 59;

[0051] (iii) a heavy chain CDRH1 consisting of the amino acid sequence as shown in SEQ ID NO: 54,

[0052] a heavy chain CDRH2 consisting of the amino acid sequence as shown in SEQ ID NO: 55,

[0053] a heavy chain CDRH3 consisting of the amino acid sequence as shown in SEQ ID NO: 56,

[0054] a light chain CDRL1 consisting of the amino acid sequence as shown in SEQ ID NO: 57,

[0055] a light chain CDRL2 consisting of the amino acid sequence as shown in SEQ ID NO: 58, and

[0056] a light chain CDRL3 consisting of an amino acid sequence derived from the amino acid sequence as shown in SEQ ID NO: 59 in which amino acid 2 is A;

[0057] (iv) a heavy chain CDRH1 consisting of the amino acid sequence as shown in SEQ ID NO: 54,

[0058] a heavy chain CDRH2 consisting of the amino acid sequence as shown in SEQ ID NO: 55,

[0059] a heavy chain CDRH3 consisting of the amino acid sequence as shown in SEQ ID NO: 56,

[0060] a light chain CDRL1 consisting of the amino acid sequence as shown in SEQ ID NO: 57,

[0061] a light chain CDRL2 consisting of the amino acid sequence as shown in SEQ ID NO: 58, and

[0062] a light chain CDRL3 consisting of an amino acid sequence derived from the amino acid sequence as shown in SEQ ID NO: 59 in which amino acid 2 is S; and

[0063] (v) a heavy chain CDRH1 consisting of the amino acid sequence as shown in SEQ ID NO: 54,

[0064] a heavy chain CDRH2 consisting of the amino acid sequence as shown in SEQ ID NO: 55,

[0065] a heavy chain CDRH3 consisting of the amino acid sequence as shown in SEQ ID NO: 56,

[0066] a light chain CDRL1 consisting of an amino acid sequence derived from the amino acid sequence as shown in SEQ ID NO: 57 in which amino acid 7 is W and amino acid 8 is K,

[0067] a light chain CDRL2 consisting of the amino acid sequence as shown in SEQ ID NO: 58, and

[0068] a light chain CDRL3 consisting of the amino acid sequence as shown in SEQ ID NO: 59.

[0069] [3] The antibody or the binding fragment thereof according to [1] comprising a heavy chain variable region consisting of an amino acid sequence having 95% or higher sequence identity to an amino acid

sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 27 or the amino acid sequence as shown in SEQ ID NO: 38 or 39 and a light chain variable region consisting of an amino acid sequence having 95% or higher sequence identity to an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 27 or SEQ ID NO: 52 or the amino acid sequence as shown in SEQ ID NO: 40.

[0070] [4] The antibody or the binding fragment thereof according to [1] comprising:

[0071] (H1) a heavy chain variable region consisting of the amino acid sequence as shown in SEQ ID NO: 6,

[0072] (H2) a heavy chain variable region consisting of the amino acid sequence as shown in SEQ ID NO: 18,

[0073] (H3) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 29,

[0074] (H4) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 26,

[0075] (H5) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 27,

[0076] (H6) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 28,

[0077] (H7) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 36,

[0078] (H8) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 47,

[0079] (H9) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 48,

[0080] (H10) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 50;

[0081] (H11) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 51,

[0082] (H12) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 52,

[0083] (H13) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 53,

[0084] (H14) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 30; or

[0085] (H15) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 156; and

[0086] (L1) a light chain variable region consisting of the amino acid sequence as shown in SEQ ID NO: 8,

[0087] (L2) a light chain variable region consisting of the amino acid sequence as shown in SEQ ID NO: 20,

[0088] (L3) a light chain variable region consisting of an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 29,

[0089] (L4) a light chain variable region consisting of an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 26,

[0090] (L5) a light chain variable region consisting of an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 27,

[0091] (L6) a light chain variable region consisting of an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 28,

[0092] (L7) a light chain variable region consisting of an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 36,

[0093] (L8) a light chain variable region consisting of an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 47,

[0094] (L9) a light chain variable region consisting of an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 48,

[0095] (L10) a light chain variable region consisting of an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 50,

[0096] (L11) a light chain variable region consisting of an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 51,

[0097] (L12) a light chain variable region consisting of an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 52,

[0098] (L13) a light chain variable region consisting of an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 53,

[0099] (L14) a light chain variable region consisting of an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 30; or

[0100] (L15) a light chain variable region consisting of an amino acid sequence of amino acids 161 to 271 of the amino acid sequence as shown in SEQ ID NO: 156.

[0101] [5] The antibody or the binding fragment thereof according to [4] comprising:

[0102] (H1L1) a heavy chain variable region consisting of the amino acid sequence as shown in SEQ ID NO: 6 and a light chain variable region consisting of the amino acid sequence as shown in SEQ ID NO: 8,

[0103] (H2L2) a heavy chain variable region consisting of the amino acid sequence as shown in SEQ ID NO: 18 and a light chain variable region consisting of the amino acid sequence as shown in SEQ ID NO: 20,

[0104] (H3L3) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 29 and a light chain variable region consisting of an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 29,

[0105] (H4L4) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 26 and a light chain variable region consisting of an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 26,

[0106] (H5L5) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 27 and a light chain variable region consisting of an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 27,

[0107] (H6L6) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 28 and a light chain variable region consisting of an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 28,

[0108] (H7L7) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 36 and a light chain variable region consisting of an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 36,

[0109] (H8L8) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 47 and a light chain variable region consisting of an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 47,

[0110] (H9L9) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 48 and a light chain variable region consisting of an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 48,

[0111] (H10L10) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 50 and a light chain variable region consisting of an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 50,

[0112] (H11L11) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 51 and a light chain variable region consisting of an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 51,

[0113] (H12L12) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 52 and a light chain variable region consisting of an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 52,

[0114] (H13L13) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 53 and a light chain variable region consisting of an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 53,

[0115] (H14L14) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 30 and a light chain variable region consisting of an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 30, or

[0116] (H15L15) a heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 156 and a light chain variable region consisting of an amino acid sequence of amino acids 161 to 271 of the amino acid sequence as shown in SEQ ID NO: 156.

[0117] [6] The antibody or the binding fragment thereof according to any of [1] to [5], which is scFv.

[0118] [7] The antibody or the binding fragment thereof according to [6], which is

[0119] (s1) scFv consisting of an amino acid sequence of amino acids 21 to 266 of the amino acid sequence as shown in SEQ ID NO: 70,

[0120] (s2) scFv comprising a heavy chain variable region consisting of the amino acid sequence as shown in SEQ ID NO: 18 and a light chain variable region consisting of the amino acid sequence as shown in SEQ ID NO: 20,

[0121] (s3) scFv consisting of an amino acid sequence of amino acids 21 to 266 of the amino acid sequence as shown in SEQ ID NO: 29,

[0122] (s4) scFv consisting of an amino acid sequence of amino acids 21 to 266 of the amino acid sequence as shown in SEQ ID NO: 26,

[0123] (s5) scFv consisting of an amino acid sequence of amino acids 21 to 266 of the amino acid sequence as shown in SEQ ID NO: 27,

- [0124] (s6) scFv consisting of an amino acid sequence of amino acids 21 to 266 of the amino acid sequence as shown in SEQ ID NO: 28,
- [0125] (s7) scFv consisting of an amino acid sequence of amino acids 21 to 266 of the amino acid sequence as shown in SEQ ID NO: 36,
- [0126] (s8) scFv consisting of an amino acid sequence of amino acids 21 to 266 of the amino acid sequence as shown in SEQ ID NO: 47,
- [0127] (s9) scFv consisting of an amino acid sequence of amino acids 21 to 266 of the amino acid sequence as shown in SEQ ID NO: 48,
- [0128] (s10) scFv consisting of an amino acid sequence of amino acids 21 to 266 of the amino acid sequence as shown in SEQ ID NO: 50,
- [0129] (s11) scFv consisting of an amino acid sequence of amino acids 21 to 266 of the amino acid sequence as shown in SEQ ID NO: 51,
- [0130] (s12) scFv consisting of an amino acid sequence of amino acids 21 to 266 of the amino acid sequence as shown in SEQ ID NO: 52,
- [0131] (s13) scFv consisting of an amino acid sequence of amino acids 21 to 266 of the amino acid sequence as shown in SEQ ID NO: 53,
- [0132] (s14) scFv consisting of an amino acid sequence of amino acids 21 to 266 of the amino acid sequence as shown in SEQ ID NO: 30, or
- [0133] (s15) scFv consisting of an amino acid sequence of amino acids 21 to 271 of the amino acid sequence as shown in SEQ ID NO: 156.
- [0134] [8] A polynucleotide encoding the antibody or the binding fragment thereof according to any of [1] to [7].
- [0135] [9] A vector comprising the polynucleotide according to [8].
- [0136] [10] A host cell comprising the polynucleotide according to [8] or the vector according to [9].
- [0137] [11] A method for producing an antibody that binds specifically to human HLA/NY-ESO or a binding fragment thereof comprising: (i) a step of culturing the host cell according to [10]; and (ii) a step of purifying an antibody or a binding fragment thereof from the culture product obtained in step (i).
- [0138] [12] An antibody that binds specifically to human HLA/NY-ESO or a binding fragment thereof, which is obtained by the method according to [11].
- [0139] [13] An antibody that has the properties (i) or (ii) below and binds to HLA-A2/NY-ESO or a binding fragment thereof:
- [0140] (i) binding to a site on HLA-A2/NY-ESO recognized by the antibody or the binding fragment thereof according to [7]; or
- [0141] (ii) binding to human HLA-A2/NY-ESO competitively with the antibody or the binding fragment thereof according to [7].
- [0142] [14] A pharmaceutical composition comprising, as an active ingredient, the antibody or the binding fragment thereof according to any of [1] to [7], [12], and [13], the polynucleotide according to [8], the vector according to [9], or the cell according to [10].
- [0143] [15] A molecule that binds specifically to human HLA/NY-ESO comprising the antibody or the binding fragment thereof according to any of [1] to [7], [12], and [13].
- [0144] [16] The molecule according to [15], which is a multispecific antibody.
- [0145] [17] The molecule according to [15], which is a bispecific antibody.
- [0146] [18] The molecule according to any of [15] to [17], which comprises an antibody that binds specifically to CD3 or a binding fragment thereof.
- [0147] [19] The molecule according to [18], wherein the antibody that binds specifically to CD3 or the binding fragment thereof comprises:
- [0148] (CCH1) a heavy chain CDRH1 consisting of the amino acid sequence as shown in SEQ ID NO: 141;
- [0149] (CCH2) a heavy chain CDRH2 consisting of the amino acid sequence as shown in SEQ ID NO: 142 or a heavy chain CDRH2 consisting of an amino acid sequence derived from the amino acid sequence as shown in SEQ ID NO: 142 in which amino acid 3 is N or S;
- [0150] (CCH3) a heavy chain CDRH3 consisting of the amino acid sequence as shown in SEQ ID NO: 143;
- [0151] (CCL1) a light chain CDRL1 consisting of the amino acid sequence as shown in SEQ ID NO: 144;
- [0152] (CCL2) a light chain CDRL2 consisting of the amino acid sequence as shown in SEQ ID NO: 145 or a light chain CDRL2 consisting of an amino acid sequence derived from the amino acid sequence as shown in SEQ ID NO: 145 in which amino acid 2 is N; and
- [0153] (CCL3) a light chain CDRL3 consisting of the amino acid sequence as shown in SEQ ID NO: 146.
- [0154] [20] The molecule according to [19], wherein the antibody that binds specifically to CD3 or the binding fragment thereof comprises:
- [0155] (CH1) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 136,
- [0156] (CH2) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 137,
- [0157] (CH3) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 147,
- [0158] (CH4) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 138,
- [0159] (CH5) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 139,
- [0160] (CH6) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 140,
- [0161] (CH7) a heavy chain variable region consisting of an amino acid sequence of amino acids 272 to 389 of the amino acid sequence as shown in SEQ ID NO: 155,

- [0162] (CH8) a heavy chain variable region consisting of an amino acid sequence of amino acids 277 to 394 of the amino acid sequence as shown in SEQ ID NO: 156; or
- [0163] (CH9) a heavy chain variable region consisting of an amino acid sequence of amino acids 277 to 394 of the amino acid sequence as shown in SEQ ID NO: 157; and
- [0164] (CL1) a light chain variable region consisting of an amino acid sequence of amino acids 135 to 243 of the amino acid sequence as shown in SEQ ID NO: 136,
- [0165] (CL2) a light chain variable region consisting of an amino acid sequence of amino acids 135 to 241 of the amino acid sequence as shown in SEQ ID NO: 137,
- [0166] (CL3) a light chain variable region consisting of an amino acid sequence of amino acids 135 to 243 of the amino acid sequence as shown in SEQ ID NO: 147,
- [0167] (CL4) a light chain variable region consisting of an amino acid sequence of amino acids 135 to 241 of the amino acid sequence as shown in SEQ ID NO: 138,
- [0168] (CL5) a light chain variable region consisting of an amino acid sequence of amino acids 135 to 243 of the amino acid sequence as shown in SEQ ID NO: 139,
- [0169] (CL6) a light chain variable region consisting of an amino acid sequence of amino acids 135 to 243 of the amino acid sequence as shown in SEQ ID NO: 140,
- [0170] (CL7) a light chain variable region consisting of an amino acid sequence of amino acids 405 to 511 of the amino acid sequence as shown in SEQ ID NO: 155,
- [0171] (CL8) a light chain variable region consisting of an amino acid sequence of amino acids 410 to 516 of the amino acid sequence as shown in SEQ ID NO: 156, or
- [0172] (CL9) a light chain variable region consisting of an amino acid sequence of amino acids 410 to 516 of the amino acid sequence as shown in SEQ ID NO: 157.
- [0173] [21] The molecule according to [20], wherein the antibody that binds specifically to CD3 or the binding fragment thereof comprises:
- [0174] (CH1CL1) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 136 and a light chain variable region consisting of an amino acid sequence of amino acids 135 to 243 of the amino acid sequence as shown in SEQ ID NO: 136;
- [0175] (CH2CL2) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 137 and a light chain variable region consisting of an amino acid sequence of amino acids 135 to 241 of the amino acid sequence as shown in SEQ ID NO: 137;
- [0176] (CH3CL3) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 147 and a light chain variable region consisting of an amino acid sequence of amino acids 135 to 243 of the amino acid sequence as shown in SEQ ID NO: 147;
- [0177] (CH4CL4) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 138 and a light chain variable region consisting of an amino acid sequence of amino acids 135 to 241 of the amino acid sequence as shown in SEQ ID NO: 138;
- [0178] (CH5CL5) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 139 and a light chain variable region consisting of an amino acid sequence of amino acids 135 to 243 of the amino acid sequence as shown in SEQ ID NO: 139;
- [0179] (CH6CL6) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 140 and a light chain variable region consisting of an amino acid sequence of amino acids 135 to 243 of the amino acid sequence as shown in SEQ ID NO: 140;
- [0180] (CH7CL7) a heavy chain variable region consisting of an amino acid sequence of amino acids 272 to 389 of the amino acid sequence as shown in SEQ ID NO: 155 and a light chain variable region consisting of an amino acid sequence of amino acids 272 to 389 of the amino acid sequence as shown in SEQ ID NO: 155;
- [0181] (CH8CL8) a heavy chain variable region consisting of an amino acid sequence of amino acids 277 to 394 of the amino acid sequence as shown in SEQ ID NO: 156 and a light chain variable region consisting of an amino acid sequence of amino acids 410 to 516 of the amino acid sequence as shown in SEQ ID NO: 156; or
- [0182] (CH9CL9) a heavy chain variable region consisting of an amino acid sequence of amino acids 277 to 394 of the amino acid sequence as shown in SEQ ID NO: 157 and a light chain variable region consisting of an amino acid sequence of amino acids 410 to 516 of the amino acid sequence as shown in SEQ ID NO: 157.
- [0183] [22] The molecule according to any of [18] to [21], wherein the antibody that binds specifically to CD3 or the binding fragment thereof is scFv.
- [0184] [23] The molecule according to [22], wherein scFv comprises:
- [0185] (CS1) scFv consisting of an amino acid sequence of amino acids 2 to 243 of the amino acid sequence as shown in SEQ ID NO: 136;
- [0186] (CS2) scFv consisting of an amino acid sequence of amino acids 2 to 241 of the amino acid sequence as shown in SEQ ID NO: 137;
- [0187] (CS3) scFv consisting of an amino acid sequence of amino acids 2 to 243 of the amino acid sequence as shown in SEQ ID NO: 147;
- [0188] (CS4) scFv consisting of an amino acid sequence of amino acids 2 to 241 of the amino acid sequence as shown in SEQ ID NO: 138;

- [0189] (CS5) scFv consisting of an amino acid sequence of amino acids 2 to 243 of the amino acid sequence as shown in SEQ ID NO: 139;
- [0190] (CS6) scFv consisting of an amino acid sequence of amino acids 2 to 243 of the amino acid sequence as shown in SEQ ID NO: 140;
- [0191] (CS7) scFv consisting of an amino acid sequence of amino acids 272 to 511 of the amino acid sequence as shown in SEQ ID NO: 155;
- [0192] (CS8) scFv consisting of an amino acid sequence of amino acids 277 to 516 of the amino acid sequence as shown in SEQ ID NO: 156; or
- [0193] (CS9) scFv consisting of an amino acid sequence of amino acids 277 to 516 of the amino acid sequence as shown in SEQ ID NO: 157.
- [0194] [24] The molecule according to any of [18] to [23] comprising: a first polypeptide comprising scFv that binds specifically to human HLA/NY-ESO, scFv that binds specifically to CD3, and an Fc region (i) in that order from the N terminus toward the C terminus; and a second polypeptide comprising an Fc region (ii), wherein the first polypeptide is preferably associated with the second polypeptide at the Fc region (i) and the Fc region (ii).
- [0195] [25] The molecule according to [24], wherein the scFv that binds specifically to human HLA/NY-ESO is the antibody that binds specifically to human HLA/NY-ESO or the binding fragment thereof according to [1], which is scFv.
- [0196] [26] The molecule according to [24], wherein the scFv that binds specifically to human HLA/NY-ESO is the antibody that binds specifically to human HLA/NY-ESO or the binding fragment thereof according to [2], which is scFv.
- [0197] [27] The molecule according to [24], wherein the scFv that binds specifically to human HLA/NY-ESO is the antibody that binds specifically to human HLA/NY-ESO or the binding fragment thereof according to [3], which is scFv.
- [0198] [28] The molecule according to [24], wherein the scFv that binds specifically to human HLA/NY-ESO is the antibody that binds specifically to human HLA/NY-ESO or the binding fragment thereof according to [4], which is scFv.
- [0199] [29] The molecule according to [24], wherein the scFv that binds specifically to human HLA/NY-ESO is the antibody that binds specifically to human HLA/NY-ESO or the binding fragment thereof according to [5], which is scFv.
- [0200] [30] The molecule according to any of [19] to [24], wherein the scFv that binds specifically to CD3 is the antibody that binds specifically to CD3 or the binding fragment thereof according to [18], which is scFv.
- [0201] [31] The molecule according to any of [19] to [24], wherein the scFv that binds specifically to CD3 is the antibody that binds specifically to CD3 or the binding fragment thereof according to [19], which is scFv.
- [0202] [32] The molecule according to any of [19] to [24], wherein the scFv that binds specifically to CD3 is the antibody that binds specifically to CD3 or the binding fragment thereof according to [20] or [21], which is scFv.
- [0203] [33] The molecule according to any of [19] to [24], wherein the scFv that binds specifically to CD3 is the antibody that binds specifically to CD3 or the binding fragment thereof according to [23], which is scFv.
- [0204] [34] The molecule according to any of [24] to [33] comprising an amino acid sequence selected from the group consisting of an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 85, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 87, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 88, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 89, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 90, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 91, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 92, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 93, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 94, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 95, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 96, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 86, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 149, and an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 150.
- [0205] [35] The molecule according to any of [24] to [33], which comprises an amino acid sequence selected from the group consisting of an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 155, an amino acid sequence of amino acids 21 to 516 of the amino acid sequence as shown in SEQ ID NO: 156, and an amino acid sequence of amino acids 21 to 516 of the amino acid sequence as shown in SEQ ID NO: 157.
- [0206] [36] The molecule according to any of [24] to [34], wherein the first polypeptide comprises an amino acid sequence of amino acids 529 to 745 of the amino acid sequence as shown in SEQ ID NO: 85, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 86, 149, or 150.
- [0207] [37] The molecule according to any of [24] to [33] and [35], wherein the first polypeptide comprises an amino acid sequence of amino acids 529 to 745 of the amino acid sequence as shown in SEQ ID NO: 155, an amino acid sequence of amino acids 534 to 750 of the amino acid sequence as shown in SEQ ID NO: 156, or an amino acid sequence of amino acids 534 to 750 of the amino acid sequence as shown in SEQ ID NO: 157.
- [0208] [38] The molecule according to [34] or [36], wherein the first polypeptide consists of an amino acid sequence selected from the group consisting of an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 85, an

- amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 87, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 88, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 89, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 90, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 91, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 92, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 93, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 94, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 95, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 96, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 86, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 149, and an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 150.
- [0209] [39] The molecule according to [35] or [37], wherein the first polypeptide consists of an amino acid sequence selected from the group consisting of an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 155, an amino acid sequence of amino acids 20 to 750 of the amino acid sequence as shown in SEQ ID NO: 156, and an amino acid sequence of amino acids 20 to 750 of the amino acid sequence as shown in SEQ ID NO: 157.
- [0210] [40] The molecule according to any of [24] to [39], wherein the second polypeptide comprises an amino acid sequence of amino acids 20 to 246 of the amino acid sequence as shown in SEQ ID NO: 84.
- [0211] [41] The molecule according to any of [18] to [23] comprising a first polypeptide, a second polypeptide, and a third polypeptide, wherein
- [0212] the first polypeptide comprises scFv that binds specifically to human HLA/NY-ESO, scFv that binds specifically to CD3, and an Fc region (i) in that order from the N terminus toward the C terminus,
- [0213] the second polypeptide comprises an immunoglobulin heavy chain comprising an Fc region (ii), and
- [0214] the third polypeptide comprises an immunoglobulin light chain; and
- [0215] the second polypeptide is preferably associated with the third polypeptide, and
- [0216] the first polypeptide is preferably associated with the second polypeptide in the Fc regions.
- [0217] [42] The molecule according to [41], wherein the scFv that binds specifically to human HLA/NY-ESO is the antibody that binds specifically to human HLA/NY-ESO or the binding fragment thereof according to [1], which is scFv.
- [0218] [43] The molecule according to [41], wherein the scFv that binds specifically to human HLA/NY-ESO is the antibody that binds specifically to human HLA/NY-ESO or the binding fragment thereof according to [2], which is scFv.
- [0219] [44] The molecule according to [41], wherein the scFv that binds specifically to human HLA/NY-ESO is the antibody that binds specifically to human HLA/NY-ESO or the binding fragment thereof according to [3], which is scFv.
- [0220] [45] The molecule according to [41], wherein the scFv that binds specifically to human HLA/NY-ESO is the antibody that binds specifically to human HLA/NY-ESO or the binding fragment thereof according to [4], which is scFv.
- [0221] [46] The molecule according to [41], wherein the scFv that binds specifically to human HLA/NY-ESO is the antibody that binds specifically to human HLA/NY-ESO or the binding fragment thereof according to [5], which is scFv.
- [0222] [47] The molecule according to any of [42] to [46], wherein the scFv that binds specifically to CD3 is the antibody that binds specifically to CD3 or the binding fragment thereof according to [18], which is scFv.
- [0223] [48] The molecule according to any of [42] to [46], wherein the scFv that binds specifically to CD3 is the antibody that binds specifically to CD3 or the binding fragment thereof according to [19], which is scFv.
- [0224] [49] The molecule according to any of [42] to [46], wherein the scFv that binds specifically to CD3 is the antibody that binds specifically to CD3 or the binding fragment thereof according to [20] or [21], which is scFv.
- [0225] [50] The molecule according to any of [42] to [46], wherein the scFv that binds specifically to CD3 is the antibody that binds specifically to CD3 or the binding fragment thereof according to [23], which is scFv.
- [0226] [51] The molecule according to any of [41] to [50], wherein the second polypeptide comprises an amino acid sequence of amino acids 20 to 242 of the amino acid sequence as shown in SEQ ID NO: 99.
- [0227] [52] The molecule according to any of [41] to [51], wherein the third polypeptide comprises the amino acid sequence as shown in SEQ ID NO: 100.
- [0228] [53] The molecule according to any of [41] to [52], wherein the first polypeptide comprises an amino acid sequence selected from the group consisting of an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 85, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 87, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 88, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 89, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 90, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 91, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 92, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 93, an

amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 94, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 95, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 96, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 86, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 149, and an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 150.

[0229] [54] The molecule according to any of [41] to [53], wherein the first polypeptide consists of an amino acid sequence selected from the group consisting of an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 85, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 87, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 88, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 89, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 90, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 91, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 92, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 93, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 94, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 95, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 96, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 86, an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 149, and an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 150.

[0230] [55] The molecule according to any one of [18] to [23] comprising a first polypeptide that comprises scFv that binds specifically to human HLA/NY-ESO, a heavy chain variable region and constant region CH1 of the antibody that binds specifically to CD3, and an immunoglobulin Fc region (i) in that order from the N terminus toward the C terminus, a second polypeptide that comprises a hinge region and an Fc region (ii) of the immunoglobulin, and a third polypeptide that comprises an antibody light chain consisting of a variable region and a constant region, wherein the first polypeptide is preferably associated with the second polypeptide at the Fc region (i) and the Fc region (ii), and the first polypeptide is preferably associated with the third polypeptide in the heavy chain variable region and the constant region CH1 of the antibody.

[0231] [56] The molecule according to [55], wherein the scFv that binds specifically to human HLA/NY-ESO is the antibody that binds specifically to human HLA/NY-ESO or the binding fragment thereof according to [1], which is scFv.

[0232] [57] The molecule according to [55], wherein the scFv that binds specifically to human HLA/NY-ESO is the antibody that binds specifically to human HLA/NY-ESO or the binding fragment thereof according to [2], which is scFv.

[0233] [58] The molecule according to [55], wherein the scFv that binds specifically to human HLA/NY-ESO is the antibody that binds specifically to human HLA/NY-ESO or the binding fragment thereof according to [3], which is scFv.

[0234] [59] The molecule according to [55], wherein the scFv that binds specifically to human HLA/NY-ESO is the antibody that binds specifically to human HLA/NY-ESO or the binding fragment thereof according to [4], which is scFv.

[0235] [60] The molecule according to [55], wherein the scFv that binds specifically to human HLA/NY-ESO is the antibody that binds specifically to human HLA/NY-ESO or the binding fragment thereof according to [5], which is scFv.

[0236] [61] The molecule according to any of [18] to [21], wherein the antibody that binds specifically to CD3 or the binding fragment thereof is Fab.

[0237] [62] The molecule according to any of [56] to [61], wherein the Fab that binds specifically to CD3 is the antibody that binds specifically to CD3 or the binding fragment thereof according to [18], which is Fab.

[0238] [63] The molecule according to any of [56] to [61], wherein the Fab that binds specifically to CD3 is the antibody that binds specifically to CD3 or the binding fragment thereof according to [19], which is Fab.

[0239] [64] The molecule according to any of [56] to [61], wherein the Fab that binds specifically to CD3 is the antibody that binds specifically to CD3 or the binding fragment thereof according to [20] or [21], which is Fab.

[0240] [65] The molecule according to any of [55] to [64], wherein the first polypeptide comprises an amino acid sequence of amino acids 21 to 394 of the amino acid sequence as shown in SEQ ID NO: 160.

[0241] [66] The molecule according to any of [55] to [65], wherein the first polypeptide comprises the amino acid sequence according to any one of (i) to (iii) below:

[0242] (i) an amino acid sequence of amino acids 20 to 724 of the amino acid sequence as shown in SEQ ID NO: 160;

[0243] (ii) an amino acid sequence of amino acids 20 to 719 of the amino acid sequence as shown in SEQ ID NO: 197; and

[0244] (iii) an amino acid sequence of amino acids 20 to 719 of the amino acid sequence as shown in SEQ ID NO: 198.

[0245] [67] The molecule according to any of [55] to [66], wherein the second polypeptide comprises an amino acid sequence of amino acids 20 to 246 of the amino acid sequence as shown in SEQ ID NO: 84.

[0246] [68] The molecule according to any of [56] to [67], wherein the third polypeptide comprises an amino acid sequence of amino acids 21 to 127 of the amino acid sequence as shown in SEQ ID NO: 161.

[0247] [69] The molecule according to any of [56] to [68], wherein the third polypeptide comprises an amino

acid sequence of amino acids 21 to 233 of the amino acid sequence as shown in SEQ ID NO: 161.

[0248] [70] The molecule according to any of [15] to [69], wherein 1 or 2 amino acids are deleted from the carboxyl terminus of the amino acid sequence of at least one polypeptide included in the molecule.

[0249] [71] A polynucleotide comprising a nucleotide sequence encoding an amino acid sequence included in the molecule according to any of [15] to [70].

[0250] [72] A vector comprising the polynucleotide according to [71].

[0251] [73] A host cell comprising the polynucleotide according to [71] or the vector according to [72].

[0252] [74] A method for producing a molecule that binds specifically to human HLA/NY-ESO and human CD3 comprising: (i) a step of culturing the host cell according to [73]; and (ii) a step of purifying an antibody or a binding fragment thereof from the culture product obtained in the step (i).

[0253] [75] A molecule that binds specifically to human HLA/NY-ESO and human CD3, which is obtained by the method according to [74].

[0254] [76] A pharmaceutical composition comprising, as an active ingredient, the molecule according to any of [15] to [70] and [75], the polynucleotide according to [71], the vector according to [72], or the host cell according to [73].

[0255] [77] The pharmaceutical composition according to [14] or [76], which is an anti-cancer agent.

[0256] [78] The pharmaceutical composition according to [77], wherein cancer is one or more cancers selected from the group consisting of renal cancer, melanoma, squamous cell carcinoma, basal cell cancer, conjunctival cancer, oral cavity cancer, laryngeal cancer, pharyngeal cancer, thyroid gland cancer, lung cancer (non-small cell lung cancer (adenocarcinoma, epidermoid cancer, large cell cancer), and small cell lung cancer), breast cancer, esophageal cancer, gastric cancer, duodenal carcinoma, small bowel cancer, large bowel cancer, rectal cancer, appendiceal cancer, anal cancer, liver cancer, gallbladder cancer, bile duct cancer, pancreatic cancer, adrenal cancer, bladder cancer, prostate cancer, uterine cancer, vaginal cancer, liposarcoma, angiosarcoma, chondrosarcoma, rhabdomyosarcoma, Ewing's sarcoma, osteosarcoma, undifferentiated pleomorphic sarcoma, myxofibrosarcoma, malignant peripheral nerve sheath tumor, retroperitoneal sarcoma, synovial sarcoma, uterine sarcoma, gastrointestinal stromal tumor, leiomyosarcoma, epithelioid sarcoma, B cell lymphoma, T/NK-cell lymphoma, Hodgkin's lymphoma, myelogenous leukemia, lymphatic leukemia, myeloproliferative disorder, myelodysplastic syndrome, multiple myeloma, testicular carcinoma, and ovarian cancer.

[0257] [79] The pharmaceutical composition according to any of [76] to [78], which is used in combination with another agent.

Advantageous Effects of Invention

[0258] According to the present invention, an antibody that binds to HLA-A2/NY-ESO and a novel bispecific antibody (bispecific molecule) that binds to HLA-A2/NY-ESO and to CD3 is obtained. Also, a novel pharmaceutical composition comprising, as an active ingredient, such an

antibody (molecule) is obtained. Such antibody or molecule has cytotoxicity and is thus useful as an agent for treatment or prevention of cancer and the like.

BRIEF DESCRIPTION OF DRAWINGS

[0259] FIG. 1 shows a table demonstrating standardized gMFI of anti-HLA/NY-ESO scFv: NYA-0001, NYA-1143, NYA-1154, NYA-1163, NYA-2023, NYA-2027, NYA-2035, NYA-2044, NYA-2045, NYA-2047, NYA-2048, NYA-2060, NYA-2061, NYA-2143, NYC-0003, and NYC-0004, relative to T2 cells supplemented with various point-mutant peptides. * Each underlined value indicates a half or lower than the standardized gMFI of each scFv relative to that of T2 cells supplemented with the NY-ESO peptide.

[0260] FIG. 2A shows a table demonstrating information concerning the selected homologous peptides. Binding affinity to HLA-A0201 predicted using NetMHCpan2.8 is indicated in terms of the 50% inhibitory concentration (IC₅₀).

[0261] FIG. 2B shows a table demonstrating standardized gMFI of anti-HLA/NY-ESO scFv: NYA-0001, NYA-1143, NYA-1154, NYA-1163, NYA-2023, NYA-2027, NYA-2035, NYA-2044, NYA-2045, NYA-2047, NYA-2048, NYA-2060, NYA-2061, NYA-2143, NYC-0003, and NYC-0004, relative to T2 cells supplemented with various homologous peptides. * Each underlined value indicates a value larger than the standardized gMFI of each scFv relative to that of T2 cells supplemented with DMSO.

[0262] FIG. 3 shows antibody formats demonstrated in the examples. (a) shows an scFv format in which the antibody H chain variable region (VH, described below) is ligated to the antibody L chain variable region (VL, described below) (both blank) with a linker. In the Examples, an anti-HLA-A2/NY-ESO scFv and an anti-CD3 scFv were used for evaluation. (b) shows an Fab format that comprises VH (blank), the antibody H chain constant region (CH1: checkered pattern), VL (blank), and the antibody L chain constant region (border lines). In the Examples, anti-HLA-A2/NY-ESO Fab and the like were used for evaluation. (c) shows a taFv format in which two types of scFv constructs (blank and positive slopes) are ligated to each other with a linker. In the Examples, taFv including anti-HLA-A2/NY-ESO scFv and anti-CD3 scFv were used for evaluation. (d) shows a taFv-heterodimer Fc-type format in which Fc (negative slopes) comprising a heterodimer-forming mutation introduced into a C-terminal site of taFv (the first polypeptide) is heterologously associated with another Fc (solid: the second polypeptide). In the example, the taFv-heterodimer Fc comprising anti-HLA-A2/NY-ESO scFv and anti-CD3 scFv was used for evaluation. (e) shows a taFv-Fab-heterodimer Fc-type format in which Fab is added to the taFv-heterodimer Fc-type format. In the example, taFv including anti-HLA-A2/NY-ESO scFv and anti-CD3 scFv and taFv-Fab-heterodimer-Fc including HLA-A2/NY-ESO Fab were used for evaluation. (f) shows the first polypeptide that is common between the taFv-heterodimer Fc-type and the taFv-Fab-heterodimer Fc-type. The first polypeptide comprises scFv that binds specifically to human HLA/NY-ESO, scFv that binds specifically to CD3, and the Fc region (i) in that order from the N terminus toward the C terminus. (g) shows the second polypeptide of the taFv-heterodimer Fc-type. The second polypeptide comprises a hinge region and the Fc region (ii). (h) shows the second polypeptide of the taFv-Fab-heterodimer Fc-type. The second polypeptide com-

prises an immunoglobulin heavy chain comprising a hinge region and the Fc region (ii). (i) shows the third polypeptide of the taFv-Fab-heterodimer Fc-type. The third polypeptide comprises an immunoglobulin light chain.

[0263] FIG. 4A demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYF-0016, NYF-0019, NYF-0022, NYF-0023, NYF-0027, NYF-0035, NYF-0044, and NYF-0047, exert cytotoxicity on human U266B1 cells endogenously expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0264] FIG. 4B demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYF-0016 and NYF-0058, exert cytotoxicity on human U266B1 cells endogenously expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0265] FIG. 4C demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYF-0023, NYF-0045, NYF-0048, NYF-0060, and NYF-0061, exert cytotoxicity on the endogenous human NY-ESO-expressing U266B1 cells endogenously expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0266] FIG. 4D demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYF-0016, NYF-0019, NYF-0022, NYF-0023, NYF-0027, NYF-0035, NYF-0044, and NYF-0047, exert cytotoxicity on human NCI-H1703 cells endogenously expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0267] FIG. 4E demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYF-0016 and NYF-0058, exert cytotoxicity on human NCI-H1703 cells endogenously expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0268] FIG. 4F demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYF-0023, NYF-0045, NYF-0048, NYF-0060, and NYF-0061, exert cytotoxicity on human NCI-H1703 cells endogenously expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0269] FIG. 4G demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYF-0016, NYF-0019, NYF-0022, NYF-0023, NYF-0027, NYF-0035, NYF-0044, and NYF-0047, do not exert cytotoxicity on human AGS cells endogenously non-expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0270] FIG. 4H demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYF-0016 and NYF-0058, do not exert cytotoxicity on human AGS cells endogenously non-expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0271] FIG. 4I demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYF-0023, NYF-0045, NYF-0048, NYF-0060, and NYF-0061, do not exert cytotoxicity on human AGS cells endogenously non-expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0272] FIG. 4J demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYF-0016, NYF-0019, NYF-0022, NYF-0023, NYF-0027, NYF-0035, NYF-0044, and NYF-0047, do not exert cytotoxicity on human CFPAC-1 cells endogenously non-expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0273] FIG. 4K demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYF-0016 and NYF-0058, do not exert cytotoxicity on human CFPAC-1 cells endogenously non-expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0274] FIG. 4L demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYF-0023, NYF-0045, NYF-0048, NYF-0060, and NYF-0061, do not exert cytotoxicity on human CFPAC-1 cells endogenously non-expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0275] FIG. 5A demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYF-0016, NYF-0019, NYF-0044, and NYF-0047, exert antitumor activity on human PBMC-transfected models. An error bar in the figure indicates the standard deviation (n=5, but only n=4 on Day 32 in the NYF-0044 group).

[0276] FIG. 5B demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYF-0016, NYF-0022, NYF-0023, NYF-0027, NYF-0035, and NYF-0058, exert antitumor activity on human PBMC-transfected models. An error bar in the figure indicates the standard deviation (n=5, n=6 only in the Vehicle control group).

[0277] FIG. 5C demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYF-0016, NYF-0045, NYF-0048, NYF-0060, and NYF-0061, exert antitumor activity on human PBMC-transfected models. An error bar in the figure indicates the standard deviation (n=5).

[0278] FIG. 6A shows antibody formats described in the examples. (a): Hybrid-type format: Fc (slopes) comprising a heterodimer-forming mutation and Fc (solid) comprising a heterodimer-forming mutation are added to the C terminal region of Fab and scFv, respectively, and the two Fc constructs are associated. In the example, a hybrid-type format comprising anti-HLA-A2/NY-ESO Fab and anti-CD3 scFv was used for evaluation. (b): Dual-type format: Fc (negative slopes) comprising a heterodimer-forming mutation and Fc (solid) comprising a heterodimer-forming mutation are added to the C terminal regions of two different types of scFv constructs and the two Fc constructs are heterologously associated. In the example, a dual-type format comprising anti-HLA-A2/NY-ESO scFv and anti-CD3 scFv was used for evaluation. (c): scFv-Fab-heterodimer Fc-type format: Fc (negative slopes) comprising a heterodimer-forming mutation is added to the C terminal region where scFv is ligated to Fab with a linker and associated with another Fc (solid). In the example, an scFv-Fab-heterodimer Fc-type format comprising anti-CD3 scFv (positive slopes) and anti-HLA-A2/NY-ESO Fab was used for evaluation. Also, the scFv-Fab-heterodimer Fc-type comprising anti-HLA-A2/NY-ESO scFv (positive slopes) and anti-CD3 Fab was used for evaluation.

[0279] FIG. 6B shows antibody formats described in the examples. (a): taFv-heterodimer Fc-type format: the same as FIG. 3 (d); Fe (negative slopes) comprising a heterodimer-forming mutation is added to the C terminal region of taFv and heterologously associated with another Fe (solid). In the example, the taFv-heterodimer Fc-type comprising anti-HLA-A2/NY-ESO scFv and anti-CD3 scFv was used for evaluation. (b): taFv (inversed)-heterodimer Fc-type format: the order of the two scFv constructs is inverted in the taFv-heterodimer Fc-type format. In the example, taFv (inversed)-heterodimer Fc-type comprising anti-CD3 scFv and anti-HLA-A2/NY-ESO scFv was used for evaluation. (c) shows the first polypeptide of the taFv (inversed)-heterodimer Fc-type. The first polypeptide comprises scFv that binds specifically to CD3, scFv that binds specifically to human HLA/NY-ESO, and the Fc region (i) in that order from the N terminus toward the C terminus. (d) shows the second polypeptide of the taFv (inversed)-heterodimer Fc-type. The second polypeptide comprises a hinge region and the Fc region (ii).

[0280] FIG. 7A demonstrates that various anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; Hybrid-type (NYG-3143), Dual-type (NYG-2143), and taFv-heterodimer Fc-type (NYF-0011), exert cytotoxicity on human U266B1 cells endogenously expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0281] FIG. 7B demonstrates that various anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; scFv-Fab-heterodimer Fc-type (NYF-0003), taFv-heterodimer Fc-type (NYF-0010), taFv (inversed)-heterodimer Fc-type (NYF-0004), exert cytotoxicity on human U266B1 cells endogenously expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0282] FIG. 8 The amino acid sequence of the peptide in NY-ESO (SEQ ID NO: 1)

[0283] FIG. 9 The amino acid sequence of the peptide in MAGEC-1 (SEQ ID NO: 2)

[0284] FIG. 10 The scFv sequence analysis primer 1 (SEQ ID NO: 3)

[0285] FIG. 11 The scFv sequence analysis primer 2 (SEQ ID NO: 4)

[0286] FIG. 12 The nucleotide sequence of the heavy chain variable region of NYA-0001 (SEQ ID NO: 5)

[0287] FIG. 13 The amino acid sequence of the heavy chain variable region of NYA-0001 (SEQ ID NO: 6)

[0288] FIG. 14 The nucleotide sequence of the light chain variable region of NYA-0001 (SEQ ID NO: 7)

[0289] FIG. 15 The amino acid sequence of the light chain variable region of NYA-0001 (SEQ ID NO: 8)

[0290] FIG. 16 The nucleotide sequence of the heavy chain variable region of NYA-0060 (SEQ ID NO: 9)

[0291] FIG. 17 The amino acid sequence of the heavy chain variable region of NYA-0060 (SEQ ID NO: 10)

[0292] FIG. 18 The nucleotide sequence of the light chain variable region of NYA-0060 (SEQ ID NO: 11)

[0293] FIG. 19 The amino acid sequence of the light chain variable region of NYA-0060 (SEQ ID NO: 12)

[0294] FIG. 20 The nucleotide sequence of the heavy chain variable region of NYA-0068 (SEQ ID NO: 13)

[0295] FIG. 21 The amino acid sequence of the heavy chain variable region of NYA-0068 (SEQ ID NO: 14)

[0296] FIG. 22 The nucleotide sequence of the light chain variable region of NYA-0068 (SEQ ID NO: 15)

[0297] FIG. 23 The amino acid sequence of the light chain variable region of NYA-0068 (SEQ ID NO: 16)

[0298] FIG. 24 The nucleotide sequence of the heavy chain variable region of NYA-0082 (SEQ ID NO: 17)

[0299] FIG. 25 The amino acid sequence of the heavy chain variable region of NYA-0082 (SEQ ID NO: 18)

[0300] FIG. 26 The nucleotide sequence of the light chain variable region of NYA-0082 (SEQ ID NO: 19)

[0301] FIG. 27 The amino acid sequence of the light chain variable region of NYA-0082 (SEQ ID NO: 20)

[0302] FIG. 28 The nucleotide sequence of the NYA-1163 tag adduct (SEQ ID NO: 21)

[0303] FIG. 29 The nucleotide sequence of the NYA-2023 tag adduct (SEQ ID NO: 22)

[0304] FIG. 30 The nucleotide sequence of the NYA-2027 tag adduct (SEQ ID NO: 23)

[0305] FIG. 31 The nucleotide sequence of the NYA-1143 tag adduct (SEQ ID NO: 24)

[0306] FIG. 32 The nucleotide sequence of the NYA-2143 tag adduct (SEQ ID NO: 25)

[0307] FIG. 33 The amino acid sequence of the NYA-1163 tag adduct (SEQ ID NO: 26); NYA-1163: amino acids 21 to 266

[0308] FIG. 34 The amino acid sequence of the NYA-2023 tag adduct (SEQ ID NO: 27); NYA-2023: amino acids 21 to 266

[0309] FIG. 35 The amino acid sequence of the NYA-2027 tag adduct (SEQ ID NO: 28); NYA-2027: amino acids 21 to 266

[0310] FIG. 36 The amino acid sequence of the NYA-1143 tag adduct (SEQ ID NO: 29); NYA-1143: amino acids 21 to 266

[0311] FIG. 37 The amino acid sequence of the NYA-2143 tag adduct (SEQ ID NO: 30); NYA-2143: amino acids 21 to 266

[0312] FIG. 38 The nucleotide sequence of the NYA-1154 tag adduct (SEQ ID NO: 31)

[0313] FIG. 39 The amino acid sequence of the NYA-1154 tag adduct (SEQ ID NO: 32); NYA-1154: amino acids 21 to 266

[0314] FIG. 40 The amino acid sequence of HLA-A*0201 (GenBank: ASA47534.1) truncate (SEQ ID NO: 33)

[0315] FIG. 41 The amino acid sequence of 02-microglobulin (SEQ ID NO: 34)

[0316] FIG. 42 The nucleotide sequence of the NYA-2035 tag adduct (SEQ ID NO: 35)

[0317] FIG. 43 The amino acid sequence of the NYA-2035 tag adduct (SEQ ID NO: 36); NYA-2035: amino acids 21 to 266

[0318] FIG. 44 The amino acid sequence of NYA-1143-VH01 (SEQ ID NO: 37)

[0319] FIG. 45 The amino acid sequence of NYA-1143-VH02 (SEQ ID NO: 38)

[0320] FIG. 46 The amino acid sequence of NYA-1143-VH03 (SEQ ID NO: 39)

[0321] FIG. 47 The amino acid sequence of NYA-1143-VL01 (SEQ ID NO: 40)

[0322] FIG. 48 The nucleotide sequence of the NYA-2044 tag adduct (SEQ ID NO: 41)

[0323] FIG. 49 The nucleotide sequence of the NYA-2045 tag adduct (SEQ ID NO: 42)

- [0324] FIG. 50 The nucleotide sequence of the NYA-2047 tag adduct (SEQ ID NO: 43)
- [0325] FIG. 51 The nucleotide sequence of the NYA-2048 tag adduct (SEQ ID NO: 44)
- [0326] FIG. 52 The nucleotide sequence of the NYA-2060 tag adduct (SEQ ID NO: 45)
- [0327] FIG. 53 The nucleotide sequence of the NYA-2061 tag adduct (SEQ ID NO: 46)
- [0328] FIG. 54 The amino acid sequence of the NYA-2044 tag adduct (SEQ ID NO: 47); NYA-2044: amino acids 21 to 266
- [0329] FIG. 55 The amino acid sequence of the NYA-2045 tag adduct (SEQ ID NO: 48); NYA-2045: amino acids 21 to 266
- [0330] FIG. 56 The amino acid sequence of the NYA-0082 (SEQ ID NO: 49)
- [0331] FIG. 57 The amino acid sequence of the NYA-2047 tag adduct (SEQ ID NO: 50); NYA-2047: amino acids 21 to 266
- [0332] FIG. 58 The amino acid sequence of the NYA-2048 tag adduct (SEQ ID NO: 51); NYA-2048: amino acids 21 to 266
- [0333] FIG. 59 The amino acid sequence of the NYA-2060 tag adduct (SEQ ID NO: 52); NYA-2060: amino acids 21 to 266
- [0334] FIG. 60 The amino acid sequence of the NYA-2061 tag adduct (SEQ ID NO: 53); NYA-2061: amino acids 21 to 266
- [0335] FIG. 61 The amino acid sequences of the CDRH1 to CDRH3 of the NYA-0001 heavy chain and CDRL1 to CDRL3 of the NYA-0001 light chain (SEQ ID NOs: 54 to 59)
- [0336] FIG. 62 The amino acid sequence of CDRL1 of NYA-2023 (SEQ ID NO: 60)
- [0337] FIG. 63 The amino acid sequence of CDRL3 of NYA-2027 (SEQ ID NO: 61)
- [0338] FIG. 64 The amino acid sequences of CDRH3 and CDRL3 of NYA-1154 (SEQ ID NOs: 62 and 63)
- [0339] FIG. 65 The amino acid sequence of CDRL1 of NYA-0035 (SEQ ID NO: 64)
- [0340] FIG. 66 The nucleotide sequence of the NYC-0003 tag adduct (SEQ ID NO: 65)
- [0341] FIG. 67 The nucleotide sequence of the NYC-0004 tag adduct (SEQ ID NO: 66)
- [0342] FIG. 68 The amino acid sequence of the NYC-0003 tag adduct (SEQ ID NO: 67); NYC-0003: amino acids 21 to 263
- [0343] FIG. 69 The amino acid sequence of the NYC-0004 tag adduct (SEQ ID NO: 68); NYC-0004: amino acids 21 to 263
- [0344] FIG. 70 The nucleotide sequence of the NYA-0001 tag adduct (SEQ ID NO: 69)
- [0345] FIG. 71 The amino acid sequence of the NYA-0001 tag adduct (SEQ ID NO: 70); NYA-0001: amino acids 21 to 266
- [0346] FIG. 72 The nucleotide sequence of HC1 (SEQ ID NO: 71)
- [0347] FIG. 73 The nucleotide sequence of NYF-0016-HC2 (SEQ ID NO: 72)
- [0348] FIG. 74 The nucleotide sequence of NYF-0019-HC2 (SEQ ID NO: 73)
- [0349] FIG. 75 The nucleotide sequence of NYF-0022-HC2 (SEQ ID NO: 74)
- [0350] FIG. 76 The nucleotide sequence of NYF-0023-HC2 (SEQ ID NO: 75)
- [0351] FIG. 77 The nucleotide sequence of NYF-0027-HC2 (SEQ ID NO: 76)
- [0352] FIG. 78 The nucleotide sequence of NYF-0035-HC2 (SEQ ID NO: 77)
- [0353] FIG. 79 The nucleotide sequence of NYF-0044-HC2 (SEQ ID NO: 78)
- [0354] FIG. 80 The nucleotide sequence of NYF-0045-HC2 (SEQ ID NO: 79)
- [0355] FIG. 81 The nucleotide sequence of NYF-0047-HC2 (SEQ ID NO: 80)
- [0356] FIG. 82 The nucleotide sequence of NYF-0048-HC2 (SEQ ID NO: 81)
- [0357] FIG. 83 The nucleotide sequence of NYF-0060-HC2 (SEQ ID NO: 82)
- [0358] FIG. 84 The nucleotide sequence of NYF-0061-HC2 (SEQ ID NO: 83)
- [0359] FIG. 85 The amino acid sequence of HC1 (SEQ ID NO: 84)
- [0360] FIG. 86 The amino acid sequence of NYF-0016-HC2 (SEQ ID NO: 85); NYA-1143: amino acids 21 to 266; C3E-7085: amino acids 272 to 511
- [0361] FIG. 87 The amino acid sequence of NYF-0019-HC2 (SEQ ID NO: 86); NYA-2143: amino acids 21 to 266; C3E-7085: amino acids 272 to 511
- [0362] FIG. 88 The amino acid sequence of NYF-0022-HC2 (SEQ ID NO: 87); NYA-1163: amino acids 21 to 266; C3E-7085: amino acids 272 to 511
- [0363] FIG. 89 The amino acid sequence of NYF-0023-HC2 (SEQ ID NO: 88); NYA-2023: amino acids 21 to 266; C3E-7085: amino acids 272 to 511
- [0364] FIG. 90 The amino acid sequence of NYF-0027-HC2 (SEQ ID NO: 89); NYA-2027: amino acids 21 to 266; C3E-7085: amino acids 272 to 511
- [0365] FIG. 91 The amino acid sequence of NYF-0035-HC2 (SEQ ID NO: 90); NYA-2035: amino acids 21 to 266; C3E-7085: amino acids 272 to 511
- [0366] FIG. 92 The amino acid sequence of NYF-0044-HC2 (SEQ ID NO: 91); NYA-2044: amino acids 21 to 266; C3E-7085: amino acids 272 to 511
- [0367] FIG. 93 The amino acid sequence of NYF-0045-HC2 (SEQ ID NO: 92); NYA-2045: amino acids 21 to 266; C3E-7085: amino acids 272 to 511
- [0368] FIG. 94 The amino acid sequence of NYF-0047-HC2 (SEQ ID NO: 93); NYA-2047: amino acids 21 to 266; C3E-7085: amino acids 272 to 511
- [0369] FIG. 95 The amino acid sequence of NYF-0048-HC2 (SEQ ID NO: 94); NYA-2048: amino acids 21 to 266; C3E-7085: amino acids 272 to 511
- [0370] FIG. 96 The amino acid sequence of NYF-0060-HC2 (SEQ ID NO: 95); NYA-2060: amino acids 21 to 266; C3E-7085: amino acids 272 to 511
- [0371] FIG. 97 The amino acid sequence of NYF-0061-HC2 (SEQ ID NO: 96); NYA-2061: amino acids 21 to 266; C3E-7085: amino acids 272 to 511
- [0372] FIG. 98 The nucleotide sequence of NYA-0001-Fab-HC1-k delete (SEQ ID NO: 97)
- [0373] FIG. 99 The nucleotide sequence of NYA-0001-LC (SEQ ID NO: 98)
- [0374] FIG. 100 The amino acid sequence of NYA-0001-Fab-HC1-k delete (SEQ ID NO: 99); NYA-0001 heavy chain variable region: amino acids 20 to 139

- [0375] FIG. 101 The amino acid sequence of NYA-0001-LC (SEQ ID NO: 100); NYA-0001 light chain variable region: amino acids 21 to 131
- [0376] FIG. 102 The nucleotide sequence of NYA-1143-Fab-HC1-k delete (SEQ ID NO: 101)
- [0377] FIG. 103 The nucleotide sequence of NYA-1143-LC (SEQ ID NO: 102)
- [0378] FIG. 104 The nucleotide sequence of C3E-7085-HC2-k deleteC (SEQ ID NO: 103)
- [0379] FIG. 105 The amino acid sequence of NYA-1143-Fab-HC1-k delete (SEQ ID NO: 104); the NYA-1143 heavy chain variable region: amino acids 20 to 139
- [0380] FIG. 106 The amino acid sequence of NYA-1143-LC (SEQ ID NO: 105); the NYA-1143 light chain variable region: amino acids 21 to 131
- [0381] FIG. 107 The amino acid sequence of C3E-7085-HC2-k delete (SEQ ID NO: 106); C3E-7085: amino acids 21 to 260
- [0382] FIG. 108 The nucleotide sequence of NYA-1143-HC1-k delete (SEQ ID NO: 107)
- [0383] FIG. 109 The amino acid sequence of NYA-1143-HC1-k delete (SEQ ID NO: 108); NYA-1143: amino acids 21 to 266
- [0384] FIG. 110 The nucleotide sequence of C3E-7085-NYA-1154-Fab-HC2-k delete (SEQ ID NO: 109)
- [0385] FIG. 111 The nucleotide sequence of NYA-1154-LC (SEQ ID NO: 110)
- [0386] FIG. 112 The nucleotide sequence of OAA-HC1-k delete (SEQ ID NO: 111)
- [0387] FIG. 113 The amino acid sequence of C3E-7085-NYA-1154-Fab-HC2-k delete (SEQ ID NO: 112); C3E-7085: amino acids 21 to 260; the NYA-1154 heavy chain variable region: amino acids 266 to 285
- [0388] FIG. 114 The amino acid sequence of NYA-1154-LC (SEQ ID NO: 113); the NYA-1154 light chain variable region: amino acids 21 to 131
- [0389] FIG. 115 The amino acid sequence of OAA-HC1-k delete (SEQ ID NO: 114)
- [0390] FIG. 116 The nucleotide sequence of NYF-0010-HC2-k delete (SEQ ID NO: 115)
- [0391] FIG. 117 The nucleotide sequence of NYF-0004-HC2-k delete (SEQ ID NO: 116)
- [0392] FIG. 118 The nucleotide sequence of NYF-0011-HC2-k delete (SEQ ID NO: 117)
- [0393] FIG. 119 The amino acid sequence of NYF-0010-HC2-k delete (SEQ ID NO: 118); NYA-1154: amino acids 21 to 266; C3E-7085: amino acids 272 to 511
- [0394] FIG. 120 The amino acid sequence of NYF-0004-HC2-k delete (SEQ ID NO: 119); C3E-7085: amino acids 21 to 260; NYA-1154: amino acids 272 to 511
- [0395] FIG. 121 The amino acid sequence of NYF-0011-HC2-k delete (SEQ ID NO: 120); NYA-1143: amino acids 21 to 266; C3E-7085: amino acids 272 to 511
- [0396] FIG. 122 The amino acid sequence of the point-mutant NY-ESO peptide 1F (SEQ ID NO: 121)
- [0397] FIG. 123 The amino acid sequence of the point-mutant NY-ESO peptide 2M (SEQ ID NO: 122)
- [0398] FIG. 124 The amino acid sequence of the point-mutant NY-ESO peptide 3A (SEQ ID NO: 123)
- [0399] FIG. 125 The amino acid sequence of the point-mutant NY-ESO peptide 4A (SEQ ID NO: 124)
- [0400] FIG. 126 The amino acid sequence of the point-mutant NY-ESO peptide 5A (SEQ ID NO: 125)
- [0401] FIG. 127 The amino acid sequence of the point-mutant NY-ESO peptide 6L (SEQ ID NO: 126)
- [0402] FIG. 128 The amino acid sequence of the point-mutant NY-ESO peptide 7F (SEQ ID NO: 127)
- [0403] FIG. 129 The amino acid sequence of the point-mutant NY-ESO peptide 8A (SEQ ID NO: 128)
- [0404] FIG. 130 The amino acid sequence of the point-mutant NY-ESO peptide 9A (SEQ ID NO: 129)
- [0405] FIG. 131 The amino acid sequence of the gp100 peptide (SEQ ID NO: 130)
- [0406] FIG. 132 The amino acid sequence of the homologous peptide DOLPP1 (SEQ ID NO: 131)
- [0407] FIG. 133 The amino acid sequence of the homologous peptide IL20RB (SEQ ID NO: 132)
- [0408] FIG. 134 The amino acid sequence of the homologous peptide PRKD2 (SEQ ID NO: 133)
- [0409] FIG. 135 The amino acid sequence of the homologous peptide CD163 (SEQ ID NO: 134)
- [0410] FIG. 136 The amino acid sequence of the homologous peptide P2RY8 (SEQ ID NO: 135)
- [0411] FIG. 137 The amino acid sequence of C3E-7034 (SEQ ID NO: 136)
- [0412] FIG. 138 The amino acid sequence of C3E-7036 (SEQ ID NO: 137)
- [0413] FIG. 139 The amino acid sequence of C3E-7085 (SEQ ID NO: 138)
- [0414] FIG. 140 The amino acid sequence of C3E-7088 (SEQ ID NO: 139)
- [0415] FIG. 141 The amino acid sequence of C3E-7093 (SEQ ID NO: 140)
- [0416] FIG. 142 The amino acid sequences of CDRH1 to CDRH3 of the C3E-7085 heavy chain and CDRL1 to CDRL3 of the C3E-7085 light chain (SEQ ID NOs: 141 to 146)
- [0417] FIG. 143 The amino acid sequence of C3E-7078 (SEQ ID NO: 147)
- [0418] FIG. 144 The nucleotide sequence of NYF-0014-HC2 (SEQ ID NO: 148)
- [0419] FIG. 145 The amino acid sequence of NYF-0014-HC2 (SEQ ID NO: 149); NYA-0001: amino acids 21 to 266; C3E-7085: amino acids 272 to 511
- [0420] FIG. 146 The amino acid sequence of NYF-0082-HC2 (SEQ ID NO: 150); NYA-0082: amino acids 21 to 266; C3E-7085: amino acids 272 to 511
- [0421] FIG. 147 The amino acid sequence of human CD3 P (SEQ ID NO: 151)
- [0422] FIG. 148 The nucleotide sequence of the entire NYZ-0038-HC2 (SEQ ID NO: 152)
- [0423] FIG. 149 The nucleotide sequence of the entire NYZ-0082-HC2 (SEQ ID NO: 153)
- [0424] FIG. 150 The nucleotide sequence of the entire NYZ-0083-HC2 (SEQ ID NO: 154)
- [0425] FIG. 151 The amino acid sequence of the entire NYZ-0038-HC2 (SEQ ID NO: 155); NYA-2061: amino acids 21 to 266; C3E-7096: amino acids 272 to 511
- [0426] FIG. 152 The amino acid sequence of the entire NYZ-0082-HC2 (SEQ ID NO: 156); NYA-3061: amino acids 21 to 271; C3E-7096: amino acids 277 to 516.
- [0427] FIG. 153 The amino acid sequence of the entire NYZ-0083-HC2 (SEQ ID NO: 157); NYA-3061: amino acids 21 to 271; C3E-7097: amino acids 277 to 516.
- [0428] FIG. 154 The nucleotide sequence of the full-length NYZ-1010-HC2 (SEQ ID NO: 158)

[0429] FIG. 155 The nucleotide sequence of the full-length C3E-7085-LC (SEQ ID NO: 159)

[0430] FIG. 156 The amino acid sequence of the full-length NYZ-1010-HC2 (SEQ ID NO: 160); NYA-3061: amino acids 21 to 271; the C3E-7085 heavy chain variable region: amino acids 277 to 394

[0431] FIG. 157 The amino acid sequence of the full-length C3E-7085-LC (SEQ ID NO: 161)

[0432] FIG. 158A FIG. 158A shows a table demonstrating standardized gMFI of the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; NYZ-0038, NYZ-0082, NYZ-0083, and NYZ-1010, relative to CD3e knockout T2 cells supplemented with various point-mutant peptides. * Each underlined value indicates a half or lower than the standardized gMFI of each Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecule, relative to CD3e knockout T2 cells supplemented with the NY-ESO peptide, and * each bold value indicates a quarter or lower of the standardized gMFI.

[0433] FIG. 158B shows a table demonstrating standardized gMFI of the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; NYZ-0038, NYZ-0082, NYZ-0083, and NYZ-1010, relative to CD3e knockout T2 cells supplemented with various homologous peptides. * Each underlined value indicates a value of each Fc-added anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecule larger than the standardized gMFI of CD3e knockout T2 cells supplemented with DMSO.

[0434] FIG. 159A FIG. 159A demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYZ-0038, NYZ-0082, and NYZ-0083, exert cytotoxicity on human U266B1 cells endogenously expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0435] FIG. 159B demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecule; i.e., NYZ-1010, exerts cytotoxicity on human U266B1 cells endogenously expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0436] FIG. 159C FIG. 159C demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYZ-0082 and NYZ-1010, exert cytotoxicity on human NCI-H1703 cells endogenously expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0437] FIG. 159D FIG. 159D demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYZ-0038 and NYZ-0083, exert cytotoxicity on human NCI-H1703 cells endogenously expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0438] FIG. 159E FIG. 159E demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYZ-0038, NYZ-0082, and NYZ-0083, do not exert cytotoxicity on human AGS cells endogenously expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0439] FIG. 159F FIG. 159F demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecule; i.e., NYZ-1010, does not exert cytotoxicity on human AGS cells endogenously non-expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0440] FIG. 159G FIG. 159G demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYZ-0082 and NYZ-1010, do not exert cytotoxicity on human CFPAC-1 cells endogenously non-expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0441] FIG. 159H FIG. 159H demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYZ-0038 and NYZ-0083, do not exert cytotoxicity on human CFPAC-1 cells endogenously non-expressing NY-ESO in the presence of human PBMC. An error bar in the figure indicates the standard deviation (n=3).

[0442] FIG. 160A FIG. 160A demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecule; i.e., NYZ-0038, exerts antitumor activity on human PBMC-transfected models. An error bar in the figure indicates the standard deviation (n=5).

[0443] FIG. 160B demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecule; i.e., NYZ-0082, exerts antitumor activity on human PBMC-transfected models. An error bar in the figure indicates the standard deviation (n=5).

[0444] FIG. 160C FIG. 160C demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecule; i.e., NYZ-0083, exerts antitumor activity on human PBMC-transfected models. An error bar in the figure indicates the standard deviation (n=5).

[0445] FIG. 160D FIG. 160D demonstrates that the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecule; i.e., NYZ-1010, exerts antitumor activity on human PBMC-transfected models. An error bar in the figure indicates the standard deviation (n=5).

[0446] FIG. 161 The amino acid sequence of the peptide linker (SEQ ID NO: 162)

[0447] FIG. 162 shows the polymer content (%) upon acid treatment of various Fc-conjugated anti-HLA/NY-ESO-anti-CD3 bispecific molecules.

[0448] FIG. 163 shows the evaluation results of solution stability of anti-HLA/NY-ESO scFv-heterodimer Fc.

[0449] FIG. 164 The nucleotide sequence of the full-length NYA-3061 (SEQ ID NO: 163)

[0450] FIG. 165 The amino acid sequence of the full-length NYA-3061 (SEQ ID NO: 164)

[0451] FIG. 166 The nucleotide sequence of the full-length NYC-0005 (SEQ ID NO: 165)

[0452] FIG. 167 The amino acid sequence of the full-length NYC-0005 (SEQ ID NO: 166)

[0453] FIG. 168 The nucleotide sequence of the full-length NYC-0006 (SEQ ID NO: 167)

[0454] FIG. 169 The amino acid sequence of the full-length NYC-0006 (SEQ ID NO: 168)

[0455] FIG. 170 The nucleotide sequence of the full-length NYC-0007 (SEQ ID NO: 169)

[0456] FIG. 171 The amino acid sequence of the full-length NYC-0007 (SEQ ID NO: 170)

[0457] FIG. 172 The nucleotide sequence of the full-length NYC-0008 (SEQ ID NO: 171)

[0458] FIG. 173 The amino acid sequence of the full-length NYC-0008 (SEQ ID NO: 172)

[0459] FIG. 174 The nucleotide sequence of the full-length NYC-0009 (SEQ ID NO: 173)

[0460] FIG. 175 The amino acid sequence of the full-length NYC-0009 (SEQ ID NO: 174)

[0461] FIG. 176 The nucleotide sequence of the full-length NYC-0010 (SEQ ID NO: 175)
 [0462] FIG. 177 The amino acid sequence of the full-length NYC-0010 (SEQ ID NO: 176)
 [0463] FIG. 178 The nucleotide sequence of the full-length HC-h (SEQ ID NO: 177)
 [0464] FIG. 179 The amino acid sequence of the full-length HC-h (SEQ ID NO: 178)
 [0465] FIG. 180 The nucleotide sequence of the full-length NYD-2047-HC-k (SEQ ID NO: 179)
 [0466] FIG. 181 The amino acid sequence of the full-length NYD-2047-HC-k (SEQ ID NO: 180)
 [0467] FIG. 182 The nucleotide sequence of the full-length NYD-2061-HC-k (SEQ ID NO: 181)
 [0468] FIG. 183 The amino acid sequence of the full-length NYD-2061-HC-k (SEQ ID NO: 182)
 [0469] FIG. 184 The nucleotide sequence of the full-length NYD-3061-HC-k (SEQ ID NO: 183)
 [0470] FIG. 185 The amino acid sequence of the full-length NYD-3061-HC-k (SEQ ID NO: 184)
 [0471] FIG. 186 The nucleotide sequence of the full-length NYC-0011-HC-k (SEQ ID NO: 185)
 [0472] FIG. 187 The amino acid sequence of the full-length NYC-0011-HC-k (SEQ ID NO: 186)
 [0473] FIG. 188 The nucleotide sequence of the full-length NYC-0012-HC-k (SEQ ID NO: 187)
 [0474] FIG. 189 The amino acid sequence of the full-length NYC-0012-HC-k (SEQ ID NO: 188)
 [0475] FIG. 190 The nucleotide sequence of the full-length NYC-0013-HC-k (SEQ ID NO: 189)
 [0476] FIG. 191 The amino acid sequence of the full-length NYC-0013-HC-k (SEQ ID NO: 190)
 [0477] FIG. 192 The nucleotide sequence of the full-length NYC-0014-HC-k (SEQ ID NO: 191)
 [0478] FIG. 193 The amino acid sequence of the full-length NYC-0014-HC-k (SEQ ID NO: 192)
 [0479] FIG. 194 The nucleotide sequence of the full-length NYC-0015-HC-k (SEQ ID NO: 193)
 [0480] FIG. 195 The amino acid sequence of the full-length NYC-0015-HC-k (SEQ ID NO: 194)
 [0481] FIG. 196 The nucleotide sequence of the full-length NYC-0016-HC-k (SEQ ID NO: 195)
 [0482] FIG. 197 The amino acid sequence of the full-length NYC-0016-HC-k (SEQ ID NO: 196)
 [0483] FIG. 198 The amino acid sequence of the full-length NYZ-1007-HC2 (SEQ ID NO: 197); NYA-2061: amino acids 21 to 266; the C3E-7085 heavy chain variable region: amino acids 272 to 389
 [0484] FIG. 199 The amino acid sequence of the full-length NYZ-1017-HC2 (SEQ ID NO: 198); NYA-2047: amino acids 21 to 266; the C3E-7085 heavy chain variable region: amino acids 277 to 389

DESCRIPTION OF EMBODIMENTS

[0485] Hereafter, the present invention is described in detail.

1. Definition

[0486] In the present invention, the term “gene” refers to a nucleotide chain including a nucleotide sequence encoding an amino acid of a protein or a complementary chain thereof. For example, a polynucleotide, oligonucleotide, DNA, mRNA, cDNA, cRNA, or the like, which is a nucleotide

chain including a nucleotide sequence encoding an amino acid of a protein or a complementary chain thereof, is within the scope of the “gene.” Such gene is a single-stranded, double-stranded, or triple or more-stranded nucleotide, and an association of a DNA chain and an RNA chain, a single nucleotide chain comprising both ribonucleotide (RNA) and deoxyribonucleotide (DNA), and a double-stranded or triple-stranded nucleotide comprising such a nucleotide chain are included within the scope of the “gene.” In the present invention, the term “base sequence” is synonymous with the term “nucleotide sequence.”

[0487] In the present invention, the terms “polynucleotide,” “nucleotide chain,” “nucleic acid,” and “nucleic acid molecule” are synonymous. For example, DNA, RNA, probe, oligonucleotide, and primer are within the scope of “polynucleotide.” Such a polynucleotide consists of 1, 2, 3, or more chains, and an association of a DNA chain and an RNA chain, a single polynucleotide chain comprising both ribonucleotide (RNA) and deoxyribonucleotide (DNA), and a double-stranded or triple-stranded nucleotide comprising such a polynucleotide chain are included within the scope of the “polynucleotide.”

[0488] In the present invention, the terms “polypeptide,” “peptide,” and “protein” are synonymous.

[0489] In the present invention, the term “antigen” may refer to the term “immunogen,” according to need.

[0490] In the present invention, the term “cell” encompasses various cells derived from animals, subculture cells, primary culture cells, cell lines, recombinant cells, and microorganisms.

[0491] In the present invention, the term “antibody” is synonymous with the term “immunoglobulin.” However, the term “antibody” in the case of the anti-HLA/NY-ESO antibody according to the present invention refers to immunoglobulin comprising a constant region and a variable region. An antibody is not particularly limited, and it may be a naturally occurring or partially or completely synthesized immunoglobulin. The anti-HLA/NY-ESO antibody according to the present invention is included in the term “molecule” described below.

[0492] In the present invention, “NY-ESO peptide” indicates a peptide consisting of the 9 amino acids 157 to 165 of NY-ESO-1 and LAGE-1 (SLLMWITQC; SEQ ID NO: 1).

[0493] In the present invention, “HLA-A2/NY-ESO” indicates a complex of NY-ESO peptide and Histocompatibility Leukocyte Antigen-A2 (HLA-A2), and it is also denoted as “HLA/NY-ESO.”

[0494] In the present invention, the term “anti-HLA-A2/NY-ESO antibody” refers to an antibody that binds to HLA-A2/NY-ESO. In other words, the term refers to an antibody that recognizes HLA-A2/NY-ESO. Also, the term “anti-HLA-A2/NY-ESO scFv” refers to scFv that binds to HLA/NY-ESO. In other words, the term refers to scFv that recognizes HLA-A2/NY-ESO. The terms “anti-HLA-A2/NY-ESO antibody” and “anti-HLA-A2/NY-ESO scFv” are also denoted as the “anti-HLA/NY-ESO antibody” and the “anti-HLA/NY-ESO scFv,” respectively.

[0495] A basic four-chain antibody structure is composed of two identical light chains (L chains) and two identical heavy chains (H chains). A light chain binds to a heavy chain via a single covalent disulfide bond. Two heavy chains are bound to each other via one or more disulfide bonds in accordance with heavy chain isotypes. A light chain and a heavy chain each have an intra-chain disulfide bond at

regular intervals. In a light chain and a heavy chain, there are a constant region exhibiting very high amino acid sequence similarity and a variable region exhibiting low amino acid sequence similarity. A light chain comprises, at its amino terminus, a variable region (VL) adjacent to a constant region (CL). A heavy chain comprises, at its amino terminus, a variable region (VH) adjacent to 3 constant regions (CH1/CH2/CH3). VL is paired with VH, and CL is aligned with a first constant region of a heavy chain (CH1). VL is paired with VH to form a single antigen-binding site.

[0496] A constant region of the antibody of the present invention is not particularly limited. The antibody of the present invention to be used for treatment or prevention of human diseases preferably comprises a constant region of a human antibody. Examples of heavy chain constant regions of a human antibody include C γ 1, C γ 2, C γ 3, C γ 4, C μ , C δ , C α 1, C α 2, and C ϵ . Examples of light chain constant regions of a human antibody include C κ and C λ .

[0497] Fab comprises a heavy chain VH, CH1 adjacent thereto, a light chain VL, and CL adjacent thereto. VH and VL each comprise a complementarity determining region (CDR).

[0498] Fc (also referred to as an “Fc region”) is a carboxyl terminal region of a heavy chain constant region, it comprises CH2 and CH3, and it is a dimer. Fc of the present invention may comprise a naturally occurring sequence or it may comprise a sequence derived from the naturally occurring sequence by mutation (referred to as “mutant Fc”). In the multispecific molecule and the bispecific molecule of the present invention, a Fc region is preferably mutant Fc, and more preferably a combination of Fc regions capable of forming a heterodimer. An example of a combination of Fc regions is a combination of Fc (i) in the first polypeptide and Fc (ii) in the second polypeptide described below. A combination is not limited thereto, provided that such combination of Fc regions is capable of association (formation of a heterodimer).

[0499] Examples of mutant Fc include, but are not limited to, a modified Fc region included in a heteropolymer with improved stability (including a heterodimer Fc region) disclosed in WO 2013/063702, Fc including an immunoglobulin CD3 region induced from the IgG antibody with a “protrusion” and a “gap” included in a heteropolymer disclosed in WO 1996/27011, Fc including a CH3 domain included in a heterodimer that becomes electrostatically advantageous via substitution of one or more amino acids with charged amino acids disclosed in WO 2009/089004, a heterodimer Fc region included in a heterodimer involving steric mutation and/or pI (isoelectric point) mutation disclosed in WO 2014/110601, and a heterodimer Fc including a CH3 domain with a modification to eliminate or reduce the binding to protein A disclosed in WO 2010/151792.

[0500] A variable region is composed of a region with an extreme variability referred to as a hypervariable region (HVR) and relatively invariable regions referred to as framework regions (FRs) divided by the HVR. Naturally occurring heavy chain and light chain variable regions comprise 4 FRs connected by 3 hypervariable regions, a hypervariable region of each chain and a hypervariable region of other chains being maintained very close thereto, and such regions contribute to formation of an antigen-binding site of an antibody.

[0501] A heavy chain and a light chain of an antibody molecule are known to comprise 3 complementarity deter-

mining regions (CDRs). A complementarity determining region is also referred to as a hypervariable region, it is present within variable regions of a heavy chain and a light chain of the antibody where variability of a primary structure is particularly high, and, in general, it is separated in 3 positions in the primary structure of a polypeptide chain of a heavy chain and a light chain. In the present invention, complementarity determining regions of a heavy chain of an antibody are denoted as CDRH1, CDRH2, and CDRH3 from the amino terminus of the heavy chain amino acid sequence, and complementarity determining regions of a light chain are denoted as CDRL1, CDRL2, and CDRL3 from the amino terminus of the light chain amino acid sequence. These regions are adjacent to each other on a steric structure and determine specificity to the antigens to which they bind.

[0502] In the present invention, the position and the length of CDR were determined in accordance with the definition of IMGT (Developmental and Comparative Immunology 27, 2003, 55-77).

[0503] FR is a variable region other than CDR. In general, a variable region comprises 4 FRs; i.e., FR1, FR2, FR3, and FR4.

[0504] CDRs and FRs included in the heavy chain and in the light chain are provided in the orders of FRH1-CDRH1-FRH2-CDRH2-FRH3-CDRH3-FRH4 and FRL1-CDRL1-FRL2-CDRL2-FRL3-CDRL3-FRL4, respectively, from the amino terminus toward the carboxyl terminus.

[0505] CDR and FR positions can be determined in accordance with various definitions well known in the art, such as the definitions of Kabat, Chothia, AbM, contact, in addition to IMGT.

[0506] In the present invention, the term “an antigen-binding fragment of the antibody” refers to a partial fragment of an antibody having the activity of binding to an antigen, which is composed of a heavy chain variable region and a light chain variable region. Examples of “an antigen-binding fragment of the antibody” include, but are not limited to, antigen-binding fragments, such as Fab, F(ab')₂, scFv, Fab', Fv, and single-domain antibody (sdAb). Such antigen-binding fragment of the antibody may be obtained by treating a full-length molecule of an antibody protein with an enzyme such as papain or pepsin or it may be a recombinant protein produced in an adequate host cell with the use of a recombinant gene. In the present invention, the term “a binding fragment of the antibody” is synonymous with the term “an antigen-binding fragment of the antibody.”

[0507] In the present invention, a “site” to which an antibody binds; i.e., a “site” that is recognized by an antibody, is a partial peptide or a partial higher-order structure of an antigen to which an antibody binds or which is recognized by the antibody.

[0508] In the present invention, such a site is referred to as an epitope or an antibody binding site. In the present invention, a “mutant antibody” refers to a polypeptide having an amino acid sequence derived from the amino acid sequence of the original antibody by substitution, deletion, or addition (“addition” encompasses “insertion”) (hereafter, collectively referred to as “mutation”) of amino acids and binding to HLA/NY-ESO of the present invention. The number of mutant amino acids in such a mutant antibody is 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 15, 20, 25, 30, 40, or 50. Such a mutant antibody is within the scope of the “antibody” of the present invention.

[0509] In the present invention, the term “several” in “one or several” indicates 2 to 10.

[0510] The term “molecule” used herein indicates a molecule comprising the antibody or the antigen-binding fragment of the antibody described above. In addition, the term “molecule” encompasses a multispecific molecule formed of an antibody or a plurality of antigen-binding fragments derived therefrom.

[0511] The term “multispecific molecule” used herein is not particularly limited, provided that such a molecule is capable of binding to a plurality of different epitopes on a molecule and/or different epitopes on two or more molecules. A multispecific molecule encompasses an antibody comprising a heavy chain variable region (VH) and a light chain variable region (VL). Examples of such a multispecific molecule include, but are not limited to, a full-length antibody molecule comprising two or more different types of heavy chains and light chains; i.e., an IgG-type multispecific molecule, and a molecule comprising an antigen-binding fragment consisting of two or more types of VLs and VHs; i.e., a molecule derived from Fab, Fab', Fv, scFv, sdAb, or the like in combination, such as tandem scFv, a diabody, a single chain diabody, or a triabody. In addition, a molecule generated by genetically or chemically ligating a protein without an immunoglobulin skeleton and capable of binding to an antigen to an antigen-binding fragment is within the scope of the multispecific molecule. In the present invention, such a multispecific molecule may be referred to as a “multispecific antibody” except for the case in which the molecule does not comprise an antibody or an antigen-binding fragment thereof, according to need.

[0512] The anti-HLA/NY-ESO antibody according to the present invention, an antigen-binding fragment of such an antibody, or the molecule according to the present invention have, for example, biological activity and physicochemical properties (which may also be referred to as physical properties). Specific examples thereof include various biological activity, such as cytotoxicity, ADCC activity, or antitumor activity (described below), and physical properties, such as antigen- or epitope-binding activity, stability during production or storage, and heat stability.

[0513] In the present invention, “hybridization under stringent conditions” is performed in a solution containing 5×SSC at 65° C., in an aqueous solution containing 2×SSC and 0.1% SDS at 65° C. for 20 minutes, in an aqueous solution containing 0.5×SSC and 0.1% SDS at 65° C. for 20 minutes, or in an aqueous solution containing 0.2×SSC and 0.1% SDS at 65° C. for 20 minutes under washing conditions or conditions equivalent thereto. SSC is an aqueous solution of 150 mM NaCl and 15 mM sodium citrate, and nx SSC is SSC of n-fold concentration.

[0514] In the present invention, “cell damage” is any form of a pathological change that has occurred in a cell. In addition to an external injury, the term “cell damage” refers to any damage caused on a cell structure or functions, such as DNA cleavage, formation of nucleotide dimers, chromosome cleavage, mitotic apparatus damage, or lowering in various enzyme activity. In the present invention, “cytotoxicity” is induction of the cell damage.

[0515] In the present invention, the term “antibody-dependent cellular cytotoxicity (ADCC)” refers to activity of NK cells to damage target cells such as tumor cells mediated by an antibody.

[0516] In the present invention, the term “T-cell-redirecting cytotoxicity” refers to induction of cytotoxicity mediated by a multispecific molecule comprising an anti-tumor antigen antibody and the anti-HLA/NY-ESO antibody. Specifically, an anti-tumor antigen antibody binds to a target tumor cell, the anti-HLA/NY-ESO antibody binds to a T cell, the distance between the target tumor cell and the T cell is shortened, and cytotoxicity is induced by T cell activation. Such a molecule can be incorporated into a pharmaceutical composition.

2. Antigen

2-1. HLA/NY-ESO Antigen

[0517] In the present invention, “HLA/NY-ESO” is used in the same sense as “HLA/NY-ESO protein.”

[0518] HLA/NY-ESO is a triple complex of HLA-A2, β 2-microglobulin, and the NY-ESO peptide. HLA-A2 is an HLA allele, which is expressed at the highest frequency in Caucasians. HLA forms a triple complex with β 2-microglobulin and a peptide fragment of an autologous protein in the cell endoplasmic reticulum; the complex is presented extracellularly, and it is recognized by the T cell receptor (TCR) of the T cell. The NY-ESO peptide (SLLMWITQC; SEQ ID NO: 1, FIG. 8) is reported to be a peptide consisting of 9 amino acids at positions 157 to 165 of NY-ESO-1 and LAGE-1, which is presented to HLA-A2.

2-2. CD3 Antigen

[0519] In the present invention, “CD3” is used in the same sense as the “CD3 protein.”

[0520] CD3 is expressed on a T cell as a part of a multimolecular T cell receptor complex, and it is a complex of 5 types of polypeptide chains γ , δ , ϵ , ζ , and η (the molecular weights thereof are 25000-28000, 21000, 20000, 16000, and 22000, respectively).

[0521] Examples of CD3 complexes include γ , δ , ϵ , ζ , and η chains, which are also referred to as subunits. When the anti-CD3 antibody binds to a T cell, cell damage is induced by T cell activation. Many anti-CD3 antibodies bind to CD3 ϵ .

[0522] The nucleotide sequence of cDNA encoding human CD3 ϵ is registered under Accession Number: NM_000733 (NM_000733.3) at NCBI/GenBank, and the amino acid sequence of human CD3 ϵ is registered under Accession Number: NP_000724 (NP_000724.1) at NCBI/GenPept. The nucleotide sequence of cDNA encoding cynomolgus monkey CD3 is registered under Accession Number: NM_001283615.1 at GenBank. The amino acid sequence of human CD3 ϵ is shown in SEQ ID NO: 151 in the Sequence Listing (FIG. 147).

2-3. Preparation of Antigen

[0523] The antigen proteins used in the present invention described above; i.e., HLA/NY-ESO and CD3 (hereafter, HLA/NY-ESO and CD3 are collectively referred to as the antigen proteins) can be prepared from animal tissue (including body fluid), cells derived from the animal tissue, or a culture product of the cells via, for example, purification, isolation, gene recombination, in vitro translation, or chemical synthesis.

[0524] cDNA of the antigen protein can be obtained by a so-called PCR method in which a polymerase chain reaction

(hereafter referred to as “PCR”) using, as a template, the cDNA library of organs expressing mRNA of the antigen protein and primers specifically amplifying cDNA of the antigen protein (Saiki, R. K., et al., Science, 1988, 239, 487-49).

[0525] The cDNA of the antigen protein encompasses a polynucleotide hybridizing, under stringent conditions, to a polynucleotide consisting of a nucleotide sequence complementary to the nucleotide sequence encoding the antigen protein expressed in a human or rat and encoding a protein having biological activity equivalent to that of the antigen protein.

[0526] In addition, the cDNA of the antigen protein encompasses a splicing variant transcribed from the antigen protein locus expressed in a human or rat and a polynucleotide hybridizing, under stringent conditions, thereto, and encoding a protein having biological activity equivalent to that of the antigen protein.

[0527] A nucleotide sequence encoding a protein comprising the amino acid sequence of the human or rat antigen protein or an amino acid sequence derived from such amino acid sequence by substitution, deletion, or addition of one to several amino acids from which the signal sequence has been deleted and having biological activity equivalent to that of the antigen protein is included in the nucleotide sequence of the antigen protein gene.

[0528] A protein comprising an amino acid sequence encoded by the splicing variant transcribed from the human or rat antigen protein gene locus or an amino acid sequence derived from such an amino acid sequence by substitution, deletion, or addition of one or several amino acids and having biological activity equivalent to that of the antigen protein is included in the antigen protein.

2-4. Binding Specificity to Antigen Protein

[0529] The anti-HLA/NY-ESO antibody according to the present invention, an antigen-binding fragment thereof, and the like recognize HLA/NY-ESO. Specifically, they bind to the HLA/NY-ESO antigen. The presence of HLA/NY-ESO is not known in non-human animals, such as mice, rats, and cynomolgus monkeys.

[0530] The anti-CD3 antibody included in the multispecific molecule of the present invention, the binding fragments thereof, and the like recognize; i.e., bind to the CD3 antigen. Such anti-CD3 antibody and the like preferably bind to, for example, human CD3 and monkey CD3 and more preferably to human CD3 and cynomolgus monkey CD3. In contrast, such a preferable anti-CD3 antibody does not bind to rat and/or mouse CD3.

[0531] The anti-tumor activity of the multispecific molecule of the present invention can be evaluated by, for example, (i) transplanting human cancer cells or human cancer tissue into nonhuman animals into which human peripheral blood lymphocytes have been transplanted, preferably into rats or mice, and more preferably to rats or mice with deficient endogenous effector functions (e.g., immunodeficient rats or mice) or (ii) transplanting mouse cancer cells into which HLA and NY-ESO genes have been transduced into human CD3 gene knock-in nonhuman animals and preferably into rats or mice. By performing evaluation with the use of such immunodeficient animals or knock-in animals, various assays, immunohistochemical assays, and the like can be performed using mouse and/or rat bodies.

This is preferable for medicines containing the multispecific molecule of the present invention, nonclinical development, and other purposes.

[0532] In the present invention, “recognition;” i.e., “binding” is binding that is not non-specific adsorption. Whether or not the antibody recognizes; i.e., binds, can be evaluated on the basis of, for example, the dissociation constant (KD). A preferable KD value of the antibody and the like according to the present invention to HLA/NY-ESO or CD3 is 1×10^{-5} M or lower, 5×10^{-6} M or lower, 2×10^{-6} M or lower, or 1×10^{-6} M or lower, a KD value thereof to HLA/NY-ESO is preferably 5×10^{-7} M or lower, 2×10^{-7} M or lower, 1×10^{-7} M or lower, 5×10^{-8} M or lower, 2×10^{-8} M or lower, 1×10^{-8} M or lower, 5×10^{-9} M or lower, or 2×10^{-9} M or lower and more preferably 1×10^{-9} M or lower. Examples of the anti-HLA/NY-ESO scFv of the present invention having excellent antigen-binding activity include NYA-1143, NYA-2023, NYA-2143, NYA-2044, NYA-2045, NYA-2060, NYA-2061, and NYA-3061, and the KD value thereof to HLA/NY-ESO, such as NYA-1143, NYA-2044, NYA-2045, and NYA-2143, is 1×10^{-9} M or lower (e.g., Example 4).

[0533] In the present invention, antigen-antibody binding can be assayed or evaluated by, for example, the system of biomolecular interaction analysis, such as SPR or BLI, ELISA, or RIA. Binding between an antigen and an antibody expressed on a cell surface can be assayed by, for example, flow cytometry.

[0534] The method of surface plasmon resonance (SPR) analysis performs reaction kinetic analysis to determine the association rate constant (Ka value) and the dissociation rate constant (Kd value) and determine the dissociation constant (KD value) serving as an affinity index. Examples of apparatuses used for SPR analysis include Biacore™ (GE Healthcare), ProteOn™ (BioRad), SPR-Navi™ (BioNavis), Spreeta™ (Texas Instruments), SPRi-PlexII™ (Horiba Ltd.), and Autolab SPR™ (Metrohm).

[0535] The method of BioLayer Interferometry (BLI) performs assaying the interaction between biomolecules using biolayer interference. An example of an apparatus used for interaction analysis by the BLI method is the Octet system (Pall ForteBio).

[0536] The method of enzyme-linked immunosorbent assay (ELISA) performs detection and quantification of the target antigen or antibody contained in a sample solution by capturing the target antigen or antibody using a specific antibody or antigen, respectively, with the utilization of an enzyme reaction. The enzyme-labeled antigen or antibody is integrated into the reaction system to detect enzymatic activity. Enzymatic activity is detected using a substrate exhibiting the absorption spectra varying depending on reactions and the enzymatic activity is quantified based on absorption assay.

[0537] Cell-ELISA performs detection and quantification of the target analyte on a cell surface by capturing the target analyte together with the cell with the utilization of an enzyme reaction.

[0538] The method of radio immunoassay (RIA) performs labeling an antibody with a radioactive substance and assaying the radioactivity of the antibody. Thus, the antibody can be quantified.

[0539] In the method of flow cytometry, cells are dispersed in fluid, the fluid is allowed to flow narrowly, and each cell is optically analyzed. The antibody labeled with a fluorescent dye is allowed to bind to a cell surface antigen by the

antigen-antibody reaction, and the intensity of fluorescence emitted by the labeled antibody bound to the cell is assayed to quantify antigen-binding capability of the antibody.

[0540] Examples of the anti-HLA/NY-ESO antibodies according to the present invention exhibiting excellent antigen-binding specificity include anti-HLA/NY-ESO scFv, such as NYA-0001, NYA-1143, NYA-1163, NYA-2023, NYA-2027, NYA-2035, NYA-2044, NYA-2045, NYA-2047, NYA-2048, NYA-2060, NYA-2061, NYA-2143, and NYA-3061 (e.g., Example 6).

3. Antibody Binding Specifically to HLA/NY-ESO or the Binding Fragment Thereof

3-1. Anti-HLA/NY-ESO or the Binding Fragment Thereof

[0541] The present invention provides an antibody that recognizes and binds to HLA/NY-ESO or a binding fragment thereof.

[0542] As described above, HLA/NY-ESO is a complex comprising HLA-A2 and a 9-mer NY-ESO peptide (SLL-MWITQC: SEQ ID NO: 1). The NY-ESO peptide is a peptide derived from NY-ESO-1 or LAGE-1, which is an intracellular protein and a cancer testis antigen. HLA/NY-ESO is expressed on a cancer cell surface.

[0543] The anti-HLA/NY-ESO antibody according to the present invention and an antigen-binding fragment of the antibody (which hereafter, may be referred to as “the antibody and the like of the present invention”) may be monoclonal or polyclonal antibodies. In the present invention, an isotype of a monoclonal antibody isotype is not particularly limited, and examples include IgG such as IgG1, IgG2, IgG3 and IgG4, IgM, IgA such as IgA1 and IgA2, IgD, and Ig. An isotype and a subclass of a monoclonal antibody can be determined by, for example, the Ouchterlony method, ELISA, or RIA. Examples of the monoclonal antibodies of the present invention include an antibody derived from a non-human animal (a non-human animal antibody), a human antibody, a chimerized antibody (also referred to as a “chimeric antibody”), and a humanized antibody, with the human antibody being preferable. The antibody of the present invention encompasses a mutant of an antibody (the “mutant antibody” described below), and, for example, the human antibody encompasses a human mutant antibody.

[0544] Examples of non-human animal antibodies include antibodies derived from vertebrates, such as mammals and birds. Examples of mammalian-derived antibodies include antibodies derived from rodents, such as mouse antibody and rat antibody. An example of a bird-derived antibody is a chicken antibody.

[0545] Examples of chimerized antibodies include, but are not limited to, antibodies comprising a variable region derived from a non-human animal antibody bound to a constant region derived from a human antibody (human immunoglobulin).

[0546] Examples of humanized antibodies include, but are not limited to, a humanized antibody prepared by transplanting CDR in a variable region of a non-human animal antibody into a human antibody (a variable region of human immunoglobulin), a humanized antibody prepared by transplanting, in addition to CDR, a part of a sequence of a framework region of a non-human animal antibody into a human antibody, and a humanized antibody prepared by

substitution of 1 or more amino acids derived from a non-human animal antibody with amino acids derived from a human antibody.

[0547] An antibody can be prepared by a variety of known techniques. For example, an antibody can be prepared by a method involving the use of a hybridoma, cell-mediated immunity, or genetic recombination. Also, a phage-display-derived human antibody selected from a human antibody library can be obtained. In a phage display method, for example, a human antibody variable region may be expressed as scFv on a phage surface, and an antigen-binding phage may then be selected. The gene of the phage selected upon its binding to the antigen may be analyzed, so that a DNA sequence encoding a human antigen variable region binding to the antigen can be determined. If a DNA sequence of the antigen-binding scFv is elucidated, an expression vector comprising such sequence may be prepared, introduced into an adequate host cell, and expressed therein. Thus, a human antibody can be obtained (WO 92/01047, WO 92/20791, WO 93/06213, WO 93/11236, WO 93/19172, WO 95/01438, WO 95/15388, Annu. Rev. Immunol., 1994, 12, 433-455).

[0548] The thus-obtained antibody with high activity may be used as a lead antibody, and a gene encoding such a lead antibody may be mutated, so that a mutant with higher activity can be prepared (“the mutant antibody” described below).

[0549] A preferable combination of CDRH1 to CDRH3 included in the heavy chain of the anti-HLA/NY-ESO antibody according to the present invention or an antigen-binding fragment thereof is CDRH1 consisting of the amino acid sequence as shown in SEQ ID NO: 54 (FIG. 61), CDRH2 consisting of the amino acid sequence as shown in SEQ ID NO: 55 (FIG. 61), and CDRH3 consisting of the amino acid sequence as shown in SEQ ID NO: 56 (FIG. 61) or an amino acid sequence derived from the amino acid sequence as shown in SEQ ID NO: 56 (FIG. 61) in which amino acid 6 is N (Asn). A more preferable combination of CDRH1 to CDRH3 is included in the NYA-0001 heavy chain variable region consisting of the amino acid sequence as shown in SEQ ID NO: 6 (FIG. 13), the NYA-0082 heavy chain variable region consisting of the amino acid sequence as shown in SEQ ID NO: 18 (FIG. 25), the NYA-2023 heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 27 (FIG. 34), the NYA-2027 heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 28 (FIG. 35), the NYA-1143 heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 29 (FIG. 36), the NYA-1163 heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 26 (FIG. 33), the NYA-2023 heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 27 (FIG. 34), the NYA-2027 heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 28 (FIG. 35), the NYA-2035 heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 36 (FIG. 43), the NYA-2044 heavy

chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 47 (FIG. 54), the NYA-2045 heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 48 (FIG. 55), the NYA-2047 heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 50 (FIG. 57), the NYA-2048 heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 51 (FIG. 58), the NYA-2060 heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 52 (FIG. 59), the NYA-2061 heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 53 (FIG. 60), or the NYA-2143 heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 30 (FIG. 37).

[0550] A further example of a combination of CDRH1 to CDRH3 is included in the NYA-3061 heavy chain variable region consisting of an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 156 (FIG. 152).

[0551] A preferable combination of CDRL1 to CDRL3 included in the light chain of the anti-HLA/NY-ESO antibody according to the present invention or an antigen-binding fragment thereof is CDRL1 consisting of the amino acid sequence as shown in SEQ ID NO: 57 (FIG. 61) or an amino acid sequence derived from the amino acid sequence as shown in SEQ ID NO: 57 (FIG. 61) in which amino acid 7 is W (Trp) or amino acid 8 is K (Lys), CDRL2 consisting of the amino acid sequence as shown in SEQ ID NO: 58 (FIG. 61), and CDRL3 consisting of the amino acid sequence as shown in SEQ ID NO: 59 (FIG. 61) or an amino acid sequence derived from the amino acid sequence as shown in SEQ ID NO: 59 (FIG. 61) in which amino acid 2 is A (Ala) or S (Ser).

[0552] A more preferable combination of CDRL1 to CDRL3 is included in the NYA-0001 light chain variable region consisting of the amino acid sequence as shown in SEQ ID NO: 8 (FIG. 15), the NYA-0082 light chain variable region consisting of the amino acid sequence as shown in SEQ ID NO: 20 (FIG. 27), the NYA-1143 light chain variable region consisting of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 29 (FIG. 36), the NYA-1163 light chain variable region consisting of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 26 (FIG. 33), the NYA-2023 light chain variable region consisting of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 27 (FIG. 34), the NYA-2027 light chain variable region consisting of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 28 (FIG. 35), the NYA-2035 light chain variable region consisting of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 36 (FIG. 43), the NYA-2044 light chain variable region consisting of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 47 (FIG. 54), the NYA-2045 light chain variable region consisting of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 48 (FIG. 55), the NYA-2047 light chain variable region consisting of

amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 50 (FIG. 57), the NYA-2048 light chain variable region consisting of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 51 (FIG. 58), the NYA-2060 light chain variable region consisting of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 52 (FIG. 59), the NYA-2061 light chain variable region consisting of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 53 (FIG. 60), or the NYA-2143 light chain variable region consisting of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 30 (FIG. 37).

[0553] A further example of a combination of CDRL1 to CDRL3 is included in the NYA-3061 light chain variable region consisting of amino acids 161 to 271 of the amino acid sequence as shown in SEQ ID NO: 156 (FIG. 152).

[0554] A preferable combination of CDRH1 to CDRH3 in the heavy chain and CDRL1 to CDRL3 in the light chain of the anti-HLA/NY-ESO antibody according to the present invention or an antigen-binding fragment thereof is a combination of CDRH1 consisting of the amino acid sequence as shown in SEQ ID NO: 54 (FIG. 61), CDRH2 consisting of the amino acid sequence as shown in SEQ ID NO: 55 (FIG. 61), and CDRH3 consisting of the amino acid sequence as shown in SEQ ID NO: 56 (FIG. 61) or an amino acid sequence derived from the amino acid sequence as shown in SEQ ID NO: 56 (FIG. 61) in which amino acid 6 is N (Asn) and a combination of CDRL1 consisting of the amino acid sequence as shown in SEQ ID NO: 57 (FIG. 61) or an amino acid sequence derived from the amino acid sequence as shown in SEQ ID NO: 57 (FIG. 61) in which amino acid 7 is N (Asn) and/or amino acid 8 is K (Lys), CDRL2 consisting of the amino acid sequence as shown in SEQ ID NO: 58 (FIG. 61), and CDRL3 consisting of the amino acid sequence as shown in SEQ ID NO: 59 (FIG. 61) or an amino acid sequence derived from the amino acid sequence as shown in SEQ ID NO: 59 in which amino acid 2 is A (Ala) or S (Ser). A more preferable combination of CDRH1 to CDRH3 and CDRL1 to CDRL3 is included in the NYA-0001 heavy chain variable region and light chain variable region consisting of the amino acid sequences as shown in SEQ ID NO: 6 and SEQ ID NO: 8, the NYA-1143 heavy chain variable region and light chain variable region consisting of amino acids 21 to 140 and amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 29 (FIG. 36), the NYA-1163 heavy chain variable region and light chain variable region consisting of amino acids 21 to 140 and amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 26 (FIG. 33), the NYA-2023 heavy chain variable region and light chain variable region consisting of amino acids 21 to 140 and amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 27 (FIG. 34), the NYA-2027 heavy chain variable region and light chain variable region consisting of amino acids 21 to 140 and amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 28 (FIG. 35), the NYA-2035 heavy chain variable region and light chain variable region consisting of amino acids 21 to 140 and amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 36 (FIG. 43), the NYA-2044 heavy chain variable region and light chain variable region consisting of amino acids 21 to 140 and amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 47 (FIG. 54), the NYA-2045 heavy chain variable region and light chain variable region con-

sisting of amino acids 21 to 140 and amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 48 (FIG. 55), the NYA-2047 heavy chain variable region and light chain variable region consisting of amino acids 21 to 140 and amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 50 (FIG. 57), the NYA-2048 heavy chain variable region and light chain variable region consisting of amino acids 21 to 140 and amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 51 (FIG. 58), the NYA-2060 heavy chain variable region and light chain variable region consisting of amino acids 21 to 140 and amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 52 (FIG. 59), the NYA-2061 heavy chain variable region and light chain variable region consisting of amino acids 21 to 140 and amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 53 (FIG. 60), or the NYA-2143 heavy chain variable region and light chain variable region consisting of amino acids 21 to 140 and amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 30 (FIG. 37).

[0555] A further example of a combination of CDRH1 to CDRH3 and CDRL1 to CDRL3 is included in the NYA-3061 heavy chain variable region and light chain variable region consisting of amino acids 21 to 140 and amino acids 161 to 271 of the amino acid sequence as shown in SEQ ID NO: 156 (FIG. 152).

[0556] Preferable examples of a heavy chain variable region of the anti-HLA/NY-ESO antibody according to the present invention or an antigen-binding fragment thereof include the heavy chain CDRs described above and variable regions including such heavy chain CDRs in adequate combination. More preferable examples include the NYA-0001 heavy chain variable region, the NYA-0082 heavy chain variable region, the NYA-1143 heavy chain variable region, the NYA-1163 heavy chain variable region, the NYA-2023 heavy chain variable region, the NYA-2027 heavy chain variable region, the NYA-2035 heavy chain variable region, the NYA-2044 heavy chain variable region, the NYA-2045 heavy chain variable region, the NYA-2047 heavy chain variable region, the NYA-2048 heavy chain variable region, the NYA-2060 heavy chain variable region, the NYA-2061 heavy chain variable region, the NYA-2143 heavy chain variable region, and the NYA-3061 heavy chain variable region. The amino acid sequence of each heavy chain variable region is as described above.

[0557] Preferable examples of a light chain variable region of the anti-HLA/NY-ESO antibody according to the present invention or an antigen-binding fragment thereof include the light chain CDRs described above and variable regions including such light chain CDRs in adequate combination. More preferable examples include the NYA-0001 light chain variable region, the NYA-0082 light chain variable region, the NYA-1143 light chain variable region, the NYA-1163 light chain variable region, the NYA-2023 light chain variable region, the NYA-2027 light chain variable region, the NYA-2035 light chain variable region, the NYA-2044 light chain variable region, the NYA-2045 light chain variable region, the NYA-2047 light chain variable region, the NYA-2048 light chain variable region, the NYA-2060 light chain variable region, the NYA-2061 light chain variable region, the NYA-2143 light chain variable region, and the NYA-3061 light chain variable region. The amino acid sequence of each light chain variable region is as described above.

[0558] Preferable examples of a heavy chain variable region and a light chain variable region of the anti-HLA/NY-ESO antibody according to the present invention or an antigen-binding fragment thereof include the heavy chain and light chain CDRs described above or those including such CDRs in adequate combination. More preferable examples include the NYA-0001 heavy chain variable region and light chain variable region, the NYA-0082 heavy chain variable region and light chain variable region, the NYA-1143 heavy chain variable region and light chain variable region, the NYA-1163 heavy chain variable region and light chain variable region, the NYA-2023 heavy chain variable region and light chain variable region, the NYA-2027 heavy chain variable region and light chain variable region, the NYA-2035 heavy chain variable region and light chain variable region, the NYA-2044 heavy chain variable region and light chain variable region, the NYA-2045 heavy chain variable region and light chain variable region, the NYA-2047 heavy chain variable region and light chain variable region, the NYA-2048 heavy chain variable region and light chain variable region, the NYA-2060 heavy chain variable region and light chain variable region, the NYA-2061 heavy chain variable region and light chain variable region, the NYA-2143 heavy chain variable region and light chain variable region, and the NYA-3061 heavy chain variable region and light chain variable region.

[0559] Preferable examples of a heavy chain of the anti-HLA/NY-ESO antibody according to the present invention or an antigen-binding fragment thereof include heavy chains including the preferable or more preferable heavy chain variable regions described above. More preferable examples include the NYA-0001 heavy chain, the NYA-0082 heavy chain, the NYA-1143 heavy chain, the NYA-1163 heavy chain, the NYA-2023 heavy chain, the NYA-2027 heavy chain, the NYA-2035 heavy chain, the NYA-2044 heavy chain, the NYA-2045 heavy chain, the NYA-2047 heavy chain, the NYA-2048 heavy chain, the NYA-2060 heavy chain, the NYA-2061 heavy chain, the NYA-2143 heavy chain, and the NYA-3061 heavy chain.

[0560] Preferable examples of a light chain of the anti-HLA/NY-ESO antibody according to the present invention or an antigen-binding fragment thereof include light chains including the preferable or more preferable light chain variable regions described above. More preferable examples include the NYA-0001 light chain, the NYA-0082 light chain, the NYA-1143 light chain, the NYA-1163 light chain, the NYA-2023 light chain, the NYA-2027 light chain, the NYA-2035 light chain, the NYA-2044 light chain, the NYA-2045 light chain, the NYA-2047 light chain, the NYA-2048 light chain, the NYA-2060 light chain, the NYA-2061 light chain, the NYA-2143 light chain, and the NYA-3061 light chain.

[0561] Preferable examples of a heavy chain and a light chain of the anti-HLA/NY-ESO antibody according to the present invention or an antigen-binding fragment thereof include heavy chains and light chains including the preferable or more preferable heavy chain variable regions and light chain variable regions described above. More preferable examples include the NYA-0001 heavy chain and light chain, the NYA-1143 heavy chain and light chain, the NYA-1163 heavy chain and light chain, the NYA-2023 heavy chain and light chain, the NYA-2027 heavy chain and light chain, the NYA-2035 heavy chain and light chain, the NYA-2044 heavy chain and light chain, the NYA-2045

heavy chain and light chain, the NYA-2047 heavy chain and light chain, the NYA-2048 heavy chain and light chain, the NYA-2060 heavy chain and light chain, the NYA-2061 heavy chain and light chain, the NYA-2143 heavy chain and light chain, and the NYA-3061 heavy chain and light chain.

[0562] An antigen-binding fragment of the antibody is a fragment that retains at least the antigen-binding ability out of the functions of the original antibody or a modified product thereof. Examples of the functions of the antibody include, in general, antigen-binding activity, the activity of regulating antigen activity, antibody-dependent cellular cytotoxicity, and complement-dependent cytotoxicity. Examples of functions of the antibody and the like of the present invention and multispecific molecules comprising the antibody and the like of the present invention include T cell redirection, T cell activation, and cancer cytotoxicity caused by T cell activation.

[0563] An antigen-binding fragment of the antibody is not particularly limited, provided that such a fragment is derived from the antibody that retains at least antigen-binding ability out of the functions of the original antibody. Examples thereof include, but are not limited to, Fab, Fab', F(ab')₂, Fv, single chain Fv (scFv) comprising heavy chain Fv ligated to light chain Fv via an adequate linker, and the single domain antibody (sdAb). A molecule comprising a region other than the antigen-binding fragment of the antibody according to the present invention as in the case of scFv comprising a linker portion is within the scope of the antigen-binding fragment of the antibody of the present invention.

[0564] A molecule of the antibody protein lacking at least one or several amino acids from the amino terminus and/or carboxyl terminus and retaining some functions of the antibody is within the scope of the antigen-binding fragment of the antibody. A modified antigen-binding fragment of the antibody is within the scope of the antibody according to the present invention, an antigen-binding fragment thereof, or a modified antibody or fragment (described below).

[0565] An embodiment of the antibody of the present invention or an antigen-binding fragment thereof is scFv. scFv is obtained by ligating the heavy chain variable region of the antibody to the light chain variable region thereof with a polypeptide linker (Pluckthun A., *The Pharmacology of Monoclonal Antibodies* 113, Rosenberg and Moore (ed.), Springer Verlag, New York, 269-315, 1994, *Nature Biotechnology*, 2005, 23, 1126-1136). Also, tandem scFv prepared by connecting two scFv constructs with a polypeptide linker can be used as a bispecific molecule. In addition, a tribody or the like comprising 3 or more scFv constructs can be used as a multispecific molecule.

[0566] Preferable examples of HLA/NY-ESO-specific scFv (also referred to as "anti-HLA/NY-ESO scFv") include scFv comprising CDRH1 to CDRH3 and CDRL1 to CDRL3 described above, more preferable examples thereof include scFv comprising the heavy chain variable region and the light chain variable region, and further preferable examples thereof include NYA-0001 (amino acids 21 to 266 of the amino acid sequence as shown in SEQ ID NO: 70 (FIG. 71)), NYA-0082 (comprising the amino acid sequence as shown in SEQ ID NO: 18 (FIG. 25) and the amino acid sequence as shown in SEQ ID NO: 20 (FIG. 27)), NYA-1143 (amino acids 21 to 266 of the amino acid sequence as shown in SEQ ID NO: 29 (FIG. 36)), NYA-1163 (amino acids 21 to 266 of the amino acid sequence as shown in SEQ ID NO: 26 (FIG. 33)), NYA-2023 (amino acids 21 to 266 of

the amino acid sequence as shown in SEQ ID NO: 27 (FIG. 34)), NYA-2027 (amino acids 21 to 266 of the amino acid sequence as shown in SEQ ID NO: 28 (FIG. 35)), NYA-2035 (amino acids 21 to 266 of the amino acid sequence as shown in SEQ ID NO: 36 (FIG. 43)), NYA-2044 (amino acids 21 to 266 of the amino acid sequence as shown in SEQ ID NO: 47 (FIG. 54)), NYA-2045 (amino acids 21 to 266 of the amino acid sequence as shown in SEQ ID NO: 48 (FIG. 55)), NYA-2047 (amino acids 21 to 266 of the amino acid sequence as shown in SEQ ID NO: 50 (FIG. 57)), NYA-2048 (amino acids 21 to 266 of the amino acid sequence as shown in SEQ ID NO: 51 (FIG. 58)), NYA-2060 (amino acids 21 to 266 of the amino acid sequence as shown in SEQ ID NO: 52 (FIG. 59)), NYA-2061 (amino acids 21 to 266 of the amino acid sequence as shown in SEQ ID NO: 53 (FIG. 60)), and NYA-2143 (amino acids 21 to 266 of the amino acid sequence as shown in SEQ ID NO: 30 (FIG. 37)). Another example is NYA-3061 (amino acids 21 to 271 of the amino acid sequence as shown in SEQ ID NO: 156 (FIG. 152)).

[0567] A preferable embodiment of anti-HLA/NY-ESO scFv is scFv comprising a FLAG-His tag fused to its carboxyl terminus (may be simply referred to as "tag adduct"). Examples of preferable tag adducts include NYA-0001 tag adduct (amino acids 20 to 292 of SEQ ID NO: 70 (FIG. 71)), NYA-1143 tag adduct (amino acids 20 to 292 of SEQ ID NO: 29 (FIG. 36)), NYA-1163 tag adduct (amino acids 20 to 292 of SEQ ID NO: 26 (FIG. 33)), NYA-2023 tag adduct (amino acids 20 to 292 of SEQ ID NO: 27 (FIG. 34)), NYA-2027 tag adduct (amino acids 20 to 292 of SEQ ID NO: 28 (FIG. 35)), NYA-2035 tag adduct (amino acids 20 to 292 of SEQ ID NO: 36 (FIG. 43)), NYA-2044 tag adduct (amino acids 20 to 292 of SEQ ID NO: 47 (FIG. 54)), NYA-2045 tag adduct (amino acids 20 to 292 of SEQ ID NO: 48 (FIG. 55)), NYA-2047 tag adduct (amino acids 20 to 292 of SEQ ID NO: 50 (FIG. 57)), NYA-2048 tag adduct (amino acids 20 to 292 of SEQ ID NO: 51 (FIG. 58)), NYA-2060 tag adduct (amino acids 20 to 292 of SEQ ID NO: 52 (FIG. 59)), NYA-2061 tag adduct (amino acids 20 to 292 of SEQ ID NO: 53 (FIG. 60)), and NYA-2143 tag adduct (amino acids 20 to 292 of SEQ ID NO: 30 (FIG. 37)).

[0568] Of these, NYA-2023 and its tag adduct, NYA-2047 and its tag adduct, NYA-2048 and its tag adduct, NYA-2060 and its tag adduct, and NYA-2061 and its tag adduct exert excellent biological activity, physical properties, and the like in Fc-added bispecific molecules and thus are more preferable.

[0569] When anti-HLA/NY-ESO scFv and its tag adduct are to be expressed in a host cell, a signal peptide can be added to its amino terminus. Examples of amino acid sequences of signal-peptide-added anti-HLA/NY-ESO scFv tag adducts include amino acid sequences as shown in SEQ ID NOS: 70, 29, 26 to 28, 36, 47, 48, 50 to 53, and 30 (FIGS. 71, 36, 33 to 35, 43, 54, 55, 57 to 60, and 37).

[0570] scFv can be obtained by the phage display method in which an antibody variable region is expressed as a single chain antibody (scFv) on a phage surface and an antigen-binding phage is then selected (*Nature Biotechnology*, 2005, 23, (9), pp. 1105-1116). The gene of the phage selected upon its binding to the antigen may be analyzed, so that a DNA sequence encoding a human antigen variable region binding to the antigen can be determined. If a DNA sequence of the antigen-binding scFv is elucidated, an expression vector comprising such sequence may be prepared, introduced into

an adequate host cell, and expressed therein. Thus, a human antibody can be obtained (WO 92/01047, WO 92/20791, WO 93/06213, WO 93/11236, WO 93/19172, WO 95/01438, WO 95/15388, *Annu. Rev. Immunol.*, 1994, 12, pp. 433-455, *Nature Biotechnology*, 2005, 23 (9), pp. 1105-1116).

[0571] The antibody of the present invention may comprise a single heavy chain variable region and may not comprise a light chain sequence. Such an antibody is referred to as a single domain antibody (sdAb) or nanobody and retains antigen-binding ability (Muylldemans S. et al., *Protein Eng.*, 1994, 7 (9), 1129-35, Hamers-Casterman C. et al., *Nature*, 1993, 363 (6428), 446-448). Such antibodies are within the scope of the antigen-binding fragment of the antibody according to the present invention.

[0572] The present invention also includes a single chain immunoglobulin comprising the full-length sequence of the heavy chain ligated to that of the light chain of the antibody with an adequate linker (Lee, H-S, et al., *Molecular Immunology*, 1999, 36, 61-71; Shirmann, T. et al., *mAbs*, 2010, 2 (1), 1-4). Such single chain immunoglobulin may be dimerized, so that it can retain the structure and activity similar to those of an antibody that is inherently a tetramer. The anti-HLA/NY-ESO antibody according to the present invention may be single chain immunoglobulin.

[0573] In scFv of the present invention, a heavy chain variable region may form a disulfide bond with a light chain variable region.

[0574] The anti-HLA/NY-ESO antibody according to the present invention may be composed of components derived from a plurality of different antibodies, provided that it binds to HLA/NY-ESO. Examples thereof include those resulting from replacement of heavy chains and/or light chains among a plurality of different antibodies, those resulting from replacement of full-length sequences of heavy chains and/or light chains, those resulting from selective replacement of either a variable region or a constant region, and those resulting from selective replacement of a part of or the entire CDR. In a chimerized antibody, a heavy chain variable region and a light chain variable region may be derived from a different anti-HLA/NY-ESO antibody according to the present invention. In the heavy chain and light chain variable regions of a humanized antibody, heavy chain CDR1 to heavy chain CDR3 and light chain CDR1 to light chain CDR3 may be derived from two or more types of the anti-HLA/NY-ESO antibodies according to the present invention. In the heavy chain and light chain variable regions of a human antibody, a combination of heavy chain CDR1 to heavy chain CDR3 and light chain CDR1 to light chain CDR3 may be derived from two or more types of the anti-HLA/NY-ESO antibodies according to the present invention.

[0575] The anti-HLA/NY-ESO antibody according to the present invention encompasses an antibody that comprises an amino acid sequence encoded by a nucleotide sequence included in a polynucleotide hybridizing, under stringent conditions, to a complementary strand of a polynucleotide comprising a nucleotide sequence encoding the amino acid sequence included in the anti-HLA/NY-ESO antibody according to the present invention and binding to HLA/NY-ESO.

[0576] It may be an antibody comprising an amino acid sequence exhibiting at least 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% identity to the amino acid

sequence included in the heavy chain variable region of the anti-HLA/NY-ESO antibody according to the present invention (preferably, an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 27, the amino acid sequence as shown in SEQ ID NO: 38, the amino acid sequence as shown in SEQ ID NO: 39, or an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 160, SEQ ID NO: 197, or SEQ ID NO: 198) and/or an amino acid sequence included in a light chain variable region thereof (preferably, an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 27, an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 52, the amino acid sequence as shown in SEQ ID NO: 40, an amino acid sequence of amino acids 161 to 271 of the amino acid sequence as shown in SEQ ID NO: 160, or an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 197 or SEQ ID NO: 198) or an antigen-binding fragment thereof.

[0577] When the position and the length of the light chain variable region are determined in accordance with a definition that is different from IMGT (e.g., Kabat, Chothia, AbM, contact), one or more amino acids, such as arginine or glycine, may further be included into the carboxyl terminus of the amino acid sequence of the light chain variable region defined in accordance with the definition of IMGT. Such antibody or the binding fragment thereof comprising such light chain variable region is within the scope of the antibody according to the present invention or the binding fragment thereof.

[0578] The antibody and the like of the present invention may be prepared by introducing a mutation into a binding fragment of the anti-HLA/NY-ESO antibody according to the present invention and optimizing the ability to bind to HLA/NY-ESO, and, in particular, to human and/or cynomolgus monkey HLA/NY-ESO. Specific examples of methods for introduction of a mutation include a random mutagenesis using error-prone PCR, site-directed amino acid introduction using NHK libraries, site-directed mutagenesis using structural information, and a combination of any thereof.

3-2. Mutant of the Anti-HLA/NY-ESO Antibody (Mutant Antibody)

[0579] A mutant antibody of the anti-HLA/NY-ESO antibody according to the present invention can be preferably provided with, for example, lowered susceptibility to protein degradation or oxidation, maintained or improved biological activity or functions, suppression of lowering or change in such activity or functions, improved or regulated antigen-binding ability, physicochemical properties, or functional properties. A protein is known to change its functions or activity upon alternation of a particular amino acid side chain on its surface, and examples include deamidation of an asparagine side chain and isomerization of an aspartic acid side chain. An antibody resulting from substitution of a particular amino acid with another amino acid so as to prevent the amino acid side chain from changing is within the scope of the mutant antibody of the present invention.

[0580] An example of the mutant antibody of the present invention is an antibody comprising an amino acid sequence derived from the amino acid sequence of the original antibody by conservative amino acid substitution. Conservative

amino acid substitution occurs within an amino acid group associated with the amino acid side chain.

[0581] Preferable amino acid groups are as follows: the acidic group: aspartic acid and glutamic acid; the basic group: lysine, arginine, and histidine; the non-polar group: alanine, valine, leucine, isoleucine, proline, phenylalanine, methionine, and tryptophan; and the uncharged polar family: glycine, asparagine, glutamine, cysteine, serine, threonine, and tyrosine. Other preferable amino acid groups are as follows: the aliphatic hydroxy group: serine and threonine; the amide-containing group: asparagine and glutamine; the aliphatic group: alanine, valine, leucine, and isoleucine; and the aromatic group: phenylalanine, tryptophan, and tyrosine. In such mutant antibody, amino acid substitution is preferably carried out by refraining from lowering the antigen-binding activity of the original antibody.

[0582] The anti-HLA/NY-ESO antibody according to the present invention, an antigen-binding fragment thereof, a mutant thereof (a mutant antibody or an antigen-binding fragment thereof), or the molecule of the present invention encompasses: a mutant antibody comprising an amino acid sequence derived from the amino acid sequence of the antibody of the present invention, such as NYA-2023, by conservative amino acid substitution and/or other mutation and binding to HLA/NY-ESO, an antigen-binding fragment thereof, and molecules comprising the same; a chimerized antibody, a humanized antibody, or a human antibody comprising CDR of an amino acid sequence derived from the amino acid sequence of any of CDRH1 to CDRH3 and CDRL1 to CDRL3 derived from the antibody of the present invention, including NYA-2023, by conservative amino acid substitution and/or other mutation and binding to HLA/NY-ESO, an antigen-binding fragment thereof, or a molecule comprising the same.

3-3. Binding Fragment of the Anti-HLA/NY-ESO Antibody

[0583] An aspect of the present invention provides an antigen-binding fragment of the anti-HLA/NY-ESO antibody according to the present invention (hereafter, it is simply referred to as a “binding fragment”). Binding fragments of the anti-HLA/NY-ESO antibody according to the present invention include binding fragments of a chimerized antibody, a humanized antibody, and a human antibody. A binding fragment of an antibody is a fragment that retains at least antigen-binding ability of functions of the original antibody or a modified product thereof. In general, examples of such antibody functions include antigen-binding activity, the activity of regulating antigen activity (e.g., agonist activity), the activity of internalizing an antigen in a cell, and the activity of inhibiting or promoting interactions between an antigen and a substance interacting therewith.

[0584] A binding fragment of an antibody is not particularly limited, provided that it is a fragment of a fragment retaining at least antigen-binding ability of the activity of the original antibody. Examples of such binding fragment of an antibody include, but are not limited to, Fab, Fab', F(ab')₂, single chain Fab (scFab) in which the carboxyl terminus of the Fab light chain is ligated to the amino terminus of the Fab heavy chain via an adequate linker, Fv, single chain Fv (scFv) comprising heavy chain Fv ligated to light chain Fv via an adequate linker, and a single domain antibody (sdAb) with a single heavy chain variable region but without a light chain sequence, which is also referred to as a nanobody (Muylldemans S. et al., *Protein Eng.*, 1994, 7 (9), 1129-35,

Hamers-Casterman C. et al., *Nature*, 1993, 363 (6428), 446-448). A molecule comprising regions other than the binding fragment of the antibody of the present invention, such as scFab or scFv comprising a linker region, is within the scope of the binding fragment of the antibody of the present invention.

3-4. Modified Anti-HLA/NY-ESO Antibody, Modified Binding Fragment Thereof, or Complex Thereof

[0585] The present invention provides a modified antibody or a modified binding fragment thereof. The modified antibody according to the present invention or the modified binding fragment thereof has been subjected to chemical or biological modification. Examples of chemical modification include a bond of a chemical portion to the amino acid skeleton and chemical modification of N-bound or O-bound carbohydrate chains. Examples of biological modification include post-translational modification (e.g., sugar chain addition to an N-bond or O-bond, processing of the amino terminal or carboxyl terminal region, deamidation, aspartic acid isomerization, and methionine oxidation), and methionine addition to the amino terminus via expression in a prokaryotic host cell. Also, labels that enable detection or isolation of the antibody or antigen according to the present invention, such as an enzyme label, a fluorescence label, and an affinity label, are within the scope of the modified antibody or antigen as described above. The modified antibody according to the present invention or the binding fragment thereof as described above is useful for improvement of stability and retentivity in blood of the original antibody according to the present invention or the binding fragment thereof, reduction of the antigenicity, detection or isolation of the antibody or antigen, and other purposes.

[0586] Examples of chemical portions included in the chemical modified antibody or fragment include water-soluble polymers, such as polyethylene glycol (PEG), ethylene glycol/propylene glycol polymer, carboxymethyl cellulose, dextran, and polyvinyl alcohol.

[0587] Examples of biological modification include enzyme treatment, cell treatment, addition of other peptides such as tags via genetic recombination, and use of host cells expressing endogenous or exogenous sugar chain modified enzymes.

[0588] Such modification may be provided at any desired position in the antibody or the binding fragment thereof, and the same or two or more different types of modification may be provided at one or more positions.

[0589] However, deletion of such a heavy chain sequence or modification of a heavy chain or light chain sequence would not significantly affect the antigen-binding ability and effector functions of the antibody (e.g., complement activation or antibody-dependent cytotoxicity).

[0590] Accordingly, the present invention encompasses the antibody subjected to such deletion or modification. Examples include a deletion mutant lacking 1 or 2 amino acids from the heavy chain carboxyl terminus (*Journal of Chromatography A*; 705: 129-134, 1995), a deletion mutant lacking 2 amino acids (glycine and lysine) from the heavy chain carboxyl terminus and subjected to amidation of proline at the carboxyl terminus (*Analytical Biochemistry*, 360: 75-83, 2007), and an antibody resulting from pyroglutamination of an amino-terminal glutamine or glutamic acid of the heavy chain or the light chain (WO 2013/147153) (they are collectively referred to as “deletion mutants”). As

long as the antigen-binding ability and effector functions are retained, the antibody of the present invention lacking the heavy chain terminus and the light chain carboxyl terminus is not limited to the deletion mutants described above. When the antibody of the present invention comprises 2 or more chains (e.g., heavy chains), either of or both of such 2 or more chains (e.g., heavy chains) may be the full-length heavy chain or a heavy chain selected from the group consisting of the deletion mutants described above. While the quantitative ratio or the number ratio of molecules of the deletion mutant would be influenced by the type and culture conditions of cultured cells of mammalian animals producing the antibody of the present invention, the main components of the antibody of the present invention can be both of the 2 heavy chains lacking an amino acid residue from the carboxyl terminus.

[0591] In addition, the antibody of the present invention or an antigen-binding fragment thereof (including those comprised in the molecule, the multispecific molecule, and the bispecific molecule of the present invention) comprising one to several amino acids derived from an expression vector and/or a signal sequence added to the amino terminus and/or carboxy terminus (and partially or entirely modified as described above) or the like are within the scope of the modified antibody of the present invention or the modified antigen-binding fragment thereof, as long as the antigen-binding activity of interest is maintained. A molecule comprising such modified antibody or modified antigen-binding fragment thereof is within the scope of the molecule of the present invention.

[0592] In the present invention, “the antibody or the binding fragment thereof” encompasses “the modified antibody or the modified antigen-binding fragment thereof” In addition, “the antibody or antigen-binding fragment thereof” included in the molecule, the multispecific molecule, and the bispecific molecule of the present invention encompasses “the modified antibody or the modified antigen-binding fragment thereof.”

[0593] Antibody dependent cellular cytotoxicity can be potentiated by regulation (glycosylation, afucosylation, and the like) of sugar chain modification bound to the antibody of the present invention. Known techniques for regulation of antibody sugar chain modification are disclosed in, for example, WO 99/54342, WO 00/61739, and WO 02/31140, although the techniques are not limited thereto.

[0594] The present invention encompasses a complex of the antibody and other molecules ligated to each other with a linker (i.e., an immunoconjugate). An example of an antibody-drug complex comprising the antibody bound to a radioactive substance or a compound having pharmacological activity is an antibody-drug conjugate (ADC) (Methods Mol. Biol., 2013, 1045: 1-27; Nature Biotechnology, 2005, 23, pp. 1137-1146).

[0595] Further, the present invention encompasses a complex comprising such an antibody bound to another functional polypeptide. An example of such antibody-peptide complex is a complex of the antibody with an albumin-bound polypeptide (Protein Eng. Des. Sel., 2012, (2): 81-8).

[0596] The modified antibody, the antibody with regulated sugar chain modification, and the complex described above are within the scope of the antibody of the present invention, and the binding fragments of the modified antibody, the antibody with regulated sugar chain modification, and the

complex are within the scope of the binding fragment of the antibody of the present invention.

4. Method for Producing Antibody

[0597] The antibody of the present invention can be produced in a cell as a recombinant antibody by, for example, inserting a DNA encoding a heavy chain variable region or a DNA encoding a light chain variable region into an expression vector, transforming a host cell for expression with the vector, and culturing the host cell.

[0598] Concerning DNAs encoding antibodies, a DNA encoding a heavy chain can be obtained by ligating a DNA encoding a heavy chain variable region to a DNA encoding a heavy chain constant region, and a DNA encoding a light chain can be obtained by ligating a DNA encoding a light chain variable region to a DNA encoding a light chain constant region.

[0599] The anti-HLA/NY-ESO antibody of the present invention can be produced by inserting the DNA encoding the heavy chain and the DNA encoding the light chain mentioned above into an expression vector, transforming a host cell with the vector, and culturing the host cell. In such a case, the DNA encoding the heavy chain and the DNA encoding the light chain mentioned above may be introduced into the same expression vector and the host cell may be transformed with the vector. Alternatively, the DNA encoding the heavy chain and the DNA encoding the light chain may be inserted into separate vectors and the host cell may be transformed with the two vectors. In this case, DNAs encoding the heavy chain variable region and the light chain variable region may be introduced into a vector into which the DNA encoding the heavy chain constant region and the DNA encoding a light chain constant region have been introduced in advance. Further, the vector may contain a DNA encoding a signal peptide, which promotes secretion of an antibody from a host cell. In this case, the DNA encoding the signal peptide and the DNA encoding the antibody are ligated in-frame in advance. An antibody can be obtained as a mature protein by removing the signal peptide after an antibody is produced.

[0600] In this case, the DNA encoding the heavy chain variable region, the DNA encoding the light chain variable region, the DNA comprising the DNA encoding the heavy chain variable region ligated to the DNA encoding the heavy chain constant region, or the DNA comprising the DNA encoding the light chain variable region ligated to the DNA encoding the light chain constant region may be operably ligated to an element, such as a promoter, an enhancer, or a polyadenylation signal. When DNA is “operably ligated” herein, DNA is ligated to an element, so that the element can exert their functions.

[0601] An expression vector is not particularly limited, as long as it can be replicated in an animal cell, a bacterial cell, a yeast cell, or other host, and examples thereof include known plasmids and phages. Examples of a vector used to construct an expression vector include pcDNA™ (Thermo Fisher Scientific), Flexi® vector (Promega), pUC19, pUEX2 (Amersham), pGEX-4T, pKK233-2 (Pharmacia), and pMAMneo (Clontech). As host cells, prokaryotic cells such as *Escherichia coli* and *Bacillus subtilis* and eukaryotic cells such as yeasts and animal cells can be used, with the use of eukaryotic cells being preferable. Examples of animal cells include the human embryonic kidney cell line HEK293 and the Chinese hamster ovary (CHO) cell. It is sufficient to

introduce an expression vector into a host cell by a known method to transform the host cell. Examples of methods include an electroporation method, a calcium phosphate precipitation method, and a DEAE-dextran transfection method. The produced antibody can be purified by usual protein isolation or purification methods. For example, affinity chromatography or other chromatography techniques, filtration, ultrafiltration, salting out, dialysis, and the like can be suitably selected and combined.

5. Molecule that Binds to HLA/NY-ESO

[0602] The molecule of the present invention comprises the anti-HLA/NY-ESO antibody according to the present invention or an antigen-binding fragment thereof. The molecule of the present invention is preferably a multispecific molecule comprising 2 or more antigen-binding sites. Specifically, the molecule of the present invention can bind to 2 or more different epitopes on a molecule or different epitopes on 2 or more different molecules and such a molecule comprises a plurality of different antigen-binding fragments. Examples of such a multispecific molecule include, but are not limited to, an IgG-type multispecific molecule and a multispecific molecule comprising two or more types of variable regions, for example, antibody fragments, such as tandem scFv (taFv), single chain diabody, a diabody, and a triabody, and an antibody fragment resulting from covalent or noncovalent binding. A multispecific molecule may comprise Fc.

[0603] The multispecific molecule of the present invention may comprise, in addition to the anti-HLA/NY-ESO antibody according to the present invention or an antigen-binding fragment thereof, one type or two or more types of antibodies or antigen-binding fragments thereof. Examples of such antigen-binding fragments of the antibody include Fab, F(ab)', Fv, scFv, and sdAb.

[0604] A preferable multispecific molecule according to the present invention further comprises an anti-CD3 antibody or an antigen-binding fragment thereof, and it binds specifically to CD3 as well.

[0605] The anti-CD3 antibody or an antigen-binding fragment thereof included in the multispecific molecule according to the present invention is not particularly limited, provided that it is a human CD3-binding antibody or a binding fragment thereof, and it preferably binds to CD3 of a non-human primate, such as a cynomolgus monkey. A more preferable example of an anti-CD3 antibody or an antigen-binding fragment thereof is the antibody or antigen-binding fragment thereof comprising the heavy chain variable region CDRH1 consisting of the amino acid sequence as shown in SEQ ID NO: 141, the heavy chain variable region CDRH2 consisting of the amino acid sequence as shown in SEQ ID NO: 142, the heavy chain variable region CDRH3 consisting of the amino acid sequence as shown in SEQ ID NO: 143, the light chain variable region CDRL1 consisting of the amino acid sequence as shown in SEQ ID NO: 144, the light chain variable region CDRL2 consisting of the amino acid sequence as shown in SEQ ID NO: 145, and the light chain variable region CDRL3 consisting of the amino acid sequence as shown in SEQ ID NO: 146 (FIG. 142).

[0606] An example of the more preferable antibody or antigen-binding fragment thereof comprising CDRH1 to CDRH3 and CDRL1 to CDRL3 is the antibody or antigen-binding fragment thereof comprising the C3E-7034 heavy

chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 136 (FIG. 137), the C3E-7036 heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 137 (FIG. 138), the C3E-7085 heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of SEQ ID NO: 138 (FIG. 139), the C3E-7088 heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 139 (FIG. 140), the C3E-7093 heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 140 (FIG. 141), the C3E-7096 heavy chain variable region consisting of an amino acid sequence of amino acids 272 to 389 of the amino acid sequence as shown in SEQ ID NO: 155 (FIG. 151), the C3E-7096 heavy chain variable region consisting of an amino acid sequence of amino acids 277 to 394 of the amino acid sequence as shown in SEQ ID NO: 156 (FIG. 152), and the C3E-7097 heavy chain variable region consisting of an amino acid sequence of amino acids 277 to 394 of the amino acid sequence as shown in SEQ ID NO: 157 (FIG. 153).

[0607] An example of the more preferable antibody or antigen-binding fragment thereof comprising CDRH1 to CDRH3 and CDRL1 to CDRL3 is the antibody or antigen-binding fragment thereof comprising the C3E-7034 light chain variable region consisting of an amino acid sequence of amino acids 135 to 243 of the amino acid sequence as shown in SEQ ID NO: 136 (FIG. 137), the C3E-7036 light chain variable region consisting of an amino acid sequence of amino acids 135 to 241 of the amino acid sequence as shown in SEQ ID NO: 137 (FIG. 138), the C3E-7085 light chain variable region consisting of an amino acid sequence of amino acids 135 to 241 of the amino acid sequence as shown in SEQ ID NO: 138 (FIG. 139), the C3E-7088 light chain variable region consisting of an amino acid sequence of amino acids 135 to 243 of the amino acid sequence as shown in SEQ ID NO: 139 (FIG. 140), the C3E-7093 light chain variable region consisting of an amino acid sequence of amino acids 135 to 243 of the amino acid sequence as shown in SEQ ID NO: 140 (FIG. 141), the C3E-7096 light chain variable region consisting of an amino acid sequence of amino acids 405 to 511 of the amino acid sequence as shown in SEQ ID NO: 155 (FIG. 151), the C3E-7096 light chain variable region consisting of an amino acid sequence of amino acids 410 to 516 of the amino acid sequence as shown in SEQ ID NO: 156 (FIG. 152), and the C3E-7097 light chain variable region consisting of an amino acid sequence of amino acids 410 to 516 of the amino acid sequence as shown in SEQ ID NO: 157 (FIG. 153).

[0608] An example of the more preferable antibody or antigen-binding fragment thereof comprising CDRH1 to CDRH3 and CDRL1 to CDRL3 is the antibody or antigen-binding fragment thereof comprising a combination of the C3E-7034 heavy chain variable region and light chain variable region consisting of amino acids 2 to 119 and amino acids 135 to 243 of SEQ ID NO: 136 (FIG. 137), a combination of the C3E-7036 heavy chain variable region and light chain variable region consisting of amino acids 2 to 119 and amino acids 135 to 241 of SEQ ID NO: 137 (FIG. 138), a combination of the C3E-7078 heavy chain variable region and light chain variable region consisting of amino acids 2 to 119 and amino acids 135 to 243 of SEQ ID NO:

147 (FIG. 143), a combination of the C3E-7085 heavy chain variable region and light chain variable region consisting of amino acids 2 to 119 and amino acids 135 to 241 of SEQ ID NO: 138 (FIG. 139), a combination of the C3E-7088 heavy chain variable region and light chain variable region consisting of amino acids 2 to 119 and amino acids 135 to 243 of SEQ ID NO: 139 (FIG. 140), a combination of the C3E-7093 heavy chain variable region and light chain variable region consisting of amino acids 2 to 119 and amino acids 135 to 243 of SEQ ID NO: 140 (FIG. 141), a combination of the C3E-7096 heavy chain variable region and light chain variable region consisting of amino acids 272 to 389 and amino acids 405 to 511 of SEQ ID NO: 155 (FIG. 151), a combination of the C3E-7096 heavy chain variable region and light chain variable region consisting of amino acids 277 to 394 and amino acids 410 to 516 of SEQ ID NO: 156 (FIG. 152), and a combination of the C3E-7097 heavy chain variable region and light chain variable region consisting of amino acids 277 to 394 and amino acids 410 to 516 of SEQ ID NO: 157 (FIG. 153).

[0609] Further, examples of the more preferable antibody or antigen-binding fragment thereof comprising CDRH1 to CDRH3 and CDRL1 to CDRL3 include C3E-7034 scFv consisting of an amino acid sequence of amino acids 2 to 243 of the amino acid sequence as shown in SEQ ID NO: 136 (FIG. 137), C3E-7036 scFv consisting of an amino acid sequence of amino acids 2 to 241 of the amino acid sequence as shown in SEQ ID NO: 137 (FIG. 138), C3E-7078 scFv consisting of an amino acid sequence of amino acids 2 to 243 of the amino acid sequence as shown in SEQ ID NO: 147 (FIG. 143), C3E-7085 scFv consisting of an amino acid sequence of amino acids 2 to 241 of the amino acid sequence as shown in SEQ ID NO: 138 (FIG. 139), C3E-7088 scFv consisting of an amino acid sequence of amino acids 2 to 243 of the amino acid sequence as shown in SEQ ID NO: 139 (FIG. 140), C3E-7093 scFv consisting of an amino acid sequence of amino acids 2 to 243 of the amino acid sequence as shown in SEQ ID NO: 140 (FIG. 141), and an antibody or a binding fragment thereof comprising any one of such scFv constructs. A preferable embodiment of CD3-specific scFv (which may be referred to as “anti-CD3 scFv”) includes scFv comprising a FLAG-His tag added to its carboxyl terminal side (may be simply referred to as “tag adduct”). Examples of preferable tag adducts include C3E-7034 (SEQ ID NO: 136; FIG. 137), C3E-7036 (SEQ ID NO: 137; FIG. 138), C3E-7085 (SEQ ID NO: 138; FIG. 139), C3E-7088 (SEQ ID NO: 139; FIG. 140), and C3E-7093 (SEQ ID NO: 140; FIG. 141), with C3E-7085 being more preferable.

[0610] A preferable example of the multispecific molecule of the present invention is a bispecific molecule. A “bispecific” molecule is capable of binding to two different epitopes on a single molecule or different epitopes on two molecules, and it includes an antibody or antigen-binding fragment having such bispecificity. The bispecific molecule of the present invention binds to HLA/NY-ESO and it further binds to CD3.

[0611] Examples of the bispecific molecule of the present invention include those having the structures (formats) described below.

[0612] In the dual scFv-type bispecific molecule, two types of scFv constructs binding to different epitopes are each allowed to bind to one Fc of a dimer with a linker or allowed to directly bind thereto without a linker. Alternatively,

two types of scFv constructs binding to different epitopes are each allowed to bind to CH and CL with linkers and are further allowed to bind to one Fc of the dimer with linkers. In such a bispecific molecule, Fc comprising a heterodimer-forming mutation downstream of one scFv is heterologously associated with another Fc comprising a heterodimer-forming mutation downstream of another scFv. The dual scFv-type bispecific molecule is referred to as a dual-type bispecific molecule or simply as a dual-type (FIG. 6A (b)).

[0613] In the present invention, for example, a dual-type bispecific molecule consisting of anti-HLA-A2/NY-ESO scFv and anti-CD3 scFv may be used.

[0614] Alternatively, the bispecific molecule of the present invention may comprise Fab and scFv binding to different epitopes, Fab of the first antibody may be bound to one Fc of a dimer, and scFv of the second antibody may be bound to another Fc with linkers. In such bispecific molecule, Fc comprising a heterodimer-forming mutation downstream of Fab is heterologously associated with Fc comprising a heterodimer-forming mutation downstream of scFv. Such bispecific molecule is referred to as a “hybrid-type bispecific molecule” or a “hybrid-type” (FIG. 6A (a)). In the present invention, for example, a hybrid type consisting of anti-HLA-A2/NY-ESO Fab and anti-CD3 scFv can be used.

[0615] In addition, a bispecific molecule may allow Fab of the first antibody and scFv of the second antibody to attach to one Fc of a dimer with linkers. In such a case, Fab may be allowed to attach to Fc and scFv may be allowed to attach to such Fab. Alternatively, scFv may be allowed to attach to Fc and Fab may be allowed to attach to such scFv. Preferably, Fab may be allowed to attach to Fc and scFv may be allowed to attach to such Fab. scFv may be allowed to attach to a variable region of Fab with a linker. In such a bispecific molecule, Fc comprising a heterodimer-forming mutation is associated with a site downstream of where scFv is ligated to Fab. Such a bispecific molecule is referred to as an scFv-Fab-heterodimer Fc-type bispecific molecule or scFv-Fab-heterodimer Fc-type (FIG. 6A (c)).

[0616] In the present invention, for example, scFv-Fab-heterodimer Fc-type consisting of anti-CD3 scFv and anti-HLA-A2/NY-ESO Fab may be used.

[0617] In addition, taFv (FIG. 3 (c)) comprising two types of scFv constructs (of the first antibody and the second antibody) ligated to each other with a linker may be attached to one Fc of a dimer with a linker or directly without a linker. Such a bispecific molecule is referred to as a taFv-heterodimer Fc-type bispecific molecule or taFv-heterodimer Fc-type (FIG. 3 (d)). In such a bispecific molecule, Fc comprising a heterodimer-forming mutation is heterologously associated with another Fc comprising a heterodimer-forming mutation in a site downstream of taFv. The order for ligating the first antibody and the second antibody to taFv is not limited. The previously mentioned bispecific molecule is referred to as taFv-heterodimer Fc-type but when the order for ligating the first antibody and the second antibody to taFv is inverted, the bispecific molecule is referred to as taFv (inversed)-heterodimer Fc-type (taFv (inversed)-Fc type).

[0618] FIG. 6 A (a) shows a structure of a hybrid-type bispecific molecule, FIG. 6 A (b) shows a structure of a dual-type bispecific molecule, and FIG. 6 A (c) shows a structure of an scFv-Fab-heterodimer Fc-type bispecific molecule. FIG. 3 (a) shows a structure of scFv, FIG. 3 (b) shows a structure of Fab, FIG. 3 (c) shows a structure of

taFv, FIG. 3 (d) shows a structure of taFv-heterodimer Fc-type bispecific molecule, and FIG. 3 (e) shows a structure of a taFv-Fab-heterodimer Fc-type bispecific molecule. FIG. 6 B (a) shows a structure of a taFv-heterodimer Fc-type bispecific molecule (the same as FIG. 3 (d)), FIG. 6 B (b) shows a structure of a taFv (inversed)-heterodimer Fc-type bispecific molecule, FIG. 6 B (c) shows a structure of the first polypeptide included in the taFv (inversed)-heterodimer Fc-type bispecific molecule, and FIG. 6 (d) shows a structure of the second polypeptide included in the taFv (inversed)-heterodimer Fc-type bispecific molecule.

[0619] In the bispecific molecule of the present invention, a plurality of polypeptides are associated.

[0620] In the present invention, a taFv, for example, one composed of an anti-HLA-A2/NY-ESO scFv and an anti-CD3 scFv may be used. The taFv-heterodimer Fc-type bispecific molecule preferably comprises (a) a first polypeptide comprising scFv that binds specifically to HLA/NY-ESO, scFv that binds specifically to CD3, and the immunoglobulin Fc region (i) in that order from the N terminus toward the C terminus and the second polypeptide comprising the hinge region and the Fc region (ii) of immunoglobulin. It more preferably comprises (b) the first polypeptide associated with a second polypeptide at the Fc region (i) and the Fc region (ii). Fc regions of the first polypeptide and of the second polypeptide may comprise a heterodimer-forming mutation. FIG. 3 (d) shows an example of the taFv-heterodimer Fc-type bispecific molecule. As shown in FIG. 3 (d), the Fc region (i) of the first polypeptide binds to the Fc region (ii) of the second polypeptide (solid), and the first polypeptide is thus associated with the second polypeptide. FIG. 3 (f) shows the first polypeptide and FIG. 3 (g) shows the second polypeptide. In FIG. 3 (d), for example, scFv shown as blank is anti-HLA-A2/NY-ESO scFv, and scFv indicated with positive slopes is anti-CD3 scFv.

[0621] The first polypeptide included in a more preferable taFv-heterodimer Fc-type bispecific molecule of the present invention comprises an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 85, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 87, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 88, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 89, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 90, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 91, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 92, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 93, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 94, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 95, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 96, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 86, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 149, an amino acid sequence of amino acids 21 to 511 of the amino acid

sequence as shown in SEQ ID NO: 150, an amino acid sequence of amino acids 20 to 511 of the amino acid sequence as shown in SEQ ID NO: 155, an amino acid sequence of amino acids 20 to 516 of the amino acid sequence as shown in SEQ ID NO: 156, or an amino acid sequence of amino acids 20 to 516 of the amino acid sequence as shown in SEQ ID NO: 157. The first polypeptide included in a further preferable taFv-heterodimer Fc-type bispecific molecule comprises an amino acid sequence of amino acids 529 to 745 of the amino acid sequence as shown in SEQ ID NO: 85, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 86, 149, 150, or 155 or an amino acid sequence of amino acids 534 to 750 of the amino acid sequence as shown in SEQ ID NO: 156 or 157. The first polypeptide included in a still further preferable taFv-heterodimer Fc-type bispecific molecule consists of an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 85, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 87, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 88, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 89, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 90, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 91, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 92, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 93, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 94, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 95, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 96, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 86, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 149, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 150, or an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 155. Alternatively, the first polypeptide consists of an amino acid sequence of amino acids 20 to 750 of the amino acid sequence as shown in SEQ ID NO: 156 or an amino acid sequence of amino acids 20 to 750 of the amino acid sequence as shown in SEQ ID NO: 157.

[0622] The second polypeptide included in the preferable taFv-heterodimer Fc-type bispecific molecule of the present invention comprises a hinge region and mutant Fc derived from a human antibody. The second polypeptide included in the more preferable taFv-heterodimer Fc-type bispecific molecule comprises an amino acid sequence of amino acids 20 to 246 of the amino acid sequence as shown in SEQ ID NO: 84.

[0623] Of these, NYF-0016 in which the first polypeptide consisting of an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 85 is associated with the second polypeptide consisting of an amino acid sequence of amino acids 21 to 246 of the amino acid sequence as shown in SEQ ID NO: 84, NYF-0022 in

which the first polypeptide consisting of an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 87 is associated with the second polypeptide consisting of an amino acid sequence of amino acids 21 to 246 of the amino acid sequence as shown in SEQ ID NO: 84, NYF-0023 in which the first polypeptide consisting of an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 88 is associated with the second polypeptide consisting of an amino acid sequence of amino acids 21 to 246 of the amino acid sequence as shown in SEQ ID NO: 84, NYF-0027 in which the first polypeptide consisting of an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 89 is associated with the second polypeptide consisting of an amino acid sequence of amino acids 21 to 246 of the amino acid sequence as shown in SEQ ID NO: 84, NYF-0035 in which the first polypeptide consisting of an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 90 is associated with the second polypeptide consisting of an amino acid sequence of amino acids 21 to 246 of the amino acid sequence as shown in SEQ ID NO: 84, NYF-0044 in which the first polypeptide consisting of an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 91 is associated with the second polypeptide consisting of an amino acid sequence of amino acids 21 to 246 of the amino acid sequence as shown in SEQ ID NO: 84, NYF-0045 in which the first polypeptide consisting of an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 92 is associated with the second polypeptide consisting of an amino acid sequence of amino acids 21 to 246 of the amino acid sequence as shown in SEQ ID NO: 84, NYF-0047 in which the first polypeptide consisting of an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 93 is associated with the second polypeptide consisting of an amino acid sequence of amino acids 21 to 246 of the amino acid sequence as shown in SEQ ID NO: 84, NYF-0048 in which the first polypeptide consisting of an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 94 is associated with the second polypeptide consisting of an amino acid sequence of amino acids 21 to 246 of the amino acid sequence as shown in SEQ ID NO: 84, NYF-0060 in which the first polypeptide consisting of an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 95 is associated with the second polypeptide consisting of an amino acid sequence of amino acids 21 to 246 of the amino acid sequence as shown in SEQ ID NO: 84, NYF-0061 in which the first polypeptide consisting of an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 96 is associated with the second polypeptide consisting of an amino acid sequence of amino acids 21 to 246 of the amino acid sequence as shown in SEQ ID NO: 84, NYF-0019 in which the first polypeptide consisting of an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 86 is associated with the second polypeptide consisting of an amino acid sequence of amino acids 21 to 246 of the amino acid sequence as shown in SEQ ID NO: 84, NYF-0014 in which the first polypeptide consisting of an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 149 is associated with the second polypeptide consisting of

an amino acid sequence of amino acids 21 to 246 of the amino acid sequence as shown in SEQ ID NO: 84, and NYF-0082 in which the first polypeptide consisting of an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 150 is associated with the second polypeptide consisting of an amino acid sequence of amino acids 21 to 246 of the amino acid sequence as shown in SEQ ID NO: 84 can be exemplified as the preferable taFv-heterodimer Fc-type bispecific molecules of the present invention.

[0624] Further examples of the preferable taFv-heterodimer Fc-type bispecific molecules of the present invention include NYZ-0038 in which the first polypeptide consisting of an amino acid sequence of amino acids 20 to 745 of the amino acid sequence as shown in SEQ ID NO: 155 is associated with the second polypeptide consisting of an amino acid sequence of amino acids 20 to 246 of the amino acid sequence as shown in SEQ ID NO: 84, NYZ-0082 in which the first polypeptide consisting of an amino acid sequence of amino acids 20 to 750 of the amino acid sequence as shown in SEQ ID NO: 156 is associated with the second polypeptide consisting of an amino acid sequence of amino acids 20 to 246 of the amino acid sequence as shown in SEQ ID NO: 84, and NYZ-0083 in which the first polypeptide consisting of an amino acid sequence of amino acids 20 to 750 of the amino acid sequence as shown in SEQ ID NO: 157 is associated with the second polypeptide consisting of an amino acid sequence of amino acids 20 to 246 of the amino acid sequence as shown in SEQ ID NO: 84.

[0625] Of these, NYF-0023, NYF-0047, NYF-0048, NYF-0060, NYF-0061, NYZ-0038, NYZ-0082, and NYZ-0083 exert excellent biological activity, physical properties, and the like and thus are particularly preferable.

[0626] In addition, taFv of the first antibody may be allowed to attach to one Fc of a dimer with a linker or directly bound thereto without a linker, and Fab of the first antibody or the second antibody may be allowed to attach to another Fc with a linker or directly bound thereto without a linker. In such a bispecific molecule, Fab is added to a site upstream of the Fc region (ii) (solid) of the second polypeptide of the taFv-heterodimer Fc-type. Such a bispecific molecule is referred to as the taFv-Fab-heterodimer Fc-type bispecific molecule or taFv-Fab-heterodimer Fc-type (FIG. 3).

[0627] In the present invention, for example, taFv of anti-HLA-A2/NY-ESO scFv, taFv of anti-CD3 scFv, and Fab of HLA/NY-ESO may be used to be included in the taFv-Fab-heterodimer Fc-type bispecific molecule.

[0628] The taFv-Fab-heterodimer Fc-type bispecific molecule preferably comprises (a) a first polypeptide comprising scFv that binds specifically to human HLA/NY-ESO, scFv that binds specifically to CD3, and the immunoglobulin Fc region (i) in that order from the N terminus toward the C terminus, a second polypeptide consisting of an immunoglobulin heavy chain including the Fc region (ii), and a third polypeptide consisting of an immunoglobulin light chain. More preferably, (b) the second polypeptide is associated with the third polypeptide, and (c) the first polypeptide is associated with the second polypeptide at the Fc region (i) and the Fc region (ii). FIG. 3 (e) shows an example of the taFv-Fab-heterodimer Fc-type bispecific molecule, FIG. 3 (f) shows the first polypeptide, FIG. 3 (h) shows the second polypeptide, and FIG. 3 (i) shows the third polypeptide. As shown in FIG. 3 (e), the Fc region (ii) (solid) of the second

polypeptide, consisting of an immunoglobulin heavy chain comprising the Fc region (ii), binds to the Fc region (i) of the first polypeptide, and an immunoglobulin light chain further attaches to the second polypeptide. Such preferable taFv-Fab-heterodimer Fc-type bispecific molecule can be composed of the taFv-heterodimer Fc-type bispecific molecule comprising taFv and the immunoglobulin Fc region and Fab bound thereto. Examples of amino acid sequences included in the first polypeptide of a preferable taFv-Fab-heterodimer Fc-type bispecific molecule include an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 85, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 87, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 88, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 89, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 90, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 91, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 92, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 93, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 94, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 95, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 96, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 86, an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 149, and an amino acid sequence of amino acids 21 to 511 of the amino acid sequence as shown in SEQ ID NO: 150. The first polypeptide included in a more preferable taFv-Fab-heterodimer Fc-type bispecific molecule consists of an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 85, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 87, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 88, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 89, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 90, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 91, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 92, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 93, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 94, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 95, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 96, an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 86, an amino acid sequence of amino acids 21 to 745 of the amino acid

sequence as shown in SEQ ID NO: 149, or an amino acid sequence of amino acids 21 to 745 of the amino acid sequence as shown in SEQ ID NO: 150.

[0629] The second polypeptide included in the preferable taFv-Fab-heterodimer Fc-type bispecific molecule of the present invention comprises the human antibody or humanized antibody heavy chain variable region, the CH1 region, a hinge region, and mutant Fc. The second polypeptide included in the more preferable taFv-Fab-heterodimer Fc-type bispecific molecule comprises an amino acid sequence of amino acids 20 to 242 of the amino acid sequence as shown in SEQ ID NO: 99.

[0630] The third polypeptide included in the preferable taFv-Fab-heterodimer Fc-type bispecific molecule of the present invention comprises the human antibody or humanized antibody light chain variable region and constant region. The third polypeptide included in the more preferable taFv-Fab-heterodimer Fc-type bispecific molecule comprises an amino acid sequence of amino acids 21 to 131 of the amino acid sequence as shown in SEQ ID NO: 100.

[0631] Of the second polypeptide included in such preferable taFv-Fab-heterodimer Fc-type bispecific molecule, the variable region, the CH1 region, and the third polypeptide constitute Fab, and Fab is preferably of the anti-HLA/NY-ESO antibody, such as Fab of NYA-0001.

[0632] In the present invention, for example, anti-HLA-A2/NY-ESO scFv, anti-CD3 scFv, HLA/NY-ESO Fab, or anti-CD3 Fab may be included in the scFv-Fab-heterodimer Fc-type bispecific molecule.

[0633] The scFv-Fab-heterodimer Fc-type bispecific molecule preferably comprises (a) a first polypeptide comprising scFv that binds specifically to human HLA/NY-ESO, the antibody heavy chain variable region and constant region CH1 binding specifically to CD3, and the immunoglobulin Fc region (i) in that order from the N terminus toward the C terminus, a second polypeptide comprising the immunoglobulin hinge region and Fc region (ii), and a third polypeptide consisting of an antibody light chain comprising a variable region and a constant region. More preferably, (b) the first polypeptide is associated with the second polypeptide via the Fc region (i) and the Fc region (ii), and the first polypeptide is associated with (the antibody light chain of) the third polypeptide via the antibody heavy chain variable region and constant region CH1 of the first polypeptide. The Fc regions of the first polypeptide and of the second polypeptide may be a wild-type or may contain a heterodimer-forming mutation. FIG. 6 A (c) shows an example of the scFv-Fab-heterodimer Fc-type bispecific molecule. In FIG. 6 A (c), the right half shows the first polypeptide and the third polypeptide and the left half shows the second polypeptide. As shown in FIG. 6 A (c), the Fc region (i) of the first polypeptide is associated with the Fc region (ii) of the second polypeptide (solid), and the first polypeptide is associated with the third polypeptide. In FIG. 6 A (c), scFv indicated with positive slopes is anti-HLA-A2/NY-ESO scFv, and Fab each indicated as blank, a checkered pattern, or horizontal lines is anti-CD3 Fab.

[0634] An example of an amino acid sequence included in the first polypeptide included in the preferable scFv-Fab-heterodimer Fc-type bispecific molecule is an amino acid sequence of amino acids 21 to 394 of the amino acid sequence as shown in SEQ ID NO: 160. A more preferable

example thereof is an amino acid sequence of amino acids 20 to 724 of the amino acid sequence as shown in SEQ ID NO: 160.

[0635] An example of another amino acid sequence included in the first polypeptide included in the preferable scFv-Fab-heterodimer Fc-type bispecific molecule is an amino acid sequence of amino acids 21 to 389 of the amino acid sequence as shown in SEQ ID NO: 197. A more preferable example thereof is an amino acid sequence of amino acids 20 to 719 of the amino acid sequence as shown in SEQ ID NO: 197. An example of another amino acid sequence included in the first polypeptide included in the preferable scFv-Fab-heterodimer Fc-type bispecific molecule is an amino acid sequence of amino acids 21 to 389 of the amino acid sequence as shown in SEQ ID NO: 198. A more preferable example thereof is an amino acid sequence of amino acids 20 to 719 of the amino acid sequence as shown in SEQ ID NO: 198.

[0636] The second polypeptide included in the preferable scFv-Fab-heterodimer Fc-type bispecific molecule of the present invention comprises a hinge region and mutant Fc derived from a human antibody. The second polypeptide included in the more preferable scFv-Fab-heterodimer Fc-type bispecific molecule comprises an amino acid sequence of amino acids 20 to 246 of the amino acid sequence as shown in SEQ ID NO: 84.

[0637] The third polypeptide included in the preferable scFv-Fab-heterodimer Fc-type bispecific molecule of the present invention comprises a human antibody-derived light chain. The third polypeptide included in the more preferable scFv-Fab-heterodimer Fc-type bispecific molecule comprises, for example, an amino acid sequence of amino acids 21 to 127 of the amino acid sequence as shown in SEQ ID NO: 161. A further preferable example thereof is an amino acid sequence of amino acids 21 to 233 of the amino acid sequence as shown in SEQ ID NO: 161.

[0638] Of these, NYZ-1010 in which the first polypeptide consisting of an amino acid sequence of amino acids 20 to 724 of the amino acid sequence as shown in SEQ ID NO: 160, the second polypeptide consisting of an amino acid sequence of amino acids 20 to 246 of the amino acid sequence as shown in SEQ ID NO: 84, and the third polypeptide consisting of an amino acid sequence of amino acids 21 to 233 of the amino acid sequence as shown in SEQ ID NO: 161 are associated with each other can be exemplified as the preferable scFv-Fab-heterodimer Fc-type bispecific molecule of the present invention.

[0639] In addition, NYZ-1007 in which the first polypeptide consisting of an amino acid sequence of amino acids 20 to 719 of the amino acid sequence as shown in SEQ ID NO: 197, the second polypeptide consisting of an amino acid sequence of amino acids 20 to 246 of the amino acid sequence as shown in SEQ ID NO: 84, and the third polypeptide consisting of an amino acid sequence of amino acids 21 to 233 of the amino acid sequence as shown in SEQ ID NO: 161 are associated with each other can be exemplified as the preferable scFv-Fab-heterodimer Fc-type bispecific molecule of the present invention.

[0640] Further, NYZ-1017 in which the first polypeptide consisting of an amino acid sequence of amino acids 20 to 719 of the amino acid sequence as shown in SEQ ID NO: 198, the second polypeptide consisting of an amino acid sequence of amino acids 20 to 246 of the amino acid sequence as shown in SEQ ID NO: 84, and the third

polypeptide consisting of an amino acid sequence of amino acids 21 to 233 of the amino acid sequence as shown in SEQ ID NO: 161 are associated with each other can be exemplified as the preferable scFv-Fab-heterodimer Fc-type bispecific molecule of the present invention.

[0641] At least 1, 2, or 3 peptides included in the bispecific molecule of the present invention may be the “deletion mutant” described above. Specifically, 1 or 2 (or more) amino acids at its carboxyl terminus (at the carboxyl terminus derived from the antibody heavy chain, in particular) may be mutated (may also be deleted). For example, the carboxyl terminus of the amino acid sequence of the first polypeptide included in NYZ-1010, which is a preferable scFv-Fab-heterodimer Fc-type bispecific molecule of the present invention, may be any of Lys at position 724 of SEQ ID NO: 160, Gly at position 723 resulting from deletion of an amino acid, or a mixture of first polypeptides including Lys and Gly at the carboxyl terminus. Also, the carboxyl terminus of the amino acid sequence of the second polypeptide included in the preferable scFv-Fab-heterodimer Fc-type bispecific molecule of the present invention may be any of Lys at position 246 of SEQ ID NO: 84, Gly at position 245 resulting from deletion of an amino acid, or a mixture of second polypeptides including Lys and Gly at the carboxyl terminus.

[0642] scFv and Fab included in the bispecific molecule of the present invention are preferably derived from a humanized antibody or human antibody, and Fc is preferably derived from a human antibody.

[0643] In a variable region in the bispecific molecule of the present invention, a heavy chain variable region and a light chain variable region may be attached in that order from the amino terminus of the antibody. Alternatively, a light chain variable region and a heavy chain variable region may be attached in that order. A linker may optionally be present between two variable regions. A glycine residue may optionally be present at the amino terminus of the amino-terminal variable region. In a tandem scFv-type bispecific molecule, a linker, a FLAG tag, and/or a His tag may optionally be attached to the carboxyl terminus of the carboxyl-terminal variable region. In a preferable embodiment, for example, a heavy chain variable region, a first linker, a light chain variable region, a second linker, a FLAG tag, and, a His tag are attached in that order from the amino terminus.

[0644] Examples of linkers include a single chain polypeptide, a single chain oligopeptide, and synthetic products, such as PEG, nucleotide, sugar chain, and compounds. Other known linkers can also be used without particular limitation, as long as they can attach 2 polypeptides to each other.

[0645] A peptide linker can comprise, for example, 5 to 30 amino acids. When a bispecific molecule comprises a plurality of linkers, all the peptide linkers may be of the same length, or peptide linkers of different lengths may be used.

[0646] An example of a peptide linker is a repetition of (Gly•Gly•Gly•Gly•Ser) (SEQ ID NO: 161). One to several amino acid residues other than Gly and Ser may be added thereto.

[0647] Among constructs (formats) of the multispecific antibody and, in particular, the bispecific antibody of the present invention as described above, taFv-heterodimer Fc-type, taFv-Fab-heterodimer Fc-type, and scFv-Fab-heterodimer Fc-type are preferable, and taFv-heterodimer Fc-type is more preferable. A format in which anti-HLA/NY-ESO scFv

and anti-CD3 scFv are positioned in that order from the N terminus toward the C terminus (the taFv-heterodimer Fc-type) is more preferable than the format in which anti-CD3 scFv and anti-HLA/NY-ESO scFv are positioned in that order (the taFv (inverted)-heterodimer Fc-type) (e.g., Example 11). Another example of a more preferable format is the scFv-Fab-heterodimer Fc-type.

[0648] The present invention also encompasses a molecule that comprises an amino acid sequence encoded by a nucleotide sequence included in a polynucleotide which hybridizes, under stringent conditions, to a complementary strand of the polynucleotide comprising a nucleotide sequence encoding an amino acid sequence included in the molecule of the present invention, and that binds to HLA/NY-ESO, and further binds to CD3.

[0649] The present invention also encompasses a molecule that comprises an amino acid sequence exhibiting at least 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% identity to the amino acid sequence included in the molecule of the present invention, and that binds to HLA/NY-ESO, and further binds to CD3.

[0650] The antibody of the present invention, the binding fragment thereof, and the multispecific antibody comprising the same have excellent biological activity, physicochemical properties (hereafter referred to as “physical properties”), safety, pharmacokinetics, and other properties. (a) Examples of biological activity or indicators thereof include antigen-binding activity, in vitro cytotoxicity, and in vivo antitumor activity. For example, the dissociation constant (KD value) to HLA/NY-ESO is 100 nM or lower or 50 nM or lower, preferably 20 nM or lower or 10 nM or lower, and more preferably 5 nM or lower. The EC₅₀ value of cytotoxicity exerted with the use of human peripheral blood mononuclear cells as effector cells to human U266B1 and/or NCI-H1703 cells endogenously expressing NY-ESO is 20 nM or lower, preferably 10 nM or lower, and more preferably 5 nM or lower (in vitro cytotoxicity can be assayed and determined in accordance with the method described in Example 8, although the method is not limited thereto). For example, 0.1 ml of a suspension of human squamous lung cancer cell lines NCI-H1703 (6×10^7 cells/ml) is injected hypodermically to NOG mice, 0.2 ml of a suspension of human peripheral blood mononuclear cells (3.75×10^7 cells/ml) is injected intravenously 4 days later, and antibody administration is initiated 14 days later and performed once a week in 3 instances. Thereafter, the tumor volume is measured. Tumor growth inhibitory activity relative to the control group to which a solvent has been administered is 50% or higher, preferably 75% or higher, and more preferably 90% or higher (in vivo antitumor activity can be assayed and determined in accordance with the method described in Example 9, although the method is not limited thereto). (b) Impurities contained in biopharmaceutical products are associated with drug safety. Thus, it is necessary to establish the adequate standard and to control the amount of impurities at the time of production and during storage. In particular, HMWS (aggregate) is a major impurity and is associated with an immunogenic risk or reduced drug efficacy. Thus, HMWS should be more strictly controlled. For the purpose of impurity control, evaluation should also be performed in respect of stability (whether or not the amount of impurities was increased) with the elapse of time during and after production, as well as at the time of production. Since the effective period of a pharmaceutical product is determined

based on the results of the long-term stability test, a longer effective period can be determined for an antibody exhibiting stability over time. Thus, long-term stability can be used as an indicator to select a preferable antibody for a biopharmaceutical product. Examples of physicochemical properties in the present invention include acid resistance (for inhibition of HMWS production) and solution stability (for inhibition of HMWS production). An example of another indicator is a high yield of a culture product of recombinant cells obtained by introducing a gene encoding the amino acid sequence of the antibody of the present invention, the binding fragment thereof, or the molecule comprising the same into a host cell suitable for production thereof, such as the Expi293F cell. The preferable antibody of the present invention, the antigen-binding fragment thereof, and the multispecific antibody comprising the same with such physicochemical properties are advantageous for the following reasons. That is, a solution containing the antibody, the binding fragment thereof, or the multispecific antibody can be exposed to acidic conditions, and production processes thereof, such as protein A or ion-exchange chromatography and virus inactivation, can be thus performed or facilitated. Also, HMWS production can be controlled at low levels even in the form of a solution of the antibody, the binding fragment thereof, or the multispecific antibody, and, thus, production thereof, drug preparation, distribution or storage of pharmaceutical products comprising the same can be performed and facilitated. Further, pharmaceutical products containing the same can be efficiently produced. Concerning acid resistance, for example, the HMWS content determined by heating at pH 3.5 at room temperature for 1 hour and then measuring HMWS by size exclusion chromatography is 5% or lower, preferably 2% or lower, and more preferably 1% or lower (the HMWS content can be measured and determined for acid resistance evaluation in accordance with the method described in Example 19-1, although the method is not limited thereto). Concerning solution stability, for example, the analyte is dissolved to a concentration of 25 mg/ml in a solution comprising 25 mM histidine and 5% sorbitol at pH 6.0, the resultant is stored at 25° C. for 6 days, and HMWS is then measured by size exclusion chromatography. The HMWS content determined thereby is 20% or lower, and preferably 10% or lower (the HMWS content can be measured and determined for evaluation of solution stability in accordance with the method described in Example 19-2, although the method is not limited thereto). Production efficiency or yield can be measured and determined in accordance with the method described in Examples 20 and 21, although the method is not limited thereto. (c) Examples of safety and indicators thereof include antigen recognition properties and observations at the time of administration. For example, an antibody of interest that recognizes a plurality of amino acids on a wild-type NY-ESO peptide but does not bind to a homologous peptide comprising an amino acid sequence similar to but not identical to the wild-type NY-ESO peptide would produce a low risk of side effects caused by off-target effects. In addition, such an antibody can be predicted to exhibit low immunogenicity in ISPRI web-based immunogenicity screening (EpiVax, Inc) and exhibit a low risk of side effects such as cytokine production. Also, NYF-0023, NYF-0045, NYF-0047, NYF-0048, NYF-0060, NYF-0061, NYZ-0082, or NYZ-1010 included in the bispecific antibody of the present invention were administered to Balb/c mice. As a

result, no significant problems were observed in the half-life in blood, and body weight loss or other apparent toxicity problems were not observed. As a result of single administration of NYZ-0082 or NYZ-1010 to cynomolgus monkeys, no significant problems were observed in the half-life in blood, and no changes resulting from administration were observed in terms of general conditions, body weight, the amount of feed intake, body temperature, and plasma cytokine levels. (d) Examples of kinetics or indicators thereof include the half-life in blood. In the present invention, no significant problems were observed in the half-life in blood as a result of administration of several bispecific antibodies to Balb/c mice or cynomolgus monkeys. The antibody of the present invention, the binding fragment thereof, and molecules provided with such excellent biological activity, physicochemical properties, safety, and kinetics can be preferably integrated into a pharmaceutical composition. Preferable examples of the antibody of the present invention or the antigen-binding fragment thereof having the antigen-binding activity described in (a) and various properties described in (b) include, but are not limited to, NYA-1143, NYA-1163, NYA-2023, NYA-2027, NYA-2035, NYA-2044, NYA-2045, NYA-2047, NYA-2048, NYA-2060, NYA-2031, NYA-2047, NYA-2061, NYA-2143, and NYA-3061. More preferable examples thereof include, but are not limited to, NYA-2047, NYA-2061, NYA-2143, and NYA-3061. Preferable examples of the multispecific antibody of the present invention having the properties (a) to (d) include, but are not limited to, NYF-0016, NYF-0019, NYF-0022, NYF-0023, NYF-0027, NYF-0035, NYF-0044, NYF-0045, NYF-0047, NYF-0048, NYF-00058, NYF-0060, NYF-0061, NYZ-0038, NYZ-0082, NYZ-0088, and NYZ-1010. More preferable examples thereof include, but are not limited to, NYF-0061, NYZ-0038, NYZ-0082, NYZ-0088, NYZ-1007, NYZ-1010, and NYZ-1017.

[0651] In the present invention, a “site” to which an antibody binds; i.e., a “site” that is recognized by an antibody, is a partial peptide or a partial higher-order structure of an antigen to which an antibody binds or which is recognized by the antibody. In the present invention, such a site is also referred to as an epitope or an antibody binding site. Examples of sites to which the anti-HLA/NY-ESO antibody according to the present invention binds or sites that are recognized thereby on HLA/NY-ESO include a plurality of amino acids and partial higher-order structures in the HLA/NY-ESO peptide.

[0652] The “antibody or the binding fragment thereof that binds to the same site” as the antibody according to the present invention or the binding fragment thereof is also within the scope of the present invention. The “antibody that binds to the same site” as a given antibody is another antibody that binds to a site on an antigen molecule recognized by such given antibody. If a second antibody binds to a partial peptide or partial conformation on an antigen molecule to which a first antibody binds, the first antibody and the second antibody can be evaluated to bind to the same site. When the first antibody of the present invention has the antigen-binding activity described in (a) above, the second antibody that binds to the same site on HLA/NY-ESO is highly likely to have similar activity, and such second antibody is thus within the scope of the present invention. An antibody that binds to HLA/NY-ESO competitively with the first antibody of the present invention having the antigen-binding activity described in (a) is within the scope of

the present invention. An antibody that binds to a site on HLA/NY-ESO recognized by the monoclonal antibody of the present invention, an antibody that binds to HLA/NY-ESO competitively with the monoclonal antibody of the present invention, and binding fragments thereof preferably have at least one of in vitro cytotoxicity and in vivo antitumor activity described in (a) and properties described in (b) to (d), more preferably 3 or more thereof, and optimally all thereof.

[0653] A binding site of the antibody can be determined by a method well known to a person skilled in the art, such as an immunoassay technique. For example, a series of peptides are prepared by removing an antigen amino acid sequence from the C or N terminus as appropriate, reactivity of an antibody thereto is examined, a recognition site is roughly determined, a shorter peptide sequence is synthesized, and reactivity of an antibody thereto is examined. Thus, the binding site can be determined. Alternatively, a particular site or region of an amino acid sequence of an antigen or an antigen fragment peptide is deleted or substituted with another amino acid sequence, or a mutation is introduced into such site or region. By examining reactivity of an antibody to such peptide, the binding site can be determined. An antigen fragment peptide can be prepared by, for example, genetic recombination or peptide synthesis.

[0654] When an antibody binds to or recognizes a partial higher-order structure of an antigen, such antigen-binding site of the antibody can be determined by identifying an amino acid residue on an antigen adjacent to the antibody via x-ray structural analysis. For example, an antibody or a fragment thereof and an antigen or a fragment thereof are bound to each other, crystallized, and subjected to structural analysis. Thus, an amino acid residue on an antigen being at an interaction distance with the antibody can be identified. The interaction distance is 8 Å or less, preferably 6 Å or less, and more preferably 4 Å or less. One or more amino acid residues being at such an interaction distance with the antibody can constitute the antigen-binding site of the antibody (epitope). When there are two or more such amino acid residues, such amino acid residues may not be adjacent to each other on a one-dimensional sequence.

[0655] The anti-HLA/NY-ESO antibody according to the present invention or the binding fragment thereof specifically recognizes a plurality of amino acids in the HLA/NY-ESO amino acid sequence. An antibody or a binding fragment thereof that recognizes a plurality of amino acids, an antibody or a binding fragment thereof that binds to HLA/NY-ESO competitively with the antibody according to the present invention or the binding fragment thereof, and an antibody or a binding fragment thereof that has an interaction distance with such a plurality of amino acids are within the scope of the present invention. A multispecific antibody comprising such antibody or binding fragment thereof is also within the scope of the present invention.

6. Pharmaceutical Composition

[0656] The present invention encompasses an anti-cancer agent comprising, as an active ingredient, the anti-HLA/NY-ESO antibody according to the present invention or a multispecific molecule that binds to HLA/NY-ESO.

[0657] The anti-cancer agent of the present invention can be used for one type or two or more types of cancer species selected from carcinoma, sarcoma, lymphoma, leukemia, myeloma, germinoma, brain tumor, carcinoid, neuroblas-

toma, retinoblastoma, and nephroblastoma. Specific examples of carcinoma include kidney cancer, melanoma, squamous cell cancer, basal cell cancer, conjunctival cancer, oral cavity cancer, laryngeal cancer, pharyngeal cancer, thyroid gland cancer, lung cancer (non-small cell lung cancer (adenocarcinoma, epidermoid cancer, large cell cancer), and small cell lung cancer), breast cancer, esophageal cancer, gastric cancer, duodenal cancer, small bowel cancer, large bowel cancer, rectal cancer, appendiceal cancer, anal cancer, liver cancer, gallbladder cancer, bile duct cancer, pancreatic cancer, adrenal cancer, bladder cancer, prostate cancer, uterine cancer, and vaginal cancer. Specific examples of sarcoma include liposarcoma, angiosarcoma, chondrosarcoma, rhabdomyosarcoma, Ewing's sarcoma, osteosarcoma, undifferentiated pleomorphic sarcoma, myxofibrosarcoma, malignant peripheral neurilemmoma, retroperitoneal sarcoma, synoviosarcoma, uterine sarcoma, gastrointestinal stromal tumor, leiomyosarcoma, and epithelioid sarcoma. Specific examples of lymphoma include B-cell lymphoma, T/NK-cell lymphoma, and Hodgkin's lymphoma. Specific examples of leukemia include myelogenic leukemia, lymphatic leukemia, myeloproliferative disorder, and myelodysplastic syndrome. Specific examples of myeloma include multiple myeloma. Specific examples of germinoma include testicular cancer and ovarian cancer. Specific examples of brain tumor include neuroglioma and meningioma.

[0658] The anti-cancer agent of the present invention can contain the anti-HLA/NY-ESO antibody or a multispecific molecule that binds to HLA/NY-ESO in an amount effective for treatment, as well as pharmaceutically acceptable carriers, diluents, solubilizers, emulsifiers, preservatives, aids, and the like. The "pharmaceutically acceptable carriers" and the like can be suitably selected from a broad range according to the type of target disease and the dosage form of a drug. An administration method for the anti-cancer agent of the present invention can be suitably selected. For example, the anti-cancer agent can be injected, and local injection, intraperitoneal injection, selective intravenous infusion, intravenous injection, subcutaneous injection, organ perfusate infusion, and the like can be employed. Further, an injection solution can be formulated using a carrier comprising a salt solution, a glucose solution, or a mixture of salt water and a glucose solution, various types of buffer solutions, or the like. Further, a powder may be formulated and mixed with a liquid carrier to prepare an injection solution before use.

[0659] Other administration methods can be suitably selected along with development of a formulation. For example, oral solutions, powders, pills, capsules, tablets, and the like can be applied for oral administration. For oral solutions, oral liquid preparations such as suspensions and syrups can be produced using water, saccharides such as sucrose, sorbitol, and fructose, glycols such as PEG, oils such as sesame oil and soybean oil, preservatives such as alkyl parahydroxybenzoates, flavors such as strawberry flavor and peppermint, and the like. Powders, pills, capsules, and tablets can be formulated using excipients such as lactose, glucose, sucrose, and mannitol, disintegrating agents such as starch and alginate soda, lubricants such as magnesium stearate and talc, binders such as polyvinyl alcohol, hydroxypropyl cellulose, and gelatin, surfactants such as fatty acid esters, plasticizers such as glycerin, and the like. Tablets and capsules are preferred unit dosage forms

for the composition of the present invention in that they are easily administered. Solid production carriers are used to produce tablets and capsules.

[0660] The effective dose of the anti-HLA/NY-ESO antibody or a multispecific molecule that binds to HLA/NY-ESO used for treatment can be changed according to the characteristics of the symptoms to be treated and the patient's age and condition and can be finally determined by a physician. For example, one dose is 0.0001 mg to 100 mg per kg of body weight. The predetermined dose may be administered once every one to 180 days, or the dose may be divided into two doses, three doses, four doses, or more doses per day at appropriate intervals.

[0661] The present invention also encompasses a polynucleotide encoding an amino acid sequence included in the anti-HLA/NY-ESO antibody according to the present invention or a multispecific molecule binding to HLA/NY-ESO, a vector comprising such a polynucleotide, a cell comprising such a polynucleotide or vector, and a pharmaceutical composition comprising, as an active ingredient, any of the polynucleotide, the vector and the cell. Such a pharmaceutical composition is preferably an anti-cancer agent.

[0662] In addition, the pharmaceutical composition of the present invention can be used in combination with other therapeutic agents or therapeutic techniques. Examples of other therapeutic agents or therapeutic techniques include, but are not particularly limited to, chemotherapeutic agents, radiation therapy, and biopharmaceutical products. The pharmaceutical composition of the present invention and other therapeutic agents can be administered simultaneously or according to a different schedule in the form of a single formulation or two or more different formulations containing the pharmaceutical composition of the present invention. Administration of the pharmaceutical composition can be performed in combination with other therapeutic techniques.

EXAMPLES

[0663] Hereafter, the present invention is described in greater detail with reference to the examples, although the present invention is not limited to these examples.

[0664] In the following examples, genetic engineering procedures were performed in accordance with the method described in Molecular Cloning (Sambrook, J., Fritsch, E. F. and Maniatis, T., Cold Spring Harbor Laboratory Press, 1989) and methods described in other experimental protocols employed by a person skilled in the art, unless otherwise specified. When a commercially available reagent or kit was to be used, the procedures in accordance with the instructions of such a commercial product were employed. Synthesis of primers required for gene synthesis or vector construction was outsourced, according to need (Fasmac Co., Ltd., Thermo Fisher Scientific, and Eurofins Genomics).

(Example 1) Acquisition of Human Antibody Phage Library-Derived Anti-HLA/NY-ESO Antibody

1)-1 Preparation of HLA/NY-ESO Antigen Protein

[0665] Inclusion bodies prepared from *E. coli* cells (BL21 (DE3), Agilent Technologies) expressing the HLA-A*0201 (GenBank: ASA47534.1) truncate (biotin ligase recognition sequence added: SEQ ID NO: 33) and β 2-microglobulin (UniProtKB-P61769: SEQ ID NO: 34) and the NY-ESO

peptide: SLLMWITQC (SEQ ID NO: 1 of Sequence Listing) were subjected to refolding at a high dilution factor, and the HLA-A*0201/ β 2-microglobulin/NY-ESO peptide complex (hereafter, referred to as “HLA/NY-ESO”) was then prepared using gel filtration columns (Superdex 200 10/300, GE Healthcare).

[0666] As a negative control antigen, the HLA-A*0201/ β 2-microglobulin/MAGEC-1 peptide complex (hereafter, referred to as “HLA/MC1”) was prepared by refolding using the MAGEC-1 peptide: ILFGISLREV (SEQ ID NO: 2 of Sequence Listing). Further, the prepared HLA/NY-ESO and HLA/MC1 were biotinylated with *E. coli* biotin ligase and then fractionated using gel filtration columns (Superdex 200 10/300, GE Healthcare) to prepare biotinylated proteins.

1)-2 Isolation of scFv Capable of Binding to HLA/NY-ESO

[0667] scFv binding to HLA/NY-ESO was isolated from the human antibody phage library. At the outset, phages were added to the biotinylated HLA/MC1-immobilized Dynabeads Streptavidin M-280 (Thermo Fisher Scientific), and unbound phages were collected. Subsequently, phages were added to the biotinylated HLA/NY-ESO-immobilized Dynabeads Streptavidin M-280Ag, and unbound phages were removed via washing using a magnet stand (DynaMag-2, Thermo Fisher Scientific).

[0668] Thereafter, *E. coli* (XL-1 Blue, Agilent Technologies) was infected with phages bound to HLA/NY-ESO, and the phages were collected and amplified. After a panning procedure was performed 3 times in total, the phages were transferred from the polyclonal phagemid to the *E. coli* expression vector comprising FLAG tag and a His tag added to the carboxyl terminus of scFv, the *E. coli* was transformed, scFv was expressed in the presence of IPTG (isopropyl- β -D-thiogalactopyranoside) (Sigma-Aldrich), and the expressed scFv was subjected to screening via ELISA.

1)-3 Screening of HLA/NY-ESO-Bound scFv Via ELISA

[0669] NeutrAvidin (Life Technologies) diluted to 1 μ g/ml in PBS (0.01 M phosphate buffered saline containing 0.138 M sodium chloride and 0.0027 M potassium chloride (pH7.4), Sigma-Aldrich) was added in an amount of 50 μ l each to a 384-well Maxi-sorp plate (Black, Nunc), and the plate was allowed to stand at 4° C. overnight for immobilization. The plate was washed 3 times with PBS (ELISA buffer) containing 0.05% Tween-20 (BioRad), biotinylated HLA/NY-ESO, which was diluted to 1 μ g/ml in PBS and used in Example 1)-2, was added, and the resultant was agitated at room temperature for 1 hour. The plate was washed 3 times with ELISA buffer, blocked with Blocker Casein (Thermo Fisher Scientific), and then washed 3 times with ELISA buffer. Thereafter, a culture solution of scFv-expressing *E. coli* was added, and the reaction was allowed to proceed at room temperature for 2 hours. After the resultant was washed 3 times with ELISA buffer, 50 μ l of the horseradish peroxidase (HRP)-labeled anti-FLAG antibody (Sigma-Aldrich) diluted to 5,000-fold with ELISA buffer was added, and the reaction was allowed to proceed at room temperature for 1 hour. After the resultant was washed 5 times with ELISA buffer, the SuperSignal Pico ELISA Chemiluminescent substrate (Thermo Fisher Scientific) was added, chemiluminescence 10 minutes later was assayed using a plate reader (Envision 2104 Multilabel Reader, Perkin Elmer), and positive clones in HLA/NY-ESO-binding ELISA were then selected.

1)-4 Determination of Nucleotide Sequence and Amino Acid Sequence of ELISA-Positive Clone NY-R119

[0670] From among the ELISA-positive clones obtained in 1)-3, NY-R119 was selected as scFv exhibiting high binding affinity to HLA/NY-ESO and excellent recognition specificity. The nucleotide sequences of the heavy chain and light chain variable regions of NY-R119 were analyzed by the Dye Terminator method (BigDye®, Terminator v3.1, Thermo Fisher Scientific). Sequences of the primers used for sequence analysis are as demonstrated below.

[0671] Primer A: 5'-CTCTTCGCTAT-TACGCCAGCTGGCGA-3' (SEQ ID NO: 3 of Sequence Listing (FIG. 10))

[0672] Primer B: 5'-ATAACAATTTACACAG-GAAACAGCTATGA-3' (SEQ ID NO: 4 of Sequence Listing (FIG. 11))

[0673] SEQ ID NO: 5 (FIG. 12) shows the determined nucleotide sequence of cDNA encoding the heavy chain variable region of NY-R119 and SEQ ID NO: 6 (FIG. 13) shows the determined amino acid sequence thereof.

[0674] SEQ ID NO: 7 (FIG. 14) shows the determined nucleotide sequence of cDNA encoding the light chain variable region of NY-R119 and SEQ ID NO: 8 (FIG. 15) shows the determined amino acid sequence thereof.

[0675] The CDR sequences of NY-R119 in accordance with the definition of CDR provided by IMGT are represented as follows: CDRH1: SEQ ID NO: 54 (FIG. 61); CDRH2: SEQ ID NO: 55 (FIG. 61); CDRH3: SEQ ID NO: 56 (FIG. 61); CDRL1: SEQ ID NO: 57 (FIG. 61); CDRL2: SEQ ID NO: 58 (FIG. 61); and CDRL3: SEQ ID NO: 59 (FIG. 61).

1)-5 Preparation of NYA-0001

1)-5-1 Construction of NYA-0001 Expression Vector

[0676] In order to prepare various evaluation samples, an NY-R119 expression vector for mammalian cells was constructed. NY-R119 expressed in a cultured mammalian cell was designated as “NYA-0001.” The amino acid sequences constituting the scFv region, the heavy chain and light chain variable regions, and CDRH1 to 3 and CDRL1 to 3 of NY-R119 are identical to those of NYA-0001. A DNA fragment encoding NYA-0001 was inserted into an expression vector for mammalian cells comprising pcDNA3.3 (Thermo Fisher Scientific) as the backbone using the In-Fusion HD cloning kit (CLONTECH) to construct the NYA-0001 expression vector.

[0677] The nucleotide sequence of the constructed NYA-0001 expression vector was reanalyzed, and the nucleotide sequence of the full-length NYA-0001 was found to be the nucleotide sequence as shown in SEQ ID NO: 69 of Sequence Listing (FIG. 70). On the basis of the nucleotide sequence, in addition, the amino acid sequence of the entire NYA-0001 encoded thereby was found to be the amino acid sequence as shown in SEQ ID NO: 70 (FIG. 71). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 266 constitute NYA-0001, and amino acids 267 to 292 constitute a Flag-His tag. Also, amino acids 46 to 53 constitute CDRH1, amino acids 71 to 78 constitute CDRH2, and amino acids 117 to 129 constitute CDRH3. Also, amino acids 181 to 188 constitute CDRL1, amino acids 206 to 208 constitute CDRL2, and amino acids 245 to 256 constitute CDRL3.

1)-5-2 Expression and Purification of NYA-0001

[0678] Expi293F cells (Thermo Fisher Scientific) were subcultured in accordance with the instructions. A culture solution of the Expi293F cells in a logarithmic growth phase was diluted to 2.5×10^6 cells/ml in the Expi293 Expression medium (Thermo Fisher Scientific) and used for NYA-0001 production. The NYA-0001 expression vector (0.3 mg) and 0.9 mg of polyethyleneimine (Polyscience #24765) were added to 20 ml of Opti-Pro SFM medium (Thermo Fisher Scientific). The mixture was stirred gently, allowed to stand for five minutes, and then added to the Expi293F cells. The culture supernatant obtained by agitation culture in an incubator at 37° C. in the presence of 8% CO₂ at 135 rpm for 6 days was filtered through a 0.2 µm-filter (Millipore). Thus, the culture supernatant of NYA-0001 was obtained. Purification was carried out by elution and concentration using Ni Sepharose excel (GE Healthcare) and then filtration through a gel filtration column (Superdex 200 Increase, GE Healthcare) equilibrated with 25 mM histidine, 300 mM NaCl, and 5% Sorbitol, pH6.0. Purified protein samples were subjected to analytical size exclusion chromatography (SEC), the degree of purification and the concentration were determined, and the samples were then subjected to various types of evaluation.

(Example 2) Preparation of NYA-0001 Mutant

2)-1 Acquisition of NYA-0001 Mutant

[0679] A phage library was constructed by a method in which the NYA-0001 gene was used as a template and a mutation was introduced via PCR (i.e., the error-prone-based library) or a method in which an oligomer was synthesized through random mutation of 20 types of amino acids for each of all CDR residues (i.e., the oligo-based library), clones with high binding affinity were screened for, NYA-0060, NYA-0068, and NYA-0082 were obtained as mutants with high binding affinity, and the nucleotide sequences thereof were then determined.

[0680] SEQ ID NO: 9 (FIG. 16) shows the nucleotide sequence of cDNA encoding the heavy chain variable region of NYA-0060 and SEQ ID NO: 10 (FIG. 17) shows the amino acid sequence thereof.

[0681] SEQ ID NO: 11 (FIG. 18) shows the nucleotide sequence of cDNA encoding the light chain variable region of NYA-0060 and SEQ ID NO: 12 (FIG. 19) shows the amino acid sequence thereof.

[0682] SEQ ID NO: 13 (FIG. 20) shows the nucleotide sequence of cDNA encoding the heavy chain variable region of NYA-0068 and SEQ ID NO: 14 (FIG. 21) shows the amino acid sequence thereof.

[0683] SEQ ID NO: 15 (FIG. 22) shows the nucleotide sequence of cDNA encoding the light chain variable region of NYA-0068 and SEQ ID NO: 16 (FIG. 23) shows the amino acid sequence thereof.

[0684] SEQ ID NO: 17 (FIG. 24) shows the nucleotide sequence of cDNA encoding the heavy chain variable region of NYA-0082 and SEQ ID NO: 18 (FIG. 25) shows the amino acid sequence thereof.

[0685] SEQ ID NO: 19 (FIG. 26) shows the nucleotide sequence of cDNA encoding the light chain variable region of NYA-0082 and SEQ ID NO: 20 (FIG. 27) shows the amino acid sequence thereof.

2)-2 Preparation of Mutants with High Binding Affinity Using the Sites of Mutation and Combination Identified in NYA-0060, NYA-0068, and NYA-0082

[0686] DNA fragments encoding NYA-1163 and NYA-2023 with sites of mutation of NYA-0060 and NYA-0068 were inserted into an expression vector for mammal cells comprising pcDNA3.3 (Thermo Fisher Scientific) as the backbone using the In-Fusion HD cloning kit (CLONTECH) to construct the scFv expression vectors for mammalian cell culture.

[0687] According to the sites of mutation and combination thereof identified in NYA-0060, NYA-0068, and NYA-0082, so NYA-2027, NYA-1143, and NYA-2143 were designed. DNA fragments encoding target scFv were inserted into the NYA-0001 expression vector constructed in 1)-5 via site-directed mutagenesis or an expression vector for mammal cells comprising pcDNA3.3 (Thermo Fisher Scientific) as the backbone using the In-Fusion HD cloning kit (CLONTECH) to construct the scFv expression vectors for mammal cell culture.

[0688] The nucleotide sequences of the constructed scFv expression vectors were reanalyzed, and the nucleotide sequence of the full-length NYA-1163 was found to be the nucleotide sequence as shown in SEQ ID NO: 21 of Sequence Listing (FIG. 28).

[0689] The nucleotide sequence of the full-length NYA-2023 was found to be the nucleotide sequence as shown in SEQ ID NO: 22 of Sequence Listing (FIG. 29).

[0690] The nucleotide sequence of the full-length NYA-2027 was found to be the nucleotide sequence as shown in SEQ ID NO: 23 of Sequence Listing (FIG. 30).

[0691] The nucleotide sequence of the full-length NYA-1143 was found to be the nucleotide sequence as shown in SEQ ID NO: 24 of Sequence Listing (FIG. 31).

[0692] The nucleotide sequence of the full-length NYA-2143 was found to be the nucleotide sequence as shown in SEQ ID NO: 25 of Sequence Listing (FIG. 32).

[0693] On the basis of the nucleotide sequences above, in addition, the amino acid sequences of the full-length NYA-1163, NYA-2023, NYA-2027, NYA-1143, and NYA-2143 encoded thereby were identified.

[0694] The amino acid sequence of the full-length NYA-1163 is the amino acid sequence as shown in SEQ ID NO: 26 of the Sequence Listing (FIG. 33). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 266 constitute NYA-1163, and amino acids 267 to 292 constitute a Flag-His tag. Also, amino acids 46 to 53 constitute CDRH1, amino acids 71 to 78 constitute CDRH2, and amino acids 117 to 129 constitute CDRH3. Also, amino acids 181 to 188 constitute CDRL1, amino acids 206 to 208 constitute CDRL2, and amino acids 245 to 256 constitute CDRL3.

[0695] The amino acid sequence of the full-length NYA-2023 is the amino acid sequence as shown in SEQ ID NO: 27 of the Sequence Listing (FIG. 34). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 266 constitute NYA-2023, and amino acids 267 to 292 constitute a Flag-His tag. Also, amino acids 46 to 53 constitute CDRH1, amino acids 71 to 78 constitute CDRH2, and amino acids 117 to 129 constitute CDRH3. Also, amino acids 181 to 188 constitute CDRL1, amino acids 206 to 208 constitute CDRL2, and amino acids 245 to 256 constitute CDRL3.

[0696] The amino acid sequence of the full-length NYA-2027 is the amino acid sequence as shown in SEQ ID NO: 28 of the Sequence Listing (FIG. 35). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 266 constitute NYA-2027, and amino acids 267 to 292 constitute a Flag-His tag. Also, amino acids 46 to 53 constitute CDRH1, amino acids 71 to 78 constitute CDRH2, and amino acids 117 to 129 constitute CDRH3. Also, amino acids 181 to 188 constitute CDRL1, amino acids 206 to 208 constitute CDRL2, and amino acids 245 to 256 constitute CDRL3.

[0697] The amino acid sequence of the full-length NYA-1143 is the amino acid sequence as shown in SEQ ID NO: 26 of the Sequence Listing (FIG. 36). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 266 constitute NYA-1143, and amino acids 267 to 292 constitute a Flag-His tag. Also, amino acids 46 to 53 constitute CDRH1, amino acids 71 to 78 constitute CDRH2, and amino acids 117 to 129 constitute CDRH3. Also, amino acids 181 to 188 constitute CDRL1, amino acids 206 to 208 constitute CDRL2, and amino acids 245 to 256 constitute CDRL3.

[0698] The amino acid sequence of the full-length NYA-2143 is the amino acid sequence as shown in SEQ ID NO: 30 of the Sequence Listing (FIG. 37). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 266 constitute NYA-2143, and amino acids 267 to 292 constitute a Flag-His tag. Also, amino acids 46 to 53 constitute CDRH1, amino acids 71 to 78 constitute CDRH2, and amino acids 117 to 129 constitute CDRH3. Also, amino acids 181 to 188 constitute CDRL1, amino acids 206 to 208 constitute CDRL2, and amino acids 245 to 256 constitute CDRL3.

[0699] All the CDR sequences of NYA-1163 are identical to the CDR sequences of NYA-0001. The CDR sequences of NYA-1143, NYA-2143, and NYA-2023 are identical to each other, and these sequences are represented as follows: CDRH1: SEQ ID NO: 54 (FIG. 61); CDRH2: SEQ ID NO: 55 (FIG. 61); CDRH3: SEQ ID NO: 56 (FIG. 61); CDRL1: SEQ ID NO: 60 (FIG. 62); CDRL2: SEQ ID NO: 58 (FIG. 61); and CDRL3: SEQ ID NO: 59 (FIG. 61). The CDR sequences of NYA-2027 are represented as follows: CDRH1: SEQ ID NO: 54 (FIG. 61); CDRH2: SEQ ID NO: 55 (FIG. 61); CDRH3: SEQ ID NO: 56 (FIG. 61); CDRL1: SEQ ID NO: 57 (FIG. 61); CDRL2: SEQ ID NO: 58 (FIG. 61); and CDRL3: SEQ ID NO: 61 (FIG. 63).

[0700] Also, NYA-1154 comprising sites of binding and mutation observed at the time of screening of clones with high binding affinity in combination was designed and introduced into the NYA-0001 expression vector via site-directed mutagenesis to construct the NYA-1154 expression vector comprising pcDNA3.3 (Thermo Fisher Scientific) as the backbone. The nucleotide sequences of the constructed scFv expression vectors were reanalyzed and found to be the nucleotide sequence as shown in SEQ ID NO: 31 (FIG. 38). On the basis of the nucleotide sequences above, the amino acid sequence of the full-length NYA-1154 encoded thereby was identified (SEQ ID NO: 32 (FIG. 39)). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 266 constitute NYA-1154, and amino acids 267 to 292 constitute a Flag-His tag. Also, amino acids 46 to 53 constitute CDRH1, amino acids 71 to 78 constitute CDRH2, and amino acids 117 to 129 constitute CDRH3. Also, amino acids 181 to 188 constitute CDRL1,

amino acids 206 to 208 constitute CDRL2, and amino acids 245 to 256 constitute CDRL3.

[0701] The CDR sequences of NYA-1154 are represented as follows: CDRH1: SEQ ID NO: 54 (FIG. 61); CDRH2: SEQ ID NO: 55 (FIG. 61); CDRH3: SEQ ID NO: 62 (FIG. 64); CDRL1: SEQ ID NO: 57 (FIG. 61); CDRL2: SEQ ID NO: 58 (FIG. 61); and CDRL3: SEQ ID NO: 63 (FIG. 64).

[0702] NYA-1163, NYA-2023, NYA-2027, NYA-1143, NYA-2143, and NYA-1154 were expressed in the same manner as in 1)-5-2 to purify the target scFv constructs. Purified protein samples were subjected to analytical SEC, the degree of purification and the concentration were determined, and the samples were then subjected to various assays.

2)-3 Preparation of NYA-1143 Mutant

2)-3-1 Preparation of NYA-2035

[0703] NYA-2035 was designed as an NYA-1143 mutant and introduced into the NYA-1143 expression vector for mammal cell culture via site-directed mutagenesis to construct an NYA-2035 expression vector. The nucleotide sequence of the constructed scFv expression vector was reanalyzed and found to be the nucleotide sequence as shown in SEQ ID NO: 35 (FIG. 42). On the basis of the nucleotide sequences above, the nucleotide sequence of the full-length NYA-2025 encoded thereby was determined (SEQ ID NO: 36 (FIG. 43)). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 266 constitute NYA-2035, and amino acids 267 to 292 constitute a Flag-His tag. Also, amino acids 46 to 53 constitute CDRH1, amino acids 71 to 78 constitute CDRH2, and amino acids 117 to 129 constitute CDRH3. Also, amino acids 181 to 188 constitute CDRL1, amino acids 206 to 208 constitute CDRL2, and amino acids 245 to 256 constitute CDRL3. The CDR sequences of NYA-2035 in accordance with the definition of CDR provided by IMGT are represented as follows: CDRH1: SEQ ID NO: 54 (FIG. 61); CDRH2: SEQ ID NO: 55 (FIG. 61); CDRH3: SEQ ID NO: 56 (FIG. 61); CDRL1: SEQ ID NO: 64 (FIG. 65); CDRL2: SEQ ID NO: 58 (FIG. 61); and CDRL3: SEQ ID NO: 59 (FIG. 61).

2)-3-2 Preparation of NYA-1143 CDR Graft Mutant

[0704] In order to improve physical properties, the NYA-1143 CDR graft mutant was designed. The VH and VL framework sequences of NYA-1143 were compared with the framework sequences of the human subgroup consensus sequence or the germline sequence defined in KABAT et al. (Sequences of Proteins of Immunological Interest, 5th Ed., Public Health Service National Institutes of Health, Bethesda, MD., 1991). As a result, a consensus sequence between the human germline sequence IGHV3_30*15 and the human γ chain subgroup 3 was selected for VH and the human germline sequence IGLV1-44*01 was selected for VL as acceptors having a high identity in the framework regions. Subsequently, amino acid residues in the framework region of an acceptor were aligned with the amino acid residues of NYA-1143 to identify different amino acid residues.

[0705] Subsequently, a three-dimensional model of NYA-1143 was used to select framework residues to be transferred onto an acceptor with reference to the criteria provided by

Queen et al. (Proc. Natl. Acad. Sci. U.S.A., 86, 10029-10033, 1989). In accordance with the technique described above, CDR graft mutant amino acid sequences of the VH regions of NYA-1143; i.e., NYA-1143-VH01, NYA-1143-VH02, and NYA-1143-VH03, were designed. Such amino acid sequences are as shown in SEQ ID NO: 37 (FIG. 44), SEQ ID NO:38 (FIG. 45), and SEQ ID NO:39 (FIG. 46). As a CDR graft mutant amino acid sequence of a VL region of NYA-1143, NYA-1143-VL01 was designed. Such an amino acid sequence is as shown in SEQ ID NO: 40 (FIG. 47).

[0706] Various scFv constructs were designed using the VH and VL sequences in combination. scFv resulting from substitution of the VH region of NYA-1143 with the amino acid sequence of NYA-1143-VH01 was designated as NYA-2044. scFv resulting from substitution of the VL region of NYA-2044 with the amino acid sequence of NYA-1143-VL01 was designated as NYA-2045.

[0707] scFv resulting from substitution of the VH region of NYA-1143 with the amino acid sequence of NYA-1143-VH02 was designated as NYA-2047. scFv resulting from substitution of the VL region of NYA-2047 with the amino acid sequence of NYA-1143-VL01 was designated as NYA-2048.

[0708] scFv resulting from substitution of the VH region of NYA-1143 with the amino acid sequence of NYA-1143-VH03 was designated as NYA-2060. scFv resulting from substitution of the VL region of NYA-2060 with the amino acid sequence of NYA-1143-VL01 was designated as NYA-2061.

[0709] The DNA fragments of the various designed NYA-1143 CDR graft mutants were fully synthesized (Fasmac Co., Ltd.) and bound to each other using the In-Fusion HD Cloning kit (CLONTECH) to construct the scFv expression vector for mammal cells comprising pcDNA3.3 (Thermo Fisher Scientific) as the backbone.

[0710] The nucleotide sequences of the constructed scFv expression vectors were reanalyzed, the nucleotide sequence of the full-length NYA-2044 was found to be the nucleotide sequence as shown in SEQ ID NO: 41 of the Sequence Listing (FIG. 48).

[0711] The nucleotide sequence of the full-length NYA-2045 was found to be the nucleotide sequence as shown in SEQ ID NO: 42 of the Sequence Listing (FIG. 49).

[0712] The nucleotide sequence of the full-length NYA-2047 was found to be the nucleotide sequence as shown in SEQ ID NO: 43 of the Sequence Listing (FIG. 50).

[0713] The nucleotide sequence of the full-length NYA-2048 was found to be the nucleotide sequence as shown in SEQ ID NO: 44 of the Sequence Listing (FIG. 51).

[0714] The nucleotide sequence of the full-length NYA-2060 was found to be the nucleotide sequence as shown in SEQ ID NO: 45 of the Sequence Listing (FIG. 52).

[0715] The nucleotide sequence of the full-length NYA-2061 was found to be the nucleotide sequence as shown in SEQ ID NO: 46 of the Sequence Listing (FIG. 53).

[0716] On the basis of the nucleotide sequences above, the amino acid sequences of the full-length NYA-2044, NYA-2045, NYA-2047, NYA-2048, NYA-2060, and NYA-2061 encoded thereby were identified. The CDR sequences of NYA-2044, NYA-2045, NYA-2047, NYA-2048, NYA-2060, and NYA-2061 are identical to those of NYA-1143.

[0717] The amino acid sequence of the full-length NYA-2044 is the amino acid sequence as shown in SEQ ID NO: 47 of the Sequence Listing (FIG. 54). In such an amino acid

sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 266 constitute NYA-2044, and amino acids 267 to 292 constitute a Flag-His tag. Also, amino acids 46 to 53 constitute CDRH1, amino acids 71 to 78 constitute CDRH2, and amino acids 117 to 129 constitute CDRH3. Also, amino acids 181 to 188 constitute CDRL1, amino acids 206 to 208 constitute CDRL2, and amino acids 245 to 256 constitute CDRL3.

[0718] The amino acid sequence of the full-length NYA-2045 is the amino acid sequence as shown in SEQ ID NO: 48 of the Sequence Listing (FIG. 55). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 266 constitute NYA-2045, and amino acids 267 to 292 constitute a Flag-His tag. Also, amino acids 46 to 53 constitute CDRH1, amino acids 71 to 78 constitute CDRH2, and amino acids 117 to 129 constitute CDRH3. Also, amino acids 181 to 188 constitute CDRL1, amino acids 206 to 208 constitute CDRL2, and amino acids 245 to 256 constitute CDRL3.

[0719] The amino acid sequence of the full-length NYA-2047 is the amino acid sequence as shown in SEQ ID NO: 50 of the Sequence Listing (FIG. 57). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 266 constitute NYA-2047, and amino acids 267 to 292 constitute a Flag-His tag. Also, amino acids 46 to 53 constitute CDRH1, amino acids 71 to 78 constitute CDRH2, and amino acids 117 to 129 constitute CDRH3. Also, amino acids 181 to 188 constitute CDRL1, amino acids 206 to 208 constitute CDRL2, and amino acids 245 to 256 constitute CDRL3.

[0720] The amino acid sequence of the full-length NYA-2048 is the amino acid sequence as shown in SEQ ID NO: 51 of the Sequence Listing (FIG. 58). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 266 constitute NYA-2048, and amino acids 267 to 292 constitute a Flag-His tag. Also, amino acids 46 to 53 constitute CDRH1, amino acids 71 to 78 constitute CDRH2, and amino acids 117 to 129 constitute CDRH3. Also, amino acids 181 to 188 constitute CDRL1, amino acids 206 to 208 constitute CDRL2, and amino acids 245 to 256 constitute CDRL3.

[0721] The amino acid sequence of the full-length NYA-2060 is the amino acid sequence as shown in SEQ ID NO: 52 of the Sequence Listing (FIG. 59). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 266 constitute NYA-2060, and amino acids 267 to 292 constitute a Flag-His tag. Also, amino acids 46 to 53 constitute CDRH1, amino acids 71 to 78 constitute CDRH2, and amino acids 117 to 129 constitute CDRH3. Also, amino acids 181 to 188 constitute CDRL1, amino acids 206 to 208 constitute CDRL2, and amino acids 245 to 256 constitute CDRL3.

[0722] The amino acid sequence of the full-length NYA-2061 is the amino acid sequence as shown in SEQ ID NO: 53 of the Sequence Listing (FIG. 60). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 266 constitute NYA-2061, and amino acids 267 to 292 constitute a Flag-His tag. Also, amino acids 46 to 53 constitute CDRH1, amino acids 71 to 78 constitute CDRH2, and amino acids 117 to 129 constitute CDRH3. Also, amino acids 181 to 188 constitute CDRL1, amino acids 206 to 208 constitute CDRL2, and amino acids 245 to 256 constitute CDRL3.

2)-3-3 Expression and Purification of NYA-1143 Mutants

[0723] NYA-2035, NYA-2044, NYA-2045, NYA-2047, NYA-2048, NYA-2060, and NYA-2061 prepared in 2)-3-1 and 2)-3-2 were expressed in the same manner as in 1)-5-2 to purify the target scFv constructs. Purified protein samples were subjected to analytical SEC, the degree of purification and the concentration were determined, and the samples were then subjected to various assays.

(Example 3) Preparation of scFv of Anti-HLA/NY-ESO Reference Antibody

[0724] The scFv constructs of the antibodies 3M4E5 and T1 with high binding affinity to HLA/NY-ESO (WO2010/106431) were designed; 3M4E5 scFv was designated as NYC-0003, and T1 scFv was designated as NYC-0004.

[0725] The DNA fragments of NYC-0003 and NYC-0004 were fully synthesized (Thermo Fisher Scientific), and the scFv expression vector for mammal cells comprising pcDNA3.3 (Thermo Fisher Scientific) as the backbone was constructed using the In-Fusion HD Cloning kit (CLONTECH).

[0726] The nucleotide sequences of the constructed scFv expression vectors were reanalyzed, and the nucleotide sequences of the full-length NYC-0003 and NYC-0004 were found to be the nucleotide sequences as shown in SEQ ID NO: 65 of the Sequence Listing (FIG. 66) and SEQ ID NO: 66 thereof (FIG. 67). On the basis of the nucleotide sequences, the amino acid sequence of the full-length NYC-0003 encoded thereby was found to be the amino acid sequence as shown in SEQ ID NO: 67 (FIG. 68), and the amino acid sequence of the full-length NYC-0004 was found to be the amino acid sequence as shown in SEQ ID NO: 68 (FIG. 69).

[0727] NYC-0003 and NYC-0004 were expressed in the same manner as in 1)-5-2 to purify the target scFv constructs. Purified protein samples were subjected to analytical SEC, the degree of purification and the concentration were determined, and the samples were then subjected to various assays.

(Example 4) Evaluation of Binding Affinity to HLA/NY-ESO Using Biacore

[0728] With the use of Biacore T200, anti-HLA/NY-ESO scFv was captured as a ligand by the immobilized anti-His antibody, and the antigen was assayed as an analyte. As the antigen, HLA/NY-ESO prepared in 1)-1 was used. The anti-His antibody (His Capture kit, GE Healthcare) was immobilized on Sensor Chip CM5 (GE Healthcare) in accordance with the instructions of the kit. The anti-HLA/NY-ESO scFv constructs diluted to 0.5 µg/ml in HBS-EP+ (GE Healthcare) to be evaluated were brought into contact therewith at 10 µl/min for 60 seconds for immobilization. Thereafter, the samples were added to the HLA/NY-ESO analytes diluted to various levels with HBS-EP+ at a flow rate of 30 µl/min for 120 seconds and dissociation was assayed for 600 seconds. The results of the calculation obtained by such single cycle kinetics analysis, K_D , are shown in Table 1. In comparison with the parent antibody NYA-0001, all the mutants thereof were found to exhibit higher binding affinity to HLA/NY-ESO. In comparison with NYC-0004 NYA-1143, NYA-2023, NYA-2143, NYA-2044, NYA-2045, NYA-2060, and NYA-2061 were, in addition, found to exhibit higher binding affinity, and, in par-

ticular, NYA-1143, NYA-2044, NYA-2045, and NYA-2143 exhibited K_D of 1 nM or lower.

TABLE 1

Clone name	K_D (nM)
NYA-0001	33.8
NYA-1143	1.0
NYA-1163	13.0
NYA-2023	2.4
NYA-2027	11.2
NYA-2035	3.2
NYA-2044	0.8
NYA-2045	0.8
NYA-2047	5.3
NYA-2048	6.3
NYA-2060	1.1
NYA-2061	1.5
NYA-2143	0.9
NYC-0003	8.1
NYC-0004	2.8

(Example 5) Analysis of Recognized Amino Acid in NY-ESO Peptide of anti-HLA/NY-ESO scFv

[0729] The concentration of the T2 human lymphoblast fusion cells (ATCC) was adjusted to an adequate level in AIM-V medium (Thermo Fisher Scientific) containing 20% FBS, a solution of the NY-ESO peptide (SEQ ID NO: 1), the point-mutant NY-ESO peptides 1F, 2M, 3A, 4A, 5A, 6L, 7F, 8A, and 9A (SEQ ID NO: 121 (FIG. 122), SEQ ID NO: 122 (FIG. 123), SEQ ID NO: 123 (FIG. 124), SEQ ID NO: 124 (FIG. 125), SEQ ID NO: 125 (FIG. 126), SEQ ID NO: 126 (FIG. 127), SEQ ID NO: 127 (FIG. 128), SEQ ID NO: 128 (FIG. 129), and SEQ ID NO: 129 (FIG. 130)), or the gp100 peptide (SEQ ID NO: 130 (FIG. 131)) (Sigma Genosys) dissolved to 5 mM in DMSO was added to result in a final concentration of 50 µM, or DMSO was added in an amount of 1/100, and the resulting mixture was incubated at 37° C. for 4 hour. The resultant was washed two times in the AIM-V medium containing 20% FBS, the concentration thereof was adjusted to an adequate level with PBS containing 5% FBS, the LIVE/DEAD Fixable Dead Cell Stain Kit (Thermo Fisher Scientific) was added thereto, and the resultant was allowed to stand at 4° C. for 30 minutes. The cells were washed 2 times in PBS containing 5% FBS, the washed cells were divided into two groups, the cells in PBS containing 5% FBS were seeded on a 96-well U-bottom microplate at 10⁵ cells/well, and the plate was subjected to centrifugation, followed by removal of the supernatant. Anti-HLA/NY-ESO scFv diluted to 100 nM with PBS containing 5% FBS was added at 25 µl/well to one of the divided cells, and the resultant was allowed to stand at 4° C. for 30 minutes. The cells were washed 2 times in PBS containing 5% FBS, Penta-His Alexa Fluor488 (QIAGEN) diluted in PBS containing 5% FBS was added at 25 µl/well, and the resultant was allowed to stand at 4° C. for 30 minutes. The cells were washed 2 times in PBS containing 5% FBS, Anti-mouse-IgG Alexa Fluor488 (Thermo Fisher Scientific) diluted in PBS containing 5% FBS was added at 25 µl/well, and the resultant was allowed to stand at 4° C. for 30 minutes. The cells were washed 2 times in PBS containing 5% FBS, immobilized with Mildform 10 N (FUJIFILM Wako Pure Chemical Corporation) overnight, and resuspended in PBS containing 5% FBS. For standardization with the amount of the HLA/peptide complex, the HLA-A2

antibody BB7.2-Alexa Fluor 488 diluted to 10 µg/ml in PBS containing 5% FBS or mouse IgG2b-Alexa Fluor 488 was added at 25 µl/well to the other group of cells, and the resultant was allowed to stand at 4° C. for 30 minutes. The cells were washed 2 times in PBS containing 5% FBS, immobilized with Mildform 10 N (FUJIFILM Wako Pure Chemical Corporation) overnight, and resuspended in PBS containing 5% FBS. The cell suspensions were subjected to detection using a flow cytometer (CantoII, Becton Dickinson). Data analysis was performed using Flowjo (Treestar), and the geometric mean fluorescence intensity (gMFI) of Alexa Fluor 488 in a T2 cell fraction from which dead cells were removed was measured. The standardized gMFI indicating the binding affinity of each scFv standardized with the amount of the HLA/peptide complex on the T2 cells was determined in accordance with the following equation.

$$\text{Standardized gMFI} = (A/B)/((C/D)/(E/F))$$

[0730] A/B=Relative gMFI
 [0731] A: gMFI of T2 cells supplemented with DMSO or peptide in the presence of antibody
 [0732] B: gMFI of T2 cells supplemented with DMSO or peptide
 [0733] (C/D)/(E/F)=Corrected amount of HLA/peptide complex of T2 cells supplemented with DMSO or peptide
 [0734] C: gMFI of T2 cells supplemented with DMSO or peptide in the presence of HLA-A2 antibody
 [0735] D: gMFI of T2 cells supplemented with DMSO or peptide in the presence of mouse IgG2b antibody or peptide
 [0736] E: gMFI of T2 cells supplemented with DMSO containing HLA-A2 antibody
 [0737] F: gMFI of T2 cells supplemented with DMSO containing mouse IgG2b antibody
 [0738] As shown in FIG. 1, anti-HLA/NY-ESO scFv: NYA-0001, 1143, 2044, 2045, 2047, 2048, 2060, and 2061, exhibited binding affinity to T2 cells supplemented with peptides comprising point mutations introduced into amino acids 1, 4, 5, and 7 decreased to a half or lower, compared with the wild-type NY-ESO peptide. This indicates that such scFv recognizes amino acids 1, 4, 5, and 7 of the NY-ESO peptide. This also indicates that NYA-1154 recognizes amino acids 1 and 5, NYA-1163 recognizes amino acids 1, 3, 4, 5, 6, and 7, NYA-2023, 2027, and 2035 recognize amino acids 1, 4, and 5, and NYA-2143 recognizes amino acids 1, 5, and 7. This further indicates that NYC-0003 and 0004 recognize amino acid 4 and 5.

(Example 6) Evaluation of Antigen-Binding Specificity of Anti-HLA/NY-ESO scFv

[0739] In order to search for a human peptide having an amino acid sequence similar but not identical to the amino acid sequence of the NY-ESO peptide: SLLMWITQC (SEQ ID NO: 1) to which the antibody may bind (hereafter such peptide is referred to as a “homologous peptide”) from among human proteomes (Swiss-Prot), a 9-mer peptide comprising amino acids 1, 4, and 5 that are consistent with those of peptides recognized by the antibody was searched. The searched 9-mer peptide was analyzed using NetMHC-Pan2.8 concerning binding affinity to HLA-A0201, and

IC50 thereof was predicted to be 500 nM or lower. The 9-mer peptide shown in FIG. 2A was selected as a homologous peptide for evaluation of binding specificity of anti-HLA/NY-ESO scFv. The concentration of the T2 cells was adjusted to an adequate level in AIM-V medium (Thermo Fisher Scientific) containing 20% FBS, a solution of the NY-ESO peptide (SEQ ID NO: 1), the homologous peptides DOLPP1, IL20RB, PRKD2, CD163, and P2RY8 (SEQ ID NO: 131 (FIG. 132), SEQ ID NO:132 (FIG. 133), SEQ ID NO:133 (FIG. 134), SEQ ID NO:134 (FIG. 135), and SEQ ID NO:135 (FIG. 136)), or the gp100 peptide (SEQ ID NO: 130 (FIG. 131)) (Sigma Genosys) dissolved to 5 mM in DMSO was added to result in a final concentration of 50 µM, or DMSO was added in an amount of 1/100, and binding affinity of the antibodies was evaluated in the same manner as in Example 5. The standardized gMFI indicating the binding affinity of each scFv standardized with the amount of the HLA/peptide complex on the T2 cells was determined in accordance with the following equation.

$$\text{Standardized gMFI} = (A/B)/((C/D)/(E/F))$$

[0740] A/B=Relative gMFI
 [0741] A: gMFI of T2 cells supplemented with DMSO or peptide in the presence of antibody
 [0742] B: gMFI of T2 cells supplemented with DMSO or peptide
 [0743] (C/D)/(E/F)=Corrected amount of HLA/peptide complex of T2 cells supplemented with DMSO or peptide
 [0744] C: gMFI of T2 cells supplemented with DMSO containing HLA-A2 antibody or peptide
 [0745] D: gMFI of T2 cells supplemented with DMSO or peptide in the presence of mouse IgG2b antibody
 [0746] E: gMFI of T2 cells supplemented with DMSO containing HLA-A2 antibody
 [0747] F: gMFI of T2 cells supplemented with DMSO containing mouse IgG2b antibody
 [0748] As shown in FIG. 2B, anti-HLA/NY-ESO scFv: NYA-0001, 1143, 1163, 2023, 2027, 2035, 2044, 2045, 2047, 2048, 2060, 2061, and 2143, did not bind to T2 cells supplemented with any homologous peptides. This indicates that such scFv has high specificity. In contrast, NYA-1154, NYC-0003, and 0004 were observed to have bound to T2 cells supplemented with some homologous peptides.

(Example 7) Preparation of Fc-Conjugated Anti-HLA/NY-ESO-Anti-CD3 Bispecific Molecule

7)-1 Preparation of Fc-Conjugated Anti-HLA/NY-ESO-Anti-CD3 Bispecific Molecule Expression Vector

7)-1-1 Preparation of taFv-Heterodimer Fc-Type Bispecific Molecule Expression Vector

[0749] In order to evaluate taFv-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecules, expression vectors for various molecules were designed. The anti-HLA/NY-ESO antibodies used were NYA-1143, NYA-2143, NYA-1163, NYA-2023, NYA-2027, NYA-2035, NYA-2044, NYA-2045, NYA-2047, NYA-2048, NYA-2060, and NYA-2061. As the anti-CD3 antibody, humanized anti-CD3 scFv; i.e., C3E-7085 (WO 2018/117237), was used. As the het-

erodimer Fc sequence, the Fc sequence comprising a mutation introduced thereinto to lower effector functions and form a heteropolymer (WO 2014/190441) was used.

[0750] A DNA fragment encoding Fc (HC1 or HC2) comprising a mutation introduced thereinto to lower effector functions and form a heteropolymer was synthesized (Fas-mac Co., Ltd.), an expression vector for mammal cells comprising pcDNA3.3 (Thermo Fisher Scientific) as the backbone was prepared using the In-Fusion HD cloning kit (CLONTECH), and the resulting vector was designated as “p_HC1.”

[0751] An expression vector for mammal cells comprising a DNA fragment encoding HC2 integrated into the carboxyl terminus of taFv comprising NYA-1143 ligated to C3E-7085 with a GGGGS linker was prepared and designated as “p_NYF-0016-HC2.”

[0752] An expression vector for mammal cells comprising a DNA fragment encoding HC2 integrated into the carboxyl terminus of taFv comprising NYA-2143 ligated to C3E-7085 with a GGGGS linker was prepared via site-directed mutagenesis into p_NYF-0016-HC2 and designated as “p_NYF-0019-HC2.”

[0753] An expression vector for mammal cells comprising a DNA fragment encoding HC2 integrated into the carboxyl terminus of taFv comprising NYA-1163 ligated to C3E-7085 with a GGGGS linker was prepared via substitution of a nucleotide sequence encoding NYA-1143 of p_NYF-0016-HC2 with a DNA fragment encoding NYA-1163 using the In-Fusion HD cloning kit (CLONTECH) and designated as “p_NYF-0022-HC2.”

[0754] An expression vector for mammal cells comprising a DNA fragment encoding HC2 integrated into the carboxyl terminus of taFv comprising NYA-2023 ligated to C3E-7085 with a GGGGS linker was prepared via site-directed mutagenesis into p_NYF-0016-HC2 and designated as “p_NYF-0023-HC2.”

[0755] An expression vector for mammal cells comprising a DNA fragment encoding HC2 integrated into the carboxyl terminus of taFv comprising NYA-2027 ligated to C3E-7085 with a GGGGS linker was prepared via site-directed mutagenesis of an expression vector for mammal cells comprising a DNA fragment encoding HC2 integrated into the carboxyl terminus of taFv comprising NYA-0001 ligated to C3E-7085 with a GGGGS linker and designated as “p_NYF-0027-HC2.”

[0756] An expression vector for mammal cells comprising a DNA fragment encoding HC2 integrated into the carboxyl terminus of taFv comprising NYA-2035 ligated to C3E-7085 with a GGGGS linker was prepared via site-directed mutagenesis into p_NYF-0016-HC2 and designated as “p_NYF-0035-HC2.”

[0757] An expression vector for mammal cells comprising a DNA fragment encoding HC2 integrated into the carboxyl terminus of taFv comprising NYA-2044 ligated to C3E-7085 with a GGGGS linker was prepared via substitution of a nucleotide sequence encoding NYA-1143 of p_NYF-0016-HC2 with a DNA fragment encoding NYA-2044 using the In-Fusion HD cloning kit (CLONTECH) and designated as “p_NYF-0044-HC2.”

[0758] An expression vector for mammal cells comprising a DNA fragment encoding HC2 integrated into the carboxyl terminus of taFv comprising NYA-2045 ligated to C3E-7085 with a GGGGS linker was prepared via substitution of a nucleotide sequence encoding NYA-1143 of p_NYF-0016-

HC2 with a DNA fragment encoding NYA-2045 using the In-Fusion HD cloning kit (CLONTECH) and designated as “p_NYF-0045-HC2.”

[0759] An expression vector for mammal cells comprising a DNA fragment encoding HC2 integrated into the carboxyl terminus of taFv comprising NYA-2047 ligated to C3E-7085 with a GGGGS linker was prepared via substitution of a nucleotide sequence encoding NYA-1143 of p_NYF-0016-HC2 with a DNA fragment encoding NYA-2047 using the In-Fusion HD cloning kit (CLONTECH) and designated as “p_NYF-0047-HC2.”

[0760] An expression vector for mammal cells comprising a DNA fragment encoding HC2 integrated into the carboxyl terminus of taFv comprising NYA-2048 ligated to C3E-7085 with a GGGGS linker was prepared via substitution of a nucleotide sequence encoding NYA-1143 of p_NYF-0016-HC2 with a DNA fragment encoding NYA-2048 using the In-Fusion HD cloning kit (CLONTECH) and designated as “p_NYF-0048-HC2.”

[0761] An expression vector for mammal cells comprising a DNA fragment encoding HC2 integrated into the carboxyl terminus of taFv comprising NYA-2060 ligated to C3E-7085 with a GGGGS linker was prepared via site-directed mutagenesis into p_NYF-0044-HC2 and designated as “p_NYF-0060-HC2.”

[0762] An expression vector for mammal cells comprising a DNA fragment encoding HC2 integrated into the carboxyl terminus of taFv comprising NYA-2061 ligated to C3E-7085 with a GGGGS linker was prepared via site-directed mutagenesis into p_NYF-0045-HC2 and designated as “p_NYF-0061-HC2.”

[0763] The nucleotide sequence of p_HC1 was reanalyzed, and the nucleotide sequence of the full-length HC1 was found to be the nucleotide sequence as shown in SEQ ID NO: 71 of the Sequence Listing (FIG. 72).

[0764] The nucleotide sequence of p_NYF-0016-HC2 was reanalyzed, and the nucleotide sequence of the full-length NYF-0016-HC2 was found to be the nucleotide sequence as shown in SEQ ID NO: 72 of the Sequence Listing (FIG. 73).

[0765] The nucleotide sequence of p_NYF-0019-HC2 was reanalyzed, and the nucleotide sequence of the full-length NYF-0019-HC2 was found to be the nucleotide sequence as shown in SEQ ID NO: 73 of the Sequence Listing (FIG. 74).

[0766] The nucleotide sequence of p_NYF-0022-HC2 was reanalyzed, and the nucleotide sequence of the full-length NYF-0022-HC2 was found to be the nucleotide sequence as shown in SEQ ID NO: 74 of the Sequence Listing (FIG. 75).

[0767] The nucleotide sequence of p_NYF-0023-HC2 was reanalyzed, and the nucleotide sequence of the full-length NYF-0023-HC2 was found to be the nucleotide sequence as shown in SEQ ID NO: 75 of the Sequence Listing (FIG. 76).

[0768] The nucleotide sequence of p_NYF-0027-HC2 was reanalyzed, and the nucleotide sequence of the full-length NYF-0027-HC2 was found to be the nucleotide sequence as shown in SEQ ID NO: 76 of the Sequence Listing (FIG. 77).

[0769] The nucleotide sequence of p_NYF-0035-HC2 was reanalyzed, and the nucleotide sequence of the full-length NYF-0035-HC2 was found to be the nucleotide sequence as shown in SEQ ID NO: 77 of the Sequence Listing (FIG. 78).

[0770] The nucleotide sequence of p_NYF-0044-HC2 was reanalyzed, and the nucleotide sequence of the full-length NYF-0044-HC2 was found to be the nucleotide sequence as shown in SEQ ID NO: 78 of the Sequence Listing (FIG. 79).

[0771] The nucleotide sequence of p_NYF-0045-HC2 was reanalyzed, and the nucleotide sequence of the full-length NYF-0045-HC2 was found to be the nucleotide sequence as shown in SEQ ID NO: 79 of the Sequence Listing (FIG. 80).

[0772] The nucleotide sequence of p_NYF-0047-HC2 was reanalyzed, and the nucleotide sequence of the full-length NYF-0047-HC2 was found to be the nucleotide sequence as shown in SEQ ID NO: 80 of the Sequence Listing (FIG. 81).

[0773] The nucleotide sequence of p_NYF-0048-HC2 was reanalyzed, and the nucleotide sequence of the full-length NYF-0048-HC2 was found to be the nucleotide sequence as shown in SEQ ID NO: 81 of the Sequence Listing (FIG. 82).

[0774] The nucleotide sequence of p_NYF-0060-HC2 was reanalyzed, and the nucleotide sequence of the full-length NYF-0060-HC2 was found to be the nucleotide sequence as shown in SEQ ID NO: 82 of the Sequence Listing (FIG. 83).

[0775] The nucleotide sequence of p_NYF-0061-HC2 was reanalyzed, and the nucleotide sequence of the full-length NYF-0061-HC2 was found to be the nucleotide sequence as shown in SEQ ID NO: 83 of the Sequence Listing (FIG. 84).

[0776] On the basis of the nucleotide sequences above, the amino acid sequences of the full-length HC1, NYF-0016-HC2, NYF-0019-HC2, NYF-0022-HC2, NYF-0023-HC2, NYF-0027-HC2, NYF-0035-HC2, NYF-0044-HC2, NYF-0045-HC2, NYF-0047-HC2, NYF-0048-HC2, NYF-0060-HC2, and NYF-0061-HC2 encoded thereby were identified.

[0777] The amino acid sequence of the full-length HC1 is the amino acid sequence as shown in SEQ ID NO: 84 of the Sequence Listing (FIG. 85). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, and amino acids 20 to 246 constitute HC.

[0778] The amino acid sequence of the full-length NYF-0016-HC2 is as shown in SEQ ID NO: 85 of the Sequence Listing (FIG. 86). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acid 21 to 511 constitute NYA-1143-C3E-7085 taFv, amino acids 512 to 513 constitute a linker, and amino acids 514 to 745 constitute HC2.

[0779] The amino acid sequence of the full-length NYF-0019-HC2 is as shown in SEQ ID NO: 86 of the Sequence Listing (FIG. 87). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acid 21 to 511 constitute NYA-2143-C3E-7085 taFv, amino acids 512 to 513 constitute a linker, and amino acids 514 to 745 constitute HC2.

[0780] The amino acid sequence of the full-length NYF-0022-HC2 is as shown in SEQ ID NO: 87 of the Sequence Listing (FIG. 88). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acid 21 to 511 constitute NYA-1163-C3E-7085 taFv, amino acids 512 to 513 constitute a linker, and amino acids 514 to 745 constitute HC2.

[0781] The amino acid sequence of the full-length NYF-0023-HC2 is as shown in SEQ ID NO: 88 of the Sequence Listing (FIG. 89). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acid 21 to 511 constitute NYA-2023-C3E-7085 taFv, amino acids 512 to 513 constitute a linker, and amino acids 514 to 745 constitute HC2.

[0782] The amino acid sequence of the full-length NYF-0027-HC2 is as shown in SEQ ID NO: 89 of the Sequence Listing (FIG. 90). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acid 21 to

511 constitute NYA-2027-C3E-7085 taFv, amino acids 512 to 513 constitute a linker, and amino acids 514 to 745 constitute HC2.

[0783] The amino acid sequence of the full-length NYF-0035-HC2 is as shown in SEQ ID NO: 90 of the Sequence Listing (FIG. 91). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acid 21 to 511 constitute NYA-2035-C3E-7085 taFv, amino acids 512 to 513 constitute a linker, and amino acids 514 to 745 constitute HC2.

[0784] The amino acid sequence of the full-length NYF-0044-HC2 is as shown in SEQ ID NO: 91 of the Sequence Listing (FIG. 92). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acid 21 to 511 constitute NYA-2044-C3E-7085 taFv, amino acids 512 to 513 constitute a linker, and amino acids 514 to 745 constitute HC2.

[0785] The amino acid sequence of the full-length NYF-0045-HC2 is as shown in SEQ ID NO: 92 of the Sequence Listing (FIG. 93). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acid 21 to 511 constitute NYA-2045-C3E-7085 taFv, amino acids 512 to 513 constitute a linker, and amino acids 514 to 745 constitute HC2.

[0786] The amino acid sequence of the full-length NYF-0047-HC2 is as shown in SEQ ID NO: 93 of the Sequence Listing (FIG. 94). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acid 21 to 511 constitute NYA-2047-C3E-7085 taFv, amino acids 512 to 513 constitute a linker, and amino acids 514 to 745 constitute HC2.

[0787] The amino acid sequence of the full-length NYF-0048-HC2 is as shown in SEQ ID NO: 94 of the Sequence Listing (FIG. 95). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acid 21 to 511 constitute NYA-2048-C3E-7085 taFv, amino acids 512 to 513 constitute a linker, and amino acids 514 to 745 constitute HC2.

[0788] The amino acid sequence of the full-length NYF-0060-HC2 is as shown in SEQ ID NO: 95 of the Sequence Listing (FIG. 96). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acid 21 to 511 constitute NYA-2060-C3E-7085 taFv, amino acids 512 to 513 constitute a linker, and amino acids 514 to 745 constitute HC2.

[0789] The amino acid sequence of the full-length NYF-0061-HC2 is as shown in SEQ ID NO: 96 of the Sequence Listing (FIG. 97). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acid 21 to 511 constitute NYA-2061-C3E-7085 taFv, amino acids 512 to 513 constitute a linker, and amino acids 514 to 745 constitute HC2.

7)-1-2 Preparation of taFv-Fab-Heterodimer Fc-Type Bispecific Molecule Expression Vector

[0790] A taFv-Fab-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecule expression vector was designed. As the anti-HLA/NY-ESO antibody, NYA-0001 was used. As the anti-CD3 antibody, humanized anti-CD3 scFv; i.e., C3E-7085 (WO 2018/117237), was used. As the heterodimer Fc sequence, the Fc sequence comprising a mutation introduced thereto to lower effector functions and form a heteropolymer (WO 2014/190441) was used.

[0791] An expression vector for mammal cells comprising DNA fragments encoding the heavy chain variable region of

NYA-0001, the human IgG-derived CH1 region, and a region comprising a mutation that lowers effector functions and encodes HC1-k delete integrated therein was prepared and designated as “p_NYA-0001-Fab-HC1-k delete.” Also, an expression vector for mammal cells comprising DNA fragments encoding the NYA-0001 light chain variable region and the human IgG-derived CL region integrated therein was prepared and designated as “p_NYA-0001-LC.”

[0792] The nucleotide sequence of p_NYA-0001-Fab-HC1-k delete was reanalyzed, and the nucleotide sequence of the full-length NYA-0001-Fab-HC1-k delete was found to be the nucleotide sequence as shown in SEQ ID NO: 97 (FIG. 98) of the Sequence Listing.

[0793] The nucleotide sequence of p_NYA-0001-LC was reanalyzed, and the nucleotide sequence of the full-length NYA-0001-LC was found to be the nucleotide sequence as shown in SEQ ID NO: 98 (FIG. 99) of the Sequence Listing.

[0794] On the basis of the nucleotide sequences above, the amino acid sequences of the full-length NYA-0001-Fab-HC1-k delete and NYA-0001-LC encoded thereby were identified.

[0795] The amino acid sequence of the full-length NYA-0001-Fab-HC1-k delete is the amino acid sequence as shown in SEQ ID NO: 99 of the Sequence Listing (FIG. 100). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 20 to 139 constitute a variable region, and amino acids 140 to 468 constitute a constant region. Also, amino acids 45 to 52 constitute CDRH1 (SEQ ID NO: 54 (FIG. 61)), amino acids 70 to 77 constitute CDRH2 (SEQ ID NO: 55 (FIG. 61)), and amino acids 116 to 128 constitute CDRH3 (SEQ ID NO: 56 (FIG. 61)).

[0796] The amino acid sequence of the full-length NYA-0001-LC is as shown in SEQ ID NO: 100 of the Sequence Listing (FIG. 101). In such an amino acid sequence, amino acids 1 to 20 constitute a signal sequence, amino acids 21 to 131 constitute a variable region, and amino acids 132 to 237 constitute a constant region. Also, amino acids 46 to 53 constitute CDRL1 (SEQ ID NO: 57 (FIG. 61)), amino acids 71 to 73 constitute CDRL2 (SEQ ID NO: 58 (FIG. 61)), and amino acids 110 to 121 constitute CDRL3 (SEQ ID NO: 59 (FIG. 61)).

7)-2 Expression of Fc-Conjugated Anti-HLA/NY-ESO-Anti-CD3 Bispecific Molecule

7)-2-1 Expression of taFv-Heterodimer Fc-Type Bispecific Molecule

[0797] Expi293F cells (Thermo Fisher Scientific) were subcultured in accordance with the instructions. A culture solution of the Expi293F cells in a logarithmic growth phase was diluted to 2.5×10^6 cells/ml in the Expi293 Expression medium (Thermo Fisher Scientific) and used for production of various bispecific molecules. A mixture (0.3 mg) comprising the p_NYF-0016-HC2 vector and p_HC1 at 1:1.5 and 0.9 mg of polyethyleneimine (Polyscience #24765) were added to 20 ml of Opti-Pro SFM medium (Thermo Fisher Scientific). The mixture was stirred gently, allowed to stand for five minutes, and then added to the Expi293F cells. The culture supernatant obtained by agitation culture in an incubator at 37° C. in the presence of 8% CO₂ at 135 rpm for 6 days was filtered through a 0.2 µm-filter (Millipore). Thus, the culture supernatant of the taFv-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecule

(NYF-0016) was obtained. The amino acid sequences obtained via expression of vectors constituting NYF-0016 are shown in SEQ ID NO: 85 of the Sequence Listing (FIG. 86) and SEQ ID NO: 84 of the Sequence Listing (FIG. 85).

[0798] In the same manner, the culture supernatant of the taFv-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecule (NYF-0019) was prepared using p_NYF-0019-HC2 and p_HC1. The amino acid sequences obtained via expression of vectors constituting NYF-0019 are shown in SEQ ID NO: 86 of Sequence Listing (FIG. 87) and SEQ ID NO: 84 of the Sequence Listing (FIG. 85).

[0799] The culture supernatant of the taFv-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecule (NYF-0022) was prepared using p_NYF-0022-HC2 and p_HC1. The amino acid sequences obtained via expression of vectors constituting NYF-0022 are shown in SEQ ID NO: 87 of the Sequence Listing (FIG. 88) and SEQ ID NO: 84 of the Sequence Listing (FIG. 85).

[0800] The culture supernatant of the taFv-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecule (NYF-0023) was prepared using p_NYF-0023-HC2 and p_HC1. The amino acid sequences obtained via expression of vectors constituting NYF-0023 are shown in SEQ ID NO: 88 of the Sequence Listing (FIG. 89) and SEQ ID NO: 84 of the Sequence Listing (FIG. 85).

[0801] The culture supernatant of the taFv-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecule (NYF-0027) was prepared using p_NYF-0027-HC2 and p_HC1. The amino acid sequences obtained via expression of vectors constituting NYF-0027 are shown in SEQ ID NO: 89 of the Sequence Listing (FIG. 90) and SEQ ID NO: 84 of the Sequence Listing (FIG. 85).

[0802] The culture supernatant of the taFv-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecule (NYF-0035) was prepared using p_NYF-0035-HC2 and p_HC1. The amino acid sequences obtained via expression of vectors constituting NYF-0035 are shown in SEQ ID NO: 90 of the Sequence Listing (FIG. 91) and SEQ ID NO: 84 of the Sequence Listing (FIG. 85).

[0803] The culture supernatant of the taFv-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecule (NYF-0044) was prepared using p_NYF-0044-HC2 and p_HC1. The amino acid sequences obtained via expression of vectors constituting NYF-0044 are shown in SEQ ID NO: 91 of the Sequence Listing (FIG. 92) and SEQ ID NO: 84 of the Sequence Listing (FIG. 85).

[0804] The culture supernatant of the taFv-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecule (NYF-0045) was prepared using p_NYF-0045-HC2 and p_HC1. The amino acid sequences obtained via expression of vectors constituting NYF-0045 are shown in SEQ ID NO: 92 of the Sequence Listing (FIG. 93) and SEQ ID NO: 84 of the Sequence Listing (FIG. 85).

[0805] The culture supernatant of the taFv-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecule (NYF-0047) was prepared using p_NYF-0047-HC2 and p_HC1. The amino acid sequences obtained via expression of vectors constituting NYF-0047 are shown in SEQ ID NO: 93 of the Sequence Listing (FIG. 94) and SEQ ID NO: 84 of the Sequence Listing (FIG. 85).

[0806] The culture supernatant of the taFv-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecule (NYF-0048) was prepared using p_NYF-0048-HC2 and p_HC1. The amino acid sequences obtained via expression

of vectors constituting NYF-0048 are shown in SEQ ID NO: 94 of the Sequence Listing (FIG. 95) and SEQ ID NO: 84 of the Sequence Listing (FIG. 85).

[0807] The culture supernatant of the taFv-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecule (NYF-0060) was prepared using p_NYF-0060-HC2 and p_HC1. The amino acid sequences obtained via expression of vectors constituting NYF-0060 are shown in SEQ ID NO: 95 of the Sequence Listing (FIG. 96) and SEQ ID NO: 84 of the Sequence Listing (FIG. 85).

[0808] The culture supernatant of the taFv-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecule (NYF-0061) was prepared using p_NYF-0061-HC2 and p_HC1. The amino acid sequences obtained via expression of vectors constituting NYF-0061 are shown in SEQ ID NO: 96 of the Sequence Listing (FIG. 97) and SEQ ID NO: 84 of the Sequence Listing (FIG. 85).

7)-2-2 Expression of taFv-Fab-heterodimer Fc-type bispecific molecules

[0809] In the same manner as in 7)-2-1, the culture supernatant of the taFv-Fab-heterodimer Fc-type bispecific molecule (NYF-0058) was prepared using a vector mixture comprising p_NYF-0023-HC2, p_NYA-0001-Fab-HC1-k delete, and p_NYA-0001-LC at 1:1:1.5. The amino acid sequences obtained via expression of vectors constituting NYF-0058 are shown in amino acids 20 to 745 of SEQ ID NO: 88 of the Sequence Listing (FIG. 89), amino acids 20 to 468 of SEQ ID NO: 99 of the Sequence Listing (FIG. 100), and amino acids 21 to 237 of SEQ ID NO: 100 of the Sequence Listing (FIG. 101).

7)-3 Purification of Fc-Conjugated Anti-HLA/NY-ESO-Anti-CD3 Bispecific Molecule

[0810] Various bispecific molecules were purified from the culture supernatants obtained in 7)-2 by 2 steps of protein A affinity chromatography and gel filtration chromatography.

[0811] The culture supernatant was applied to the Mab-SelectSuRe column equilibrated with PBS at pH 7.4 (GE Healthcare Bioscience, also referred to simply as "GE Healthcare") to allow the target bispecific molecules to adsorb thereto. After the non-adsorbed components were removed by PBS, the adsorbed components were eluted using acetate buffer (pH 3.5). The elution fraction was neutralized with the aid of Tris buffer (pH 9.5), concentrated, and then applied to the gel filtration column Superdex 200 10/300 (GE Healthcare Bioscience) equilibrated with 25 mM histidine, 300 mM NaCl, 5% Sorbitol at pH 5.5 in advance. From the peak fraction obtained via gel filtration chromatography, fractions equivalent to the target heterodimer were collected, and generation of the target anti-HLA/NY-ESO-anti-CD3 bispecific molecule was confirmed via SDS-polyacrylamide electrophoresis (SDS-PAGE). Purified protein samples were subjected to analytical SEC, the degree of purification and the concentration were determined, and the samples were then subjected to various types of evaluation.

(Example 8) Evaluation of Cytotoxicity of Fc-Conjugated Anti-HLA-A2/NY-ESO-Anti-CD3 Bispecific Molecule

8)-1 Preparation of Target Cells

[0812] The endogenous human NY-ESO-expressing cells (U266B1 and NCI-H1703) and the endogenous human

NY-ESO-non-expressing cells (AGS and CFPAC-1) were adjusted to 1×10^6 cells/ml in RPMI 1640 medium containing 10% FBS (FUJIFILM Wako Pure Chemical Corporation), 100 μ l of Chromium-51 Radionuclide (PerkinElmer) was added per 1 ml of each cell suspension, and the resultants were cultured at 37° C. in the presence of 5% CO₂ for 2 hours. The cells were washed 2 times in RPMI 1640 medium containing 10% FBS, resuspended to 1×10^5 cells/ml in RPMI 1640 medium containing 10% FBS, and then used as the target cells.

8)-2 Preparation of Effector Cells

[0813] Commercially available frozen human PBMCs (Cellular Technology Limited) were thawed at 37° C., transferred to a solution comprising RPMI 1640 medium containing 10% FBS supplemented with the Anti-aggregate Wash reagent (Cellular Technology Limited), washed 2 times, and adjusted to 1×10^6 cells/ml in RPMI 1640 medium containing 10% FBS to prepare effector cells.

8)-3 Cytotoxicity Assay

[0814] The target cells obtained in 8)-1 were seeded on a 96-well U-bottom microplate at 50 μ l/well. Various Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules prepared in Example 7 adjusted to various concentrations were added thereto at 50 μ l/well, the effector cells prepared in Example 8)-2 were added thereto at 100 l/well, centrifugation was carried out at room temperature and 1,000 rpm for 1 minute, and then they were incubated at 37° C. in the presence of 5% CO₂ for 20 to 24 hours. The supernatant (50 μ l) was collected on a LumaPlate (PerkinElmer), dried at 50° C. for approximately 2 hours, and then assayed using a plate reader (TopCount, PerkinElmer). The test was performed in triplicate, and the rate of cell lysis was determined in accordance with the following equation.

$$\text{Rate of cell lysis (\%)} = (A - B)/(C - B) \times 100$$

[0815] A: Sample well count

[0816] B: Mean of background (wells not supplemented with antibody) count (n=3).

[0817] When the antibody was added, 50 μ l of an assay medium was added. Other procedures were the same as those for the sample wells.

[0818] C: Mean of maximal release (wells in which target cells were lysed with a surfactant) count (n=3).

[0819] When the antibody was added, 50 μ l of an assay medium was added. A surfactant was added in an amount of 100 μ l, a fraction of 50 μ l was transferred to the LumaPlate as in the case of the sample wells, and assays were then performed.

[0820] As shown in FIG. 4A to FIG. 4F, various anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules exerted cytotoxicity on the endogenous human NY-ESO-expressing cells (U266B1 and NCI-H1703). The results of calculation of EC₅₀ obtained using the Four Parameter Logistic Curve equation of analytical software (Sigmaplot, version 12.0) concerning the U266B1 cells are shown in Table 2, and the results concerning the NCI-H1703 cells are shown in Table 3. "Not Applicable (NA)" indicates that it is impossible to perform curve fitting because the correlation coefficient (R)

was not calculated. As shown in FIG. 4G to FIG. 4L, in contrast, no cytotoxicity was observed on the endogenous human NY-ESO-non-expressing cells (AGS and CFPAC-1).

TABLE 2

	Name	EC ₅₀ (nM)
1	NYF-0016	0.30
2	NYF-0019	0.20
3	NYF-0022	1.30
4	NYF-0023	0.90
5	NYF-0027	2.68
6	NYF-0035	0.97
7	NYF-0044	0.10
8	NYF-0045	0.18
9	NYF-0047	0.93
10	NYF-0048	1.10
11	NYF-0058	0.95
12	NYF-0060	0.39
13	NYF-0061	0.40

TABLE 3

	Name	EC ₅₀ (nM)
1	NYF-0016	0.90
2	NYF-0019	4.28
3	NYF-0022	NA*
4	NYF-0023	1.16
5	NYF-0027	6.40
6	NYF-0035	2.10
7	NYF-0044	1.11
8	NYF-0045	5.76
9	NYF-0047	7.63
10	NYF-0048	4.21
11	NYF-0058	1.03
12	NYF-0060	0.74
13	NYF-0061	1.04

*NA = Not Applicable

(Example 9) Evaluation of In Vivo Activity of Fc-Conjugated Anti-HLA-A2/NY-ESO-Anti-CD3 Bispecific Molecules in Human PBMC-Transfected Models

[0821] The human squamous lung cancer cell lines NCI-H1703 (ATCC) were adjusted to 6×10^7 cells/ml in PBS containing 50% Matrigel (Corning) and 0.1 ml thereof was injected hypodermically into NOG mice (female, 6- to 7-week-old) (Day 0). On Day 4, human PBMCs were adjusted to 3.75×10^7 cells/ml in PBS, and 0.2 ml thereof was injected intravenously. Approximately 1 week thereafter (Day 6 to Day 7), measurement of the major diameter (mm) and the minor diameter (mm) of the tumor was initiated, measurement was performed using an electronic digital caliper with the elapse of time, and the estimated tumor volume was calculated in accordance with the equation below.

$$\text{Estimated tumor volume (mm}^3\text{)} = \text{mean of estimated tumor volume among individuals} \times \frac{\text{Estimated tumor volume of an individual} = \text{major diameter} \times [\text{minor diameter}]^2 / 2}{2}$$

[0822] On Day 14, mice were divided into groups each consisting of 5 or 6 mice on the basis of tumor volumes, and various anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules were administered intravenously (1 mg/kg, 1.5 mg/kg of NYF-0058 for comparison at the same molar weight).

Administration was performed on Day 14, Day 21, and Day 28. Antitumor effects were observed in the treatment groups to which various anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules had been administered (FIG. 5A to FIG. 5C). The tumor growth inhibition (%) on Day 31 to Day 32 was calculated in accordance with the following equation and shown in Table 4.

Tumor growth inhibition (%) =

$$100 - \left(\frac{\text{estimated tumor volume of treatment group}}{\text{estimated tumor volume of vehicle control}} \times 100 \right)$$

TABLE 4

	Name	Tumor Growth Inhibition (%)
1	NYF-0016	100.0
2	NYF-0019	98.8
3	NYF-0022	90.3
4	NYF-0023	98.6
5	NYF-0027	99.1
6	NYF-0035	92.5
7	NYF-0044	100.0
8	NYF-0045	95.4
9	NYF-0047	78.9
10	NYF-0048	96.3
11	NYF-0058	88.5
12	NYF-0060	88.8
13	NYF-0061	100.0

(Example 10) Preparation of Various Fc-Conjugated Anti-HLA/NY-ESO-Anti-CD3 Bispecific Molecules

10)-1 Preparation of Various Fc-Conjugated Anti-HLA/NY-ESO-anti-CD3 Bispecific Molecule Expression Vectors

10)-1-1 Preparation of Hybrid-Type Bispecific Molecule Expression Vector

[0823] In order to evaluate hybrid-type anti-HLA/NY-ESO-anti-CD3 bispecific molecules, expression vectors for various molecules were designed. As the anti-HLA/NY-ESO antibody, NYA-1143 was used. As the anti-CD3 antibody, humanized anti-CD3 scFv; i.e., C3E-7085 (PCT/JP2017/046006), was used. As the heterodimer Fc sequence, the Fc sequence comprising a mutation introduced thereto to lower effector functions and form a heteropolymer (WO 2014/190441) was used.

[0824] An expression vector for mammal cells comprising DNA fragments encoding the heavy chain variable region of NYA-1143, the human JgG-derived CH1 region, and the Fc region into which a mutation has been introduced therein to lower effector functions and form a heteropolymer was prepared and designated as “p_NYA-1143-Fab-HC1-k delete.” Also, an expression vector for mammal cells comprising DNA fragments encoding the NYA-1143 light chain variable region and the human IgG-derived CL region was prepared and designated as “p_NYA-1143-LC.” Further, an expression vector for mammal cells comprising a DNA fragment encoding the Fc region comprising a mutation to lower effector functions and form a heteropolymer intro-

duced into the carboxyl terminus of humanized anti-CD3 scFv (C3E-7085) was prepared and designated as “p_C3E-7085-HC2-k delete.”

[0825] The nucleotide sequence of p_NYA-1143-Fab-HC1-k delete was reanalyzed, and the nucleotide sequence of the full-length NYA-1143-Fab-HC1-k delete was found to be the nucleotide sequence as shown in SEQ ID NO: 101 of the Sequence Listing (FIG. 102).

[0826] The nucleotide sequence of p_NYA-1143-LC was reanalyzed, and the nucleotide sequence of the full-length NYA-1143-LC was found to be the nucleotide sequence as shown in SEQ ID NO: 102 of the Sequence Listing (FIG. 103).

[0827] The nucleotide sequence of p_C3E-7085-HC2-k delete was reanalyzed, and the nucleotide sequence of the full-length C3E-7085-HC2-k delete was found to be the nucleotide sequence as shown in SEQ ID NO: 103 of the Sequence Listing (FIG. 104).

[0828] On the basis of the nucleotide sequences above, the amino acid sequences of the full-length NYA-1143-Fab-HC1-k delete, NYA-1143-LC, and C3E-7085-HC2-k delete encoded thereby were identified.

[0829] The amino acid sequence of the full-length NYA-1143-Fab-HC1-k delete is the amino acid sequence as shown in SEQ ID NO: 104 of the Sequence Listing (FIG. 105). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 139 constitute a variable region, and amino acids 140 to 468 constitute a constant region. Also, amino acids 45 to 52 constitute CDRH1 (SEQ ID NO: 54 (FIG. 61)), amino acids 70 to 77 constitute CDRH2 (SEQ ID NO: 55 (FIG. 61)), amino acids 116 to 128 constitute CDRH3 (SEQ ID NO: 56 (FIG. 61)).

[0830] The amino acid sequence of the full-length NYA-1143-LC is the amino acid sequence as shown in SEQ ID NO: 105 of the Sequence Listing (FIG. 106). In such an amino acid sequence, amino acids 1 to 20 constitute a signal sequence, amino acids 21 to 131 constitute a variable region, and amino acids 132 to 237 constitute a constant region. Also, amino acids 46 to 53 constitute CDRL1 (SEQ ID NO: 60 (FIG. 62)), amino acids 71 to 73 constitute CDRL2 (SEQ ID NO: 58 (FIG. 61)), and amino acids 110 to 121 constitute CDRL3 (SEQ ID NO: 59 (FIG. 61)).

[0831] The amino acid sequence of the full-length C3E-7085-HC2-k delete is the amino acid sequence as shown in SEQ ID NO: 106 of the Sequence Listing (FIG. 107). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 260 constitute C3E-7085, amino acids 261 to 262 constitute a linker, and amino acids 263 to 493 constitute Fc comprising a mutation introduced thereinto to lower effector functions and form a heterodimer.

10)-1-2 Preparation of Dual-Type Bispecific Molecule Expression Vector

[0832] In order to evaluate the dual-type anti-HLA/NY-ESO-anti-CD3 bispecific molecules, the expression vector described below was designed. An expression vector for mammal cells comprising a DNA fragment encoding the Fc region comprising a mutation to lower effector functions and form a heteropolymer integrated into the carboxyl terminus of NYA-1143 was prepared and designated as “p_NYA-1143-HC1-k delete.”

[0833] The nucleotide sequence of p_NYA-1143-HC1-k delete was reanalyzed, and the nucleotide sequence of the

full-length NYA-1143-HC1-k delete was found to be the nucleotide sequence as shown in SEQ ID NO: 107 of the Sequence Listing (FIG. 108). On the basis of the nucleotide sequences above, the amino acid sequence of the full-length NYA-1143-HC1-k delete encoded was identified. The amino acid sequence of the full-length NYA-1143-HC1-k delete is the amino acid sequence as shown in SEQ ID NO: 108 (FIG. 109) of the Sequence Listing. In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 266 constitute NYA-1143, amino acids 267 to 268 constitute a linker, amino acids 269 to 499 constitute Fc comprising a mutation introduced thereinto to lower effector functions and form a heterodimer.

10)-1-3 Preparation of scFv-Fab-Heterodimer Fc-Type Bispecific Molecule Expression Vector

[0834] In order to evaluate scFv-Fab-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecules, expression vectors for various molecules were designed. As the anti-HLA/NY-ESO antibody, NYA-1154 was used, and, as the anti-CD3 antibody, C3E-7085 was used. As the heterodimer Fc sequence, the Fc sequence comprising a mutation introduced thereinto to lower effector functions and form a heteropolymer was used.

[0835] An expression vector for mammal cells comprising DNA fragments encoding the heavy chain variable region of NYA-1154, the human JgG-derived CH1 region, and the Fc region comprising a mutation to lower effector functions and form a heteropolymer integrated into the carboxyl terminus of C3E-7085 was prepared and designated as “p_C3E-7085-NYA-1154-Fab-HC2-k delete.” Also, an expression vector for mammal cells comprising DNA fragments encoding the light chain variable region of NYA-1154 and the human IgG-derived CL region integrated therein was prepared and designated as “p_NYA-1154-LC.” Further, an expression vector for mammal cells comprising a DNA fragment encoding HC1-k delete integrated therein was prepared and designated as “p_OAA-HC1-k delete.”

[0836] The nucleotide sequence of p_C3E-7085-NYA-1154-Fab-HC2-k delete was reanalyzed, and the nucleotide sequence of the full-length C3E-7085-NYA-1154-Fab-HC2-k delete was found to be the nucleotide sequence as shown in SEQ ID NO: 109 of the Sequence Listing (FIG. 110).

[0837] The nucleotide sequence of p_NYA-1154-LC was reanalyzed, and the nucleotide sequence of the full-length NYA-1154-LC was found to be the nucleotide sequence as shown in SEQ ID NO: 110 (FIG. 111) of the Sequence Listing.

[0838] The nucleotide sequence of p_OAA-HC1-k delete was reanalyzed, and the nucleotide sequence of the full-length OAA-HC1-k delete was found to be the nucleotide sequence as shown in SEQ ID NO: 111 of the Sequence Listing (FIG. 112).

[0839] On the basis of the nucleotide sequences above, the amino acid sequences of the full-length C3E-7085-NYA-1154-Fab-HC2-k delete, NYA-1154-LC, and OAA-HC1-k delete encoded thereby were identified.

[0840] The amino acid sequence of the full-length C3E-7085-NYA-1154-Fab-HC2-k delete is the amino acid sequence as shown in SEQ ID NO: 112 of the Sequence Listing (FIG. 113). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 260 constitute C3E-7085, amino acids 261 to 385 constitute a linker, and amino acids 266 to 385 constitute the heavy

chain variable region of NYA-1154. Also, amino acids 386 to 714 constitute HC2-k delete.

[0841] The amino acid sequence of the full-length NYA-1154-LC is the amino acid sequence as shown in SEQ ID NO: 113 of the Sequence Listing (FIG. 114). In such an amino acid sequence, amino acids 1 to 20 constitute a signal sequence, amino acids 21 to 131 constitute a variable region, and amino acids 132 to 237 constitute a constant region. Also, amino acids 46 to 53 constitute CDRL1 (SEQ ID NO: 57 (FIG. 61)), amino acids 71 to 73 constitute CDRL2 (SEQ ID NO: 58 (FIG. 61)), and amino acids 110 to 121 constitute CDRL3 (SEQ ID NO: 63 (FIG. 64)).

[0842] The amino acid sequence of the full-length OAA-HC1-k delete is the amino acid sequence as shown in SEQ ID NO: 114 of the Sequence Listing (FIG. 115). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, and amino acids 20 to 245 constitute OAA-HC1-k delete.

10)-1-4 Preparation of taFv-Heterodimer Fc-Type Bispecific Molecule Expression Vectors for Evaluation of Various Formats

[0843] In order to evaluate taFv-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecules for evaluation of various formats, expression vectors for various molecules were designed. As the anti-HLA/NY-ESO antibodies, NYA-1143 and NYA-1154 were used. As the anti-CD3 antibody, humanized anti-CD3 scFv; i.e., C3E-7085, was used. As the heterodimer Fc sequence, the Fc sequence comprising a mutation introduced thereinto to lower effector functions and form a heteropolymer was used.

[0844] An expression vector for mammal cells comprising a DNA fragment encoding HC2-k delete integrated into the carboxyl terminus of taFv comprising NYA-1154 ligated to C3E-7085 with a GGGGS linker was prepared and designated as “p_NYF-0010-HC2-k delete.”

[0845] An expression vector for mammal cells comprising a DNA fragment encoding HC2-k delete integrated into the carboxyl terminus of taFv comprising C3E-7085 ligated to NYA-1154 with a GGGGS linker was prepared and designated as “p_NYF-0004-HC2-k delete.”

[0846] An expression vector for mammal cells comprising a DNA fragment encoding HC2-k delete integrated into the carboxyl terminus of taFv comprising NYA-1143 ligated to C3E-7085 with a GGGGS linker was prepared and designated as “p_NYF-0011-HC2-k delete.”

[0847] The nucleotide sequence of p_NYF-0010-HC2-k delete was reanalyzed, and the nucleotide sequence of the full-length NYF-0010-HC2-k delete was found to be the nucleotide sequence as shown in SEQ ID NO: 115 of the Sequence Listing (FIG. 116).

[0848] The nucleotide sequence of p_NYF-0004-HC2-k delete was reanalyzed, and the nucleotide sequence of the full-length NYF-0004-HC2-k delete was found to be the nucleotide sequence as shown in SEQ ID NO: 116 of the Sequence Listing (FIG. 117).

[0849] The nucleotide sequence of p_NYF-0011-HC2-k delete was reanalyzed, and the nucleotide sequence of the full-length NYF-0011-HC2-k delete was found to be the nucleotide sequence as shown in SEQ ID NO: 117 of the Sequence Listing (FIG. 118).

[0850] On the basis of the nucleotide sequences above, the amino acid sequences of the full-length NYF-0010-HC2-k delete, NYF-0004-HC2-k delete, and NYF-0011-HC2-k delete encoded thereby were identified.

[0851] The amino acid sequence of the full-length NYF-0010-HC2-k delete is the amino acid sequence as shown in SEQ ID NO: 118 (FIG. 119) of the Sequence Listing. In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acid 21 to 511 constitute NYA-1154-C3E-7085 taFv, amino acids 512 to 513 constitute a linker, and amino acids 514 to 744 constitute HC2-k delete.

[0852] The amino acid sequence of the full-length NYF-0004-HC2-k delete is the amino acid sequence as shown in SEQ ID NO: 119 (FIG. 120) of the Sequence Listing. In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acid 21 to 511 constitute C3E-7085-NYA-1154 taFv, amino acids 512 to 513 constitute a linker, and amino acids 514 to 744 constitute HC2-k delete.

[0853] The amino acid sequence of the full-length NYF-0011-HC2-k delete is the amino acid sequence as shown in SEQ ID NO: 120 of the Sequence Listing (FIG. 121). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acid 21 to 511 constitute NYA-1143-C3E-7085 taFv, amino acids 512 to 513 constitute a linker, and amino acids 514 to 745 constitute HC2-k delete.

10)-2 Expression of Fc-Conjugated Anti-HLA/NY-ESO-Anti-CD3 Bispecific Molecules of Various Formats

10)-2-1 Expression of Hybrid-Type Bispecific Molecules

[0854] In the same manner as in 7)-2-1, the culture supernatant of the hybrid-type bispecific molecule (NYG-3143) was prepared using a vector mixture comprising p_NYA-1143-Fab-HC1-k delete, p_C3E-7085-HC2-k delete, and p_NYA-1143-LC at 1:1:1.5. The amino acid sequences obtained via expression of vectors constituting NYG-3143 are shown in SEQ ID NO: 104 of the Sequence Listing (FIG. 105), SEQ ID NO: 105 of the Sequence Listing (FIG. 106), and SEQ ID NO: 106 of the Sequence Listing (FIG. 107).

10)-2-2 Expression of Dual-Type Bispecific Molecules

[0855] In the same manner as in 7)-2-1, the culture supernatant of the dual-type bispecific molecule (NYG-2143) was prepared using a vector mixture comprising p_NYA-1143-HC1-k delete and p_C3E-7085-HC2-k delete at 2:1. The amino acid sequences obtained via expression of vectors constituting NYG-2143 are shown in SEQ ID NO: 108 of the Sequence Listing (FIG. 109) and SEQ ID NO: 106 of the Sequence Listing (FIG. 107).

10)-2-3 Expression of scFv-Fab-Heterodimer Fc-Type Bispecific Molecules

[0856] In the same manner as in 7)-2-1, the culture supernatant of the scFv-Fab-heterodimer Fc-type (scFv-Fab-Fc-type) bispecific molecule (NYF-0003) was prepared using a vector mixture comprising p_C3E-7085-NYA-1154-Fab-HC2-k delete, p_OAA-HC1-k delete, and p_NYA-1154-LC at 1:1:1.5. The amino acid sequences obtained via expression of vectors constituting NYF-0003 are shown in SEQ ID NO: 112 of the Sequence Listing (FIG. 113), SEQ ID NO: 113 of the Sequence Listing (FIG. 114), and SEQ ID NO: 114 of the Sequence Listing (FIG. 115).

10)-2-4 Expression of taFv-Heterodimer Fc-Type Bispecific Molecules for Evaluation of Various Formats

[0857] In the same manner as in 7)-2-1, the culture supernatant of the taFv-heterodimer Fc-type (taFv-Fc-type) anti-HLA/NY-ESO-anti-CD3 bispecific molecule (NYF-0010)

was prepared using p_NYF-0010-HC2-k delete and p_OAA-HC1-k delete. The amino acid sequences obtained via expression of vectors constituting NYF-0010 are shown in SEQ ID NO: 118 of the Sequence Listing (FIG. 119) and SEQ ID NO: 114 of the Sequence Listing (FIG. 115).

[0858] The culture supernatant of the taFv (inverted)-heterodimer Fc-type (taFv (inverted)-Fc-type) anti-HLA/NY-ESO-anti-CD3 bispecific molecule (NYF-0004) was prepared using p_NYF-0004-HC2-k delete and p_OAA-HC1-k delete. The amino acid sequences obtained via expression of vectors constituting NYF-0004 are shown in SEQ ID NO: 119 of the Sequence Listing (FIG. 120) and SEQ ID NO: 114 of the Sequence Listing (FIG. 115).

[0859] The culture supernatant of the taFv-Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecule (NYF-0011) was prepared using p_NYF-0011-HC2-k delete and p_OAA-HC1-k delete. The amino acid sequences obtained via expression of vectors constituting NYF-0011 are shown in SEQ ID NO: 120 of the Sequence Listing (FIG. 121) and SEQ ID NO: 114 of the Sequence Listing (FIG. 115).

10)-3 Purification of Fc-Conjugated Anti-HLA/NY-ESO-Anti-CD3 Bispecific Molecules of Various Formats

[0860] Various bispecific molecules were purified from the culture supernatants obtained in 10)-2-1 and 10)-2-2 by 2 steps of protein A affinity chromatography and ceramic hydroxyapatite chromatography.

[0861] The culture supernatant was applied to the Mab-SelectSuRe column equilibrated with PBS at pH 7.4 (GE Healthcare Bioscience) to allow the target bispecific molecules to adsorb thereto. After the non-adsorbed components were removed by PBS, the adsorbed components were eluted using acetate buffer (pH 3.6). The elution fraction was neutralized with the aid of Tris buffer (pH 9.5), and a buffer was exchanged with 25 mM histidine, 150 mM NaCl, 5% Sorbitol, at pH 5.5. The solution of the target fraction diluted to 5-fold with a buffer comprising 10 mM potassium phosphate and 50 mM MES (pH 6.5) was applied to the ceramic hydroxyapatite column (Bio-Scale CHT Type-1 Hydroxyapatite Column, BioRad Japan) equilibrated with 10 mM potassium phosphate and 50 mM MES (pH 6.5). Linear concentration gradient elution was performed using sodium chloride, and fractions equivalent to the target heterodimer were collected. The fractions were applied to the gel filtration column Superdex 200 10/300 (GE Healthcare Bioscience) equilibrated with 25 mM histidine, 300 mM NaCl, 5% Sorbitol at pH 6.0 in advance. From the peak fraction obtained via gel filtration chromatography, fractions equivalent to the target heterodimer were collected, and the buffer was exchanged with 25 mM histidine, 300 mM NaCl, 5% Sorbitol at pH 5.5. Generation of the target anti-HLA/NY-ESO-anti-CD3 bispecific molecule was confirmed via SDS-polyacrylamide electrophoresis (SDS-PAGE). Purified protein samples were subjected to analytical SEC, the degree of purification and the concentration were determined, and the samples were then subjected to various types of evaluation.

[0862] Various bispecific molecules were purified from the culture supernatants obtained in 10)-2-3 and 10)-2-4 by 2 steps of protein A affinity chromatography and gel filtration chromatography in the same manner as in 7)-3. Generation of the target anti-HLA/NY-ESO-anti-CD3 bispecific molecule was confirmed via SDS-polyacrylamide electrophoresis (SDS-PAGE). Purified protein samples were sub-

jected to analytical SEC, the degree of purification and the concentration were determined, and the samples were then subjected to various types of evaluation.

(Example 11) Evaluation of In Vitro Activity of Fc-Conjugated Anti-HLA-A2/NY-ESO-Anti-CD3 Bispecific Molecules of Various Formats

11)-1 Evaluation of In Vitro Activity of Hybrid-Type, Dual-Type, and taFv-Fc-Type Anti-HLA-A2/NY-ESO-anti-CD3 Bispecific Molecules

11)-1-1 Preparation of Target Cells

[0863] U266B1 cells prepared in the same manner as in Example 8)-1 were used as the target cells.

11)-1-2 Preparation of Effector Cells

[0864] Commercially available frozen human PBMCs (Cellular Technology Limited) prepared in the same manner as in Example 8)-2 were used as the effector cells.

11)-1-3 Cytotoxicity Assay

[0865] The target cells obtained in Example 11)-1-1 were seeded on a 96-well U-bottom microplate at 50 μ l/well. The anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules of various formats prepared in Example 10 and adjusted to various concentration levels were added thereto at 50 μ l/well, the effector cells prepared in Example 11)-1-2 were added thereto at 100 l/well, centrifugation was carried out at room temperature and 1,000 rpm for 1 minute, and then they were incubated at 37° C. in the presence of 5% CO₂ for 20 to 24 hours. The supernatant (50 μ l) was collected on a LumaPlate (PerkinElmer), dried at 50° C. for approximately 2 hours, and then assayed using a plate reader (TopCount, PerkinElmer). The test was performed in triplicate, and the rate of cell lysis was determined in accordance with the following equation.

$$\text{Rate of cell lysis (\%)} = (A - B)/(C - B) \times 100$$

[0866] A: Sample well count

[0867] B: Mean of background (wells not supplemented with antibody) count (n=3).

[0868] When the antibody was added, 50 μ l of an assay medium was added. Other procedures were the same as those for the sample wells.

[0869] C: Mean of maximal release (wells in which target cells were lysed with a surfactant) count (n=3).

[0870] When the antibody was added, 50 μ l of an assay medium was added. A surfactant was added in an amount of 100 μ l, a fraction of 50 μ l was transferred to the LumaPlate as in the case of the sample wells, and assays were then performed. As shown in FIG. 7A, the taFv-Fc type exerted cytotoxicity. The results of calculations of EC₅₀ obtained using the Four Parameter Logistic Curve equation of analytical software (Sigmaplot, version 12.0) are shown in Table 5. "Not Applicable (NA)" indicates that it is impossible to perform curve fitting because the correlation coefficient (R) was not calculated. Neither Hybrid-type nor Dual-type exerted cytotoxicity.

TABLE 5

	Format	Name	EC ₅₀ (nM)
1	Hybrid	NYG-3143	NA*
2	Dual	NYG-2143	NA*
3	taFv-Fc	NYF-0011	0.27

*NA = Not Applicable

11)-2 Evaluation of In Vitro Activity of scFv-Fab-Fc-Type, taFv-Fc-Type, and taFv (Inversed)-Fc-Type Anti-HLA-A2/NY-ESO-Anti-CD3 Bispecific Molecules

11)-2-1 Preparation of Target Cells

[0871] U266B1 cells prepared in the same manner as in Example 8)-1 were used as the target cells.

11)-2-2 Preparation of Effector Cells

[0872] Commercially available frozen human PBMCs (Cellular Technology Limited) prepared in the same manner as in Example 8)-2 were used as the effector cells.

11)-2-3 Cytotoxicity Assay

[0873] The target cells obtained in Example 11)-2-1 were seeded on a 96-well U-bottom microplate at 50 µl/well. The anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules of various formats prepared in Example 10 and adjusted to various concentration levels were added thereto at 50 µl/well, the effector cells prepared in Example 11)-1-2 were added thereto at 100 l/well, centrifugation was carried out at room temperature and 1,000 rpm for 1 minute, and then they were incubated at 37° C. in the presence of 5% CO₂ for 20 to 24 hours. The supernatant (50 µl) was collected on a LumaPlate (PerkinElmer), dried at 50° C. for approximately 2 hours, and then assayed using a plate reader (TopCount, PerkinElmer). The test was performed in triplicate, and the rate of cell lysis was determined in accordance with the following equation.

$$\text{Rate of cell lysis (\%)} = (A - B)/(C - B) \times 100$$

[0874] A: Sample well count

[0875] B: Mean of background (wells not supplemented with antibody) count (n=3).

[0876] When the antibody was added, 50 µl of an assay medium was added. Other procedures were the same as those for the sample wells.

[0877] C: Mean of maximal release (wells in which target cells were lysed with a surfactant) count (n=3).

[0878] When the antibody was added, 50 µl of an assay medium was added. A surfactant was added in an amount of 100 µl, a fraction of 50 µl was transferred to the LumaPlate as in the case of the sample wells, and assays were then performed. As shown in FIG. 7B, the taFv-Fc type exerted cytotoxicity. The results of calculations of EC₅₀ obtained using the Four Parameter Logistic Curve equation of analytical software (Sigmaplot, version 12.0) are shown in Table 6. "Not Applicable (NA)" indicates that it is impossible to perform curve fitting because the correlation coefficient (R) was not calculated. In comparison with the

taFv-heterodimer Fc-type, the scFv-Fab-heterodimer Fc-type and the taFv (inversed)-heterodimer Fc-type exhibited lower activity.

TABLE 6

	Format	Name	EC ₅₀ (nM)
1	scFv-Fab-heterodimer Fc	NYF-0003	NA*
2	taFv-heterodimer Fc	NYF-0010	0.13
3	taFv (inversed)-heterodimer Fc	NYF-0004	0.88

*NA = Not Applicable

(Example 12) Preparation of Various Fc-Conjugated Anti-HLA/NY-ESO-Anti-CD3 Bispecific Molecules Via Mutagenesis and Format Modification of NYF-0061

12)-1 Preparation of Various Fc-Conjugated Anti-HLA/NY-ESO-Anti-CD3 Bispecific Molecule Expression Vectors Via Mutagenesis and Format Modification of NYF-0061

[0879] In order to improve physical properties, NYF-0061 mutants were designed and the vectors constituting the molecules described below were prepared.

[0880] A mutation was introduced into the C3E-7085 sequence of the p_NYF-0061-HC2 vector containing the nucleotide sequence encoding the amino acid sequence of NYF-0061 and the resultant was designated as "p_NYZ-0038-HC2."

[0881] A mutation was introduced into the NYA-2061 sequence and the C3E-7085 sequence of the p_NYF-0061-HC2 vector containing the nucleotide sequence encoding the amino acid sequence of NYF-0061 and the resultants were designated as "p_NYZ-0082-HC2" and "p_NYZ-0083-HC2," respectively.

[0882] The nucleotide sequence of p_NYZ-0038-HC2 was reanalyzed, and the nucleotide sequence of the full-length NYZ-0038-HC2 was found to be the nucleotide sequence as shown in SEQ ID NO: 152 of the Sequence Listing (FIG. 148).

[0883] The nucleotide sequence of p_NYZ-0082-HC2 was reanalyzed, and the nucleotide sequence of the full-length NYZ-0082-HC2 was found to be the nucleotide sequence as shown in SEQ ID NO: 153 of the Sequence Listing (FIG. 149).

[0884] The nucleotide sequence of p_NYZ-0083-HC2 was reanalyzed, and the nucleotide sequence of the full-length NYZ-0083-HC2 was found to be the nucleotide sequence as shown in SEQ ID NO: 154 of the Sequence Listing (FIG. 150).

[0885] On the basis of the nucleotide sequences above, the amino acid sequences of the full-length NYZ-0038-HC2, NYZ-0082-HC2, and NYZ-0083-HC2 encoded thereby were identified.

[0886] The amino acid sequence of the full-length NYZ-0038-HC2 is the amino acid sequence as shown in SEQ ID NO: 155 of the Sequence Listing (FIG. 151). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acid 21 to 511 constitute NYA-2061-C3E-7096 taFv, amino acids 512 to 513 constitute a linker, and amino acids 514 to 745 constitute HC2.

[0887] The amino acid sequence of the full-length NYZ-0082-HC2 is the amino acid sequence as shown in SEQ ID

NO: 156 of the Sequence Listing (FIG. 152). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 516 constitute NYA-3061-C3E-7096 taFv, amino acids 517 to 518 constitute a linker, and amino acids 519 to 750 constitute HC2.

[0888] The amino acid sequence of the full-length NYZ-0083-HC2 is the amino acid sequence as shown in SEQ ID NO: 157 of the Sequence Listing (FIG. 153). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 516 constitute NYA-3061-C3E-7097 taFv, amino acids 517 to 518 constitute a linker, and amino acids 519 to 750 constitute HC2.

[0889] In order to examine format modification of NYF-0061 into the scFv-Fab-heterodimer Fc-type, the vectors constituting the molecules described below were designed.

[0890] An expression vector for mammal cells comprising DNA fragments encoding an amino acid sequence included in a polypeptide comprising (i) the heavy chain variable region of C3E-7085, (ii) the human IgG-derived CH1 region, and (iii) the Fc region comprising a mutation to lower effector functions and form a heteropolymer integrated in that order into the carboxyl terminus of NYA-3061 was prepared and designated as "p_NYZ-1010-HC2." Also, an expression vector for mammal cells comprising a DNA fragment encoding an amino acid sequence included in a polypeptide comprising the human IgG-derived CL region added to the carboxyl terminus of the light chain variable region of C3E-7085 was prepared and designated as "p_C3E-7085-LC."

[0891] The nucleotide sequence of p_NYZ-1010-HC2 was reanalyzed, and the nucleotide sequence of the full-length NYZ-1010-HC2 was found to be the nucleotide sequence as shown in SEQ ID NO: 158 of the Sequence Listing (FIG. 154).

[0892] The nucleotide sequence of p_C3E-7085-LC was reanalyzed, and the nucleotide sequence of the full-length C3E-7085-LC was found to be the nucleotide sequence as shown in SEQ ID NO: 159 of the Sequence Listing (FIG. 155).

[0893] On the basis of the nucleotide sequences above, the amino acid sequences of the full-length NYZ-1010-HC2 and C3E-7085-LC encoded thereby were identified.

[0894] The amino acid sequence of the full-length NYZ-1010-HC2 is the amino acid sequence as shown in SEQ ID NO: 160 of the Sequence Listing (FIG. 156). In such an amino acid sequence, amino acids 1 to 19 constitute a signal sequence, amino acids 21 to 271 constitute NYA-3061, amino acids 272 to 276 constitute a linker, and amino acids 277 to 394 constitute the heavy chain variable region of C3E-7085. Also, amino acids 395 to 724 constitute a constant region.

[0895] The amino acid sequence of the full-length C3E-7085-LC is the amino acid sequence as shown in SEQ ID NO: 161 of the Sequence Listing (FIG. 157). In such an amino acid sequence, amino acids 1 to 20 constitute a signal sequence, amino acids 21 to 127 constitute a variable region, and amino acids 128 to 233 constitute a constant region. Also, amino acids 46 to 53 constitute CDRL1 (SEQ ID NO: 144 (FIG. 142)), amino acids 71 to 73 constitute CDRL2 (SEQ ID NO: 145 (FIG. 142)), and amino acids 110 to 117 constitute CDRL3 (SEQ ID NO: 146 (FIG. 142)).

12)-2 Expression and Purification of Various Fc-Conjugated Anti-HLA/NY-ESO-Anti-CD3 Bispecific Molecules Via Mutagenesis and Format Modification or NYF-0061

[0896] In the same manner as in 7)-2-1, the culture supernatant of the taFv-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecule (NYZ-0038) was prepared using p_NYZ-0038-HC2 and p_HC1. The amino acid sequences obtained via expression of vectors constituting NYZ-0038 are shown in SEQ ID NO: 155 of the Sequence Listing (FIG. 151) and SEQ ID NO: 84 of the Sequence Listing (FIG. 85).

[0897] The culture supernatant of the taFv-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecule (NYZ-0082) was prepared using p_NYZ-0082-HC2 and p_HC1. The amino acid sequences obtained via expression of vectors constituting NYZ-0082 are shown in SEQ ID NO: 156 of the Sequence Listing (FIG. 152) and SEQ ID NO: 84 of the Sequence Listing (FIG. 85).

[0898] The culture supernatant of the taFv-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecule (NYZ-0083) was prepared using p_NYZ-0083-HC2 and p_HC1. The amino acid sequences obtained via expression of vectors constituting NYZ-0083 are shown in SEQ ID NO: 157 of the Sequence Listing (FIG. 153) and SEQ ID NO: 84 of the Sequence Listing (FIG. 85).

[0899] The culture supernatant of the scFv-Fab-heterodimer Fc-type anti-HLA/NY-ESO-anti-CD3 bispecific molecule (NYZ-1010) was prepared using p_NYZ-1010-HC2, p_C3E-7085-LC, and p_HC1. The amino acid sequences obtained via expression of vectors constituting NYZ-1010 are shown in SEQ ID NO: 160 of the Sequence Listing (FIG. 156), SEQ ID NO: 161 of the Sequence Listing (FIG. 157), and SEQ ID NO: 84 of the Sequence Listing (FIG. 85).

[0900] The culture supernatants prepared above were purified in the same manner as in 7)-3. Purified protein samples were subjected to analytical SEC, the degree of purification and the concentration were determined, and the samples were then subjected to various types of evaluation.

(Example 13) Evaluation of Binding Affinity to HLA/NY-ESO Using Biacore

[0901] With the use of Biacore T200, various Fc-conjugated anti-HLA/NY-ESO-anti-CD3 bispecific molecules were captured as ligands to the immobilized anti-human IgG (Fc) antibody, and the antigen was assayed as an analyte. As the antigen, HLA/NY-ESO prepared in 1)-1 was used. The anti-human IgG (Fc) antibody (Human antibody Capture kit, GE Healthcare) was immobilized on Sensor Chip CM5 (GE Healthcare) in accordance with the instructions of the kit. The Fc-conjugated anti-HLA/NY-ESO-anti-CD3 bispecific molecules diluted to 0.5 µg/ml in HBS-EP+ (GE Healthcare) to be evaluated were brought into contact therewith at 10 µl/min for 60 seconds for immobilization. Thereafter, the samples were added to the HLA/NY-ESO analytes diluted to various levels with HBS-EP+ at a flow rate of 30 µl/min for 120 seconds and dissociation was assayed for 600 seconds. The results of calculations obtained by such single cycle kinetics analysis, K_D , are shown in Table 7. NYZ-0038, NYZ-0082, NYZ-0083, and NYZ-1010 retained binding affinity equivalent to that of NYF-0061.

TABLE 7

Clone name	K _D (nM)
NYF-0061	1.5
NYZ-0038	1.5
NYZ-0082	1.8
NYZ-0083	1.7
NYZ-1010	1.9

(Example 14) Preparation of CD3e Knockout T2 Human Lymphoblast Fusion Cells

[0902] In order to knockout CD3e in the genome sequence of the T2 human lymphoblast fusion cells (ATCC) by the technique of CRISPR, the Cas9 expression plasmid (GE Healthcare) and the sgRNA expression plasmid were introduced via electroporation (LONZA). Thereafter, cell cloning was performed by limited dilution. The cells in which deletion of the target gene fragment from the introduced cells and CD3e gene expression were not observed as a result of genome analysis and RT-PCR analysis of total RNA, respectively, were subjected to the subsequent experiments.

(Example 15) Analysis of Recognition Amino Acid in NY-ESO Peptide of Fc-Conjugated Anti-HLA-A2/NY-ESO-Anti-CD3 Bispecific Molecule

[0903] The concentration of the CD3e knockout T2 cells prepared in Example 14 was adjusted to an adequate level in AIM-V medium (Thermo Fisher Scientific) containing 20% FBS, the NY-ESO peptide (SEQ ID NO: 1), the point-mutant NY-ESO peptides 1F, 2M, 3A, 4A, 5A, 6L, 7F, 8A, and 9A (SEQ ID NO: 121 (FIG. 122), 122 (FIG. 123), 123 (FIG. 124), 124 (FIG. 125), 125 (FIG. 126), 126 (FIG. 127), 127 (FIG. 128), 128 (FIG. 129), and 129 (FIG. 130)), and the gp100 peptide (SEQ ID NO: 130 (FIG. 131)) (Sigma Genosys) were added in the same manner as in Example 5, cells stained with the use of the LIVE/DEAD Fixable Dead Cell Stain Kit (Thermo Fisher Scientific) were divided into two groups, the cells in PBS containing 5% FBS were seeded on a 96-well U-bottom microplate at 10⁵ cells/well, and the plate was subjected to centrifugation, followed by removal of the supernatant. The various Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules prepared in Example 12 and diluted to 100 nM with PBS containing 5% FBS were added at 25 μl/well to one of the two groups, and the resultant was allowed to stand at 4° C. for 30 minutes. The cells were washed 2 times in PBS containing 5% FBS, PE conjugated F(ab')₂ Fragment Anti-Human IgG (Jackson Immuno Research Laboratories) diluted with PBS containing 5% FBS was added at 25 l/well, and the resultant was allowed to stand at 4° C. for 30 minutes. The cells were washed 2 times in PBS containing 5% FBS, immobilized with Mildform 10 N (FUJIFILM Wako Pure Chemical Corporation) overnight, and resuspended in PBS containing 5% FBS. For standardization with the amount of the HLA/peptide complex, the HLA-A2 antibody BB7.2-Alexa Fluor 488 diluted to 10 μg/ml in PBS containing 5% FBS or mouse IgG2b-Alexa Fluor 488 (BioRAD) was added at 25 μl/well, and the resultant was allowed to stand at 4° C. for 30 minutes. The cells were washed 2 times in PBS containing 5% FBS, immobilized with Mildform 10 N (FUJIFILM

Wako Pure Chemical Corporation) overnight, and resuspended in PBS containing 5% FBS. The cell suspensions were subjected to detection using a flow cytometer (CantoII, Becton Dickinson). Data analysis was performed using Flowjo (Treestar), and the geometric mean fluorescence intensity (gMFI) of PE or Alexa Fluor 488 in the CD3e knockout T2 cells from which dead cells were removed was measured. The standardized gMFI indicating the binding affinity of each Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecule standardized with the amount of the HLA/peptide complex on the CD3e knockout T2 cells was determined in accordance with the following equation.

Standardized gMFI = (A/B)/((C/D)/(E/F))

- [0904] A/B=Relative gMFI
- [0905] A: gMFI of PE in CD3e knockout T2 cells supplemented with DMSO or peptide in the presence of antibody
- [0906] B: gMFI of PE in CD3e knockout T2 cells supplemented with DMSO or peptide
- [0907] (C/D)/(E/F)=Corrected amount of HLA/peptide complex in CD3e knockout T2 cells supplemented with DMSO or peptide
- [0908] C: gMFI of Alexa488 in CD3e knockout T2 cells supplemented with DMSO or peptide in the presence of HLA-A2 antibody
- [0909] D: gMFI of Alexa488 in CD3e knockout T2 cells supplemented with DMSO or peptide in the presence of mouse IgG2b antibody
- [0910] E: gMFI of Alexa488 in CD3e knockout T2 cells supplemented with DMSO containing HLA-A2 antibody
- [0911] F: gMFI of Alexa488 in CD3e knockout T2 cells supplemented with DMSO containing mouse IgG2b antibody
- [0912] As shown in FIG. 158A, the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYZ-0038, NYZ-0082, NYZ-0083, and NYZ-1010, exhibited binding affinity to the CD3e knockout T2 cells, supplemented with peptides comprising point mutations introduced into amino acids 1, 4, 5, and 7, that had been lowered to a quarter of the binding affinity for the wild-type NY-ESO peptide or lower. This indicates that such constructs strongly recognize amino acids 1, 4, 5, and 7 of the NY-ESO peptide.

(Example 16) Evaluation of Antigen-Binding Specificity of Fc-Conjugated Anti-HLA-A2/NY-ESO-Anti-CD3 Bispecific Molecules

[0913] In the same manner as in Example 6), the 9-mer peptide shown in FIG. 2A exhibiting identity to the sequence of the NY-ESO peptide: SLLMWITQC (SEQ ID NO: 1) at amino acids 1, 4, and 5, which are indicated to be strongly recognized by the four types of antibodies, was selected as a homologous peptide, and binding specificity thereto was evaluated. The concentration of the CD3e knockout T2 cells was adjusted to an adequate level in AIM-V medium (Thermo Fisher Scientific) containing 20% FBS, a solution of the NY-ESO peptide (SEQ ID NO: 1), the homologous peptides DOLPP1, IL20RB, PRKD2, CD163, and P2RY8 (SEQ ID NO: 131 (FIG. 132), SEQ ID NO: 132 (FIG. 133),

SEQ ID NO: 133 (FIG. 134), SEQ ID NO: 134 (FIG. 135), and SEQ ID NO: 135 (FIG. 136)), or the gp100 peptide (SEQ ID NO: 130 (FIG. 131)) (Sigma Genosys) dissolved to 5 mM in DMSO was added to result in a final concentration of 50 μ M, or DMSO was added in an amount of 1/100, and binding affinity of the antibodies was evaluated in the same manner as in Example 15. The standardized gMFI indicating the binding affinity of each Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecule standardized with the amount of the HLA/peptide complex on the CD3e knockout T2 cells was determined in accordance with the following equation.

$$\text{Standardized gMFI} = (A/B)/((C/D)/(E/F))$$

- [0914] A: gMFI of PE in CD3e knockout T2 cells supplemented with DMSO or peptide in the presence of antibody
- [0915] B: gMFI of PE in CD3e knockout T2 cells supplemented with DMSO or peptide
- [0916] $(C/D)/(E/F)$ =Corrected amount of HLA/peptide complex in CD3e knockout T2 cells supplemented with DMSO or peptide
- [0917] C: gMFI of Alexa488 in CD3e knockout T2 cells supplemented with DMSO or peptide of the presence of HLA-A2 antibody
- [0918] D: gMFI of Alexa488 in CD3e knockout T2 cells supplemented with DMSO or peptide containing mouse IgG2b antibody
- [0919] E: gMFI of Alexa488 in CD3e knockout T2 cells supplemented with DMSO containing HLA-A2 antibody
- [0920] F: gMFI of Alexa488 in CD3e knockout T2 cells supplemented with DMSO containing mouse IgG2b antibody
- [0921] As shown in FIG. 158B, none of the Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules; i.e., NYZ-0038, NYZ-0082, NYZ-0083, and NYZ-1010, had bound to cells supplemented with homologous peptides, and these molecules exhibited very high specificity.

(Example 17) Evaluation of Cytotoxicity of
Fc-Conjugated Anti-HLA-A2/NY-ESO-Anti-CD3
Bispecific Molecules

17)-1 Preparation of Target Cells

[0922] In the same manner as in Example 8)-1, suspensions of the endogenous human NY-ESO-expressing cells (U266B1 and NCI-H1703) and the endogenous human NY-ESO-non-expressing cells (AGS and CFPAC-1) were prepared and used as the target cells.

17)-2 Preparation of Effector Cells

[0923] In the same manner as in Example 8)-2, a suspension of commercially available frozen human PBMCs (Cellular Technology Limited) was prepared and used as the effector cells.

17)-3 Cytotoxicity Assay

[0924] The target cells obtained in Example 17)-1 were seeded on a 96-well U-bottom microplate at 50 μ l/well. The

various Fc-conjugated anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules prepared in Example 12 and adjusted to various concentration levels were added thereto at 50 μ l/well, the effector cells prepared in Example 17)-2 were added thereto at 100 l/well, centrifugation was carried out at room temperature and 1,000 rpm for 1 minute, and then they were incubated at 37° C. in the presence of 5% CO₂ for 20 to 24 hours. The supernatant (50 μ l) was collected on a LumaPlate (PerkinElmer), dried at 50° C. for approximately 2 hours, and then assayed using a plate reader (TopCount, PerkinElmer). The test was performed in triplicate, and the rate of cell lysis was determined in accordance with the following equation.

$$\text{Rate of cell lysis (\%)} = (A - B)/(C - B) \times 100$$

[0925] A: Sample well count

[0926] B: Mean of background (wells not supplemented with antibody) count (n=3).

[0927] When the antibody was added, 50 μ l of an assay medium was added. Other procedures were the same as those for the sample wells.

[0928] C: Mean of maximal release (wells in which target cells were lysed with a surfactant) count (n=3).

[0929] When the antibody was added, 50 μ l of an assay medium was added. A surfactant was added in an amount of 100 μ l, a fraction of 50 μ l was transferred to the LumaPlate as in the case of the sample wells, and assays were then performed.

[0930] As shown in FIG. 159A to FIG. 159D, various anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules exerted cytotoxicity on the endogenous human NY-ESO-expressing cells (U266B1 and NCI-H1703). The results of calculations of EC₅₀ obtained using the Four Parameter Logistic Curve equation of analytical software (Sigmaplot, version 12.0) concerning the U266B1 cells are shown in Table 8, and the results concerning the NCI-H1703 cells are shown in Table 9. As shown in FIG. 159E to FIG. 159H, in contrast, no cytotoxicity was observed on the endogenous human NY-ESO-non-expressing cells (AGS and CFPAC-1).

TABLE 8

	Name	EC ₅₀ (nM)
1	NYZ-0038	1.01
2	NYZ-0082	0.45
3	NYZ-0083	0.25
4	NYZ-1010	0.96

TABLE 9

	Name	EC ₅₀ (nM)
1	NYZ-0038	0.61
2	NYZ-0082	1.00
3	NYZ-0083	0.71
4	NYZ-1010	2.56

(Example 18) Evaluation of In Vivo Activity of Fc-Conjugated Anti-HLA-A2/NY-ESO-Anti-CD3 Bispecific Molecules in Human PBMC-Transfected Models

[0931] The human squamous lung cancer cell lines NCI-H1703 (ATCC) were adjusted to 6×10^7 cells/ml in PBS containing 50% Matrigel (Corning) and 0.1 ml thereof was injected hypodermically to NOG mice (female, 6- to 7-week-old) (Day 0). On Day 4, human PBMCs were adjusted to 3.75×10^7 to 5×10^7 cells/ml in PBS, and 0.2 ml thereof was injected intravenously. Approximately 1 week thereafter (Day 6 to Day 7), measurement of the major diameter (mm) and the minor diameter (mm) of the tumor was initiated, measurement was performed using an electronic digital caliper with the elapse of time, and the estimated tumor volume was calculated in accordance with the equation below.

Estimated tumor volume (mm³)=mean of estimated tumor volume among individuals Estimated tumor volume of an individual=major diameter×[minor diameter]²/2

[0932] On Day 14, mice were divided into groups each consisting of 5 or 6 mice on the basis of tumor volumes, and various anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules were administered intravenously (NYZ-1010 at 1 mg/kg, NYZ-1010 at 1.2 mg/kg for comparison at the same molar weight). Administration was performed on Day 14 and Day 21. Antitumor effects were observed in the treatment groups to which various anti-HLA-A2/NY-ESO-anti-CD3 bispecific molecules had been administered (FIG. 160A to FIG. 160D). The tumor growth inhibition (%) on Day 28 to Day 29 was calculated in accordance with the following equation and shown in Table 10.

Tumor growth inhibition (%) = $100 - \left(\frac{\text{estimated tumor volume of treatment group}}{\text{estimated tumor volume of vehicle control}} \times 100 \right)$

TABLE 10

	Name	Tumor growth inhibition (%)
1	NYZ-0038	100.0
2	NYZ-0082	100.0
3	NYZ-0083	99.8
4	NYZ-1010	99.9

(Example 19) Comparison of Physicochemical Properties of Various Fc-Conjugated Anti-HLA/NY-ESO-Anti-CD3 Bispecific Molecules

19)-1 Acid Treatment Evaluation

[0933] NYF-0016 and various NYA-1143 CDR graft mutants (NYF-0044, NYF-0045, NYF-0047, NYF-0048, NYF-0060, and NYF-0061) prepared in 7)-2-1 were concentrated by centrifugation using Amicon (Millipore), the buffer was exchanged with 25 mM sodium acetate, 5% sorbitol (pH 5.5), and the concentration was adjusted to 15 mg/ml (NYF-0016 at 10 mg/ml). Subsequently, the samples

were dialyzed using Xpress Micro Dialyzer (Scienova) against 100 mM sodium acetate (pH 3.5) or 25 mM sodium acetate, 5% sorbitol (pH 5.5) as a control, and the samples were collected and allowed to stand at room temperature for 1 hour. Subsequently, pH levels of the samples were adjusted to 5.0 using 500 mM Tris-HCl (pH 9.0). The samples were analyzed via size exclusion chromatography (SEC) using ACQUITY UPLC BEH200 SEC 1.7 m 4.6*150 mm (Waters). As the mobile phase, 0.2 M Ki/200 mM KCl/pH 7.0 was employed, and analysis was performed at a flow rate of 0.2 ml/min (detection wavelength: 280 nm). The peak contents (%) of polymers contained in the samples were analyzed and calculated by the area percentage method. The results are shown in FIG. 162. As a result of acid treatment, the polymer content in NYF-0016 was increased to 96%. In various NYA-1143 CDR graft mutants, in contrast, the amount of polymer formation upon acid treatment was decreased. In NYF-0047, NYF-0048, NYF-0060, and NYF-0061, in particular, the amount of polymer formation upon acid treatment was decreased to 1% or lower. This indicates that acid resistance was improved to a significant extent.

19)-2 Evaluation of Solution Stability

[0934] NYF-0061 prepared in 7)-2-1 and NYZ-0038, NYZ-0082, NYZ-0083, and NYZ-1010 prepared in 12)-2 were concentrated by centrifugation using Amicon Ultra-4 (Millipore), the buffer was exchanged with 25 mM histidine, 5% sorbitol (pH 6.0), 25 mM histidine and 5% sorbitol (pH 6.0) were added thereto, and the sample concentration was adjusted to 25 mg/ml to prepare evaluation samples. In the beginning, the high molecular weight species (HMWS) of each evaluation sample was determined by the area percentage method via size exclusion chromatography using AdvanceBio SEC 300A 2.7 m 4.6*300 mm (Agilent). As the mobile phase, 0.2 M Ki/200 mM KCl/pH 7.0 was employed, and analysis was performed at a flow rate of 0.2 ml/min (detection wavelength: 280 nm). The solution stability test of each sample was performed by storage at 25° C. for 6 days (NYF-0061: 7 days), size exclusion chromatography under the conditions as described above, and calculation of HMWS (%) of each sample by the area percentage method. The results are shown in Table 11. HMWS (%) of NYF-0061 was increased to approximately 10.6% with the elapse of storage time. In contrast, an increase in HMWS (%) of NYZ-0038, NYZ-0082, NYZ-0083, and NYZ-1010 was approximately 2% or lower. This demonstrates that solution stability of NYZ-0038, NYZ-0082, NYZ-0083, and NYZ-1010 is superior to that of NYF-0061.

TABLE 11

Clone name	HMWS (%)
NYF-0061	10.6
NYZ-0038	2.1
NYZ-0082	1.7
NYZ-0083	1.9
NYZ-1010	0.8

(Example 20) Preparation of Anti-HLA/NY-ESO scFv

20)-1 Construction of Anti-HLA/NY-ESO scFv Expression Vector

[0935] scFv constructs of antibodies that may have high binding affinity to HLA-A2/NY-ESO-1; i.e., mAb24955N, mAb24956N, mAb28075P, mAb28105P, mAb28113P, and mAb29822P2 (WO2021/003357), were designed, and scFv of mAb24955N was designated as NYC-0005, scFv of mAb24956N was designated as NYC-0006, scFv of mAb28075P was designated as NYC-0007, scFv of mAb28105P was designated as NYC-0008, scFv of mAb28113P was designated as NYC-0009, and scFv of mAb29822P2 was designated as NYC-0010.

[0936] Also, scFv expression vectors for NYC-0005, NYC-0006, NYC-0007, NYC-0008, NYC-0009, and NYC-0010 each comprising pcDNA3.4 (ThermoFisher Scientific) as the backbone in mammalian cells were designed. In order to compare expression levels of the same vector backbones and formats in mammalian cells, in addition, scFv expression vectors for NYA-2047, NYA-2061, and NYA-3061 in mammalian cells were designed.

[0937] The nucleotide sequences of the constructed scFv expression vectors were reanalyzed, and the nucleotide sequences of the full-length NYA-2047, NYA-2061, NYA-3061, NYC-0005, NYC-0006, NYC-0007, NYC-0008, NYC-0009, and NYC-0010 were found to be the nucleotide sequences as shown in SEQ ID NO: 43 (FIG. 50), SEQ ID NO: 46 (FIG. 53), SEQ ID NO: 163 (FIG. 164), SEQ ID NO: 165 (FIG. 166), SEQ ID NO: 167 (FIG. 168), SEQ ID NO: 169 (FIG. 170), SEQ ID NO: 171 (FIG. 172), SEQ ID NO: 173 (FIG. 174), and SEQ ID NO: 175 (FIG. 176) of Sequence Listing. On the basis of the nucleotide sequences indicated above, the amino acid sequences encoded thereby are as follows: the amino acid sequence of the full-length NYA-2047 is as shown in SEQ ID NO: 50 (FIG. 57); the amino acid sequence of the full-length NYA-2061 is as shown in SEQ ID NO: 53 (FIG. 60); the amino acid sequence of the full-length NYA-3061 is as shown in SEQ ID NO: 164 (FIG. 165); the amino acid sequence of the full-length NYC-0005 is as shown in SEQ ID NO: 166 (FIG. 167); the amino acid sequence of the full-length NYC-0006 is as shown in SEQ ID NO: 168 (FIG. 169); the amino acid sequence of the full-length NYC-0007 is as shown in SEQ ID NO: 170 (FIG. 171); the amino acid sequence of the full-length NYC-0008 is as shown in SEQ ID NO: 172 (FIG. 173); the amino acid sequence of the full-length NYC-0009 is as shown in SEQ ID NO: 174 (FIG. 175); and the amino acid sequence of the full-length NYC-0010 is as shown in SEQ ID NO: 176 (FIG. 177).

20)-2 Expression and Purification of Anti-HLA/NY-ESO scFv

[0938] Culture of and gene transfection into the Expi293F cells (ThermoFisher Scientific) were performed in the same manner as described in 1)-5-2. For evaluation of the amount of production, the cells were subjected to agitation culture in an incubator at 37° C. in the presence of 8% CO₂ at 135 rpm for 4 days after gene introduction, the culture supernatant was filtered through a 0.2-μm filter (ThermoFisher Scientific), and the culture supernatant of scFv of anti-HLA/NY-ESO was obtained. Purification was performed using Ni Sepharose excel (Cytiva), the eluate was subjected to analytical size exclusion chromatography (SEC), and the concentration was determined. Table 12 shows the amount of

production. The amounts of production of NYA-2047, NYA-2061, and NYA-3061 determined with the use of Ni Sepharose excel relative to 1 liter of the culture supernatant in the purified eluate were higher than those of NYC-0005, NYC-0006, NYC-0007, NYC-0008, NYC-0009, and NYC-0010.

TABLE 12

Clone name	Amount of production per 1 liter (mg)
NYA-2047	53.3
NYA-2061	45.7
NYA-3061	52.5
NYC-0005	2.5
NYC-0006	1.1
NYC-0007	10.8
NYC-0008	0.5
NYC-0009	2.4
NYC-0010	1.8

[0939] The eluate of each scFv obtained with the use of Ni Sepharose excel was concentrated and purified through a gel filtration column (Superdex 200 Increase, Cytiva) equilibrated with 25 mM histidine, 300 mM NaCl, 5% Sorbitol (pH 5.5). Purified protein samples were subjected to analytical SEC, the degree of purification and the concentration were determined, and the samples were then subjected to various types of evaluation.

(Example 21) Expression and Purification of scFv-Heterodimer-Fc of Anti-HLA/NY-ESO

21)-1 Construction of scFv-Heterodimer-Fc Expression Vector of Anti-HLA/NY-ESO

[0940] For comparison of productivity resulting from different formats, an expression vector was constructed by format conversion of scFv of anti-HLA/NY-ESO into an Fc fusion type. As the heterodimer Fc sequences (hereafter, referred to as HC-h and HC-k), those reported in Nat. Biotechnol., July 1998, 16 (7), 677-81 were used.

[0941] scFv-heterodimer-Fc constructs of antibodies that may have high binding affinity to HLA-A2/NY-ESO-1; i.e., mAb24955N, mAb24956N, mAb28075P, mAb28105P, mAb28113P, and mAb29822P2 (WO2021/003357), were designed, and scFv-heterodimer-Fc of mAb24955N was designated as NYC-0011, scFv-heterodimer-Fc of mAb24956N was designated as NYC-0012, scFv-heterodimer-Fc of mAb28075P was designated as NYC-0013, scFv-heterodimer-Fc of mAb28105P was designated as NYC-0014, scFv-heterodimer-Fc of mAb28113P was designated as NYC-0015, and scFv-heterodimer-Fc of mAb29822P2 was designated as NYC-0016. Also, scFv-heterodimer-Fc of NYA-2047 was designated as NYD-2047, scFv-heterodimer-Fc of NYA-2061 was designated as NYD-2061, and scFv-heterodimer-Fc of NYA-3061 was designated as NYD-3061.

[0942] Expression vectors for HC-h, NYC-0011-HC-k, NYC-0012-HC-k, NYC-0013-HC-k, NYC-0014-HC-k, NYC-0015-HC-k, and NYC-0016-HC-k each comprising pcDNA3.3 or pcDNA3.4 (ThermoFisher Scientific) as the backbone in mammalian cells were designed. Further, expression vectors for NYD-2047-HC-k, NYD-2061-HC-k, and NYD-3061-HC-k in mammalian cells were designed to compare expression levels in mammalian cells with the same vector backbones and formats.

[0943] The nucleotide sequences of the constructed scFv-heterodimer-Fc expression vectors were reanalyzed, and the nucleotide sequences of the full-length HC-h, NYD-2047-HC-k, NYD-2061-HC-k, NYD-3061-HC-k, NYC-0011-HC-k, NYC-0012-HC-k, NYC-0013-HC-k, NYC-0014-HC-k, NYC-0015-HC-k, and NYC-0016-HC-k were the nucleotide sequences as shown in SEQ ID NO: 177 (FIG. 178), SEQ ID NO: 179 (FIG. 180), SEQ ID NO: 181 (FIG. 182), SEQ ID NO: 183 (FIG. 184), SEQ ID NO: 185 (FIG. 186), SEQ ID NO: 187 (FIG. 188), SEQ ID NO: 189 (FIG. 190), SEQ ID NO: 191 (FIG. 192), SEQ ID NO: 193 (FIG. 194), and SEQ ID NO: 195 (FIG. 196) of Sequence Listing. On the basis of the nucleotide sequences indicated above, the full-length amino acid sequences encoded thereby are as follows: the amino acid sequence of the entire HC-h is as shown in SEQ ID NO: 178 (FIG. 179); the amino acid sequence of the full-length NYD-2047-HC-k is as shown in SEQ ID NO: 180 (FIG. 181); the amino acid sequence of the full-length NYD-2061-HC-k is as shown in SEQ ID NO: 182 (FIG. 183); the amino acid sequence of the full-length NYD-3061-HC-k is as shown in SEQ ID NO: 184 (FIG. 185); the amino acid sequence of the full-length NYC-0011-HC-k is as shown in SEQ ID NO: 186 (FIG. 187); the amino acid sequence of the full-length NYC-0012-HC-k is as shown in SEQ ID NO: 188 (FIG. 189); the amino acid sequence of the full-length NYC-0013-HC-k is as shown in SEQ ID NO: 190 (FIG. 191); the amino acid sequence of the full-length NYC-0014-HC-k is as shown in SEQ ID NO: 192 (FIG. 193); the amino acid sequence of the full-length NYC-0015-HC-k is as shown in SEQ ID NO: 194 (FIG. 195); and the amino acid sequence of the full-length NYC-0016-HC-k is as shown in SEQ ID NO: 196 (FIG. 197).

21)-2 Expression and Purification of scFv-Heterodimer-Fc of Anti-HLA/NY-ESO

[0944] Culture of and gene introduction into the Expi293F cells (ThermoFisher Scientific) were performed in the same manner as described in 1)-5-2. Plasmids used for preparation of clones are shown in Table 13 below.

TABLE 13

Clone name	HC-k	HC-h
NYD-2047	NYD-2047-HC-k	HC-h
NYD-2061	NYD-2061-HC-k	HC-h
NYD-3061	NYD-3061-HC-k	HC-h
NYC-0011	NYC-0011-HC-k	HC-h
NYC-0012	NYC-0012-HC-k	HC-h
NYC-0013	NYC-0013-HC-k	HC-h
NYC-0014	NYC-0014-HC-k	HC-h
NYC-0015	NYC-0015-HC-k	HC-h
NYC-0016	NYC-0016-HC-k	HC-h

[0945] For evaluation of the amount of production, the cells were subjected to agitation culture in an incubator at 37° C. in the presence of 8% CO₂ at 135 rpm for 4 days after gene introduction, the culture supernatant was filtered through a 0.2-μm filter (ThermoFisher Scientific), and the culture supernatant of scFv-heterodimer-Fc of anti-HLA/NY-ESO was obtained. For purification, the culture supernatant was applied to MabSelectSuRe resin (Cytiva) equilibrated with PBS (pH 7.4) to allow the target scFv-heterodimer-Fc of anti-HLA/NY-ESO to adsorb thereto. After the non-adsorbed components were removed by PBS, the adsorbed components were eluted using acetate buffer, the eluate was neutralized with the aid of Tris buffer, and the

eluate was subjected to analytical size exclusion chromatography (SEC) to determine the purity and the concentration. Table 14 shows the amount of production relative to 1 liter of the culture supernatant. The amounts of production of NYA-2047, NYA-2061, and NYA-3061 determined with the use of MabSelectSuRe resin relative to 1 liter of the culture supernatant in the purified eluate were higher than those of NYC-0011, NYC-0012, NYC-0013, NYC-0014, NYC-0015, and NYC-0016.

TABLE 14

Clone name	Amount of production per 1 liter (mg)
NYD-2047	93.3
NYD-2061	112.2
NYD-3061	127.5
NYC-0011	17.2
NYC-0012	4.2
NYC-0013	54.1
NYC-0014	1.8
NYC-0015	12.2
NYC-0016	16.6

[0946] The eluate of each scFv-heterodimer-Fc obtained with the use of MabSelectSuRe resin was concentrated and purified through a gel filtration column (Superdex 200 Increase, Cytiva) equilibrated with 25 mM histidine, 300 mM NaCl, 5% Sorbitol (pH 5.5). Purified protein samples were subjected to analytical size exclusion chromatography (SEC), and NYD-2047, NYD-2061, NYD-3061, NYC-0011, NYC-0013, and NYC-0015 with sufficient degrees of purity and amounts were used as samples for evaluation of solution stability. A target object was not detected in NYC-0016 within the elution time deduced by SEC. It was thus determined that a target object was not expressed in the culture supernatant.

(Example 22) Evaluation of Solution Stability of scFv-Heterodimer-Fc of Anti-HLA/NY-ESO

[0947] NYD-2047, NYD-2061, NYD-3061, NYC-0011, NYC-0013, and NYC-0015 prepared in Example 21 were concentrated by centrifugation using Amicon Ultra-4 (Millipore), the buffer was exchanged with PBS, PBS was added thereto, and the sample concentration was adjusted to 5 mg/ml to prepare evaluation samples. In the beginning, HMWS (%) of each evaluation sample was determined by the area percentage method via size exclusion chromatography using AdvanceBio SEC 300A 2.7 m 4.6×150 mm (Agilent). As the mobile phase, 0.2 M Ki/200 mM KCl/pH 7.0 was employed, and analysis was performed at a flow rate of 0.2 ml/min (detection wavelength: 280 nm). The accelerated aging test of each sample was performed by storage at 40° C. for 1 day and 7 days, size exclusion chromatography under the conditions as described above, and calculation of HMWS (%) of each sample by the area percentage method. FIG. 163 shows the results demonstrating changes occurred in each sample upon storage at 40° C. with the elapse of time.

[0948] HMWS (%) of NYD-2047, NYD-2061, and NYD-3061 was increased to 8.7%, 8.1%, and 4.4%, respectively, with the elapse of storage time. In contrast, HMWS (%) of NYC-0011, NYC-0013, and NYC-0015 was increased to a significant extent; i.e., 71.8%, 42.4%, and 97.3%, respectively, with the elapse of storage time.

[0949] The results demonstrate that NYD-2047, NYD-2061 and NYD-3061 are superior to NYC-0011, NYC-0013, and NYC-0015 in terms of solution stability. In addition, HMWS (%) of NYD-3061 was the lowest among the evaluation samples. That is, NYD-3061 was excellent in terms of solution stability.

(Example 23) Evaluation of Binding Affinity to HLA/NY-ESO Using Biacore

[0950] NYC-0005, NYC-0007, and NYC-0008 equivalent to scFv constituting anti-HLA/NY-ESO of NYC-0011, NYC-0013, and NYC-0015 were used to evaluate binding affinity to HLA/NY-ESO

[0951] With the use of Biacore T200, the anti-HLA/NY-ESO scFv was captured as a ligand to the immobilized anti-His antibody, and the antigen was assayed as an analyte. As the antigen, HLA/NY-ESO prepared in 1)-1 was used. The anti-His antibody (His Capture kit, Cytiva) was immobilized on Sensor Chip CM5 (Cytiva) in accordance with the instructions of the kit. The anti-HLA/NY-ESO scFv constructs diluted to 0.5 pg/ml in HBS-EP+(Cytiva) to be evaluated were brought into contact therewith at 10 μ l/min for 60 seconds for immobilization. Thereafter, the samples were added to the HLA/NY-ESO analytes diluted to various levels with HBS-EP+ at a flow rate of 30 μ l/min for 120 seconds and dissociation was assayed for 600 seconds. The results of calculations obtained by such single cycle kinetics analysis, K_D , are shown in Table 15. It was confirmed that NYC-0005, NYC-0007, and NYC-0008 would bind to HLA/NY-ESO under the conditions in which the binding affinity of NYA-2047 used as a positive control to HLA/NY-ESO was stably confirmed.

TABLE 15

Clone name	K_D
NYC-0005	9.3 nM
NYC-0007	1.0 μ M
NYC-0008	7.9 nM

INDUSTRIAL APPLICABILITY

[0952] The bispecific antibody of the present invention can be used as a therapeutic or preventive agent for cancer and the like.

[0953] All publications, patents, and patent applications cited herein are incorporated herein by reference in their entirety.

SEQUENCE LISTING FREE TEXT

- [0954] SEQ ID NO 1: The amino acid sequence of the peptide in NY-ESO (FIG. 8)
- [0955] SEQ ID NO 2: The amino acid sequence of the peptide in MAGEC-1 (FIG. 9)
- [0956] SEQ ID NO 3: The scFv sequence analysis primer 1 (FIG. 10)
- [0957] SEQ ID NO 4: The scFv sequence analysis primer 2 (FIG. 11)
- [0958] SEQ ID NO 5: The nucleotide sequence of the heavy chain variable region of NYA-0001 (FIG. 12)
- [0959] SEQ ID NO 6: The amino acid sequence of the heavy chain variable region of NYA-0001 (FIG. 13)

- [0960] SEQ ID NO 7: The nucleotide sequence of the light chain variable region of NYA-0001 (FIG. 14)
- [0961] SEQ ID NO 8: The amino acid sequence of the light chain variable region of NYA-0001 (FIG. 15)
- [0962] SEQ ID NO 9: The nucleotide sequence of the heavy chain variable region of NYA-0060 (FIG. 16)
- [0963] SEQ ID NO 10: The amino acid sequence of the heavy chain variable region of NYA-0060 (FIG. 17)
- [0964] SEQ ID NO 11: The nucleotide sequence of the light chain variable region of NYA-0060 (FIG. 18)
- [0965] SEQ ID NO 12: The amino acid sequence of the light chain variable region of NYA-0060 (FIG. 19)
- [0966] SEQ ID NO 13: The nucleotide sequence of the heavy chain variable region of NYA-0068 (FIG. 20)
- [0967] SEQ ID NO 14: The amino acid sequence of the heavy chain variable region of NYA-0068 (FIG. 21)
- [0968] SEQ ID NO 15: The nucleotide sequence of the light chain variable region of NYA-0068 (FIG. 22)
- [0969] SEQ ID NO 16: The amino acid sequence of the light chain variable region of NYA-0068 (FIG. 23)
- [0970] SEQ ID NO 17: The nucleotide sequence of the heavy chain variable region of NYA-0082 (FIG. 24)
- [0971] SEQ ID NO 18: The amino acid sequence of the heavy chain variable region of NYA-0082 (FIG. 25)
- [0972] SEQ ID NO 19: The nucleotide sequence of the light chain variable region of NYA-0082 (FIG. 26)
- [0973] SEQ ID NO 20: The amino acid sequence of the light chain variable region of NYA-0082 (FIG. 27)
- [0974] SEQ ID NO 21: The nucleotide sequence of the NYA-1163 tag adduct (FIG. 28)
- [0975] SEQ ID NO 22: The nucleotide sequence of the NYA-2023 tag adduct (FIG. 29)
- [0976] SEQ ID NO 23: The nucleotide sequence of the NYA-2027 tag adduct (FIG. 30)
- [0977] SEQ ID NO 24: The nucleotide sequence of the NYA-1143 tag adduct (FIG. 31)
- [0978] SEQ ID NO 25: The nucleotide sequence of the NYA-2143 tag adduct (FIG. 32)
- [0979] SEQ ID NO 26: The amino acid sequence of the NYA-1163 tag adduct (FIG. 33); NYA-1163: amino acids 21 to 266
- [0980] SEQ ID NO 27: The amino acid sequence of the NYA-2023 tag adduct (FIG. 34); NYA-2023: amino acids 21 to 266
- [0981] SEQ ID NO 28: The amino acid sequence of the NYA-2027 tag adduct (FIG. 35); NYA-2027: amino acids 21 to 266
- [0982] SEQ ID NO 29: The amino acid sequence of the NYA-1143 tag adduct (FIG. 36); NYA-1143: amino acids 21 to 266
- [0983] SEQ ID NO 30: The amino acid sequence of the NYA-2143 tag adduct (FIG. 37); NYA-2143: amino acids 21 to 266
- [0984] SEQ ID NO 31: The nucleotide sequence of the NYA-1154 tag adduct (FIG. 38)
- [0985] SEQ ID NO 32: The amino acid sequence of the NYA-1154 tag adduct (FIG. 39); NYA-1154: amino acids 21 to 266
- [0986] SEQ ID NO 33: The amino acid sequence of the HLA-A*0201 (GenBank: ASA47534.1) truncate (FIG. 40)
- [0987] SEQ ID NO 34: The amino acid sequence of 02-microglobulin (FIG. 41)

- [0988] SEQ ID NO 35: The nucleotide sequence of the NYA-2035 tag adduct (FIG. 42)
- [0989] SEQ ID NO 36: The amino acid sequence of the NYA-2035 tag adduct (FIG. 43); NYA-2035: amino acids 21 to 266
- [0990] SEQ ID NO 37: The amino acid sequence of NYA-1143-VH01 (FIG. 44)
- [0991] SEQ ID NO 38: The amino acid sequence of NYA-1143-VH02 (FIG. 45)
- [0992] SEQ ID NO 39: The amino acid sequence of NYA-1143-VH03 (FIG. 46)
- [0993] SEQ ID NO 40: The amino acid sequence of NYA-1143-VL01 (FIG. 47)
- [0994] SEQ ID NO 41: The nucleotide sequence of the NYA-2044 tag adduct (FIG. 48)
- [0995] SEQ ID NO 42: The nucleotide sequence of the NYA-2045 tag adduct (FIG. 49)
- [0996] SEQ ID NO 43: The nucleotide sequence of the NYA-2047 tag adduct (FIG. 50)
- [0997] SEQ ID NO 44: The nucleotide sequence of the NYA-2048 tag adduct (FIG. 51)
- [0998] SEQ ID NO 45: The nucleotide sequence of the NYA-2060 tag adduct (FIG. 52)
- [0999] SEQ ID NO 46: The nucleotide sequence of the NYA-2061 tag adduct (FIG. 53)
- [1000] SEQ ID NO 47: The amino acid sequence of the NYA-2044 tag adduct (FIG. 54); NYA-2044: amino acids 21 to 266
- [1001] SEQ ID NO 48: The amino acid sequence of the NYA-2045 tag adduct (FIG. 55); NYA-2045: amino acids 21 to 266
- [1002] SEQ ID NO 49: The amino acid sequence of NYA-0082 (FIG. 56)
- [1003] SEQ ID NO 50: The amino acid sequence of the NYA-2047 tag adduct (FIG. 57); NYA-2047: amino acids 21 to 266
- [1004] SEQ ID NO 51: The amino acid sequence of the NYA-2048 tag adduct (FIG. 58); NYA-2048: amino acids 21 to 266
- [1005] SEQ ID NO 52: The amino acid sequence of the NYA-2060 tag adduct (FIG. 59); NYA-2060: amino acids 21 to 266
- [1006] SEQ ID NO 53: The amino acid sequence of the NYA-2061 tag adduct (FIG. 60); NYA-2061: amino acids 21 to 266
- [1007] SEQ ID NO 54: The heavy chain CDRH1 of NYA-0001 (FIG. 61)
- [1008] SEQ ID NO 55: The heavy chain CDRH2 of NYA-0001 (FIG. 61)
- [1009] SEQ ID NO 56: The heavy chain CDRH3 of NYA-0001 (FIG. 61)
- [1010] SEQ ID NO 57: The light chain CDRL1 of NYA-0001 (FIG. 61)
- [1011] SEQ ID NO 58: The light chain CDRL2 of NYA-0001 (FIG. 61)
- [1012] SEQ ID NO 59: The light chain CDRL3 of NYA-0001 (FIG. 61)
- [1013] SEQ ID NO 60: The amino acid sequence of CDRL1 of NYA-2023 (FIG. 62)
- [1014] SEQ ID NO 61: The amino acid sequence of CDRL3 of NYA-2027 (FIG. 63)
- [1015] SEQ ID NO 62: The heavy chain CDRH3 of NYA-1154 (FIG. 64)
- [1016] SEQ ID NO 63: The light chain CDRL3 of NYA-1154 (FIG. 64)
- [1017] SEQ ID NO 64: The amino acid sequence of CDRL1 of NYA-0035 (FIG. 65)
- [1018] SEQ ID NO 65: The nucleotide sequence of the NYC-0003 tag adduct (FIG. 66)
- [1019] SEQ ID NO 66: The nucleotide sequence of the NYC-0004 tag adduct (FIG. 67)
- [1020] SEQ ID NO 67: The amino acid sequence of NYC-0003 tag adduct (FIG. 68); NYC-0003: amino acids 21 to 263
- [1021] SEQ ID NO 68: The amino acid sequence of NYC-0004 tag adduct (FIG. 69); NYC-0004: amino acids 21 to 263
- [1022] SEQ ID NO 69: The nucleotide sequence of the NYA-0001 tag adduct (FIG. 70)
- [1023] SEQ ID NO 70: The amino acid sequence of the NYA-0001 tag adduct (FIG. 71); NYA-0001: amino acids 21 to 266
- [1024] SEQ ID NO 71: The nucleotide sequence of HC1 (FIG. 72)
- [1025] SEQ ID NO 72: The nucleotide sequence of NYF-0016-HC2 (FIG. 73)
- [1026] SEQ ID NO 73: The nucleotide sequence of NYF-0019-HC2 (FIG. 74)
- [1027] SEQ ID NO 74: The nucleotide sequence of NYF-0022-HC2 (FIG. 75)
- [1028] SEQ ID NO 75: The nucleotide sequence of NYF-0023-HC2 (FIG. 76)
- [1029] SEQ ID NO 76: The nucleotide sequence of NYF-0027-HC2 (FIG. 77)
- [1030] SEQ ID NO 77: The nucleotide sequence of NYF-0035-HC2 (FIG. 78)
- [1031] SEQ ID NO 78: The nucleotide sequence of NYF-0044-HC2 (FIG. 79)
- [1032] SEQ ID NO 79: The nucleotide sequence of NYF-0045-HC2 (FIG. 80)
- [1033] SEQ ID NO 80: The nucleotide sequence of NYF-0047-HC2 (FIG. 81)
- [1034] SEQ ID NO 81: The nucleotide sequence of NYF-0048-HC2 (FIG. 82)
- [1035] SEQ ID NO 82: The nucleotide sequence of NYF-0060-HC2 (FIG. 83)
- [1036] SEQ ID NO 83: The nucleotide sequence of NYF-0061-HC2 (FIG. 84)
- [1037] SEQ ID NO 84: The amino acid sequence of HC1 (FIG. 85)
- [1038] SEQ ID NO 85: The amino acid sequence of NYF-0016-HC2 (FIG. 86); NYA-1143: amino acids 21 to 266; and C3E-7085: amino acids 272 to 511
- [1039] SEQ ID NO 86: The amino acid sequence of NYF-0019-HC2 (FIG. 87); NYA-2143: amino acids 21 to 266; and C3E-7085: amino acids 272 to 511
- [1040] SEQ ID NO 87: The amino acid sequence of NYF-0022-HC2 (FIG. 88); NYA-1163: amino acids 21 to 266; and C3E-7085: amino acids 272 to 511
- [1041] SEQ ID NO 88: The amino acid sequence of NYF-0023-HC2 (FIG. 89); NYA-2023: amino acids 21 to 266; and C3E-7085: amino acids 272 to 511
- [1042] SEQ ID NO 89: The amino acid sequence of NYF-0027-HC2 (FIG. 90); NYA-2027: amino acids 21 to 266; and C3E-7085: amino acids 272 to 511

- [1043] SEQ ID NO 90: The amino acid sequence of NYF-0035-HC2 (FIG. 91); NYA-2035: amino acids 21 to 266; and C3E-7085: amino acids 272 to 511
- [1044] SEQ ID NO 91: The amino acid sequence of NYF-0044-HC2 (FIG. 92); NYA-2044: amino acids 21 to 266; and C3E-7085: amino acids 272 to 511
- [1045] SEQ ID NO 92: The amino acid sequence of NYF-0045-HC2 (FIG. 93); NYA-2045: amino acids 21 to 266; and C3E-7085: amino acids 272 to 511
- [1046] SEQ ID NO 93: The amino acid sequence of NYF-0047-HC2 (FIG. 94); NYA-2047: amino acids 21 to 266; and C3E-7085: amino acids 272 to 511
- [1047] SEQ ID NO 94: The amino acid sequence of NYF-0048-HC2 (FIG. 95); NYA-2048: amino acids 21 to 266; and C3E-7085: amino acids 272 to 511
- [1048] SEQ ID NO 95: The amino acid sequence of NYF-0060-HC2 (FIG. 96); NYA-2060: amino acids 21 to 266; and C3E-7085: amino acids 272 to 511
- [1049] SEQ ID NO 96: The amino acid sequence of NYF-0061-HC2 (FIG. 97); NYA-2061: amino acids 21 to 266; and C3E-7085: amino acids 272 to 511
- [1050] SEQ ID NO 97: The nucleotide sequence of NYA-0001-Fab-HC1-k delete (FIG. 98)
- [1051] SEQ ID NO 98: The nucleotide sequence of NYA-0001-LC (FIG. 99)
- [1052] SEQ ID NO 99: The amino acid sequence of NYA-0001-Fab-HC1-k delete (FIG. 100); the NYA-0001 heavy chain variable region: amino acids 20 to 139
- [1053] SEQ ID NO 100: The amino acid sequence of NYA-0001-LC (FIG. 101); the NYA-0001 light chain variable region: amino acids 21 to 131
- [1054] SEQ ID NO 101: The nucleotide sequence of NYA-1143-Fab-HC1-k delete (FIG. 102)
- [1055] SEQ ID NO 102: The nucleotide sequence of NYA-1143-LC (FIG. 103)
- [1056] SEQ ID NO 103: The nucleotide sequence of C3E-7085-HC2-k deleteC (FIG. 104)
- [1057] SEQ ID NO 104: The amino acid sequence of NYA-1143-Fab-HC1-k delete (FIG. 105); the NYA-1143 heavy chain variable region: amino acids 20 to 139
- [1058] SEQ ID NO 105: The amino acid sequence of NYA-1143-LC (FIG. 106); the NYA-1143 light chain variable region: amino acids 21 to 131
- [1059] SEQ ID NO 106: The amino acid sequence of C3E-7085-HC2-k delete (FIG. 107); C3E-7085: amino acids 21 to 260
- [1060] SEQ ID NO 107: The nucleotide sequence of NYA-1143-HC1-k delete (FIG. 108)
- [1061] SEQ ID NO 108: The amino acid sequence of NYA-1143-HC1-k delete (FIG. 109); NYA-1143: amino acids 21 to 266
- [1062] SEQ ID NO 109: The nucleotide sequence of C3E-7085-NYA-1154-Fab-HC2-k delete (FIG. 110)
- [1063] SEQ ID NO 110: The nucleotide sequence of NYA-1154-LC (FIG. 111)
- [1064] SEQ ID NO 111: The nucleotide sequence of OAA-HC1-k delete (FIG. 112)
- [1065] SEQ ID NO 112: The amino acid sequence of C3E-7085-NYA-1154-Fab-HC2-k delete (FIG. 113); C3E-7085: amino acids 21 to 260; the NYA-1154 heavy chain variable region: amino acids 266 to 285
- [1066] SEQ ID NO 113: The amino acid sequence of NYA-1154-LC (FIG. 114); the NYA-1154 light chain variable region: amino acids 21 to 131
- [1067] SEQ ID NO 114: The amino acid sequence of OAA-HC1-k delete (FIG. 115)
- [1068] SEQ ID NO 115: The nucleotide sequence of NYF-0010-HC2-k delete (FIG. 116)
- [1069] SEQ ID NO 116: The nucleotide sequence of NYF-0004-HC2-k delete (FIG. 117)
- [1070] SEQ ID NO 117: The nucleotide sequence of NYF-0011-HC2-k delete (FIG. 118)
- [1071] SEQ ID NO 118: The amino acid sequence of NYF-0010-HC2-k delete (FIG. 119); NYA-1154: amino acids 21 to 266; and C3E-7085: amino acids 272 to 511
- [1072] SEQ ID NO 119: The amino acid sequence of NYF-0004-HC2-k delete (FIG. 120); C3E-7085: amino acids 21 to 260; and NYA-1154: amino acids 272 to 511
- [1073] SEQ ID NO 120: The amino acid sequence of NYF-0011-HC2-k delete (FIG. 121); NYA-1143: amino acids 21 to 266; and C3E-7085: amino acids 272 to 511
- [1074] SEQ ID NO 121: The amino acid sequence of the point-mutant NY-ESO peptide 1F (FIG. 122)
- [1075] SEQ ID NO 122: The amino acid sequence of the point-mutant NY-ESO peptide 2M (FIG. 123)
- [1076] SEQ ID NO 123: The amino acid sequence of the point-mutant NY-ESO peptide 3A (FIG. 124)
- [1077] SEQ ID NO 124: The amino acid sequence of the point-mutant NY-ESO peptide 4A (FIG. 125)
- [1078] SEQ ID NO 125: The amino acid sequence of the point-mutant NY-ESO peptide 5A (FIG. 126)
- [1079] SEQ ID NO 126: The amino acid sequence of the point-mutant NY-ESO peptide 6L (FIG. 127)
- [1080] SEQ ID NO 127: The amino acid sequence of the point-mutant NY-ESO peptide 7F (FIG. 128)
- [1081] SEQ ID NO 128: The amino acid sequence of the point-mutant NY-ESO peptide 8A (FIG. 129)
- [1082] SEQ ID NO 129: The amino acid sequence of the point-mutant NY-ESO peptide 9A (FIG. 130)
- [1083] SEQ ID NO 130: The amino acid sequence of the gp100 peptide (FIG. 131)
- [1084] SEQ ID NO 131: The amino acid sequence of the homologous peptide DOLPP1 (FIG. 132)
- [1085] SEQ ID NO 132: The amino acid sequence of the homologous peptide IL20RB (FIG. 133)
- [1086] SEQ ID NO 133: The amino acid sequence of the homologous peptide PRKD2 (FIG. 134)
- [1087] SEQ ID NO 134: The amino acid sequence of the homologous peptide CD163 (FIG. 135)
- [1088] SEQ ID NO 135: The amino acid sequence of the homologous peptide P2RY8 (FIG. 136)
- [1089] SEQ ID NO 136: The amino acid sequence of C3E-7034 (FIG. 137)
- [1090] SEQ ID NO 137: The amino acid sequence of C3E-7036 (FIG. 138)
- [1091] SEQ ID NO 138: The amino acid sequence of C3E-7085 (FIG. 139)
- [1092] SEQ ID NO 139: The amino acid sequence of C3E-7088 (FIG. 140)
- [1093] SEQ ID NO 140: The amino acid sequence of C3E-7093 (FIG. 141)
- [1094] SEQ ID NO 141: The heavy chain CDRH1 of C3E-7085 (FIG. 142)

- [1095] SEQ ID NO 142: The heavy chain CDRH2 of C3E-7085 (FIG. 142)
- [1096] SEQ ID NO 143: The heavy chain CDRH3 of C3E-7085 (FIG. 142)
- [1097] SEQ ID NO 144: The light chain CDRL1 of C3E-7085 (FIG. 142)
- [1098] SEQ ID NO 145: The light chain CDRL2 of C3E-7085 (FIG. 142)
- [1099] SEQ ID NO 146: The light chain CDRL3 of C3E-7085 (FIG. 142)
- [1100] SEQ ID NO 147: The amino acid sequence of C3E-7078 (FIG. 143)
- [1101] SEQ ID NO 148: The nucleotide sequence of NYF-0014-HC2 (FIG. 144)
- [1102] SEQ ID NO 149: The amino acid sequence of NYF-0014-HC2 (FIG. 145); NYA-0001: amino acids 21 to 266; and C3E-7085: amino acids 272 to 511
- [1103] SEQ ID NO 150: The amino acid sequence of NYF-0082-HC2 (FIG. 146); NYA-0082: amino acids 21 to 266; and C3E-7085: amino acids 272 to 511
- [1104] SEQ ID NO 151: The amino acid sequence of human CD3 P (FIG. 147)
- [1105] SEQ ID NO 152: The nucleotide sequence of the full-length NYZ-0038-HC2 (FIG. 148)
- [1106] SEQ ID NO 153: The nucleotide sequence of the full-length NYZ-0082-HC2 (FIG. 149)
- [1107] SEQ ID NO 154: The nucleotide sequence of the full-length NYZ-0083-HC2 (FIG. 150)
- [1108] SEQ ID NO 155: The amino acid sequence of the full-length NYZ-0038-HC2 (FIG. 151)
- [1109] SEQ ID NO 156: The amino acid sequence of the full-length NYZ-0082-HC2 (FIG. 152)
- [1110] SEQ ID NO 157: The amino acid sequence of the full-length NYZ-0083-HC2 (FIG. 153)
- [1111] SEQ ID NO 158: The nucleotide sequence of the full-length NYZ-1010-HC2 (FIG. 154)
- [1112] SEQ ID NO 159: The nucleotide sequence of the full-length C3E-7085-LC (FIG. 155)
- [1113] SEQ ID NO 160: The amino acid sequence of the full-length NYZ-1010-HC2 (FIG. 156)
- [1114] SEQ ID NO 161: The amino acid sequence of the full-length C3E-7085-LC (FIG. 157)
- [1115] SEQ ID NO 162: The amino acid sequence of the peptide linker (FIG. 161)
- [1116] SEQ ID NO 163: The nucleotide sequence of the full-length NYA-3061 (FIG. 164)
- [1117] SEQ ID NO 164: The amino acid sequence of the full-length NYA-3061 (FIG. 165)
- [1118] SEQ ID NO 165: The nucleotide sequence of the full-length NYC-0005 (FIG. 166)
- [1119] SEQ ID NO 166: The amino acid sequence of the full-length NYC-0005 (FIG. 167)
- [1120] SEQ ID NO 167: The nucleotide sequence of the full-length NYC-0006 (FIG. 168)
- [1121] SEQ ID NO 168: The amino acid sequence of the full-length NYC-0006 (FIG. 169)
- [1122] SEQ ID NO 169: The nucleotide sequence of the full-length NYC-0007 (FIG. 170)
- [1123] SEQ ID NO 170: The amino acid sequence of the full-length NYC-0007 (FIG. 171)
- [1124] SEQ ID NO 171: The nucleotide sequence of the full-length NYC-0008 (FIG. 172)
- [1125] SEQ ID NO 172: The amino acid sequence of the full-length NYC-0008 (FIG. 173)
- [1126] SEQ ID NO 173: The nucleotide sequence of the full-length NYC-0009 (FIG. 174)
- [1127] SEQ ID NO 174: The amino acid sequence of the full-length NYC-0009 (FIG. 175)
- [1128] SEQ ID NO 175: The nucleotide sequence of the full-length NYC-0010 (FIG. 176)
- [1129] SEQ ID NO 176: The amino acid sequence of the full-length NYC-0010 (FIG. 177)
- [1130] SEQ ID NO 177: The nucleotide sequence of the full-length HC-h (FIG. 178)
- [1131] SEQ ID NO 178: The amino acid sequence of the full-length HC-h (FIG. 179)
- [1132] SEQ ID NO 179: The nucleotide sequence of the full-length NYD-2047-HC-k (FIG. 180)
- [1133] SEQ ID NO 180: The amino acid sequence of the full-length NYD-2047-HC-k (FIG. 181)
- [1134] SEQ ID NO 181: The nucleotide sequence of the full-length NYD-2061-HC-k (FIG. 182)
- [1135] SEQ ID NO 182: The amino acid sequence of the full-length NYD-2061-HC-k (FIG. 183)
- [1136] SEQ ID NO 183: The nucleotide sequence of the full-length NYD-3061-HC-k (FIG. 184)
- [1137] SEQ ID NO 184: The amino acid sequence of the full-length NYD-3061-HC-k (FIG. 185)
- [1138] SEQ ID NO 185: The nucleotide sequence of the full-length NYC-0011-HC-k (FIG. 186)
- [1139] SEQ ID NO 186: The amino acid sequence of the full-length NYC-0011-HC-k (FIG. 187)
- [1140] SEQ ID NO 187: The nucleotide sequence of the full-length NYC-0012-HC-k (FIG. 188)
- [1141] SEQ ID NO 188: The amino acid sequence of the full-length NYC-0012-HC-k (FIG. 189)
- [1142] SEQ ID NO 189: The nucleotide sequence of the full-length NYC-0013-HC-k (FIG. 190)
- [1143] SEQ ID NO 190: The amino acid sequence of the full-length NYC-0013-HC-k (FIG. 191)
- [1144] SEQ ID NO 191: The nucleotide sequence of the full-length NYC-0014-HC-k (FIG. 192)
- [1145] SEQ ID NO 192: The amino acid sequence of the full-length NYC-0014-HC-k (FIG. 193)
- [1146] SEQ ID NO 193: The nucleotide sequence of the full-length NYC-0015-HC-k (FIG. 194)
- [1147] SEQ ID NO 194: The amino acid sequence of the full-length NYC-0015-HC-k (FIG. 195)
- [1148] SEQ ID NO 195: The nucleotide sequence of the full-length NYC-0016-HC-k (FIG. 196)
- [1149] SEQ ID NO 196: The amino acid sequence of the full-length NYC-0016-HC-k (FIG. 197)
- [1150] SEQ ID NO 197: The amino acid sequence of the full-length NYZ-1007-HC (FIG. 198)
- [1151] SEQ ID NO 198: The amino acid sequence of the full-length NYZ-1017-HC (FIG. 199)

SEQUENCE LISTING

Sequence total quantity: 198
 SEQ ID NO: 1 moltype = AA length = 9

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FEATURE	Location/Qualifiers	
REGION	1..9	
source	note = Peptide sequence of NY-ESO 1..9 mol_type = protein organism = synthetic construct	
SEQUENCE: 1		
SLLMWITQC		9
SEQ ID NO: 2	moltype = AA length = 10	
FEATURE	Location/Qualifiers	
REGION	1..10	
source	note = Peptide sequence of MAGEC-1 1..10 mol_type = protein organism = synthetic construct	
SEQUENCE: 2		
ILFGISLREV		10
SEQ ID NO: 3	moltype = DNA length = 26	
FEATURE	Location/Qualifiers	
misc_feature	1..26 note = Primer for analyzing scFv sequence	
source	1..26 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 3		
ctcttcgcta ttacgccagc tggcga		26
SEQ ID NO: 4	moltype = DNA length = 30	
FEATURE	Location/Qualifiers	
misc_feature	1..30 note = Primer for analyzing scFv sequence	
source	1..30 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 4		
ataacaattt cacacaggaa acagctatga		30
SEQ ID NO: 5	moltype = DNA length = 360	
FEATURE	Location/Qualifiers	
misc_feature	1..360 note = Nucleotide sequence of a heavy chain variable region of NYA-0001	
source	1..360 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 5		
gaggtgcagc tgggtggagtc tgggggaggc gtgggtccagc ctgggaggtc cctgacaccc	60	
tctgtttcag cgtctggatt ctccatccgt agttatgata tacactgggt ccgccaggct	120	
ccaggcaagg ggctagagtg ggtggccact atatcatatg atggaagtca gaagtacttc	180	
gcagactccg tgaagggccg atttaccatc ttcagagacg aatcggagaa catggtgtat	240	
ctgcaaatga gcggcctgag agttgaggac acggctgttt atcactgtgc gagagggagt	300	
agtggtcatt atgaggcttt tgatatctgg ggccaaggga caatggtcac cgtctcttca	360	
SEQ ID NO: 6	moltype = AA length = 120	
FEATURE	Location/Qualifiers	
REGION	1..120 note = Amino acid sequence of a heavy chain variable region of NYA-0001	
source	1..120 mol_type = protein organism = synthetic construct	
SEQUENCE: 6		
EVQLVESGGG VVQPGRLTP SCSASGFSIR SYDIHWVRQA PGKGLEWVAT ISYDGSQKYF	60	
ADSVKGRFTI FRDESENMYV LQMSGLRVED TAVYHCARGG SGHYEAFDIW GQGTMTVTVSS	120	
SEQ ID NO: 7	moltype = DNA length = 333	
FEATURE	Location/Qualifiers	
misc_feature	1..333 note = Nucleotide sequence of a light chain variable region of NYA-0001	
source	1..333 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 7		

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cagctctgtgt tgacgcagcc gccctcagtg tctgcggccc caggacagaa ggtcaccatc 60
tcctgctctg gaagcagctc caacattggg aataattatg tatcctggta ccagcagctc 120
ccaggaacag ccccaaaact cctcatttat gacaataata agcgaccctc agggattcct 180
gaccgattct ctggctccaa gtctggcacg tcagccaccc tgggcatcac cggactccag 240
actggggacg aggcgatta ttactgcgga acatgggata gcagcctgag tgccccttgg 300
gtgttcggcg gagggacca ggtcaccgtc cta 333

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SEQ ID NO: 8      moltype = AA  length = 111
FEATURE          Location/Qualifiers
REGION          1..111
                note = Amino acid sequence of a light chain variable region
                of NYA-0001
source          1..111
                mol_type = protein
                organism = synthetic construct

```

```

SEQUENCE: 8
QSVLTQPPSV SAAPGQKVTI SCSGSSSNIG NNYVSWYQQL PGTAPKLLIY DNNKRPSGIP 60
DRFSGSKSGT SATLGITGLQ TGDEADYYCG TWDSSLAPW VFGGGTKVTV L 111

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SEQ ID NO: 9      moltype = DNA  length = 360
FEATURE          Location/Qualifiers
misc_feature     1..360
                note = Nucleotide sequence of a heavy chain variable region
                of NYA-0060
source          1..360
                mol_type = other DNA
                organism = synthetic construct

```

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SEQUENCE: 9
gaggtgcagc tgggtggagtc tgggggaggc gtgggtccagc ctgggaggtc cctgacaccc 60
tcctgttcag cgtctggatt ctccatccgt agctatgata tacactgggt ccgcctggct 120
ccaggcaagg ggctagagtg ggtggccact atatcatatg atggaagtca gaagtacttc 180
gcagactccg tgaagggcgc atttaccatc ttcagagacg aatcgagaaa catgggtgat 240
ctgcaaatga gcgcctgag agttgaggac acggctgttt atcactgtgc gagagggagt 300
agtggtcatt atgaggcttt tgatatctgg ggccaaggga caatggtcac cgtctcttca 360

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SEQ ID NO: 10     moltype = AA  length = 120
FEATURE          Location/Qualifiers
REGION          1..120
                note = Amino acid sequence of a heavy chain variable region
                of NYA-0060
source          1..120
                mol_type = protein
                organism = synthetic construct

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```

SEQUENCE: 10
EVQLVESGGG VVQPGRLTP SCSASGFSIR SYDIHWVRLA PGKGLEWVAT ISYDGSQKYF 60
ADSVKGRFTI FRDESENMYV LQMSGLRVED TAVYHCARGS SGHYEAFDIW GQGTMTVTVSS 120

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SEQ ID NO: 11     moltype = DNA  length = 333
FEATURE          Location/Qualifiers
misc_feature     1..333
                note = Nucleotide sequence of a light chain variable region
                of NYA-0060
source          1..333
                mol_type = other DNA
                organism = synthetic construct

```

```

SEQUENCE: 11
cagctctgtgt tgacgcagcc gccctcagtg tctgcggccc caggacagaa ggtcaccatc 60
tcctgctctg gaagcagctc caacattggg aataattatg tatcctggta ccagcagctc 120
ccaggaacag ccccaaaact cctcatttat gacaataata agcgaccctc agggattcct 180
gaccgattct ctggctccaa gtctggcacg tcagccaccc tgggcatcac cggactccag 240
actggggacg aggcgatta ttactgcgga acatgggata gcagcctgag tgccccttgg 300
gtgttcggcg gagggacca ggtcaccgtc cta 333

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SEQ ID NO: 12     moltype = AA  length = 111
FEATURE          Location/Qualifiers
REGION          1..111
                note = Amino acid sequence of a light chain variable region
                of NYA-0060
source          1..111
                mol_type = protein
                organism = synthetic construct

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SEQUENCE: 12
QSVLTQPPSV SAAPGQKVTI SCSGSSSNIG NNYVSWYQQL PGTAPKLLIY DNNKRPSGIP 60
DRFSGSKSGT SATLGITGLQ TGDEADYYCG TWDSSLAPW VFGGGTKVTV L 111

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SEQ ID NO: 13     moltype = DNA  length = 360

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FEATURE Location/Qualifiers
misc_feature 1..360
note = Nucleotide sequence of a heavy chain variable region
of NYA-0068

source 1..360
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 13
gaggtgcagc tgggtggagtc tgggggaggc gtggtccagc ctgggaggtc cctgacaccc 60
tcctgttcag cgtctggatt ctccatccgt agttatgata tacactgggt ccgccaggtc 120
ccaggcaagg ggctagagtg ggtggccact atatcatatg atggaagtca gaagtacttc 180
gcagactccg tgaagggccg atttaccatc ttcagagacg aatcggagaa catggtgtat 240
ctgcaaatga gcggcctgag agttgaggac acggctgttt atcactgtgc gagagggagt 300
agtggtcatt atgaggtttt tgatatctgg ggccaaggga caatggtcac cgtctcttca 360

SEQ ID NO: 14 moltype = AA length = 120
FEATURE Location/Qualifiers
REGION 1..120
note = Amino acid sequence of a heavy chain variable region
of NYA-0068

source 1..120
mol_type = protein
organism = synthetic construct

SEQUENCE: 14
EVQLVESGGG VVQPGKSLTP SCSASGFSIR SYDIHWVRQA PGKGLEWVAT ISYDGSQKYF 60
ADSVKGRFTI PRDESENMYV LQMSGLRVED TAVYHCARGS SGHYEAFDIW GQGMVTIVSS 120

SEQ ID NO: 15 moltype = DNA length = 333
FEATURE Location/Qualifiers
misc_feature 1..333
note = Nucleotide sequence of a light chain variable region
of NYA-0068

source 1..333
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 15
cagtcgtgtg tgacgcagcc gccctcagtg tctgcggccc caggacagaa ggtcaccatc 60
tcctgctctg gaagcagctc caacattggg aattgggatg tatcctggta ccagcagctc 120
ccaggcaacag ccccaaaact cctcatatga gacaataata agcgaccctc agggattcct 180
gaccgattct ctgggtccaa gtctggcagc tcagccaccc tgggcacac cggactccag 240
actggggagc aggcgatta ttactgcgga acatgggata gcagcctgag tgcctcttgg 300
gtgttcggcg gagggaccaa ggtcaccgtc cta 333

SEQ ID NO: 16 moltype = AA length = 111
FEATURE Location/Qualifiers
REGION 1..111
note = Amino acid sequence of a light chain variable region
of NYA-0068

source 1..111
mol_type = protein
organism = synthetic construct

SEQUENCE: 16
QSVLTQPPSV SAAPGQKVTI SCSGSSSNIG NWYVSWYQQL PGTAPKLLIY DNNKRPSGIP 60
DRFSGSKSGT SATLGITGLQ TGDEADYYCG TWDSLSAPW VFGGGRKTV L 111

SEQ ID NO: 17 moltype = DNA length = 360
FEATURE Location/Qualifiers
misc_feature 1..360
note = Nucleotide sequence of a heavy chain variable region
of NYA-0082

source 1..360
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 17
gaggtgcagc tgggtggagtc tgggggaggc gtggtccagc ctgggaggtc cctgacaccc 60
tcctgttcag cgtctggatt ctccatccgt agttatgata tacactgggt ccgccaggtc 120
ccaggcaagg ggctagagtg ggtggccact atatcatatg atggaagtca gaagtacttc 180
gcagactccg tgaagggccg atttaccatc ttcagagacg aatcggagaa catggtgtat 240
ctgcaaatga gcggcctgag agttgaggac acggctgttt atcactgtgc gagagggagt 300
agtggtcatt atgaggtttt tgatatctgg ggccaaggga caatggtcac cgtctcttca 360

SEQ ID NO: 18 moltype = AA length = 120
FEATURE Location/Qualifiers
REGION 1..120
note = Amino acid sequence of a heavy chain variable region
of NYA-0082

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source                1..120
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 18
EVQLVESGGG VVQGRSLTP SCSASGFSIR SYDIHWVRQA PGKGLEWVAT ISYDGSQKYP 60
ADSVKGRFTI FRDESENMVY LQMSGLRVED TAVYHCARGS SGHYEAFDIW GQGTMTVTVSS 120

SEQ ID NO: 19         moltype = DNA length = 333
FEATURE               Location/Qualifiers
misc_feature          1..333
                      note = Nucleotide sequence of a light chain variable region
                      of NYA-0082

source                1..333
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 19
cagctctgtg tgacgcagcc gccctcagtg tctgcggccc caggacagaa ggtcaccatc 60
tcctgctctg gaagcagctc caacattggg aataattatg tatcctggta ccagcggctc 120
ccaggaaacag ccccaaaact cctcatttat gacaataata agcgaccctc agggattcct 180
gaccgattct cttggctccaa gtctggcacg tcagccaccc tgggcatcac cggactccag 240
actggggacg aggcgatta ttactgcgga tcatgggata gcagcctgag tgccccttgg 300
gtgttcggcg gagggacca ggtcaccgtc cta 333

SEQ ID NO: 20         moltype = AA length = 111
FEATURE               Location/Qualifiers
REGION               1..111
                      note = Amino acid sequence of a light chain variable region
                      of NYA-0082

source                1..111
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 20
QSVLTQPPSV SAAPGQKVTI SCSGSSSNIG NNYVSWYQRL PGTAPKLLIY DNNKRPSGIP 60
DRFSGSKSGT SATLGITGLQ TGDEADYYCG SWDSSLAPFW VFGGGTKVTV L 111

SEQ ID NO: 21         moltype = DNA length = 876
FEATURE               Location/Qualifiers
misc_feature          1..876
                      note = Full-length nucleotide sequence of NYA-1163

source                1..876
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 21
atgaaacacc tgtggttctt cctcctgctg gtggcagctc ccagatgggt gctgagcgga 60
gaggtgcagc tgggtggagtc tgggggaggc gtggtccagc ctgggaggtc cctgacaccc 120
tcctgttcag cgtctggatt ctccatccgt agctatgata tacactgggt ccgcctggct 180
ccaggcaagg ggctagagtg ggtggccact atatcatatg atggaagtca gaagtacttc 240
gcagactccg tgaagggcgc atttaccatc ttcagagacg aatcggagaa catggtgtat 300
ctgcaaatga gcgcctgag agttgaggac acggctgttt atcactgtgc gagagggagt 360
agtggtcatt atgaggcttt tgatatctgg ggccaaggga caatggtcac cgtctcttca 420
ggtggagcgc gttcaggcgc aggtggcagc ggcggtggcg ggagtcagtc tgtgttgacg 480
cagccgcctc cagtgtctgc ggccccagga cagaagggtc ccatctcctg ctctggaagc 540
agctccaaca ttgggaataa ttatgtatcc tggtagccag agctcccagg aacagccccc 600
aaactcctca tttatgacaa taataagcga ccctcaggga ttctgaccc attctctggc 660
tccaagtctg gcacgtcagc caccctgggc atcacgggac tccagactgg ggacgaggcc 720
gattattact gcggaacatg ggatagcagc ctgagtgccc cttgggtgtt cggcggaggg 780
accaaggtca ccgtcctagg cgcggccgca ggtgcaggtg gtgattacaa agatgatgac 840
gataaagggtg cagcggcgca tcaccatcat caccac 876

SEQ ID NO: 22         moltype = DNA length = 876
FEATURE               Location/Qualifiers
misc_feature          1..876
                      note = Full-length nucleotide sequence of NYA-2023

source                1..876
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 22
atgaaacacc tgtggttctt cctcctgctg gtggcagctc ccagatgggt gctgagcgga 60
gaggtgcagc tgggtggagtc tgggggaggc gtggtccagc ctgggaggtc cctgacaccc 120
tcctgttcag cgtctggatt ctccatccgt agttatgata tacactgggt ccgccagggt 180
ccaggcaagg ggctagagtg ggtggccact atatcatatg atggaagtca gaagtacttc 240
gcagactccg tgaagggcgc atttaccatc ttcagagacg aatcggagaa catggtgtat 300
ctgcaaatga gcgcctgag agttgaggac acggctgttt atcactgtgc gagagggagt 360
agtggtcatt atgaggcttt tgatatctgg ggccaaggga caatggtcac cgtctcttca 420
ggtggagcgc gttcaggcgc aggtggcagc ggcggtggcg ggagtcagtc tgtgttgacg 480
cagccgcctc cagtgtctgc ggccccagga cagaagggtc ccatctcctg ctctggaagc 540

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agctccaaca ttgggaattg gtatgtatcc tgggtaccagc agctcccagg aacagccccc 600
aaactcctca tttatgacaa taataagcga ccctcaggga ttcttgaccg attctctggc 660
tccaagtctg gcacgtcagc caccctgggc atcaccggac tccagactgg ggacgaggcc 720
gattattact gcggaacatg gtagatgcagc ctgagtgccc cttgggtgtt cggcggaggg 780
accaaggtca ccgtcctagg cgcggccgca ggtgcagggtg gtgattacaa agatgatgac 840
gataaagggtg cagcggcgca tcaccatcat caccac 876

SEQ ID NO: 23      moltype = DNA length = 876
FEATURE           Location/Qualifiers
misc_feature       1..876
                   note = Full-length nucleotide sequence of NYA-2027
source            1..876
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 23
atgaaacacc tgtggttctt cctcctgctg gtggcagctc ccagatgggt gctgagcgga 60
gaggtgcagc tggtagagtc tgggggaggc gtgggtccagc ctgggagggt cctgacaccc 120
tcctgttcag cgtctggatt ctccatccgt agttatgata tacactgggt ccgccagggt 180
ccaggcaagg ggctagagtg ggtggccact atatcatatg atggaagtca gaagtacttc 240
gcagactccg tgaaggggcg atttaccatc ttcagagacg aatcggagaa catggtgtat 300
ctgcaaatga gcggcctgag agttgaggac acggctgttt atcactgtgc gagagggagt 360
agtggtcatt atgaggcttt tgatatctgg ggccaaggga caatggtcac cgtctcttca 420
ggtggaggcg gttcaggcgg aggtggcagc ggcggtggcg ggagtcagtc tgtgttgacg 480
cagccgcctc cagtgtctgc ggcgccagga cagaaggcca ccatctcctg ctctggaagc 540
agctccaaca ttgggaataa ttatgtatcc tgggtaccagc agctcccagg aacagccccc 600
aaactcctca tttatgacaa taataagcga ccctcaggga ttcttgaccg attctctggc 660
tccaagtctg gcacgtcagc caccctgggc atcaccggac tccagactgg ggacgaggcc 720
gattattact gcggaacatg gtagatgcagc ctgagtgccc cttgggtgtt cggcggaggg 780
accaaggtca ccgtcctagg cgcggccgca ggtgcagggtg gtgattacaa agatgatgac 840
gataaagggtg cagcggcgca tcaccatcat caccac 876

SEQ ID NO: 24      moltype = DNA length = 876
FEATURE           Location/Qualifiers
misc_feature       1..876
                   note = Full-length nucleotide sequence of NYA-1143
source            1..876
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 24
atgaaacacc tgtggttctt cctcctgctg gtggcagctc ccagatgggt gctgagcgga 60
gaggtgcagc tggtagagtc tgggggaggc gtgggtccagc ctgggagggt cctgacaccc 120
tcctgttcag cgtctggatt ctccatccgt agctatgata tacactgggt ccgccctgggt 180
ccaggcaagg ggctagagtg ggtggccact atatcatatg atggaagtca gaagtacttc 240
gcagactccg tgaaggggcg atttaccatc ttcagagacg aatcggagaa catggtgtat 300
ctgcaaatga gcggcctgag agttgaggac acggctgttt atcactgtgc gagagggagt 360
agtggtcatt atgaggcttt tgatatctgg ggccaaggga caatggtcac cgtctcttca 420
ggtggaggcg gttcaggcgg aggtggcagc ggcggtggcg ggagtcagtc tgtgttgacg 480
cagccgcctc cagtgtctgc ggcgccagga cagaaggcca ccatctcctg ctctggaagc 540
agctccaaca ttgggaattg gtatgtatcc tgggtaccagc agctcccagg aacagccccc 600
aaactcctca tttatgacaa taataagcga ccctcaggga ttcttgaccg attctctggc 660
tccaagtctg gcacgtcagc caccctgggc atcaccggac tccagactgg ggacgaggcc 720
gattattact gcggaacatg gtagatgcagc ctgagtgccc cttgggtgtt cggcggaggg 780
accaaggtca ccgtcctagg cgcggccgca ggtgcagggtg gtgattacaa agatgatgac 840
gataaagggtg cagcggcgca tcaccatcat caccac 876

SEQ ID NO: 25      moltype = DNA length = 876
FEATURE           Location/Qualifiers
misc_feature       1..876
                   note = Full-length nucleotide sequence of NYA-2143
source            1..876
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 25
atgaaacacc tgtggttctt cctcctgctg gtggcagctc ccagatgggt gctgagcgga 60
gaggtgcagc tggtagagtc tgggggaggc gtgggtccagc ctgggagggt cctgacaccc 120
tcctgttcag cgtctggatt ctccatccgt agctatgata tacactgggt ccgccctgggt 180
ccaggcaagg ggctagagtg ggtggccact atatcatatg atggaagtca gaagtacttc 240
gcagactccg tgaaggggcg atttaccatc ttcagagacg aatcggagaa catggtgtat 300
ctgcaaatga gcggcctgag agttgaggac acggctgttt atcactgtgc gagagggagt 360
agtggtcatt atgaggcttt tgatatctgg ggccaaggga caatggtcac cgtctcttca 420
ggtggaggcg gttcaggcgg aggtggcagc ggcggtggcg ggagtcagtc tgtgttgacg 480
cagccgcctc cagtgtctgc ggcgccagga cagaaggcca ccatctcctg ctctggaagc 540
agctccaaca ttgggaattg gtatgtatcc tgggtaccagc ggctcccagg aacagccccc 600
aaactcctca tttatgacaa taataagcga ccctcaggga ttcttgaccg attctctggc 660
tccaagtctg gcacgtcagc caccctgggc atcaccggac tccagactgg ggacgaggcc 720
gattattact gcggaacatg gtagatgcagc ctgagtgccc cttgggtgtt cggcggaggg 780

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accaaggtca cgcctcctagg cgcggccgca ggtgcaggtg gtgattacaa agatgatgac 840
gataaaggtg cagcggcgca tcaccatcat caccac 876

SEQ ID NO: 26      moltype = AA length = 292
FEATURE           Location/Qualifiers
REGION           1..292
                  note = Full-length amino acid sequence of NYA-1163
source           1..292
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 26
MKHLWFFLLL VAAPRWVLSG EVQLVESGGG VVQPGKSLTP SCSASGFSIR SYDIHWVRLA 60
PGKGLEWVAT ISYDGSQKYF ADSVKGRFTI FRDESENVMY LQMSGLRVED TAVYHCARGS 120
SGHYEAFDIW GQGTMTVTSV GGGGSGGGGS GGGGSQSVLT QPPSVSAAPG QKVTISCSGS 180
SSNIGNNIVS WYQQLPGTAP KLLIYDNNKR PSGIPDRFSG SKSGTSATLG ITGLQTGDEA 240
DYCYGTWDS LSAFWVFGGG TKVTVLGAAA GAGGDYKDDD DKGAAHHHH HH 292

SEQ ID NO: 27      moltype = AA length = 292
FEATURE           Location/Qualifiers
REGION           1..292
                  note = Full-length amino acid sequence of NYA-2023
source           1..292
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 27
MKHLWFFLLL VAAPRWVLSG EVQLVESGGG VVQPGKSLTP SCSASGFSIR SYDIHWVRQA 60
PGKGLEWVAT ISYDGSQKYF ADSVKGRFTI FRDESENVMY LQMSGLRVED TAVYHCARGS 120
SGHYEAFDIW GQGTMTVTSV GGGGSGGGGS GGGGSQSVLT QPPSVSAAPG QKVTISCSGS 180
SSNIGNNIVS WYQQLPGTAP KLLIYDNNKR PSGIPDRFSG SKSGTSATLG ITGLQTGDEA 240
DYCYGTWDS LSAFWVFGGG TKVTVLGAAA GAGGDYKDDD DKGAAHHHH HH 292

SEQ ID NO: 28      moltype = AA length = 292
FEATURE           Location/Qualifiers
REGION           1..292
                  note = Full-length amino acid sequence of NYA-2027
source           1..292
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 28
MKHLWFFLLL VAAPRWVLSG EVQLVESGGG VVQPGKSLTP SCSASGFSIR SYDIHWVRQA 60
PGKGLEWVAT ISYDGSQKYF ADSVKGRFTI FRDESENVMY LQMSGLRVED TAVYHCARGS 120
SGHYEAFDIW GQGTMTVTSV GGGGSGGGGS GGGGSQSVLT QPPSVSAAPG QKVTISCSGS 180
SSNIGNNIVS WYQQLPGTAP KLLIYDNNKR PSGIPDRFSG SKSGTSATLG ITGLQTGDEA 240
DYCYGTWDS LSAFWVFGGG TKVTVLGAAA GAGGDYKDDD DKGAAHHHH HH 292

SEQ ID NO: 29      moltype = AA length = 292
FEATURE           Location/Qualifiers
REGION           1..292
                  note = Full-length amino acid sequence of NYA-1143
source           1..292
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 29
MKHLWFFLLL VAAPRWVLSG EVQLVESGGG VVQPGKSLTP SCSASGFSIR SYDIHWVRLA 60
PGKGLEWVAT ISYDGSQKYF ADSVKGRFTI FRDESENVMY LQMSGLRVED TAVYHCARGS 120
SGHYEAFDIW GQGTMTVTSV GGGGSGGGGS GGGGSQSVLT QPPSVSAAPG QKVTISCSGS 180
SSNIGNNIVS WYQQLPGTAP KLLIYDNNKR PSGIPDRFSG SKSGTSATLG ITGLQTGDEA 240
DYCYGTWDS LSAFWVFGGG TKVTVLGAAA GAGGDYKDDD DKGAAHHHH HH 292

SEQ ID NO: 30      moltype = AA length = 292
FEATURE           Location/Qualifiers
REGION           1..292
                  note = Full-length amino acid sequence of NYA-2143
source           1..292
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 30
MKHLWFFLLL VAAPRWVLSG EVQLVESGGG VVQPGKSLTP SCSASGFSIR SYDIHWVRLA 60
PGKGLEWVAT ISYDGSQKYF ADSVKGRFTI FRDESENVMY LQMSGLRVED TAVYHCARGS 120
SGHYEAFDIW GQGTMTVTSV GGGGSGGGGS GGGGSQSVLT QPPSVSAAPG QKVTISCSGS 180
SSNIGNNIVS WYQQLPGTAP KLLIYDNNKR PSGIPDRFSG SKSGTSATLG ITGLQTGDEA 240
DYCYGTWDS LSAFWVFGGG TKVTVLGAAA GAGGDYKDDD DKGAAHHHH HH 292

SEQ ID NO: 31      moltype = DNA length = 876
FEATURE           Location/Qualifiers
misc_feature     1..876

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source          note = Full-length nucleotide sequence of NYA-1154
                1..876
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 31
atgaaacacc tgtggttctt cctcctgctg gtggcagctc ccagatgggt gctgagcgga 60
gaggtgcagc tgggtggagtc tgggggaggc gtggtccagc ctgggaggtc cctgacaccc 120
tcctgttcag cgtctggatt ctccatccgt agttatgata tgcactgggt ccgccaggct 180
ccaggcaagg ggctagagtg ggtggccact atatcatatg atggaagtca gaagtaacttc 240
gcagactccg tgaaggggcg atttaccatc ttcagagacg aatcggagaa catggtgtat 300
ctgcaaatga gcggcctgag agttgaggac acggccgttt atcactgtgc gagagggagt 360
agtaatcatt atgaggcttt tgatatctgg ggccaaggga caatggtcac cgtctcttca 420
ggtggaggcg gttcaggcgg aggtggcagc ggccgtggcg ggagtcagtc tgtgttgacg 480
cagccgcctt cagtgtctgc gggcccagga cagaaggcca ccatctcttg ctctggaagc 540
agctccaaca ttgggaataa ttaatgtatc ttgtaccagc ggctcccagg aacagccccc 600
aaactcctca tttatgacaa taataagcga ccctcaggga ttcctgacgc attctctggc 660
tccaagtctg gcacgtcagc caccctgggc atcacccgac tccagactgg ggacgaggcc 720
gattactact gcggagcatg ggatagcagc ctgagtgccc cttgggtgtt cggcggaggg 780
accaaggtca ccgtcctagg cgcggccgca ggtgcaggtg gtgattacaa agatgatgac 840
gataaagggt cagcggcgca tcaccatcat caccac 876

SEQ ID NO: 32      moltype = AA length = 292
FEATURE           Location/Qualifiers
REGION           1..292
                  note = Full-length amino acid sequence of NYA-1154
source           1..292
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 32
MKHLWFFLLL VAAPRWVLSG EVQLVESGGG VVQPGRSLTP SCSASGFSIR SYDMHWVRQA 60
PGKGLEWVAT ISYDGSQKYF ADSVKGRFTI FRDESENVMY LQMSGRLVED TAVYHCARGS 120
SNHYEAFDIW QQGTMTVTSV GGGGSGGGGS GGGGSQSVLT QPPSVSAAPG QKVTISCSGS 180
SSNIGNNVYS WYQRLPGTAP KLLIYDNMKR PSGIPDRFSG SKSGTSATLG ITGLQTGDEA 240
DYCYGAWDSS LSAFWVFGGG TKVTVLGAAA GAGGDYKDDD DGAAAHHHH HH 292

SEQ ID NO: 33      moltype = AA length = 290
FEATURE           Location/Qualifiers
REGION           1..290
                  note = Amino acid sequence of truncated form of
                  HLA-A*0201(UniProtKB-P01892)
source           1..290
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 33
MSHSMRYFFT SVSRPGRGEP RFIAVGYVDD TQFVRFDSDA ASQRMEPRAP WIEQEGPEYW 60
DGETRKVKAH SQTHRVDLGT LRGYYNQSEA GSHTVQRMYG CDVGSDFRFL RGYHQYAYDG 120
KDYIALKEDL RSWTAADMAA QTTKHKWEAA HVAEQLRAYL EGTCTVEWLR YLENGKETLQ 180
RTDAPKTHMT HHAUSDHEAT LRCWALSFYP AEITLTWQRD GEDQTQDTEL VETRPAGDGT 240
FQKWAAVVVP SGQEQRYTCH VQHEGLPKPL TLRWEGLNDI FEAQKIEWHE 290

SEQ ID NO: 34      moltype = AA length = 100
FEATURE           Location/Qualifiers
REGION           1..100
                  note = Amino acid sequence of beta 2-microglobulin
source           1..100
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 34
MIQRTPKIQV YSRHPAENGK SNFLNCYVSG FHPSDIEVDL LKNGERIEKV EHSDFSFSKD 60
WSFYLLYYTE FTPTEKDEYA CRVNHVTLNQ PKIVKWDRDM 100

SEQ ID NO: 35      moltype = DNA length = 876
FEATURE           Location/Qualifiers
misc_feature     1..876
                  note = Full-length nucleotide sequence of NYA-2035
source           1..876
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 35
atgaaacacc tgtggttctt cctcctgctg gtggcagctc ccagatgggt gctgagcgga 60
gaggtgcagc tgggtggagtc tgggggaggc gtggtccagc ctgggaggtc cctgacaccc 120
tcctgttcag cgtctggatt ctccatccgt agctatgata tacactgggt ccgcctggct 180
ccaggcaagg ggctagagtg ggtggccact atatcatatg atggaagtca gaagtaacttc 240
gcagactccg tgaaggggcg atttaccatc ttcagagacg aatcggagaa catggtgtat 300
ctgcaaatga gcggcctgag agttgaggac acggctgttt atcactgtgc gagagggagt 360
agtggtcatt atgaggcttt tgatatctgg ggccaaggga caatggtcac cgtctcttca 420

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ggtggaggcg gttcaggcgg aggtggcagc ggcggtggcg ggagtcagtc tgtgttgacg 480
cagccgcctt cagtgtctgc gggcccagga cagaagggtc ccatctcctg ctctggaagc 540
agctccaaca ttgggaattg gaaggtatcc tggtagcagc agctcccagg aacagccccc 600
aaactctcca tttatgacaa taataagcga ccctcaggga ttcttgaccg attctctggc 660
tccaagtctg gcacgtcagc caccctgggc atcacgggac tccagactgg ggacgaggcc 720
gattattact gcggaacatg ggatagcagc ctgagtgccc cttgggtgtt cggcggaggg 780
accaagggtc ccgtcctagg cgcggccgca ggtgcagggt gtgattacaa agatgatgac 840
gataaagggt cagcggcgca tcaccatcat caccac 876

SEQ ID NO: 36      moltype = AA  length = 292
FEATURE           Location/Qualifiers
REGION            1..292
                  note = Full-length amino acid sequence of NYA-2035
source            1..292
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 36
MKHLWFFLLL VAAPRWVLSG EVQLVESGGG VVQPGKSLTP SCSASGFSIR SYDIHWVRLA 60
PGKGLEWVAT ISYDGSQKYF ADSVKGRFTI FRDESENMVY LQMSGLRVED TAVYHCARGS 120
SGHYEAFDIW GQGTMTVTVSS GGGGSGGGGS GGGGSQSVLT QPPSVSAAPG QKVTISCSGS 180
SSNIGNWKVS WYQLPGTAP KLLIYDNMKR PSGIPDRFSG SKSGTSATLG ITGLQTGDEA 240
DYCYGTWDS LSAFWVFGG TKVTVLGAAA GAGGDYKDD DKGAAHHHH HH 292

SEQ ID NO: 37      moltype = AA  length = 120
FEATURE           Location/Qualifiers
REGION            1..120
                  note = Amino acid sequence of NYA-1143-VH01
source            1..120
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 37
EVQLVESGGG LVQPGGSLRL SCAASGFSIR SYDIHWVRLA PGKGLEWVAT ISYDGSQKY 60
ADSVKGRFTI SRDESKNTLY LQMSLRAED TAVYHCARGS SGHYEAFDIW GQGLTVTVSS 120

SEQ ID NO: 38      moltype = AA  length = 120
FEATURE           Location/Qualifiers
REGION            1..120
                  note = Amino acid sequence of NYA-1143-VH02
source            1..120
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 38
QVQLVESGGG VVQPGKSLRL SCAASGFSIR SYDMHWVRQA PGKGLEWVAT ISYDGSQKY 60
ADSVKGRFTI SRDNSKNTLY LQMSSLRAED TAVYHCARGS SGHYEAFDIW GQGLTVTVSS 120

SEQ ID NO: 39      moltype = AA  length = 120
FEATURE           Location/Qualifiers
REGION            1..120
                  note = Amino acid sequence of NYA-1143-VH03
source            1..120
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 39
EVQLVESGGG LVQPGGSLRL SCAASGFSIR SYDIHWVRQA PGKGLEWVAT ISYDGSQKY 60
ADSVKGRFTI SRDESKNTLY LQMSLRAED TAVYHCARGS SGHYEAFDIW GQGLTVTVSS 120

SEQ ID NO: 40      moltype = AA  length = 111
FEATURE           Location/Qualifiers
REGION            1..111
                  note = Amino acid sequence of NYA-1143-VL01
source            1..111
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 40
QSVLTQPPSA SGTGQQRVTI SCSGSSSNIG NWYVSWYQQL PGTAPKLLIY DNNKRPSGVP 60
DRFSGSKSGT SASLAISGLQ SEDEADYYCG TWDSSLSAPW VFGGGTKLTV L 111

SEQ ID NO: 41      moltype = DNA  length = 876
FEATURE           Location/Qualifiers
misc_feature      1..876
                  note = Full-length nucleotide sequence of NYA-2044
source            1..876
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 41
atgaaacacc tgtgtgtctt cctcctgctg gtggcagctc ccagatgggt gctgagcgtt 60

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gaggtagacagc tgggtggaatc cgggtggaggc cttgtgcaac cgggagggtc cctgcgactc 120
tcttgcgccg catctgggctt ttccatacga tctacgata ttcattgggt ccgactcgcc 180
cctggcaaaag gtcttgaatg ggttgccact atctcctacg acgggtctca gaaatattat 240
gcggtattcag tgaagggggcg gttcaccaatt tcacgagacg agtcaaagaa tacactgtac 300
ctccaaatga attcactgag agccgaggat actgcagtct atcattgtgc aagagggtcc 360
tcaggccact acgaggcctt tgatatatgg ggccaaggca ccttggtaac cgttagtagc 420
ggaggtggag gaagcggagg cggcggttcc ggaggaggtg gaagtcaatc cgttctgact 480
cagccacctt ccgtgagtgc agcgctgga cagaagggtc caatatcatg ttctggctct 540
tcaagcaaca tcggaaactg gtacgtgagc tggtagcagc agtcccccg cagggcgccc 600
aagcttctga tatatgacaa caacaaacgg ccagtgga tccctgacag attcagtggtg 660
tctaaaagtgt gtacaagcgc taccctgggt atcaccggat tgcagaccgg agatgaggcg 720
gactattatt gcggaacctg ggactccagc ctgagcgctc cctgggtttt cggcggagggt 780
actaagggtta ccgtcttggg cgcggccgca ggtgcaggtg gtgattacaa agatgatgac 840
gataaagggtg cagcggcgca tcaccatcat caccac 876

SEQ ID NO: 42      moltype = DNA length = 876
FEATURE            Location/Qualifiers
misc_feature        1..876
                    note = Full-length nucleotide sequence of NYA-2045
source              1..876
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 42
atgaaacacc tgtggttctt cctcctgctg gtggcagctc ccagatgggt gctgagcggg 60
gaggtagacagc tgggtggaatc cgggtggaggc cttgtgcaac cgggagggtc cctgcgactc 120
tcttgcgccg catctgggctt ttccatacga tctacgata ttcattgggt ccgactcgcc 180
cctggcaaaag gtcttgaatg ggttgccact atctcctacg acgggtctca gaaatattat 240
gcggtattcag tgaagggggcg gttcaccaatt tcacgagacg agtcaaagaa tacactgtac 300
ctccaaatga attcactgag agccgaggat actgcagtct atcattgtgc aagagggtcc 360
tcaggccact acgaggcctt tgatatatgg ggccaaggca ccttggtaac cgttagtagc 420
ggaggtggag gaagcggagg cggcggttcc ggaggaggtg gaagtcaatc cgttctgact 480
cagccacctt ccgctctctg cacccggtgc caacgggtc caatatcatg ttctggctct 540
tcaagcaaca tcggaaactg gtacgtgagc tggtagcagc agtcccccg cagggcgccc 600
aagcttctga tatatgacaa caacaaacgg ccagtgga tccctgacag attcagtggtg 660
tctaaaagtgt gtacaagcgc tagcctgggt ataagtggtc tgcagagtga agatgaggcg 720
gactattatt gcggaacctg ggactccagc ctgagcgctc cctgggtttt cggcggagggt 780
actaagctga ccgtcttggg cgcggccgca ggtgcaggtg gtgattacaa agatgatgac 840
gataaagggtg cagcggcgca tcaccatcat caccac 876

SEQ ID NO: 43      moltype = DNA length = 876
FEATURE            Location/Qualifiers
misc_feature        1..876
                    note = Full-length nucleotide sequence of NYA-2047
source              1..876
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 43
atgaaacacc tgtggttctt cctcctgctg gtggcagctc ccagatgggt gctgagcggg 60
caagtacagc tgggtggaatc cgggtggaggc gtggtccagc cgggacgcag cttgagactg 120
tcttgcgctg catctgggctt ttccatacga tctacgata tgcactgggt tcgccaagcc 180
cctggcaaaag gtcttgaatg ggttgccact atctcctacg acgggtctca gaaatattac 240
gcggtattcag tgaagggggcg gttcaccaatt tcacgggaca attcaaagaa taccttgtat 300
ctccagatgt cttcactgag agccgaggat actgcagtct atcattgtgc aagagggtcc 360
tcaggccact acgaggcctt tgatatatgg ggccaaggca ccttggtaac cgttagtagc 420
ggaggtggag gaagcggagg cggcggttcc ggaggaggtg gaagtcaatc cgttctgact 480
cagccacctt ccgtgagtgc agcgctgga cagaagggtc caatatcatg ttctggctct 540
tcaagcaaca tcggaaactg gtacgtgagc tggtagcagc agtcccccg cagggcgccc 600
aagcttctga tatatgacaa caacaaacgg ccagtgga tccctgacag attcagtggtg 660
tctaaaagtgt gtacaagcgc taccctgggt atcaccggat tgcagaccgg agatgaggcg 720
gactattatt gcggaacctg ggactccagc ctgagcgctc cctgggtttt cggcggagggt 780
actaaggtta ccgtcttggg cgcggccgca ggtgcaggtg gtgattacaa agatgatgac 840
gataaagggtg cagcggcgca tcaccatcat caccac 876

SEQ ID NO: 44      moltype = DNA length = 876
FEATURE            Location/Qualifiers
misc_feature        1..876
                    note = Full-length nucleotide sequence of NYA-2048
source              1..876
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 44
atgaaacacc tgtggttctt cctcctgctg gtggcagctc ccagatgggt gctgagcggg 60
caagttcaac tcgtggaatc tgggtggagg gtggtacaac caggccgggtc acttagactg 120
tcttgcgctg cgagtggatt ttccatcaga tcttacgaca tgcactgggt tcgccaagca 180
cccgaaaagg gtttgggaatg gttggctacg atctcctacg atggatccca gaagtattac 240
gccgacagcg tcaagggccg atttacaata tcacgggaca attcaaagaa taccttgtat 300

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ctccagatgt cttcactgag agccgaggat actgcagtct atcattgtgc aagagggtcc 360
tcaggccact acgaggcctt tgatatatgg ggccaaggca ccttggtaac cgttagtagc 420
ggaggtggag gaagcggagg cggcgggtcc ggaggagggtg gaagtcaatc cgttctgact 480
cagccaccct ccgcctctgg caccocgggc caacgggtca caatatcatg ttctggctct 540
tcaagcaaca tcggaacctg gtacgtgagc tggtaaccagc agctccccgg caccggcgccc 600
aagcttctga tatatgacaa caacaaacgg ccagtgagg ttcttgacag attcagtgagg 660
tctaaaagtg gtacaagcgc tagcctggcc ataagtggtc tgcagagtga agatgaggcg 720
gactattatt gcggaacctg ggactccagc ctgagcgctc cctgggtttt cggcggagggt 780
actaagctga ccgtcttggg cgcggccgca ggtgcagggtg gtgattacaa agatgatgac 840
gataaagggtg cagcggcgca tcaccatcat caccac 876

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SEQ ID NO: 45      moltype = DNA length = 876
FEATURE
misc_feature       1..876
                    note = Full-length nucleotide sequence of NYA-2060
source             1..876
                    mol_type = other DNA
                    organism = synthetic construct

```

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SEQUENCE: 45
atgaaacacc tgtggttctt cctcctgctg gtggcagctc ccagatgggtg gctgagcggt 60
gaggtacagc tgggtggaatc cgggtggaggc cttgtgcaac cgggagggtc cctgcgactc 120
tcttgccgag catctggctt ttccatacga tctacagata ttcattgggt cgcacaagcc 180
cctggcaaaag gtcttgaatg ggttgccact atctcctacg acgggtctca gaaatattat 240
gcggtattcag tgaaggggagc gttcaccaatt tcacagagac agtcaaaaga tacactgtac 300
ctccaaatga attcactgag agccgaggat actgcagtct atcattgtgc aagagggtcc 360
tcaggccact acgaggcctt tgatatatgg ggccaaggca ccttggtaac cgttagtagc 420
ggaggtggag gaagcggagg cggcgggtcc ggaggagggtg gaagtcaatc cgttctgact 480
cagccaccct ccgtgagtgc agcgcctgga cagaagggtca caatatcatg ttctggctct 540
tcaagcaaca tcggaacctg gtacgtgagc tggtaaccagc agctccccgg caccggcgccc 600
aagcttctga tatatgacaa caacaaacgg ccagtgagg ttcttgacag attcagtgagg 660
tctaaaagtg gtacaagcgc tagcctgggt atcaccggat tgcagaccgg agatgaggcg 720
gactattatt gcggaacctg ggactccagc ctgagcgctc cctgggtttt cggcggagggt 780
actaaggtta ccgtcttggg cgcggccgca ggtgcagggtg gtgattacaa agatgatgac 840
gataaagggtg cagcggcgca tcaccatcat caccac 876

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SEQ ID NO: 46      moltype = DNA length = 876
FEATURE
misc_feature       1..876
                    note = Full-length nucleotide sequence of NYA-2061
source             1..876
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 46
atgaaacacc tgtggttctt cctcctgctg gtggcagctc ccagatgggtg gctgagcggt 60
gaggtacagc tgggtggaatc cgggtggaggc cttgtgcaac cgggagggtc cctgcgactc 120
tcttgccgag catctggctt ttccatacga tctacagata ttcattgggt cgcacaagcc 180
cctggcaaaag gtcttgaatg ggttgccact atctcctacg acgggtctca gaaatattat 240
gcggtattcag tgaaggggagc gttcaccaatt tcacagagac agtcaaaaga tacactgtac 300
ctccaaatga attcactgag agccgaggat actgcagtct atcattgtgc aagagggtcc 360
tcaggccact acgaggcctt tgatatatgg ggccaaggca ccttggtaac cgttagtagc 420
ggaggtggag gaagcggagg cggcgggtcc ggaggagggtg gaagtcaatc cgttctgact 480
cagccaccct ccgcctctgg caccocgggc caacgggtca caatatcatg ttctggctct 540
tcaagcaaca tcggaacctg gtacgtgagc tggtaaccagc agctccccgg caccggcgccc 600
aagcttctga tatatgacaa caacaaacgg ccagtgagg ttcttgacag attcagtgagg 660
tctaaaagtg gtacaagcgc tagcctggcc ataagtggtc tgcagagtga agatgaggcg 720
gactattatt gcggaacctg ggactccagc ctgagcgctc cctgggtttt cggcggagggt 780
actaagctga ccgtcttggg cgcggccgca ggtgcagggtg gtgattacaa agatgatgac 840
gataaagggtg cagcggcgca tcaccatcat caccac 876

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SEQ ID NO: 47      moltype = AA length = 292
FEATURE
REGION            1..292
                    note = Full-length amino acid sequence of NYA-2044
source            1..292
                    mol_type = protein
                    organism = synthetic construct

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SEQUENCE: 47
MKHLWFFLLL VAAPRWVLSG EVQLVESGGG LVQPGGSLRL SCAASGFSIR SYDIHWVRLA 60
PGKGLEWVAT ISYDGSQKYY ADSVKGRFTI SRDESKNTLY LQMNSLRAED TAVYHCARGS 120
SGHYEAFDIW GQGTLLVTSS GGGGSGGGGS GGGGSGSVLT QPPSVSAAPG QKVTISCSGS 180
SSNIGNWVYS WYQLPGTAP KLLIYDNMKR PSGIPDRFSG SKSGTSATLG ITGLQTGDEA 240
DYCYGTWDSS LSAPWVFGGG TKVTVLGAAA GAGGDYKDD DKGAAHHHH HH 292

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SEQ ID NO: 48      moltype = AA length = 292
FEATURE
REGION            1..292

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source note = Full-length amino acid sequence of NYA-2045
1..292
mol_type = protein
organism = synthetic construct

SEQUENCE: 48
MKHLWFFLLL VAAPRWVLSG EVQLVESGGG LVQPGGSLRL SCAASGFSIR SYDIHWVRLA 60
PGKGLEWVAT ISYDGSQKYY ADSVKGRFTI SRDESKNTLY LQMNSLRAED TAVYHCARGS 120
SGHYEAFDIW GQGTLLVTSS GGGGSGGGGS GGGGSQSVLT QPPSASGTPG QRVITSCSGS 180
SSNIGNWYVS WYQQLPGTAP KLLIYDNNKR PSGVPDRFSG SKSGTSASLA ISGLQSEDEA 240
DYYCGTWDSS LSAPWVFGGG TKLTVLGAAA GAGGDYKDDD DKGAAHHHH HH 292

SEQ ID NO: 49 moltype = AA length = 246
FEATURE Location/Qualifiers
REGION 1..246
note = Full-length amino acid sequence of NYA-0082
1..246
source mol_type = protein
organism = synthetic construct

SEQUENCE: 49
EVQLVESGGG VVQPGRLTP SCSASGFSIR SYDIHWVRQA PGKGLEWVAT ISYDGSQKYP 60
ADSVKGRFTI FRDESENMYV LQMSGLRVED TAVYHCARGS SGHYEAFDIW GQGTMTVTVSS 120
GGGSGGGGS GGGGSQSVLT QPPSVSAAPG QKVTISCSGS SSNIGNWYVS WYQRLPGTAP 180
KLLIYDNNKR PSGIPDRFSG SKSGTSATLG ITGLQTGDEA DYYCGSWDSS LSAPWVFGGG 240
TKVTVL 246

SEQ ID NO: 50 moltype = AA length = 292
FEATURE Location/Qualifiers
REGION 1..292
note = Full-length amino acid sequence of NYA-2047
1..292
source mol_type = protein
organism = synthetic construct

SEQUENCE: 50
MKHLWFFLLL VAAPRWVLSG QVQLVESGGG VVQPGRLRL SCAASGFSIR SYDMHWVRQA 60
PGKGLEWVAT ISYDGSQKYY ADSVKGRFTI SRDNSKNTLY LQMSSLRAED TAVYHCARGS 120
SGHYEAFDIW GQGTLLVTSS GGGGSGGGGS GGGGSQSVLT QPPSVSAAPG QKVTISCSGS 180
SSNIGNWYVS WYQQLPGTAP KLLIYDNNKR PSGIPDRFSG SKSGTSATLG ITGLQTGDEA 240
DYYCGTWDSS LSAPWVFGGG TKVTVLGAAA GAGGDYKDDD DKGAAHHHH HH 292

SEQ ID NO: 51 moltype = AA length = 292
FEATURE Location/Qualifiers
REGION 1..292
note = Full-length amino acid sequence of NYA-2048
1..292
source mol_type = protein
organism = synthetic construct

SEQUENCE: 51
MKHLWFFLLL VAAPRWVLSG QVQLVESGGG VVQPGRLRL SCAASGFSIR SYDMHWVRQA 60
PGKGLEWVAT ISYDGSQKYY ADSVKGRFTI SRDNSKNTLY LQMSSLRAED TAVYHCARGS 120
SGHYEAFDIW GQGTLLVTSS GGGGSGGGGS GGGGSQSVLT QPPSASGTPG QRVITSCSGS 180
SSNIGNWYVS WYQQLPGTAP KLLIYDNNKR PSGVPDRFSG SKSGTSASLA ISGLQSEDEA 240
DYYCGTWDSS LSAPWVFGGG TKLTVLGAAA GAGGDYKDDD DKGAAHHHH HH 292

SEQ ID NO: 52 moltype = AA length = 292
FEATURE Location/Qualifiers
REGION 1..292
note = Full-length amino acid sequence of NYA-2060
1..292
source mol_type = protein
organism = synthetic construct

SEQUENCE: 52
MKHLWFFLLL VAAPRWVLSG EVQLVESGGG LVQPGGSLRL SCAASGFSIR SYDIHWVRQA 60
PGKGLEWVAT ISYDGSQKYY ADSVKGRFTI SRDESKNTLY LQMNSLRAED TAVYHCARGS 120
SGHYEAFDIW GQGTLLVTSS GGGGSGGGGS GGGGSQSVLT QPPSVSAAPG QKVTISCSGS 180
SSNIGNWYVS WYQQLPGTAP KLLIYDNNKR PSGIPDRFSG SKSGTSATLG ITGLQTGDEA 240
DYYCGTWDSS LSAPWVFGGG TKVTVLGAAA GAGGDYKDDD DKGAAHHHH HH 292

SEQ ID NO: 53 moltype = AA length = 292
FEATURE Location/Qualifiers
REGION 1..292
note = Full-length amino acid sequence of NYA-2061
1..292
source mol_type = protein
organism = synthetic construct

SEQUENCE: 53
MKHLWFFLLL VAAPRWVLSG EVQLVESGGG LVQPGGSLRL SCAASGFSIR SYDIHWVRQA 60

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SEQ ID NO: 61      moltype = AA  length = 12
FEATURE            Location/Qualifiers
REGION            1..12
                  note = Amino acid sequence of NYA-2027-CDRL3
source            1..12
                  mol_type = protein
                  organism = synthetic construct

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SEQUENCE: 61
GSWDSSLSAP WV 12

SEQ ID NO: 62 moltype = AA length = 13
FEATURE Location/Qualifiers
REGION 1..13
 note = Amino acid sequence of NYA-1154-CDRH3
source 1..13
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 62
ARGSSNHYEAFDI 13

SEQ ID NO: 63 moltype = AA length = 12
FEATURE Location/Qualifiers
REGION 1..12
 note = Amino acid sequence of NYA-1154-CDRL3
source 1..12
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 63
GAWDSSLSAP WV 12

SEQ ID NO: 64 moltype = AA length = 8
FEATURE Location/Qualifiers
REGION 1..8
 note = Amino acid sequence of NYA-0035-CDRL1
source 1..8
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 64
SSNIGNWK 8

SEQ ID NO: 65 moltype = DNA length = 867
FEATURE Location/Qualifiers
misc_feature 1..867
 note = Full-length nucleotide sequence of NYC-0003
source 1..867
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 65
atgaaacacc tgtggttctt cctcctgctg gtggcagctc ccagatgggt gctgagcggc 60
gaagtccagc tgttgagctc aggtggcgga ttggtgcaac ctggcgggtc actgaggctg 120
agttgtgcag ctagcggctt cacattctct acgtatcaga tgagctgggt gagacaggct 180
ccaggaaaagg gtctggaatg ggtcagcggg attgtgtcta gcggtggctc cactgcctat 240
gccgatatcg taaaaggccg ctttacgata tctcgggaca actctaagaa cacactctat 300
ctgcagatga attcccttag agccgaagat accgccgtgt actactgtgc tggggaactg 360
ctcccgtatt acggtatgga cgtttggggc caaggcacca ctgtcacagt gatttccggt 420
ggaggcgggt caggcggagg cggtagtgga ggtggaggat cacaaagcga gctgacacag 480
cctaggtccg tatccggaag tccagggcag agcgtcacca tcagctgcac tggcacctct 540
cgagatgtcg gccgatacaa ctacgtgtct tggatcagc agcatcccgg caaagcgccc 600
aaactcatca tacacgacgt gattgagcgg tccagtgggg ttcccgatcg ttccagcggg 660
tcaaagtcgg gaaataccgc aagcctgacc atttctgggc ttcaggcaga ggatgaggct 720
gactactact gctggtcctt tgcggggagg tattacgtgt ttgggacagg gactgacgtg 780
actgttctgg gcgcggccgc aggtgcaggt ggtgattaca aagatgatga cgataaagg 840
gcagcggcgc atcaccatca tcaccac 867

SEQ ID NO: 66 moltype = DNA length = 867
FEATURE Location/Qualifiers
misc_feature 1..867
 note = Full-length nucleotide sequence of NYC-0004
source 1..867
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 66
atgaaacacc tgtggttctt cctcctgctg gtggcagctc ccagatgggt gctgagcggc 60
gaagtccagc tgttgagctc aggtggcgga ttggtgcaac ctggcgggtc actgaggctg 120
agttgtgcag ctagcggctt cacattctct acgtatcaga tgagctgggt gagacaggct 180
ccaggaaaagg gtctggaatg ggtcagcggg attgtgtcta gcggtggctc cactgcctat 240
gccgatatcg taaaaggccg ctttacgata tctcgggaca actctaagaa cacactctat 300
ctgcagatga attcccttag agccgaagat accgccgtgt actactgtgc tggggaactg 360
ctcccgtatt acggtatgga cgtttggggc caaggcacca ctgtcacagt gatttccggt 420
ggaggcgggt caggcggagg cggtagtgga ggtggaggat cacaaagcga gctgacacag 480
cctaggtccg tatccggaag tccagggcag agcgtcacca tcagctgcac tggcacccag 540
cgagatgtcg gccgatacaa ctacgtgtct tggatcagc agcatcccgg caaagcgccc 600
aaactcatca tacacgacgt gattgagcgg tccagtgggg ttcccgatcg ttccagcggg 660

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tcaaagtcgc gaaataccgc aagcctgacc atttctgggc ttcaggcaga ggatgaggct 720
gactactact gctggtcctt tgccgggggc tattacgtgt ttgggacagg gactgacgtg 780
actgttcttg gcgcgggcgc aggtgcaggt ggtgattaca aagatgatga cgataaaggt 840
gcagcggcgc atcaccatca tcaccac 867

SEQ ID NO: 67      moltype = AA  length = 289
FEATURE           Location/Qualifiers
REGION            1..289
                  note = Full-length amino acid sequence of NYC-0003
source            1..289
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 67
MKHLWFFLLL VAAPRWVLSG EVQLLES GGG LVQPGGSLRL SCAASGFTFS TYQMSWVRQA 60
PGKGLEWVSG IVSSGGSTAY ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCAGEL 120
LPYYGMDVWG QGTTVTYVSS GGGSGGGSG GGGSQSELTQ PRSVSGSPGQ SVTISCTGTS 180
RDVGGYNYVS WYQQHPGKAP KLIHDVIER SSGVPDRFSG SKSGNTASLT ISGLQAEDEA 240
DYCWSFAGS YYVFGTGTDV TVLGAAAGAG GDYKDDDDKG AAAHHHHHHH 289

SEQ ID NO: 68      moltype = AA  length = 289
FEATURE           Location/Qualifiers
REGION            1..289
                  note = Full-length amino acid sequence of NYC-0004
source            1..289
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 68
MKHLWFFLLL VAAPRWVLSG EVQLLES GGG LVQPGGSLRL SCAASGFTFS TYQMSWVRQA 60
PGKGLEWVSG IVSSGGSTAY ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCAGEL 120
LPYYGMDVWG QGTTVTYVSS GGGSGGGSG GGGSQSELTQ PRSVSGSPGQ SVTISCTGTS 180
RDVGGYNYVS WYQQHPGKAP KLIHDVIER SSGVPDRFSG SKSGNTASLT ISGLQAEDEA 240
DYCWSFAGS YYVFGTGTDV TVLGAAAGAG GDYKDDDDKG AAAHHHHHHH 289

SEQ ID NO: 69      moltype = DNA  length = 876
FEATURE           Location/Qualifiers
misc_feature      1..876
                  note = Full-length nucleotide sequence of NYA-0001
source            1..876
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 69
atgaaacacc tgtggttctt cctcctgctg gtggcagctc ccagatgggt gctgagcgga 60
gaaggtgcagc tgggtgagtc tgggggagggc gtgggtccagc ctgggaggtc cctgacaccc 120
tctgtgtcag cgtctggatt ctccatccgt agttatgata tacactgggt ccgccaggct 180
ccaggcaagg ggctagagtg ggtggccact atatcatatg atggaagtca gaagtacttc 240
gcagactccg tgaagggcgc atttaccatc ttcagagacg aatcggagaa catggtgtat 300
ctgcaaatga gcggcctgag agttgaggac acggtctgtt atcactgtgc gagagggagt 360
agtggtcatt atgaggcttt tgatatctgg ggccaaggga caatggtcac cgtctcttca 420
ggtggaggcg gttcaggcgg aggtggcagc ggcggtggcg ggagtcagtc tgtgttgacg 480
cagcgcgcct cagtgtctgc ggccccagga cagaagggtc ccattctctg ctctggaagc 540
agctccaaca ttgggaataa ttatgtatcc tggtagcagc agctcccagg aacagcccc 600
aaactctca tttatgacaa taataagcga ccctcaggga ttcctgaccg attctctggc 660
tccaagtctg gcacgtcagc caccctgggc atcaccggac tccagactgg ggacgaggcc 720
gattattact gcggaacatg ggatagcagc ctgagtgccc cttgggtgtt cggcgaggag 780
accaaggtca cgcctctagg cgcggccgca ggtgcaggtg gtgattacaa agatgatgac 840
gataaagggtg cagcggcgca tcaccatcat caccac 876

SEQ ID NO: 70      moltype = AA  length = 292
FEATURE           Location/Qualifiers
REGION            1..292
                  note = Full-length amino acid sequence of NYA-0001
source            1..292
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 70
MKHLWFFLLL VAAPRWVLSG EVQLVES GGG VVQPGRSLTP SCSASGFSIR SYDIHWVRQA 60
PGKGLEWVAT ISYDGSQKYF ADSVKGRFTI FRDESENMYV LQMSGLRVED TAVYHCARGS 120
SGHYEAFDIW GQGTMTVTSS GGGSGGGSG GGGGSQSVLT QPPSVSAAPG QKVTISCSGS 180
SSNIGNNYVS WYQQLPGTAP KLLIYDNNKR PSGIPDRFSG SKSGTSATLG ITGLQTGDEA 240
DYCYGTWDS LSAPWVFGG TKVTVLGAAA GAGGDYKDDD DKGAAHHHHH HH 292

SEQ ID NO: 71      moltype = DNA  length = 738
FEATURE           Location/Qualifiers
misc_feature      1..738
                  note = Full-length nucleotide sequence of HCl
source            1..738

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mol_type = other DNA
organism = synthetic construct

SEQUENCE: 71
atgaaacacc tgtggttctt cctcctgctg gtggcagctc ccagatgggt gctgagcgac 60
aaaactcaca catgcccacc ctgcccagca cctgaagccg cagggggacc ctcagtcttc 120
ctcttccccc caaaacccaa ggacaccctc atgatctccc ggacccctga ggtcacatgc 180
gtggtggtgt cagtgaacca cgaagaccct gaggtcaagt tcaactggta cgtggacggc 240
gtggagggtg ataattgcaa gacaaagccc cgggaggagc agtacaacag cacgtaccgg 300
gtggtcagcg tctcaccgt cctgcaccag gactggctga atggcaagga gtacaagtgc 360
aaggtctcca acaaagccct ccagccccc atcgagaaaa ccattctcaa agccaaaggc 420
cagcccccgg aaccacaggt gtacgtgctg ccccatccc gggacgagct gaccaagaac 480
caggtcagcc tgetgtgctt ggtcaaaggc ttctatccca gcgacatcgc cgtggagtgg 540
gagagcaatg gccagccgga gaacaactac ctgacctggc ctcccgctgt ggactccgac 600
ggctccttct tctctacag caagtcacc gtggacaaga gcagggtggc gcagggcaac 660
gtcttctcat gctcctgat gcatgaggct ctgcacaacc actacacca gaagagcctc 720
tcctgtcttc ccggcaag 738

SEQ ID NO: 72      moltype = DNA length = 2235
FEATURE            Location/Qualifiers
misc_feature        1..2235
                    note = Full-length nucleotide sequence of NYF-0016-HC2
source              1..2235
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 72
atgaaacacc tgtggttctt cctcctgctg gtggcagctc ccagatgggt gctgagcgga 60
gaggtgcagc tgggtggagtc tgggggaggg gtggtccagc ctgggaggtc cctgacaccc 120
tcctgttcag cgtctggatt ctccatccgt agctatgata tacactgggt ccgcctggct 180
ccaggcaagg ggctagatg ggtggccact atatcatatg atggaagtca gaagtacttc 240
gcagactccg tgaagggccg atttaccatc ttcagagacg aatcggagaa catggtgtat 300
ctgcaaatga cggcctgag agttgaggac acggctgttt atcactgtgc gagaggaggt 360
agtgttcatt atgagctttt tgatatctgg ggccaaggga caatggtcac cgtctcttca 420
ggtggaggcg gttcaggcgg aggtggcagc ggcggtggcg ggagtcactc tgtgttgacg 480
cagccgcctc cagtgtctgc ggcaccagga cagaagggtc ccatctcctg ctctggaagc 540
agctccaaca ttgggaattg gtatgtatcc tggtagccag agctcccagg aacagccccc 600
aaaactcctc tttatgacaa taataagcga cctcaggga ttctgacccg attctctggc 660
tccaaagtct gcacgtcagc caccctgggc atcacccgac tccagactgg ggaaggagcc 720
gattattact gcggaacatg ggatagcagc ctgagtgccc cttgggtgtt cggcggaggg 780
accaagggtc ccgtcctagg gggaggcggg tcagaagtgc agctggtgga atccgggggg 840
ggcctggtgc agcctggggg gagcctgaga ctgagttgtg ccgcctctgg ggtgacattt 900
aaactactat gcatgtcttg gatccgccag gcacctggaa agggcctgga gtgggtggcc 960
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gaaggtgcata atgccaagac aaagcccccg gaggagcagt acaacagcac gtaccgggtg 1800
gtcagcgtcc tcaccgtcct gcaccaggac tggctgaatg gcaaggagta caagtgcagg 1860
gtctccaaca aagccctccc agcccccctc gagaaaacca tctccaaagc caaaggccag 1920
ccccgggaac cacagggtga cgtgtacccc ccattccggg acgagctgac caagaaccag 1980
gtcagcctga cctgcctggt caaaggcttc tatccagcgc acatcgccgt ggagtgggag 2040
agcaatggcc agcccgagaa caactacaag accaccctc ccgtgctgga ctccgacggc 2100
tccttcgccc tcgtgagcaa gctcacctg gacaagagca ggtggcagca gggcaacgtc 2160
ttctcatgct ccgtgatgca tgaggctctg cacaaccact acaccagaa gagcctctcc 2220
ctgtctcccg gcaag 2235

SEQ ID NO: 73      moltype = DNA length = 2235
FEATURE            Location/Qualifiers
misc_feature        1..2235
                    note = Full-length nucleotide sequence of NYF-0019-HC2
source              1..2235
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 73
atgaaacacc tgtggttctt cctcctgctg gtggcagctc ccagatgggt gctgagcgga 60
gaggtgcagc tgggtggagtc tgggggaggg gtggtccagc ctgggaggtc cctgacaccc 120
tcctgttcag cgtctggatt ctccatccgt agctatgata tacactgggt ccgcctggct 180
ccaggcaagg ggctagatg ggtggccact atatcatatg atggaagtca gaagtacttc 240

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gcagactccg tgaagggccg atttaccatc ttcagagacg aatcggagaa catggtgtat 300
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agtggtcatt atgaggcttt tgatatctgg ggccaaggga caatggtcac cgtctcttca 420
ggtggaggcg gttcaggccg aggtggcagc ggcggtggcg ggagtcagtc tgtgttgacg 480
cagccgcctc cagtgtctgc gggcccagga cagaagggtca ccatctcctg ctctggaagc 540
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aaactcctca tttatgacaa taataagcga ccctcaggga ttctgaccg attctctggc 660
tccaagtctg gcaagtcagc caccctgggc atcacccgac tccagactgg ggaagaggcc 720
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gtcagcgtcc tcaccgtcct gcaccaggac tggctgaatg gcaaggagta caagtgaag 1860
gtctccaaca aagccctccc agcccccatc gagaaaacca tctccaaagc caaaggccag 1920
ccccgggaac cacaggtgta cgtgtacccc ccatcccggg acgagctgac caagaaccag 1980
gtcagcctga cctgcctggt caaaggcttc tatcccagc acatcccggt ggagtgagg 2040
agcaatggcc agcccgagaa caactacaag accaccctc cgtgtctgga ctccagcgc 2100
tccttcgccc tctgtgacaa gctcacccgtg gacaagagca ggtggcagca gggcaacgtc 2160
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ctgtctcccg gcaag 2235

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SEQ ID NO: 74      moltype = DNA length = 2235
FEATURE
misc_feature       1..2235
                    note = Full-length nucleotide sequence of NYF-0022-HC2
source             1..2235
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 74
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tcctgttcag cgtctggatt ctccatccgt agctatgata tacactgggt ccgcctggct 180
ccaggcaagg ggctagagtg ggtggccact atatcatatg atggaaagtc gaagtacttc 240
gcagactccg tgaagggccg atttaccatc ttcagagacg aatcggagaa catggtgtat 300
ctgcaaatga gcggcctgag agttgaggac acggctgttt atcactgtgc gagagggagt 360
agtggtcatt atgaggcttt tgatatctgg ggccaaggga caatggtcac cgtctcttca 420
ggtggaggcg gttcaggccg aggtggcagc ggcggtggcg ggagtcagtc tgtgttgacg 480
cagccgcctc cagtgtctgc gggcccagga cagaagggtca ccatctcctg ctctggaagc 540
agctccaaca ttgggaataa ttatgtatcc tggtagcagc agctcccagg aacagccccc 600
aaactcctca tttatgacaa taataagcga ccctcaggga ttctgaccg attctctggc 660
tccaagtctg gcaagtcagc caccctgggc atcacccgac tccagactgg ggaagaggcc 720
gattattact gcggaacatg ggatagcagc ctgagtgccc cttgggtgtt cggcggaggg 780
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atcttcggag gcggaactaa gctgacagtg ttggcagccg agcccaaatc ttctgacaaa 1560
actcacacat gccacacctg ccagcacct gaagccgcag ggggaccctc agtcttctc 1620
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gtcagcgtcc tcaccgtcct gcaccaggac tggctgaatg gcaaggagta caagtgaag 1860
gtctccaaca aagccctccc agcccccatc gagaaaacca tctccaaagc caaaggccag 1920
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gtcagcctga cctgctctgt caaaggcttc tatcccagcg acatcgccgt ggagtgggag 2040
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tccttcgccc tcgtgagcaa gctcaccgtg gacaagagca ggtggcagca gggcaacgtc 2160
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ctgtctcccg gcaag                2235

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SEQ ID NO: 75      moltype = DNA length = 2235
FEATURE            Location/Qualifiers
misc_feature       1..2235
                    note = Full-length nucleotide sequence of NYF-0023-HC2
source             1..2235
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 75
atgaaacacc tgtggttctt cctcctgctg gtggcagctc ccagatgggt gctgagcgga 60
gaggtgcagc tgggtggagtc tgggggaggc gtggtccagc ctgggaggtc cctgacaccc 120
tcctgttcag cgtctggatt tcctatccgt agttatgata tacactgggt ccgccagggt 180
ccaggcaagg ggctagagtg ggtggccact atatcatatg atggaagtca gaagtaactt 240
gcagactccg tgaaggcccg atttaccatc ttcagagacg aatcggagaa catggtgtat 300
ctgcaaatga gcgccctgag agttgaggac acggctgttt atcactgtgc gagagggagt 360
agtggtcatt atgaggcttt tgatatctgg ggccaaggga caatgggtcac cgtctcttca 420
ggtggaggcg gttcaggccg aggtggcagc ggcggtggcg ggagtcagtc tgtgttgacg 480
cagccgcccc cagtgtctgc gggcccagga cagaaggta ccatctctct ctctggaagc 540
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aaactctcca tttatgacaa taataagcga cctcaggga ttctgacgg attctctggc 660
tccaagtctg gcacgtcagc caccctgggc atcacgggac tccagactgg ggacgaggcc 720
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gatacagctg tgtactattg cactctggac ggcagggatg ggtgggtcgc ctattggggg 1140
cagggaaccc ttggtgacag agctccgga ggaggaggat ctggcggagg aggcagtgagg 1200
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gttccaaca aagccctccc agcccccatc agaaaaacca tctccaaagc caaaggccag 1920
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gtcagcctga cctgcctggt caaaggcttc tatcccagcg acatcgccgt ggagtgggag 2040
agcaatggcc agcccgagaa caactacaag accacccctc ccgtgctgga ctccgacggc 2100
tccttcgccc tcgtgagcaa gctcaccgtg gacaagagca ggtggcagca gggcaacgtc 2160
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ctgtctcccg gcaag                2235

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SEQ ID NO: 76      moltype = DNA length = 2235
FEATURE            Location/Qualifiers
misc_feature       1..2235
                    note = Full-length nucleotide sequence of NYF-0027-HC2
source             1..2235
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 76
atgaaacacc tgtggttctt cctcctgctg gtggcagctc ccagatgggt gctgagcgga 60
gaggtgcagc tgggtggagtc tgggggaggc gtggtccagc ctgggaggtc cctgacaccc 120
tcctgttcag cgtctggatt tcctatccgt agttatgata tacactgggt ccgccagggt 180
ccaggcaagg ggctagagtg ggtggccact atatcatatg atggaagtca gaagtaactt 240
gcagactccg tgaaggcccg atttaccatc ttcagagacg aatcggagaa catggtgtat 300
ctgcaaatga gcgccctgag agttgaggac acggctgttt atcactgtgc gagagggagt 360
agtggtcatt atgaggcttt tgatatctgg ggccaaggga caatgggtcac cgtctcttca 420
ggtggaggcg gttcaggccg aggtggcagc ggcggtggcg ggagtcagtc tgtgttgacg 480
cagccgcccc cagtgtctgc gggcccagga cagaaggta ccatctctct ctctggaagc 540
agctccaaca ttgggaataa ttatgtatcc tggtagcagc agctcccagg aacagccccc 600
aaactctcca tttatgacaa taataagcga cctcaggga ttctgacgg attctctggc 660
tccaagtctg gcacgtcagc caccctgggc atcacgggac tccagactgg ggacgaggcc 720
gattattact cgggaacatg gtagtagcagc ctgagtgccc cttgggtgtt cggcggaggg 780
accaaggtca ccgctctagg gggaggcggg tcagaagtgc agctggtgga atccgggggg 840
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aactactatg gcatgtcttg gatccgccag gcacctggaa agggcctgga gtgggtggcc 960
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gtcagcgtcc tcaccgtctc gcaccaggac tggtgtaatg gcaaggagta caagtgcaag 1860
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gtcagcctga cctgctgggt caaaggcttc tatccacgag acatcgccgt ggagtgggag 2040
agcaatggcc agcccgagaa caactacaag accaccctc ccgtgctgga ctccgacggc 2100
tccttcgccc tcgtgagcaa gctcaccgtg gacaagagca ggtggcagca gggcaacgtc 2160
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ctgtctcccg gcaag                                     2235

SEQ ID NO: 77      moltype = DNA length = 2235
FEATURE           Location/Qualifiers
misc_feature       1..2235
                   note = Full-length nucleotide sequence of NYF-0035-HC2
source             1..2235
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 77
atgaaacacc tgtggttctt cctcctgctg gtggcagctc ccagatgggt gctgagcgga 60
gaggtgcagc tggtagagtc ttgggggaggc gtgggtccagc ctgggaggtc cctgacaccc 120
tcctgttcag cgtctggatt tccatccgtg agctatgata tacactgggt ccgcctggct 180
ccaggcaagg ggttagagtg ggtggccact atatcatatg atggaagtca gaagtaactc 240
gcagactccg tgaaggcgcc atttaccatc ttcagagacg aatcggagaa catggtgtat 300
ctgcaaatga gcggcctgag agttgaggac acgggtgttt atcactgtgc gagaggaggt 360
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aaactcctca tttatgacaa taataagcga cctcagggga ttcctgacgg attctctggc 660
tccaagtctg gcacgtcagc caccctgggc atcaccggac tccagactgg ggacgaggcc 720
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gtcagcgtcc tcaccgtctc gcaccaggac tggtgtaatg gcaaggagta caagtgcaag 1860
gtctccaaca aagccctccc agccccatc gagaaaacca tctccaaagc caaaggccag 1920
ccccgggaac cacagggtga cgtgtacccc ccatcccggg acgagctgac caagaaccag 1980
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agcaatggcc agcccgagaa caactacaag accaccctc ccgtgctgga ctccgacggc 2100
tccttcgccc tcgtgagcaa gctcaccgtg gacaagagca ggtggcagca gggcaacgtc 2160
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ctgtctcccg gcaag                                     2235

SEQ ID NO: 78      moltype = DNA length = 2235
FEATURE           Location/Qualifiers
misc_feature       1..2235
                   note = Full-length nucleotide sequence of NYF-0044-HC2
source             1..2235

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mol_type = other DNA
organism = synthetic construct

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SEQ ID NO: 79      moltype = DNA      length = 2235
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misc_feature        1..2235
                    note = Full-length nucleotide sequence of NYF-0045-HC2
source              1..2235
                    mol_type = other DNA
                    organism = synthetic construct

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SEQ ID NO: 80      moltype = DNA length = 2235
FEATURE            Location/Qualifiers
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                    note = Full-length nucleotide sequence of NYF-0047-HC2
source             1..2235
                    mol_type = other DNA
                    organism = synthetic construct

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SEQ ID NO: 81      moltype = DNA length = 2235
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                    note = Full-length nucleotide sequence of NYF-0048-HC2
source             1..2235
                    mol_type = other DNA
                    organism = synthetic construct

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SEQ ID NO: 82      moltype = DNA length = 2235
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                   note = Full-length nucleotide sequence of NYF-0060-HC2
source             1..2235
                   mol_type = other DNA
                   organism = synthetic construct

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SEQ ID NO: 83 moltype = DNA length = 2235
 FEATURE Location/Qualifiers
 misc_feature 1..2235
 note = Full-length nucleotide sequence of NYF-0061-HC2
 source 1..2235
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 83

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tatcagcagc	ttccagggac	agctcccaag	ttgctgatct	atcgcgacga	caaaagaccc	1380
tcagggggtcc	atgaccgatt	tagtggcagc	aaaagcggta	cttcgcgttc	cctggcgata	1440
accggcttct	aggccgaaga	tgaggcagac	tactattgcc	agtcataatc	cagcggcttc	1500
atcttcggag	gcggaactaa	gctgacagt	ttggcagccg	agcccaaatc	ttctgacaaa	1560
actcacacat	gccaccctc	cccagcacct	gaagccgcag	ggggaccctc	agtcctctc	1620
ttccccccaa	aaacccaagga	caccctcatg	atctcccgga	cccctgaggt	cacatgcgtg	1680
gtggtgtcag	tgagccacga	agaccctgag	gtcaagttca	actggtacgt	ggacggcgtg	1740
gaggtgcata	atgccaaagac	aaagcccg	gaggagcagt	acaacagcac	gtaccgggtg	1800
gtcagcgtcc	tcaccgtcct	gcaccaggac	tggtggaatg	gcaaggagta	caagtgcgaag	1860
gtctccaaca	aagccctccc	agcccccatc	gagaaaacca	tctccaaagc	caaaggccag	1920
ccccgggaac	cacaggtgta	ctgtgtacc	ccatcccg	acgagctgac	caagaaccag	1980
gtcagcctga	cctgcctgg	caaaggcttc	tatcccagcg	acatgcgcgt	ggagtgagg	2040
agcaatggcc	agcccgagaa	caactacaag	accaccctc	ccgtgctgga	ctccgacggc	2100
tccttcgccc	tcgtgagcaa	gctcaccgtg	gacaagagca	ggtggcagca	gggcaacgtc	2160
ttctcatgct	ccgtgatgca	tgaggctctg	cacaaccact	acaccagaa	gagcctctcc	2220
ctgtctcccg	gcaag					2235

SEQ ID NO: 84 moltype = AA length = 246
 FEATURE Location/Qualifiers
 REGION 1..246
 note = Full-length amino acid sequence of HCl
 source 1..246
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 84

MKHLWFFLLL	VAAPRWVLS	KTHTCPPCP	PEAAGGPSV	LFPPKPKD	TL	MISRTPEV	TC	60
VVSVSHEDP	EVKFNWYVD	VEVHNAKTK	REEQYNSTY	R				120
KVSNKALPAP	IEKTISKAKG	QPREPQVYV	PPSRDELTK	N				180
ESNGQPENNY	LWTPPVLDS	GSFFLYSKLT	VDKSRWQQN	V				240
SLSPGK								246

SEQ ID NO: 85 moltype = AA length = 745
 FEATURE Location/Qualifiers
 REGION 1..745
 note = Full-length amino acid sequence of NYF-0016-HC
 source 1..745
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 85

MKHLWFFLLL	VAAPRWVLS	EVQLVESGG	VVQPGRS	LTP	SCSASGFS	SIR	SYDIHWVRLA	60
PGKGLEWVAT	ISYDGSQKY	F						120
SGHYEAFDIW	QGTMTVTSS	GGGSGGGGS	GGGGSQSV	LTP	QPPSVS	AAPG	QKVTISCSGS	180
SSNIGNWVYS	WYQLPGTAP	KLLIYDNMK	R					240
DYYCGTWDSS	LSAPWVFGG	TKVTVLGGG	SEVQLVESG	G	GLVQPGG	SLR	LSCAASGVTF	300

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NYGMSWIRQ	APGKGLEWVA	SITRSGGRIY	YPSVKGRFT	ISRENTQKTL	YLQMNSLRAE	360
DTAVYYCTLD	GRDGWVAYWG	QGTLLTVSSG	GGGSGGGGSG	GGGSNFMLTQ	PSSVSGVPGQ	420
RVTISCTGNT	GNIGSNYVNW	YQQLPGTAPK	LLIYRDDKRP	SGVPDRFSGS	KSGTSASLAI	480
TGFQAEDEAD	YYCQSYSSGF	IFGGGTKLTV	LAAEPKSSDK	THTCPPCPAP	EAAGGPSVFL	540
FPPKPKDTLM	ISRTEVTCV	VVSVSHEDPE	VKFNWYVDGV	EVHNAKTKPR	EEQYNSTYRV	600
VSVLTVLHQD	WLNKEYKCK	VSNKALPAPI	EKTISKAKGQ	PREPQVYVYP	PSRDELTKNQ	660
VSLTCLVKGF	YPSDIAVEWE	SNGQPENNYK	TTPPVLDSDG	SFALVSKLTV	DKSRWQQGNV	720
FSCSVMEAL	HNHYTQKSLS	LSPGK				745

SEQ ID NO: 86 moltype = AA length = 745
FEATURE Location/Qualifiers
REGION 1..745
 note = Full-length amino acid sequence of NYF-0019-HC2
source 1..745
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 86

MKHLWFFLLL	VAAPRWVLSG	EVQLVESGGG	VVQPGRLTP	SCSASGFSIR	SYDIHWVRLA	60
PGKGLEWVAT	ISYDGSQKYF	ADSVKGRFTI	FRDESENMYV	LQMSGRLVED	TAVYHCARGS	120
SGHYEAFDIW	GQGTMTVTVSS	GGGSGGGGSG	GGGGSQSVLT	QPPSVSAAPG	QKVTISCSGS	180
SSNIGNWYVS	WYQRLPGTAP	KLLIYDNMKR	PSGIPDRFSG	SKSGTSATLG	ITGLQTGDEA	240
DYYCGTWDS	LSAPWVFGGG	TKVTVLGGGG	SEVQLVESGG	GLVQPGGSLR	LSCAASGVTF	300
NYGMSWIRQ	APGKGLEWVA	SITRSGGRIY	YPSVKGRFT	ISRENTQKTL	YLQMNSLRAE	360
DTAVYYCTLD	GRDGWVAYWG	QGTLLTVSSG	GGGSGGGGSG	GGGSNFMLTQ	PSSVSGVPGQ	420
RVTISCTGNT	GNIGSNYVNW	YQQLPGTAPK	LLIYRDDKRP	SGVPDRFSGS	KSGTSASLAI	480
TGFQAEDEAD	YYCQSYSSGF	IFGGGTKLTV	LAAEPKSSDK	THTCPPCPAP	EAAGGPSVFL	540
FPPKPKDTLM	ISRTEVTCV	VVSVSHEDPE	VKFNWYVDGV	EVHNAKTKPR	EEQYNSTYRV	600
VSVLTVLHQD	WLNKEYKCK	VSNKALPAPI	EKTISKAKGQ	PREPQVYVYP	PSRDELTKNQ	660
VSLTCLVKGF	YPSDIAVEWE	SNGQPENNYK	TTPPVLDSDG	SFALVSKLTV	DKSRWQQGNV	720
FSCSVMEAL	HNHYTQKSLS	LSPGK				745

SEQ ID NO: 87 moltype = AA length = 745
FEATURE Location/Qualifiers
REGION 1..745
 note = Full-length amino acid sequence of NYF-0022-HC2
source 1..745
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 87

MKHLWFFLLL	VAAPRWVLSG	EVQLVESGGG	VVQPGRLTP	SCSASGFSIR	SYDIHWVRLA	60
PGKGLEWVAT	ISYDGSQKYF	ADSVKGRFTI	FRDESENMYV	LQMSGRLVED	TAVYHCARGS	120
SGHYEAFDIW	GQGTMTVTVSS	GGGSGGGGSG	GGGGSQSVLT	QPPSVSAAPG	QKVTISCSGS	180
SSNIGNWYVS	WYQRLPGTAP	KLLIYDNMKR	PSGIPDRFSG	SKSGTSATLG	ITGLQTGDEA	240
DYYCGTWDS	LSAPWVFGGG	TKVTVLGGGG	SEVQLVESGG	GLVQPGGSLR	LSCAASGVTF	300
NYGMSWIRQ	APGKGLEWVA	SITRSGGRIY	YPSVKGRFT	ISRENTQKTL	YLQMNSLRAE	360
DTAVYYCTLD	GRDGWVAYWG	QGTLLTVSSG	GGGSGGGGSG	GGGSNFMLTQ	PSSVSGVPGQ	420
RVTISCTGNT	GNIGSNYVNW	YQQLPGTAPK	LLIYRDDKRP	SGVPDRFSGS	KSGTSASLAI	480
TGFQAEDEAD	YYCQSYSSGF	IFGGGTKLTV	LAAEPKSSDK	THTCPPCPAP	EAAGGPSVFL	540
FPPKPKDTLM	ISRTEVTCV	VVSVSHEDPE	VKFNWYVDGV	EVHNAKTKPR	EEQYNSTYRV	600
VSVLTVLHQD	WLNKEYKCK	VSNKALPAPI	EKTISKAKGQ	PREPQVYVYP	PSRDELTKNQ	660
VSLTCLVKGF	YPSDIAVEWE	SNGQPENNYK	TTPPVLDSDG	SFALVSKLTV	DKSRWQQGNV	720
FSCSVMEAL	HNHYTQKSLS	LSPGK				745

SEQ ID NO: 88 moltype = AA length = 745
FEATURE Location/Qualifiers
REGION 1..745
 note = Full-length amino acid sequence of NYF-0023-HC2
source 1..745
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 88

MKHLWFFLLL	VAAPRWVLSG	EVQLVESGGG	VVQPGRLTP	SCSASGFSIR	SYDIHWVRQA	60
PGKGLEWVAT	ISYDGSQKYF	ADSVKGRFTI	FRDESENMYV	LQMSGRLVED	TAVYHCARGS	120
SGHYEAFDIW	GQGTMTVTVSS	GGGSGGGGSG	GGGGSQSVLT	QPPSVSAAPG	QKVTISCSGS	180
SSNIGNWYVS	WYQRLPGTAP	KLLIYDNMKR	PSGIPDRFSG	SKSGTSATLG	ITGLQTGDEA	240
DYYCGTWDS	LSAPWVFGGG	TKVTVLGGGG	SEVQLVESGG	GLVQPGGSLR	LSCAASGVTF	300
NYGMSWIRQ	APGKGLEWVA	SITRSGGRIY	YPSVKGRFT	ISRENTQKTL	YLQMNSLRAE	360
DTAVYYCTLD	GRDGWVAYWG	QGTLLTVSSG	GGGSGGGGSG	GGGSNFMLTQ	PSSVSGVPGQ	420
RVTISCTGNT	GNIGSNYVNW	YQQLPGTAPK	LLIYRDDKRP	SGVPDRFSGS	KSGTSASLAI	480
TGFQAEDEAD	YYCQSYSSGF	IFGGGTKLTV	LAAEPKSSDK	THTCPPCPAP	EAAGGPSVFL	540
FPPKPKDTLM	ISRTEVTCV	VVSVSHEDPE	VKFNWYVDGV	EVHNAKTKPR	EEQYNSTYRV	600
VSVLTVLHQD	WLNKEYKCK	VSNKALPAPI	EKTISKAKGQ	PREPQVYVYP	PSRDELTKNQ	660
VSLTCLVKGF	YPSDIAVEWE	SNGQPENNYK	TTPPVLDSDG	SFALVSKLTV	DKSRWQQGNV	720
FSCSVMEAL	HNHYTQKSLS	LSPGK				745

SEQ ID NO: 89 moltype = AA length = 745

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FEATURE Location/Qualifiers
REGION 1..745
note = Full-length amino acid sequence of NYF-0027-HC2
source 1..745
mol_type = protein
organism = synthetic construct

SEQUENCE: 89

MKHLWFFLLL	VAAPRWVLSG	EVQLVESGGG	VVQPGRSITP	SCSASGFSIR	SYDIHWVRQA	60
PGKGLEWVAT	ISYDGSQKYF	ADSVKGRFTI	FRDESENVMY	LQMSGLRVED	TAVYHCARGS	120
SGHYEAFDIW	GQGTMTVTSS	GGGSGGGGGS	GGGGSQSVLT	QPPSVSAAPG	QKVTISCSGS	180
SSNIGNNIVS	WYQQLPGTAP	KLLIYDNMKR	PSGIPDRFSG	SKSGTSATLG	ITGLQTGDEA	240
DYYCGSWDSS	LSAPWVFGGG	TKVTVLGGGG	SEVQLVESGG	GLVQPGGSLR	LSCAASGVTF	300
NYYGMSWIRQ	APGKGLEWVA	SITRSGGRIY	YPSVKGFRFT	ISRENTQKTL	YLQMNSLRAE	360
DTAVYYCTLD	GRDGWVAYWG	QGTLLVTSSG	GGGSGGGGSG	GGGSNFMILTQ	PSSVSGVPGQ	420
RVTISCTGNT	GNIGSNYVNW	YQQLPGTAPK	LLIYRDDKRP	SGVPDRFSGS	KSGTSASLAI	480
TGFQAEDEAD	YYCQSYSSGF	IFGGGTKLTV	LAAEPKSSDK	THTCPPCPAP	EAAGGPSVFL	540
FPPKPKDTLM	ISRTPEVTCV	VVSVSHEDPE	VKFNWYVDGV	EVHNAKTKPR	EEQYNSTYRV	600
VSVLTVLHQD	WLNKEYKCKK	VSNKALPAPI	EKTISKAKGQ	PREPQVYVYP	PSRDELTKNQ	660
VSLTCLVKGF	YPSDIAVEWE	SNGQPENNYK	TTPPVLDSDG	SFALVSKLTV	DKSRWQQGNV	720
FSCSVMEAL	HNHYTQKSLS	LSPGK				745

SEQ ID NO: 90 moltype = AA length = 745
FEATURE Location/Qualifiers
REGION 1..745
note = Full-length amino acid sequence of NYF-0035-HC2
source 1..745
mol_type = protein
organism = synthetic construct

SEQUENCE: 90

MKHLWFFLLL	VAAPRWVLSG	EVQLVESGGG	VVQPGRSITP	SCSASGFSIR	SYDIHWVRLA	60
PGKGLEWVAT	ISYDGSQKYF	ADSVKGRFTI	FRDESENVMY	LQMSGLRVED	TAVYHCARGS	120
SGHYEAFDIW	GQGTMTVTSS	GGGSGGGGGS	GGGGSQSVLT	QPPSVSAAPG	QKVTISCSGS	180
SSNIGNWVVS	WYQQLPGTAP	KLLIYDNMKR	PSGIPDRFSG	SKSGTSATLG	ITGLQTGDEA	240
DYYCGTWDSS	LSAPWVFGGG	TKVTVLGGGG	SEVQLVESGG	GLVQPGGSLR	LSCAASGVTF	300
NYYGMSWIRQ	APGKGLEWVA	SITRSGGRIY	YPSVKGFRFT	ISRENTQKTL	YLQMNSLRAE	360
DTAVYYCTLD	GRDGWVAYWG	QGTLLVTSSG	GGGSGGGGSG	GGGSNFMILTQ	PSSVSGVPGQ	420
RVTISCTGNT	GNIGSNYVNW	YQQLPGTAPK	LLIYRDDKRP	SGVPDRFSGS	KSGTSASLAI	480
TGFQAEDEAD	YYCQSYSSGF	IFGGGTKLTV	LAAEPKSSDK	THTCPPCPAP	EAAGGPSVFL	540
FPPKPKDTLM	ISRTPEVTCV	VVSVSHEDPE	VKFNWYVDGV	EVHNAKTKPR	EEQYNSTYRV	600
VSVLTVLHQD	WLNKEYKCKK	VSNKALPAPI	EKTISKAKGQ	PREPQVYVYP	PSRDELTKNQ	660
VSLTCLVKGF	YPSDIAVEWE	SNGQPENNYK	TTPPVLDSDG	SFALVSKLTV	DKSRWQQGNV	720
FSCSVMEAL	HNHYTQKSLS	LSPGK				745

SEQ ID NO: 91 moltype = AA length = 745
FEATURE Location/Qualifiers
REGION 1..745
note = Full-length amino acid sequence of NYF-0044-HC2
source 1..745
mol_type = protein
organism = synthetic construct

SEQUENCE: 91

MKHLWFFLLL	VAAPRWVLSG	EVQLVESGGG	LVQPGGSLRL	SCAASGFSIR	SYDIHWVRLA	60
PGKGLEWVAT	ISYDGSQKYF	ADSVKGRFTI	SRDESKNTLY	LQMNSLRAED	TAVYHCARGS	120
SGHYEAFDIW	GQGTLLVTSS	GGGSGGGGGS	GGGGSQSVLT	QPPSVSAAPG	QKVTISCSGS	180
SSNIGNWVVS	WYQQLPGTAP	KLLIYDNMKR	PSGIPDRFSG	SKSGTSATLG	ITGLQTGDEA	240
DYYCGTWDSS	LSAPWVFGGG	TKVTVLGGGG	SEVQLVESGG	GLVQPGGSLR	LSCAASGVTF	300
NYYGMSWIRQ	APGKGLEWVA	SITRSGGRIY	YPSVKGFRFT	ISRENTQKTL	YLQMNSLRAE	360
DTAVYYCTLD	GRDGWVAYWG	QGTLLVTSSG	GGGSGGGGSG	GGGSNFMILTQ	PSSVSGVPGQ	420
RVTISCTGNT	GNIGSNYVNW	YQQLPGTAPK	LLIYRDDKRP	SGVPDRFSGS	KSGTSASLAI	480
TGFQAEDEAD	YYCQSYSSGF	IFGGGTKLTV	LAAEPKSSDK	THTCPPCPAP	EAAGGPSVFL	540
FPPKPKDTLM	ISRTPEVTCV	VVSVSHEDPE	VKFNWYVDGV	EVHNAKTKPR	EEQYNSTYRV	600
VSVLTVLHQD	WLNKEYKCKK	VSNKALPAPI	EKTISKAKGQ	PREPQVYVYP	PSRDELTKNQ	660
VSLTCLVKGF	YPSDIAVEWE	SNGQPENNYK	TTPPVLDSDG	SFALVSKLTV	DKSRWQQGNV	720
FSCSVMEAL	HNHYTQKSLS	LSPGK				745

SEQ ID NO: 92 moltype = AA length = 745
FEATURE Location/Qualifiers
REGION 1..745
note = Full-length amino acid sequence of NYF-0045-HC2
source 1..745
mol_type = protein
organism = synthetic construct

SEQUENCE: 92

MKHLWFFLLL	VAAPRWVLSG	EVQLVESGGG	LVQPGGSLRL	SCAASGFSIR	SYDIHWVRLA	60
PGKGLEWVAT	ISYDGSQKYF	ADSVKGRFTI	SRDESKNTLY	LQMNSLRAED	TAVYHCARGS	120
SGHYEAFDIW	GQGTLLVTSS	GGGSGGGGGS	GGGGSQSVLT	QPPASGTPG	QKVTISCSGS	180

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SSNIGNWYVS WYQQLPGTAP KLLIYDNMKR PSGVPDRFSG SKSGTSASLA ISGLQSEDEA 240
DYYCGTWDSS LSAPWVFGGG TKLTVLGGGG SEVQLVESGG GLVQPGGSLR LSCAASGVTF 300
NYYGMSWIRQ APGKGLEWVA SITRSGGRIY YPDSVKGRFT ISRENTQKTL YLQMNSLRAE 360
DTAVYYCTLD GRDGWVAYWG QGTLVTVSSG GGGSGGGGSG GGGSNFMLTQ PSSVSGVPGQ 420
RVTISCTGNT GNIGSNYVNW YQQLPGTAPK LLIYRDDKRP SGVPDRFSGS KSGTSASLAI 480
TGFQAEDEAD YYCQSYSSGF IFGGGTKLTV LAAEPKSSDK THTCPPCPAP EAAGGPSVFL 540
FPPKPKDTLM ISRTPEVTCV VVSVSHEDPE VKFNWYVDGV EVHNAKT KPR EEQYNSTYRV 600
VSVLTVLHQD WLNKEYKCK VSNKALPAPI EKTISKAKGQ PREPQVYVYP PSRDELTKNQ 660
VSLTCLVKGF YPSDIAVEWE SNGQPENNYK TTPPVLDSDG SFALVSKLTV DKSRWQQGNV 720
FSCSVMHEAL HNHYTQKSL LSPGK 745

SEQ ID NO: 93      moltype = AA length = 745
FEATURE           Location/Qualifiers
REGION            1..745
                  note = Full-length amino acid sequence of NYF-0047-HC2
source            1..745
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 93
MKHLWFFLLL VAAPRWVLSG QVQLVESGGG VVQPGRSLRL SCAASGFSIR SYDMHWVRQA 60
PGKGLEWVAT ISYDGSQKYY ADSVKGRFTI SRDNSKNTLY LQMSSLRAED TAVYHCARGS 120
SGHYEAFDIW GQGLTVTVSS GGGSGGGGSG GGGGQSQVLT QPPSVSAAPG QKVTISCSGS 180
SSNIGNWYVS WYQQLPGTAP KLLIYDNMKR PSGIPDRFSG SKSGTSATLG ITGLQTGDEA 240
DYYCGTWDSS LSAPWVFGGG TKVTVLGGGG SEVQLVESGG GLVQPGGSLR LSCAASGVTF 300
NYYGMSWIRQ APGKGLEWVA SITRSGGRIY YPDSVKGRFT ISRENTQKTL YLQMNSLRAE 360
DTAVYYCTLD GRDGWVAYWG QGTLVTVSSG GGGSGGGGSG GGGSNFMLTQ PSSVSGVPGQ 420
RVTISCTGNT GNIGSNYVNW YQQLPGTAPK LLIYRDDKRP SGVPDRFSGS KSGTSASLAI 480
TGFQAEDEAD YYCQSYSSGF IFGGGTKLTV LAAEPKSSDK THTCPPCPAP EAAGGPSVFL 540
FPPKPKDTLM ISRTPEVTCV VVSVSHEDPE VKFNWYVDGV EVHNAKT KPR EEQYNSTYRV 600
VSVLTVLHQD WLNKEYKCK VSNKALPAPI EKTISKAKGQ PREPQVYVYP PSRDELTKNQ 660
VSLTCLVKGF YPSDIAVEWE SNGQPENNYK TTPPVLDSDG SFALVSKLTV DKSRWQQGNV 720
FSCSVMHEAL HNHYTQKSL LSPGK 745

SEQ ID NO: 94      moltype = AA length = 745
FEATURE           Location/Qualifiers
REGION            1..745
                  note = Full-length amino acid sequence of NYF-0048-HC2
source            1..745
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 94
MKHLWFFLLL VAAPRWVLSG QVQLVESGGG VVQPGRSLRL SCAASGFSIR SYDMHWVRQA 60
PGKGLEWVAT ISYDGSQKYY ADSVKGRFTI SRDNSKNTLY LQMSSLRAED TAVYHCARGS 120
SGHYEAFDIW GQGLTVTVSS GGGSGGGGSG GGGGQSQVLT QPPSASGTPG QRVITISCSGS 180
SSNIGNWYVS WYQQLPGTAP KLLIYDNMKR PSGVPDRFSG SKSGTSASLA ISGLQSEDEA 240
DYYCGTWDSS LSAPWVFGGG TKLTVLGGGG SEVQLVESGG GLVQPGGSLR LSCAASGVTF 300
NYYGMSWIRQ APGKGLEWVA SITRSGGRIY YPDSVKGRFT ISRENTQKTL YLQMNSLRAE 360
DTAVYYCTLD GRDGWVAYWG QGTLVTVSSG GGGSGGGGSG GGGSNFMLTQ PSSVSGVPGQ 420
RVTISCTGNT GNIGSNYVNW YQQLPGTAPK LLIYRDDKRP SGVPDRFSGS KSGTSASLAI 480
TGFQAEDEAD YYCQSYSSGF IFGGGTKLTV LAAEPKSSDK THTCPPCPAP EAAGGPSVFL 540
FPPKPKDTLM ISRTPEVTCV VVSVSHEDPE VKFNWYVDGV EVHNAKT KPR EEQYNSTYRV 600
VSVLTVLHQD WLNKEYKCK VSNKALPAPI EKTISKAKGQ PREPQVYVYP PSRDELTKNQ 660
VSLTCLVKGF YPSDIAVEWE SNGQPENNYK TTPPVLDSDG SFALVSKLTV DKSRWQQGNV 720
FSCSVMHEAL HNHYTQKSL LSPGK 745

SEQ ID NO: 95      moltype = AA length = 745
FEATURE           Location/Qualifiers
REGION            1..745
                  note = Full-length amino acid sequence of NYF-0060-HC2
source            1..745
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 95
MKHLWFFLLL VAAPRWVLSG EVQLVESGGG LVQPGGSLRL SCAASGFSIR SYDIHWVRQA 60
PGKGLEWVAT ISYDGSQKYY ADSVKGRFTI SRDESNTLY LQMNSLRAED TAVYHCARGS 120
SGHYEAFDIW GQGLTVTVSS GGGSGGGGSG GGGGQSQVLT QPPSVSAAPG QKVTISCSGS 180
SSNIGNWYVS WYQQLPGTAP KLLIYDNMKR PSGIPDRFSG SKSGTSATLG ITGLQTGDEA 240
DYYCGTWDSS LSAPWVFGGG TKVTVLGGGG SEVQLVESGG GLVQPGGSLR LSCAASGVTF 300
NYYGMSWIRQ APGKGLEWVA SITRSGGRIY YPDSVKGRFT ISRENTQKTL YLQMNSLRAE 360
DTAVYYCTLD GRDGWVAYWG QGTLVTVSSG GGGSGGGGSG GGGSNFMLTQ PSSVSGVPGQ 420
RVTISCTGNT GNIGSNYVNW YQQLPGTAPK LLIYRDDKRP SGVPDRFSGS KSGTSASLAI 480
TGFQAEDEAD YYCQSYSSGF IFGGGTKLTV LAAEPKSSDK THTCPPCPAP EAAGGPSVFL 540
FPPKPKDTLM ISRTPEVTCV VVSVSHEDPE VKFNWYVDGV EVHNAKT KPR EEQYNSTYRV 600
VSVLTVLHQD WLNKEYKCK VSNKALPAPI EKTISKAKGQ PREPQVYVYP PSRDELTKNQ 660
VSLTCLVKGF YPSDIAVEWE SNGQPENNYK TTPPVLDSDG SFALVSKLTV DKSRWQQGNV 720
FSCSVMHEAL HNHYTQKSL LSPGK 745

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SEQ ID NO: 96 moltype = AA length = 745
 FEATURE Location/Qualifiers
 REGION 1..745
 note = Full-length amino acid sequence of NYF-0061-HC2
 source 1..745
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 96
 MKHLWFFLLL VAAPRWVLSG EVQLVESGGG LVQPGGSLRL SCAASGFSIR SYDIHWVRQA 60
 PGKGLEWVAT ISYDGSQKYY ADSVKGRFTI SRDESKNTLY LQMNSLRAED TAVYHCARGS 120
 SGHYEAFDIW GQGLLVTVSS GGGGSGGGGS GGGGQSQSVLT QPPSASGTPG QRVTIISCSGS 180
 SSNIGNWVYS WYQQLPGTAP KLLIYDNNKR PSGVPDRFSG SKSGTSASLA ISGLQSEDEA 240
 DYYCGTWDS LSAFWVFGG TKLTVLGGGG SEVQLVESGG GLVQPGGSLR LSCAASGVTF 300
 NYYGMSWIRQ APGKGLEWVA SITRSGGRIY YPDSVKGRFT ISRENTQKTL YLQMNSLRAE 360
 DTAVYYCTLD GRDGVVAYWG QGTLVTVSSG GGGSGGGGSG GGSNFMILTQ PSSVSGVPGQ 420
 RVTISCTGNT GNIGSNYVNW YQQLPGTAPK LLIYRDDKRP SGVPDRFSGS KSGTSASLAI 480
 TGFQAEDEAD YYCQSYSSGF IFGGGTKLTV LAAEPKSSDK THTCPPCPAP EAAGGPSVFL 540
 FPPKPKDTLM ISRTPEVTCV VVSVSHEDPE VKFNWYVDGV EVHNAKTKPR EEQYNSTYRV 600
 VSVLTVLHQD WLNGKEYKCK VSNKALPAPI EKTISKAKGQ PREPQVYVYP PSRDELTKNQ 660
 VSLTCLVKGF YPSDIAVEWE SNGQPENNYK TTPVPLDSGD SFALVSKLTV DKSRWQQGNV 720
 FSCVMHEAL HNHYTQKSL SLPKGK 745

SEQ ID NO: 97 moltype = DNA length = 1404
 FEATURE Location/Qualifiers
 misc_feature 1..1404
 note = Full-length nucleotide sequence of
 NYA-0001-Fab-HC1-k delete
 source 1..1404
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 97
 atgaaacacc tgtggttctt cctcctgctg gtggcagctc ccagatgggt gctgagcgag 60
 gtgcagctgg tggagctctgg gggaggcgtg gtccagcctg ggaggtccct gacaccctcc 120
 tgttcagcgt ctggattctc catccgtagt tatgatatac actgggtccg ccaggtctca 180
 ggcaaggggc tagagtggtt ggcacatata tcatatgatg gaagtacaga gtacttcgca 240
 gactccgtga agggccgatt taccatcttc agagacgaat cggagaacat ggtgtatctg 300
 caaatgagcg gccctgagagt tgaggacacg gctgtttatc actgtgcgag agggagtagt 360
 ggtcattatg aggtctttga tatctggggc caagggacaa tggtcaccgt ctcttcagcc 420
 tccaccaagg gcccaagcgt ctccccctg gcaccctcct ccaagagcac ctctggcggc 480
 acagccgccc tgggtctgct ggtcaaggac tacttccccg aaccctgac cgtgagctgg 540
 aactcaggcg cctcgaccag cggcgtgcac accttccccg ctgtcctgca gtccctagga 600
 ctctactccc tcagcagcgt ggtgaccgtg cctccagca gcttgggcac ccagacctac 660
 atctgcaacg tgaatcacaa gccccagcaac accaaggtgg acaagagagt tgagcccaaa 720
 tcttgtgaca aaactcacac atgcccaccc tgcccagcac ctgaagccgc agggggagcc 780
 tcagtcttcc tcttcccccc aaaacccaag gacacctca tgatctcccg gacctctgag 840
 gtcacatgcg tgggtggtgc agtgagccac gaagacctg aggtcaagt caactggtac 900
 gtggacggcg tggaggtgca taatgccaa gcaaaagccc gggaggagca gtacaacagc 960
 acgtaccggg tggtcagcgt cctcaccgtc ctgaccagg actggctgaa tggcaaggag 1020
 tacaagtcca aggtctccaa caaagccctc ccagccccc tcgagaaaac catctccaaa 1080
 gccaaaggcc agccccggga accacaggtg tacgtgtgct ccccatcccc ggacgagctg 1140
 accaagaacc aggtcagcgt gctgtgcctg gtcaaaggct tctatccag cgacatcgcc 1200
 gtggagtggt agagcaatgg ccagcccgag aacaactacc tgacctggcc tcccgtgctg 1260
 gactccgacg gctcctctct cctctacagc aagctcaccg tggacaagag caggtggcag 1320
 cagggcaacg tcttctcatg ctccgtgatg catgaggctc tgcacaacca ctacaccag 1380
 aagagcctct cctctgtctc cggc 1404

SEQ ID NO: 98 moltype = DNA length = 711
 FEATURE Location/Qualifiers
 misc_feature 1..711
 note = Full-length nucleotide sequence of NYA-0001-LC
 source 1..711
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 98
 atggtgctgc agaccaggt gtctcatctc ctgctgctgt ggatctccgg cgcgtacggc 60
 cagctctgtg tgacgcagcc gccctcagtg tctgcccggc caggacagaa ggtcaccatc 120
 tctgctctgc gaagcagctc caacatttgg aataattatg tatcctggta ccagcagctc 180
 ccaggacagc cccccaaact cctcatttat gacaataata agcgaccctc agggattcct 240
 gaccgattct ctggctccaa gtctggcagc tcagccaccc tgggcatcac cggactccag 300
 actggggagc agggccgata ttactcgga acatgggata gcagcctgag tgccccttgg 360
 gtgttcggcg gaggagccaa ggtcacgctc ctaggccagc ctaaggctgc ccctagcgtg 420
 accctgttcc ctcttccag caggagctt caagctaaca aggccacctt ggtgtgtctt 480
 atctctgact tctacctgag cgctgtgacc gtggcctgga aggtgacag ctcccctgtg 540
 aaggccggag tggagaccac cacacctagc aagcagctca acaacaagta cgctgccagc 600
 tcctacttga gccttaccct tgagcagtggt aagtctcaca gaagctactc ctgtcaagtg 660

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acccacgagg gcagcaccgt ggagaagacc gtggtccta ccgagtgttc c 711

SEQ ID NO: 99      moltype = AA length = 468
FEATURE           Location/Qualifiers
REGION            1..468
                  note = Full-length amino acid sequence of
                  NYA-0001-Fab-HC1-k delete
source            1..468
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 99
MKHLWFFLLL VAAPRWLSE VQLVESGGGV VQPGKSLTSP CSASGFSIRS YDIHWVRQAP 60
GKGLEWVATI SYDGSQKYFA DSVKGRFTIF RDESENMVYL QMSGLRVEDT AVYHCARGSS 120
GHYEAFDIWG QGTMVTVSSA STKGPSVFPL APSSKSTSGG TAALGCLVKD YFPEPVTVSW 180
NSGALTSGVH TFPVAVLQSSG LYSLSVTV PSSSLGTQTY ICNVNHKPSN TKVDKRVPEK 240
SCDKTHTCPP CPAPEAAGGP SVFLPPPKPK DTLMSRTPE VTCVVVSVSH EDPEVKFNWY 300
VDGVEVHNAK TKPREEQYNS TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK 360
AKGQPREPQV YVLPFSRDEL TKNQVSLCL VKGFYPSDIA VEWESNGQPE NNYLTWPPVL 420
DSGDSFFLYS KLTVDKSRWQ QGNVFSCSVM HEALHNHYTQ KSLSLSPG 468

SEQ ID NO: 100     moltype = AA length = 237
FEATURE           Location/Qualifiers
REGION            1..237
                  note = Full-length amino acid sequence of NYA-0001-LC
source            1..237
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 100
MVLQTQVFIS LLLWISGAYG QSVLTQPPSV SAAPGQKVTI SCSGSSSNIG NNYVSWYQQL 60
PGTAPKLLIY DNNKRPSGIP DRFSGSKSGT SATLGITGLQ TGDEADYYCG TWDSSLSAPV 120
VFGGGTKVTV LGQKAAPSV TLFPPSSEEL QANKATLVCL ISDFYPGAVT VANKADSSPV 180
KAGVETTTPS KQSNKNYAAS SYLSLTPEQW KSHRSYSCQV THEGSTVEKT VAPTECS 237

SEQ ID NO: 101     moltype = DNA length = 1404
FEATURE           Location/Qualifiers
misc_feature      1..1404
                  note = Full-length nucleotide sequence of
                  NYA-1143-Fab-HC1-k delete
source            1..1404
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 101
atgaaacacc tgtggttctt cctcctgctg gtggcagctc ccagatgggt gctgagcgag 60
gtgcagctgg tggagctctg gggaggcgtg gtccagcctg ggaggtccct gacacccctc 120
tgttcagcgt ctggattctc catccgtagc tatgatatac actgggtccg cctggctcca 180
ggcaaggggc tagagtgggt gccactata tcatatgatg gaagtacaga gtacttcgca 240
gactccgtga agggccgatt taccatcttc agagacgaat cggagaacat ggtgtatctg 300
caaatgagcg gcctgagagt tgaggacacg gctgtttatc actgtgcgag agggagtagt 360
ggtcattatg aggttttga tatctggggc caaggacaaa tggtcacctg ctcttcagcc 420
tcaccaagg gcccaagcgt ctcccccctg gcacctcct ccaagagcac ctctggcgcc 480
acagccgccc tgggctgcct ggtcaaggac tacttccccg aaccctgac cgtgagctgg 540
aactcaggcg cctgaccagc cggcgtgcac accttccccg ctgtcctgca gtccctcagg 600
ctctactccc tcagcagcgt ggtgaccgtg cctccacgac gcttgggcac ccagacctac 660
atctgcaacg tgaatcacaa gccacgcaac accaagggtg acaagagagt tgagcccaaa 720
tcttgtgaca aaactcacac atgccaccct tgcccagcac ctgaagccgc agggggaccc 780
tcagttcttc tcttcccccc aaaacccaag gacacctca tgatctcccc gacctctgag 840
gtcacatgcg tgggtggtgc agtgagccac gaagacctg aggtcaagtt caactggtac 900
gtggacggcg tggaggtgca taatgccaag acaaagcccc gggaggagca gtacaacagc 960
acgtaccggg tggtcagcgt cctcaccgtc ctgcaccagg actggctgaa tggcaaggag 1020
tacaagtgca aggtctccaa caaagccctc ccagccccc tgcagaaaaa catctccaaa 1080
gccaaaggcc acccccggga accacaggtg tacgtgctgc ccccatcccc ggacgagctg 1140
accaagaacc aggtcagcct gctgtgctcg gtcaaaggct tctatcccag cgacatcgcc 1200
gtggagtggt agagcaatgg ccagcccgag aacaactacc tgacctggcc tcccgtgctg 1260
gactccgacg gctcctctt cctctacagc aagctcaccg tggacaagag caggtggcag 1320
cagggaacg tcttctcatg ctccgtgatg catgaggctc tgcacaacca ctacacccag 1380
aagagcctct cctgtctctc cggc 1404

SEQ ID NO: 102     moltype = DNA length = 711
FEATURE           Location/Qualifiers
misc_feature      1..711
                  note = Full-length nucleotide sequence of NYA-1143-LC
source            1..711
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 102
atggtgctgc agaccaggt gttcatctcc ctgctgctgt ggatctccg cgctacggc 60

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cagctctgtgt tgaocgcagcc gccctcagtg tctgcggccc caggacagaa ggtcaccatc 120
tcctgctctg gaagcagctc caacattggg aattgggatg tatcctggta ccagcagctc 180
ccaggaaacag ccccaaaact cctcatthtat gacaataata agcgaccctc agggattcct 240
gaccgattct ctggctccaa gtctggcacg tcagccaccc tgggcacac cggactccag 300
actggggacg aggcogatta ttactgcgga acatgggata gcagcctgag tgccccttgg 360
gtgttcggcg gaggagccaa ggccaccgtc ctaggccagc ctaaggctgc ccctagcgtg 420
accctgttcc ctcttccag cgaggagctt caagctaaca agggccacct ggtgtgtctt 480
atctctgact tctaccctgg cgctgtgacc gtggcctgga aggctgacag ctcccctgtg 540
aaggccggag tggagaccac cacacctagc aagcagtcta acaacaagta cgctgccagc 600
tcctacctga gccttaccct tgagcagtggt aagtctcaca gaagctactc ctgtcaagtg 660
acccacgagg gcagcaccgt ggagaagacc gtggctccta ccgagtgttc c 711

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SEQ ID NO: 103      moltype = DNA length = 1476
FEATURE            Location/Qualifiers
misc_feature        1..1476
                    note = Full-length nucleotide sequence of C3E-7085-HC2-k
                    delete
source              1..1476
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 103
atgaaacacc tgtggtttctt cctcctgctg gtggcagctc ccagatgggt gctgagcgga 60
gaagtgcagc tggtggaatc cggggggggc ctgggtgcagc ctgggggggag cctgagactg 120
agttgtgcgc cctctggggt gacatttaac tactatggca tgtcttggat ccgccaggca 180
cctggaaagg gcctggagtg ggtggccagc atcactaggt ccggcgggag aatctactat 240
cccgacagcg tcaagggcag gttcaccaatt tcccgcgaga acacacagaa aactctgtac 300
ctgcagatga atagcctgag agccgaagat acagctgtgt actattgcac tctggacggc 360
agggatgggt gggtcgccta ttggggggcag ggaacccttg tgacagtcag ctccggagga 420
ggagatctg gcggaggagg cagtggggga ggcgggtcaa actttatgct cactcagccg 480
tcctctgttt ctggcgctac ttggccaacgg gtgaccatta gctgtacggg taataccggg 540
aatatcgggt ctaactacgt gaactgggtat cagcagcttc cagggacagc tcccaagttg 600
ctgatctatc gcgacgacaa aagaccctca ggggtccctg accgatttag tggcagcaaa 660
agcggctact ccgcttccct ggcgataaac ggctttcagg ccgaagatga ggcagactac 720
tattgccagt catattccag cgcttccatc ttcggaggcg gaactaagct gacagtgttg 780
gcagccgagc ccaaatcttc tgacaaaact cacacatgcc caccctgccc agcacctgaa 840
gccgagggg gacctcagct ctctctcttc ccccaaaaac ccaaggacac cctcatgatc 900
tcccggaccc ctgaggtcac atgcgtgggt gtgtcagtg gccacgaaga cctgaggttc 960
aagtccaact ggtacgtgga cggcgtggag gtgcataatg ccaagacaaa gccccgggag 1020
gagcagtaca acagcacgta ccgggtgggt agcgtcctca ccgtcctgca ccaggactgg 1080
ctgaatggca aggagtacaa gtgcaagggt tccaacaaag ccctccagc ccccatcgag 1140
aaaaccatct ccaaaagccaa aggccagccc cgggaaccac aggtgtacgt gtacccccca 1200
tcccgggacg agctgaccaa gaaccagggt agcctgacct gcctggtcaa aggtctctat 1260
cccagcgaca tcgccgtgga gtgggagagc aatggccagc ccgagaacaa ctacaagacc 1320
acccctcccc tgctggactc cgacggctcc ttcgccctcg tgagcaagct caccgtggac 1380
aagagcagggt ggcagcaggg caacgtcttc tcatgctccg tgatgcatga ggctctgcac 1440
aaccactaca cccagaagag cctctccctg tctccc 1476

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SEQ ID NO: 104      moltype = AA length = 468
FEATURE            Location/Qualifiers
REGION            1..468
                    note = Full-length amino acid sequence of
                    NYA-1143-Fab-HC1-k delete
source            1..468
                    mol_type = protein
                    organism = synthetic construct

```

```

SEQUENCE: 104
MKHLWFFLLL VAAPRWLSE VQLVESGGGV VQPGKSLTPS CSASGFSIRS YDIHWVRLAP 60
GKLEWVATI SYDGSQKYFA DSVKGRFTIF RDESENMVYL QMSGLRVEDT AVYHCARGSS 120
GHYEAFDIWG QGTMVTSSA STKGPSVFPL APSSKSTSGG TAALGCLVKD YFPEPTVSW 180
NSGALTSGVH TFPVAVLQSSG LYSLSVTV PSSSLGTQTY ICNVNHKPSN TKVDKRVPEK 240
SCDKTHTCPP CPAPEAAGGP SVFLFPPKPK DTLMISRTPE VTCVVSFVSH EDPEVKFNWY 300
VDGVEVHNAK TKPREQYNS TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK 360
AKGQPREPQV YVLPSPDEL TKNQVSLCL VKGFYPSDIA VEWESNGQPE NNYLTWPPVL 420
DSDGSFFLYS KLTVDKSRWQ QGNVFSCSVM HEALHNHYTQ KSLSLSPG 468

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SEQ ID NO: 105      moltype = AA length = 237
FEATURE            Location/Qualifiers
REGION            1..237
                    note = Full-length amino acid sequence of NYA-1143-LC
source            1..237
                    mol_type = protein
                    organism = synthetic construct

```

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SEQUENCE: 105
MVLQTQVFIS LLLWISGAYG QSVLTQPPSV SAAPGQKVTI SCSGSSSNIG NWYVSWYQQL 60
PGTAPKLLIY DNNKRPSGIP DRFSGSKSGT SATLGITGLQ TGDEADYYCG TWDSSLSAPW 120
VFGGGTKVTV LGQPKAAPSV TLPPPSSEEL QANKATLVCL ISDFYPGAVT VAWKADSSPV 180

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KAGVETTTTPS KQSNKYAAS SYLSLTPEQW KSHRSYSCQV THEGSTVEKT VAPTECS 237

SEQ ID NO: 106 moltype = AA length = 493
 FEATURE Location/Qualifiers
 REGION 1..493
 note = Full-length amino acid sequence of C3E-7085-HC2-k
 delete
 source 1..493
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 106
 MKHLWFFLLL VAAPRWVLSG EVQLVESGGG LVQPGGSLRL SCAASGVTFN YYGMSWIRQA 60
 PGKGLEWVAS ITRSGGRIYY PDSVKGRFTI SRENTQKTLY LQMNSLRAED TAVYYCTLDG 120
 RDGWVAYWQG GTLTVVSSGG GSGGGGGSGG GGSNFMILTQP SSVSGVPGQR VTISCTGNTG 180
 NIGSNYVNWY QQLPGTAPKL LIYRDDKRPS GVPDRFSGSK SGTSASLAIT GFQAEDEADY 240
 YQSYSSSGFI FGGGTLKTLV AAEPKSSDKT HTCPCPCAPE AAGGPSVFLF PPKPKDTLMI 300
 SRTPEVTCVV VSVSHEDPEV KFNWYVDGVE VHNAKTKPRE EQYNSTYRVV SVLTVLHQDW 360
 LNGKEYKCKV SNKALPAPIE KTIKAKGQP REPQVYVYP SRDELTKNQV SLTCLVKGFY 420
 PSDIAVEWES NGQPENNYKT TPPVLDSDGS FALVSKLTVD KSRWQQGNVF SCSVMHEALH 480
 NHYTQKSLSL SPG 493

SEQ ID NO: 107 moltype = DNA length = 1497
 FEATURE Location/Qualifiers
 misc_feature 1..1497
 note = Full-length nucleotide sequence of NYA-1143-HC1-k
 delete
 source 1..1497
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 107
 atgaaacacc tgtggttctt cctcctgctg gtggcagctc ccagatgggt gctgagcgga 60
 gaggtgcagc tgggtggagc tgggggaggc gtgggccagc ctgggaggtc cctgacaccc 120
 tcctgttcag cgtctggatt ctccatccgt agctatgata tacactgggt ccgcctggct 180
 ccaggcaagg ggctagagtg ggtggccact atatcatatg atggaagtca gaagtacttc 240
 gcagactccg tgaagggccg atttaccatc ttcagagacg aatcggagaa catggtgtat 300
 ctgcaaatga gcgcctgag agttgaggac acggctgttt atcactgtgc gagagggagt 360
 agtggtcatt atgaggcttt tgatatctgg ggccaaggga caatggtcac cgtctcttca 420
 ggtggaggcg gttcaggcgg aggtggcagc ggcgggtggcg ggaagtcagc tgtgttgacg 480
 cagccgccct cagtgtctgc ggccccagga cagaaggcca ccatctcctg ctctggaagc 540
 agctccaaca ttgggaattg gtatgtatcc tggtaaccag agctccagg aacagccccc 600
 aaactctca tttatgacaa taataagcga cctcaggga ttctgaccg attctctggc 660
 tccaagtctg gcaagtcagc caccctgggc atcaccggac tccagactgg ggacgaggcc 720
 gattattact cgggaacatg ggaagcagc ctgagtgccc cttgggtgtt cggcggaggg 780
 accaaggcca ccgtcctagc agccgagccc aaatcttctg acaaaaactc cacatgccc 840
 ccctgccag cacctgaagc cgcaggggga cctcagctc tcctcttccc cccaaaaccc 900
 aaggacaccc tcatgatctc ccggaccctc gaggtcacat gcgtgggtgt gtcagtgcgc 960
 cacgaagacc ctgaggtcaa gttcaactgg tacgtggacg gcgtggaggt gcataatgcc 1020
 aagacaaaag cccgggagga gcagtacaac agcacgtacc ggggtggtcag cgtcctcacc 1080
 gtctgtcacc ccattggctg aatggccaag gagtacaagt gcaaggtctc caacaaagcc 1140
 ctcccagccc ccactcgagaa aaccatctcc aaagccaaag gccagccccg ggaaccacag 1200
 gtgtacgtgc tgccccatc ccgggacgag ctgaccaaga accaggtcag cctgctgtgc 1260
 ctggtcaaaag gcttctatcc cagcgacatc gccgtggagt gggagagcaa tggccagccc 1320
 gagaacaact acctgacctg gcctccctgt ctggactccg acggctcctt ctctctctac 1380
 agcaagctca ccgtggacaa gagcagggtg cagcagggca acgtctctc atgctccgtg 1440
 atgcatgagg ctctgcacaa ccactacacc cagaagagcc tctcctgtc tcccgcc 1497

SEQ ID NO: 108 moltype = AA length = 499
 FEATURE Location/Qualifiers
 REGION 1..499
 note = Full-length amino acid sequence of NYA-1143-HC1-k
 delete
 source 1..499
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 108
 MKHLWFFLLL VAAPRWVLSG EVQLVESGGG VVQPGRSLTP SCSASGFSIR SYDIHWVRLA 60
 PGKGLEWVAT ISYDGSQKYP ADSVKGRFTI FRDESENMYV LQMSGLRVED TAVYHCARGS 120
 SGHYEAFDIW GQGTMTVTVSS GGGGSGGGGS GGGGQSFLT QPPSVSAAPG QKVTISCSGS 180
 SSNIGNWYVS WYQQLPGTAP KLLIYDNNKR PSGIPDRFSG SKSGTSATLG ITGLQTGDEA 240
 DYYCGTWDSS LSAPWVFGG TKVTVLAAEP KSSDKTHTCP PCPAPEAAGG PSVFLPPPKP 300
 KDTLMSRTP EVTCVVSVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYN STYRVVSLT 360
 VLHQDWLNGK EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYVLPSPRDE LTKNQVSLLC 420
 LVKGFYPSDI AVEWESNGQP ENNYLTWPPV LDSDGSFFLY SKLTVDKSRW QQGNVFSCSV 480
 MHEALHNHYT QKSLSLSPG 499

SEQ ID NO: 109 moltype = DNA length = 2142

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FEATURE                               Location/Qualifiers
misc_feature                           1..2142
                                         note = Full-length nucleotide sequence of
                                         C3E-7085-NYA-1154-Fab-HC2-k delete

source                                1..2142
                                         mol_type = other DNA
                                         organism = synthetic construct

SEQUENCE: 109
atgaaacacc tgtggttctt cctcctgctg gtggcagctc ccagatgggt gctgagcgga 60
gaagtgcagc tgggtggaatc cggggggggc ctggtgcagc ctggggggag cctgagactg 120
agttgtgccg cctctggggg gacatttaac tactatggca tgtcttggat ccgccaggca 180
cctggaaagg gcctggagtg ggtggccagc atcactaggt ccggcgggcg aatctactat 240
cccgacagcg tcaagggcag gtccacaatt tcccgcgaga acacacagaa aactctgtac 300
ctgcagatga atagcctgag agccgaagat acagctgtgt actattgcac tctggacggc 360
agggatgggt gggctgccta ttgggggcag ggaaccctgg tgacagtcag ctccggagga 420
ggaggatctg gcggagggag cagtggggga ggccgggtcaa actttatgct cactcagccg 480
tcctctgttt ctggcgtacc tggccaacgg gtgaccatta gctgtacggg taataccggg 540
aatatcgggt ctaactacgt gaactggtat cagcagcttc cagggacagc tcccaggtg 600
ctgatctatc gcgacgacaa aagaccctca ggggtccctg accgatttag tggcagcaaa 660
agcgttactt ccgcttccct ggcgataacc ggctttcagg ccgaagatga ggcagactac 720
tattgccagt catattccag cggcttccatc ttcggaggcg gaactaagct gacagtgttg 780
gggggagggc gttcagaggt gcagctggtg gagtctgggg gaggcgtggt ccagcctggg 840
aggtccctga caccctcctg ttcagcgtct ggattctcca tccgtagtta tgatatgcac 900
tgggtccgcc aggtccagg caaggggcta gagtgggttg ccactatata atatgatgga 960
agtcagaagt acttcgcaga atccgtgaag ggccgattta ccattctcag agacgaatcg 1020
gagaacatgg tgtatctgca aatgagcggc ctgagagttg aggcacagcg cgtttatcac 1080
tgtgcgagag ggagtagtaa tcattatgag gcttttgata tctggggcca agggacaatg 1140
gtcaccgctc cttcagcctc caccaagggc ccaagcgtct tcccctggc accctcctcc 1200
aagagcacct ctggcggcac agccgcctcg ggctgccttg tcaaggacta cttccccgaa 1260
cccgtgacgg tgagctggaa ctacggcgcc ctgaccagcg gcgtgcacac cttccccgct 1320
gtcctgcagt cctcaggact ctactccctc agcagcgtgg tgaccgtgcc ctccagcagc 1380
ttgggcaccc agacctacat ctgcaacgtg aatcacaaag ccagcaaac caaggtggac 1440
aagagagttg agcccaaatc ttgtgacaaa actcacacat gccacacctg cccagcacct 1500
gaagccgcag ggggaccctc agtcttctct tcccccccaa aacccaagga caccctcatg 1560
atctccccga cccctgaggt cacatgcgtg gtggtgtcag tgagccacga agaccctgag 1620
gtcaagttca actggtacgt ggcgcggctg gaggtgcata atgccaagac aaagccccgg 1680
gaggagcagt acaacagcac gtaccgggtg gtcagcgtcc tcacgcgtct gcccaggac 1740
tggctgaatg gcaaggagta caagtgcagg gtctccaaca aagccctccc agcccccatc 1800
gagaaaaacca tctccaaagc caaaggccag ccccgggaac cacaggtgta cgtgtacccc 1860
ccatccccgg acagagctgac caagaaccag gtcagcctga cctgcctggt caaaggcttc 1920
tatccccagc acatgcacct ggagtgggag agcaatggcc agcccgagaa caactacaag 1980
accacccctc ccgtgctgga ctccgacggc tccttcgccc tcgtgagcaa gctcacccgtg 2040
gacaagagca ggtggcagca gggcaacgtc ttctcatgct ccgtgatgca tgaggctctg 2100
cacaccactt aaccccagaa gagcctctcc ctgtctcccg gc 2142

SEQ ID NO: 110                       moltype = DNA length = 711
FEATURE                               Location/Qualifiers
misc_feature                           1..711
                                         note = Full-length nucleotide sequence of NYA-1154-LC

source                                1..711
                                         mol_type = other DNA
                                         organism = synthetic construct

SEQUENCE: 110
atggtgctgc agaccaggt gtctcatctc ctgctgctgt ggatctccgg cgcgtacggc 60
cagtcgtgtg tgacgcagcc gccctcagtg tctgcggccc caggacagaa ggtcaccatc 120
tcctgctctg gaagcagctc caacattggg aataattatg tatcctggta ccagcggctc 180
ccaggaaacg cccccaaact cctcatttat gacaataata agcgaccctc agggattcct 240
gaccgattct ctggtctcaa gtctggcacg tcagccaccc tgggcatcac cggactccag 300
actggggacg aggcgatta ctactcggga gcatgggata gcagcctgag tgccccttgg 360
gtgttcggcg gagggaacca ggctaccgtc ctaggccagc ctaaggctgc cctagcgtg 420
accctgttcc ctctctccag cgaggagctt caagctaaca agccaccctt ggtgtgtctt 480
atctctgact tctaccctgg cgctgtgacc gtggcctgga aggctgacag ctcccctgtg 540
aagggccggg tggagaccac cacacctagc aagcagctca acaacaagta cgctgccagc 600
tcctacctga gccttaccct tgagcagtgg aagcttcaca gaagctactc ctgtcaagtg 660
accacagagg gcagcacctg ggagaagacc gtggctccta ccgagtgttc c 711

SEQ ID NO: 111                       moltype = DNA length = 735
FEATURE                               Location/Qualifiers
misc_feature                           1..735
                                         note = Full-length nucleotide sequence of OAA-HC1-k delete

source                                1..735
                                         mol_type = other DNA
                                         organism = synthetic construct

SEQUENCE: 111
atgaaacacc tgtggttctt cctcctgctg gtggcagctc ccagatgggt gctgagcgac 60
aaaactcaca catgccacc ctgccagca cctgaagccg cagggggacc ctcaagtctc 120

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ctcttccccc caaaacccaa ggacaccctc atgatctccc ggacccctga ggtcacatgc 180
gtgggtgggtg cagtgagcca cgaagaccct gaggtcaagt tcaactggta cgtggacggc 240
gtggaggtgc ataatgccc gacaaagccc cgggaggagc agtacaacag cacgtaccgg 300
gtggtcacgc tctcaccgt cctgcaccag gactggctga atggcaagga gtacaagtgc 360
aaggtctcca acaaaagcct ccagccccc atcgagaaaa ccatctccaa agccaaaggc 420
cagcccccgg aaccacaggt gtacgtgctg cccccatccc gggacgagct gaccaagaac 480
caggtcagcc tgctgtgctt ggctcaaaggc ttctatccca gcgacatcgc cgtggagtgg 540
gagagcaatg gccagcccca gaacaactac ctgacctggc ctcccgtgct ggactccgac 600
ggctccttct tcctctacag caagctcacc gtggacaaga gcaggtggca gcagggcaac 660
gtcttctcat gctccgtgat gcatgaggct ctgcacaacc actacaccca gaagagcctc 720
tccctgtctc ccggc 735

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SEQ ID NO: 112      moltype = AA length = 714
FEATURE            Location/Qualifiers
REGION             1..714
                   note = Full-length amino acid sequence of
                   C3E-7085-NYA-1154-Fab-HC2-k delete
source             1..714
                   mol_type = protein
                   organism = synthetic construct

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SEQUENCE: 112
MKHLWFFLLL VAAPRWVLSG EVQLVESGGG LVQPGGSLRL SCAASGVTFN YYGMSWIRQA 60
PGKGLEWVAS ITRSGGRIYY PDSVKGRFTI SRENTQKTLY LQMNSLRAED TAVYYCTLDG 120
RDGWVAYWQ GTLVTVSSGG GSGGGGSGG GGSNFMLTQP SSVSGVPGQR VTISCTGNTG 180
NIGSNVWNWY QQLPGTAPKL LIYRDDKRPS GVPDRFSGSK SGTASLAIT GFQAEDEADY 240
YCQSYSSGFI FGGGKLTIVL GGGGSEVQLV ESGGGVVQPG RSLTPSCSAS GFSIRSYDMH 300
WVRQAPGKGL EWVATISYDG SQKYFADSVK GRFTIFRDES ENMVYLQMSG LRVEDTAVYH 360
CARGSSNHYE AFDIWGGTGM VTVSSASTKG PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE 420
PVTVSWNSGA LTSGVHTFPA VLQSSGLYSL SSVVTVPPSS LGTQTYICNV NHKPSNTKVD 480
KRVEPKSCDK THTCPPCPAP EAAGGSPSVFL FPPKPKDTLM ISRTPEVTCV VVSVSHEDPE 540
VKFNWYVDGV EVHNAKTKPR EEQYNSTYRV VSVLTVLHQD WLNKEYKCK VSNKALPAPI 600
EKTISKAKGQ PREPQVYVYP PSRDELTKNQ VSLTCLVKGF YPSDIAVEWE SNGQPENNYK 660
TPPVLDSGD SFALVSKLTV DKSRWQQGNV FSCSVMEHAL HNHYTQKSLS LSPG 714

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SEQ ID NO: 113      moltype = AA length = 237
FEATURE            Location/Qualifiers
REGION             1..237
                   note = Full-length amino acid sequence of NYA-1154-LC
source             1..237
                   mol_type = protein
                   organism = synthetic construct

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SEQUENCE: 113
MVLQTQVFIS LLLWISGAYG QSVLTQPPSV SAAPGQKVTI SCSGSSSNIG NNVVSWYQRL 60
PGTAPKLLIY DNNKRPSPGIP DRFSGSKSGT SATLGITGLQ TGDEADYYCG AWDSSLSAPW 120
VFGGGTKVTV LGQPKAAPSV TLFPPSSEEL QANKATLVCL ISDFYPGAVT VANKADSSPV 180
KAGVETTTPS KQSNKNYAAS SYLSLTPEQW KSHRSYSCQV THEGSTVEKT VAPTECS 237

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SEQ ID NO: 114      moltype = AA length = 245
FEATURE            Location/Qualifiers
REGION             1..245
                   note = Full-length amino acid sequence of OAA-HC1-k delete
source             1..245
                   mol_type = protein
                   organism = synthetic construct

```

```

SEQUENCE: 114
MKHLWFFLLL VAAPRWVLSG KTHTCPPCPA PEAAGGPSVF LFPPKPKDTL MISRTPEVTC 60
VVVSVSHEDP EVKFNWYVDG VEVHNAKTKP REEQYNSTYR VVSVLTVLHQ DWLNKEYKCK 120
KVSNAKALPAP IEKTISKAKG QPREPQVYVL PPSRDELTKN QVSLCLVKG FYPSDIAVEW 180
ESNGQPENNY LTWPPVLDSG GSFFLYSKLT VDKSRWQQGN VFSCSVMEHA LHNHYTQKSL 240
SLSPG 245

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SEQ ID NO: 115      moltype = DNA length = 2232
FEATURE            Location/Qualifiers
misc_feature        1..2232
                   note = Full-length nucleotide sequence of NYF-0010-HC2-k
                   delete
source             1..2232
                   mol_type = other DNA
                   organism = synthetic construct

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SEQUENCE: 115
atgaaacacc tgtggttctt cctcctgctg gtggcagctc ccagatgggt gctgagcgga 60
gaggtgcagc tggtaggagc tgggggaggc gtggtcagc ctgggaggtc cctgacaccc 120
tcctgttcag cgtctgatt ctccatccgt agttatgata tgcactgggt ccgccaggct 180
ccaggcaagg ggctagagtg ggtggccact atatcatatg atggaagtca gaagtacttc 240
gcagactccg tgaagggccg atttaccatc ttcagagacg aatcgagaaa catgggtgat 300
ctgcaaatga gcggcctgag agttgaggac acggccggtt atcactgtgc gagagggagt 360

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agtaatcatt atgagggcttt tgatatctgg ggccaaggga caatgggtcac cgtctcttca 420
gggtggaggcg gttcaggcgagg aggtgggcagc ggcggtggcg ggagtcagtc tgtgttgacg 480
cagccgcccc cagtgctctgc ggccccaggga cagaagggtca ccatctctctg ctctggaagc 540
agctccaaca ttgggaataa ttatgtatcc tggtagccagc ggctcccagg aacagccccc 600
aaactcctca tttatgacaa taataagcga cctcaggga ttectgaccg attctctggc 660
tccaagtcctg gcaagtcagc caccctgggc atcaccggac tccagactgg ggacgaggcc 720
gattactact cgggagcatg gtagtagcagc ctgagtgccc cttgggtgtt cggcggaggcg 780
accaaggtca ccgtcctagg gggaggcggt tcagaagtgc agctgggtgga atccgggggg 840
ggcctgggtgc agcctggggg gagcctgaga ctgagttgtg ccgctctctg ggtgacattt 900
aactactatg gcatgtcttg gatccgccag gcacctggaa agggcctgga gtgggtggcc 960
agcatcacta ggtccggcgg gcgaatctac tatcccgaca gcgtcaaggg cagggttcaca 1020
atttcccgcg agaacacaca gaaaactctg tacctgcaga tgaatagcct gagagccgaa 1080
gatacagctg tgtactattg cactctggac ggcagggatg ggtgggtcgc ctattggggg 1140
cagggaaaccc tgggtgacagt cagctccgga ggaggaggat ctggcggagg aggcagtggg 1200
ggaggcgggt caaactttat gctcactcag ccgtcctctg tttctggcgt acctggccaa 1260
cgggtgacca ttagctgtac gggtaatacc ggaataatcg ggtctaacta cgtgaactgg 1320
tatcagcagc ttccagggac agctccaaag ttgctgatct atcgcgaca caaaagacc 1380
tcagggggtcc ctgaccgatt tagtggcagc aaaagcggtta cttccgcttc cctggcgata 1440
accggctttc aggcgaaga gtaggcagac tactattgcc agtcatattc cagcggcttc 1500
atcttcggag gcggaactaa gctgacagtg ttggcagccg agcccaaatc ttctgacaaa 1560
actcacacat gcccaccctg ccagcacct gaagcccgag ggggaccctc agtcttctc 1620
ttcccccaa aaccacaagg accctcatg atctcccgga cccctgaggt cacatgcgtg 1680
gtggtgtcag tgagccacga agacctgag gtcaagttca actggtacgt ggacggcgtg 1740
gaggtgcata atgccaagac aaagcccgcg gaggagcagt acaacagcac gtaccgggtg 1800
gtcagcgtcc tcaccgtcct gcaccaggac tggctgaatg gcaaggagta caagtgcag 1860
gtctccaaca aagccctccc agcccccac gagaaaaaca tctccaaagc caaaggccag 1920
ccccgggaac cacaggtgta cgtgtacccc ccatcccggg acgagctgac caagaaccag 1980
gtcagcctga cctgctgggt caaaggcttc tatccagcg acatcgccgt ggagtgggag 2040
agcaatggcc agcccgagaa caactacaag accaccctc ccgtgctgga ctccgacggc 2100
tccttcgccc tcgtgagcaa gctcacctg gacaagagca ggtggcagca gggcaacgtc 2160
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ctgtctcccg gc 2232

SEQ ID NO: 116      moltype = DNA length = 2232
FEATURE             Location/Qualifiers
misc_feature         1..2232
                     note = Full-length nucleotide sequence of NYF-0004-HC2-k
                     delete
source               1..2232
                     mol_type = other DNA
                     organism = synthetic construct

SEQUENCE: 116
atgaaacacc tgtggtttctt cctcctgctg gtggcagctc ccagatgggt gctgagcgga 60
gaagtgcagc tgggtggaatc cggggggggc ctgggtgcagc ctgggggggag cctgagactg 120
agttgtgccg cctctgggggt gacatttaac tactatggca tgtcttgat cccgccaggca 180
cctggaaagg gccctggagtg ggtggccagc atcactaggt ccggcggggc aatctactat 240
cccgacagcg tcaagggcag gttcacaaat tcccgcgaga acacacagaa aactctgtac 300
ctgcagatga atagcctgag agccgaagat acagctgtgt actattgcac tctggacggc 360
agggatgggt gggtcgccta ttggggggcag ggaaccctgg tgacagtcag ctccggagga 420
ggaggatctg gcggaggagg cagtggggga ggcgggtcaa actttatgct cactcagccg 480
tcctctgttt ctggcgtaac tggccaacgg gtgaccatta gctgtacggg taataccggg 540
aatatcgggt ctaactacgt gaactggat cagcagcttc cagggacagc tcccaagtgt 600
ctgatctatc gcgacgacaa aagaccctca ggggtccctg accgatttag tggcagcaaa 660
agcgtgactt ccgtctccct ggcgataacc ggctttcagg ccgaagatga ggcagactac 720
tattgcagct catattccag cggcttcac ttcggaggcg gaactaagct gacagtgtt 780
gggggaggcg gttcagaggt gcagctggtg gagtctgggg gaggcgtggt ccagcctggg 840
aggtccctga caccctcctg ttcagcgtct ggattctcca tccgtagtta tgatagcac 900
tgggtcccg ccggtccagg caagggggta gagtgggtgg ccaactatc atatgatgga 960
agtcagaagt acttcgcaga ctccgtgaag ggcgattta ccatcttcag agacgaatcg 1020
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tgtcgagag ggagtagtaa tcattatgag gcttttgata tctggggcca agggacaatg 1140
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ccaggaacag cccccaact cctcatttat gacaataata agcgaccctc agggattcct 1380
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gtcagcgtcc tcaccgtcct gcaccaggac tggctgaatg gcaaggagta caagtgcag 1860
gtctccaaca aagccctccc agcccccac gagaaaaaca tctccaaagc caaaggccag 1920
ccccgggaac cacaggtgta cgtgtacccc ccatcccggg acgagctgac caagaaccag 1980
gtcagcctga cctgctgggt caaaggcttc tatccagcg acatcgccgt ggagtgggag 2040

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agcaatggcc agcccgagaa caactacaag accacccctc ccgtgctgga ctccgacggc 2100
tccttcgccc tcgtgagcaa gctcacccgtg gacaagagca ggtggcagca gggcaacgtc 2160
ttctcatgct ccgtgatgca tgaggctctg cacaaccact acaccagaa gagcctctcc 2220
ctgtctcccg gc 2232

SEQ ID NO: 117      moltype = DNA length = 2232
FEATURE            Location/Qualifiers
misc_feature        1..2232
                    note = Full-length nucleotide sequence of NYF-0011-HC2-k
                    delete
source              1..2232
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 117
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gaggtgcagc tggtggagtc tgggggaggg ctggtccagc ctgggaggtc cctgacaccc 120
tcctgttcag cgtctggatt ctccatccgt agctatgata tacactgggt ccgcctggct 180
ccaggccaagg ggctagagtg ggtggccact atatcatatg atggaagtca gaagtaacttc 240
gcagactccg tgaaggcgcc atttaccatc ttcagagacg aatcggagaa catggtgtat 300
ctgcaaatga gcggcctgag agttgaggac acggctgttt atcactgtgc gagagggagt 360
agtggtcatt atgaggcttt tgatatctgg ggccaaggga caatgggtcac cgtctcttca 420
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cagccgcccc cagtgtctgc gggcccagga cagaaggcca ccatctctgc ctctggaagc 540
agctccaaca ttgggaattg gtatgtatcc tggtagcagc agctcccagg aacagccccc 600
aaactctcca tttatgacaa taataagcga ccctcaggga ttcctgaccg attctctggc 660
tccaagtctg gcacgtcagc caccctgggc atcacccgac tccagactgg ggacgaggcc 720
gattattact gcggaacatg ggatagcagc ctgagtgccc cttgggtgtt cggcggaggg 780
accaaggtca ccgctctagg gggaggcggt tcagaagtgc agctggtgga atccgggggg 840
ggcctgtgtc agcctggggg gagcctgaga ctgagtttgt ccgcctctgg ggtgacattt 900
aactactatg gcattgtctg gatccgccag gcacctggaa agggcctgga gtgggtggcc 960
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gatacagctg tgtactattg cactctggac ggcagggatg ggtgggtcgc ctattggggg 1140
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gtggtgtcag tgagccacga agaccctgag gtcaagtcca actggtacgt ggacggcggtg 1740
gaggtgcata tgaccaagac aaagcccgcg gagggagcag acaacagcac gtaccgggtg 1800
gtcagcgctc tcaccgtcct gcaccaggac tggtgtaatg gcaaggagta caagtgcaag 1860
gttcccaaca aagccctccc agcccccatc gagaaaaacca tctccaaagc caaaggccag 1920
ccccgggaac cacaggtgta cgtgtacccc ccatcccggg acgagctgac caagaaccag 1980
gtcagcctga cctgcctggt caaaggcttc tatccacagc acatcgccgt ggagtgaggag 2040
agcaatggcc agcccgagaa caactacaag accacccctc ccgtgctgga ctccgacggc 2100
tccttcgccc tcgtgagcaa gctcaccgtg gacaagagca ggtggcagca gggcaacgtc 2160
ttctcatgct ccgtgatgca tgaggctctg cacaaccact acaccagaa gagcctctcc 2220
ctgtctcccg gc 2232

SEQ ID NO: 118      moltype = AA length = 744
FEATURE            Location/Qualifiers
REGION              1..744
                    note = Full-length amino acid sequence of NYF-0010-HC2-k
                    delete
source              1..744
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 118
MKHLWFFLLL VAAPRWVLSG EVQLVESGGG VVQPGKSLTP SCSASGFSIR SYDMHWVRQA 60
PGKGLEWVAT ISYDGSQKYF ADSVKGRFTI FRDESENMYV LQMSGRLVED TAVYHCARGS 120
SNHYEAFDIW QGQTMVTVSS GGGGSGGGGS GGGGSGSVLT QPPSVSAAPG QKVTISCSGS 180
SSNIGNNYVS WYQRLPGTAP KLLIYDNMKR PSIGIDRFSG SKSGTSATLG ITGLQTGDEA 240
DYYCGAWDSS LSAPIWVFGG TKTVTLGGGG SEVQLVESGG GLVQPGGSLR LSCAASGVTF 300
NYYGMSWIRQ APGKLEWVA SITRSGGRIY YPDSVKGRFT ISRENTQKTL YLQMNSLRAE 360
DTAVYYCTLD GRDGVVAYWG QGTLVTVSSG GGGSGGGGSG GGGSNFMLTQ PSSVSGVPGQ 420
RVTISCTGNT GNIGSNVYNW YQQLPGTAPK LLIYRDDKRP SGVPDRFSGS KSGTSASLAI 480
TGFQAEDEAD YYCQSYSSGF IFGGGTKLTV LAAEPKSSDK THTCPPCPAP EAAGGPSVFL 540
FPPPKPDITL ISRTPEVTCV VVSVSHEDPE VKFNWYVDGV EVHNAKTKPR EEQYNSTYRV 600
VSVLTVLHQD WLNKEYKCK VSNKALPAPI EKTISKAKGQ PREPQVYVYP PSRDELTKNQ 660
VSLTCLVKGF YPSDIAVEWE SNGQPENNYK TTPPVLDSDG SFALVSKLTV DKSRWQQGNV 720
FSCSVMEHAL HNHYTQKSL SPSG 744

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SEQ ID NO: 119 moltype = AA length = 744
 FEATURE Location/Qualifiers
 REGION 1..744
 note = Full-length amino acid sequence of NYF-0004-HC2-k
 delete
 source 1..744
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 119
 MKHLWFFLLL VAAPRWVLSG EVQLVESGGG LVQPGGSLRL SCAASGVTFN YYGMSWIRQA 60
 PGKGLEWVAS ITRSGGRIYY PDSVKGRFTI SRENTQKTLY LQMNSLRAED TAVYCYCTLDG 120
 RDGWVAYWQ GTLVTVSSGG GSGGGGGSGG GGSNFMLTQP SSVSGVPGQR VTISCTGNTG 180
 NIGSNYVNWY QQLPGTAPKL LIYRDDKRPS GVPDRFSGSK SGTASLAI GFQAEDEADY 240
 YCQSYSSGFI PGGGTKLTVL GGGGSEVQLV ESGGGVVQPG RSLTPSCSAS GFSIRSYDMH 300
 WVRQAPGKGL EWVATISYDG SQKYFADSVK GRFTIFRDES ENMVYLQMSG LRVEDTAVYH 360
 CARGSSNHYE AFDIWQGTMT VTVSSGGGGG GGGSGGGGGS QSVLTQPPSV SAAPGQKVTI 420
 SCSGSSSNIG NNYVSWYQRL PGTAPKLLIY DNNKRPSGIP DRFSGSKSGT SATLGITGLQ 480
 TGDEADYYCG AWDSSLSAPW VFGGGTKVTV LAAEPKSSDK THTCPPCPAP EAAGGPSVFL 540
 FPPKPKDTLM ISRTPEVTCV VVSVSHEDPE VKFNWYVDGV EVHNATKPR EEQYNSTYRV 600
 VSVLTVLHQD WLNGKEYKCK VSNKALPAPI EKTISKAKGQ PREPQVYVYP PSRDELTKNQ 660
 VSLTCLVKGF YPSDIAVEWE SNGQPENNYK TTPPVLDSDG SFALVSKLTV DKSRWQQGNV 720
 FSCSVMHEAL HNHYTQKSLS LSPG 744

SEQ ID NO: 120 moltype = AA length = 744
 FEATURE Location/Qualifiers
 REGION 1..744
 note = Full-length amino acid sequence of NYF-0011-HC2-k
 delete
 source 1..744
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 120
 MKHLWFFLLL VAAPRWVLSG EVQLVESGGG VVQPGRSLTP SCSASGFSIR SYDIHWVRLA 60
 PGKGLEWVAT ISYDGSQKYF ADSVKGRFTI FRDESENMYV LQMSGLRVED TAVYHCARGS 120
 SGHYEAFDIW GQGTMTVTVSS GGGSGGGGGS GGGGQSQVLT QPPSVSAAPG QKVTISCSGS 180
 SSNIGNWYVS WYQQLPGTAP KLLIYDNNKR PSGIPDRFSG SKSGTSATLG ITGLQTGDEA 240
 DYYCGTWDSS LSAPWVFGGG TKVTVLGGGG SEVQLVESGG GLVQPGGSLR LSCAASGVTF 300
 NYYGMSWIRQ APGKGLEWVA SITRSGGRIY YPDSVKGRFT ISRENTQKTL YLQMNSLRAE 360
 DTAVYYCTLD GRDGWVAYWG QGTLVTVSSG GGGSGGGGGS GGSNFMLTQ PSSVSGVPGQ 420
 RVTISCTGNT GNIGSNYVNW YQQLPGTAPK LLIYRDDKRPS GVPDRFSGS KSGTSASLAI 480
 TGFQAEDEAD YYCQSYSSGF IFGGGKTLTV LAAEPKSSDK THTCPPCPAP EAAGGPSVFL 540
 FPPKPKDTLM ISRTPEVTCV VVSVSHEDPE VKFNWYVDGV EVHNATKPR EEQYNSTYRV 600
 VSVLTVLHQD WLNGKEYKCK VSNKALPAPI EKTISKAKGQ PREPQVYVYP PSRDELTKNQ 660
 VSLTCLVKGF YPSDIAVEWE SNGQPENNYK TTPPVLDSDG SFALVSKLTV DKSRWQQGNV 720
 FSCSVMHEAL HNHYTQKSLS LSPG 744

SEQ ID NO: 121 moltype = AA length = 9
 FEATURE Location/Qualifiers
 REGION 1..9
 note = Amino acid sequence of point mutated NY-ESO peptide
 1F
 source 1..9
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 121
 FLLMWITQC 9

SEQ ID NO: 122 moltype = AA length = 9
 FEATURE Location/Qualifiers
 REGION 1..9
 note = Amino acid sequence of point mutated NY-ESO peptide
 2M
 source 1..9
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 122
 SMLMWITQC 9

SEQ ID NO: 123 moltype = AA length = 9
 FEATURE Location/Qualifiers
 REGION 1..9
 note = Amino acid sequence of point mutated NY-ESO peptide
 3A
 source 1..9
 mol_type = protein
 organism = synthetic construct

-continued

SEQUENCE: 123	
SLAMWITQC	9
SEQ ID NO: 124	moltype = AA length = 9
FEATURE	Location/Qualifiers
REGION	1..9
	note = Amino acid sequence of point mutated NY-ESO peptide
	4A
source	1..9
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 124	
SLLAWITQC	9
SEQ ID NO: 125	moltype = AA length = 9
FEATURE	Location/Qualifiers
REGION	1..9
	note = Amino acid sequence of point mutated NY-ESO peptide
	5A
source	1..9
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 125	
SLLMAITQC	9
SEQ ID NO: 126	moltype = AA length = 9
FEATURE	Location/Qualifiers
REGION	1..9
	note = Amino acid sequence of point mutated NY-ESO peptide
	6L
source	1..9
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 126	
SLLMWLTQC	9
SEQ ID NO: 127	moltype = AA length = 9
FEATURE	Location/Qualifiers
REGION	1..9
	note = Amino acid sequence of point mutated NY-ESO peptide
	7F
source	1..9
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 127	
SLLMWIFQC	9
SEQ ID NO: 128	moltype = AA length = 9
FEATURE	Location/Qualifiers
REGION	1..9
	note = Amino acid sequence of point mutated NY-ESO peptide
	8A
source	1..9
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 128	
SLLMWITAC	9
SEQ ID NO: 129	moltype = AA length = 9
FEATURE	Location/Qualifiers
REGION	1..9
	note = Amino acid sequence of point mutated NY-ESO peptide
	9A
source	1..9
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 129	
SLLMWITQA	9
SEQ ID NO: 130	moltype = AA length = 9
FEATURE	Location/Qualifiers
REGION	1..9
	note = Amino acid sequence of gp100 peptide
source	1..9
	mol_type = protein
	organism = synthetic construct

-continued

SEQUENCE: 130
IMDQVPFSV 9

SEQ ID NO: 131 moltype = AA length = 9
FEATURE Location/Qualifiers
REGION 1..9
note = Amino acid sequence of homologous peptide DOLPP1
source 1..9
mol_type = protein
organism = synthetic construct

SEQUENCE: 131
SQFMWFFSV 9

SEQ ID NO: 132 moltype = AA length = 9
FEATURE Location/Qualifiers
REGION 1..9
note = Amino acid sequence of homologous peptide IL20RB
source 1..9
mol_type = protein
organism = synthetic construct

SEQUENCE: 132
SLFMWFFYA 9

SEQ ID NO: 133 moltype = AA length = 9
FEATURE Location/Qualifiers
REGION 1..9
note = Amino acid sequence of homologous peptide PRKD2
source 1..9
mol_type = protein
organism = synthetic construct

SEQUENCE: 133
SLDMWSVGV 9

SEQ ID NO: 134 moltype = AA length = 9
FEATURE Location/Qualifiers
REGION 1..9
note = Amino acid sequence of homologous peptide CD163
source 1..9
mol_type = protein
organism = synthetic construct

SEQUENCE: 134
SIPMWVDNV 9

SEQ ID NO: 135 moltype = AA length = 9
FEATURE Location/Qualifiers
REGION 1..9
note = Amino acid sequence of homologous peptide P2RY8
source 1..9
mol_type = protein
organism = synthetic construct

SEQUENCE: 135
SVAMWAVFL 9

SEQ ID NO: 136 moltype = AA length = 269
FEATURE Location/Qualifiers
REGION 1..269
note = Amino acid sequence of C3E-7034
source 1..269
mol_type = protein
organism = synthetic construct

SEQUENCE: 136
GEVQLVESGG GLVQPGGSLR LSCAASGVTF NYYGMSWIRQ APGKGLEWVA SITNSGGRIY 60
YDPSVKGRFT ISRENTQKTL YLQMNSLRAE DTAVYYCTLD GRDGVVAYWG QGTLVTVSSG 120
GGSGGGGGSG GGSNFMILTQ PHSVSESPGK TVTISCKRNT GNIGSNYVNW YQHEGSSPT 180
TIIYRDDKRP DGVSDRFSGS IDRSSKSASL TISNLKTEDE ADYFCQSYSS GFIFGGGTKL 240
TVLGAAAGAG GDYKDDDDKG AAHHHHHHH 269

SEQ ID NO: 137 moltype = AA length = 267
FEATURE Location/Qualifiers
REGION 1..267
note = Amino acid sequence of C3E-7036
source 1..267
mol_type = protein
organism = synthetic construct

SEQUENCE: 137
GEVQLVESGG GLVQPGGSLR LSCAASGVTF NYYGMSWIRQ APGKGLEWVA SITNSGGRIY 60

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YPDSVKGRFT ISRENTQKTL YLQMNSLRAE DTAVYYCTLD GRDGWVAYWG QGTLVTVSSG 120
 GGGSGGGGSG GGGSNFMLTQ PSSVSGVPGQ RVTISCTGNT GNIGSNYVNW YQQLPGTAPK 180
 LLIYRDDKRP SGVPDRFSGS KSGTSASLAI TGFQAEDEAD YYCQSYSSGF IFGGGTKLTV 240
 LGAAAGAGGD YKDDDDKGAA AHHHHHH 267

SEQ ID NO: 138 moltype = AA length = 267
 FEATURE Location/Qualifiers
 REGION 1..267
 note = Amino acid sequence of C3E-7085
 source 1..267
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 138
 GEVQLVESGG GLVQPGGSLR LSCAASGVTF NYYGMSWIRQ APGKGLEWVA SITRSGGRIY 60
 YPDSVKGRFT ISRENTQKTL YLQMNSLRAE DTAVYYCTLD GRDGWVAYWG QGTLVTVSSG 120
 GGGSGGGGSG GGGSNFMLTQ PSSVSGVPGQ RVTISCTGNT GNIGSNYVNW YQQLPGTAPK 180
 LLIYRDDKRP SGVPDRFSGS KSGTSASLAI TGFQAEDEAD YYCQSYSSGF IFGGGTKLTV 240
 LGAAAGAGGD YKDDDDKGAA AHHHHHH 267

SEQ ID NO: 139 moltype = AA length = 269
 FEATURE Location/Qualifiers
 REGION 1..269
 note = Amino acid sequence of C3E-7088
 source 1..269
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 139
 GEVQLVESGG GLVQPGGSLR LSCAASGVTF NYYGMSWIRQ APGKGLEWVA SITRSGGRIY 60
 YPDSVKGRFT ISRENTQKTL YLQMNSLRAE DTAVYYCTLD GRDGWVAYWG QGTLVTVSSG 120
 GGGSGGGGSG GGGSNFMLTQ PHSVSESPGK TVTISCKRNT GNIGSNYVNW YQQHEGSSPT 180
 TTIYRNDKRP DGVSDRFSGS IDRSSKSASL TISNLKTEDE ADYFCQSYSS GFIFGGGTKL 240
 TVLGAAAGAG GDYKDDDDKG AAHHHHHHH 269

SEQ ID NO: 140 moltype = AA length = 269
 FEATURE Location/Qualifiers
 REGION 1..269
 note = Amino acid sequence of C3E-7093
 source 1..269
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 140
 GEVQLVESGG GLVQPGGSLR LSCAASGVTF NYYGMSWIRQ APGKGLEWVA SITSSGGRIY 60
 YPDSVKGRFT ISRENTQKTL YLQMNSLRAE DTAVYYCTLD GRDGWVAYWG QGTLVTVSSG 120
 GGGSGGGGSG GGGSNFMLTQ PHSVSESPGK TVTISCKRNT GNIGSNYVNW YQQHEGSSPT 180
 TTIYRNDKRP DGVSDRFSGS IDRSSKSASL TISNLKTEDE ADYFCQSYSS GFIFGGGTKL 240
 TVLGAAAGAG GDYKDDDDKG AAHHHHHHH 269

SEQ ID NO: 141 moltype = AA length = 8
 FEATURE Location/Qualifiers
 REGION 1..8
 note = Amino acid sequence of heavy chain CDR1 of C3E-7085
 source 1..8
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 141
 GVTFNYYG 8

SEQ ID NO: 142 moltype = AA length = 8
 FEATURE Location/Qualifiers
 REGION 1..8
 note = Amino acid sequence of heavy chain CDR2 of C3E-7085
 source 1..8
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 142
 ITRSGGRI 8

SEQ ID NO: 143 moltype = AA length = 11
 FEATURE Location/Qualifiers
 REGION 1..11
 note = Amino acid sequence of heavy chain CDR3 of C3E-7085
 source 1..11
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 143
 TLDGRDGWVA Y 11

SEQUENCE: 144
TGNIGSNY

```

SEQ ID NO: 146      moltype = AA  length = 8
FEATURE            Location/Qualifiers
REGION             1..8
                  note = Amino acid sequence of light chain CDR3 of C3E-7085
source             1..8
                  mol_type = protein
                  organism = synthetic construct

```

SEQUENCE: 146
QSYSSGFI 8

```
SEQ ID NO: 147      moltype = AA  length = 269
FEATURE            Location/Qualifiers
REGION            1..269
                  note = Amino acid sequence of C3E-7078
source            1..269
                  mol_type = protein
                  organism = synthetic construct
```

SEQUENCE: 147

GPEVLVSESSG	GLVPQGGSRLR	LSCAASGVTF	NYYGMSWIRQ	APKGKLEWA	SITRSGGRIY	60
YDVSVKGRFTI	ISRENTQKITL	YLQMNSLRAE	D'AVICYCTLD	GRDGWAYWG	QGTILTVTSSG	120
GGSGGGGGSG	GGSNFMILTQ	PHSVSES PGK	TVTISCRRNT	GNIGSNIYNW	YQQHEGSSPT	180
TI1YRDDKR P	DGVSDFRFSG	IDRSSKSALS	TISNLKTDEE	ADYFCQS YSS	GFIPGGGTKL	240
TVLGAAGAA G	GDYKDNDDBK	AAAHHHHHS				269

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SEQ ID NO: 148      multype = DNA   length = 2235
FEATURE            Location/Qualifiers
misc_feature        1..2235
                    note = Full-length nucleotide sequence of NYF-0014-HC2
source              1..2235
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 148						
atgaaacacac	tgtggttctt	ctctctgtctg	gtggcagctc	ccagatgggt	gctgagcgga	60
ggaggtgcagc	tgtgtggagtc	tggggggagcg	gtgtgtccagc	tctgggaggtc	ctcgacaccc	120
tctctgttcag	cgtctggatt	ctccatccgt	agttatgata	tacactgggt	cccgccaggct	180
ccagggaagg	ggctagaggt	ggtgcccact	tatccatatg	atggaagtca	gaagtaacttc	240
gcagactccg	tgaaggccgc	atttaccatc	atccagagac	aatccggagaa	catggtgtat	300
ctgc aaatga	gcggccctgag	agt tagaac	acggctgttt	atcactgtgc	gagagggaagt	360
agttggtcatt	atgagctctt	tgatatactg	ggccaagga	caatggctcac	cgctctctca	420
gggtgaggcg	tgtcaggcgg	agggtggcag	ggccgtgtgcg	ggagtcagtc	tgtgttgacg	480
cagccgccct	cagtgtctgc	ggccccagga	cagaaggtca	ccatctctcg	ctctggaagg	540
cgctccaaca	tttggaataa	tatgtatcc	tgttaccagc	agctcccagg	aacagcccc	600
aaactctctca	tttatgacaa	taataaacga	ccctcagga	tctctgaccg	attctctggc	660
tccaagttct	gcacgtcagc	caccctgggc	atcacggcag	ctcagactgt	ggaacgagcc	720
gattatttact	gcggaacatg	ggatagcagc	ctgaagtgcc	cttgggtgtt	cgccggaggcg	780
accaaggtca	ccgtctcagg	gggaggcggt	tcagaagtgc	agctgtgtga	atccggggggg	840
ggcctggtgc	agcctggggg	gagcctgaga	ctgaattgtg	cgccctctgg	ggtgacattt	900
aaactactatg	gcattgtctg	gattccgcag	gcacctggaa	agggcctgga	tggggtgtgcc	960
agcatcacta	ggctccggcg	gcgaatctac	tatcccgaca	cgctcaaggg	caggttcaca	1020
atttccgcgc	agaacacaca	gaaaacctgt	tacctcgaga	tgaatagctc	gagagccgaa	1080
gatacagctg	tgtactattg	cactctggag	ggcagggatg	gggtggctcgc	ctattggggg	1140
cagggaaccc	tgtgtcagct	cagctccgga	ggaggaggat	tgcgcgagg	aggcagtggt	1200
ggaggccgggt	caaactttat	gctcaactac	ccgtctctcg	ttctggcgt	acctggccaa	1260
ccggtgacca	ttagctgtat	gggtaataac	gggaatatcg	gggtcacta	cgtgaactgg	1320
tctgacagc	ttcaggggac	agctcccaag	tgtcgtatct	atccgcagca	caaaagacc	1380
tcagggggtcc	ctgaccgatt	tagtggcagc	aaaagcggta	cttccgcttc	cttcgcgata	1440
acggcgttct	aggcccgaga	tagggcagac	tactattggc	agtcatactc	cagcggcttc	1500
atctccggag	gcggaactaa	gctcagactg	tgggcaggcg	agcccaaatc	tcttgacaaa	1560
actcacacat	gccaccctgt	cccagcactc	gaagcccgag	ggggaccctc	agtcctctct	1620
ttccccccaa	aaaccaagga	caccctcatg	atctcccgca	ccctcagggt	ccatcgcgtg	1680
gttgtgttcag	tgaqccacaa	aqacctctaa	gtcqaagtca	actqqtacat	gqaacqccgt	1740

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gagggtgcata atgccaagac aaagcccccgg gaggagcagt acaacagcac gtaccgggtg 1800
gtcagcgctcc tcacgcgtct gcaccaggac tggtctgaatg gcaaggagta caagtgcaag 1860
gtctccaaca aagccctccc agcccccata gagaaaacca tctccaaagc caaaggccag 1920
ccccgggaac cacaggtgta cgtgtacccc ccatcccggt acgagctgac caagaaccag 1980
gtcagcctga cctgcctggt caaaggcttc tatcccagcg acatcgccgt ggagtgggag 2040
agcaatggcc agcccgagaa caactacaag accaccctc ccgtgctgga ctccgacggc 2100
tccttcgccc tcgtgagcaa gctcaccgtg gacaagagca ggtggcagca gggcaacgtc 2160
ttctcatgct ccgtgatgca tgaggctctg cacaaccact acaccagaa gagcctctcc 2220
ctgtctcccg gcaag 2235

SEQ ID NO: 149      moltype = AA  length = 745
FEATURE            Location/Qualifiers
REGION            1..745
                  note = Full-length amino acid sequence of NYF-0014-HC2
source            1..745
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 149
MKHLWFFLLL VAAPRWVLSG EVQLVESGGG VVQPGKSLTP SCSASGFSIR SYDIHWVRQA 60
PGKGLEWVAT ISYDGSQKYF ADSVKGRFTI FRDESENMYV LQMSGRLVED TAVYHCARGS 120
SGHYEAFDIW GQGTMTVTSS GGGGSGGGGS GGGGSQSVLT QPPSVSAAPG QKVTISCSGS 180
SENIGNNYVS WYQLPGTAP KLLIYDNMKR PSGIPDRFSG SKSGTSATLG ITGLQTGDEA 240
DYYCGTWDS LSAWVVFSGG TKVTVLGGGG SEVQLVESGG GLVQPGGSLR LSCAASGVTF 300
NYYGMSWIRQ APGKLEWVA SITRSGGRIY YPDSVKGRFT ISRENTQKTL YLQMNSLRAE 360
DTAVYYCTLD GRDGWVAYWG QGTLVTSSG GGGSGGGGSG GGSNFMLTQ PSSVSGVPGQ 420
RVTISCTGNT GNIGSNYVNW YQQLPGTAPK LLIYRDDKRP SGVPDRFSGS KSGTSASLAI 480
TGFQAEDEAD YYCQSYSSGF IFGGGTKLTV LAAEPKSSDK THTCPPCPAP EAAGGPSVFL 540
FPPKPKDTLM ISRTPEVTCV VVSVSHEDPE VKFNWYVDGV EVHNAKTKPR EEQYNSTYRV 600
VSVLTVLHQD WLNGKEYKCK VSNKALPAPI EKTISKAKGQ PREPQVYVYP PSRDELTKNQ 660
VSLTCLVKGF YPSDIAVEWE SNGQPENNYK TTPPVLDSDG SFALVSKLTV DKSRWQQGNV 720
FSCSVMEAL HNHYTQKSL LSPGK 745

SEQ ID NO: 150      moltype = AA  length = 745
FEATURE            Location/Qualifiers
REGION            1..745
                  note = Full-length amino acid sequence of NYF-0082-HC2
source            1..745
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 150
MKHLWFFLLL VAAPRWVLSG EVQLVESGGG VVQPGKSLTP SCSASGFSIR SYDIHWVRQA 60
PGKGLEWVAT ISYDGSQKYF ADSVKGRFTI FRDESENMYV LQMSGRLVED TAVYHCARGS 120
SGHYEAFDIW GQGTMTVTSS GGGGSGGGGS GGGGSQSVLT QPPSVSAAPG QKVTISCSGS 180
SSNIGNNYVS WYQLPGTAP KLLIYDNMKR PSGIPDRFSG SKSGTSATLG ITGLQTGDEA 240
DYYCGSWDS LSAWVVFSGG TKVTVLGGGG SEVQLVESGG GLVQPGGSLR LSCAASGVTF 300
NYYGMSWIRQ APGKLEWVA SITRSGGRIY YPDSVKGRFT ISRENTQKTL YLQMNSLRAE 360
DTAVYYCTLD GRDGWVAYWG QGTLVTSSG GGGSGGGGSG GGSNFMLTQ PSSVSGVPGQ 420
RVTISCTGNT GNIGSNYVNW YQQLPGTAPK LLIYRDDKRP SGVPDRFSGS KSGTSASLAI 480
TGFQAEDEAD YYCQSYSSGF IFGGGTKLTV LAAEPKSSDK THTCPPCPAP EAAGGPSVFL 540
FPPKPKDTLM ISRTPEVTCV VVSVSHEDPE VKFNWYVDGV EVHNAKTKPR EEQYNSTYRV 600
VSVLTVLHQD WLNGKEYKCK VSNKALPAPI EKTISKAKGQ PREPQVYVYP PSRDELTKNQ 660
VSLTCLVKGF YPSDIAVEWE SNGQPENNYK TTPPVLDSDG SFALVSKLTV DKSRWQQGNV 720
FSCSVMEAL HNHYTQKSL LSPGK 745

SEQ ID NO: 151      moltype = AA  length = 207
FEATURE            Location/Qualifiers
source            1..207
                  mol_type = protein
                  organism = Homo sapiens

SEQUENCE: 151
MQSGTHWRVL GLCLLSVGWV GQDGNEMGG ITQTPYKVISI SGTTVILTCP QYPGSEILWQ 60
HNDKNIGGDE DDKNIGSDED HLSLKEFSEL EQSGYYVCYP RGSKPEDANF YLYLRARVCE 120
NCMEMDVMSV ATIVIVDICI TGGLLLLVY WSKNRKAKAK PVTRGAGAGG RQRGQNKERP 180
PPVPNPDIYEP IRKGQRLYS GLNQRI 207

SEQ ID NO: 152      moltype = DNA  length = 2235
FEATURE            Location/Qualifiers
misc_feature      1..2235
                  note = Full-length nucleotide sequence of NYZ-0038-HC2
source            1..2235
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 152
atgaaacacc tgtggttctt cctcctgctg gtggcagctc ccagatgggt gctgagcggt 60
gaggtagcagc ttggtggaatc cggtaggagc cttgtgcaac cgggaggctc cctgcgactc 120
tcttgccgag catctggctt ttccatacga tcctacgata ttcatgggt cgcacaagcc 180

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cctggcaaa gttctgaatg ggttgccact atctcctacg acgggtctca gaaatattat 240
gcggattcag tgaaggggag gttcaccaatt tcacgagacg agtcaaagaa tacactgtac 300
ctccaaatga attcactgag agccgaggat actgcagctc atcattgtgc aagaggggtcc 360
tcaggccact acgaggcctt tgatatatgg ggccaaggca ccttggtaac cgttagtagc 420
ggaggtggag gaagcggagg cgccgggttc ggaggaggtg gaagtcaatc cgttctgact 480
cagccaccct ccgcctctgg caccgccggc caacgggtca caatatcatg ttctggtctc 540
tcaagcaaca tcggaaactg gtacgtgagc tggtagcagc agctcccccg cacggcgccc 600
aagcttctga tatatgacaa caacaaacgg ccagtgagg ttcctgacag attcagtgagg 660
tctaaaagtg gtacaagcgc tagcctggcc ataagtggtc tgcagagtga agatgaggcg 720
gactattatt gcggaacctg ggactccagc ctgagcgcct cctgggtttt cggcggagggt 780
actaagctga ccgtcttggg agggagcggt tcagaagtgc agctgggtgga atccgggggg 840
ggcctggtgc agcctggggg gagcctgaga ctgagttgtg ccgcctctgg ggtgacattt 900
aactactatg gcatgtcttg gatccgccag gcacctggaa agggcctgga gtgggtggcc 960
agcaccacta ggcctcgagg gcgaatctac tatcccgaca gcgtcaaggg cagggtccaca 1020
atttcccgcg agaacacaca gaaaactctg tacctgcaga tgaatagcct gagagccgaa 1080
gatacagctg tgtactattg cactctggac ggcaaggatg ggtgggtcgc ctattggggg 1140
tgtggaaccc tggtagcagt cagctccgga ggaggaggat ctggcggagg aggcagtgagg 1200
ggaggcgggt caaactttat gctcactcag ccgtcctctg ttctggtcgt acctggccaa 1260
cgggtgacca ttagctgtac gggtaatacc gggaatatcg ggtctaacta cgtgaactgg 1320
tatcagcagc ttccagggac atgtcccaag ttgctgatct atcgcgacga caaaagaccc 1380
tcagggggtc ctgaccgatt tagtggcagc aaaaagcggt cttccgcttc cctggcgata 1440
accggcttcc aggcggaaga tgaggcagac tactattgcc agtcatattc cagcggcttc 1500
atcttcggag gcggaactaa gctgacagtg ttggcagccg agcccaaatc ttctgacaaa 1560
actcacacat gccaccctcg ccagcacctc gaagcccgag ggggaccctc agtcttctc 1620
ttcccccaa aaccocaggc caccctcatg atctcccgga ccctgaggtg cacatgcgtg 1680
gtggtgtcag tgagccacga agaccctgag gtcaagtcca actggtagct ggacggcgctg 1740
gaggtgcata atgccaagac aaagcccgcg gaggagcagt acaacagcac gtaccgggtg 1800
gtcagcgtcc tcaccgtcct gcaccaggac tggctgaatg gcaaggagta caagtgcag 1860
gtctccaaca agccctctcc agcccccac gagaaaaaca tctccaaag caaaggccag 1920
ccccgggaac cacagggtga cgtgtacccc ccatcccggg acgagctgac caagaaccag 1980
gtcagcctga cctgctgggt caaaggcttc tatcccgcg acatcgccgt ggagtgggag 2040
agcaatggcc agcccgagaa caactacaag accaccctc ccgtgctgga ctccgacggc 2100
tccttcgccc tcgtgagcaa gctcaccgtg gacaagagca ggtggcagca gggcaacgtc 2160
ttctcatgct ccgtgatgca tgaggctctg cacaaccact acaccagaa gagcctctcc 2220
ctgtctcccg gcaag 2235

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SEQ ID NO: 153      moltype = DNA length = 2250
FEATURE             Location/Qualifiers
misc_feature         1..2250
                     note = Full-length nucleotide sequence of NYZ-0082-HC2
source               1..2250
                     mol_type = other DNA
                     organism = synthetic construct

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SEQUENCE: 153
atgaaacacc tgtggttctt cctcctgctg gtggcagctc ccagatgggt gctgagcgggt 60
gaggtacagc tgggtggaatc cgttggaaggc cttgtgcaac cgggagggctc cctgcgactc 120
tcttgccgag cactctggctt ttccatacga tctacgata ttctattgggt ccgacaagcc 180
cctggcaaa gttctgaatg ggttgccact atctcctacg acgggtctca gaaatattat 240
gcggattcag tgaaggggag gttcaccaatt tcacgagacg agtcaaagaa tacactgtac 300
ctccaaatga attcactgag agccgaggat actgcagctc atcattgtgc aagaggggtcc 360
tcaggccact acgaggcctt tgatatatgg ggccaaggca ccttggtaac cgttagtagc 420
ggaggtggag gaagcggagg cgccgggttc ggaggaggtg gaagtggagg aggtggatct 480
caatccgttc tgactcagcc accctccgcc tctggcacc cgggccaacg ggtcacataa 540
tcattgtctg gctcttcaag caacatcgga aactggtagc tgagctggta ccagcagctc 600
cccgccacgg cgcccaagct tctgatatat gacaacaaca aacggcccgag tggagtctct 660
gacagattca gtgggtctaa aagtgggtaca agcgtagcc tggccataag tgggtctgag 720
agtgaagatg aggcggacta ttattcgga accctgggact ccagcctgag cgctccctgg 780
gttttcggcg gaggtactaa gctgaccgtc ttgggaggag gcggttcaga agtgacagctg 840
gtggaatccg gggggggcct ggtgcagcct ggggggagcc tgagactgag ttgtgccgcc 900
tctgggggtg catttaacta ctatggcatg tcttggtacc gccaggcacc tggaaaaggcc 960
ctggagtggtg tggccagcat cactaggtcc ggccggcgaa tctactatcc cgacagcgtc 1020
aagggcaggt tcacaatttc ccgcgagaa acacagaaaa ctctgtacct gcagatgaat 1080
agcctgagag ccgaagatag agctgtgtac tattgcactc tggacggcag ggaatgggtg 1140
gtcgccattt ggggggtgtg aacctgggtg acagtcagct ccggaggagg aggatctggc 1200
ggaggaggca gtgggggagg cgggtcaaac ttatgctca ctcagccgtc ctctgtttct 1260
ggcgtacctg gccaacgggt gaccattagc tgtacgggta ataccgggaa tatcggtct 1320
aactacgtga actggtatca gcagcttcca gggacatgtc ccaagttgct gatctatcgc 1380
gacgacaaaa gacctcagg ggtccctgac cgatttagtg gcagcaaaa cggtacttcc 1440
gcttccctcg cgataaccgg ctttcaggcc gaagatgagg cagactacta ttgccagtca 1500
tattccagcg gcttcattct cggaggcgga actaagctga cagtggtggc agccgagccc 1560
aaatctctg acaaaactca cacatgccca ccctgccag cacctgaagc cgcaggggga 1620
ccctcagctc tctctctccc cccaaaaccc aaggacaccc tcatgatctc ccggacccct 1680
gaggtcacat gcgtgggtgt gtacgtgagc cacgaagacc ctgaggtcaa gttcaactgg 1740
tacgtggagc gcgtggaggt gcataatgcc aagacaaaag cccgggagga gcagtacaac 1800
agcacgtacc ggggtggctcag cgtcctcacc gtcctgcacc aggactggct gaatggcaag 1860
gagtacaagt gcaaggctctc caacaaagcc ctcccagccc ccatcgagaa aacctctccc 1920

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aaagccaaag gccagccccc ggaaccacag gtgtacgtgt acccccccac ccgggacgag 1980
ctgaccaaga accagggtcag cctgacctgc ctgggtcaaa gcttctatcc cagcgacatc 2040
gccgtggagt gggagagcaa tggccagccc gagaacaact acaagaccac ccctcccgtg 2100
ctggactccg acggctcctt cgccctcgtg agcaagctca ccgtggacaa gagcagggtg 2160
cagcagggca acgtctcttc atgctccgtg atgcatgagg ctctgcacaa ccactacacc 2220
cagaagagcc tctccctgtc tcccggcaag                2250

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SEQ ID NO: 154      moltype = DNA length = 2250
FEATURE            Location/Qualifiers
misc_feature        1..2250
                    note = Full-length nucleotide sequence of NYZ-0083-HC2
source              1..2250
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 154
atgaaacacc tgtggttctt cctcctgctg gtggcagctc ccagatgggt gctgagcggg 60
gaggtacagc tgggtggaatc cgggtggaggc cttgtgcaac cgggagggtc cctgcgactc 120
tcttcgcgcg catctggctt ttccatacga tctctacgata ttcatgtggg ccgacaagcc 180
cctggcaaaag gtcttgaatg ggttgccact atctcctacg acgggtctca gaaatattat 240
gcggtattcag tgaagggggcg gttcaccaat tcacgagacg agtcaaaaga tacactgtac 300
ctccaaatga attcactgag agccgaggat actgcagtct atcattgtgc aagagggttc 360
tcaggccact acgaggcctt tgatatatgg ggccaaggca ccttggtaac cgtagtagc 420
ggaggtggag gaagcggagg cgccggttcc ggaggagggt gaagtggagg aggtggatct 480
caatccgttc tgactcagcc accctccgcc tctggcaccg cgggccaacg ggtcacaata 540
tcatgttctg gctcttcaag caacatcgga aactggtagc tgagctggta ccagcagctc 600
cccgccacgg cgcccaagct tctgatataa gacaacaaca aacggcccag tggagtctct 660
gacagattca gtgggtctaa aagtgttaca agcgtagcc ttggccataa tggctctgcag 720
agtgaagatg aggcgggact ttattgcgga acctgggact ccagcctgag cgctccctgg 780
gttttcggcg gaggtactaa gctgaccgtc ttgggaggag gcggttcaga agtgcagctg 840
tgggaatccg gggggggcct ggtgcagcct ggggggagcc tgagactgag ttgtccgcgc 900
tctgggggtg catttaacta ctatggcatg tcttggatcc gccaggcacc tggaaagtgt 960
ctggagtggt tggccagcat cactaggtcc ggccggcgaa tctactatcc cgacagcgtc 1020
aagggcaggt tcacaatttc ccgcgagaa acacagaaaa ctctgtacct gcagatgaat 1080
agcctgagag ccgaagatag agctgtgtac tattgcactc tggacggcag ggatgggtgg 1140
gtcgccattt gggggcaggg aaccttggtg acagtcagct ccggaggagg aggatctggc 1200
ggaggaggca gtgggggagg cgggtcaaac ttatgtctca ctacgcctgc ctctgtttct 1260
ggcgtacctg gccaacgggt gaccattagc tgtacgggta ataccgggaa tatcggtctc 1320
aactacgtga actggttatc gcagcttcca gggacagctc ccaagttgct gatctatcgc 1380
gacgacaaaa gacctcagg ggctccctgac cgatttagtg gcagcaaaag cggtaacttc 1440
gcttccctgg cgttcattct cggatgtgga actaagctga cagtgttgcc agccgagccc 1500
tattccagcg gcttcattct cggatgtgga actaagctga cagtgttgcc agccgagccc 1560
aaatcttctg acaaaaactc cacatgcccc cctgccccag cacctgaagc cgcaggggga 1620
ccctcagctc tctctctccc cccaaaaacc aaggacaccc tcatgatctc ccggacccct 1680
gaggtcacat gcgtggtggt gtcagtgagc cacgaagacc ctgaggtcaa gttcaactgg 1740
tacgtggagc gcgtggagggt gcataatgcc aagacaaaag cccgggaggga gcagtaaac 1800
agcacgtacc ggggtggtcag gtcctcacc gtcctgcacc aggaactggc gaatggcaag 1860
gagtacaagt gcaagggtctc caacaaagcc ctcccagccc ccacgcagaa aacctctctc 1920
aaagccaaag gccagccccc ggaaccacag gtgtacgtgt acccccccac ccgggacgag 1980
ctgaccaaga accaggtcag cctgacctgc ctgggtcaaa gcttctatcc cagcgacatc 2040
gccgtggagt gggagagcaa tggccagccc gagaacaact acaagaccac ccctcccgtg 2100
ctggactccg acggctcctt cgccctcgtg agcaagctca ccgtggacaa gagcagggtg 2160
cagcagggca acgtctcttc atgctccgtg atgcatgagg ctctgcacaa ccactacacc 2220
cagaagagcc tctccctgtc tcccggcaag                2250

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SEQ ID NO: 155      moltype = AA length = 745
FEATURE            Location/Qualifiers
REGION            1..745
                    note = Full-length amino acid sequence of NYZ-0038-HC2
source              1..745
                    mol_type = protein
                    organism = synthetic construct

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SEQUENCE: 155
MKHLWFFLLL VAAPRWVLSG EVQLVESGGG LVQPGGSLRL SCAASGFSIR SYDIHWVRQA 60
PGKGLEWVAT ISYDGSQKYY ADSVKGRFTI SRDESKNTLY LQMNSLRAED TAVYHCARGS 120
SGHYEAFDIW QGGTLVTVSS GGGGSGGGGS GGGGSGSVLT QPPSASGTPG QRVITSCSGS 180
SSNIGNWVYS WYQQLPGTAP KLLIYDNMKR PSGVPDRFSG SKSGTSASLA ISGLQSEDEA 240
DYYCGTWDS SLPAPWVFGG TKLTVLGGGG SEVQLVESGG GLVQPGGSLR LSCAASGVTF 300
NYYGMSWIRQ APGKGLWVA SITRSGGRIY YPDSVKGRFT ISRENTQKTL YLQMNSLRAE 360
DTAVYYCTLD GRDGVVAYWG CGLVTVVSSG GGGSGGGGSG GGSNFMFLTQ PSSVSGVPGQ 420
RVTISCTGNT GNIGSNVYNW YQQLPGTCTPK LLIYRDDKRP SGVPDRFSGS KSGTSASLAI 480
TGFQAEDEAD YYCQSYSSGF IFGGGTKLTV LAAEPKSSDK THTCPPCPAP EAAGGPSVFL 540
FPPPKDPTLM ISRTPEVTCV VVSVSHEDPE VKFNWYVDGV EVHNAKTKPR EEQYNSTYRV 600
VSVLTVLHQD WLNKEYKCK VSNKALPAPI EKTISKAKGQ PREPQVYVYP PSRDELTKNQ 660
VSLTCLVKGF YPSDIAVEWE SNGQPENNYK TTPPVLDSDG SFALVSKLTV DKSRWQQGNV 720
FSCSVMEHAL HNHYTQKSL SLPKGK                745

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SEQ ID NO: 156 moltype = AA length = 750
 FEATURE Location/Qualifiers
 REGION 1..750
 note = Full-length amino acid sequence of NYZ-0082-HC2
 source 1..750
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 156
 MKHLWFFLLL VAAPRWVLSG EVQLVESGGG LVQPGGSLRL SCAASGFSIR SYDIHWVRQA 60
 PGKGLEWVAT ISYDGSQKYY ADSVKGRFTI SRDESKNTLY LQMNSLRAED TAVYHCARGS 120
 SGHYEAFDIW GQGTLLVTSS GGGGSGGGGS GGGGSGGGGS QSVLTQPPSA SGTPGQQRVTI 180
 SCSGSSSNIG NWYVSWYQQL PGTAPKLLIY DNNKRPSGVP DRFSGSKSGT SASLAISGLQ 240
 SEDEADYYCG TWDSSLSAPW VFGGGTKLTV LGGGGSEVQL VESGGGLVQP GGSRLRLSCAA 300
 SGVTFNYYGM SWIRQAPGKG LEWVASITRS GGRIYYPDSV KGRFTISREN TQKTLYLQMN 360
 SLRAEDTAVY YCTLDGRDGW VAYWGCGTLV TVSSGGGGSG GGGSGGGGSN FMLTQPSSVS 420
 GVPGQQRVTIS CTGNTGNIGS NYVNWYQQLP GTCPKLLIYR DDKRPSGVPD RFSGSKSGTS 480
 ASLAITGFQA EDEADYYCQS YSSGFIFGGG TKLTVLAAEP KSSDKTHTCP PCPAPEAAGG 540
 PSVFLFPPKP KDTLMISRTPEVTCVVVSVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYN 600
 STYRVVSVLT VLHQDWLNGK EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VVYPPSRDE 660
 LTKNQVSLTC LVKGFYPSDI AVEWESNGQP ENNYKTPPVV LDSDGSFALV SKLTVDKSRW 720
 QQGNVFSCSV MHEALHNHYT QKSLSLSPGK 750

SEQ ID NO: 157 moltype = AA length = 750
 FEATURE Location/Qualifiers
 REGION 1..750
 note = Full-length amino acid sequence of NYZ-0083-HC2
 source 1..750
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 157
 MKHLWFFLLL VAAPRWVLSG EVQLVESGGG LVQPGGSLRL SCAASGFSIR SYDIHWVRQA 60
 PGKGLEWVAT ISYDGSQKYY ADSVKGRFTI SRDESKNTLY LQMNSLRAED TAVYHCARGS 120
 SGHYEAFDIW GQGTLLVTSS GGGGSGGGGS GGGGSGGGGS QSVLTQPPSA SGTPGQQRVTI 180
 SCSGSSSNIG NWYVSWYQQL PGTAPKLLIY DNNKRPSGVP DRFSGSKSGT SASLAISGLQ 240
 SEDEADYYCG TWDSSLSAPW VFGGGTKLTV LGGGGSEVQL VESGGGLVQP GGSRLRLSCAA 300
 SGVTFNYYGM SWIRQAPGKG LEWVASITRS GGRIYYPDSV KGRFTISREN TQKTLYLQMN 360
 SLRAEDTAVY YCTLDGRDGW VAYWGCGTLV TVSSGGGGSG GGGSGGGGSN FMLTQPSSVS 420
 GVPGQQRVTIS CTGNTGNIGS NYVNWYQQLP GTAPKLLIYR DDKRPSGVPD RFSGSKSGTS 480
 ASLAITGFQA EDEADYYCQS YSSGFIFGCG TKLTVLAAEP KSSDKTHTCP PCPAPEAAGG 540
 PSVFLFPPKP KDTLMISRTPEVTCVVVSVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYN 600
 STYRVVSVLT VLHQDWLNGK EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VVYPPSRDE 660
 LTKNQVSLTC LVKGFYPSDI AVEWESNGQP ENNYKTPPVV LDSDGSFALV SKLTVDKSRW 720
 QQGNVFSCSV MHEALHNHYT QKSLSLSPGK 750

SEQ ID NO: 158 moltype = DNA length = 2172
 FEATURE Location/Qualifiers
 misc_feature 1..2172
 note = Full-length nucleotide sequence of NYZ-1010-HC2
 source 1..2172
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 158
 atgaaacacc tgtggttctt cctcctgctg gtggcagctc ccagatgggt gctgagcggt 60
 gaggtacagc tgggtggaatc cgggtggaggc cttgtgcaac cgggaggctc cctgcgactc 120
 tcttgccgcc catctggctt ttccatacga tcctacgata ttcattgggt cgcacaagcc 180
 cctggcaaaag gtcttgaatg ggttgccact atctcctacg acgggtctca gaaatattat 240
 gcggattcag tgaagggggcg gttcaccaat tcacgagacg agtcaaagaa tacactgtac 300
 ctccaaatga attcactgag agccgaggat actgcagctc atcattgtgc aagaggggtc 360
 tcaggccact acgaggcctt tgatatatgg ggccaaggca ccttggtaac cgtagtagc 420
 ggaggtggag gaagcggagg cggcggttcc ggaggagggt gaagtggagg aggtggatct 480
 caatccgttc tgactcagcc accctccgcc tctggcacc cgggccaacg ggtcacaata 540
 tcatgttctg gctcttcaag caacatcgga aactggtacg tgagctggta ccagcagctc 600
 cccggcacgg cgcccaagct tctgatatat gacaacaaca aacggccacg tggagttcct 660
 gacagattca gtgggtctaa aagtgtgata agcgctagcc tggccataag tggctctgcag 720
 agtgaagatg agcgggacta ttattgcgga acctgggact ccagcctgag cgctccctgg 780
 gttttcggcg gaggtactaa gctgaccgtc ttgggaggag gcggttcaga agtgcagctg 840
 gtggaatccg gggggggcct ggtgcagcct ggggggagcc tgagactgag ttgtgccgcc 900
 tctggggtga catttaacta ctatggcatg tcttggatcc gccaggcacc tggaaagggc 960
 ctggagtggtg tggccagcat cactaggtcc ggccggcgaa tctactatcc cgacagcgct 1020
 aagggcaggt tcacaatttc ccgcgagaa acacagaaaa ctctgtacct gcagatgaat 1080
 agcctgagag ccgaagatac agctgtgtac tattgcactc tggacggcag ggatgggtgg 1140
 gtgcctattt ggggtcaggg aaccctgggt acagtcagct ccgcctccac caagggccca 1200
 agcgtcttcc cctggcacc ctctccaag agcacctctg gcggcacagc cgccctgggc 1260
 tgctgtgtca aggactactt ccccgaaacc gtgaccgtga gctggaactc aggcgcctgt 1320
 accagcggcg tgcacacctt ccccgctgtc ctgcagtcct caggactcta ctccctcagc 1380
 agcgtggtga ccgtgccctc cagcagcttg ggcaccacca cctacatctg caacgtgaat 1440

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cacaagccca gcaacaccaa ggtggacaag agagttgagc ccaagtcttg cgacaaaact 1500
cacacatgcc caccctgccc agcacctgaa gccgcagggg gaccctcagt ctctctcttc 1560
cccccaaaac ccaaggacac cctcatgatc tcccggaccc ctgaggtcac atgcgtgggtg 1620
gtgtcagtga gccacgaaga ccttgaggtc aagttcaact ggtacgtgga cggcgtggag 1680
gtgcataatg ccaagacaaa gccccgggag gagcagtaga acagcacgta ccgggtgggtc 1740
agcgtctcta ctgtctctga ccaggactgg ctgaatggca aggagtacaa gtgcaaggtc 1800
tccaacaaag cctctccagc ccccatcgag aaaaccatct ccaaagccaa aggccagccc 1860
cgggaaccac aggtgtacgt gtacccccca tcccgggacg agctgaccaa gaaccaggtc 1920
agcctgacct gccctggtaa aggtctctat cccagcgaca tcgcccgtga gtgggagagc 1980
aatggccagc ccgagaacaa ctacaagacc acccctcccg tgctggactc cgacggctcc 2040
ttcgccctcg tgagcaagct caccgtggac aagagcaggt gccagcaggg caacgtcttc 2100
tcatgctccg tgatgcatga ggctctgcac aaccactaca ccagaagag cctctccctg 2160
tctcccgcca ag                                     2172

SEQ ID NO: 159      moltype = DNA length = 699
FEATURE            Location/Qualifiers
misc_feature       1..699
                    note = Full-length nucleotide sequence of C3E-7085-LC
source             1..699
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 159
atggtgtctgc agaccacaggt gttcatctcc ctgctgtgtg ggatctccgg cgcgtacggc 60
aacttttatgc tcactcagcc gtctctgtgt tctggcgtag ctggccaacg ggtgaccatt 120
agctgtacggg gtaataccggg gaatatcggg tctaactacg tgaactggta tcagcagctt 180
ccaggggacag ctcccaagtt gctgatctat cgcgacgaca aaagaccctc aggggtccct 240
gaccgatatta gtggcagcaa aagcgttact tccgcttccc tggcgataac cggtcttcag 300
gccgaagatg aggcagacta ctattgccag tcatattcca gcggtctcat ctctggaggc 360
ggaaactaagc tgacagtgtc gggccagcct aaggtgtccc ctacgctgac cctgttccct 420
ccttccagcg aggaagctca agctaacaag gccaccctgg tgtgtcttat ctctgacttc 480
taccctggcg ctgtgaccgt ggcctggaag gctgacagct cccctgtgaa ggcggagtg 540
gagaccacca cacctagcaa gcagctctaac aacaagtagc ctgccagctc ctacctgagc 600
cttaccctcg agcagtgga gtcctacaga agtactcctc gtcaagttag ccacgagggc 660
agcaccgtgg agaagaccgt ggctctacc gagtggtcc 699

SEQ ID NO: 160      moltype = AA length = 724
FEATURE            Location/Qualifiers
REGION            1..724
                    note = Full-length amino acid sequence of NYZ-1010-HC2
source             1..724
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 160
MKHLWFFLLL VAAPRWVLSG EVQLVESGGG LVQPGGSLRL SCAASGFSIR SYDIHWVRQA 60
PGKLEWVAT ISYDGSQKYY ADSVKGRFTI SRDESKNTLY LQMNSLRAED TAVYHCARGS 120
SGHYEAFDIW GQGLVTVYSS GGGSGGGGGS GGGSGGGGGS QSVLTQPPSA SGTGQQRVTI 180
SCSGSSSNIG NWYVSWYQQL PGTAPKLLIY DNNKRPSGVP DRFSGSKSGT SASLAISGLQ 240
SEDEADYYCG TWDSSLAPW VFGGGTKLTV LGGGGSEVQL VESGGGLVQP GGSRLRLSCAA 300
SGVTFFNYGM SWIRQAPGKG LEWVASITRS GGRIYYPDSV KGRFTISREN TQKTLVLMQN 360
SLRAEDTAVY YCTLDGRDGV VAYWGQGLTV TVSSASTKGP SVFPLAPSSK STSGGTAALG 420
CLVKDYFPEP VTVSWNSGAL TSGVHTFPAV LQSSGLYSLS SVVTVPSSSL GTQTYICNVN 480
HKPSNTKVDK RVEPKSCDKT HTPCPCPAPE AAGGPSVFLF PPKPKDTLMI SRTPEVTCVV 540
VSVSHEDPEV KFNWYVDGVE VHNATKPRE EQYNSTYRVV SVLTVLHQDW LNGKEYCKKV 600
SNKALPAPIE KTISKAKGQP REPQVYVYPP SRDELTKNQV SLTCLVKGFY PSDIAVEWES 660
NGQPENNYKT TPPVLDSDGS PALVSKLTVD KSRWQQGNVF SCSVMHEALH NHYTQKSLSL 720
SPGK                                     724

SEQ ID NO: 161      moltype = AA length = 233
FEATURE            Location/Qualifiers
REGION            1..233
                    note = Full-length amino acid sequence of C3E-7085-LC
source             1..233
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 161
MVLQTQVFIS LLLWISGAYG NFMLTQPSSV SGVPGQQRVTI SCTGNTGNIG SNYVNWYQQL 60
PGTAPKLLIY RDDKRPSGVP DRFSGSKSGT SASLAITGFQ AEDEADYYCQ SYSSGFIFGG 120
GTLKTVLGGP KAAPSVLTFP PSSEELQANK ATLVCLISDF YPGAIVTAWK ADSSPVKAGV 180
ETTPSKQSN NKYAASSYLS LTPEQWKSHR SYSCQVTHEG STVEKTVAPT ECS 233

SEQ ID NO: 162      moltype = AA length = 5
FEATURE            Location/Qualifiers
REGION            1..5
                    note = Peptide linker
source             1..5
                    mol_type = protein

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organism = synthetic construct

SEQUENCE: 162
GGGGS 5

SEQ ID NO: 163      moltype = DNA length = 891
FEATURE            Location/Qualifiers
misc_feature        1..891
                    note = Full-length nucleotide sequence of NYA-3061
source              1..891
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 163
atgaaacacc tgtggttctt cctcctgctg gtggcagctc ccagatgggt gctgagcggg 60
gaggtacagc tgggtggaatc cgggtggaggc cttgtgcaac cgggaggctc cctgcgactc 120
tcttcgcgcg catctggctt ttccatacga tcctacgata ttcatctgggt ccgacaagcc 180
cctggcaaaag gtcttgaatg ggttgccact atctcctacg acgggtctca gaaatattat 240
gcggattcag tgaagggggcg gttcaccaatt tcacgagacg agtcaaagaa tacactgtac 300
ctocaaatga attcactgag agccgaggat actgcagtct atcattgtgc aagagggtcc 360
tcaggccact acgaggcctt tgatatatgg ggccaaggca ccttggtaac cgttagtagc 420
ggaggtggag gaagcggagg cggcggttcc ggaggagggt gaagtggagg aggtggatct 480
caatccgttc tgactcagcc accctccgcc tctggcaccg cgggccaacg ggtcacaata 540
tcatgttctg gctcttcaag caacatcgga aactggtagc tgagctggta ccagcagctc 600
cccgccacgg cgcccaagct tctgatatat gacaacaaca aacggcccag tggagtctct 660
gacagattca gtgggtctaa aagtgtgata agcgtagacc tggccataag tggctctgcag 720
agtgaagatg aggcgggacta ttattgcgga acctgggact ccagcctgag cgctccctgg 780
gttttcggcg gaggtactaa gctgaccgtc ttgggcgcgg ccgcaggtgc aggtggtgat 840
tacaagatg atgacgataa aggtgcagcg gcgcacacac atcatcacca c 891

SEQ ID NO: 164      moltype = AA length = 297
FEATURE            Location/Qualifiers
REGION            1..297
                    note = Full-length amino acid sequence of NYA-3061
source            1..297
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 164
MKHLWFFLLL VAAPRWVLSG EVQLVESGGG LVQPGGSLRL SCAASGFSIR SYDIHWVRQA 60
PGKGLEWVAT ISYDGSQKYY ADSVKGRFTI SRDESKNTLY LQMNSLRAED TAVYHCARGS 120
SGHYEAFDIW GQGTFLVTVSS GGGGSGGGGS GGGGSGGGGS QSVLTQPPSA SGTTPGQRVTI 180
SCSGSSSNIQ NWYVSWYQQL PGTAPKLLIY DNNKRPSGVP DRFSGSKSGT SASLAISGLQ 240
SEDEADYYCG TWDSSLSAPW VFGGGTKLTV LGAAAGAGGD YKDDDDKGAA AHHHHHH 297

SEQ ID NO: 165      moltype = DNA length = 903
FEATURE            Location/Qualifiers
misc_feature        1..903
                    note = Full-length nucleotide sequence of NYC-0005
source              1..903
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 165
atgaagcacc tgtggttctt tctgctgctg gtggccgctc ctatagtggt gttgtctggc 60
gaagtgcagc tgggtggaatc tgggtggcga gttgttcagc ctggcggctc tctgagactg 120
agctgtgtgg ccagcgggctt caccctcgac gactacgcca tgtactgggt ccgacagggt 180
ccaggcaagg acctggaatg ggtgtccctg atttctggcg acggcgacat caccctactc 240
gtggactctg tgaagggcag attcaccgtg tccagagaca acaacaagaa cagcctgtac 300
ctgcagatga agtccctgcg cgtggaagat acagccctgt actactgcgc caaggacatg 360
atctactacg cctcttggag cggctacggc agcagcgatt actactacta tgtgatggac 420
gtgtggggcc agggcaccac cgttacagtt tctagcggag gcggaggaag tggcggcgga 480
ggatctggcg gtgtgtgttc tgatatccag atgacacaga gcccagcag cctgtctgcc 540
tctgtgggag acagagtgac catcacctgt agagccagcc agagcatcag cagctacctg 600
aactggatc agcagaagcc cggcaaggcc cctaaactgc tgatctatgc cgctccagt 660
ctgcagagcg gagtgcctag cagattttct ggcagcggct ccggcaccga ttaccacctg 720
accatattca gcctgcagcc tgaggacttc gccacctact attgccagca gagctacagc 780
acccctccta tcacctttgg ccagggaacc agactggaaa tcaaaggcgc tgctgcagcg 840
gtcggcggcg actacaaaga cgatgatgat aagggcgctg ccgctcacca ccaccatcac 900
cat 903

SEQ ID NO: 166      moltype = AA length = 301
FEATURE            Location/Qualifiers
REGION            1..301
                    note = Full-length amino acid sequence of NYC-0005
source            1..301
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 166
MKHLWFFLLL VAAPRWVLSG EVQLVESGGG VQPGGSLRL SCVASGFTFD DYAMYWVRQV 60

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PKKLEWVSL ISGDDGITYY VDSVKGRFTV SRDNNKNSLY LQMKSLRVED TALYYCAKDM 120
 IYYASWSGYG SSDYYYVMD VWGQGTTVTV SSGGGGSGGG GSGGGGSDIQ MTQSPSSLSA 180
 SVGDRVTITC RASQSISSYL NWYQQKPGKA PKLLIYAASS LQSGVPSRFS GSGSGTDFTL 240
 TISSLQPEDF ATYYCQSYS TPPITFGQGT RLEIKGAAAG AGGDYKDDDD KGAAHHHHH 300
 H 301

SEQ ID NO: 167 moltype = DNA length = 903
 FEATURE Location/Qualifiers
 misc_feature 1..903
 note = Full-length nucleotide sequence of NYC-0006
 source 1..903
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 167
 atgaagcacc tgtggttctt tctgtctgtg gtggccgctc ctataggggt gttgtctggc 60
 gaagtgcagc tgggtggaatc tgggtggcga gttgttcagc ctggcggctc tctgagactg 120
 tcttgtgccg ccagcggctt cacttcgac gactacgcca tgactgggt cgcacaggcc 180
 cctggcaaaag gccttgaatg ggtgtccctg atcagcggag atggcggcag catgtactac 240
 gccgatagcg tgaagggcag attcaccatc agccgggaca acagcaagaa cagcctgtac 300
 ctgcagatga actccctgcg gaccgaggac acagccctgt actactgcgc caaggatatg 360
 atcttctacg ccttttggag cggctacggc agcagcgatt actactacta cgtgatggac 420
 gtgtggggcc agggcaccac agtgacagtt tattctggcg gcggaggatc tggcggaggt 480
 ggaagcggag gcggtggatc tgatatccag atgacacaga gccccagcag cctgtctgcc 540
 tctgtgggag acagagtgc catcacctgt cgggccagcc agagaatcag cactacctg 600
 aactggatc agcagaagcc cggcaaggcc cctaagctgc tgatctatgc tgctccagt 660
 ctgcagagcg gcgtgccaag cagatcttct ggcagcggct ctggcacgca cttcaccctg 720
 accatatcta gcctgcagcc tgaggacttc gccacctact actgccagca gagctacagc 780
 accctccta tcaccttgg ccagggaacc agactggaaa tcaaaggcgc tgctgcaggc 840
 gcaggcggcg actacaaaga cgatgatgat aagggcgctg ccgctcacca ccaccatcac 900
 cat 903

SEQ ID NO: 168 moltype = AA length = 301
 FEATURE Location/Qualifiers
 REGION 1..301
 note = Full-length amino acid sequence of NYC-0006
 source 1..301
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 168
 MKHLWFFLLL VAAPRWVLSG EVQLVESGGG VVQPGGSLRL SCAASGTFD DYAMYWRQA 60
 PGKLEWVSL ISGDDGSMY ADVKGRFTI SRDNNKNSLY LQMNSLRTD TALYYCAKDM 120
 IFYAFWSGYG SSDYYYVMD VWGQGTTVTV YSGGGGSGGG GSGGGGSDIQ MTQSPSSLSA 180
 SVGDRVTITC RASQRISTYL NWYQQKPGKA PKLLIYAASS LQSGVPSRFS GSGSGTDFTL 240
 TISSLQPEDF ATYYCQSYS TPPITFGQGT RLEIKGAAAG AGGDYKDDDD KGAAHHHHH 300
 H 301

SEQ ID NO: 169 moltype = DNA length = 888
 FEATURE Location/Qualifiers
 misc_feature 1..888
 note = Full-length nucleotide sequence of NYC-0007
 source 1..888
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 169
 atgaagcacc tgtggttctt tctgtctgtg gtggccgctc ctataggggt gttgtctggc 60
 gaagtgcagc tgggtggaatc tgggtggcga gttgttcagc ctggcggctc tctgagactg 120
 tcttgtgccg ccagcggctt cacttcgac gattatgcca tgcactgggt cgcacaggcc 180
 cctggcaaaag gacttgaatg ggtgtccctg atcagcggcg acggcgacaa tacctactac 240
 gccgatagcg tgaagggcag attcaccatc agccgggaca acaacaagaa cagcctgtac 300
 ctgcagatga actccctgcg gaccgaggac accgccttct actactgtgc caaagagctg 360
 atcttcggca aggtgctgca cgacttttac tactacgtga tggacgtgtg gggccagggc 420
 accacagtga cagtttctag cggaggcgga ggaagtggcg gcggaggatc tggcgggtgt 480
 ggttctgata tccagatgac acagagcccc agcagcctgt ctgcctctgt gggagacaga 540
 gtgaccatca cctgtagagc cagccagagc atcagcagct acctgaactg gtatcagcag 600
 aagcccgcca aggtccctaa actgctgac tatgcgcct ccagtctgca gagcggagtg 660
 cctagcagat tttctggcag cggctccggc accgatttca ccctgacct atctagcctg 720
 cagcctgagg acttcgccac ctactactgc cagcagagct acagcacccc tcctatcacc 780
 tttggccagg gaaccagact ggaatcaaaa ggcgctgctg caggcgagcg cggcgactac 840
 aaagacgatg atgataaggg cgctgccgct caccaccacc atccacct 888

SEQ ID NO: 170 moltype = AA length = 296
 FEATURE Location/Qualifiers
 REGION 1..296
 note = Full-length amino acid sequence of NYC-0007
 source 1..296
 mol_type = protein

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                                organism = synthetic construct
SEQUENCE: 170
MKHLWFFLLL VAAPRWVLSG EVQLVESGGG VVQPGGSLRL SCAASGFTFD DYAMHWVRQA 60
PGKGLEWVSL ISGDDNTYY ADSVKGRTI SRDNNKNSLY LQMNSLRTE TAFYYCAKEL 120
IFGKVLHDFY YYVMDVWQGG TTVTVSSGGG GSGGGGSGGG GSDIQMTQSP SSLASVGD R 180
VTITCRASQS ISSYLNWYQQ KPGKAPKLLI YAASSLQSGV PSRFSGSGSG TDFTLTISSL 240
QPEDFATYYC QQSYSTPPIT FGQGRLEIK GAAAGAGGDY KDDDDKGAAA HHHHHH 296

SEQ ID NO: 171      moltype = DNA length = 888
FEATURE            Location/Qualifiers
misc_feature        1..888
                    note = Full-length nucleotide sequence of NYC-0008
source              1..888
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 171
atgaagcacc tgtggttctt tctgtgctg gtggcgcgtc ctataggggt gttgtctggc 60
gaagtgcagc tgggtggaatc tgggtggcga gttgttcagc ctggcggctc tctgagactg 120
agctgtgaag cctccggctt catcttcgac gactacgcca tgcactgggt ccgacaggca 180
cctggcaaa gctctgaatg ggtgtccctg atctctggcg acggcgacat catctactac 240
gccgactctg tgaagggcag attcaccatc agccgggaca acagcaagaa cagcctgtac 300
ctgcagatga actccctgat catcaggagc acagccctgt actactgcgc caaggattgg 360
gtgttcggcg tcgtgatgac ccactactgg tacttcggcc tggatgtgtg gggccaggga 420
accacagtg aagtttctag cggaggcgga ggaagtggcg gcggaggatc tggcgggtgt 480
ggttctgata tccagatgac acagagcccc agcagcctgt ctgcctctga aggcgacaga 540
gtgacctca cctgtagagc cagccagagc atcagcacct acctgaactg gtatcagcag 600
aagcccgcca agggccctaa gctgctgata tatggtgcct ccagtctgca gagcggcgctg 660
ccaagcagat tttctggcag cggctctggc accgacttca cctgaccat atctagcctg 720
cagcctgagg acttcgccac ctactactgc cagcagtcct acagcacccc tctatcacca 780
tttgccagg gcaccaaggt ggaatcaaaa ggcgctgctg caggcgctgg cgcgactac 840
aaagacgatg atgataaggg cgctgccgct caccaccacc atccaccat 888

SEQ ID NO: 172      moltype = AA length = 296
FEATURE            Location/Qualifiers
REGION              1..296
                    note = Full-length amino acid sequence of NYC-0008
source              1..296
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 172
MKHLWFFLLL VAAPRWVLSG EVQLVESGGG VVQPGGSLRL SCEASGFIFD DYAMHWVRQA 60
PGKGLEWVSL ISGDDNTYY ADSVKGRTI SRDNNKNSLY LQMNSLIIE TALIYYCAKDW 120
VFGVVMTHYW YFGLDVWQGG TTVTVSSGGG GSGGGGSGGG GSDIQMTQSP SSLASSEGDR 180
VTITCRASQS ISTYLNWYQQ KPGKAPKLLI YGASSLQSGV PSRFSGSGSG TDFTLTISSL 240
QPEDFATYYC QQSYSTPPIT FGQGRKVEIK GAAAGAGGDY KDDDDKGAAA HHHHHH 296

SEQ ID NO: 173      moltype = DNA length = 885
FEATURE            Location/Qualifiers
misc_feature        1..885
                    note = Full-length nucleotide sequence of NYC-0009
source              1..885
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 173
atgaagcacc tgtggttctt tctgtgctg gtggcgcgtc ctataggggt gttgtctggc 60
gaagtgcagc tgggtggaatc tgggtggcga gttgttcagc ctggcggctc tctgagactg 120
tcttgtgccg ccagcggctt cacttcgac gattatgcca tgcactgggt ccgacaggcc 180
cctggcaaa gacttgaatg ggtgtccctg atctctggcg gagcggcgcg aacctactac 240
agcgattctg tgaagggcag attcaccatc agccgggaca ccagcaagga cagcctgtac 300
ctgcagatga acagcctgcg gaccgaggac acagccctgt actactgcgc caaggacatg 360
gtgttcggcg tggtaacccc ttactactac ttgcacctgg atgtgtgggg ccagggcaca 420
acagtgcagc tctcttctgg cggcggagga agcggaggcg gaggatccgg tgggtggtgga 480
tctgacatcc agatgacaca gagccccagc agcctgtctg cctctgtggg agacagagtg 540
accatcacct gtagagccag ccagagcacc aacagctacc tgaactggta tcagcagaag 600
ccggcaagg cccctaagct gctgatctat gctgcctcca gtctgcagag cgcgctgcca 660
agcagatttt ctgacagcgg ctctggcacc gacttcaccc tgacatatac tagcctgcag 720
cctgaggact tcgccacctc ctactgccag cagagctaca gcgccccctc tatcacattt 780
ggccagggaa ccagactgga aatcaaaagg gctgctgcag gcgacggcgg cgactacaaa 840
gacgatgatg ataagggcgc tgccgctcac caccatcacc atccat 885

SEQ ID NO: 174      moltype = AA length = 295
FEATURE            Location/Qualifiers
REGION              1..295
                    note = Full-length amino acid sequence of NYC-0009
source              1..295
                    mol_type = protein

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                                organism = synthetic construct
SEQUENCE: 174
MKHLWFFLLL VAAPRWVLSG EVQLVESGGG VVQPGGSLRL SCAASGFTFD DYAMHWVRQA 60
PGKLEWVSL  ISGGGGTTY  SDSVKGRFTI SRDTSKDSLY LQMNSLRTEF TALYYCAKDM 120
VFGVVTPIYY FALDVWGQGT TVTVSSGGGG SGGGGSGGGG SDIQMTQSPS SLSASVGDRV 180
TITCRASQSI NSYLNWYQQK PGKAPKLLIY AASSLQSGVP SRFSGSGSGT DFTLTISSLQ 240
PEDFATYYCQ QSYSPAPITF GQGTRLEIKG AAAGAGGDYK DDDDKGAAA HHHHHH 295

SEQ ID NO: 175      moltype = DNA  length = 888
FEATURE            Location/Qualifiers
misc_feature        1..888
                    note = Full-length nucleotide sequence of NYC-0010
source              1..888
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 175
atgaagcacc tgtggttctt tctgtgctg gtggcgcctc ctataggggt gttgtctggc 60
gaagtgcagc tgggtggaatc tgggtggcga gttgttcagc ctggcggctc tctgagactg 120
agctgtgctg tgctcggctt cacttcgac gactacgcca tgcactgggt ccgacaggca 180
cctggcaaaag gccttgaatg ggtgtccctg atcagcggag atggcggcag cacacactac 240
gccgattctg tgaagggcag attcaccatc agccgggaca acagcaagaa cagcctgtac 300
ctgcagatga actccctgcg gacaggcgac acagccctgt actactgcgc caaggacatg 360
atcttcgcgc tgggtcatcac cgactaccac tactacggca tggacgtgtg gggccaggga 420
accacagatg cagtttctag cggaggcgga ggaagtggcg gcggaggatc tggcgggtgt 480
ggttctgata tccagatgac acagagcccc agcagcctgt ctgcctctgt gggagacaga 540
gtgaccatca cctgtagagc cagccagagc atcagcagct acctgaactg gtatcagcag 600
aagcccggca agggccctaa actgctgata tatgccgcct ccagtctgca gagcggagtg 660
ctagcagatg tttctggcag cggtccggcg accgatttca cctgaccat atctagcctg 720
cagcctgagg acttcgccac ctactactgc cagcagagct acagcacccc tcctatcacc 780
tttgccaggg gcaccagact ggaatcaaaa ggcgctgctg caggcgctgg cggcgactac 840
aaagacgatg atgataaggg cgctgccgct caccaccacc atcaccat 888

SEQ ID NO: 176      moltype = AA  length = 296
FEATURE            Location/Qualifiers
REGION            1..296
                    note = Full-length amino acid sequence of NYC-0010
source            1..296
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 176
MKHLWFFLLL VAAPRWVLSG EVQLVESGGG VVQPGGSLRL SCAVSGFTFD DYAMHWVRQA 60
PGKLEWVSL  ISGDDGTHY  ADSVKGRFTI SRDTSKNSLY LQMNSLRTEF TALYYCAKDM 120
IFAVITDHYH YYGMDVWQGG TTVTVSSGGG SGGGGSGGGG GSDIQMTQSP SLSASVGDR 180
VTITCRASQS ISSYLNWYQQ KPGKAPKLLI YAASSLQSGV PSRFSGSGSG TDFTLTISSL 240
QPEDFATYYC QSYSTPPITF FGQTRLEIKG GAAAGAGGDY KDDDDKGAAA HHHHHH 296

SEQ ID NO: 177      moltype = DNA  length = 738
FEATURE            Location/Qualifiers
misc_feature        1..738
                    note = Full-length nucleotide sequence of HC-h
source              1..738
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 177
atgaaacacc tgtggttctt cctcctgctg gtggcagctc ccagatgggt gctgagcgac 60
aaaactcaca catgcccacc ctgcccagca cctgaagccg ccggcggacc ctcagtcttc 120
ctcttcccc  caaaacccaa ggacaccctc atgatctccc ggaccctga ggtcacatgc 180
gtggtgtgtg acgtgagcca cgaagaccct gaggtcaagt tcaactggta cgtggacggc 240
gtggaggtgc ataagccaa  gaccaagcct agagagggaac agtacaacag cacctaccgg 300
gtggtgtccg tgctgacagt gctgcaccag gactggctga acggcaaaag gtacaagtgc 360
aaggtgtcca acaaggccct gcctgcccc  atcgagaaaa ccatcagcaa ggccaagggc 420
cagccccgcg aacctcaagt gtgcaccctg ccaccctccc gggatgagct gaccaagaac 480
caggtgtccc tgagctgtgc cgtgaagggc ttctaccctc ccgatatgc cgtggaatgg 540
gagagcaacg gccagcccg  gaacaactac aagaccaccc ccctgtgtgt ggacagcgac 600
ggctcattct tctgggtgtc caagctgacc gtggacaagt ccagatggca gcagggaac 660
gtcttctcat gctcgtgat  gcatgaggct ctgcacaacc actacaccca gaagagcctc 720
tcctgtcttc ccggcaaa  738

SEQ ID NO: 178      moltype = AA  length = 246
FEATURE            Location/Qualifiers
REGION            1..246
                    note = Full-length amino acid sequence of HC-h
source            1..246
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 178

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MKHLWFFLLL	VAAPRWLSD	KTHTCPPCPA	PEAAGGPSVF	LFPKPKDIL	MISRTPEVTC	60
VVDVSHEDP	EVKNWYVDG	VEVHNAKTKP	REEQYNSTYR	VSVLTVLHQ	DWLNGKEYKC	120
KVSNKALPAP	IEKTISKAKG	QPPEQVQCTL	PPSRDELTKN	QVSLSCAVKG	FYPSPDIAVEW	180
ESNGQPENNY	KTPPVLDSD	GSFPLVSKLT	VDKSRWQGN	VFSCSVMHEA	LHNHYTQKSL	240
LSLSPGK						246

SEQ ID NO: 179 moltype = DNA length = 1500
 FEATURE Location/Qualifiers
 misc_feature 1..1500
 note = Full-length nucleotide sequence of NYD-2047-HC-k
 source 1..1500
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 179

atgaagcacc	tgtggttctt	tctgctgctg	gtggccgctc	ctagatgggt	gctgtctggt	60
caagtacagc	tggtggaatc	cgggtggaggc	gtggtccagc	cgggacgcag	cttgagactg	120
tcttgccgtg	catctggctt	ttccatacga	tcctacgata	tgcaactgggt	tcgccaagcc	180
cctggcaaa	gtcttgaatg	ggttgccact	atctcctacg	acgggtctca	gaaatattac	240
gcggattcag	tgaaggggag	gttcacaatt	tcacgggaca	attcaaaaga	taccttgat	300
ctccagatgt	cttcaactgag	agccgaggat	actgcagtct	atcattgtgc	aagaggggtc	360
tcaggccact	acgaggcctt	tgatatatgg	ggccaaggca	ccttggtaac	cgttagtagc	420
ggaggtggag	gaagcggagg	cggcggttcc	ggaggagggtg	gaagtcaatc	cgttctgact	480
cagccaccct	cgtgagtg	agcgctgga	cagaaggcca	caatatcatg	ttctggctct	540
tcaagcaaca	tcggaaactg	gtacgtgagc	tggtaccagc	agctccccgg	cacggcgccc	600
aagcttctga	tatatgacaa	caacaacagg	cccagtgga	tcctgacag	attcagtggt	660
tctaaaagt	gtacaagcgc	taccctgggt	atcacccgat	tgcaagccgg	agatgaggcg	720
gactattatt	gcggaacctg	ggactccagc	ctgagcgctc	cctgggtttt	cggcggagggt	780
actaaggtta	ccgtcttggg	cggagaaccc	aagagcgcag	acaagaccca	cacctgtcct	840
ccatgtcctg	ctccagaagc	tgccagcgcc	ccttcctgtg	ttctgttccc	tccaaagcct	900
aaggacaccc	tgatgatcag	cgggacacct	gaagtgcact	gcgtgggtgt	ggatgtgtcc	960
cacgaggatc	ccgaagtga	gttcaattgg	tacgtggacg	gcgtggaaagt	gcacaacgcc	1020
aagaccaagc	ctagagagga	acagtagaca	agcacctaca	gagtggtgtc	tgtgtctgaca	1080
gtgctgcacc	aggactggct	gaacggcaaa	gagtagaagt	gcaaggtgtc	caacaaggcc	1140
ctgcctgctc	ctatcgagaa	aacctcagc	aaggccaagg	gccagcctag	ggaacccag	1200
gtttacaccc	tgccctccatg	cagggatgag	ctgaccaaga	accaggtgtc	cctgtggtgc	1260
ctggttaagg	gcttctaccc	ctccgatatc	gccgtggaat	gggagagcaa	tgccagacca	1320
gagaacaact	acaagacaac	cctctctgtg	ctggactccg	acggtcatt	cttctgtgac	1380
agcaagctga	ccgtggacaa	gagcagatgg	cagcagggca	acgtgttcag	ctgcagcgtg	1440
atgcacgagg	ccctgcacaa	ccactacaca	cagaagtccc	tgtctctgag	ccccggcaaa	1500

SEQ ID NO: 180 moltype = AA length = 500
 FEATURE Location/Qualifiers
 REGION 1..500
 note = Full-length amino acid sequence of NYD-2047-HC-k
 source 1..500
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 180

MKHLWFFLLL	VAAPRWLSG	QVQLVESGGG	VVQPGRLRL	SCAASGFSIR	SYDMHWVRQA	60
PGKGLEWVAT	ISYDGSQKYY	ADSVKGRFTI	SRDNSKNTLY	LQMSSLRAED	TAVYHCARGS	120
SGHYEAFDIW	QGGTLVTVSS	GGGGSGGGGS	GGGGSQSVLT	QPPSVSAAPG	QKVTISCSGS	180
SSNIGNWYVS	WYQLPGTAP	KLLIYDNMKR	PSGIPDRFSG	SKSGTSATLG	ITGLQTGDEA	240
DYYCGTWDS	LSAPWVFGG	TKVTVLGGEP	KSADKTHTCP	PCPAPEAAGG	PSVFLFPPKP	300
KDTLMISRT	EVTCVVVDVS	HEDPEVKFNW	YVDGVEVHNA	KTKPREEQYN	STYRVVSVLT	360
VLHQDLWNGK	EYCKVSNKA	LPAPIEKTIS	KAKGPREPQ	VYTLPPCRDE	LTKNQVSLWC	420
LKGFYPSDI	AVEWESNGQ	ENNYKTPPV	LDSGGSFFLY	SKLTVDKSRW	QQGNVFSCSV	480
MHEALHNHYT	QKSLSLSPGK					500

SEQ ID NO: 181 moltype = DNA length = 1500
 FEATURE Location/Qualifiers
 misc_feature 1..1500
 note = Full-length nucleotide sequence of NYD-2061-HC-k
 source 1..1500
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 181

atgaagcacc	tgtggttctt	tctgctgctg	gtggccgctc	ctagatgggt	gctgtctggt	60
gaggtacagc	tggtggaatc	cgggtggaggc	cttggtgcaac	cgggaggctc	cctgcgactc	120
tcttgccgag	catctggctt	ttccatacga	tcctacgata	ttcattgggt	ccgacaagcc	180
cctggcaaa	gtcttgaatg	ggttgccact	atctcctacg	acgggtctca	gaaatattat	240
gcggattcag	tgaaggggag	gttcacaatt	tcacgagacg	agtcaaaaga	tacactgtac	300
ctccaaatga	attcaactgag	agccgaggat	actgcagtct	atcattgtgc	aagaggggtc	360
tcaggccact	acgaggcctt	tgatatatgg	ggccaaggca	ccttggtaac	cgttagtagc	420
ggaggtggag	gaagcggagg	cggcggttcc	ggaggagggtg	gaagtcaatc	cgttctgact	480
cagccaccct	ccgcctctgg	cacccggggc	caacgggtca	caatatcatg	ttctggctct	540
tcaagcaaca	tcggaaactg	gtacgtgagc	tggtaccagc	agctccccgg	cacggcgccc	600

-continued

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aagcttctga tatatgacaa caacaaacgg cccagtggag ttcttgacag attcagtggg 660
tctaaaagtg gtacaagcgc tagcctggcc ataagtggtc tgcagagtga agatgaggcg 720
gactattatt gcggaacctg ggactccagc ctgagcgctc cctgggtttt cggcggagggt 780
actaagctga ccgtcttggg cggagaaccc aagagcgcag acaagaccca cacctgtcct 840
ccatgtcctg ctccagaagc tgcaggcggc ccttcctgtg ttctgttccc tccaaagcct 900
aaggacaccc tgatgatcag ccggacacct gaagtgaacct gcgtgggtgt ggatgtgtcc 960
cacgaggatc ccgaagtga gttcaattgg tacgtggacg gcgtggaagt gcacaacgcc 1020
aagaccaagc cttagagagg acagtacaac agcacctaca gagtgggtgc tgtgtgaca 1080
gtgtgtcacc aggactggct gaacggcaaa gagtacaagt gcaaggtgtc caacaaggcc 1140
ctgctgtctc ctatcgagaa aaccatcagc aaggccaagg gccagcctag ggaaccccag 1200
gtttacaccc tgctccatg cagggatgag ctgaccaaga accaggtgtc cctgtgggtg 1260
ctgggttaag gcttctaccc ctccgatc gccgtggaat gggagagcaa tggccagcca 1320
gagaacaact acaagacaaa cctcctgtg ctggactccg acggctcatt ctctctgtac 1380
agcaagctga ccgtggacaa gacgagtgag cagcagggca acgtgttcag ctgcagcgtg 1440
atgcacgagg ccctgcacaa ccactacaca cagaagtccc tgtctctgag ccccgcaaaa 1500

SEQ ID NO: 182      moltype = AA  length = 500
FEATURE            Location/Qualifiers
REGION              1..500
                    note = Full-length amino acid sequence of NYD-2061-HC-k
source              1..500
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 182
MKHLWFFLLL VAAPRWLVSG EVQLVESGGG LVQPGGSLRL SCAASGFSIR SYDIHWVRQA 60
PGKGLEWVAT ISYDGSQKYY ADSVKGRFTI SRDESKNTLY LQMNSLRAED TAVYHCARGS 120
SGHYEAFDIW QGGTLVTVSS GGGGSGGGGS GGGGSGSVLT QPPSASGTPG QRVTIISCSGS 180
SNIGNWVYVS WYQLPGTAP KLLIYDNMKR PSGVPDRFSG SKSGTSASLA ISGLQSEDEA 240
DYYCGTWDS SLSAPWVFGG TKLTVLGGE PCADKTHTCP PCPAPEAAGG PSVFLFPPKP 300
KDTLMISRT ETVCVVDVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYN STYRVVSVLT 360
VLHQDWLNGK EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPCRDE LTKNQVSLWC 420
LVKGFYPSDI AVEWESNGQP ENNYKTTTPV LDSDGSFFLY SKLTVDKSRW QQGNVFSCSV 480
MHEALHNHYT QKSLSLSPGK                                     500

SEQ ID NO: 183      moltype = DNA  length = 1515
FEATURE            Location/Qualifiers
misc_feature        1..1515
                    note = Full-length nucleotide sequence of NYD-3061-HC-k
source              1..1515
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 183
atgaagcacc tgtgtgttct tctgtgtctg gtggccgctc ctatagtggt gctgtcttgt 60
gaggtagacg tgggtgaatc cgggtggagg cttgtgcaac cgggaggctc cctgcgactc 120
tcttgcgcgc catctggctt ttccatacga tcctacgata ttcatgtggg cgcgacaagcc 180
cctggcaaa gttctgaatg ggttgccact atctcctacg acgggtctca gaaatattat 240
gcggtattcg tgaagggggc gttcaccaat tcacgagacg agtcaaaaga tacactgtac 300
ctccaaatga attcactgag agccgaggat actgcagctc atcattgtgc aagaggggtc 360
tcaggccact acgaggccct tgatatatgg ggccaaggca ccttggtaac cgttagtagc 420
ggaggtggag gaagcggagg cggcggttcc ggaggagggt gaagtggagg aggtggatct 480
caatccgttc tgactcagcc accctccgcc tctggcacc cgggcccaac ggtcacataa 540
tcatgttctg gctcttcaag caacatcgga aactggtagc tgagctggta ccagcagctc 600
cccgccacgg gcctccaaagt tctgatatat gacaacaaca aacggcccag tggagtctct 660
gacagattca gtgggtctaa aagtgtgata agcgctagcc tggccataag tggctctgag 720
agtgaagatg aggcggacta ttattgcgga acctgggact ccagcctgag cgctccctgg 780
gttttcggcg gaggtactaa gctgacccgtc ttgggaggag aacccaagag cgcagacaag 840
acccacacct gtcctccatg tctgtctcca gaagctgcag gcggcccttc cgtgtttctg 900
ttccctccaa agcctaagga caccctgatg atcagccgga cacctgaagt gacctgcgtg 960
gtggtggatg tgtcccacga ggatcccga gtagaagttc attggtagct ggacggcgtg 1020
gaagtgcaca acgccaagac caagcctaga gaggaacagt acaacagcac ctacagagtg 1080
gtgtctgtgc tgacagtgtc gcaccaggac tgggtgaacg gcaaaagata caagtgcag 1140
gtgtccaaca aggcctctgc tgcctctatc gagaaaacca tcagcaaggc caagggccag 1200
cctagggaac cccaggttta caccctgcct ccatgcaggg atgagctgac caagaaccag 1260
gtgtccctgt ggtgcctggt taagggtctc taacctccg atatcgccgt ggaatgggag 1320
agcaatggcc agccagagaa caactacaag acaacccctc ctgtgtgga ctcgcagcgc 1380
tcattcttcc tgtacagcaa gctgaccgtg gacaagagca gatggcagca gggcaacgtg 1440
ttcagctgca gcgtgatgca cgaggccctg cacaaccact acacacagaa gtccctgtct 1500
ctgagccccc gcaaaa                                     1515

SEQ ID NO: 184      moltype = AA  length = 505
FEATURE            Location/Qualifiers
REGION              1..505
                    note = Full-length amino acid sequence of NYD-3061-HC-k
source              1..505
                    mol_type = protein
                    organism = synthetic construct

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-continued

SEQUENCE: 184

MKHLWFFLLL	VAAPRWVLSG	EVQLVESGGG	LVQPGGSLRL	SCAASGFSIR	SYDIHWVRQA	60
PGKGLEWVAT	ISYDGSQKYY	ADSVKGRFTI	SRDESKNTLY	LQMNSLRAED	TAVYHCARGS	120
SGHYEAFDIW	GQGLTVTVSS	GGGSGGGGGS	GGGSGGGGGS	QSVLTQPPSA	SGTPGQRVTI	180
SCSGSSSNIG	NWYVSWYQQL	PGTAPKLLIY	DNNKRPSGVP	DRFSGSKSGT	SASLAISGLQ	240
SEDEADYYCG	TWDSLSAPW	VFGGGTKLTV	LGGEPKSADK	THTCPPCPAP	EAAGGPSVFL	300
FPPKPKDTLM	ISRTPEVTCV	VVDVSHEDPE	VKFNWYVDGV	EVHNAKTKPR	EEQYNSTYRV	360
VSVLTVLHQD	WLNGKEYKCK	VSNKALPAPI	EKTISKAKGQ	PREPQVYTL	PCRDELTKNQ	420
VSLWCLVKGF	YPSDIAVEWE	SNGQPENNYK	TTTPVLDSDG	SFFLYSKLTV	DKSRWQQGNV	480
FSCSVMHEAL	HNHYTQKSLS	LSPGK				505

SEQ ID NO: 185 moltype = DNA length = 1527

FEATURE Location/Qualifiers

misc_feature 1..1527

 note = Full-length nucleotide sequence of NYC-0011-HC-k

source 1..1527

 mol_type = other DNA

 organism = synthetic construct

SEQUENCE: 185

atgaagcacc	tgtggttctt	tctgctgctg	gtggccgctc	ctagatgggt	gctgtctggc	60
gaagtgcagc	tggtggaatc	tggtggcgga	gttggttcagc	ctggcggctc	tctgagactg	120
agctgtgtgg	ccagcggtct	caccttcgac	gactacgcca	tgtactgggt	ccgacagggt	180
ccaggcaagg	acctggaatg	ggtgtccctg	attttctggc	acggcgacat	cacctactac	240
gtggactctg	tgaagggcag	attcaccgtg	tccagagaca	acaacaagaa	cagcctgtac	300
ctgcagatga	agtcctctgc	cgtggaagat	acagccctgt	actactgcgc	caaggacatg	360
atctactacg	cctcttggag	cggctacggc	agcagcgatt	actactacta	tgtgatggac	420
gtgtggggcc	agggcaccac	cgttacagtt	tctagcggag	gcggaggaa	tgccggcgga	480
ggatctggcg	gtggtggttc	tgatatccag	atgacacaga	gccccagcag	cctgtctgcc	540
tctgtgggag	acagagtgac	catcacctgt	agagccagcc	agagcatcag	cagctacctg	600
aactgggtatc	agcagaagcc	cggcaaggcc	cctaaactgc	tgatctatgc	cgctccagct	660
ctgcagagcg	gagtgctcag	cagattttct	ggcagcggct	ccggcaccga	tttcacctg	720
accatatcta	gcctgcagcc	tgaggacttc	gccacctact	attgccagca	gagctacagc	780
acccctccta	tcacctttgg	ccagggaacc	agactggaaa	tcaaggcg	agaacccaag	840
agcgacagca	agaccacac	ctgtcctcca	tgctctgctc	cagaagctgc	aggcgccct	900
tccgtgttcc	tggtccctcc	aaagcctaag	gacacctga	tgatcagccg	gacacctgaa	960
gtgacctcgc	tggtggtgga	tgtgtccccc	gaggatcccc	aagtgaagtt	caattggtac	1020
gtggacggcg	tgggaagtga	ccaagccca	gagaggaaca	gtacaacagc		1080
acctacagag	tggtgtctgt	gctgacagtg	ctgcaccagg	actggctgaa	cggaacagag	1140
tacaagtcca	aggtgtccaa	caaggccctg	cctgtctcta	tcgagaaaa	catcagcaag	1200
gccaagggcc	agcctagggg	accctagggt	tacaccctgc	ctccatgcag	ggatgagctg	1260
accaagaacc	aggtgtccct	gtggtgcctg	gttaagggct	tctacccttc	cgatatcgcc	1320
gtggaatggg	agagcaatgg	ccagccagag	aacaactaca	agacaacccc	tcctgtgctg	1380
gactccgacg	gctcattctt	cctgtacagc	aagctgaccg	tggacaagag	cagatggcag	1440
cagggcaacg	tggtcagctg	cagcgtgatg	cacgaggccc	tgacacaaca	ctacacacag	1500
aagtccctgt	ctctgagccc	cggcaaaa				1527

SEQ ID NO: 186 moltype = AA length = 509

FEATURE Location/Qualifiers

REGION 1..509

 note = Full-length amino acid sequence of NYC-0011-HC-k

source 1..509

 mol_type = protein

 organism = synthetic construct

SEQUENCE: 186

MKHLWFFLLL	VAAPRWVLSG	EVQLVESGGG	VVQPGGSLRL	SCVASGFTFD	DYAMYWVRQV	60
PGKLEWVSL	ISGDGDITYY	VDSVKGRFTV	SRDNNKNSLY	LQMKSLRVED	TALYYCAKDM	120
IYYASWSGYG	SSDYVVVMD	VWQGTITVTV	SSGGGSGGGG	GSGGGSDIQ	MTQSPSSLSA	180
SVGDRVITIC	RASQSISSYL	NWYQQKPGKA	PKLLIYAASS	LQSGVPSRFS	SGSGTDFTL	240
TISSLQPEDF	ATYYCQQSYS	TPPITFGQGT	RLEIKGGEPK	SADKTHTCPP	CPAPEAAGGP	300
SVFLFPPKPK	DTLMISRTPE	VTCVVVDVSH	EDPEVKFNWY	VDGVEVHNAK	TKPREEQYNS	360
TYRVVSVLTV	LHQVDLNGKE	YKCKVSNKAL	PAPIETISK	AKGQPREPQV	YTLPPCRDEL	420
TKNQVSLWCL	VKGFPSPDIA	VEWESNGQPE	NNYKTTTPVL	DSGQSFPLY	KLTVDKSRWQ	480
QGNVFSVSM	HEALHNHYTQ	KSLSLSPGK				509

SEQ ID NO: 187 moltype = DNA length = 1527

FEATURE Location/Qualifiers

misc_feature 1..1527

 note = Full-length nucleotide sequence of NYC-0012-HC-k

source 1..1527

 mol_type = other DNA

 organism = synthetic construct

SEQUENCE: 187

atgaagcacc	tgtggttctt	tctgctgctg	gtggccgctc	ctagatgggt	gctgtctggc	60
gaagtgcagc	tggtggaatc	tggtggcgga	gttggttcagc	ctggcggctc	tctgagactg	120
tcttgtgccg	ccagcggtct	caccttcgac	gactacgcca	tgtactgggt	ccgacaggcc	180
cctggcaaa	gccttgaatg	ggtgtccctg	atcagcggag	atggcggcag	catgtactac	240

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gccgatagcg tgaagggcag attcaccatc agccgggaca acagcaagaa cagcctgtac 300
ctgcagatga actccttgcg gaccgaggac acagccctgt actactgcgc caaggatatg 360
atcttctacg ccttttgtag cggctacggc agcagcgatt actactacta cgtgatggac 420
gtgtggggcc agggcaccac agtgacagtt tattctggcg gcggaggatc tggcggagggt 480
ggaagcggag gcggtggatc tgatatccag atgacacaga gcccagcag cctgtctgcc 540
tctgtgggag acagagtgc catcacctgt cgggccagcc agagaatcag cacctacctg 600
aactggatc agcagaagcc cggcaaggcc cctaagctgc tgatctatgc tgcctccagt 660
ctgcagagcg gcgtgccaa gcatgtttct ggcagcggct ctggcacoga cttcacctg 720
accatatcta gccctgcagc tgaggacttc gccacctact actgccagca gagctacagc 780
acccctccta tcacctttgg ccagggaacc agactggaaa tcaaaggcgg agaaccacag 840
agcgcagaca agacccacac ctgtcctcca tgtcctgctc cagaagctgc aggcggccct 900
tccgtgttct tgttccctcc aaagcctaag gacacctga tgatcagccg gacacctgaa 960
gtgacctgcy tgggtgtgga tgtgtccccc gaggatcccg aagtgaagtt caattggtac 1020
gtggacggcg tggagtgca caacgccta accaagccta gagaggaca gtacaacagc 1080
acctacagag tgggtgtctgt gctgacagtg ctgacaccag actggctgaa cggcaaaag 1140
tacaagtga aggtgtccaa caaggccctg cctgctccta tcgagaaaac catcagcaag 1200
gccaaggccc agcctaggga accccaggtt tacacctgc ctccatgca ggtatgagct 1260
accaagaacc aggtgtccct gtggtgctct gtttaagggt tctacctctc cgatatcgcc 1320
gtggaatggg agagcaatgg ccagccagag aacaactaca agacaacccc tctgtgtctg 1380
gactccgacg gctcattctt cctgtacagc aagctgacgg tggacaagag cagatggcag 1440
cagggcaacg tgttcagctg cagcgtgatg cagcaggccc tgcacaacca ctacacacag 1500
aagtcctctg ctctgagccc cggcaaa 1527

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SEQ ID NO: 188      moltype = AA length = 509
FEATURE
REGION              1..509
                    note = Full-length amino acid sequence of NYC-0012-HC-k
source              1..509
                    mol_type = protein
                    organism = synthetic construct

```

```

SEQUENCE: 188
MKHLWFFLLL VAAPRWVLSG EVQLVESGGG VQPGGSLRL SCAASGFTFD DYAMYWVRQA 60
PKGLEWVSL ISGDSGSMYY ADSVKGRFTI SRDNSKNSLY LQMNSLRTE TALYYCAKDM 120
IFYAFWSGYG SSDYVVYVMD VWGQGTITV YSGGGSGGGG GSGGGSDIQ MTQSPSSLSA 180
SVGDRVTITC RASQRISTYL NWYQQKPGKA PKLLIYAASS LQSGVPSRFS GSGSGTDFTL 240
TISSLQPEDF ATYYCQSYS TPPIFGQGT RLEIKGGEPK SADKTHCTPP CPAPEAAGGP 300
SVFLFPKPKF DTLMISRTPE VTCVVVDVSH EDPEVKFNWY VDGVEVHNAK TKPREQYNS 360
TYRVVSVLTV LHQDNLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPCRDEL 420
TKNQVSLWCL VKGFYPSDIA VEWESNGQPE NNYKTPPVL DSDGSFFLYS KLTVDKSRWQ 480
QGNVFSQSVH HEALHNYHTQ KSLSLSPGK 509

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SEQ ID NO: 189      moltype = DNA length = 1512
FEATURE
misc_feature        1..1512
                    note = Full-length nucleotide sequence of NYC-0013-HC-k
source              1..1512
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 189
atgaagcacc tgtggttctt tctgctgctg gtggccgctc ctataggggt gctgtctggc 60
gaagtgcagc tgggtgaatc ttgtggcgga gttgttcagc ctggcggctc tctgagactg 120
tcttgtgcgc ccagcggtt cacttcgac gattatgcca tgcactgggt ccgacaggcc 180
cctggcaag gacttgaatg ggtgtccctg atcagcggcg acggcgacaa tacctactac 240
gccgatagcg tgaagggcag attcaccatc agccgggaca acaacaagaa cagcctgtac 300
ctgcagatga actccttcgc gaccgaggac accgccttct actactgtgc caaagagctg 360
atcttcggca aggtgctgca cgacttttac tactacgtga tggacgtgtg gggccagggc 420
accacagtg cagtttctag cggaggcgga ggaagtggcg gcggaggatc tggcgggtgt 480
ggttctgata tccagatgac acagagcccc agcagcctgt ctgcctctgt gggagacaga 540
gtgaccatca cctgtagagc cagccagagc atcagcagct acctgaactg gtatcagcag 600
aagcccgga agggccctaa actgctgctc tatgcgcctt ccagtctgca gagcggagtg 660
cctagcagat tttctggcag cggctccggc accgatttca cctgacctat atctagcctg 720
cagcctgagg acttcgccac ctactactgc cagcagagct acagaccccc tctatcaccc 780
tttgccagg gaaccagact ggaatcaaaa ggcggagaac ccaagagcgc agacaagacc 840
cacacctgtc ctccatgtcc tgtccagaa gctgcaggcg gcccttccgt gtttctgttc 900
cctccaaagc ctaaggacac cctgatgatc agccggacac ctgaagtgtg ctgcgtgtgtg 960
gtggatgtgt cccacagga tcccgaagtg aagttcaatt ggtacgtgga cggcgtggaa 1020
gtgcacaacg ccaagaccaa gccatagagag gaacagtaca acagcaccta cagagtgggt 1080
tctgtgctga cagtgtctga ccaggactcg ctgaacggca aagagtacaa gtgcaagggt 1140
tccaacaagg cctgtcctgc tctatcgag aaaaccatca gcaaggccaa gggccagcct 1200
agggaaaccc aggtttacac cctgcctcca tgcagggatg agctgaccaa gaaccagggt 1260
tccctgttgt gccctgttaa gggcttctac cctccgata tgcctgtgga atgggagagc 1320
aatggccagc cagagaacaa ctacaagaca acccctcctg tgcgtggact cgacggctca 1380
ttcttctgt acagcaagct gaccgtggac aagagcagat ggcagcagg caactgtgtc 1440
agctgcagcg tgatgcacga ggcctgcac aaccactaca cacagaagtc cctgtctctg 1500
agcccgga aa 1512

```

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SEQ ID NO: 190 moltype = AA length = 504
 FEATURE Location/Qualifiers
 REGION 1..504
 note = Full-length amino acid sequence of NYC-0013-HC-k
 source 1..504
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 190
 MKHLWFFLLL VAAPRWVLSG EVQLVESGGG VVQPGGSLRL SCAASGFTFD DYAMHWVRQA 60
 PGKGLEWVSL ISGDDNTYY ADSVKGRFTI SRDNNKNSLY LQMNSLRTE TAFYYCAKEL 120
 IFGKVLHDFY YYVMDVWQGG TTVTVSSGGG GSGGGGSGGG GSDIQMTQSP SSLASVGDGR 180
 VTITCRASQS ISSYLNWYQQ KPGKAPKLLI YAASSLQSGV PSRFSGSGSG TDFTLTISSL 240
 QPEDFATYYC QQSSTPPIIT FGQGRLEIK GGEPKSADKT HTCPCCPAPE AAGGPSVFLF 300
 PPKPKDTLMI SRTPEVTCVV VDVSHEDPEV KFNWYVDGVE VHNAKTKPRE EQYNSTYRVV 360
 SVLTVLHQDW LNGKEYCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP CRDELTKNQV 420
 SLWCLVKGFY PSDIAVEWES NGQPENNYKT TPPVLDSGGS FFLYSLKLTVD KSRWQQGNVF 480
 SCSVMHEALH NHYTQKSLSL SPGK 504

SEQ ID NO: 191 moltype = DNA length = 1512
 FEATURE Location/Qualifiers
 misc_feature 1..1512
 note = Full-length nucleotide sequence of NYC-0014-HC-k
 source 1..1512
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 191
 atgaagcacc tgtggttctt tctgctgctg gtggccgctc ctagatgggt gctgtctggc 60
 gaagtgcagc tgggtggaatc tgggtggcga gttgttcagc ctggcggctc tctgagactg 120
 agctgtgaag cctccggctt catcttcgac gactacgcca tgcactgggt ccgacaggca 180
 cctggcaaaag gccttgaatg ggtgtccctg atctctggcg acggcgacat catctactac 240
 gccgactctg tgaagggcag attcaccatc agccgggaca acagcaagaa cagcctgtac 300
 ctgcagatga actccctgat catcgaggac acagccctgt actactgcgc caaggattgg 360
 gtgttcggcg tcgtgatgac ccactactgg tacttcggcc tggatgtgtg gggccaggga 420
 accacagtga cagtttctag cggaggcgga ggaagtggcg gcggaggatc tggcgggtgt 480
 ggttctgata tccagatgac acagagcccc agcagcctgt ctgcctctga aggcgacaga 540
 gtgaccatca cctgtagagc cagccagagc atcagcacct acctgaactg gtatcagcag 600
 aagcccgcca agccccccta gctgctgac tatggtgcct ccagtctgca gagcggcgtg 660
 ccaagcagat tttctggcag gcgctctggc accgacttca cctcgacct atctagcctg 720
 cagcctgagg acttcgccac ctactactgc cagcagtcct acagcacccc tcctatcaca 780
 tttggccagg gcaccaaggt ggaatatcaaa ggccggagaac ccaagagcgc agacaagacc 840
 cacacctgtc tccctatgtc tgcctcagaa gctgcaggcg gcccttcctg gtttctgttc 900
 cctccaaagc ctaaggacac cctgatgatc agccgggacac ctgaagtgcac ctgcgtggtg 960
 gtggatgtgt cccacgagga tcccgaaagt aagttcaatt ggtacgtgga cggcgtggaa 1020
 gtgcacaaag ccaagaccaa gcctagagag gaacagtaca acagcaccta cagagtgggt 1080
 tctgtgtgta cagtgtgtga ccaggactgg ctgaacggca aagagtacaa gtgcaagggt 1140
 tccaacaagg cctgcctgcg tcctatcgag aaaaccatca gcaaggccaa gggccagcct 1200
 agggaaaccc aggtttacac cctgcctcca tgcagggatg agctgaccaa gaaccagggt 1260
 tccctgtggt gcctggttaa gggcttctac cctccgata tcgcccgtga atgggagagc 1320
 aatggccagc cagagaacaa ctacaagaca acccctcctg tgcctggaact cgacggctca 1380
 ttcttctgt acagcaagct gaccgtggac aagagcagat ggcagcagg caactgtgtc 1440
 agctgcagcg tgatgcacga ggcctgcac aaccactaca cacagaagtc cctgtctctg 1500
 agccccgcca aa 1512

SEQ ID NO: 192 moltype = AA length = 504
 FEATURE Location/Qualifiers
 REGION 1..504
 note = Full-length amino acid sequence of NYC-0014-HC-k
 source 1..504
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 192
 MKHLWFFLLL VAAPRWVLSG EVQLVESGGG VVQPGGSLRL SCEASGFIFD DYAMHWVRQA 60
 PGKGLEWVSL ISGDDIIYY ADSVKGRFTI SRDNNKNSLY LQMNSLIIED TALYYCAKDW 120
 VFGVVMTHYW YFGLDVWQGG TTVTVSSGGG GSGGGGSGGG GSDIQMTQSP SSLASSEGDR 180
 VTITCRASQS ISTYLNWYQQ KPGKAPKLLI YGASSLQSGV PSRFSGSGSG TDFTLTISSL 240
 QPEDFATYYC QQSSTPPIIT FGQGTKVEIK GGEPKSADKT HTCPCCPAPE AAGGPSVFLF 300
 PPKPKDTLMI SRTPEVTCVV VDVSHEDPEV KFNWYVDGVE VHNAKTKPRE EQYNSTYRVV 360
 SVLTVLHQDW LNGKEYCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP CRDELTKNQV 420
 SLWCLVKGFY PSDIAVEWES NGQPENNYKT TPPVLDSGGS FFLYSLKLTVD KSRWQQGNVF 480
 SCSVMHEALH NHYTQKSLSL SPGK 504

SEQ ID NO: 193 moltype = DNA length = 1509
 FEATURE Location/Qualifiers
 misc_feature 1..1509
 note = Full-length nucleotide sequence of NYC-0015-HC-k
 source 1..1509

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mol_type = other DNA
organism = synthetic construct

SEQUENCE: 193
atgaagcacc tgtggttctt tctgctgctg gtggccgctc ctataggggt gctgtctggc 60
gaagtgcagc tgggtggaatc tgggtggcgga gttgttcagc ctggcggctc tctgagactg 120
tcttgtgcgc ccagcggctt cacttcgcac gattatgcca tgcactgggt ccgacaggcc 180
cctggcaaa gacttgaatg ggtgtccctg atctctggcg gaggcggcgg aacctactac 240
agcgattctg tgaagggcag attcaccatc agccgggaca ccagcaagga cagcctgtac 300
ctgcagatga acagcctgcg gaccgaggac acagccctgt actactgcgc caaggacatg 360
gtgttcggcg tggtcacccc ttactactac ttcgccctgg atgtgtgggg ccagggcaca 420
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aactgtcttc catgtctctg tccagaagct gcaggcggcc ctcccggtt tctgttccct 900
ccaaagccta aggaacacct gatgatcagc cggacacctg aagtgcacct cgtggtggtg 960
gatgtgtccc acgagatcc cgaagtgaag ttcaattggt acgtggacgg cgtggaagtg 1020
cacaacgcca agaccaagcc tagagaggaa cagtacaaca gcacctacag agtggtgtct 1080
gtgtgcagc tgcctgcaca ggactggctg aacggcaaa agtacaagt caaggtgttc 1140
aacaaggccc tggctgtctc tatcgagaaa accatcagca aggccaaagg ccagcctagg 1200
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ttcctgtaca gcaagctgac cgtggacaag agcagatggc agcagggcaa cgtgttcagc 1440
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cccgcaaaa                                     1509

SEQ ID NO: 194      moltype = AA length = 503
FEATURE            Location/Qualifiers
REGION              1..503
note = Full-length amino acid sequence of NYC-0015-HC-k
source              1..503
mol_type = protein
organism = synthetic construct

SEQUENCE: 194
MKHLWFFLLL VAAPRWVLSG EVQLVESGGG VQPGGSLRL SCAASGFTFD DYAMHWVRQA 60
PGKLEWVSL ISGGGGTTY SDSVKGRFTI SRDTSKDSLY LQMNSLRLED TALYYCAKDM 120
VFGVVTPIYY FALDVWGQGT TVTVSSGGGG SGGGGSGGGG SDIQMTQSPS SLSASVGDRV 180
TITCRASQSI NSYLNWYQOK PGKAPKLLIY AASSLQSGVP SRFGSGSGST DFTLTISLQ 240
PEDFATYYCQ QSYSAPIITF GQGTRLEIKG GEPKSADKTH TCPPCPAPEA AGGPSVFLFP 300
PKPKDTLMIS RTPEVTCVVV DVSHEDPEVK FNWYVDGVEV HNAKTKPRE QYNSTYRVVS 360
VLTVLHQDWL NGKEYKCKVS NKALPAPIEK TISKAKGQPR EPQVYTLPPC RDELTKNQVS 420
LWCLVKGFYP SDIAVEWESN GQPENNYKTT PPVLDSGGSF FLYSKLTVDK SRWQQGNVFS 480
CSVMHEALHN HYTKSLSLSL PGK                                     503

SEQ ID NO: 195      moltype = DNA length = 1512
FEATURE            Location/Qualifiers
misc_feature        1..1512
note = Full-length nucleotide sequence of NYC-0016-HC-k
source              1..1512
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 195
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gccgattctg tgaagggcag attcaccatc agccgggaca ccagcaagaa cagcctgtac 300
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aagcccgga agggccctaa actgctgata tatgccgctt ccagtctgca gagcggagtg 660
cctagcagat tttctggcag cggctccggc accgatttca cctgaccatt atctagcctg 720
cagcctgagg acttcgccac ctactactgc cagcagagct acagacccc tcctatcacc 780
tttgccagg gcaccagact ggaatcaaaa ggcggagaac ccaagagcgc agacaagacc 840
cacacctgtc ctccatgtcc tgctccagaa gctgcaggcg gcccttccgt gttctgttc 900
cctccaaagc ctaaggacac cctgatgata agccggacac ctgaagtgtc ctgctggtgt 960
gtggatgtgt cccacgagga tcccgaaagt aagttcaatt ggtacgtgga cggcgtggaa 1020
gtgcacaacg ccaagaccaa gcctagagag gaacagtaca acagcaccta cagagtgggt 1080
tctgtgctga cagtgtctga ccaggactcg ctgaacggca aagagtacaa gtgcaagggt 1140
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-continued

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aggggaaccccc aggttttacac cctgcctcca tgcaggggatg agctgaccaa gaaccagggtg 1260
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aatggccagc cagagaacaa ctacaagaca acccctcctg tgcctggactc cgacggctca 1380
ttcttctctgt acagcaagct gaccgtggac aagagcagat ggcagcaggg caacgtgttc 1440
agctgcagcg tgatgcacga ggcctgcac aaccactaca cacagaagtc cctgtctctg 1500
agccccgcga aa 1512

SEQ ID NO: 196      moltype = AA length = 504
FEATURE            Location/Qualifiers
REGION             1..504
                    note = Full-length amino acid sequence of NYC-0016-HC-k
source             1..504
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 196
MKHLWFFLLL VAAPRWVLSG EVQLVESGGG VVQPGGSLRL SCAVSGFTFD DYAMHWVRQA 60
PGKGLEWVSL ISGDDGSTHY ADSVKGRFTI SRDNSKNSLY LQMNSLRTGD TALYYCAKDM 120
IFAVITDYH YYGMDVWQGG TTVTVSSGGG GSGGGSGGGG GSDIQMTQSP SLSASVGD 180
VTITCRASQS ISSYLNWYQQ KPGKAPKLLI YAASSLQSGV PSRFSGSGSG TDFTLTISSL 240
QPEDFATYYC QSYSTPPIT FGQGRLEIK GGEPKSADKT HTCPPCPAPE AAGGSPVFLF 300
PPKPKDTLMI SRTPEVTCVV VDVSHEDPEV KFNWYVDGVE VHNATKPRE EQYNSTYRVV 360
SVLTVLHQDW LNGKEYKCKV SNKALPAPIE KTISKAKGQP REPQVYTLPP CRDELTKNQV 420
SLWCLVKGFP PSDIAVEWES NGQPENNYKT TPPVLDSDGS PFLYSKLTVD KSRWQQGNVF 480
SCSVMEALH NHYQKSLSL SPGK 504

SEQ ID NO: 197      moltype = AA length = 719
FEATURE            Location/Qualifiers
REGION             1..719
                    note = Full-length amino acid sequence of NYZ-1007-HC2
source             1..719
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 197
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PGKGLEWVAT ISYDGSQKYY ADSVKGRFTI SRDESNTLY LQMNSLRAED TAVYHCARG 120
SGHYEAFDIW GQGLTVTVSS GGGGSGGGGS GGGGQSVLT QPPSASGTPG QRVTISSCGS 180
SSNIGNWYVS WYQQLPGTAP KLLIYDNNKR PSGVPDRFSG SKSGTSASLA ISGLQSEDEA 240
DYYCGTWDS LSAWVFGGG TKLTVLGGGG SEVQLVESGG GLVQPGGSLR LSCAASGVTF 300
NYYGMSWIRQ APGKLEWVA SITRSGGRIY YPDSVKGRFT ISRENTQKTL YLQMNSLRAE 360
DTAVYYCTLD GRDGVVAYWG QGTLTVTVSSA STKGPSVFLP APSKSTSGG TAALGCLVKD 420
YFPEPVTVSW NSGALTSGVH TFPVQLQSSG LYSLSVTVV PSSSLGTQTY ICNVNHKPSN 480
TKVDKRVEPK SCDKTHCTPP CPAPEAAGGP SVFLFPPKPK DTLMISTRPE VTCVVVSVSH 540
EDPEVKFNWY VDGVEVHNAK TKPREEQYNS TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL 600
PAPIEKTISK AKGQPREPQV YVYPPSRDEL TKNQVSLTCL VKGFYPSDIA VEWESNGQPE 660
NNYKTPPVVL DSDGSFALVS KLTVDKSRWQ QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 719

SEQ ID NO: 198      moltype = AA length = 719
FEATURE            Location/Qualifiers
REGION             1..719
                    note = Full-length amino acid sequence of NYZ-1017-HC2
source             1..719
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 198
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SGHYEAFDIW GQGLTVTVSS GGGGSGGGGS GGGGQSVLT QPPSVSAAPG QKVTISSCGS 180
SSNIGNWYVS WYQQLPGTAP KLLIYDNNKR PSGIPDRFSG SKSGTSATLG ITGLQTGDEA 240
DYYCGTWDS LSAWVFGGG TKVTVLGGGG SEVQLVESGG GLVQPGGSLR LSCAASGVTF 300
NYYGMSWIRQ APGKLEWVA SITRSGGRIY YPDSVKGRFT ISRENTQKTL YLQMNSLRAE 360
DTAVYYCTLD GRDGVVAYWG QGTLTVTVSSA STKGPSVFLP APSKSTSGG TAALGCLVKD 420
YFPEPVTVSW NSGALTSGVH TFPVQLQSSG LYSLSVTVV PSSSLGTQTY ICNVNHKPSN 480
TKVDKRVEPK SCDKTHCTPP CPAPEAAGGP SVFLFPPKPK DTLMISTRPE VTCVVVSVSH 540
EDPEVKFNWY VDGVEVHNAK TKPREEQYNS TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL 600
PAPIEKTISK AKGQPREPQV YVYPPSRDEL TKNQVSLTCL VKGFYPSDIA VEWESNGQPE 660
NNYKTPPVVL DSDGSFALVS KLTVDKSRWQ QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 719

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1. A polynucleotide comprising a nucleotide sequence encoding a bispecific antibody that comprises a site which binds specifically to human HLA/NY-ESO and a site which binds specifically to human CD3, wherein the site which binds specifically to human HLA/NY-ESO comprises:

a heavy chain CDRH1 consisting of the amino acid sequence as shown in SEQ ID NO: 54,

a heavy chain CDRH2 consisting of the amino acid sequence as shown in SEQ ID NO: 55,

a heavy chain CDRH3 consisting of the amino acid sequence as shown in SEQ ID NO: 56,

a light chain CDRL1 consisting of an amino acid sequence derived from the amino acid sequence as shown in SEQ ID NO: 57 in which amino acid 7 is W,

- a light chain CDRL2 consisting of the amino acid sequence represented by DNN, and
- a light chain CDRL3 consisting of the amino acid sequence as shown in SEQ ID NO: 59.

2. The polynucleotide according to claim 1, wherein the site which binds specifically to human HLA/NY-ESO comprises:

- a heavy chain variable region consisting of an amino acid sequence having 95% or higher sequence identity to an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 53 and a light chain variable region consisting of an amino acid sequence having 95% or higher sequence identity to an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 53, or

- a heavy chain variable region consisting of an amino acid sequence having 95% or higher sequence identity to an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 156 and a light chain variable region consisting of an amino acid sequence having 95% or higher sequence identity to an amino acid sequence of amino acids 161 to 271 of the amino acid sequence as shown in SEQ ID NO: 156.

3. The polynucleotide according to claim 1, wherein the site which binds specifically to human HLA/NY-ESO comprises:

- a heavy chain variable region consisting of an amino acid sequence having 98% or higher sequence identity to an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 53 and a light chain variable region consisting of an amino acid sequence having 98% or higher sequence identity to an amino acid sequence of amino acids 156 to 266 of the amino acid sequence as shown in SEQ ID NO: 53, or

- a heavy chain variable region consisting of an amino acid sequence having 98% or higher sequence identity to an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 156 and a light chain variable region consisting of an amino acid sequence having 98% or higher sequence identity to an amino acid sequence of amino acids 161 to 271 of the amino acid sequence as shown in SEQ ID NO: 156.

4. The polynucleotide according to claim 1 wherein the bispecific antibody comprises:

- a heavy chain variable region consisting of an amino acid sequence derived from an amino acid sequence as shown in SEQ ID NO: 53 by substitution, deletion, addition or insertion of one or more amino acids and a light chain variable region consisting of an amino acid sequence derived from an amino acid sequence as shown in SEQ ID NO: 53 by substitution, deletion, addition or insertion of one or more amino acids, or

- a heavy chain variable region consisting of an amino acid sequence derived from an amino acid sequence of amino acids 21 to 140 of the amino acid sequence as shown in SEQ ID NO: 156 by substitution, deletion, addition or insertion of one or more amino acids and a light chain variable region consisting of an amino acid sequence derived from an amino acid sequence of amino acids 161 to 271 of the amino acid sequence as

shown in SEQ ID NO: 156 by substitution, deletion, addition or insertion of one or more amino acids.

5. The polynucleotide according to claim 1, wherein the site which binds specifically to human HLA/NY-ESO is scFv.

6. The polynucleotide according to claim 1, wherein the site which binds specifically to human CD3 comprises:

- (CCH1) a heavy chain CDRH1 consisting of the amino acid sequence as shown in SEQ ID NO: 141;

- (CCH2) a heavy chain CDRH2 consisting of the amino acid sequence as shown in SEQ ID NO: 142 or a heavy chain CDRH2 consisting of an amino acid sequence derived from the amino acid sequence as shown in SEQ ID NO: 142 in which amino acid 3 is N or S;

- (CCH3) a heavy chain CDRH3 consisting of the amino acid sequence as shown in SEQ ID NO: 143;

- (CCL1) a light chain CDRL1 consisting of the amino acid sequence as shown in SEQ ID NO: 144;

- (CCL2) a light chain CDRL2 consisting of the amino acid sequence represented by RDD or a light chain CDRL2 consisting of an amino acid sequence derived from the amino acid sequence represented by RDD in which amino acid 2 is N; and

- (CCL3) a light chain CDRL3 consisting of the amino acid sequence as shown in SEQ ID NO: 146.

7. The polynucleotide according to claim 1, wherein the site which binds specifically to human CD3 comprises:

- (CH1) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 136,

- (CH2) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 137,

- (CH3) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 147,

- (CH4) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 138,

- (CH5) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 139,

- (CH6) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 140,

- (CH7) a heavy chain variable region consisting of an amino acid sequence of amino acids 272 to 389 of the amino acid sequence as shown in SEQ ID NO: 155,

- (CH8) a heavy chain variable region consisting of an amino acid sequence of amino acids 277 to 394 of the amino acid sequence as shown in SEQ ID NO: 156; or

- (CH9) a heavy chain variable region consisting of an amino acid sequence of amino acids 277 to 394 of the amino acid sequence as shown in SEQ ID NO: 157; and

- (CL1) a light chain variable region consisting of an amino acid sequence of amino acids 135 to 243 of the amino acid sequence as shown in SEQ ID NO: 136,

- (CL2) a light chain variable region consisting of an amino acid sequence of amino acids 135 to 241 of the amino acid sequence as shown in SEQ ID NO: 137,

- (CL3) a light chain variable region consisting of an amino acid sequence of amino acids 135 to 243 of the amino acid sequence as shown in SEQ ID NO: 147,

- (CL4) a light chain variable region consisting of an amino acid sequence of amino acids 135 to 241 of the amino acid sequence as shown in SEQ ID NO: 138;
- (CL5) a light chain variable region consisting of an amino acid sequence of amino acids 135 to 243 of the amino acid sequence as shown in SEQ ID NO: 139;
- (CL6) a light chain variable region consisting of an amino acid sequence of amino acids 135 to 243 of the amino acid sequence as shown in SEQ ID NO: 140;
- (CL7) a light chain variable region consisting of an amino acid sequence of amino acids 405 to 511 of the amino acid sequence as shown in SEQ ID NO: 155;
- (CL8) a light chain variable region consisting of an amino acid sequence of amino acids 410 to 516 of the amino acid sequence as shown in SEQ ID NO: 156; or
- (CL9) a light chain variable region consisting of an amino acid sequence of amino acids 410 to 516 of the amino acid sequence as shown in SEQ ID NO: 157.
- 8.** The polynucleotide according to claim 1, wherein the site which binds specifically to human CD3 comprises:
- (CH1CL1) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 136 and a light chain variable region consisting of an amino acid sequence of amino acids 135 to 243 of the amino acid sequence as shown in SEQ ID NO: 136;
- (CH2CL2) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 137 and a light chain variable region consisting of an amino acid sequence of amino acids 135 to 241 of the amino acid sequence as shown in SEQ ID NO: 137;
- (CH3CL3) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 147 and a light chain variable region consisting of an amino acid sequence of amino acids 135 to 243 of the amino acid sequence as shown in SEQ ID NO: 147;
- (CH4CL4) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 138 and a light chain variable region consisting of an amino acid sequence of amino acids 135 to 241 of the amino acid sequence as shown in SEQ ID NO: 138;
- (CH5CL5) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 139 and a light chain variable region consisting of an amino acid sequence of amino acids 135 to 243 of the amino acid sequence as shown in SEQ ID NO: 139;
- (CH6CL6) a heavy chain variable region consisting of an amino acid sequence of amino acids 2 to 119 of the amino acid sequence as shown in SEQ ID NO: 140 and a light chain variable region consisting of an amino acid sequence of amino acids 135 to 243 of the amino acid sequence as shown in SEQ ID NO: 140;
- (CH7CL7) a heavy chain variable region consisting of an amino acid sequence of amino acids 272 to 389 of the amino acid sequence as shown in SEQ ID NO: 155 and a light chain variable region consisting of an amino acid sequence of amino acids 405 to 511 of the amino acid sequence as shown in SEQ ID NO: 155;
- (CH8CL8) a heavy chain variable region consisting of an amino acid sequence of amino acids 277 to 394 of the amino acid sequence as shown in SEQ ID NO: 156 and a light chain variable region consisting of an amino acid sequence of amino acids 410 to 516 of the amino acid sequence as shown in SEQ ID NO: 156; or
- (CH9CL9) a heavy chain variable region consisting of an amino acid sequence of amino acids 277 to 394 of the amino acid sequence as shown in SEQ ID NO: 157 and a light chain variable region consisting of an amino acid sequence of amino acids 410 to 516 of the amino acid sequence as shown in SEQ ID NO: 157.
- 9.** The polynucleotide according to claim 1, wherein the site which binds specifically to human CD3 is Fab.
- 10.** The polynucleotide according to claim 1, wherein the bispecific antibody comprises: a first polypeptide comprising scFv that binds specifically to human HLA/NY-ESO, a variable region of an antibody that binds specifically to human CD3, and an Fc region (i) in that order from the N terminus toward the C terminus, a second polypeptide comprising a hinge region and an Fc region (ii) of the immunoglobulin, and a third polypeptide comprising an antibody light chain consisting of a variable region and a constant region, wherein the first polypeptide is associated with the second polypeptide at the Fc region (i) and the Fc region (ii), and the first polypeptide is associated with the third polypeptide in the heavy chain variable region and the constant region CH1 of the antibody.
- 11.** The polynucleotide according to claim 10, wherein the first polypeptide comprises an amino acid sequence of amino acids 21 to 394 of the amino acid sequence as shown in SEQ ID NO: 160.
- 12.** The polynucleotide according to claim 10, wherein the first polypeptide comprises the amino acid sequence according to (i) or (ii) below:
- (i) an amino acid sequence of amino acids 20 to 724 of the amino acid sequence as shown in SEQ ID NO: 160; and
- (ii) an amino acid sequence of amino acids 20 to 719 of the amino acid sequence as shown in SEQ ID NO: 197.
- 13.** The polynucleotide according to claim 10, wherein the second polypeptide comprises an amino acid sequence of amino acids 20 to 246 of the amino acid sequence as shown in SEQ ID NO: 84.
- 14.** The polynucleotide according to claim 10, wherein the third polypeptide comprises an amino acid sequence of amino acids 21 to 127 of the amino acid sequence as shown in SEQ ID NO: 161.
- 15.** A vector comprising the polynucleotide according to claim 1.
- 16.** A host cell comprising the polynucleotide according to claim 1 or a vector comprising the polynucleotide according to claim 1.
- 17.** A vector comprising the polynucleotide according to claim 10.
- 18.** A host cell comprising the polynucleotide according to claim 10 or a vector comprising the polynucleotide according to claim 10.
- 19.** A method for producing a bispecific antibody that binds specifically to human HLA/NY-ESO and human CD3 comprising: (i) a step of culturing the host cell according to claim 16; and (ii) a step of purifying an antibody or a binding fragment thereof from the culture product obtained in the step (i).

20. A bispecific antibody that binds specifically to human HLA/NY-ESO and human CD3, which is obtained by the method according to claim **19**.

* * * * *